Pre-Calculus 12 safe working copy

1. Compared to the graph of the base function, the graph of the function is translated 9 units down. Rendered ExamView reference:

$$f(x) = x^2$$

$$g(x) = x^2 - 9$$

(A) 9 units down (B) 9 units up (C) 9 units left (D) 9 units right

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

2. Compared to the graph of the base function, the graph of the function is translated 5 units down. Rendered ExamView reference:

$$f(x) = x^2$$

$$g(x) = x^2 - 9$$

(A) 5 units down (B) 5 units up (C) 5 units left (D) 5 units right

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

3. Compared to the graph of the base function, the graph of the function is translated 4 units right. Rendered ExamView reference:

$$f(x) = x^2$$

$$g(x) = (x-4)^2$$

(A) 4 units right (B) 4 units left (C) 4 units down (D) 4 units up

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

4. Compared to the graph of the base function, the graph of the function is translated 9 units left. Rendered ExamView reference:

$$f(x) = x^2$$

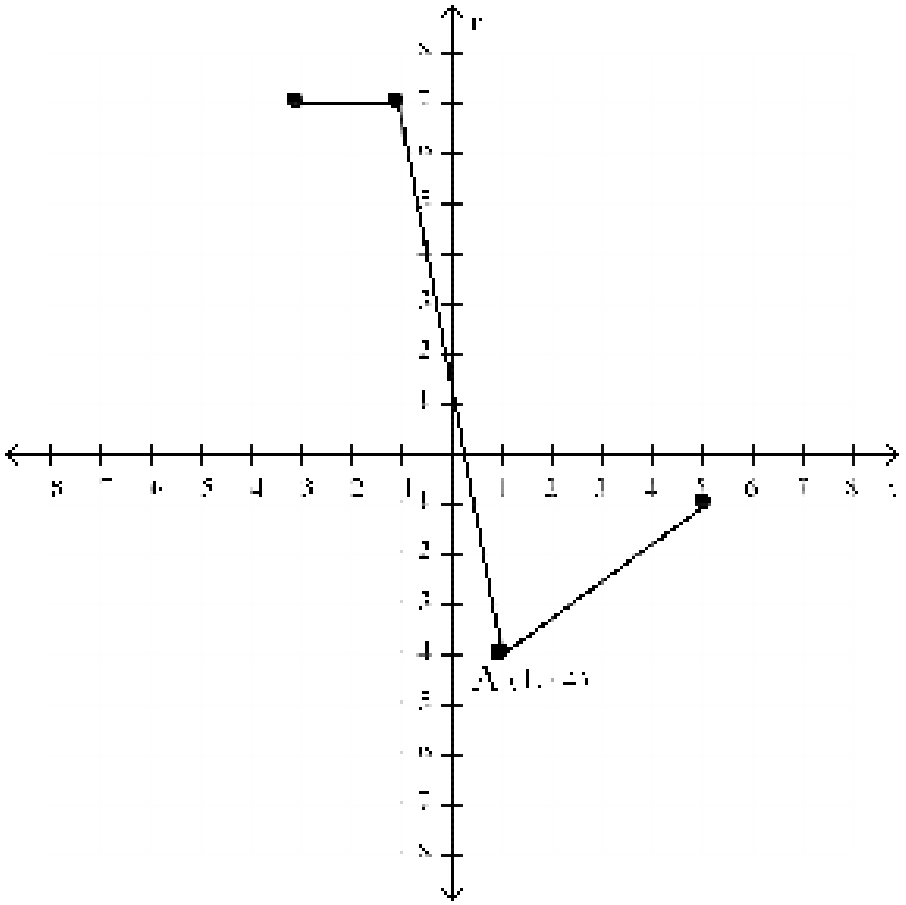
$$g(x) = (x-4)^2$$

(A) 9 units left (B) 9 units right (C) 9 units down (D) 9 units up

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

5. Given the graph of $f(x)$ shown below, what are the coordinates of point A if the transformed graph is represented by 3? 3 Rendered ExamView reference:

$$g(x) = f(x-2)$$



$(3, -4)$

$(-1, -2)$

$(-1, -4)$

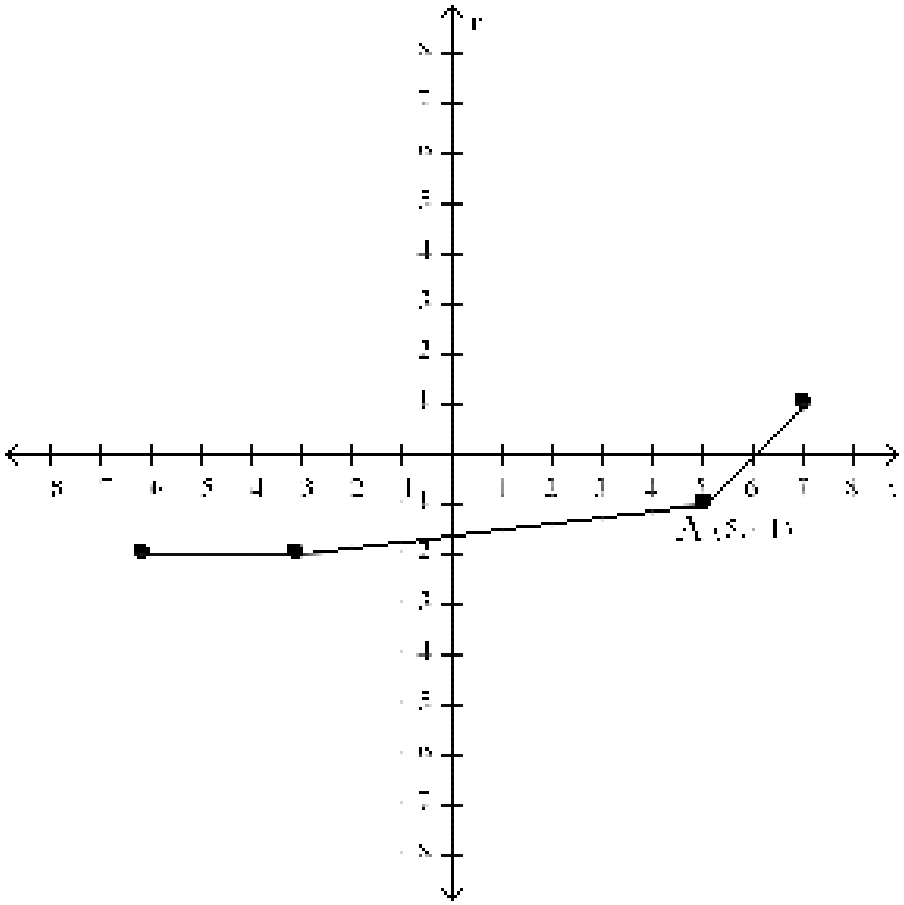
$(3, -6)$

(A) – 2 (B) – 2 (C) 3 (D) $–6$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

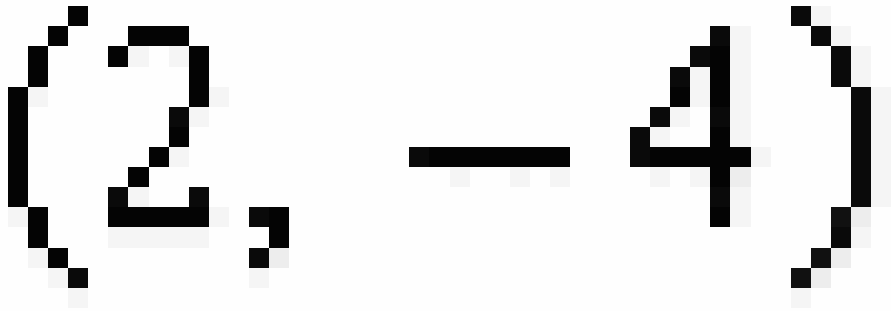
6. Given the graph of $f(x)$ shown below, what are the coordinates of point A if the transformed graph is represented by 2? $+ 3$ Rendered ExamView reference:

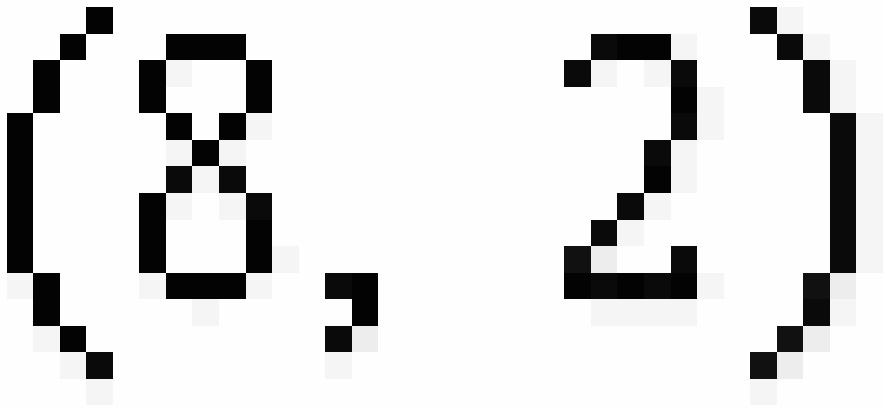
$$g(x) = f(x) + 3$$



(5, 2)

(2, −1)

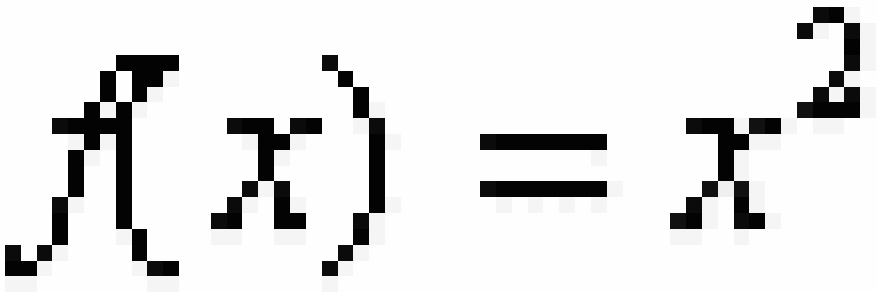


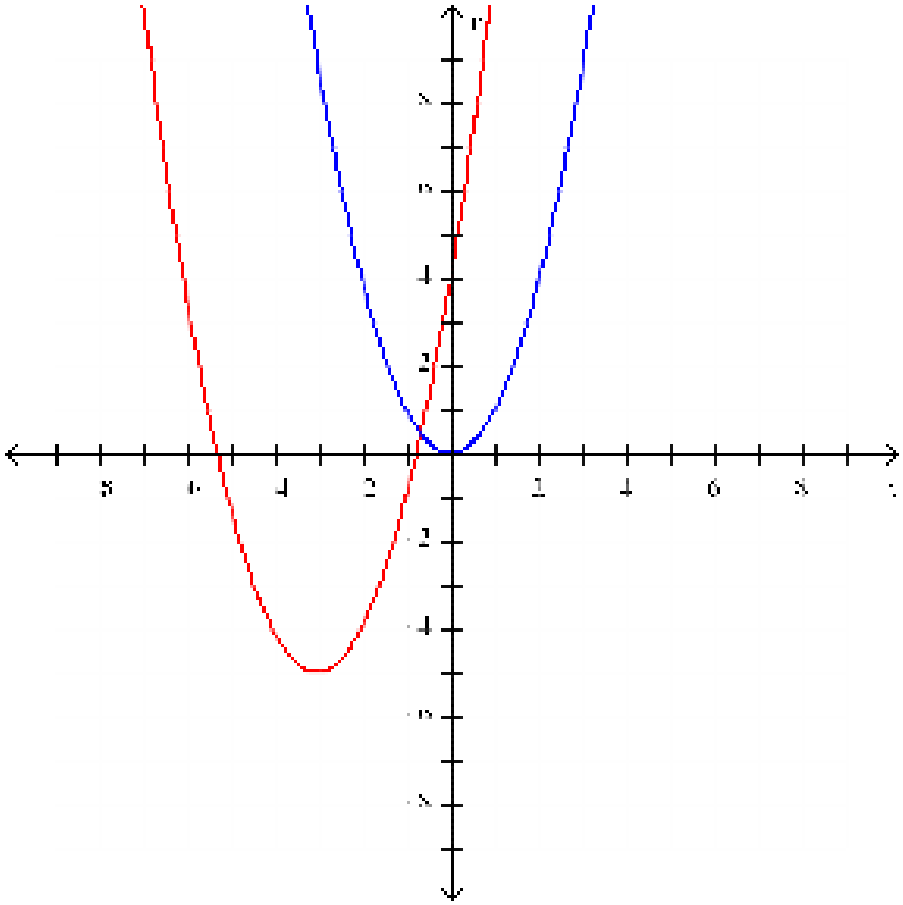


(A) 2 (B) $−4$ (C) 8 (D) 2

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

7. What is the equation of the transformed function, $g(x),$ after the transformations are applied to the graph of the base function $+ 3,$ shown in blue, to obtain the graph of $g(x),$ shown in red? – 5 Rendered ExamView reference:





$$g(x)+3=(x-5)^2$$

$$g(x)-5=(x+3)^2$$

$$g(x)=(x+3)^2-5$$

$$g(x)=(x-5)^2+3$$

(A) + 3 (B) – 5 (C) – 5 (D) + 3

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

8. The function y_axis represents a transformation that can best be described as

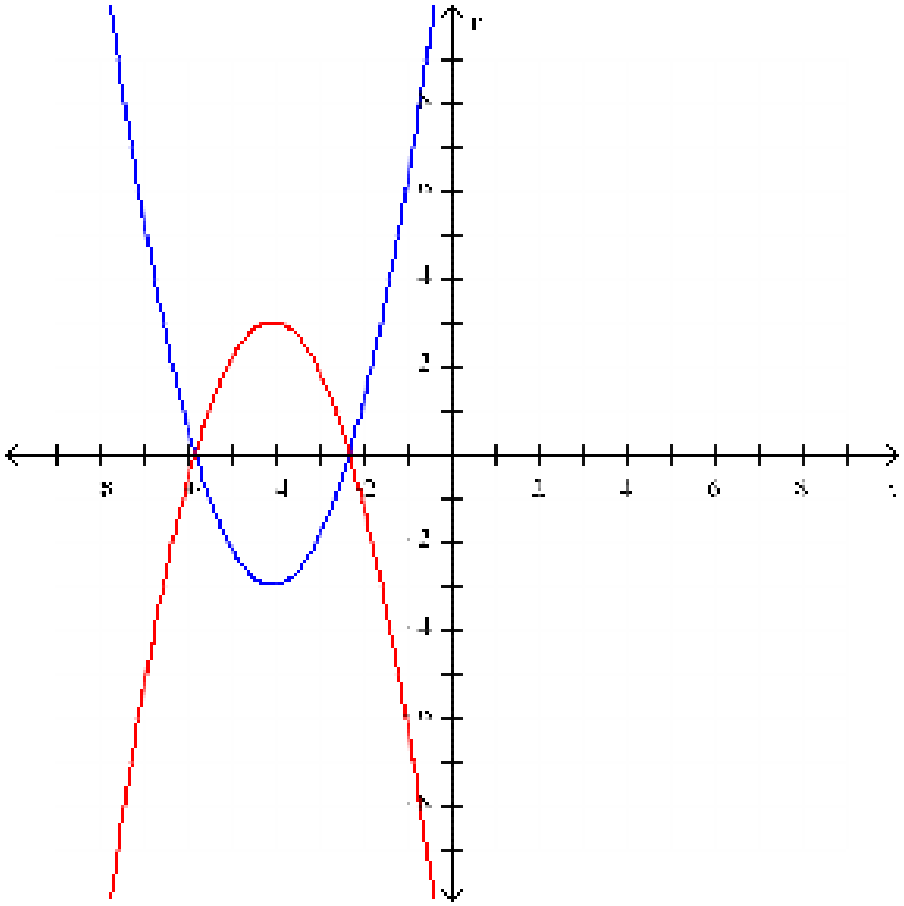
(A) a reflection in the x-axis

(B) a reflection in the y-axis

(C) a reflection in the x-axis and the y-axis

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

9. The two functions in the graph shown are reflections of each other. Select the type of reflection(s). Rendered ExamView reference:



(A) y-axis

(B) x-axis and the y-axis

(C) x-axis

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

10. When a function is reflected in the x $-$axis, the coordinates of point (x, y) become

(A) x, –y

(B) –x, –y

(C) –x, y

(D) x, y

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

11. When a $> 0,$ the function $g(x) = ax^2$ has what relationship to the base function $f(x) = x^2$?

(A) stretched vertically by a factor of |a| and reflected in the x-axis

(B) stretched horizontally by a factor of 1/|a|

(C) stretched vertically by a factor of |a|

(D) stretched horizontally by a factor of 1/|a| and reflected in the y-axis

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

12. When b $> 0,$ the function $g(x) = bx$ has what relationship to the base function $f(x) = x$?

(A) stretched vertically by a factor of |b| and reflected in the x-axis

(B) stretched vertically by a factor of |b|

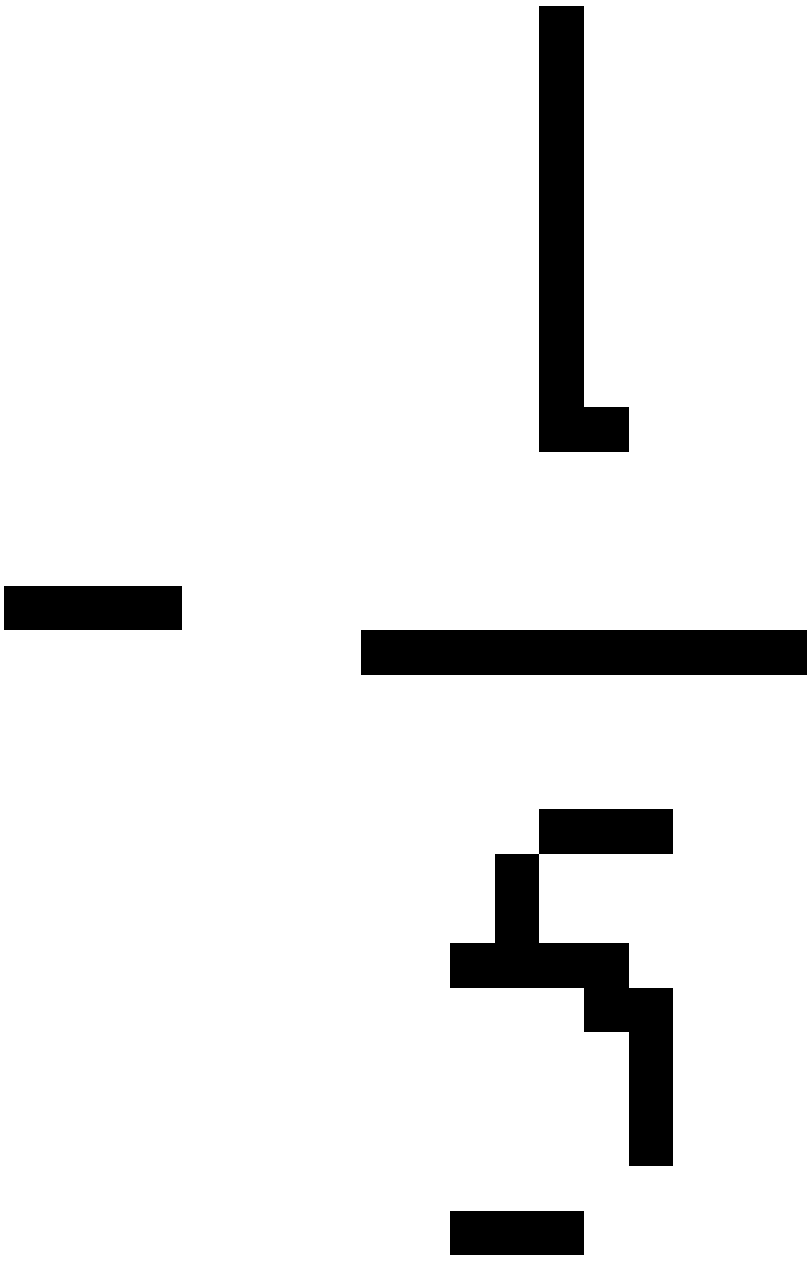
(C) stretched horizontally by a factor of 1/|b| and reflected in the y-axis

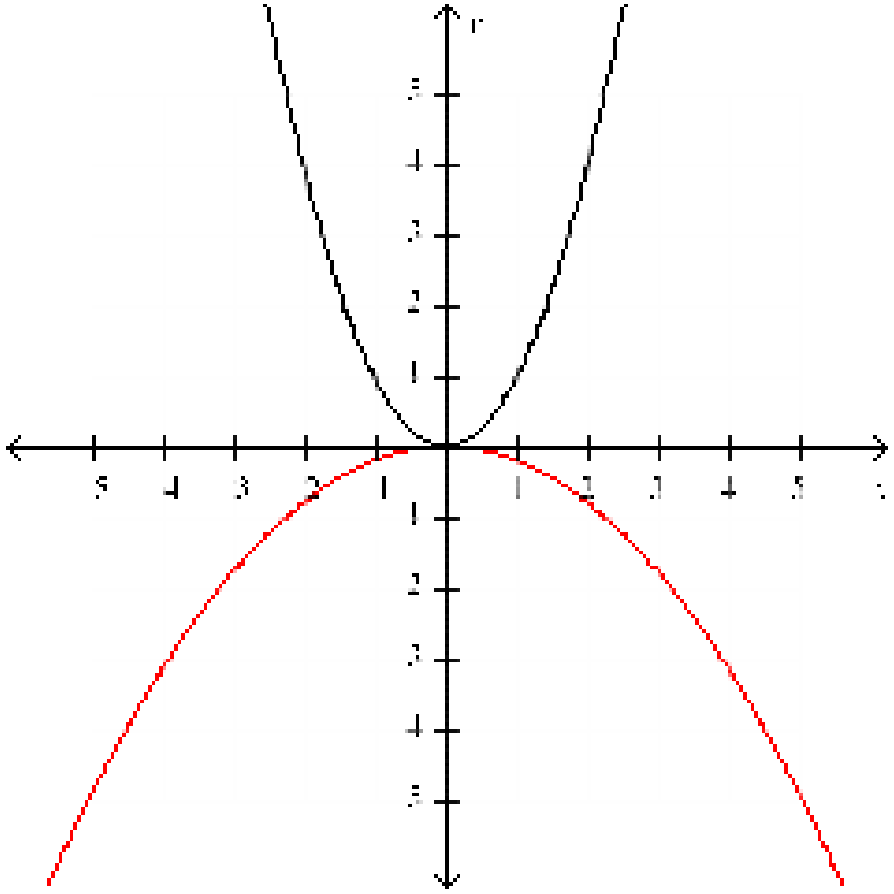
(D) stretched horizontally by a factor of 1/|b|

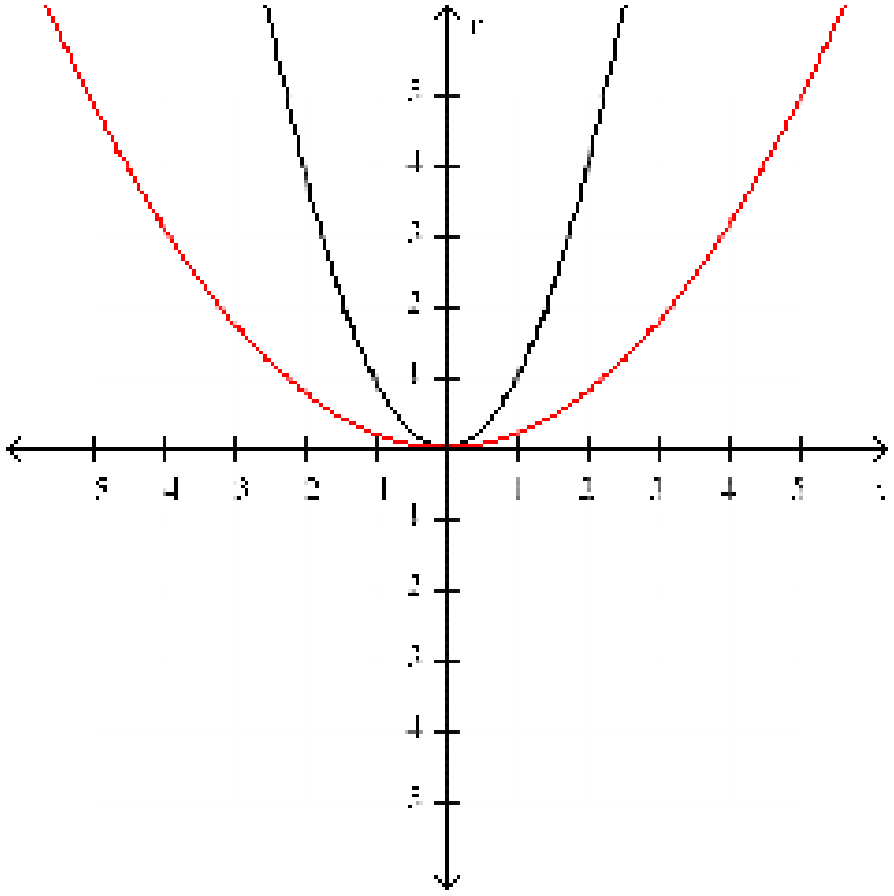
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

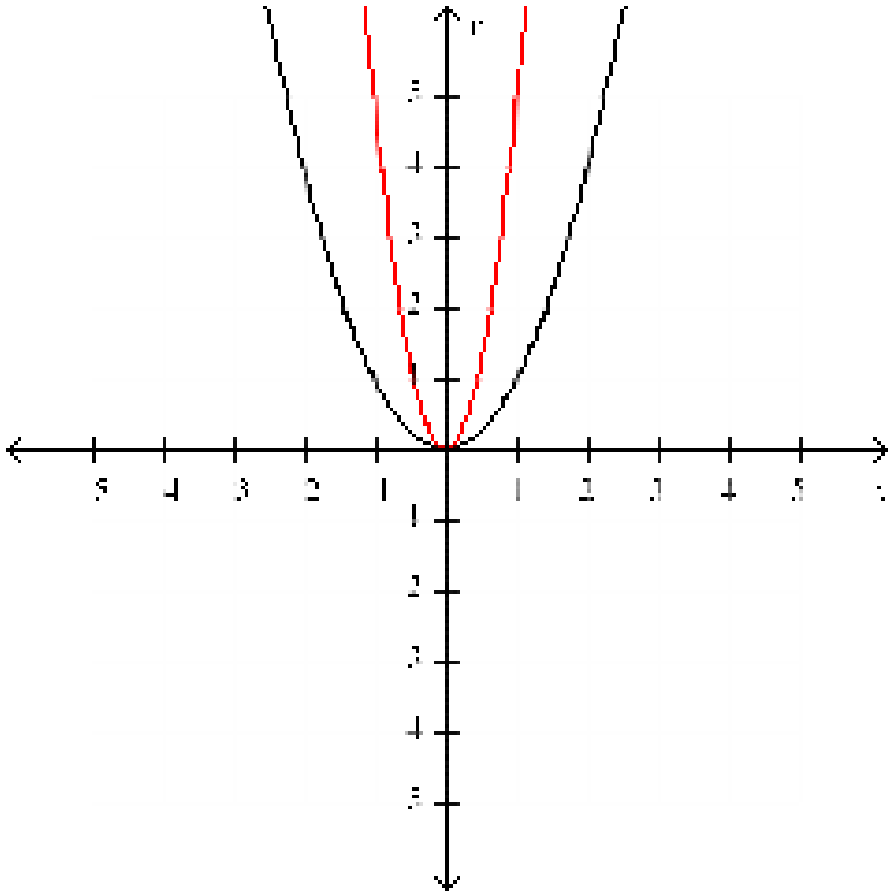
13. Which of the graphs shown below represents the base function $−1/5$ and the stretched function $g(x) = 5 x_2$? Recovered graph functions: $x^2; ((−1)/(5))*x^2; (0.2)*x^2; (5)*x^2; (−5)*x^2$. Rendered ExamView reference:

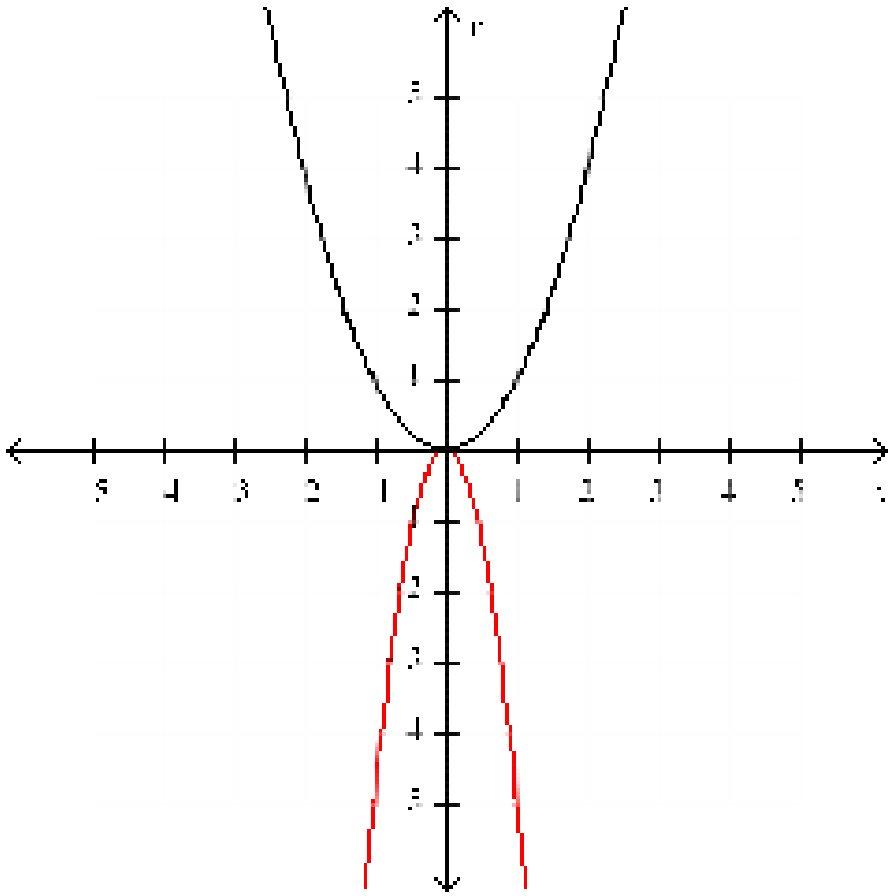
$$f(x) = x^2$$







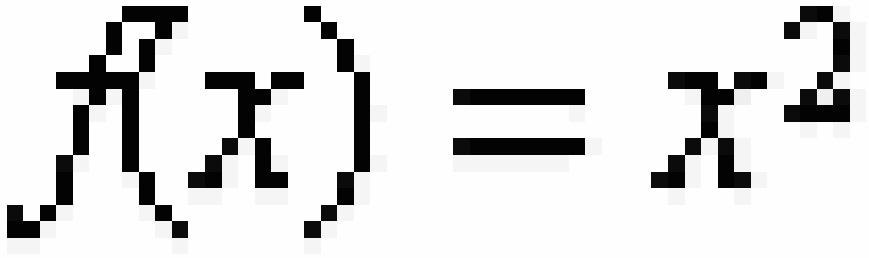


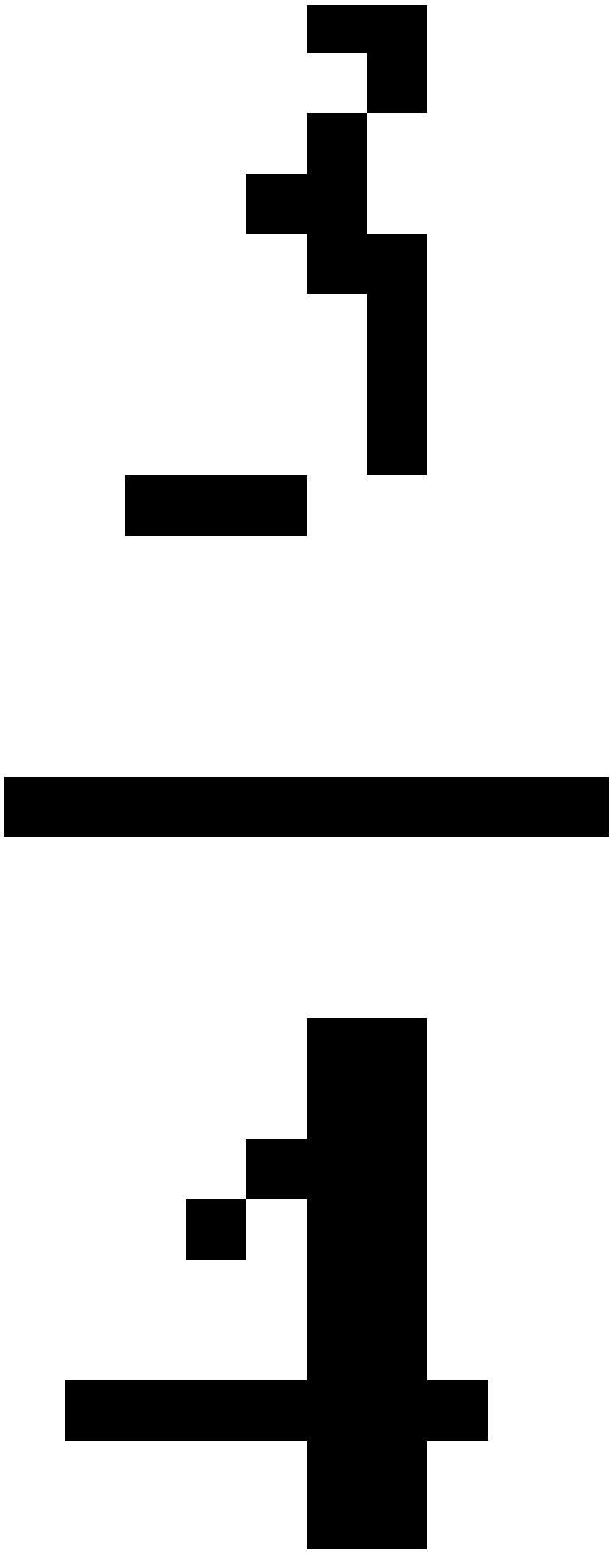


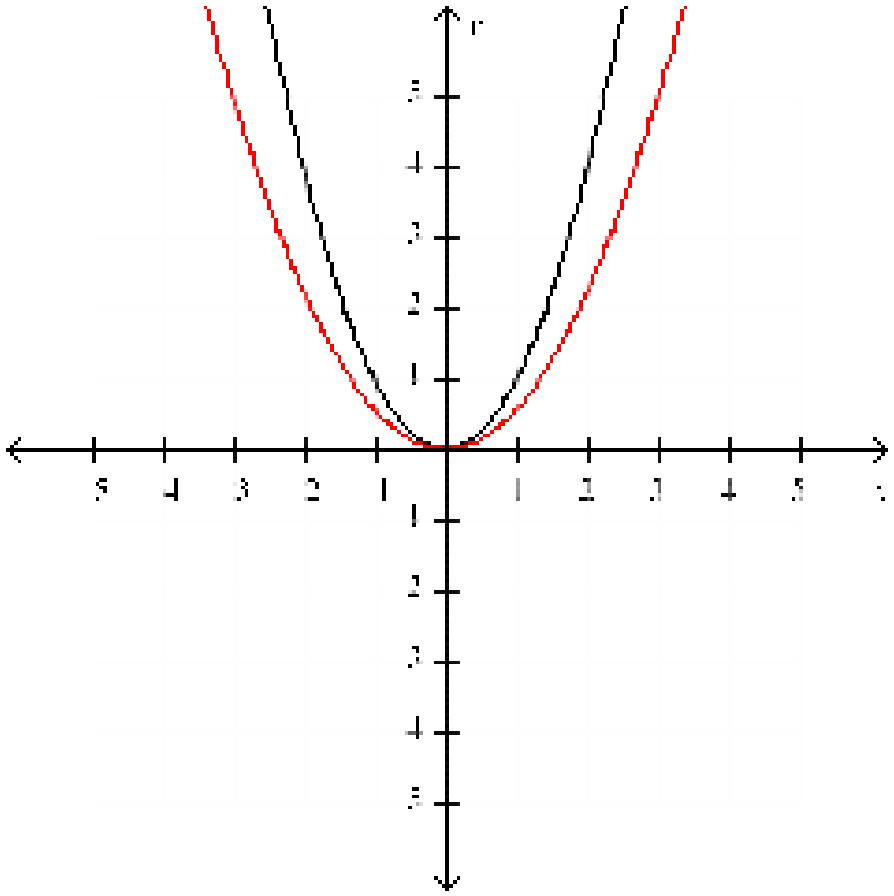
(A) 0.2 (B) -5 (C) 5

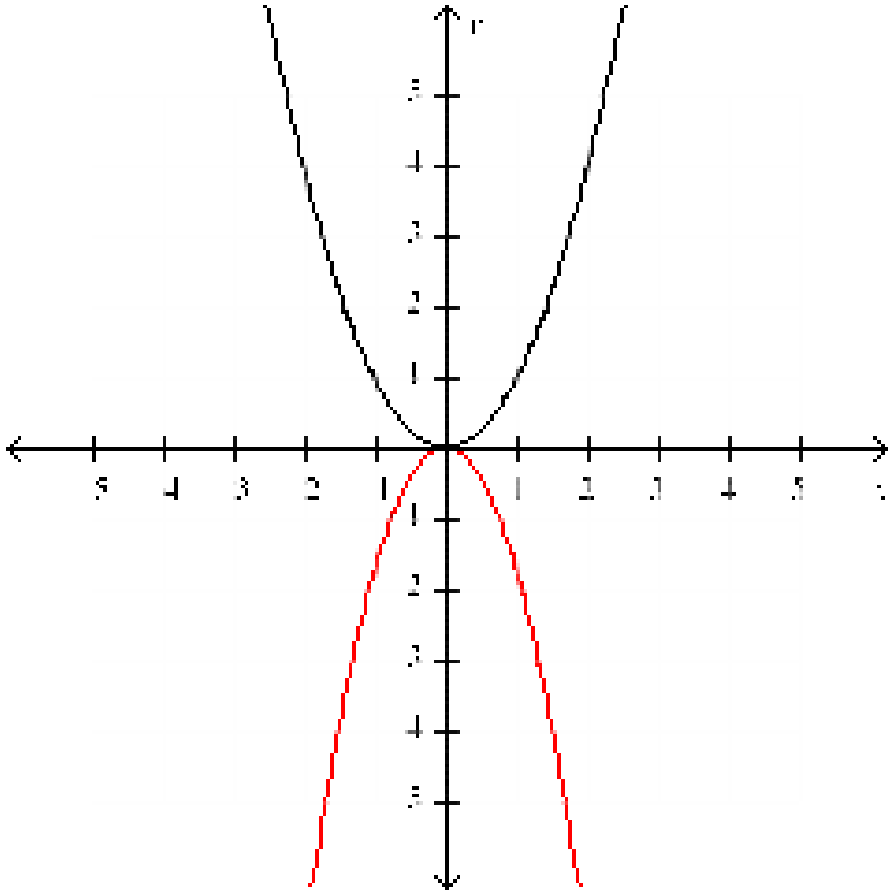
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

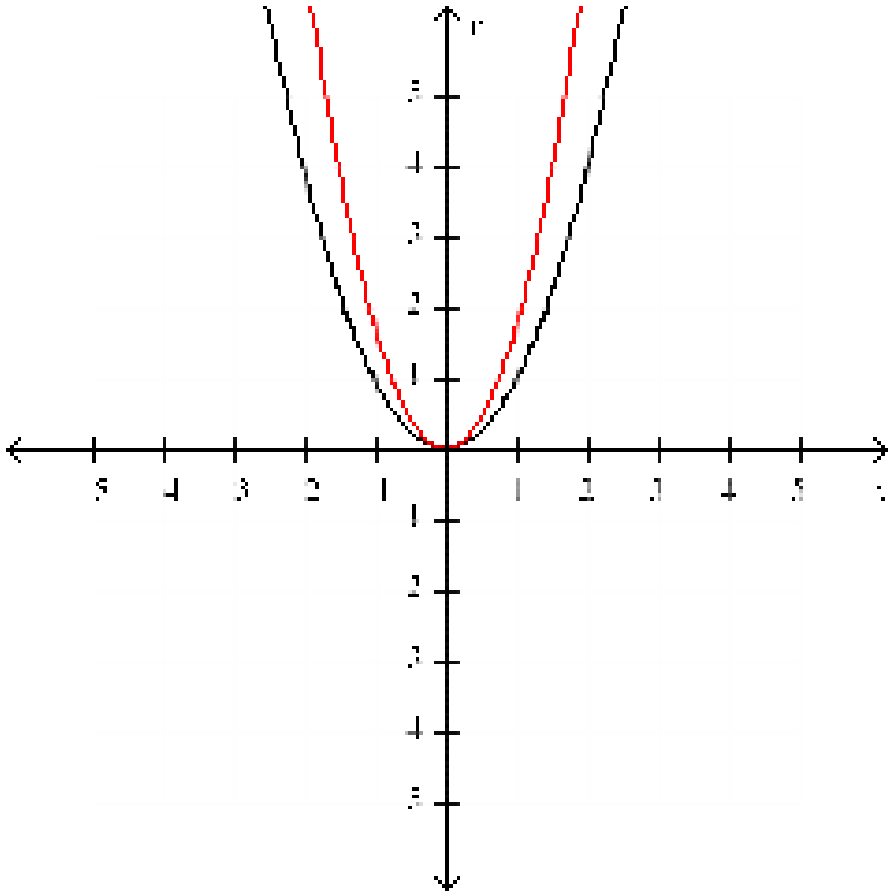
14. Which of the graphs shown below represents the base function $3/4$ and the stretched function $g(x) = (4 x)2$? Recovered graph functions: $x^2; (((3)/(4))*x)^2; −1*((−1.333333333333)*x)^2; ((1.333333333333)*x)^2; −1*((−0.75)*x)^2$. Rendered ExamView reference:

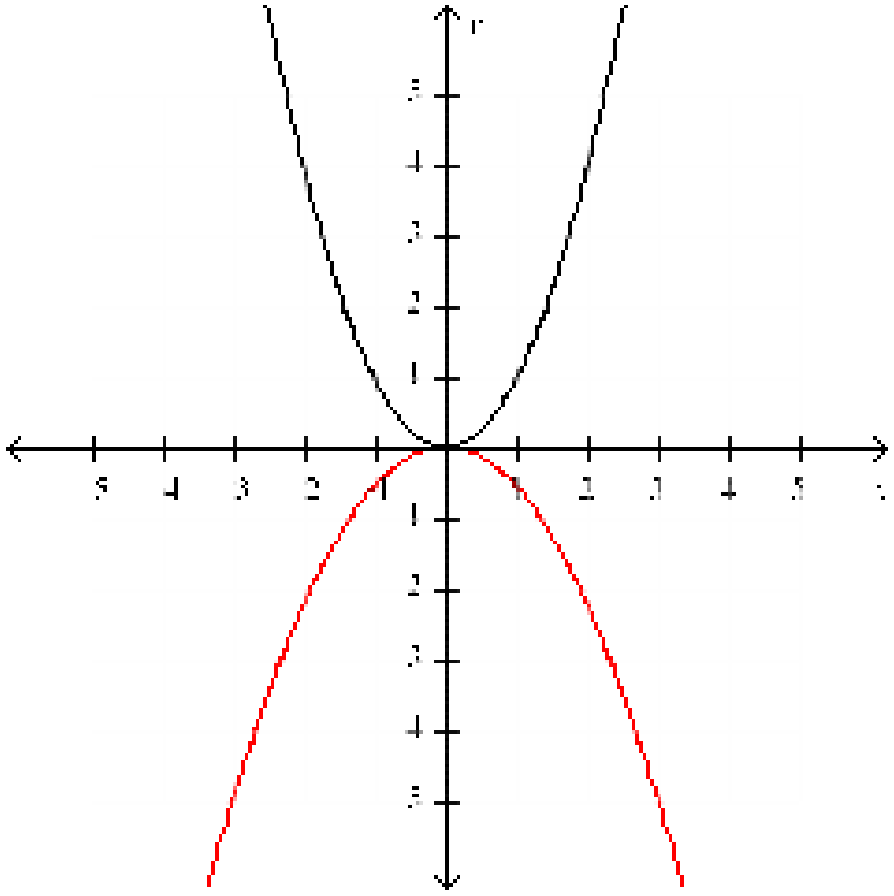












(A) -0.75

(B) 1.333333333333

(C) -1.333333333333

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

15. What are the coordinates of the invariant point(s) when the function –2 is reflected in the –$2 -$axis?

(A) (2, –2) (B) (0, –2) (C) (–2, 0) and (2, 0) (D) (0, 2)

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

16. When the value of a is less than –$1,$ the function $g(x) = ax^2$ has what relationship to the base function $f(x) = x^2$?

(A) compressed vertically (B) reflected and compressed vertically (C) stretched vertically (D) reflected and stretched vertically

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

17. Compared to the graph of the base function $f(x) = x^2$, the graph of $g(x) = (x + 4)^2 + 9$ is translated 4 units right and 9 units up. Rendered ExamView reference:

$$f(x) = x^2$$

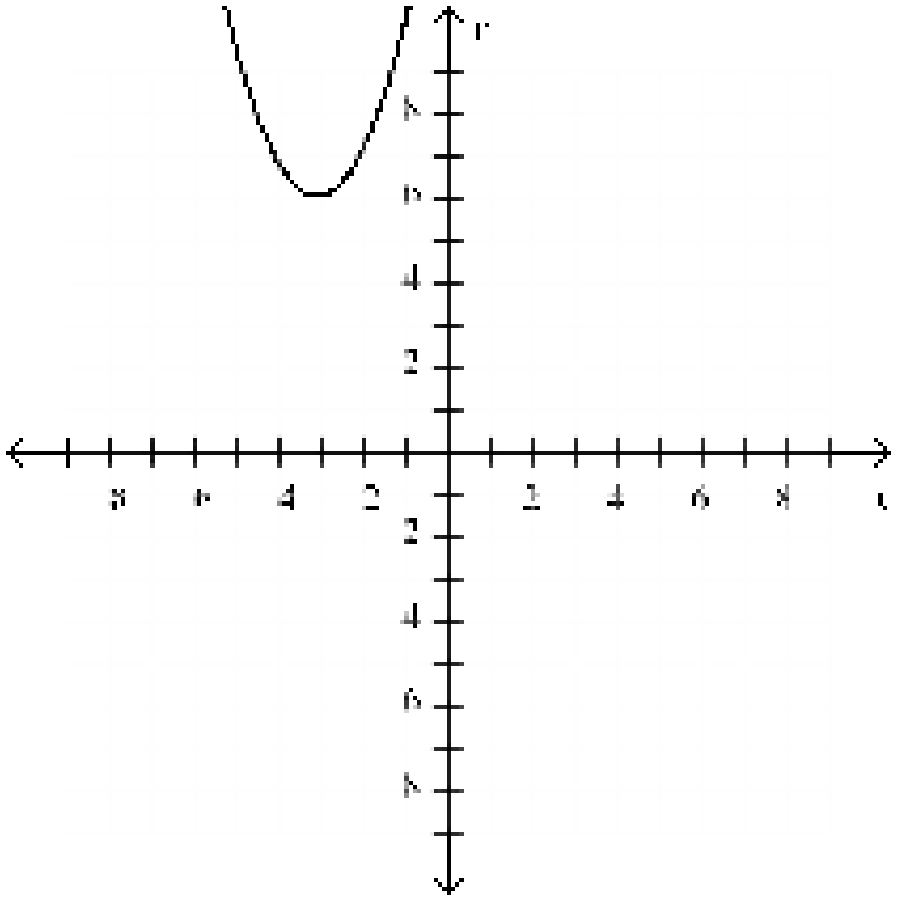
$$g(x) = (x - 4)^2 + 9$$

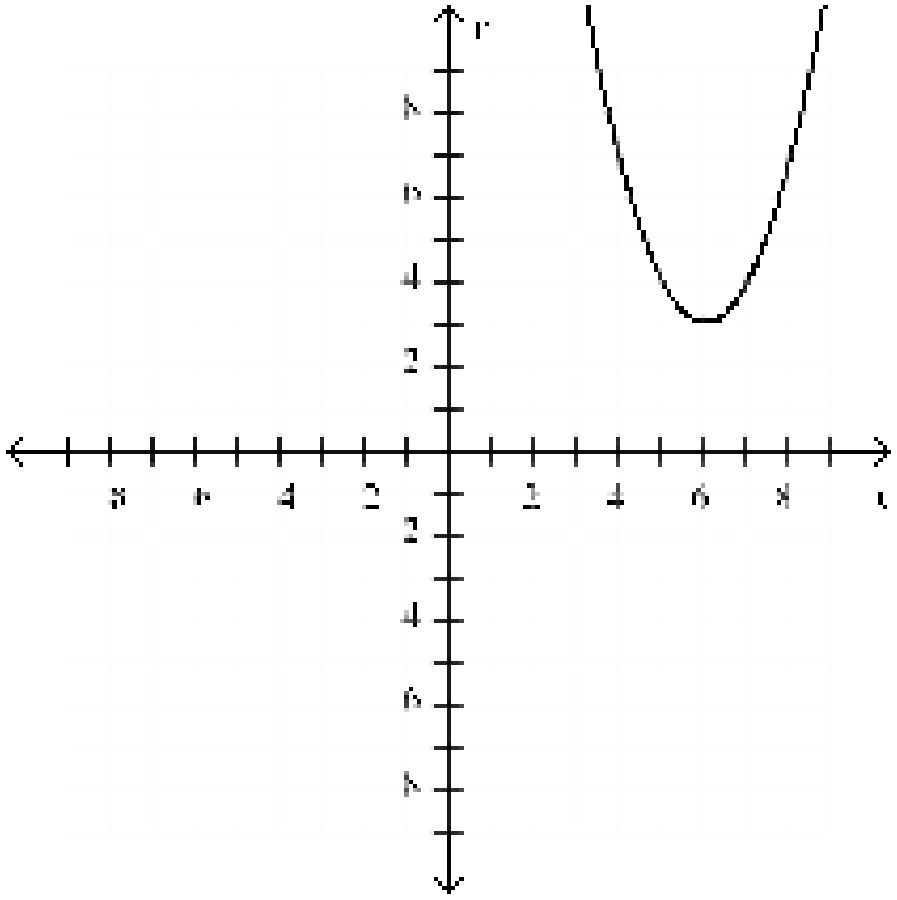
(A) 9 units to the left and 4 units down (B) 4 units to the left and 9 units down (C) 9 units to the right and 4 units up (D) 4 units to the right and 9 units up

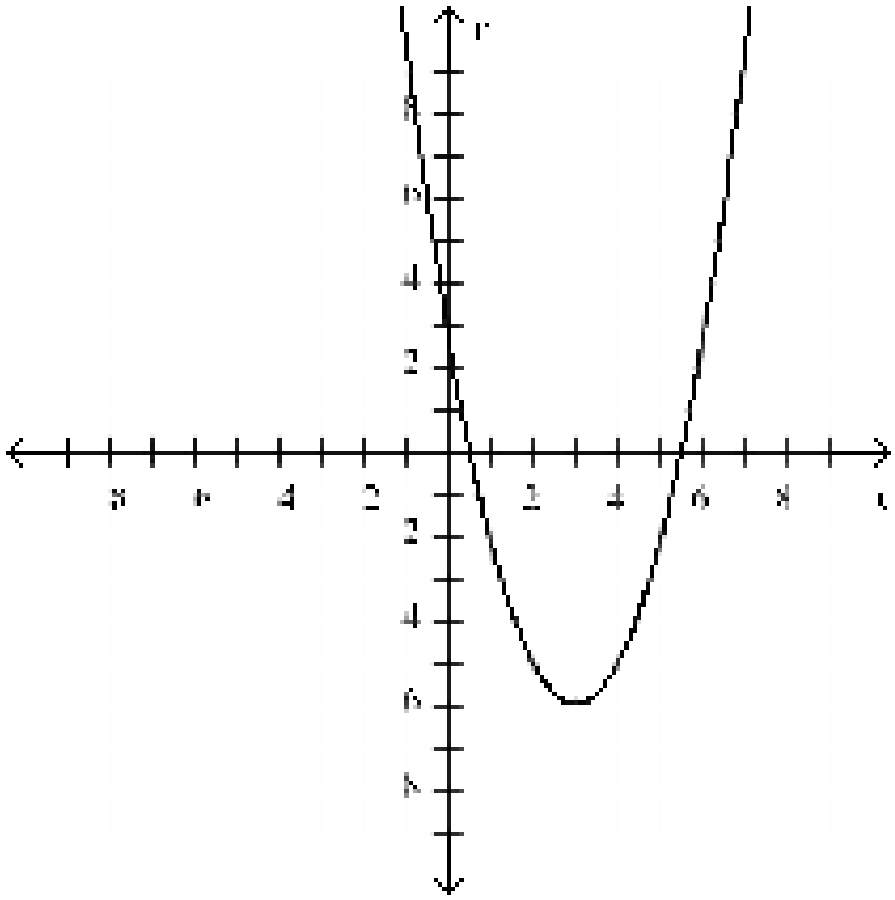
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

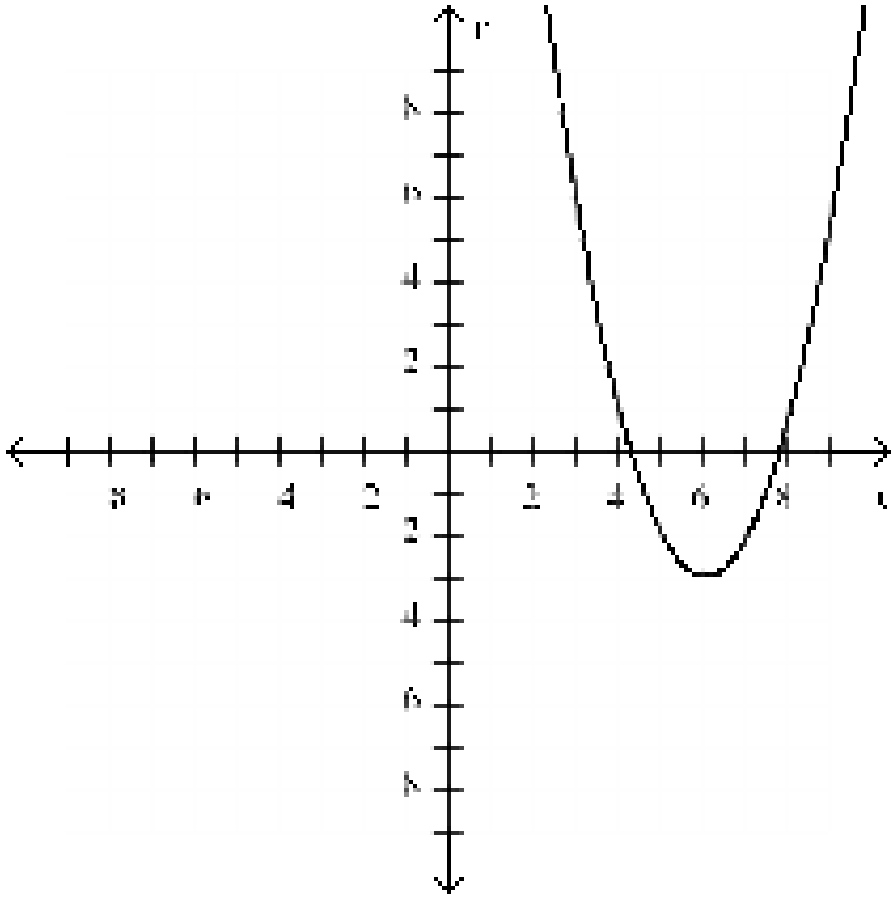
18. Which graph represents $g(x) = f(x + 6) + 3$? Rendered ExamView reference:

$$f(x) = (x - 6)^2 + 3$$









(A) graph of $g(x) = f(x - 6) + 3$; translated 6 units right and 3 units up

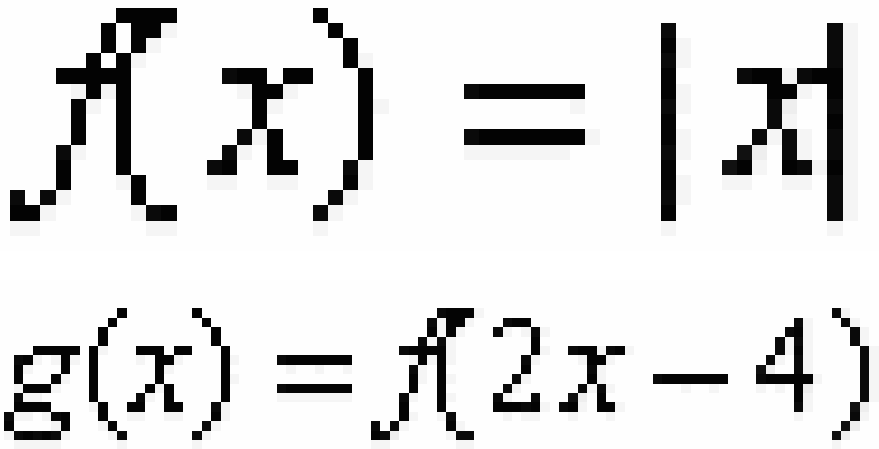
(B) graph of $g(x) = f(x + 6) - 3$; translated 6 units left and 3 units down

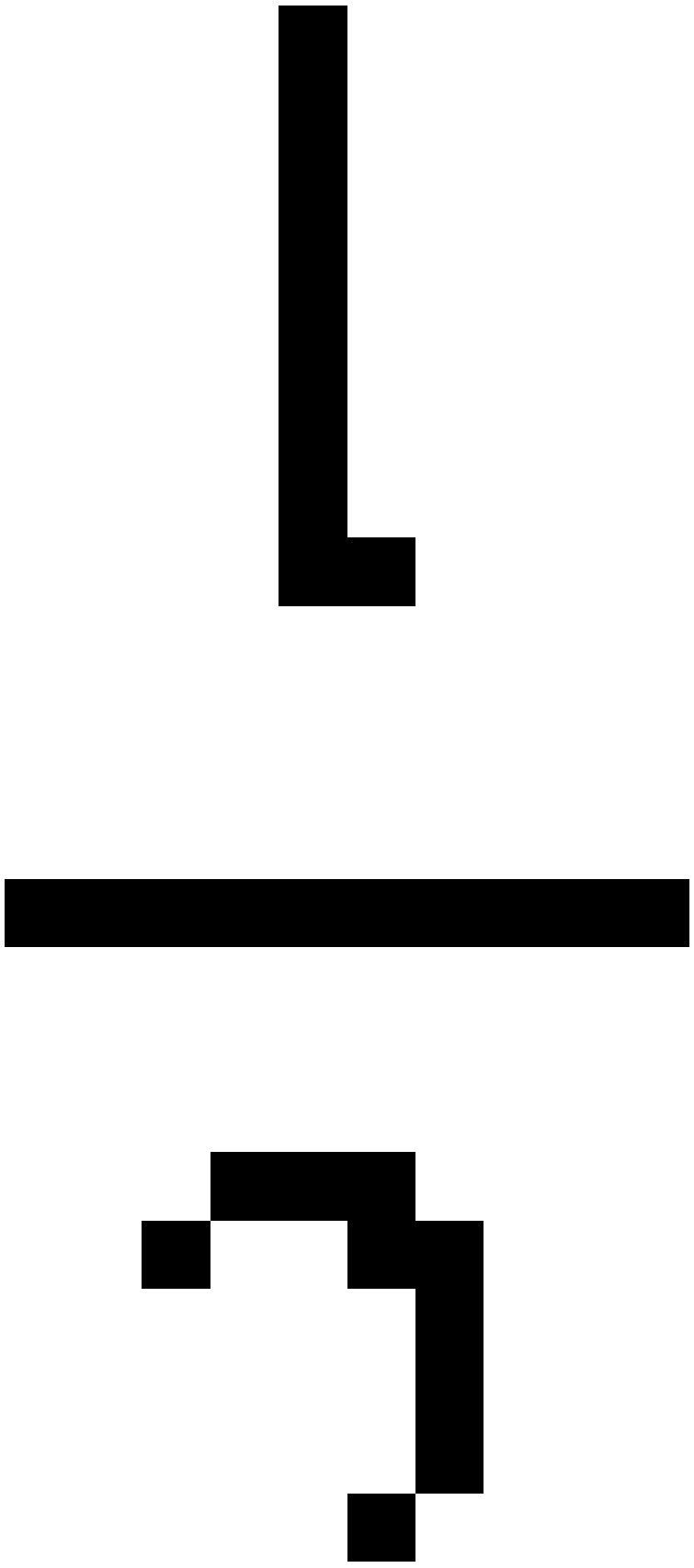
(C) graph of $g(x) = f(x + 6) + 3$; translated 6 units left and 3 units up

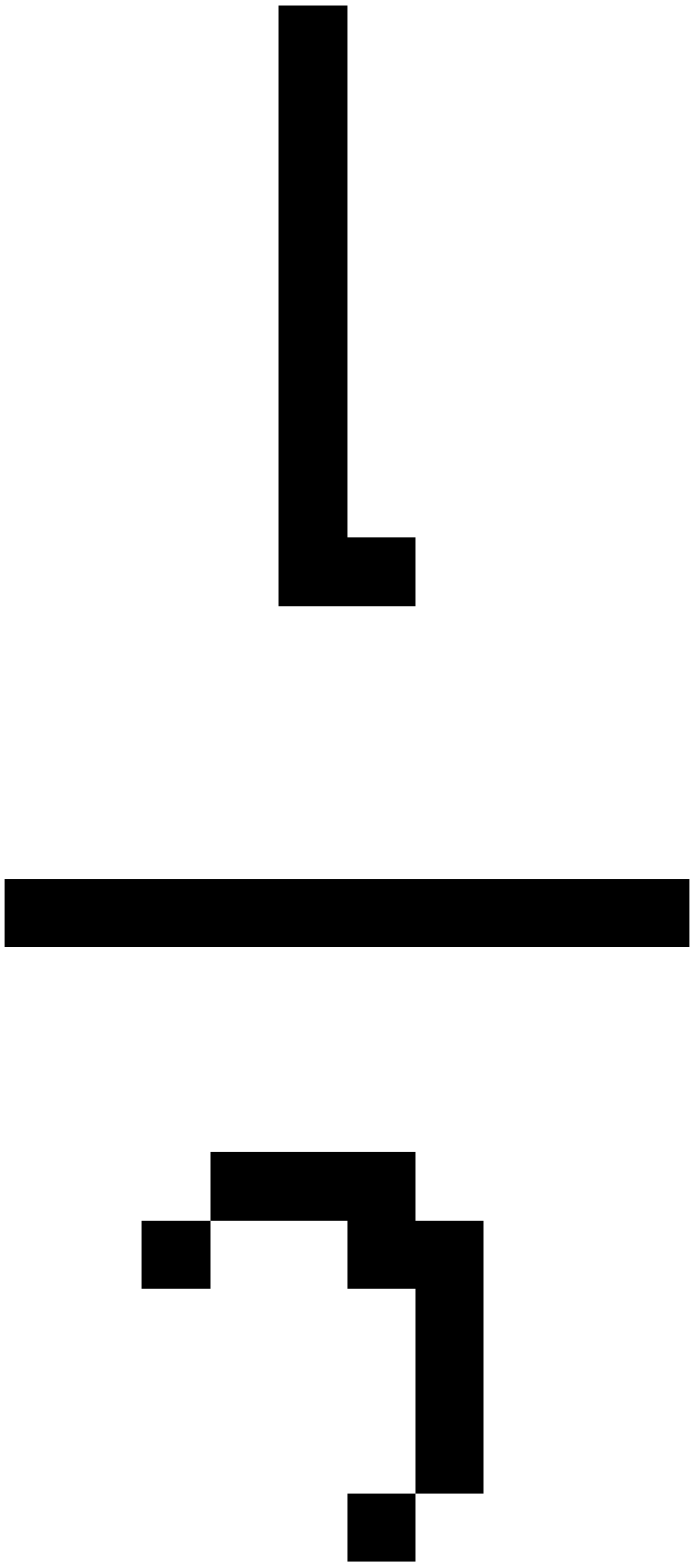
(D) graph of $g(x) = f(x - 6) - 3$; translated 6 units right and 3 units down

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

19. Which choice best describes the combination of transformations that must be applied to the graph of $f(x)$ to obtain the graph of $g(x) = f(2x - 4)$? Rendered ExamView reference:







(A) a horizontal stretch by a factor of 2 and a horizontal translation of 2 units to the left

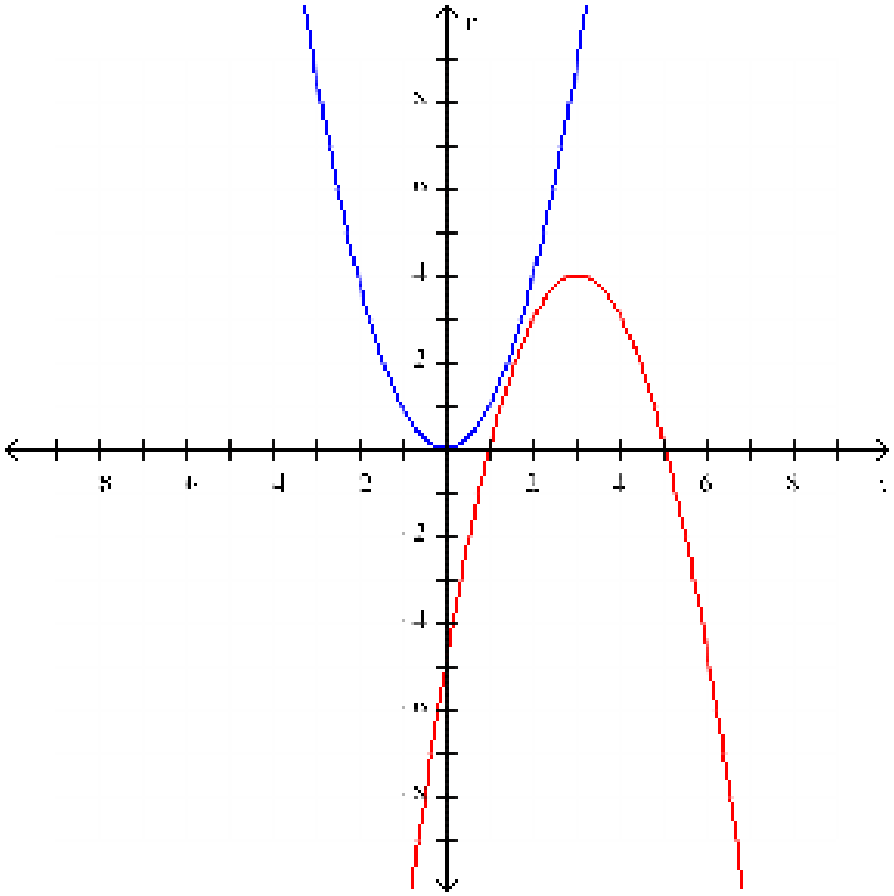
(B) a horizontal stretch by a factor of 1/2 and a horizontal translation of 4 units to the right

(C) a horizontal stretch by a factor of 1/2 and a horizontal translation of 2 units to the right

(D) a horizontal stretch by a factor of –2 and a horizontal translation of 2 units to the right

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

20. Which graph represents $g(x) = f(x - 3) + 4$? Rendered ExamView reference:



(A) a reflection in the x-axis, a vertical translation of 4 units up, and a horizontal translation of 3 units to the right

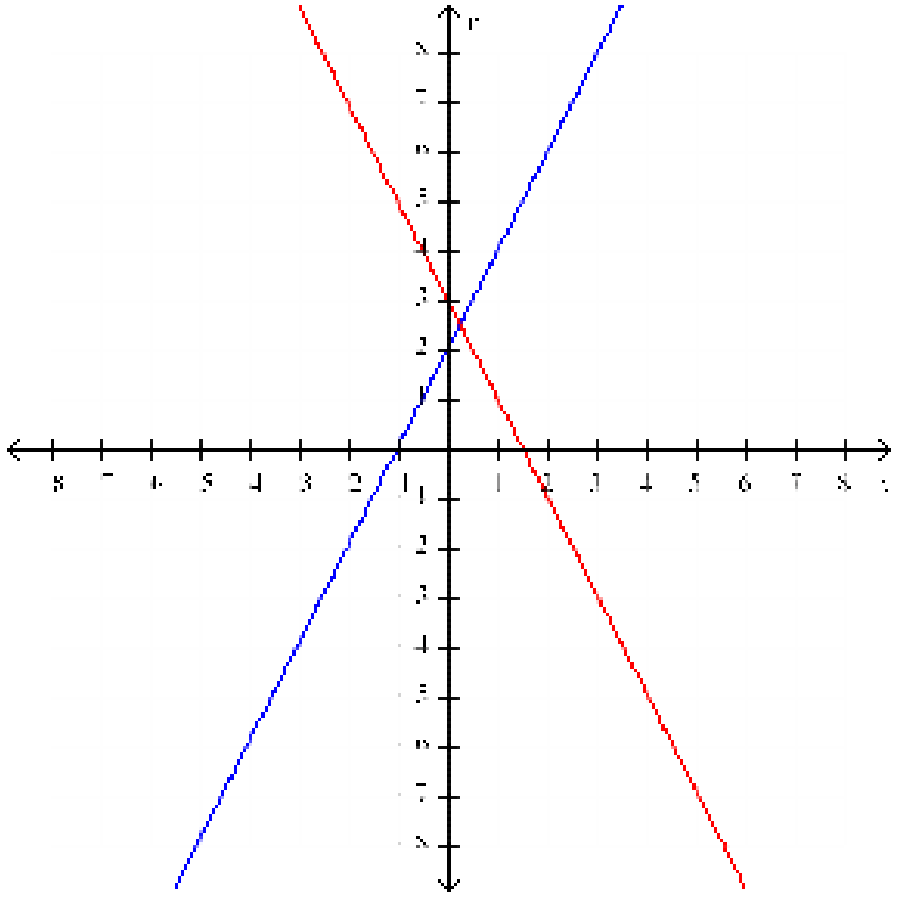
(B) a reflection in the x-axis, a vertical translation of 3 units up, and a horizontal translation of 4 units to the right

(C) a reflection in the x-axis, a vertical translation of 3 units down, and a horizontal translation of 4 units to the right

(D) a reflection in the x-axis, a vertical translation of 4 units up, and a horizontal translation of 3 units to the left

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

21. In the graph shown, which transformations must be applied to the blue curve to obtain the red curve? $2*(x+1)$ Rendered ExamView reference:



(A) a reflection in the x-axis and a translation of down units 5

(B) a reflection in the y-axis and a translation of up units 5

(C) a reflection in the x-axis and a translation of up units 5

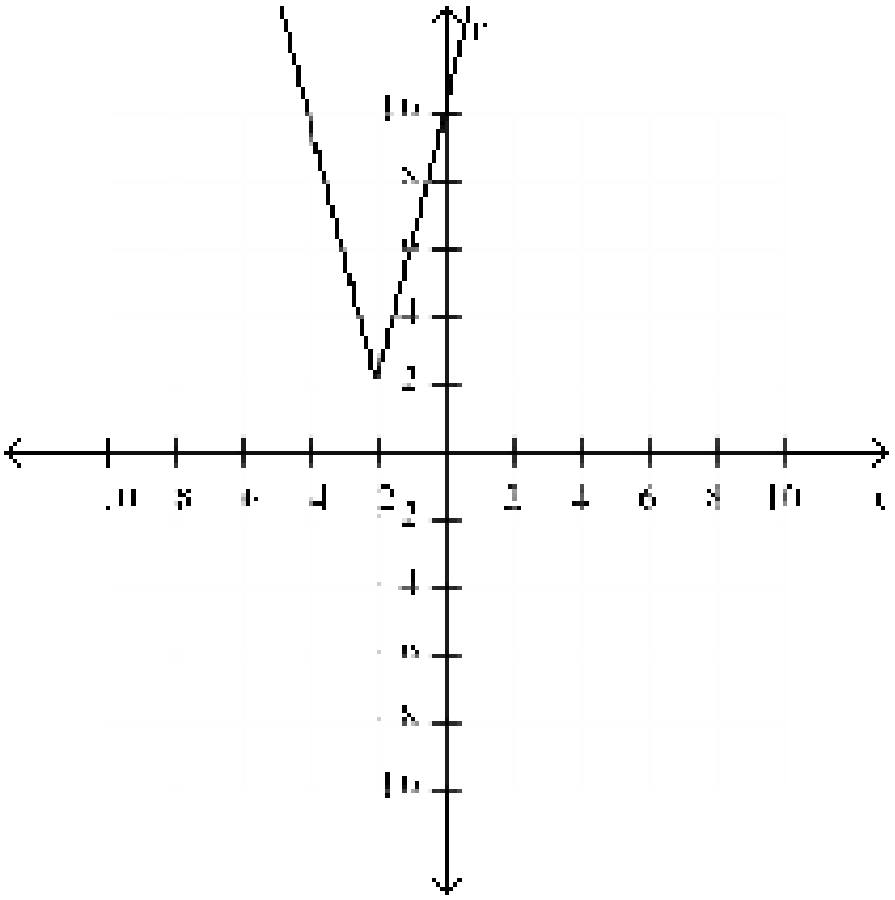
(D) a reflection in the y-axis and a translation of down units 5

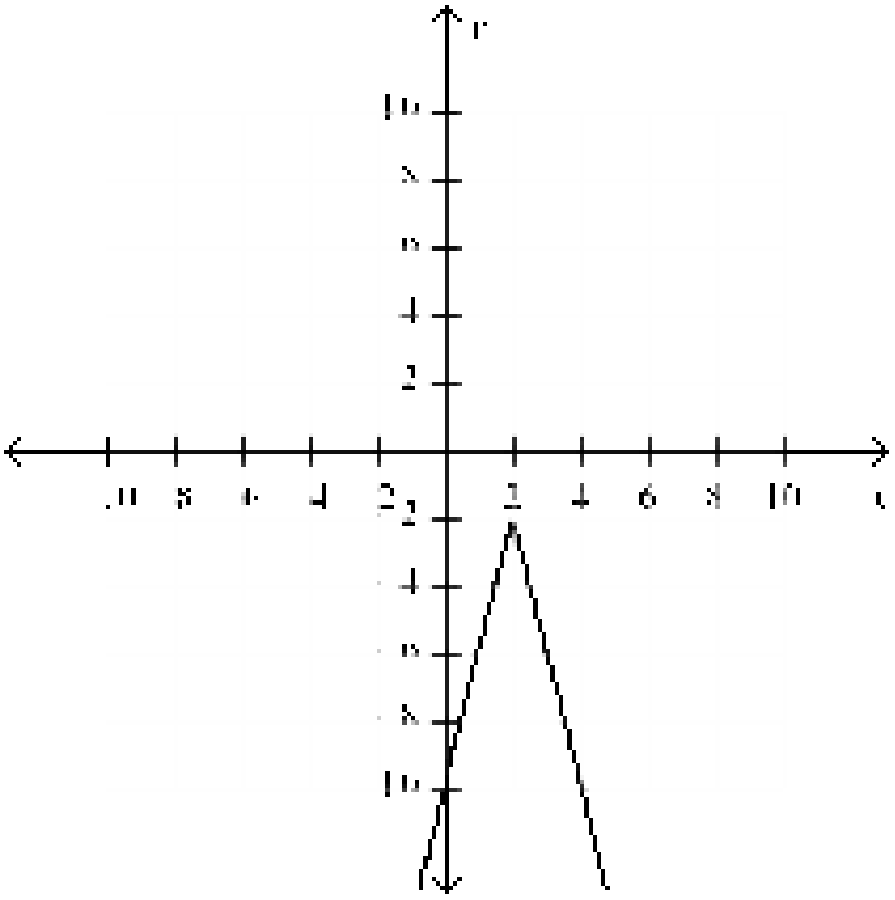
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

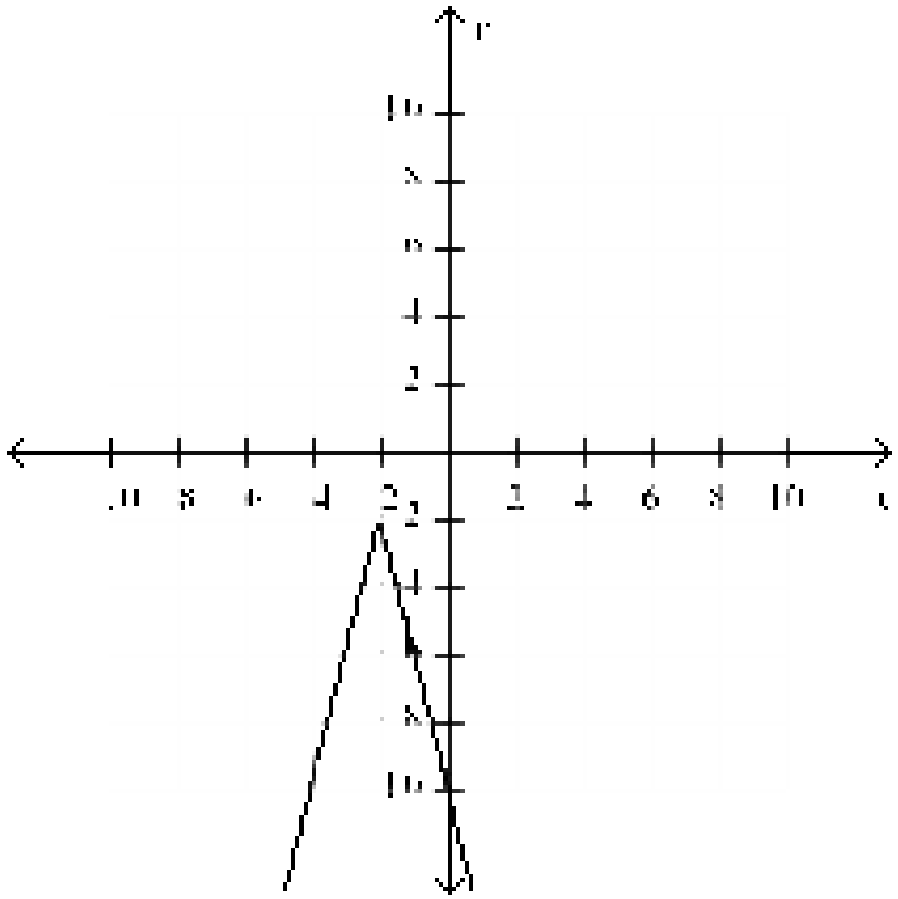
22. Which of the following graphs represents the graph of the function –$2 + 2$ transformed to $1 0$? Recovered graph functions: ($2)*abs((−2)*x-(4))+(2); (2)*((−2)*x-(4))^2+(2); -(2)*abs((−2)*x+(4))-(2); -(2)*((−2)*x+(4))^2-(2); -(2)*abs((−2)*x-(4))-(2); -(2)*((−2)*x-(4))^2-(2); (2)*abs((−2)*x+(4))+(2); (2)*((−2)*x+(4))^2+(2)$. Rendered ExamView reference:

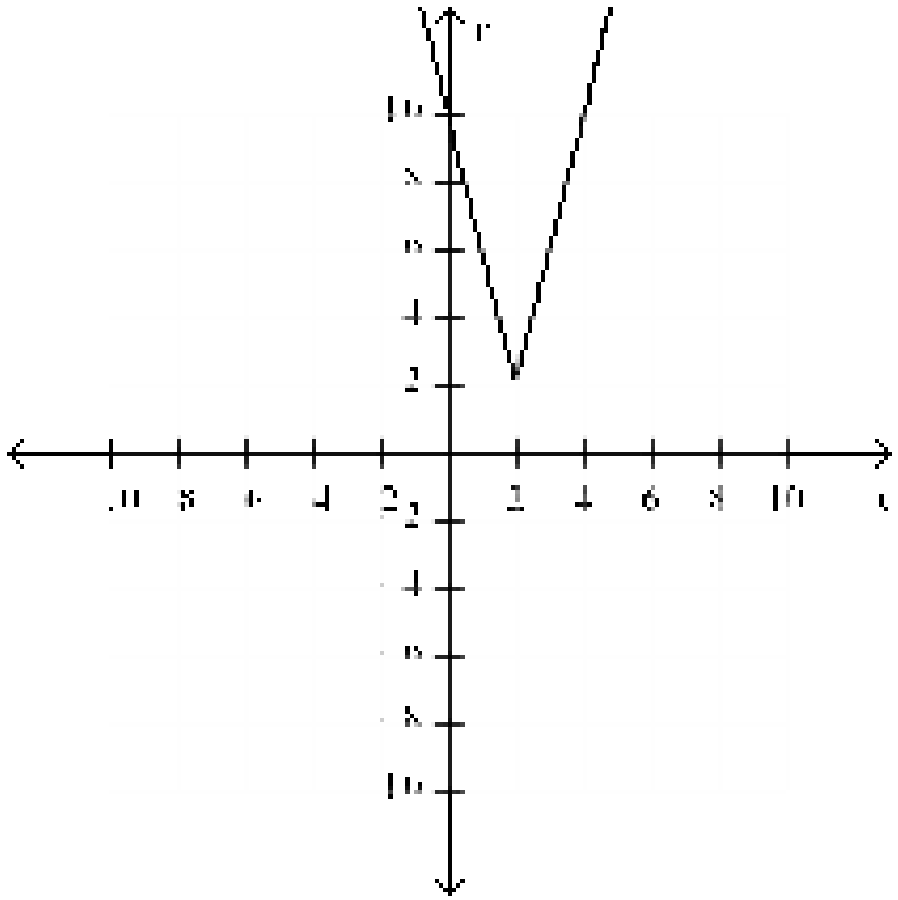
$$f(x) = |x|$$

$$f(x) = 2|-2x+4| + 2$$









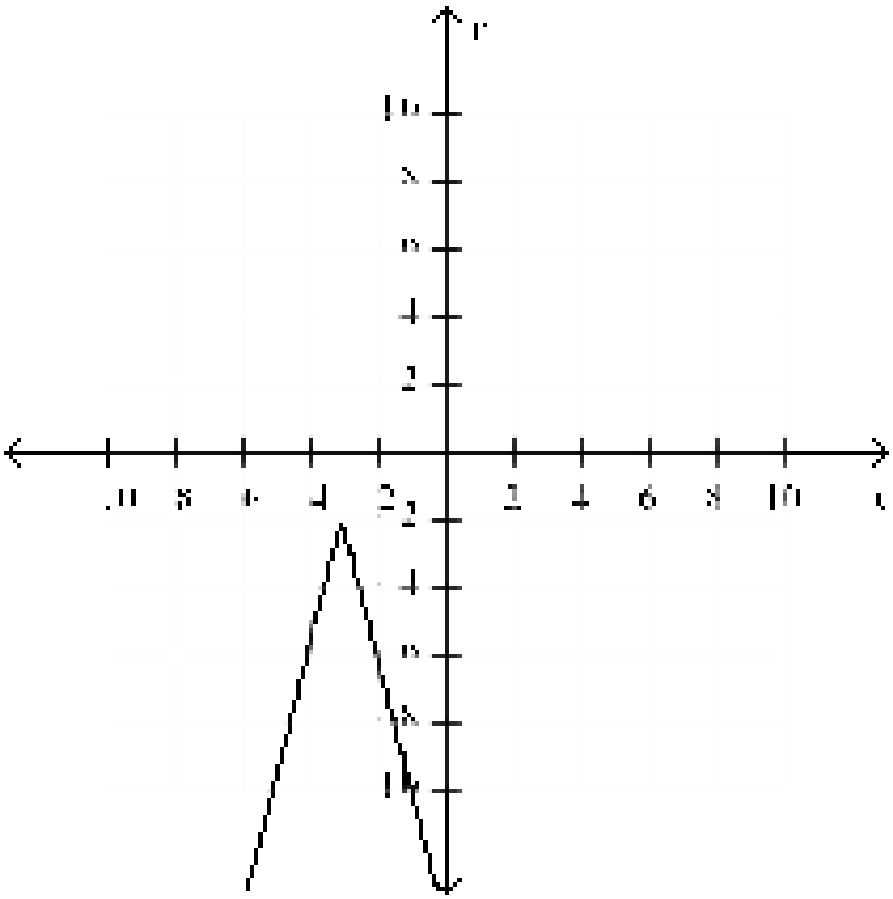
(A) $x >= + 2$ (B) $x >= – 2$ (C) $x >= –2$ (D) $x <= + 2$

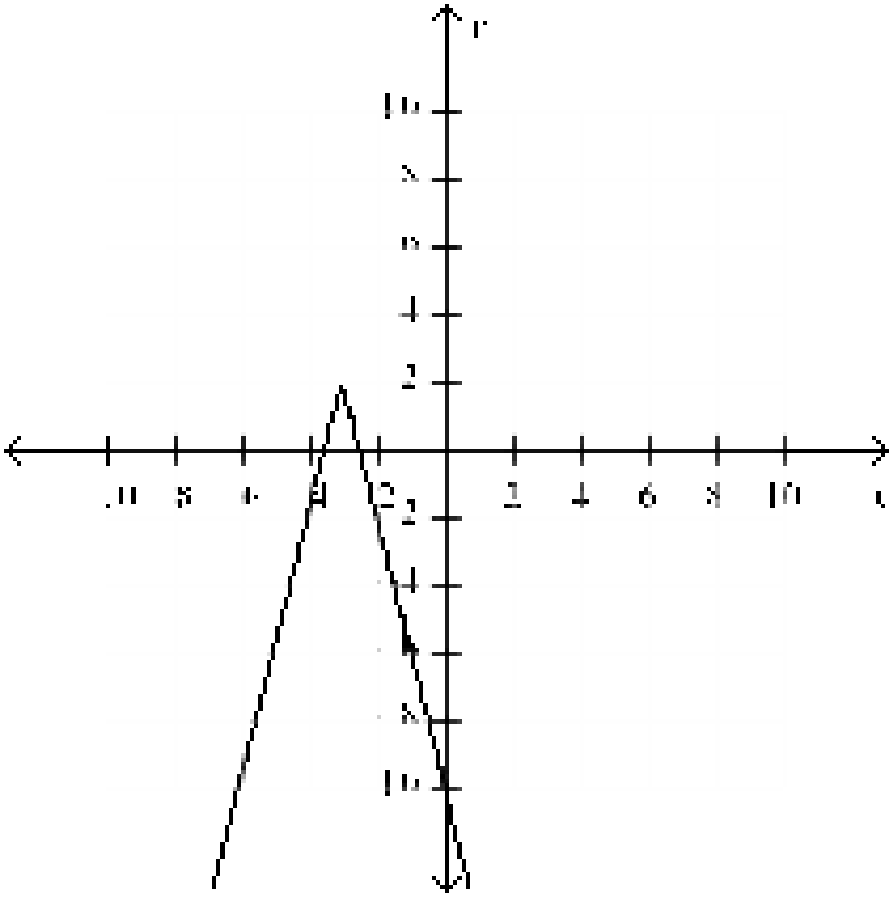
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

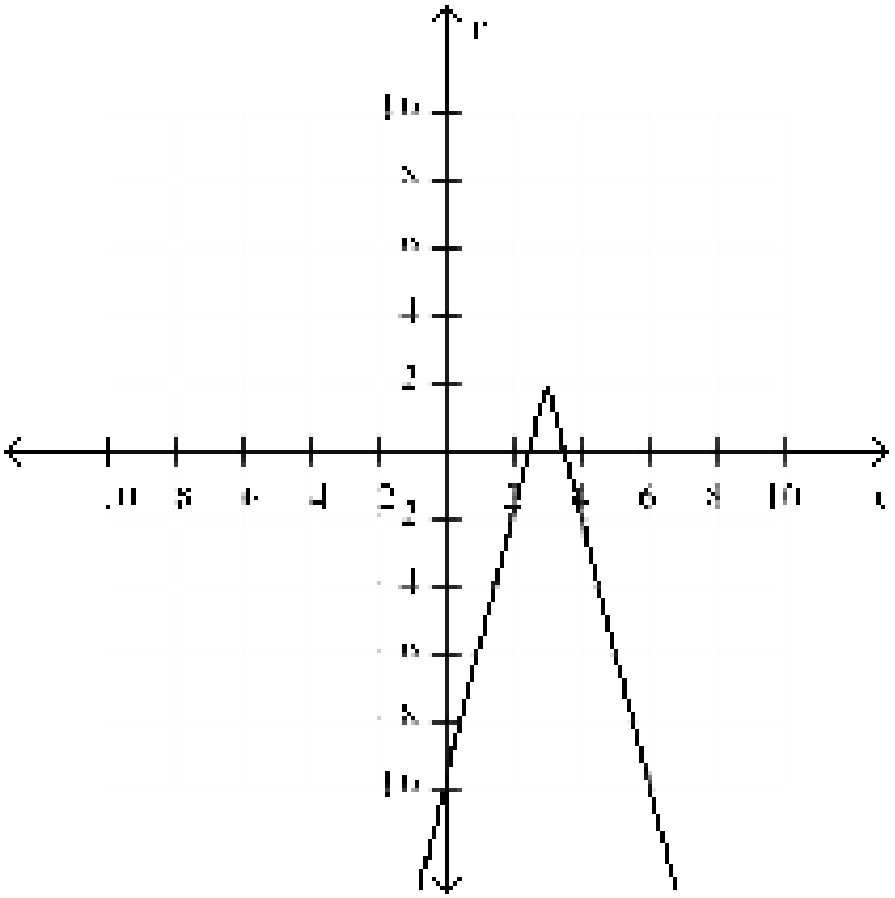
23. When the function $1 0$ is transformed to –$4 + 3,$ the graph looks like Recovered graph functions: ($−4)*abs(x+(3))-(2); (−4)*(x+(3))^2-(2); (−4)*abs(x+(3))+(2); (−4)*(x+(3))^2+(2); (−4)*abs(x-(3))+(2); (−4)*(x-(3))^2+(2); (−4)*abs(x-(3))-(2); (−4)*(x-(3))^2-(2)$. Rendered ExamView reference:

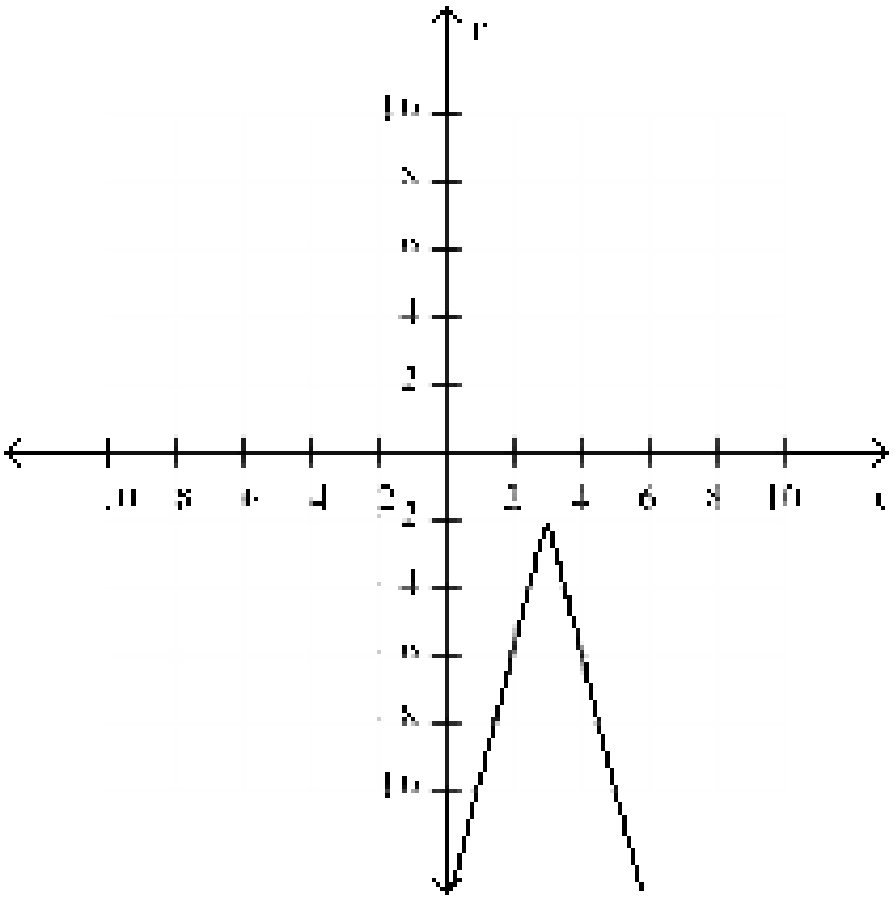
$$f(x) = |x|$$

$$f(x) = -4|x+3| + 2$$









(A) $x >= + 2$ (B) $x >= + 3$ (C) $x >= 1$ (D) $x <= + 2$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

24. Which of the following functions is the correct inverse for the function $f(x) = 3x + 5$?

(A) $f^(−1)(x) = 1/3x - 5/3$ (B) $f^(−1)(x) = 3x + 5$ (C) $f^(−1)(x) = −1/3x + 5/3$ (D) $f^(−1)(x) = 1/3x + 5/3$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

25. Which of the following functions is the correct inverse for the function $f(x) = −9/2x + 6$?

(A) −9/2 = + 6 x –2 (B) –2 = –2 x –2 (C) –2 = –2 x –2 (D) –2 = −2/9 x 4/3

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

26. Which of the following functions is the correct inverse for the function $f(x) = sqrt(x) + 7$, {x | x $>= 0, x$ in R}?

(A) $f^(−1)(x) = (x + 7)^2, {x | x >= −7, x in R}$ (B) $f^(−1)(x) = sqrt(x) - 7$ (C) $f^(−1)(x) = (x - 7)^2, {x | x >= 7, x in R}$ (D) $f^(−1)(x) = sqrt(x) + 7$

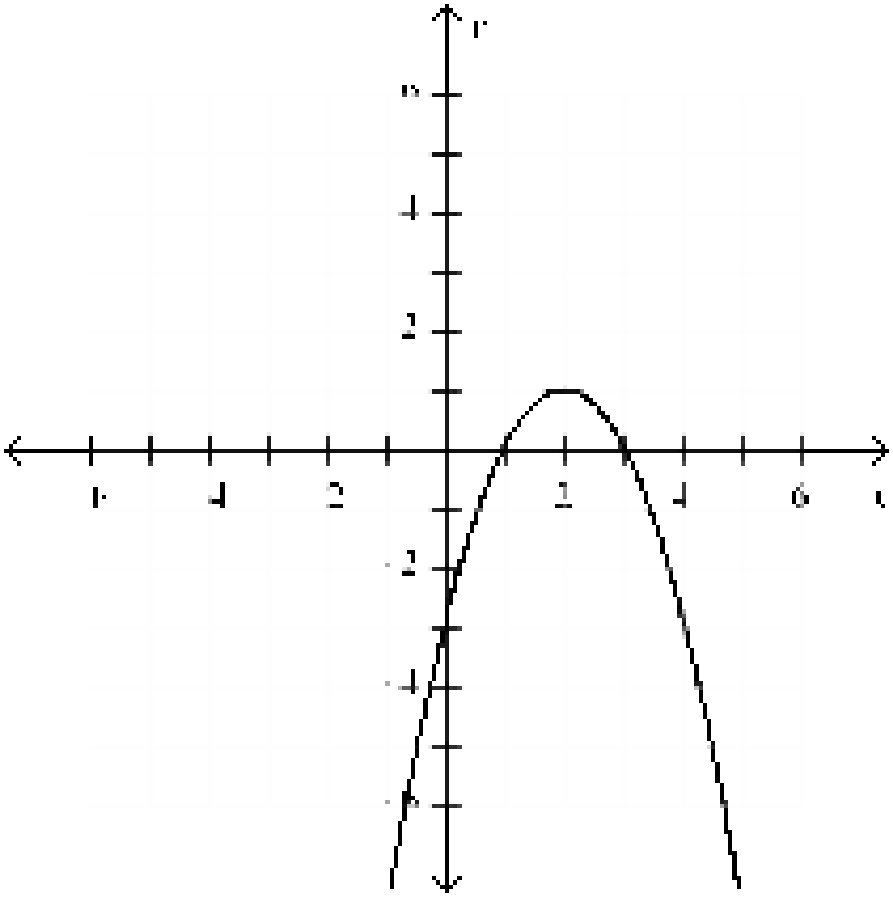
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

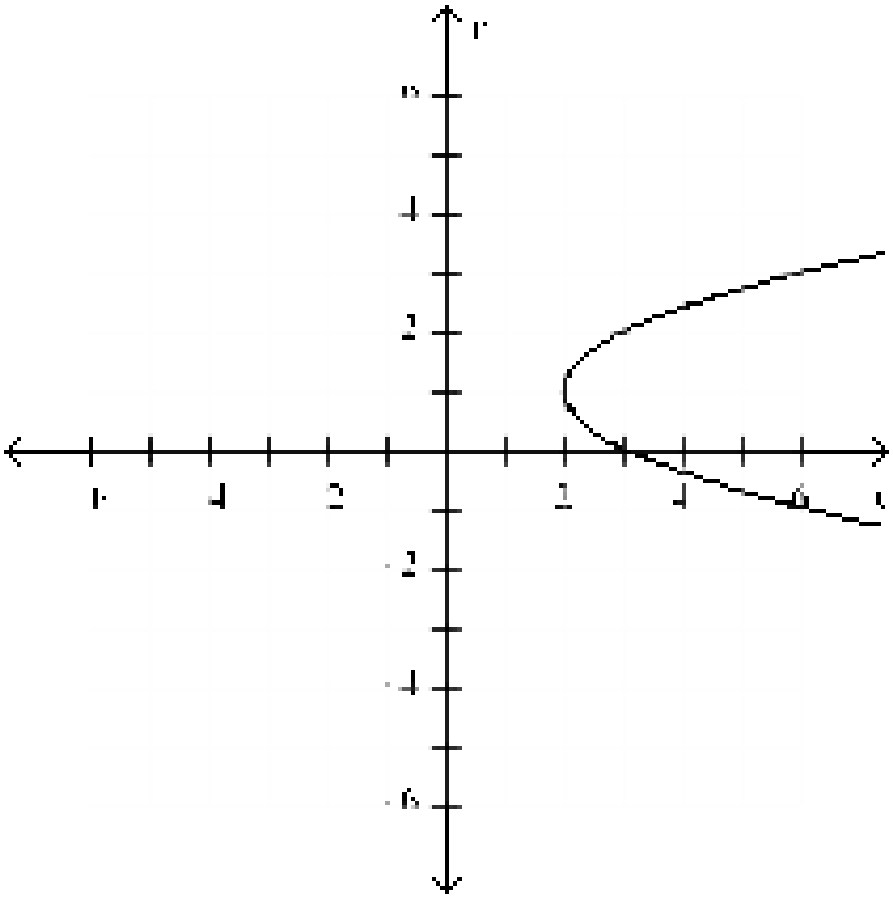
27. Which of the following relations is the correct inverse for the function $f(x) = x^2 + 6x + 2$?

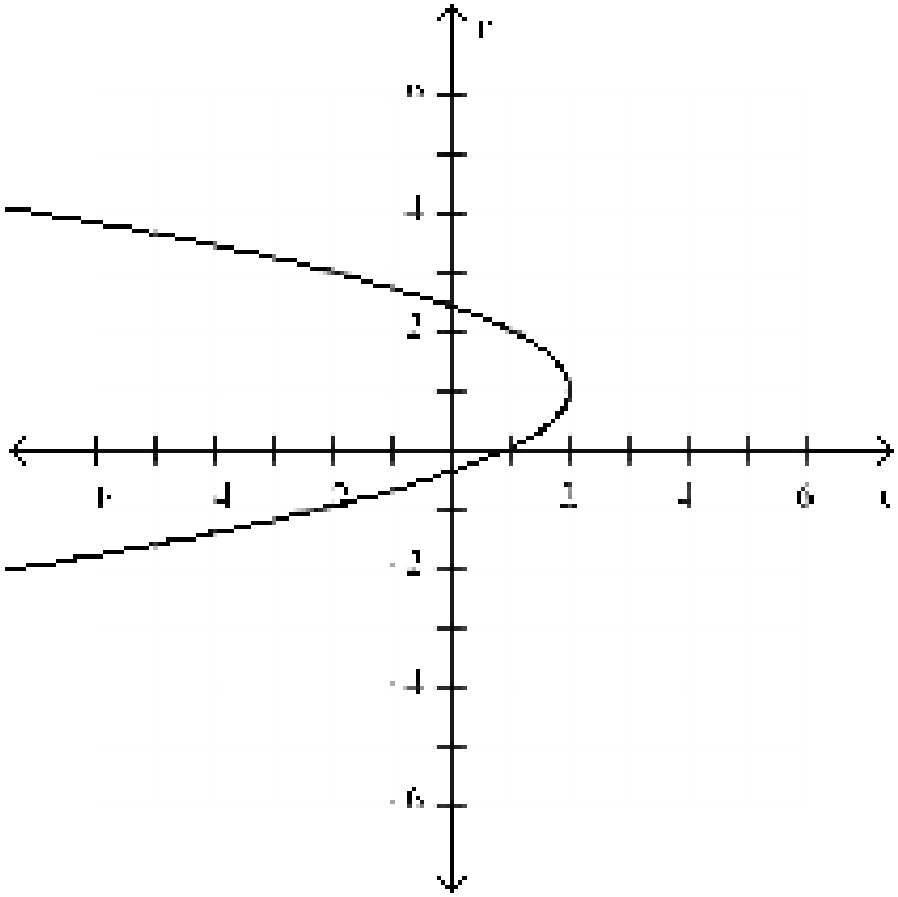
(A) $y = 3 +/- sqrt(x + 7)$ (B) $y = −3 +/- sqrt(x - 7)$ (C) $y = −3 +/- sqrt(x + 7)$ (D) $y = 3 +/- sqrt(x + 3)$

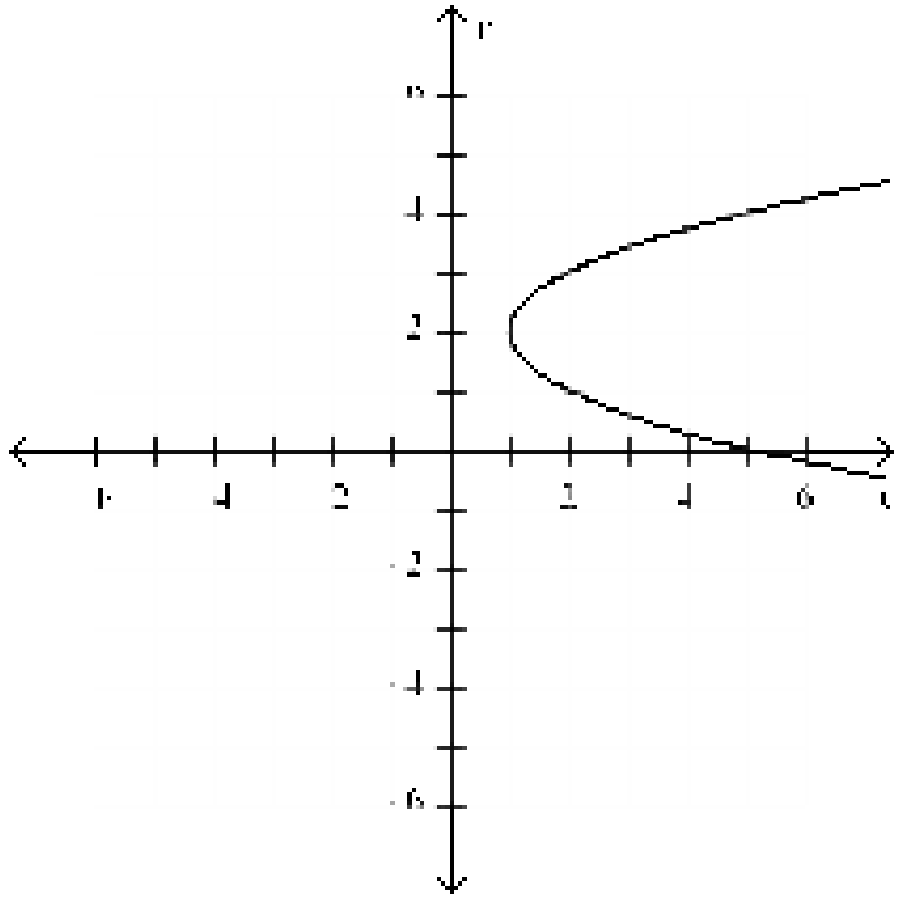
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

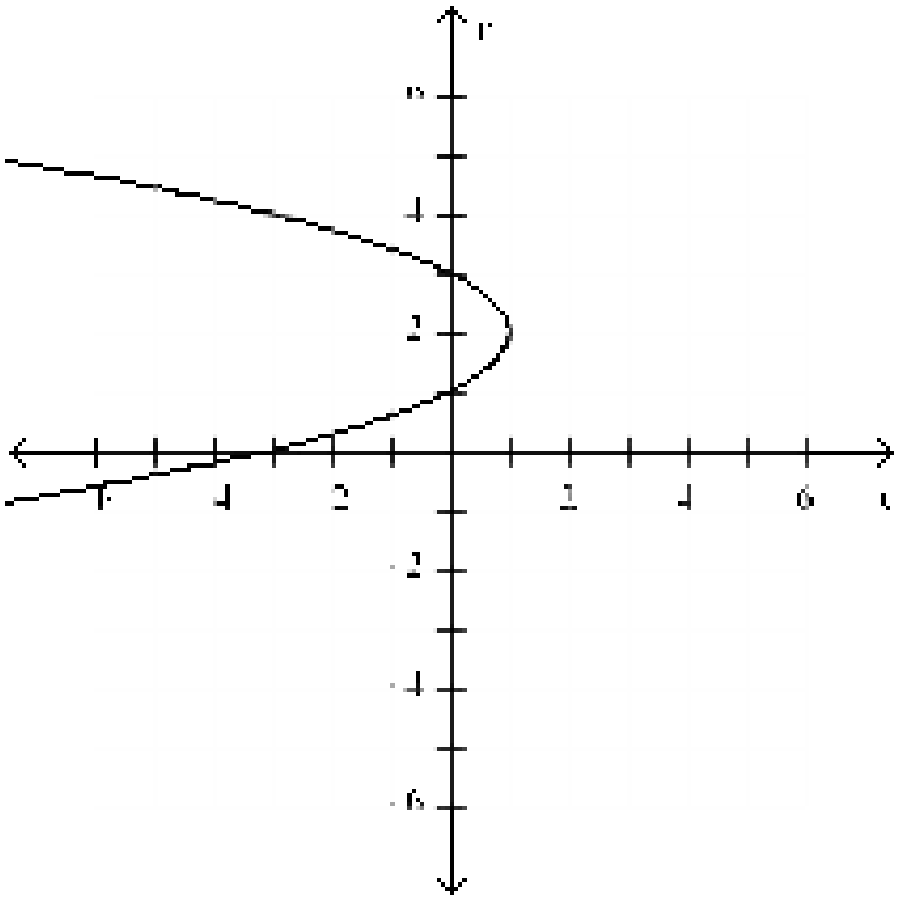
28. Which graph represents the inverse of the graph shown? Original graph: $-(x-2)^2+1$. Inverse relation: $(y-1)^2+2$. Rendered ExamView reference:


y
x
6
4
2
-2
-4
-6
-4
-2
2
4
6


y
x
6
4
2
-2
-4
-6
-4
-2
2
4
6


y
x
6
4
2
-2
-4
-6
-8
-6
-4
-2
2
4
6

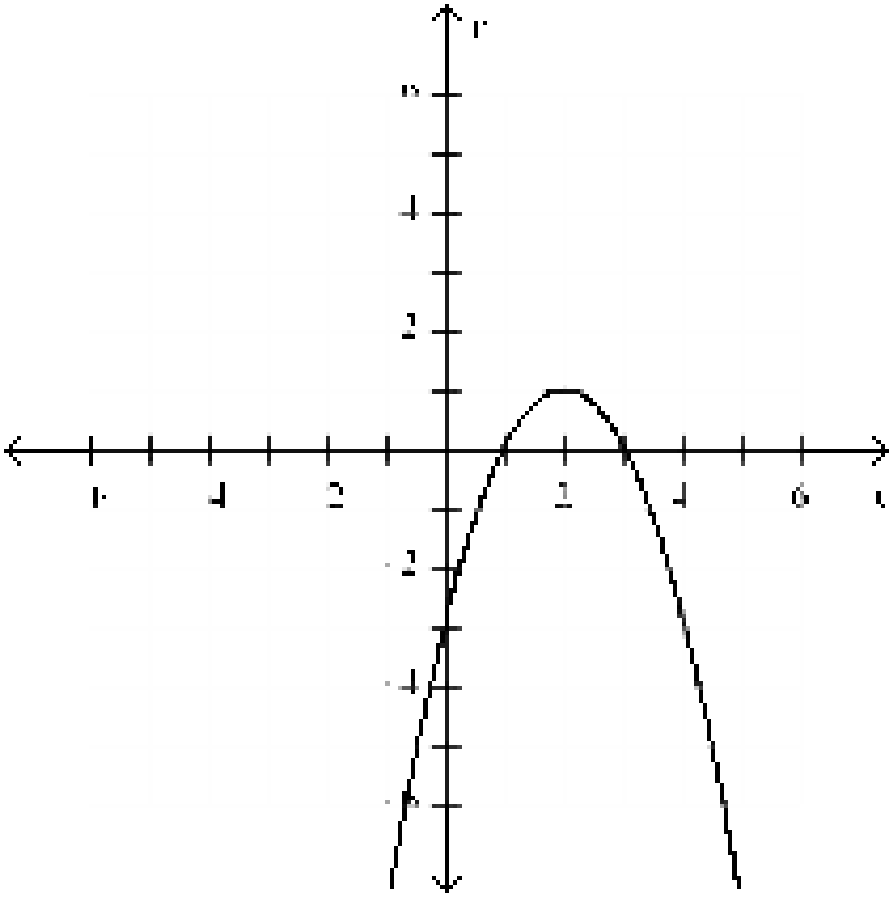

y
x
6
4
2
-2
-4
-6
-8
-6
-4
-2
2
4
6

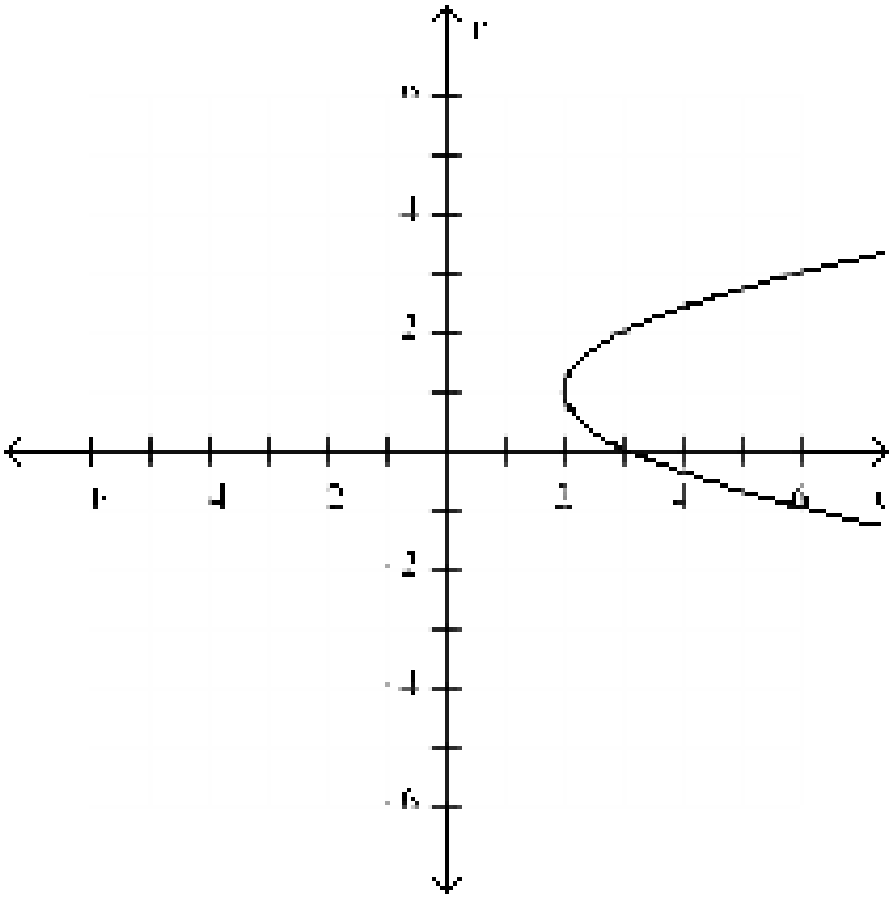


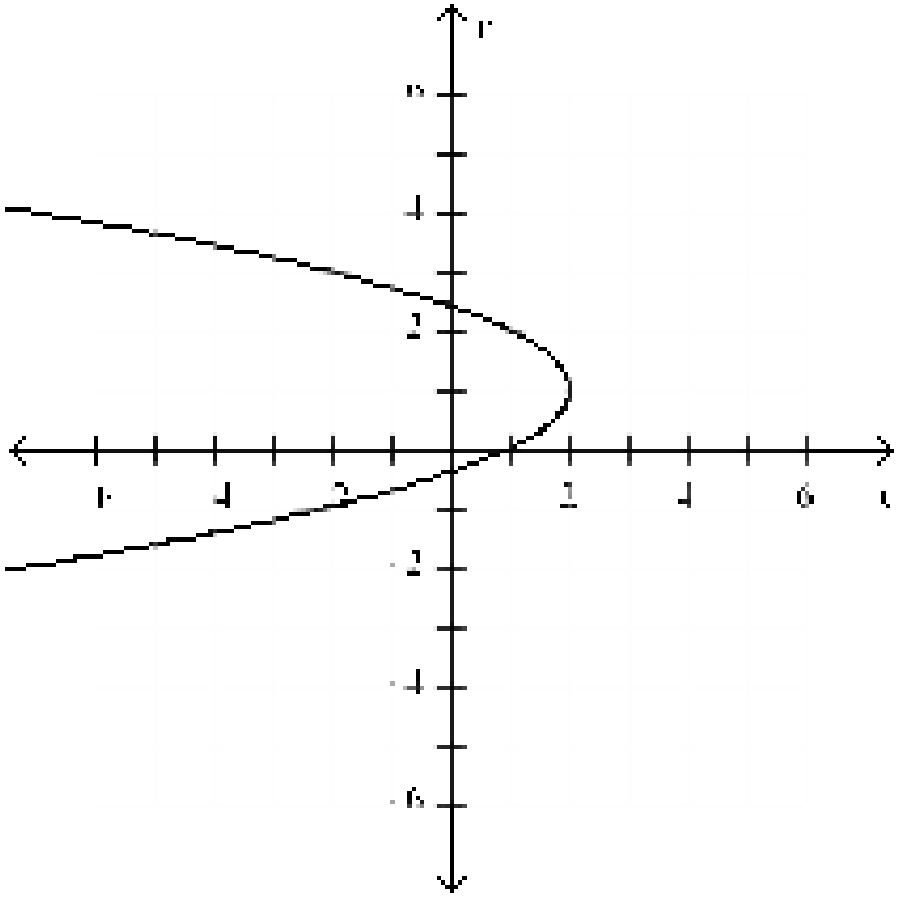
(A) graph of $-(x-2)^2+1$ (B) graph of $(y-1)^2+2$ (C) graph of $-(y-1)^2+2$ (D) graph of $(y-2)^2+1$

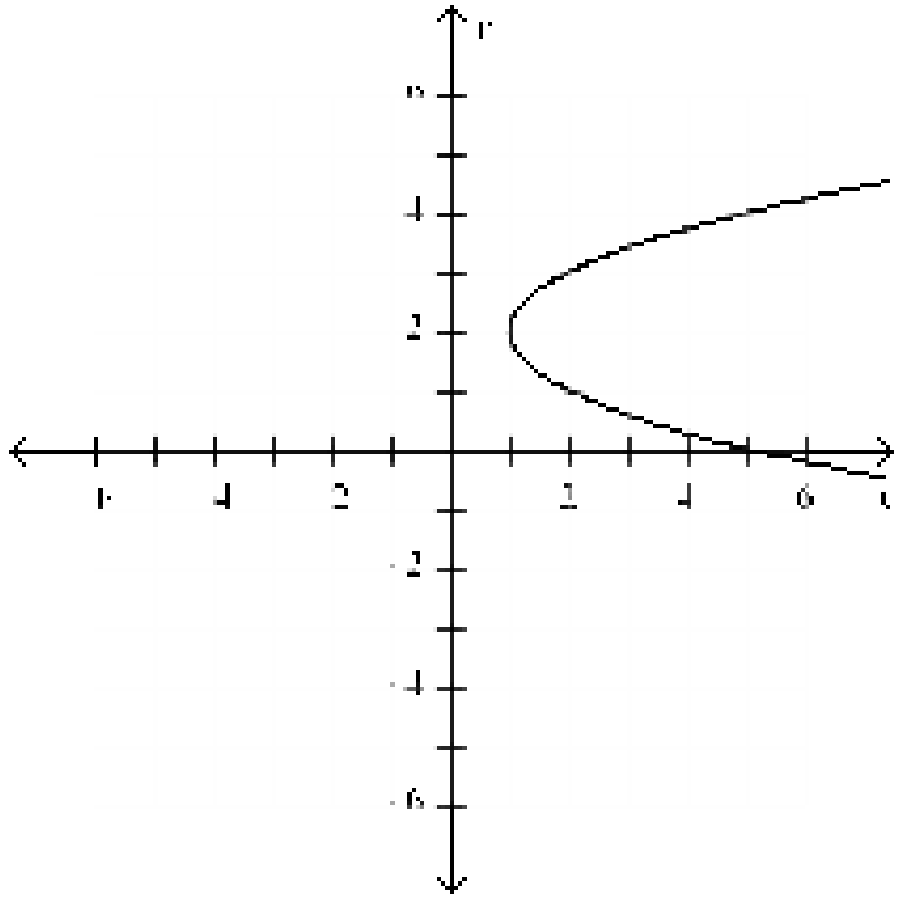
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

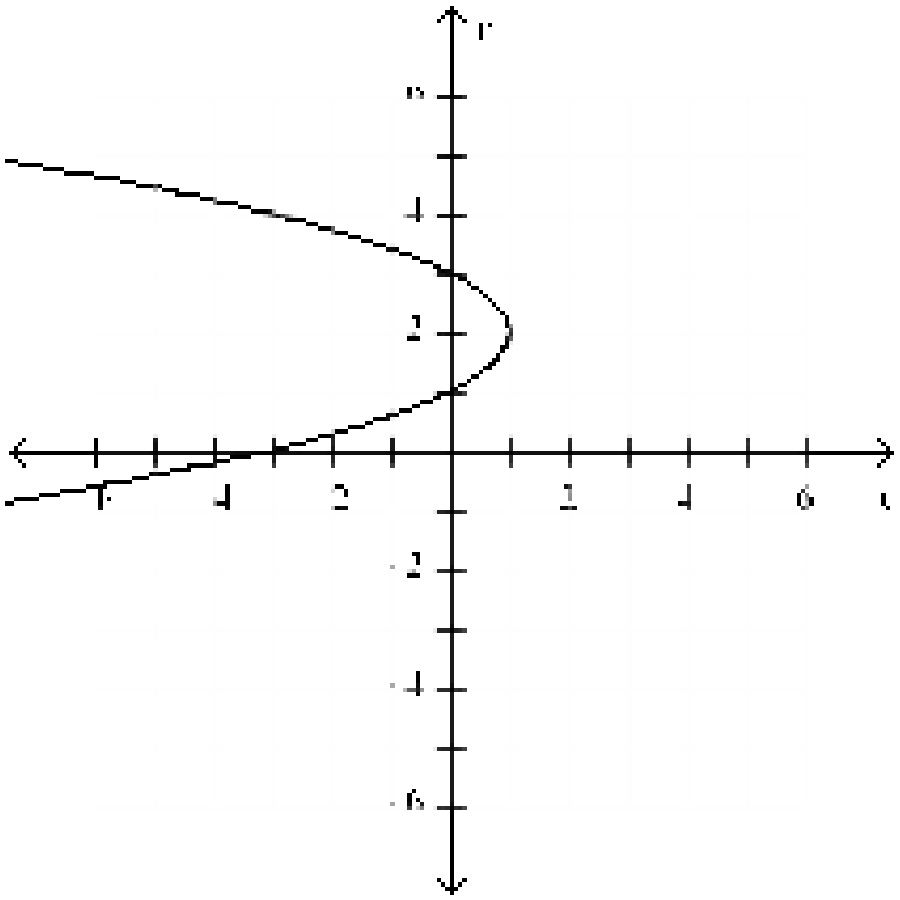
29. Which graph represents the inverse of the graph shown? Original graph: $-(x3)/5$. Inverse relation: $-(x-3)/5$. Rendered ExamView reference:


y
x
6
4
2
-2
-4
-6
-4
-2
2
4
6


y
x
6
4
2
-2
-4
-6
-4
-2
2
4
6


y
x
6
4
2
-2
-4
-6
-8
-6
-4
-2
2
4
6


y
x
6
4
2
-2
-4
-6
-8
-6
-4
-2
2
4
6



(A) graph of $-(x3)/5$ (B) graph of $-(x-3)/5$ (C) graph of $(x-3)/5$ (D) graph of $(x3)/5$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

30. The equation of the inverse of –9 is

(A) $−9$ (B) 9 (C) 9 (D) $−9$

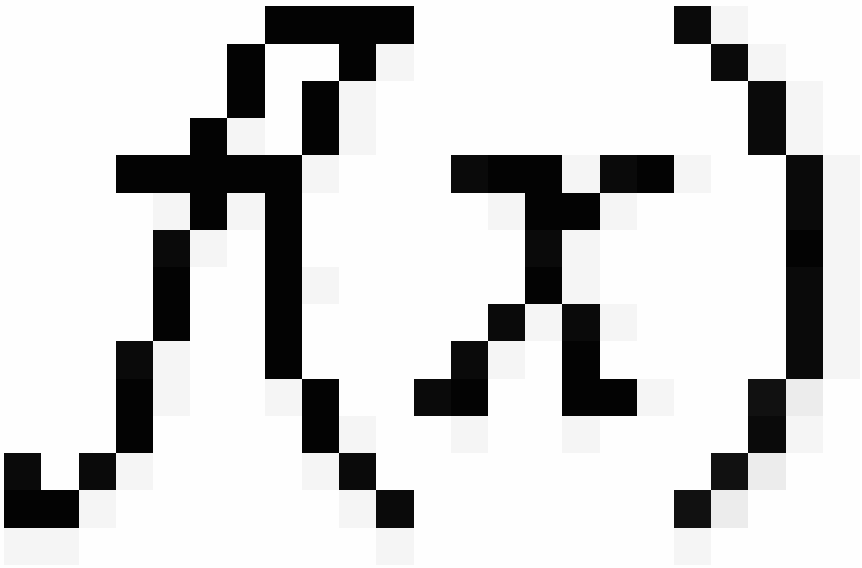
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

31. _____ are transformations that cause the graph of a function to change shape without changing the orientation of the graph.

TOP: bnk | completion **TYPE:** completion

32. The graph of – 2 – 2 is the graph of 0 translated 2 units _____. Rendered ExamView reference:

$$g(x) = f(x - 2)$$

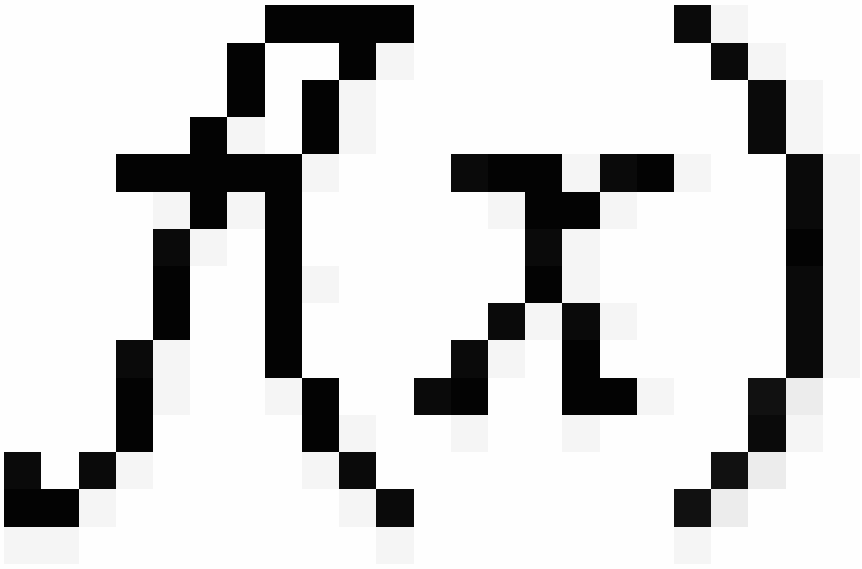


TOP: bnk | true_false **TYPE:** true_false

33. The _____ is (are) invariant on a function after a reflection in the y $-$axis.

TOP: bnk | completion **TYPE:** completion

34. The graph of y is reflected in the $8 -$axis, translated left units to the $7,$ and translated down units if(which$=1,-1,1)$. The equation of the transformed function is $g(x) =$ _____. Rendered ExamView reference:



TOP: bnk | true_false **TYPE:** true_false

35. The inverse of the function $f(x) = 5/4 x$ – 5 is _____.

TOP: bnk | completion **TYPE:** completion

36. the point that is the result of a transformation of the original point on the graph

TOP: bnk | short_answer **TYPE:** short_answer

37. a point that is mapped to itself by the function

TOP: bnk | short_answer **TYPE:** short_answer

38. a rule of correspondence established between sets that associates each element of a set with an element in the same or another set

TOP: bnk | short_answer **TYPE:** short_answer

39. a relation that undoes another relation

TOP: bnk | short_answer **TYPE:** short_answer

40. a change made to a figure or graph of a relation that results in a shift or change in shape of the figure or graph

TOP: bnk | short_answer **TYPE:** short_answer

41. Create a graph of x^2 for each base function given, using transformations. a) [Graph diagram: Graph: x^2 | x: [$ −10.0, 10.0$], y: [$0.0, 100.0$]] b) [Graph diagram: Graph: $abs(x)$ | x: [$−10.0, 10.0$], y: [$0.0, 10.0$]] Rendered ExamView reference:

$$g(x) = f(x - 1) + 2$$

$$f(x) = x^2$$

$$f(x) = |x|$$

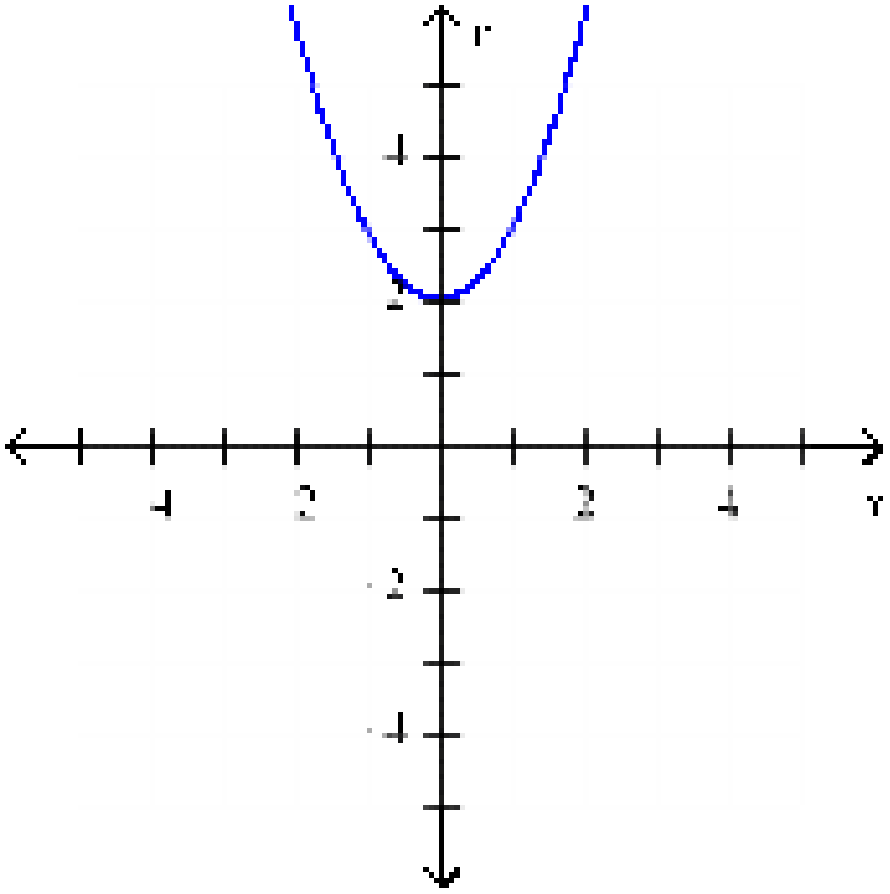
TOP: bnk | problem **TYPE:** problem

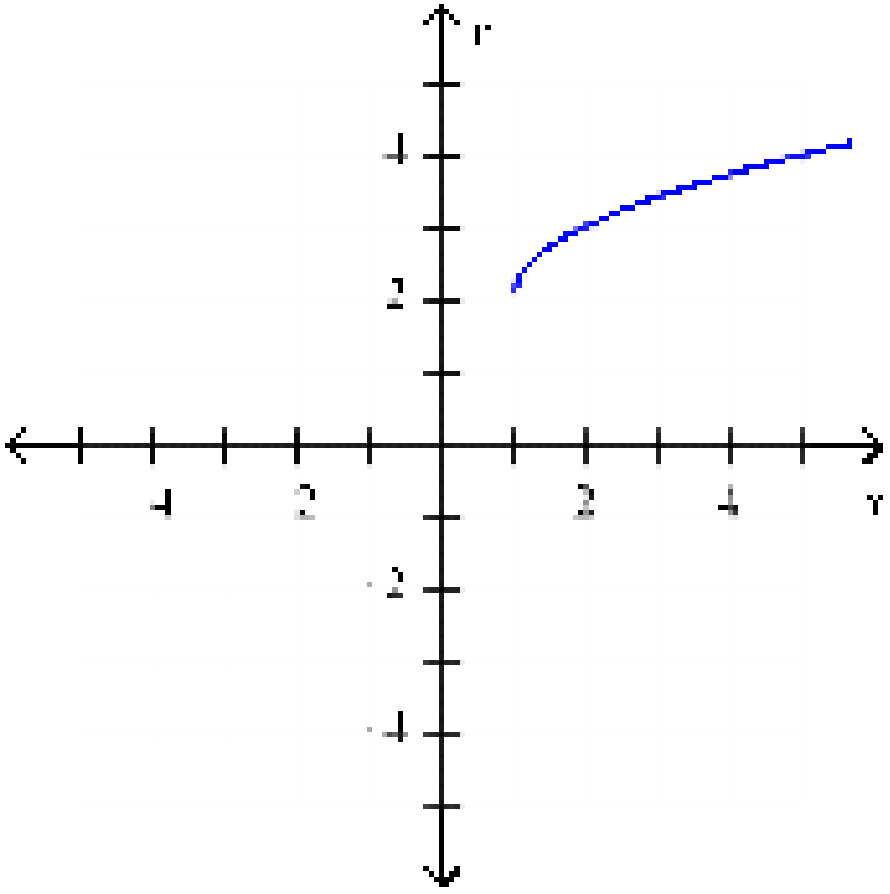
42. Determine the equation, in standard form, of each parabola after being transformed from $f(x) = x^2$ by the given translations.

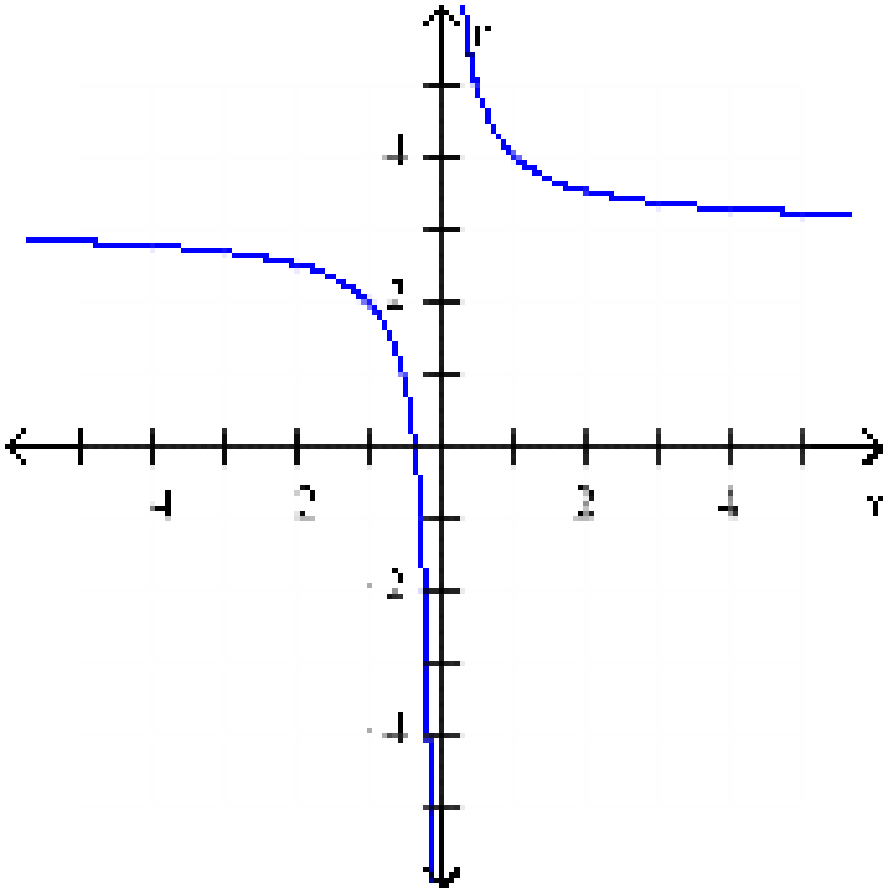
(a) 4 units to the right and 3 units up
(b) 2 units to the left and 1 unit up
(c) 2 units down and 7 units to the left

TOP: bnk | problem **TYPE:** problem

43. Given the graph of a function, sketch the resulting graph after the specified transformation. a) reflection in the $x-$axis $x^2+2 b)$ reflection in the $y-$axis [Graph diagram: Graph: x^2+2 | x: [$−10.0, 10.0$], y: [$2.0, 102.0$]] c) reflection in the $x-$axis and the $y-$axis [Graph diagram: Graph: $-x^2-2$ | x: [$−10.0, 10.0$], y: [$−102.0, −2.0$]] Rendered ExamView reference:


y
4
2
-4
-2
2
4
x
-2
-4


y
4
2
-4
-2
2
4
x
-2
-4



TOP: bnk | problem **TYPE:** problem

44. Determine the equation of the function $g(x)$ after the indicated reflection.

(a) $f(x) = x - 1 2 + 2$, in the $x-$axis
(b) $f(x) = x + 1$, in the $y-$axis

TOP: bnk | problem **TYPE:** problem

45. (a) Sketch the graph of $4*x$ for each base function.
(b) [Graph diagram: Graph: x^2 | x: [$−10.0, 10.0$], y: [$0.0, 100.0$]] ii) [Graph diagram: Graph: $4*x$ | x: [$−10.0, 10.0$], y: [$−40.0, 40.0$]] iii) [Graph diagram: Graph: $8*x^2$ | x: [$−10.0, 10.0$], y: [$0.0, 800.0$]] b) Write the equation for $g(x)$ to represent a single stretch that results in the same graph as in each function in part a).
(c) Describe how each stretch affects the domain and range for each function.

TOP: bnk | problem **TYPE:** problem

46. For each $g(x),$ describe the transformation(s) from the base function $f(x) = x^2$. a) $g(x) = 1 / 2 x^2$ b) $g(x) = x^3 2$ c)

$$f(x) = x^2$$

Rendered ExamView reference:

$$g(x) = \frac{1}{2}x^2$$

$$g(x) = \left(\frac{x}{3}\right)^2$$

$$g(x) = 5\left(\frac{x}{2}\right)^2$$

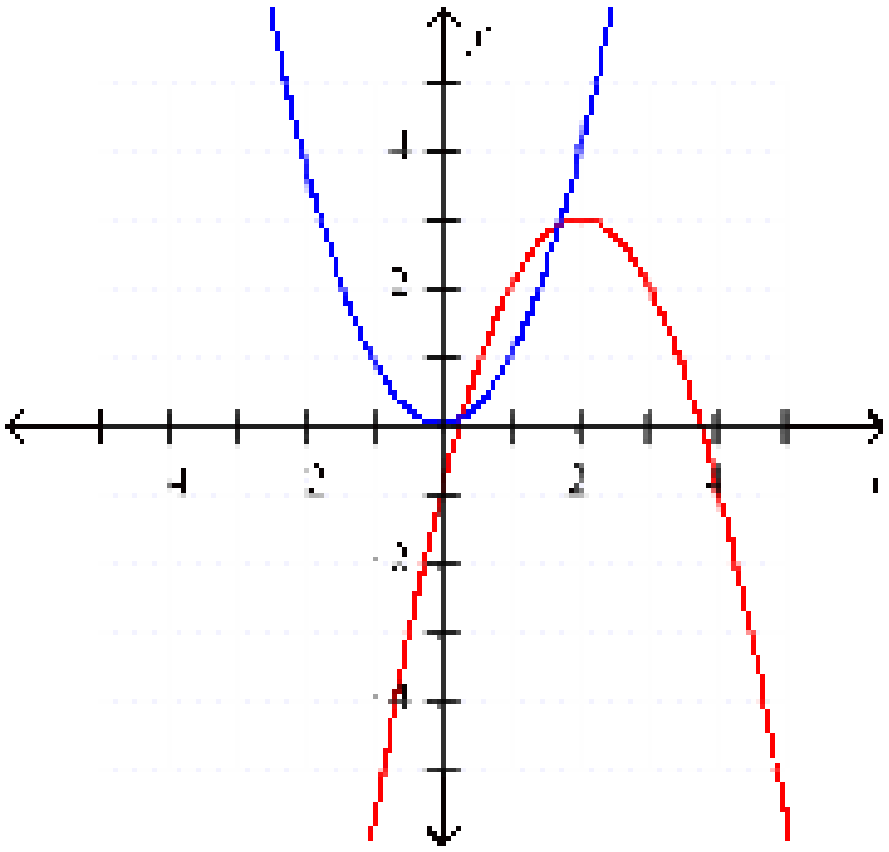
TOP: bnk | problem **TYPE:** problem

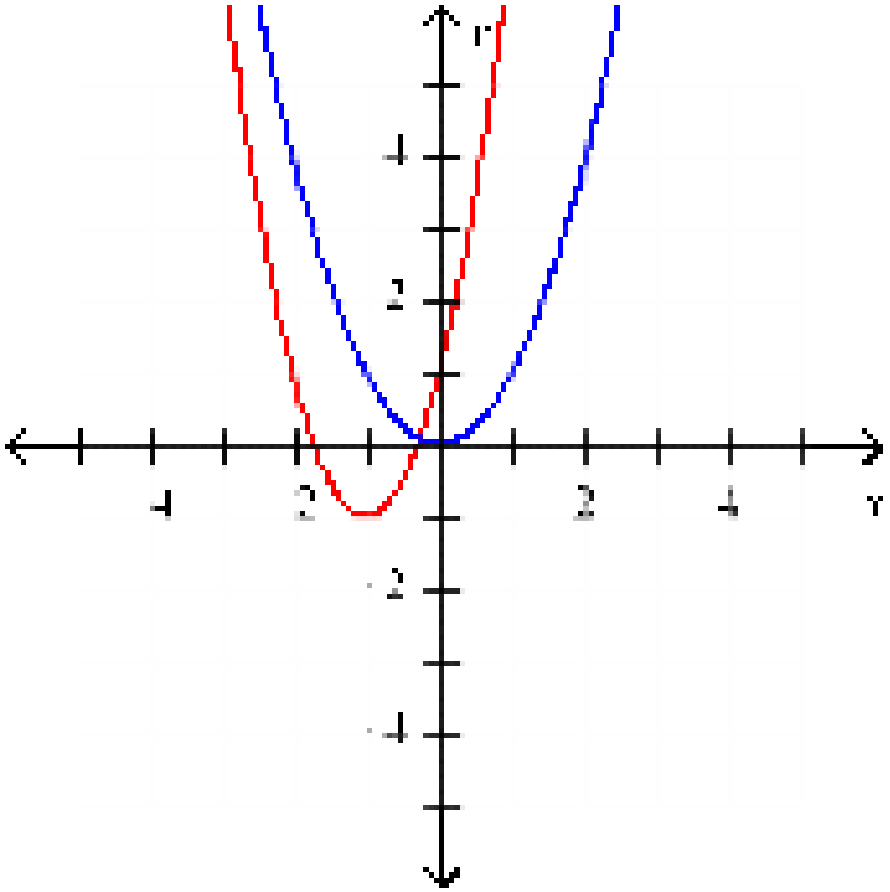
47. For each $g(x)$, describe, in the appropriate order, the combination of transformations that must be applied to the base function $f(x) = x$. a) $g(x) = f(−2(x + 1)) - 2$ b) $g(x) = 2f(x - 3) - 4$ c) $g(x) = −1/2 f(5 - x) + 1$

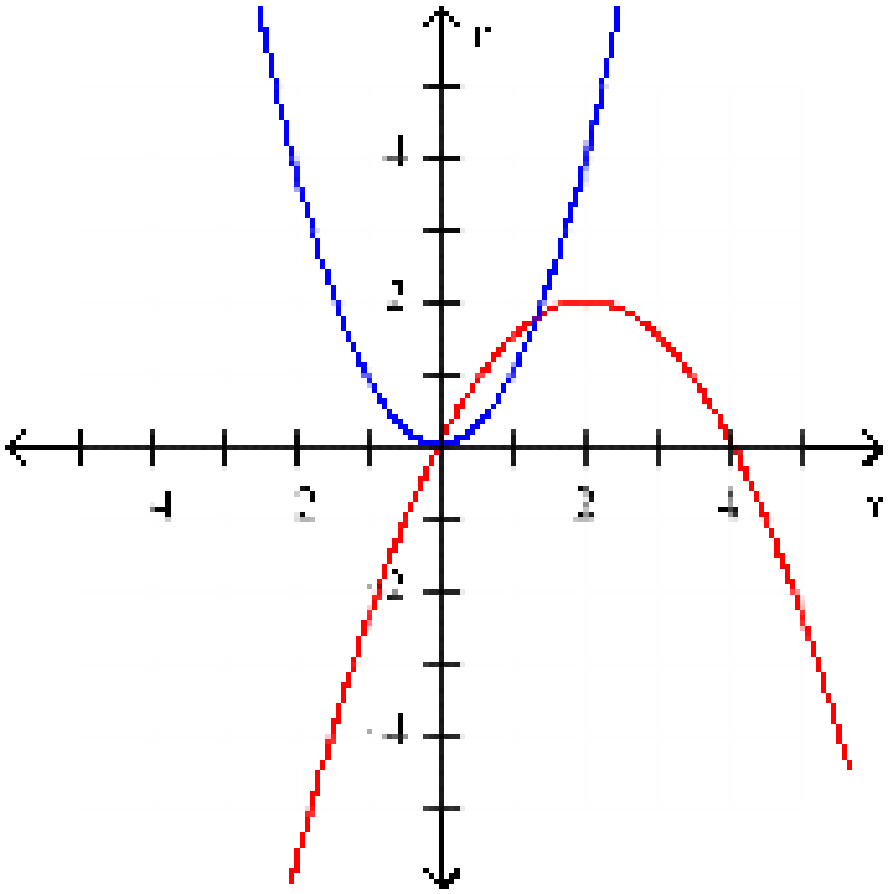
TOP: bnk | problem **TYPE:** problem

48. For each of the following, describe the combination of transformations that must be applied to the graph of $x^2 ($shown in blue) to obtain the graph of $g(x) ($shown in red). a) [Graph diagram: Graph: x^2 | x: [$−10.0, 10.0$], y: [$0.0, 100.0$]] b) [Graph diagram: Graph: x^2 | x: [$−10.0, 10.0$], y: [$0.0, 100.0$]] c) [Graph diagram: Graph: x^2 | x: [$ −10.0, 10.0$], y: [$0.0, 100.0$]] Rendered ExamView reference:

$$f(x) = x^2$$







TOP: bnk | problem **TYPE:** problem

49. For each function $f(x), i)$ determine $2.5*x-3$ ii) graph $f(x)$ and its inverse a) [Graph diagram: Graph: $2.5*x-3$ | x: [$−10.0, 10.0$], y: [$−28.0, 22.0$]] b) [Graph diagram: Graph: $0.4*x+1.2$ | x: [$−10.0, 10.0$], y: [$−2.8, 5.2$]] Rendered ExamView reference:

$$f(x) = 3x + 7$$

TOP: bnk | problem **TYPE:** problem

50. Which graph represents the inverse of the graph shown? Original graph: $3*x+2$. Inverse relation: $(x-2)/3$. Rendered ExamView reference:

$$f(x) = 2x + 4$$

$$f^{-1}(x)$$

$$f^{-1}(x)$$

TOP: bnk | problem **TYPE:** problem

51. (a) Determine the inverse of the function $f(x) = x^2 - 4x + 5$.
(b) Determine the domain and range of the function and its inverse.
(c) Is the inverse a function? Explain.

TOP: bnk | problem **TYPE:** problem

52. The approximate height, h, in metres, of an object above the ground after it is dropped from a distance, d, in metres, can be modelled by the function $400,$ where t is the time, in seconds, after being dropped.

(a) Determine the function for an object being dropped from each height.
(b) $550 m$

(i) $850 m$
(ii) $400 m

(a) $ Graph all three functions on the same set of axes.
(b) Describe the transformation that relates the highest curve to the lowest curve.
(c) Describe the transformation that relates the highest curve to the curve in the middle.

TOP: bnk | problem **TYPE:** problem

53. To rent a cottage for the months of July and August, the renter must pay a flat fee of $150 plus a daily amount of $$100$. a) Write a function that represents the total cost, C, in dollars, of the rental as a function of the number of days, d, that the cottage is rented.

(a) Determine the domain and range of the function.
(b) In the months of November and December, the renter only needs to pay a flat fee of $50, but the daily fee is still $ 100. Write a function that represents the total cost, T, in dollars, as a function of the number of days, d, of the rental.
(c) Describe a transformation that relates the function $C(d)$ to the function $T(d)$.

TOP: bnk | problem **TYPE:** problem

54. Consider the function $f(x) = 2^(x) - 1 2 - 3$. a) Determine the equation of each function.

(a) $f(x)$ ii) $f($–x)
(b) From the graph, determine the pairs of equations that can be represented as translations of each other.
(c) Describe the translation that can be applied to each pair of functions you determined in part c) to generate the same graph.

TOP: bnk | true_false **TYPE:** true_false

55. For the base function x^2, a reflection in the $y-$axis is followed by a reflection in the $x-$axis.

(a) Discuss the effect of each reflection on the base function.
(b) Is there a point that you expect to be invariant during these reflections? If so, state its coordinates.
(c) Graph the function after each reflection.
(d) Use your graph in part c) to verify your answer for part b).
(e) Make a general statement regarding invariant points after reflections in the $x-$axis and the $y-$axis.

TOP: bnk | problem **TYPE:** problem

56. For the base function $abs(x), a$ reflection in the $y-$axis is followed by a reflection in the $x-$axis.

(a) Discuss the effect of each reflection on the base function.
(b) Is there a point that you expect to be invariant during these reflections? If so, state its coordinates.
(c) Graph each function after the reflections.
(d) Use this graph to verify your answer for part b).
(e) Make a general statement regarding invariant points after reflections in the $x-$axis and the $y-$axis.

TOP: bnk | problem **TYPE:** problem

57. The kinetic energy (energy of motion), E, in joules, of an object is given by the equation $7.5*x^2,$ where m represents the mass of the object, in kilograms, and v represents its speed, in metres per second.

(a) Determine the equation for the kinetic energy of a mass of 15 kg as a function of its speed.
(b) Find the kinetic energy of an object of mass 15 kg moving at
(c) $5 m/s$

(i) $10 m/s$
(ii) $15 m/s$
(iii) $20 m/s

(a) $ Determine the equation for the kinetic energy of a mass of 20 kg as a function of its speed.
(b) Find the kinetic energy of an object of mass 20 kg moving at i) $5 m/s$

(i) $10 m/s$
(ii) $15 m/s$
(iii) $20 m/s

(a) $ Graph the functions in parts b) and d) on the same set of axes.
(b) Explain how the two functions are vertical stretches of each other, and determine the value of the vertical stretch factor.

TOP: bnk | problem **TYPE:** problem

58. An object falls to the ground from a height of $25 m$. The height, h, in metres, of the object above the ground can be modelled by the function $−4.9*x^2+25,$ where a is the acceleration due to gravity, in metres per second squared, and t is the time, in seconds.

(a) Write an equation for the height of the object on Earth given a $= 9.8 m/s_2$.
(b) Write an equation for the height of the object on Mars given a $= 3.7 m/s_2$.
(c) Graph both functions on the same set of axes.
(d) What scale factor can be applied to the Earth function to transform it to the Mars function?

TOP: bnk | problem **TYPE:** problem

59. The base function $g (x) .$ is reflected in the $x-$axis, stretched horizontally by a factor of $2,$ compressed vertically by a factor of $f(x) = x$, and translated 3 units to the left and 5 units down.

(a) Write the equation of the transformed function $g(x)$.
(b) Graph the original function and the transformed function on the same set of axes.
(c) Which transformations must be done first but in any order?
(d) Which transformations must be done last but in any order?

TOP: bnk | problem **TYPE:** problem

60. The height, h, in metres, of a flare fired into the air from a boat can be modelled using the function $x^2,$ where t is the time, in seconds, after the flare is fired.

(a) Explain how this function can be obtained from the function x^2 using a series of transformations.
(b) Graph the base function and the transformed function on the same set of axes.
(c) Use the graph to estimate i) how long the flare is in the air

TOP: bnk | problem **TYPE:** problem

61. Which graph represents the inverse of the graph shown? Original graph: $0.5*x-2$.

TOP: bnk | problem **TYPE:** problem

62. Consider the function $x - 1, x$.

(a) State the domain and range of the function.
(b) Use this information to determine the domain and range of the inverse of the function.
(c) Determine the inverse of the function.
(d) Graph the function and its inverse on the same set of axes (include the line y $= x$ to verify the inverse).

TOP: bnk | problem **TYPE:** problem

63. The cost of renting a car for a day is a flat fee of $50 plus $$0.12$ for each kilometre driven. Let C represent the total cost of renting a car for a day if it is driven a distance, x, in kilometres.

(a) Write the total cost function for the car rental.
(b) Determine the inverse of this function.
(c) What does this inverse function represent?
(d) Give an example of how this function can be used.

TOP: bnk | problem **TYPE:** problem

64. The graph of $f(x) = x^2$ is transformed to the graph of [DIAGRAM NEEDED].

(a) Describe the two translations represented by this transformation.
(b) Determine three points on the base function. Horizontally translate and then vertically translate the points. What are the three resulting image points?

TOP: bnk | problem **TYPE:** problem

65. Graph $f(x)$ and then $f(x)$ and $f(x)$ to determine the invariant points. $2 a)$ From the graph, the invariant points are ($1, 0)$ and ($5, 0)$. b) From the graph, the invariant point is ($0, 5)$. Recovered graph functions: ($x-2)^2-9; (-x-2)^2-9; -((x-2)^2-9)$. Rendered ExamView reference:

$$f(x) = (x-2)^2 - 9$$

TOP: bnk | problem **TYPE:** problem

66. The first skydiver s motion can be modelled by the function $4000-5*x^2$. The second skydiver jumps $10 s$ later with a goal of catching up to the first skydiver. The motion of the second skydiver can be modelled by [Graph diagram: Graph: $4000-5*x^2$ | x: [$−10.0, 10.0$], y: [$3500.0, 4000.0$]]. For both functions, the height is measured in metres and the time is the number of seconds after the second skydiver jumps.

(a) Graph the functions on the same set of axes.
(b) Will the second skydiver catch up to the first before they have to open their parachutes at $800 m$?
(c) State the domain and range of these functions in this context.

TOP: bnk | problem **TYPE:** problem

67. The value, V, in thousands of dollars, of a certain car after t years can be modelled by the equation $35/(x+3)$.

(a) Sketch the graph of this relation using transformations of the function $35/(x+3)$.
(b) What is the initial value of this car?
(c) What is the projected value of this car after 2 years?

TOP: bnk | problem **TYPE:** problem

68. In ballistics, the paths of projectiles are studied. It is known that the greater the speed at which an object is fired, the farther it will travel. For a particular spring$-$loaded cannon, this relationship can be modelled by $x^2/10,$ where d is the distance, in metres, travelled by the projectile and v is the muzzle speed, in metres per second.

(a) Graph the distance function and state its domain and range.
(b) Graph the inverse of the distance function. What does it represent? State its domain and range.

TOP: bnk | problem **TYPE:** problem

69. The equation $9*x/5+32$ can be used to convert between Celsius and Fahrenheit temperatures, where x is the temperature in degrees Celsius and y is the temperature in degrees Fahrenheit.

(a) Determine the inverse of this equation. What does it represent? What do the variables represent?
(b) Graph the original and inverse functions on the same set of axes.
(c) Which temperature is the same in Celsius and Fahrenheit? Explain how you know.

TOP: bnk | problem **TYPE:** problem

70. Jai works at an electronics store. She earns $600 a week, plus commission of 5% of her sales.

(a) Write a function to describe Jai s total weekly earnings as a function of her sales.
(b) Determine the inverse of this function.
(c) One week Jai earned $$775$. What were her sales that week?

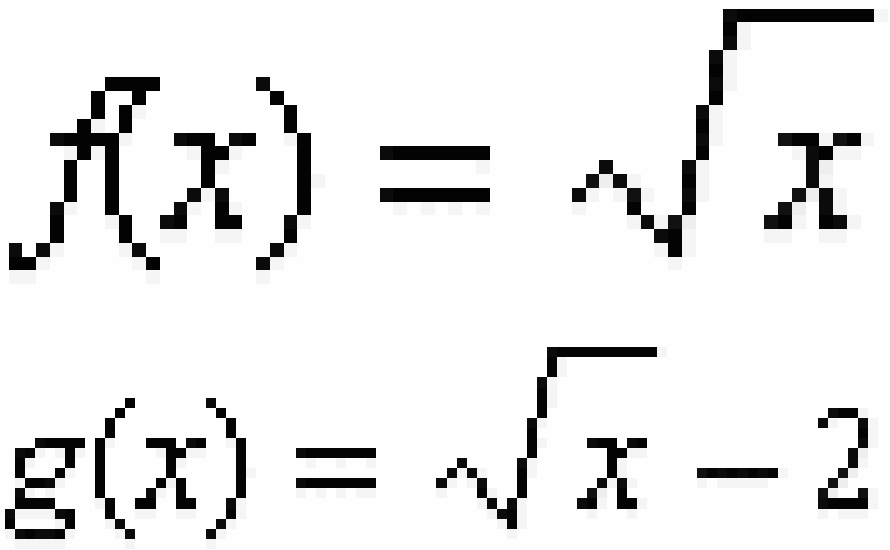
TOP: bnk | problem **TYPE:** problem

71. Given the function $3*x+7,

(a) $ determine the inverse of the function
(b) graph the function and its inverse on the same set of axes

TOP: bnk | problem **TYPE:** problem

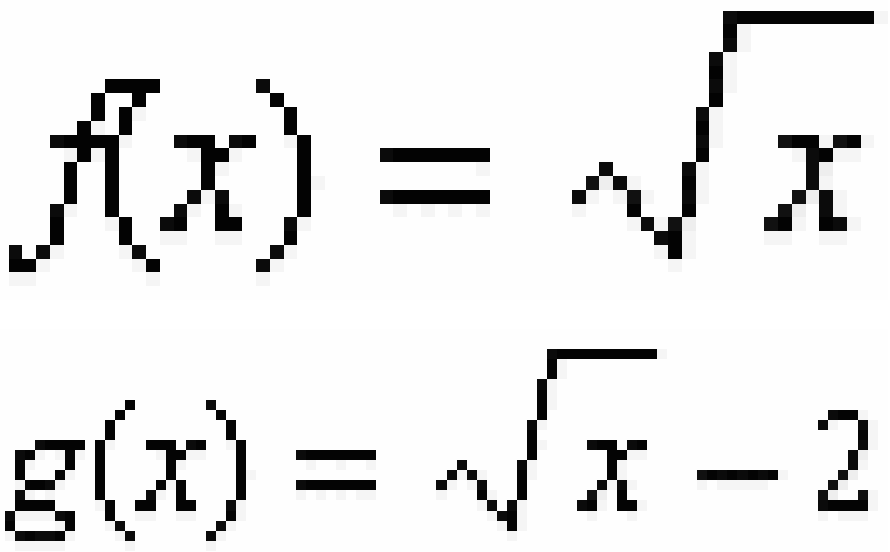
72. Compared to the graph of the base function, the graph of the function is translated 2 units down. Rendered ExamView reference:



(A) 2 units down (B) 2 units up (C) 2 units left (D) 2 units right

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

73. Compared to the graph of the base function, the graph of the function is translated 8 units down. Rendered ExamView reference:



(A) 8 units down (B) 8 units up (C) 8 units left (D) 8 units right

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

74. Compared to the graph of the base function, the graph of the function is translated 5 units right. Rendered ExamView reference:

$$f(x) = \sqrt{x}$$

$$g(x) = \sqrt{x - 5}$$

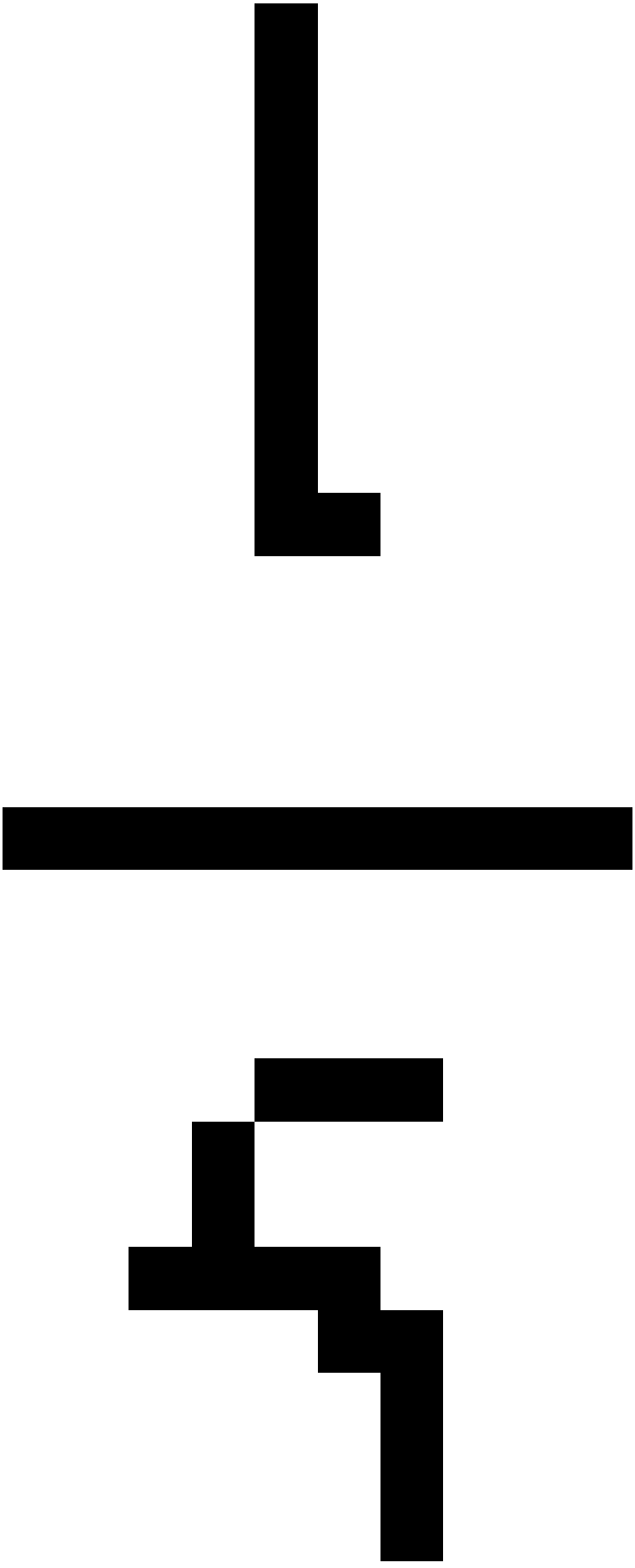
(A) 5 units right (B) 5 units left (C) 5 units down (D) 5 units up

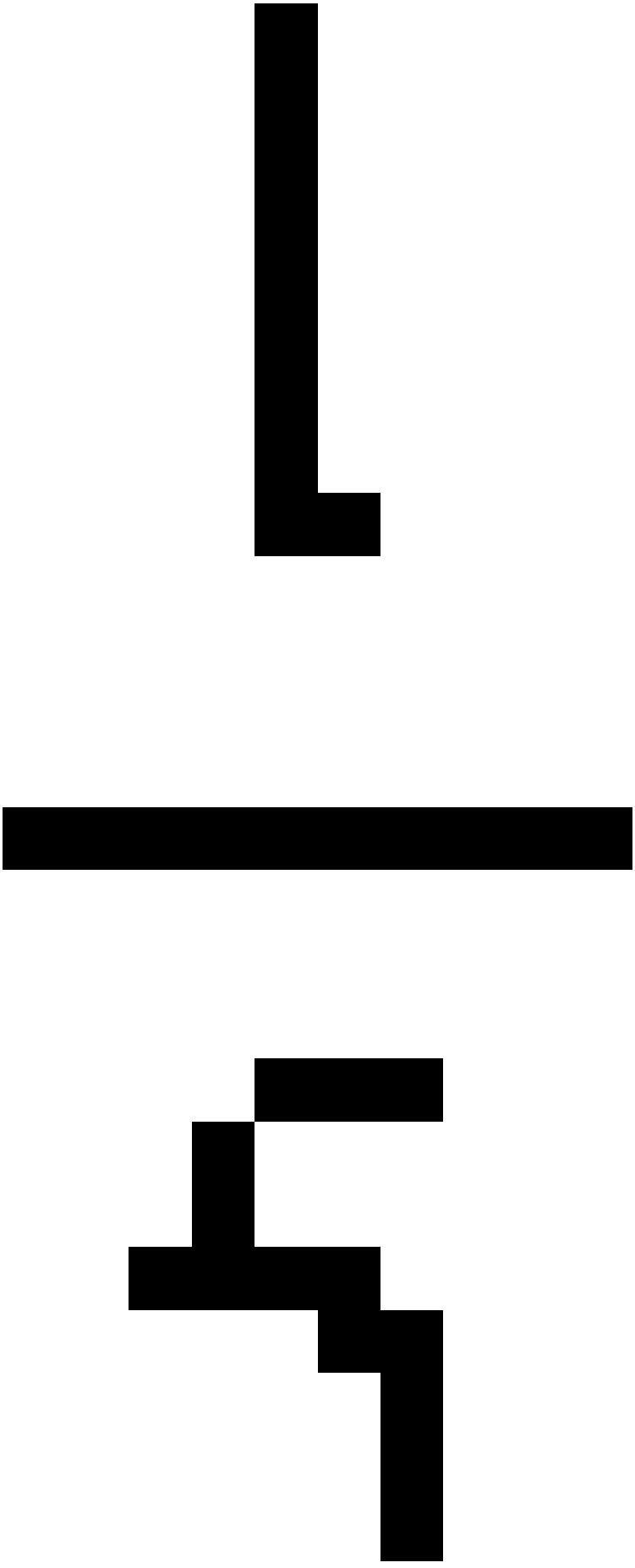
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

75. Compared to the graph of the base function $f(x) = x$, the graph of the function $g(x) = b x$ is Rendered ExamView reference:

$$f(x) = \sqrt{x}$$

$$g(x) = \sqrt{5x}$$





(A) compressed by a factor of 5 and 1/5

(B) stretched by a factor of not reflected and 5

(C) compressed by a factor of reflected in the y-axis and 1/5

(D) stretched by a factor of reflected in the y-axis and 5

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

76. When b $< 0,$ the function $g(x) = bx$ has what relationship to the base function $f(x) = x$?

(A) stretched horizontally by a factor of 1/|b|

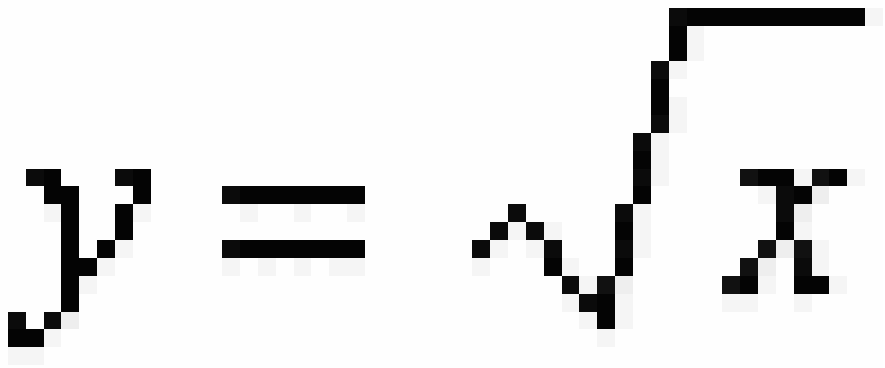
(B) stretched horizontally by a factor of 1/|b| and reflected in the y-axis

(C) stretched vertically by a factor of |b|

(D) stretched vertically by a factor of |b| and reflected in the x-axis

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

77. When the graph of a function is vertically stretched by a factor of 3, the point $(9, -10)$ is transformed to the point $(9, -30)$. Rendered ExamView reference:



(A) $(-10, -10)$

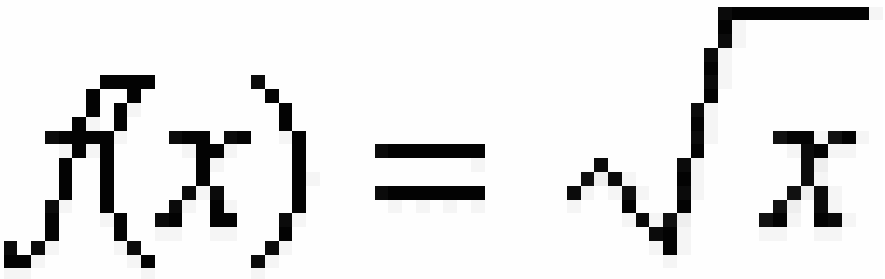
(B) $(9, 3)$

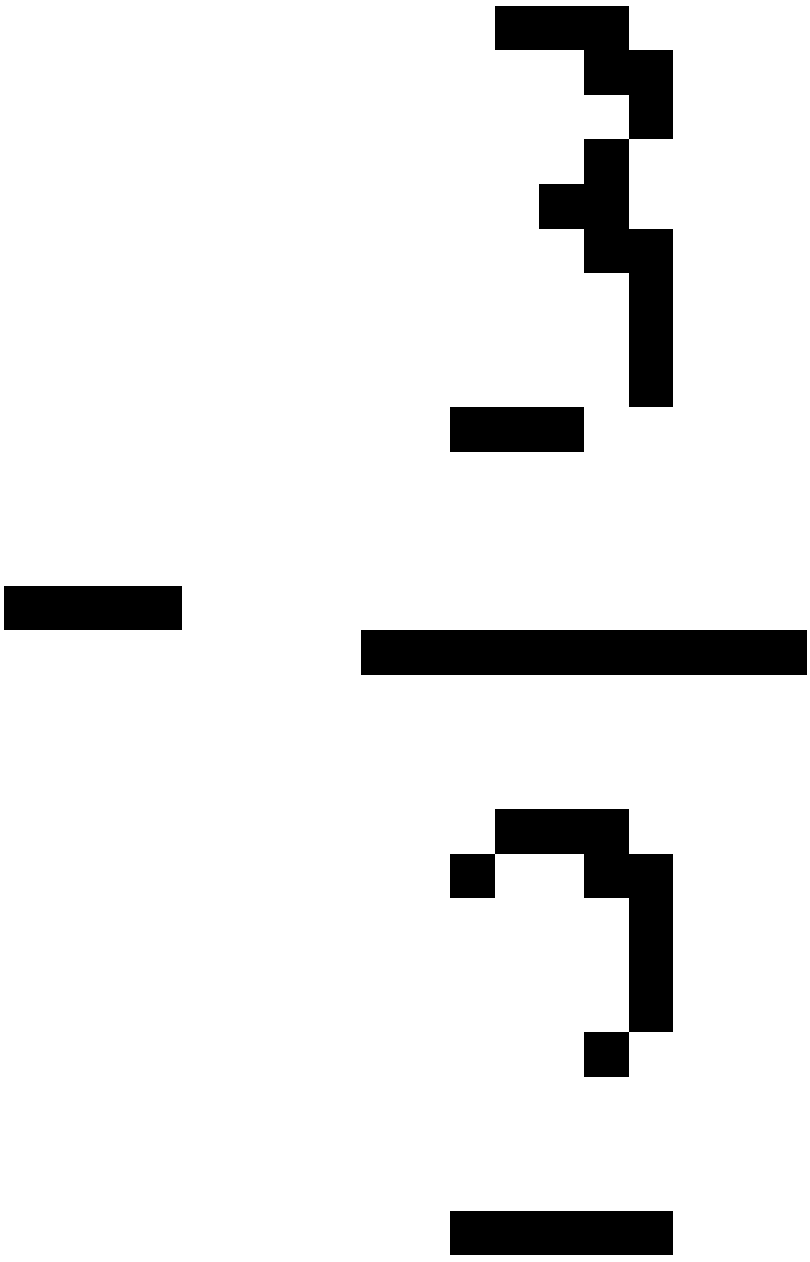
(C) $(-30, -30)$

(D) $(9, -30)$

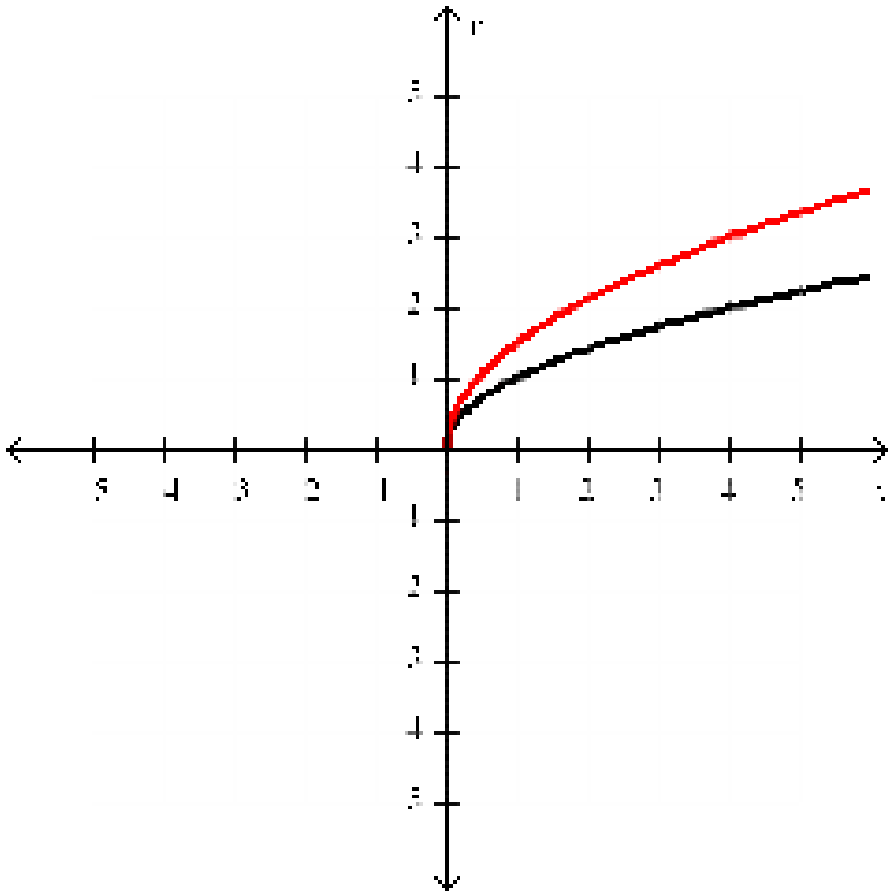
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

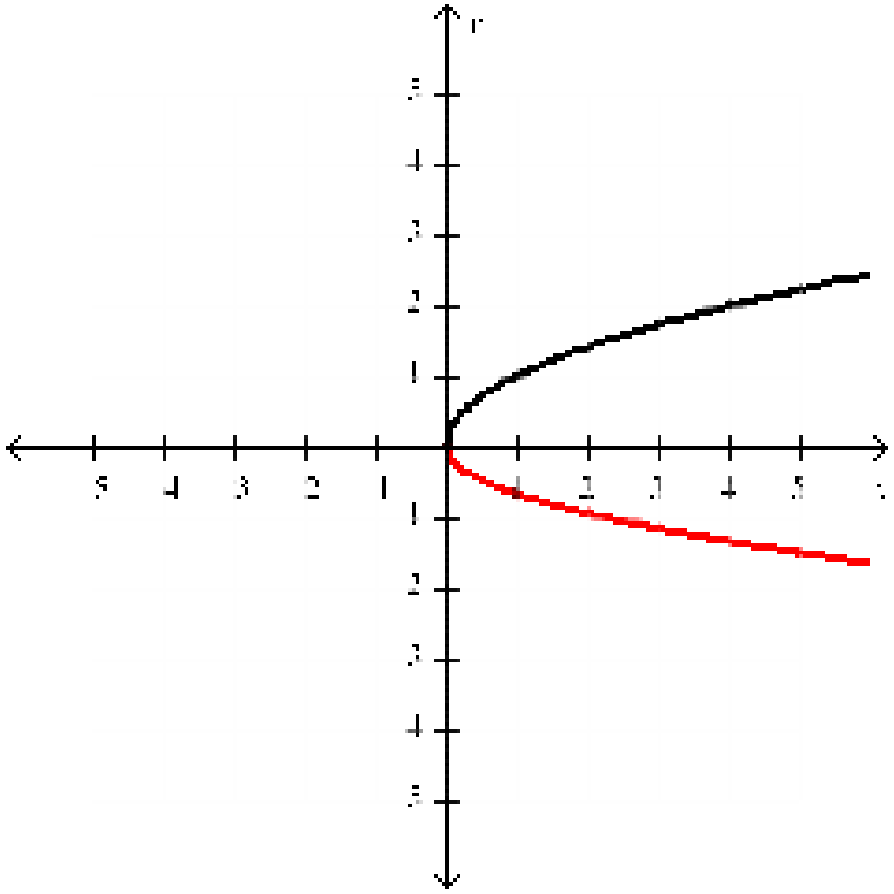
78. Which of the graphs shown below represents the base function $6 ($in black) and the stretched function $g(x) = 6 6 ($in red)? Recovered graph functions: $sqrt(x); (1.5)*sqrt(x); (−0.666666666667)*sqrt(x); ((−9)/(6))*sqrt(x); (0.666666666667)*sqrt(x)$. Rendered ExamView reference:

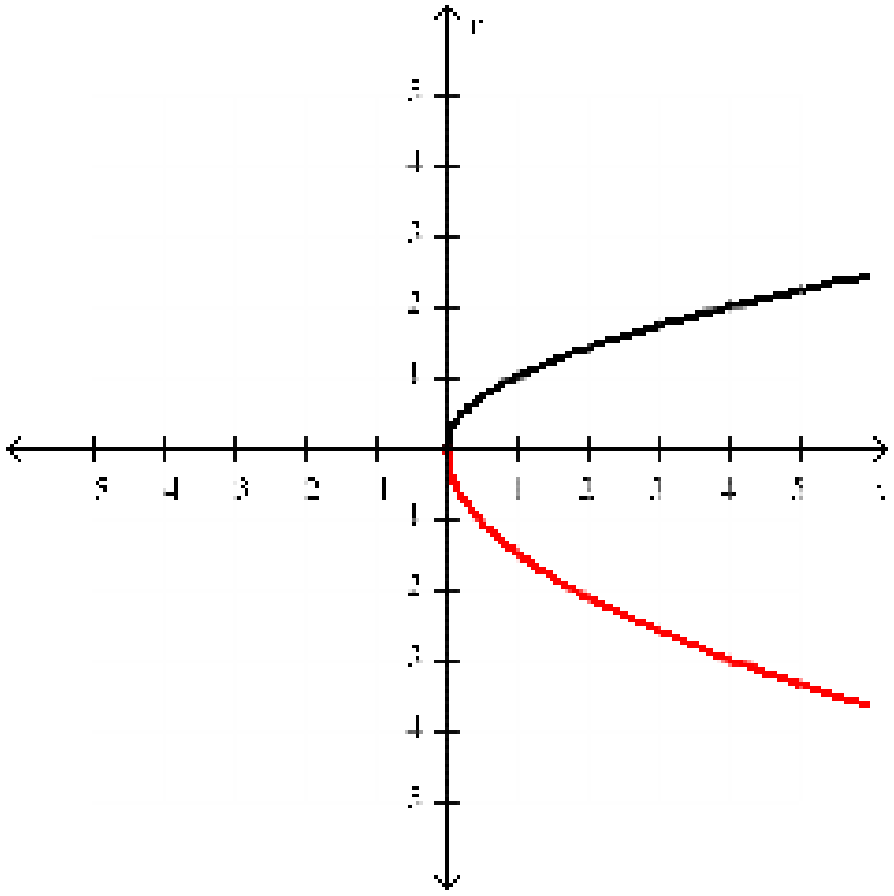


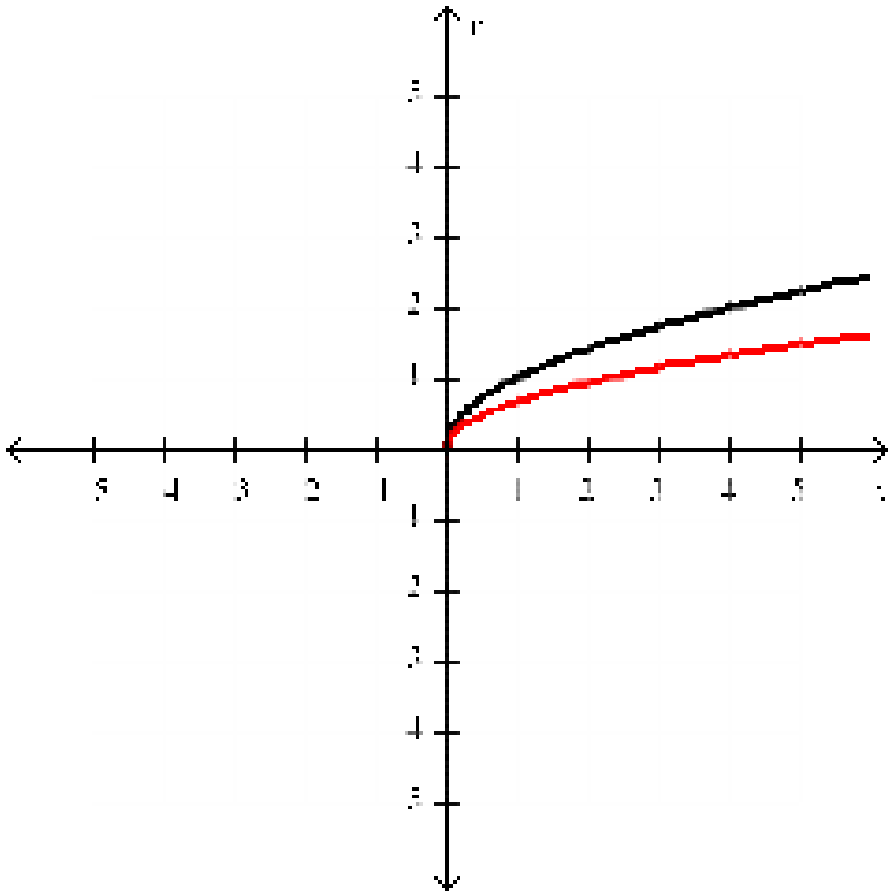


$\sqrt{x}$









(A) 1.5 (B) $−0.6666666666667$ (C) 0.666666666667

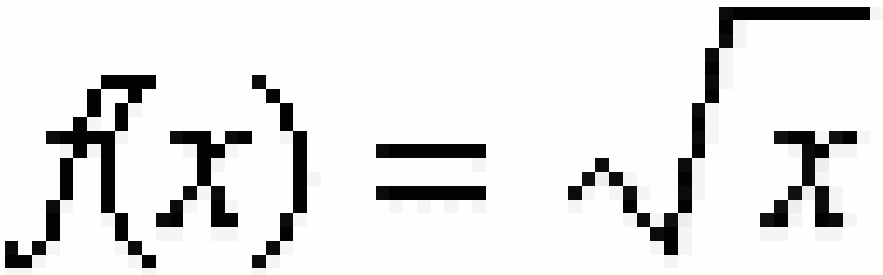
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

79. Given the function $f(x) = sqrt(x + 5) + 8$ with domain $x >= −5$ and range $y >= 8$, which of the following best describes the translations with respect to the base graph? Rendered ExamView reference:

$$f(x) = \sqrt{x - h} + k$$

$$\{x | x \geq -5,\ x \in R\}$$

$$\{y | y \geq 8,\ y \in R\}$$



(A) 5 units to the left and 8 units up

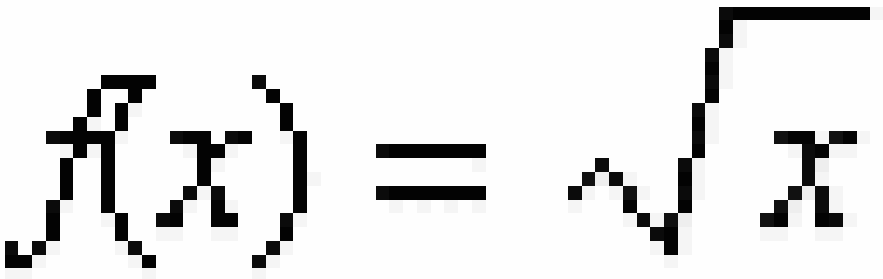
(B) 8 units to the left and 5 units down

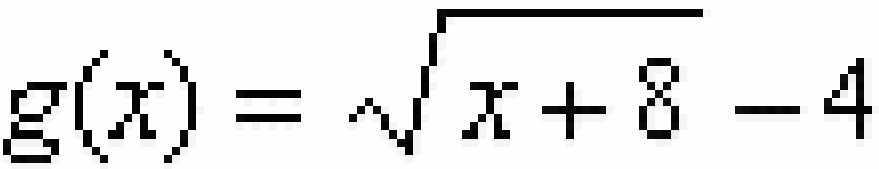
(C) 8 units to the left and 5 units up

(D) 5 units to the left and 8 units down

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

80. Compared to the graph of the base function $f(x) = x^2$, the graph of $g(x) = (x - 8)^2 - 4$ is translated 8 units left and 4 units down. Rendered ExamView reference:





(A) 4 units to the right and 8 units up

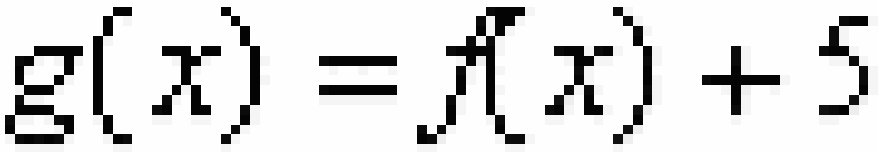
(B) 4 units to the left and 8 units down

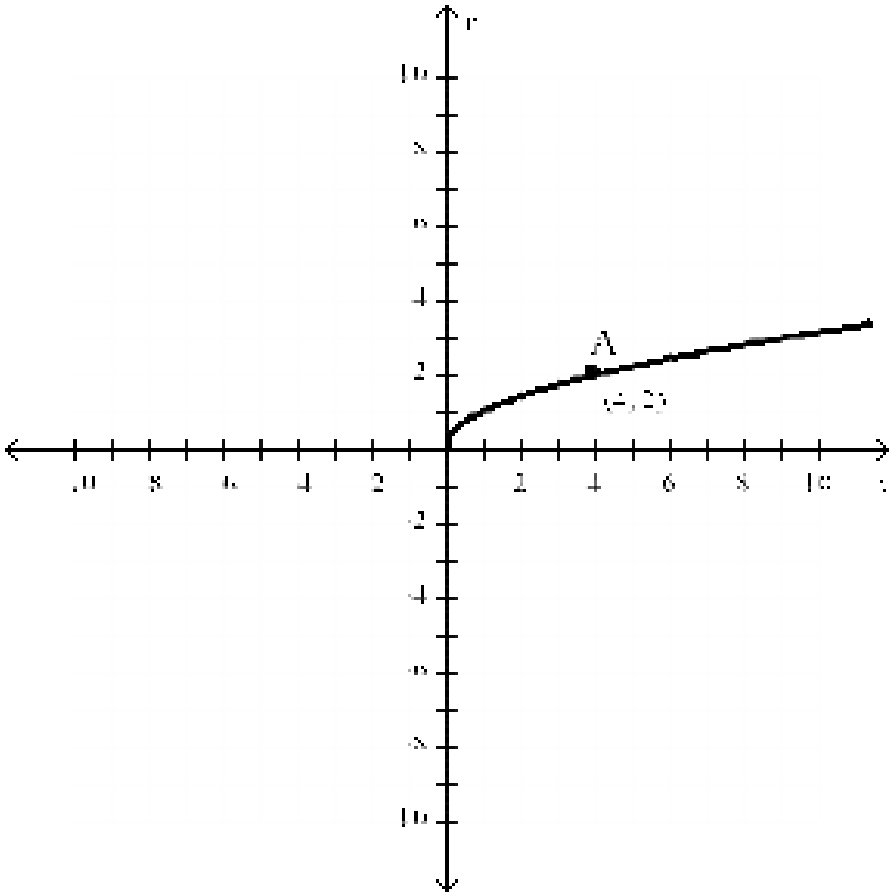
(C) 8 units to the left and 4 units down

(D) 8 units to the right and 4 units up

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

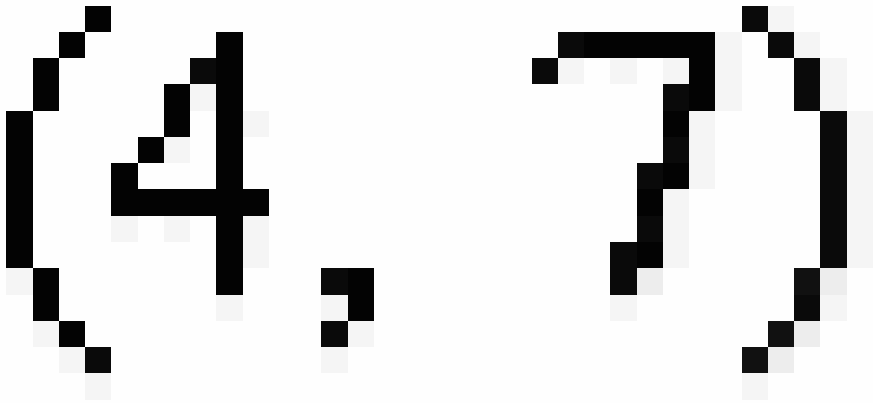
81. Given the graph of $f(x)$ shown below, point A is ($4, 2)$. What are the coordinates of point A after the transformation? Rendered ExamView reference:

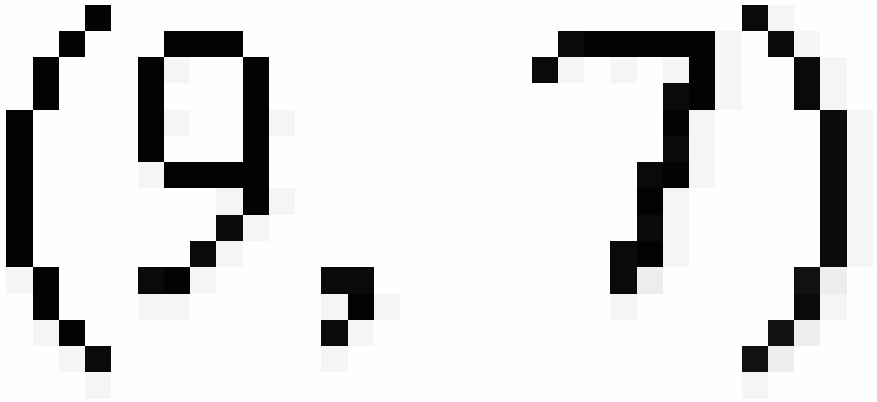




$(-1, 7)$

$(4, -3)$

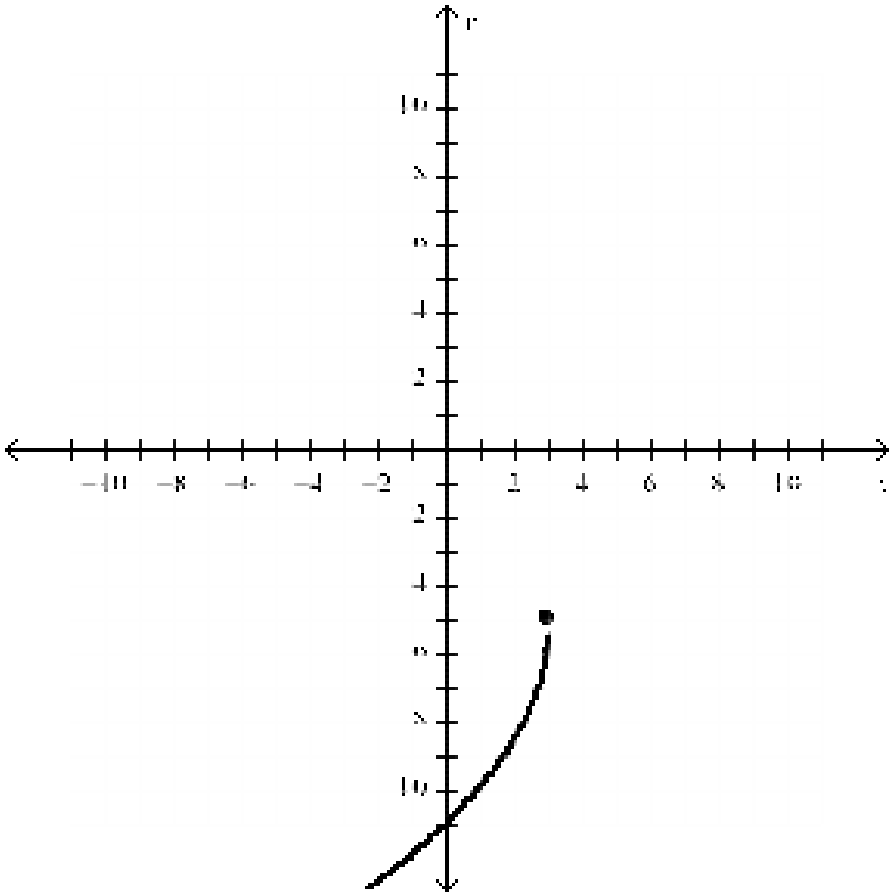




(A) (4, 7) (B) (4, –3) (C) (–1, 7) (D) (9, 7)

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

82. What is the equation of the radical function shown in the graph below? – 5 Rendered ExamView reference:



$$f(x) = -2\sqrt{\frac{1}{-3}(x-5)} - 3$$

$$f(x) = -2\sqrt{-3(x-3)} - 5$$

$$f(x) = -2\sqrt{-3(x+3)} - 5$$

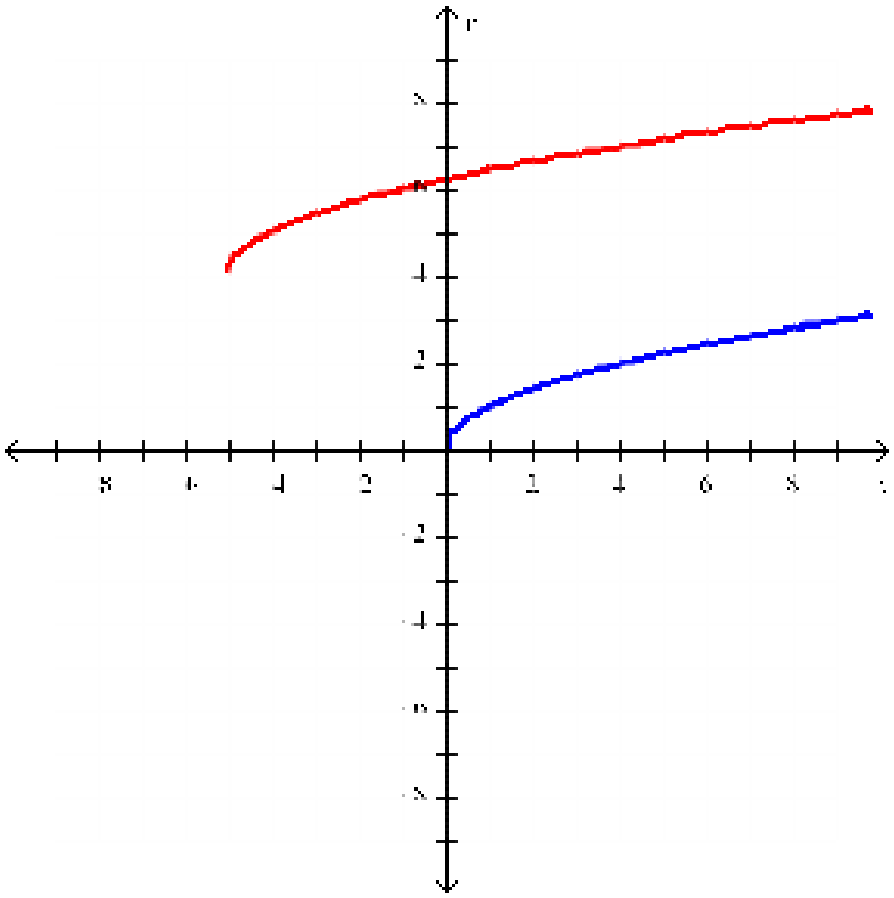
$$f(x) = -2\sqrt{\frac{1}{-3}(x-3)} - 5$$

(A) $x >= – 5$ (B) $x >= – 3$ (C) $x >= –3$ (D) $x <= – 5$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

83. What is the equation of the transformed function, $g(x),$ after the transformations are applied to the graph of the base function $+ 4,$ shown in blue, to obtain the graph of $g(x),$ shown in red? $+ 4$ Rendered ExamView reference:

$$f(x) = \sqrt{x}$$



$$g(x) + 4 = \sqrt{x+4}$$

$$g(x) + 5 = \sqrt{x+4}$$

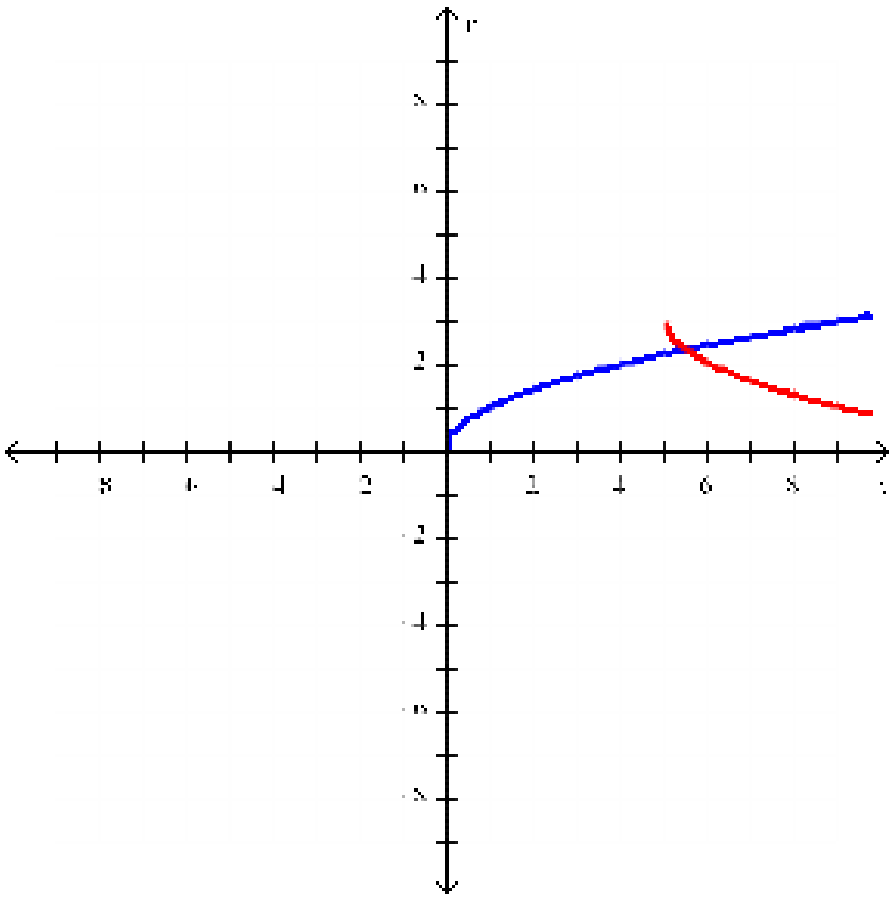
$$g(x) = \sqrt{x+4} + 5$$

$$g(x) = \sqrt{x+5} + 4$$

(A) + 4 (B) + 5 (C) + 5 (D) + 4

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

84. Which graph represents $g(x) = f(x - 5) + 3$? Rendered ExamView reference:



(A) a reflection in the x-axis, a vertical translation 5 units up, and a horizontal translation 3 units to the right

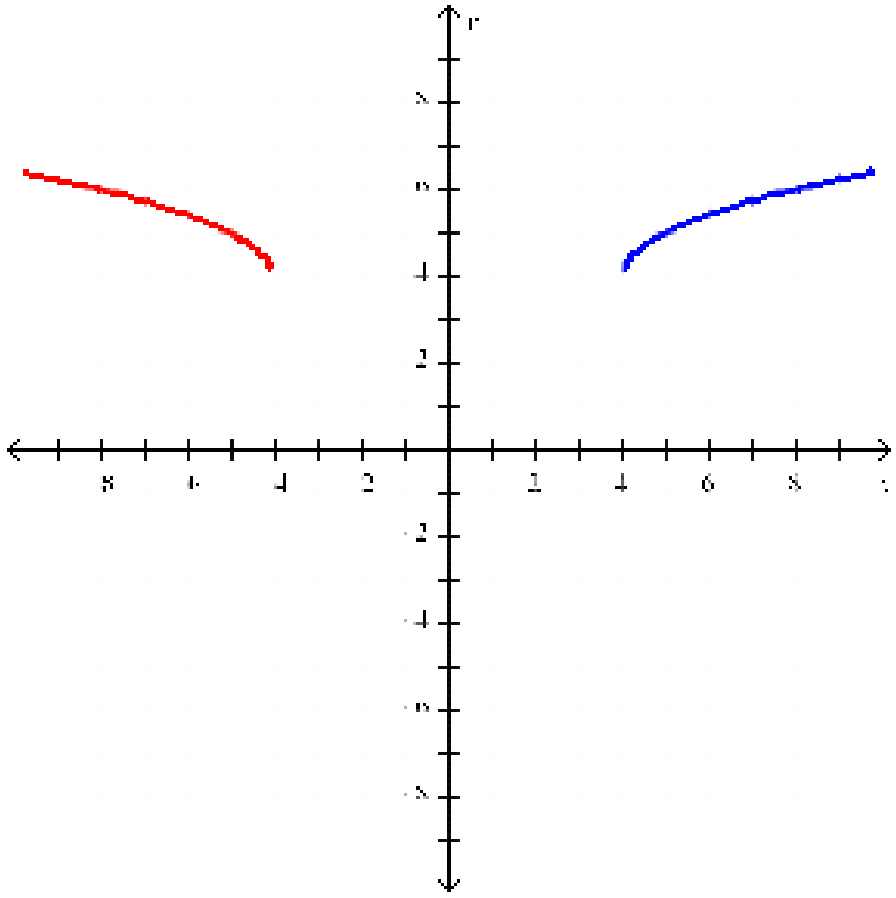
(B) a reflection in the x-axis, a vertical translation 5 units down, and a horizontal translation 3 units to the right

(C) a reflection in the x-axis, a vertical translation 3 units up, and a horizontal translation 5 units to the left

(D) a reflection in the x-axis, a vertical translation 3 units up, and a horizontal translation 5 units to the right

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

85. The two functions in the graph shown are reflections of each other. Select the type of reflection(s). Rendered ExamView reference:



(A) y-axis (B) x-axis and the y-axis (C) x-axis

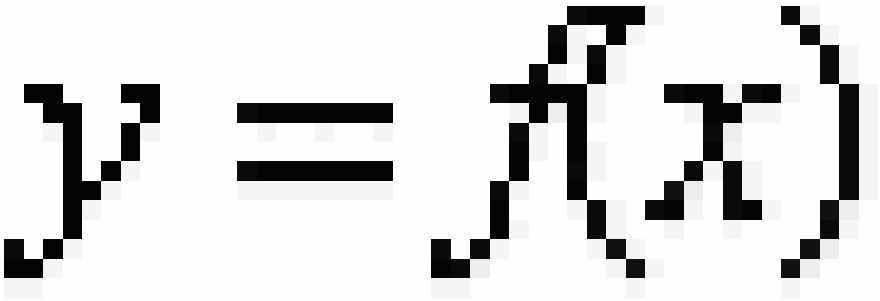
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

86. Given the function $4,$ which of the following is a restriction on the inverse function, – 4?

(A) $x >= – 4$ (B) $x >= 8$ (C) $x >= 4$ (D) $x <= – 4$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

87. Which point on the graph 3 does not exist on the graph of 4? Rendered ExamView reference:

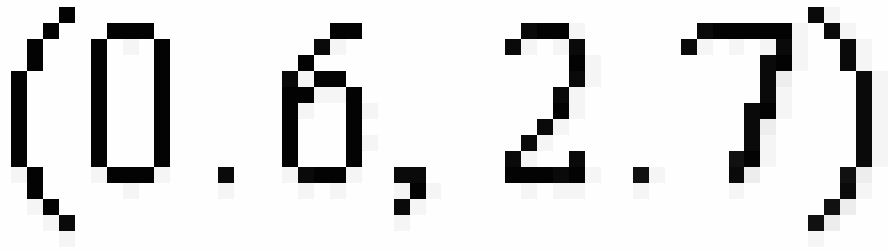


$$y = \sqrt{f(x)}$$

$$(3, 4)$$

$$(-8, -3)$$

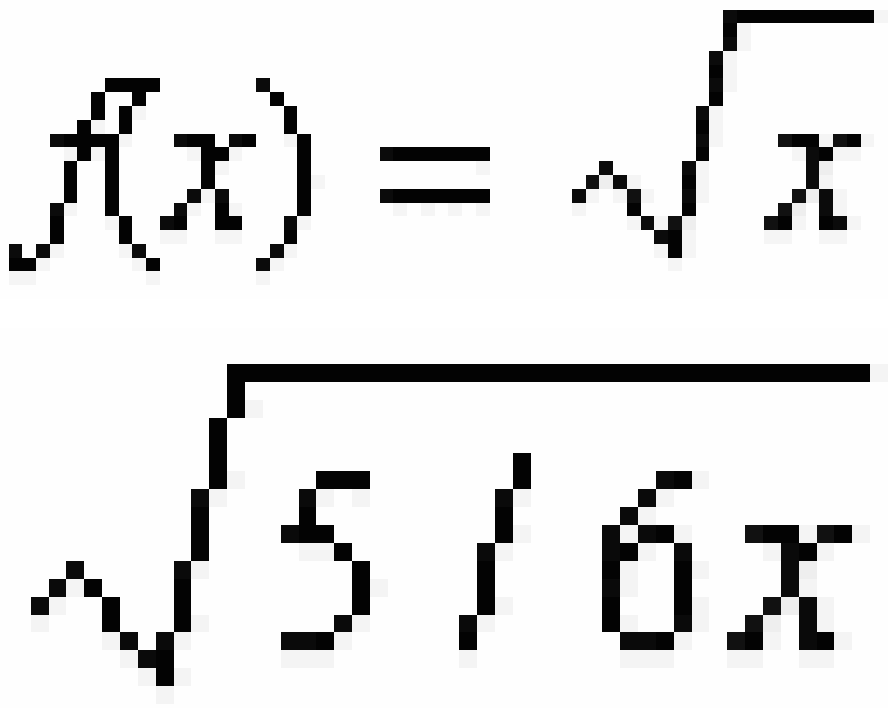
$$(-4, 4)$$

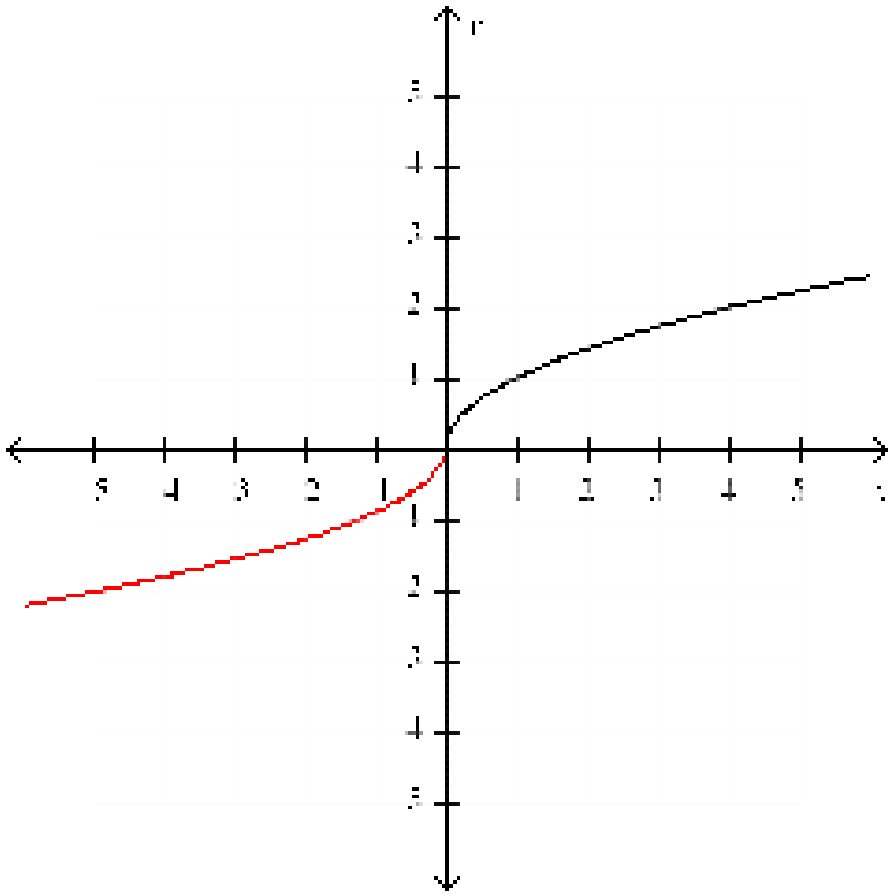


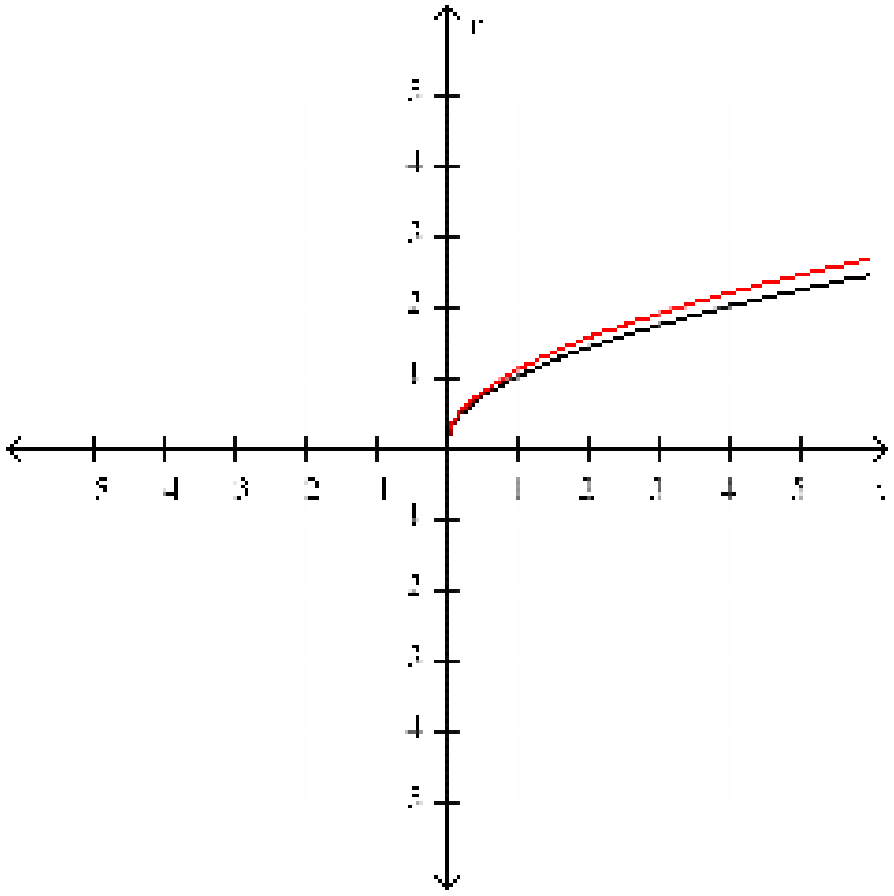
(A) (3, 4) (B) (0.6, 2.7) (C) (–4, 4) (D) (–8, –3)

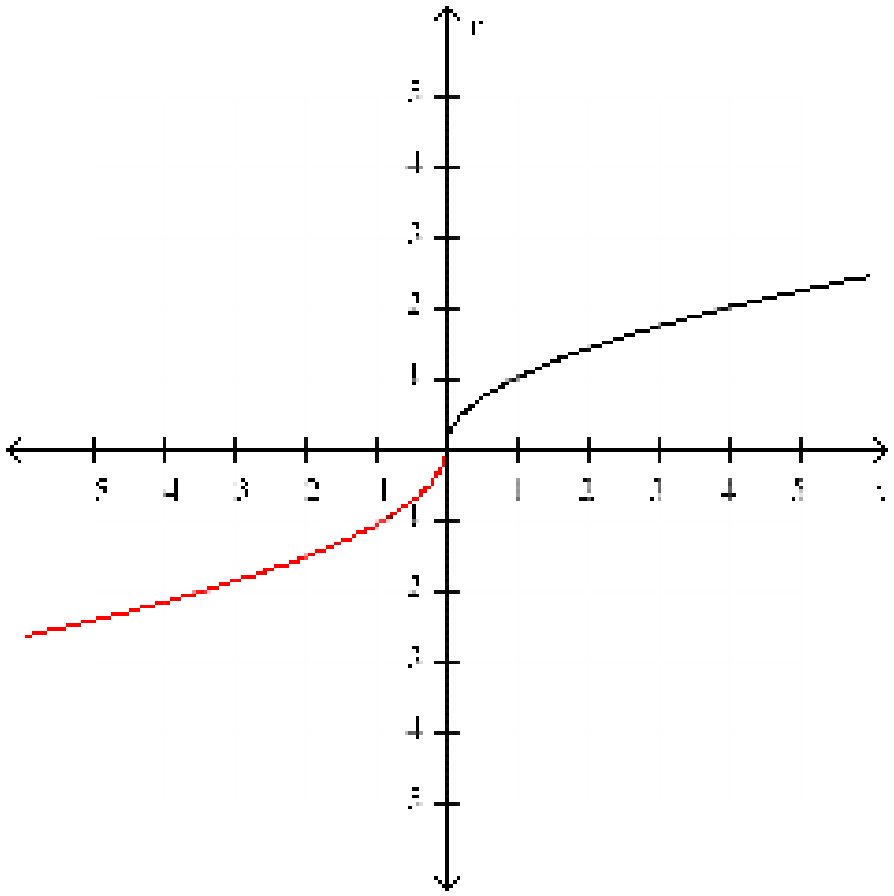
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

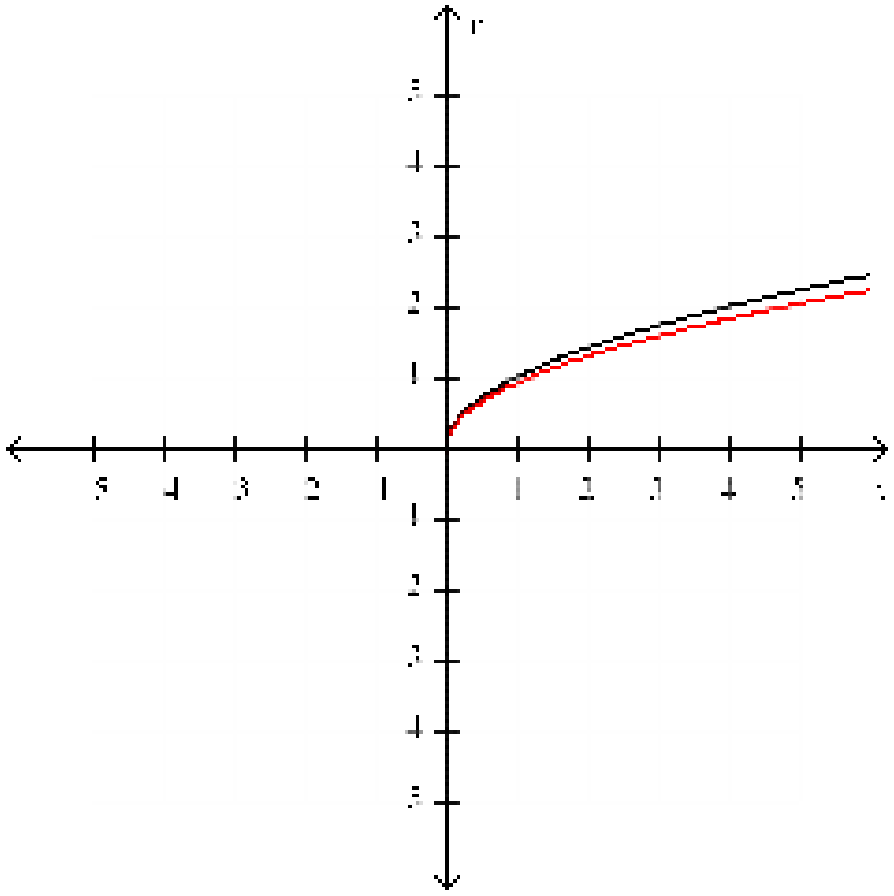
88. Which of the graphs shown below represents the base function –$6 ($in black) and the stretched function $g(x) =$ –$6 ($in red)? Recovered graph functions: $sqrt(x); −1*sqrt((−0.833333333333)*x); sqrt((1.2)*x); −1*sqrt((−1.2)*x); sqrt((−5)/(−6)*x)$. Rendered ExamView reference:











(A) -0.833333333333

(B) 1.2

(C) -1.2

ANS: –5 **TOP:** bnk | mcq **TYPE:** mcq

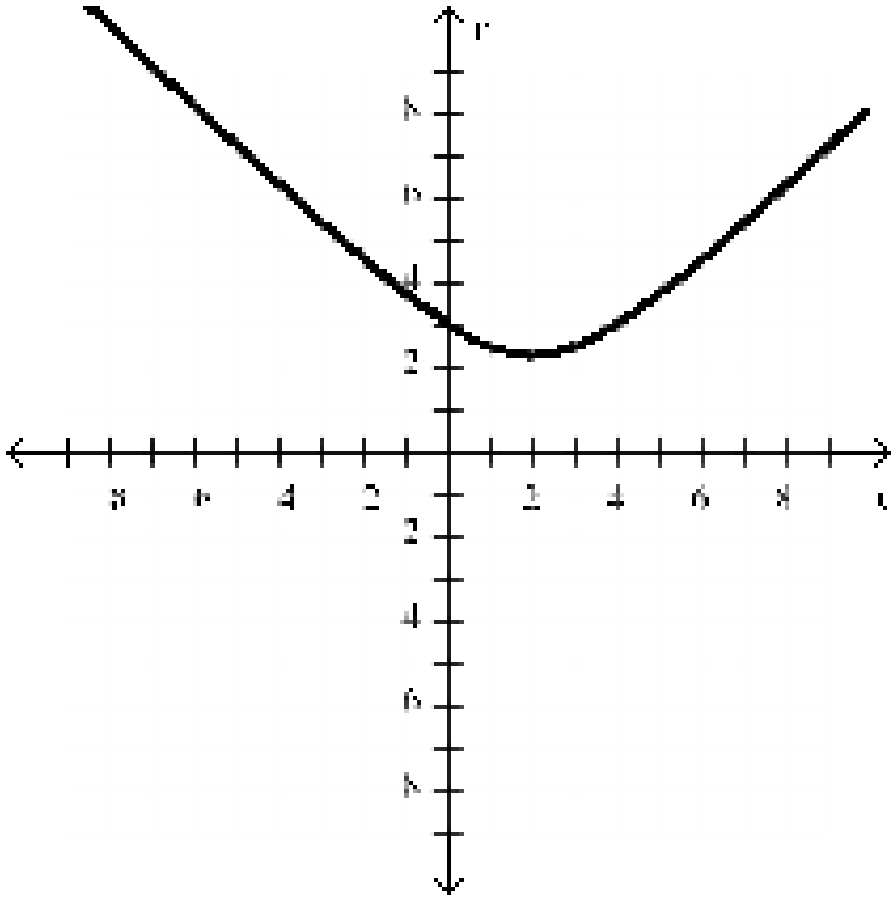
89. What are the coordinates of the invariant point(s) when the function –3 is reflected in the –$3 -$axis?

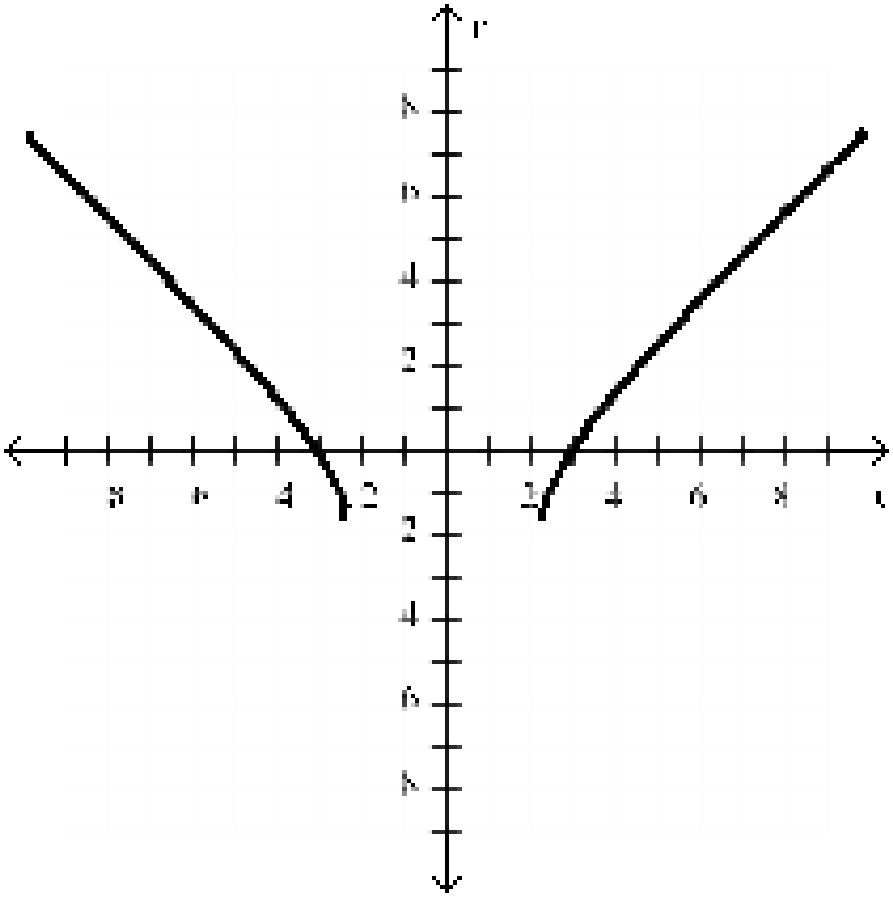
(A) (9, –3) (B) (0, –3) (C) (–3, 0) and (9, 0) (D) (0, 9)

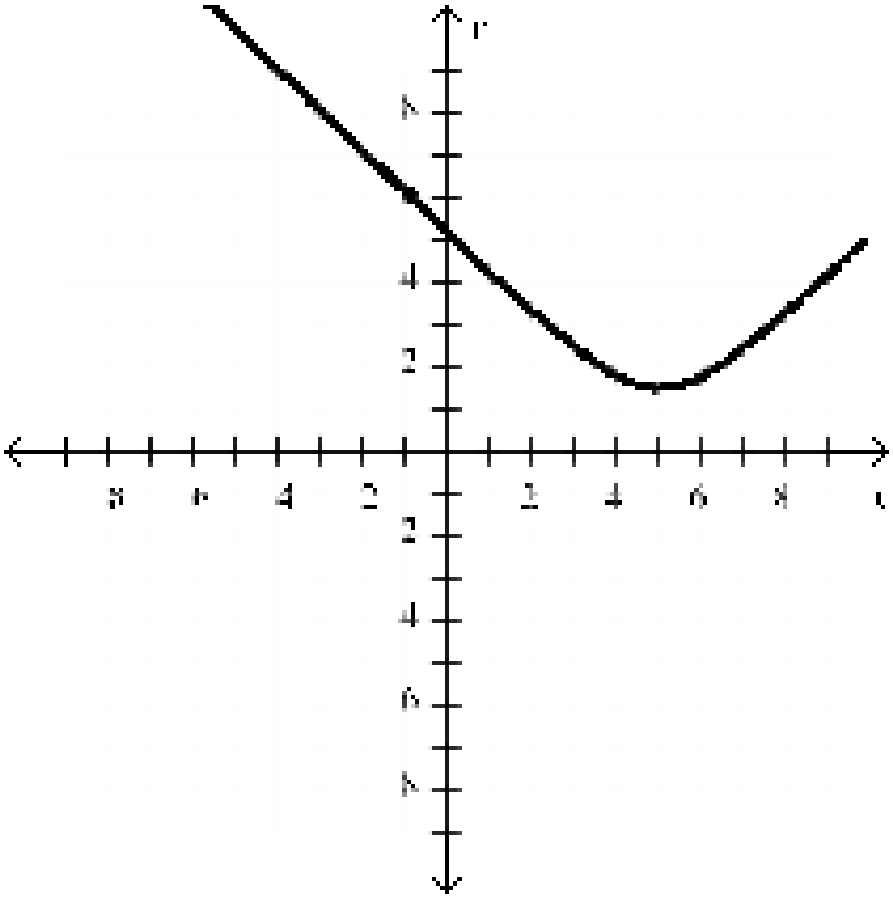
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

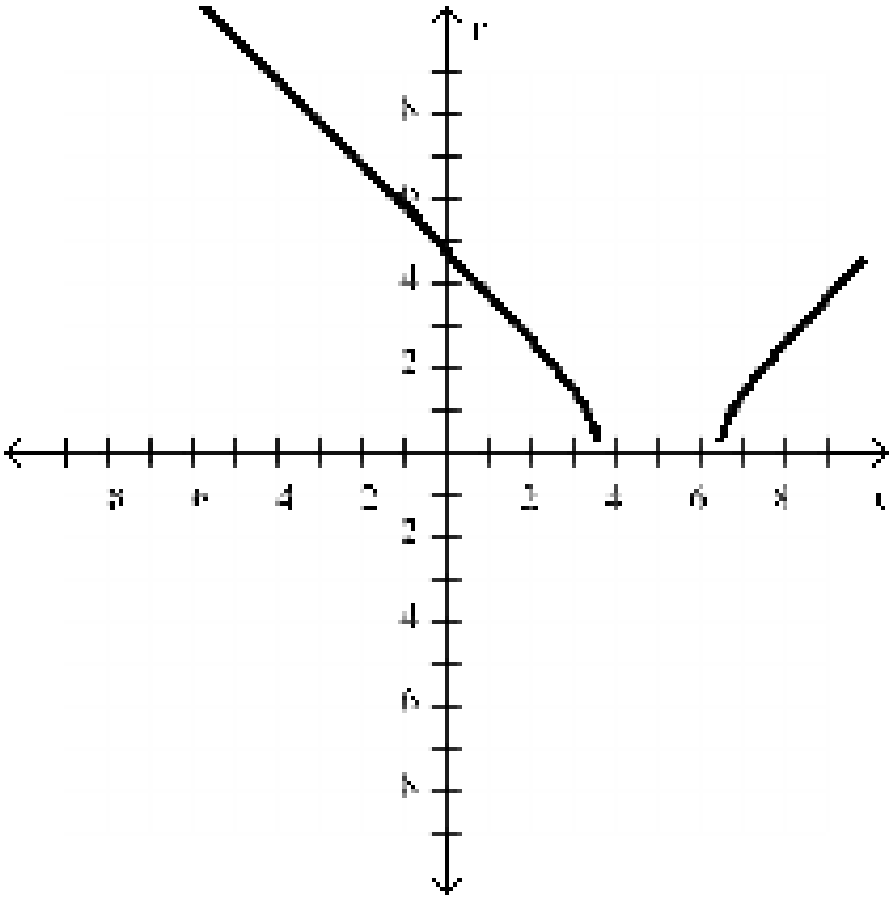
90. Which is the graph of the square root of the function – 5? Recovered graph functions: $sqrt(((x+(−2))^2)-(−5)); sqrt(((x+(−5))^2)-(−2)); sqrt(((x+(−5))^2)+(−2))$. Rendered ExamView reference:

$$f(x) = (x-5)^2 - 2$$




r
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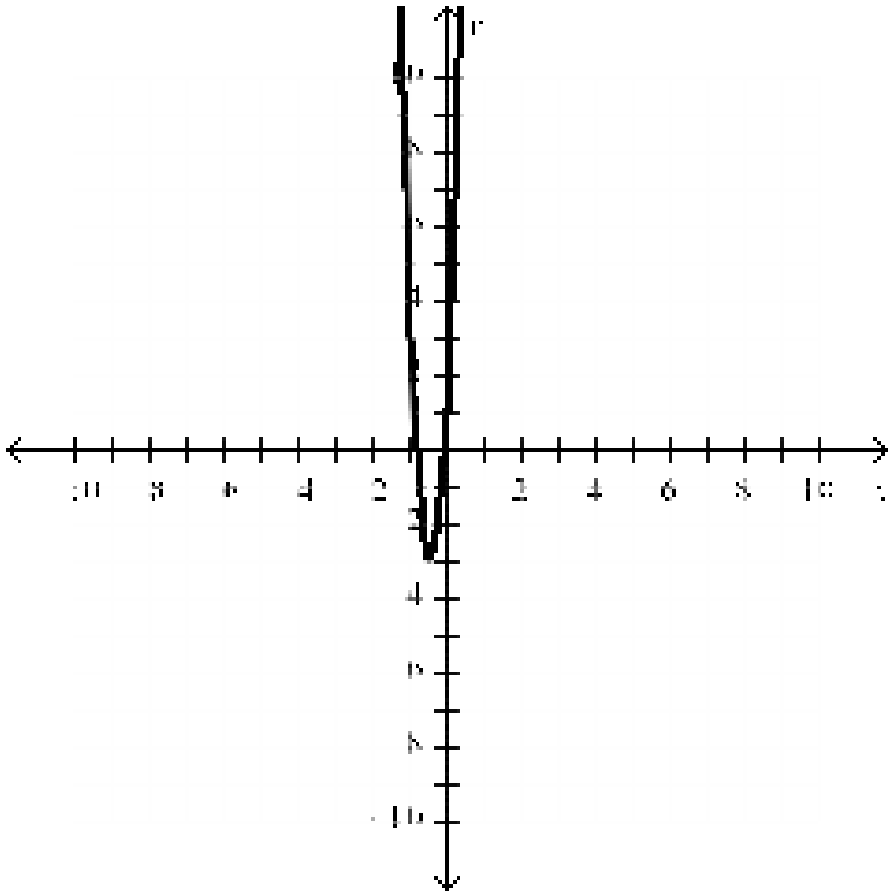

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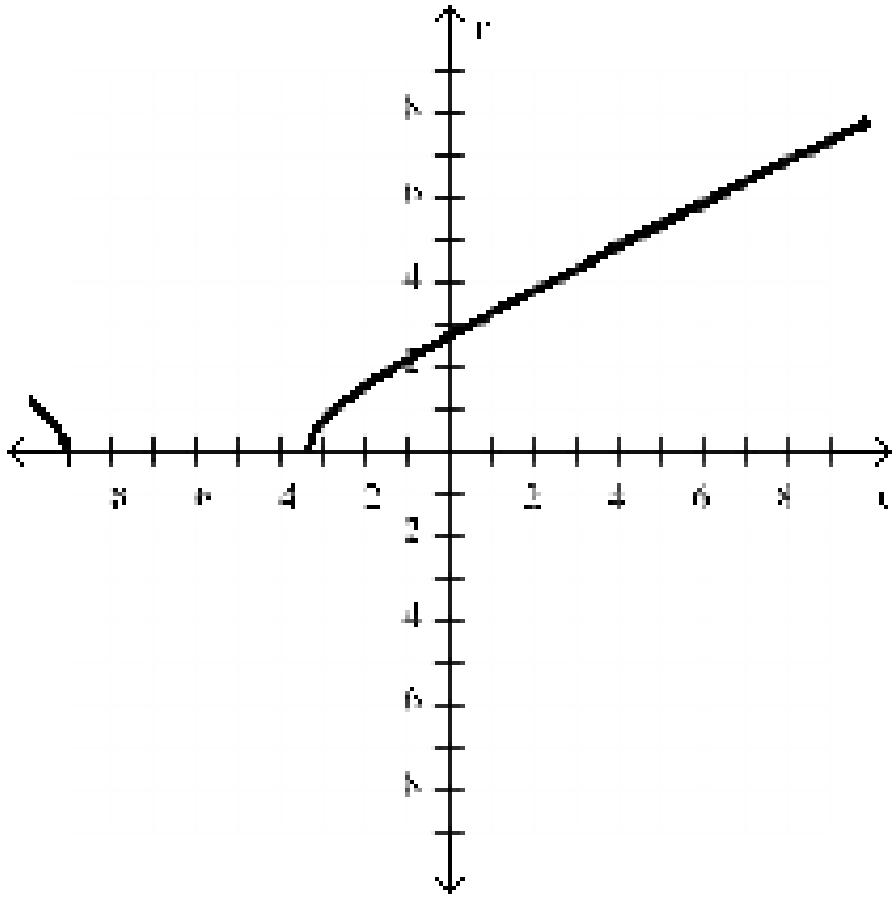


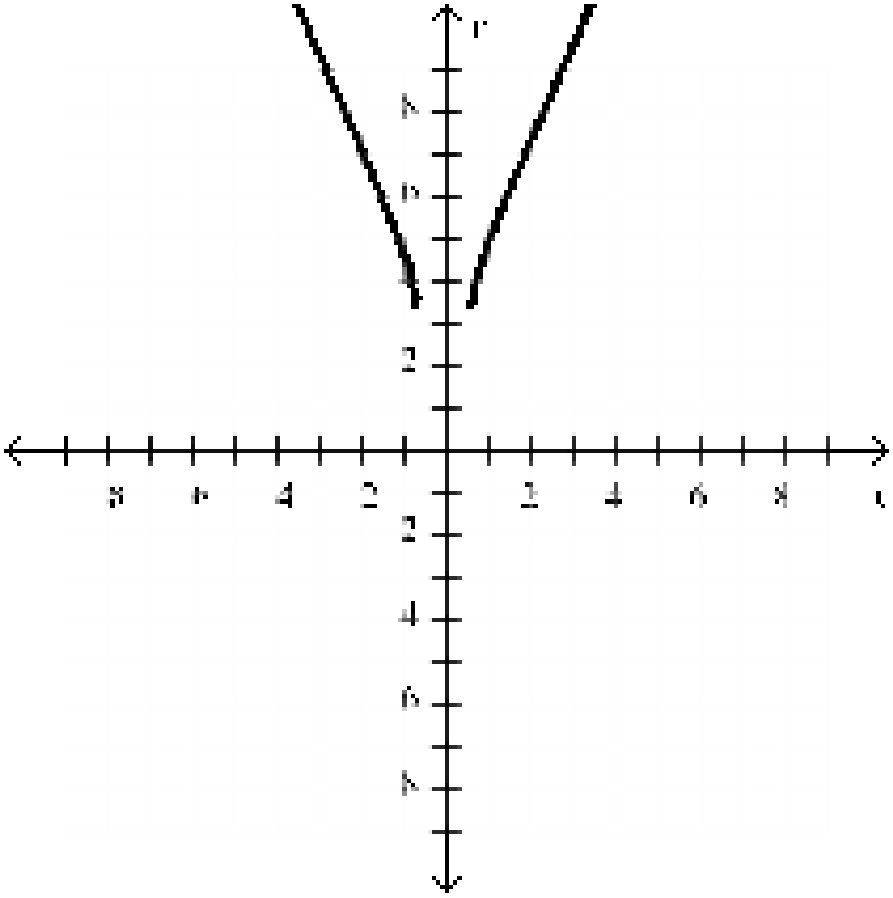
(A) graph of sqrt(((x+(−2))^2)-(−5)) (B) graph of sqrt(((x+(−5))^2)-(−2)) (C) graph of sqrt(((x+(−5))^2)+(−2))

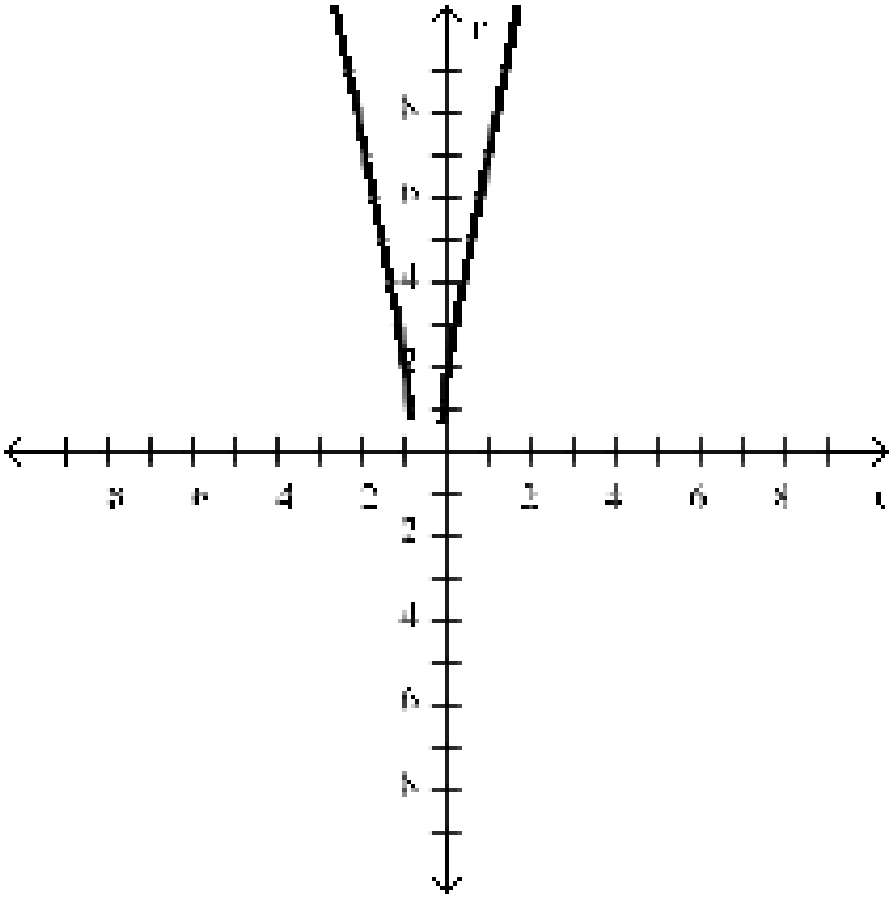
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

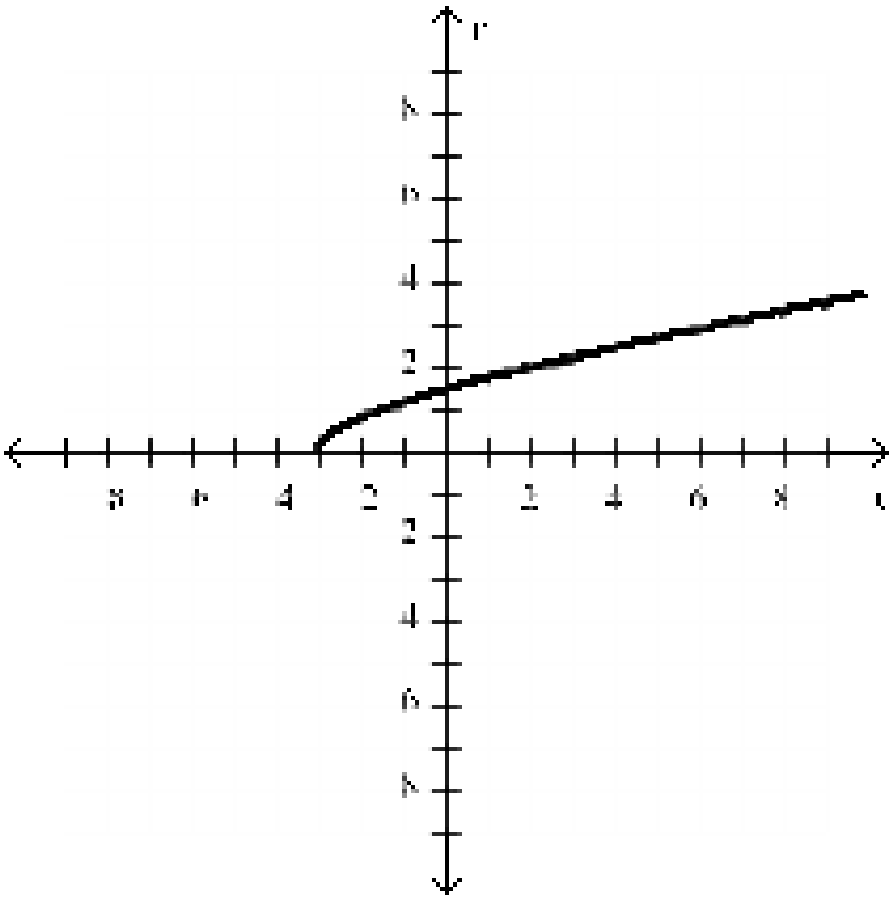
91. Which graph represents the square root of the function shown in the graph? (((\$5)*x-(−2))^2)+(−3)\$ Recovered graph functions: (((\$5)*x-(−2))^2)+(−3); sqrt((((x/2)-(−3))^2)+(−2)); sqrt((((5)*x-(−2))^2)+(−3)); sqrt((((x/(5))-(−2))^2)+(−2))\$. Rendered ExamView reference:











(A) $x >= –3$ (B) $x >= – 2$ (C) $x >= 5$ (D) $x <= –3$

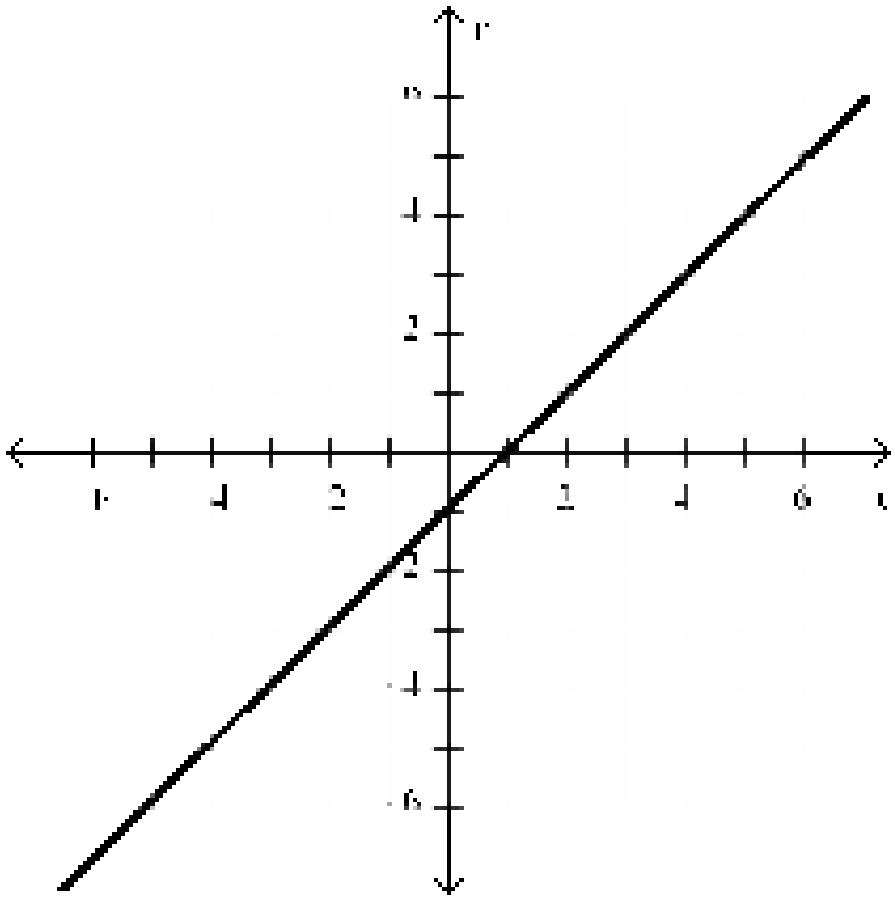
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

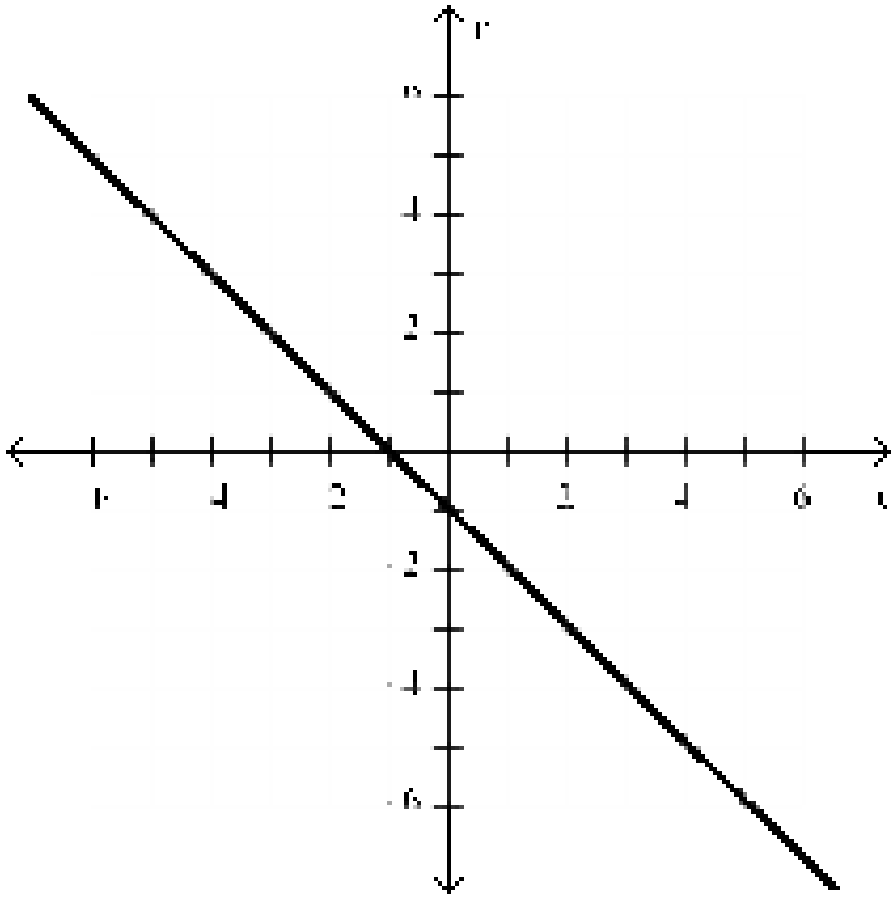
92. Which of the following functions is the correct inverse for the function $f(x) = sqrt(x) + 2$, {x | x $>= 0, x$ in R}?

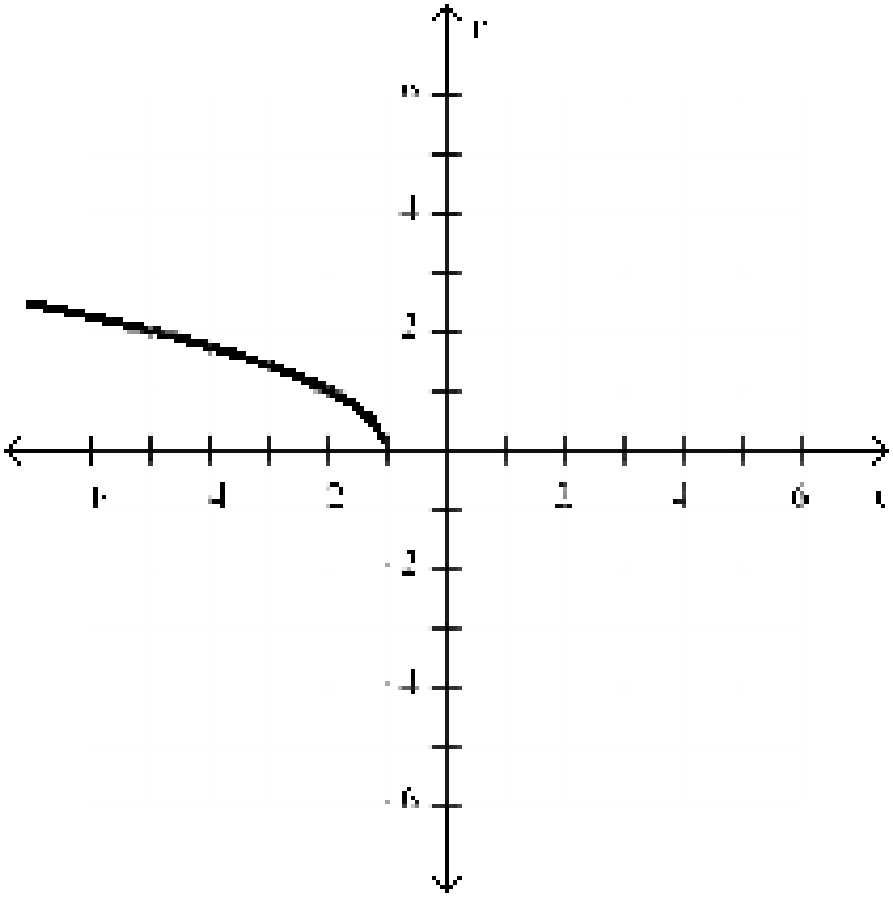
(A) $f^(−1)(x) = sqrt(x) - 2$ (B) $f^(−1)(x) = (x + 2)^2$ (C) $f^(−1)(x) = (x - 2)^2$ (D) $f^(−1)(x) = sqrt(x + 2)$

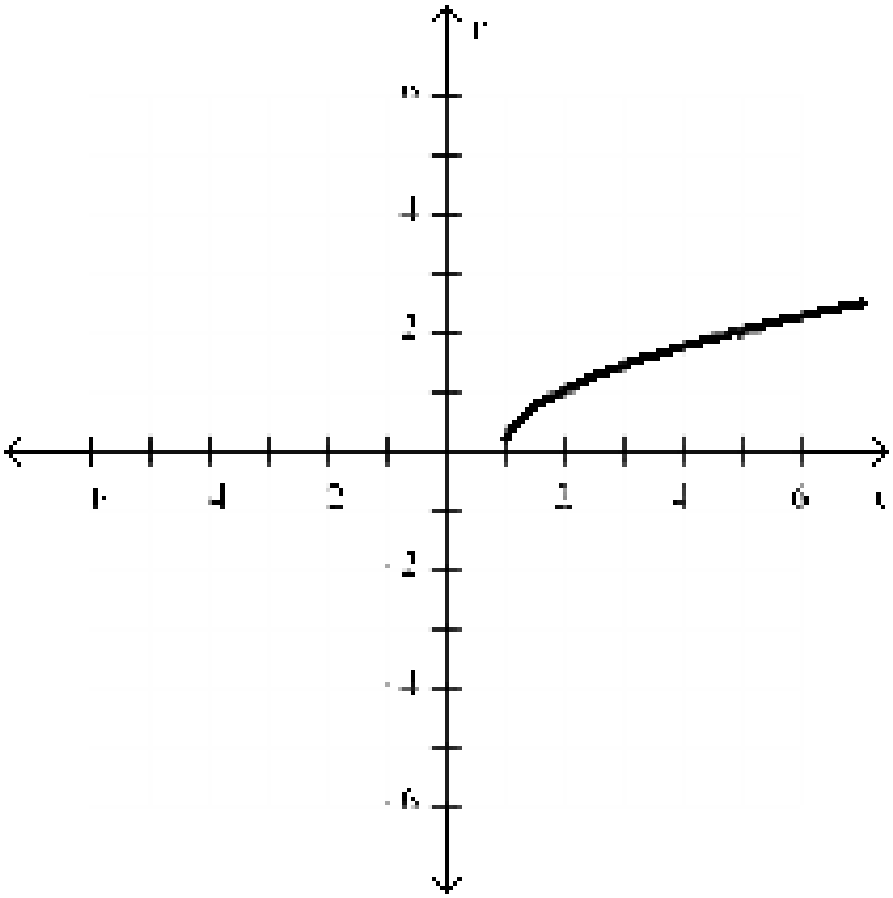
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

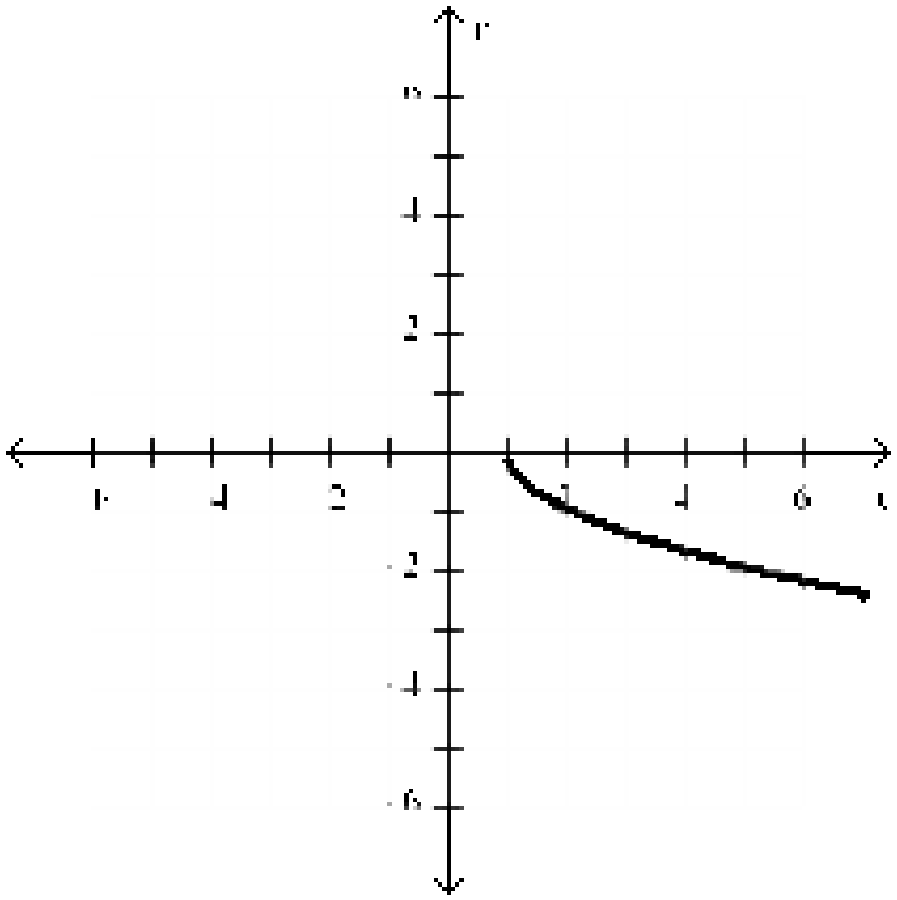
93. Which graph represents the square root of the graph shown? Original graph: $x-1$. Comparison graph: $-x-1$. Rendered ExamView reference:


y
x
6
4
2
-4
-2
2
4
6
-2
-4
-6


y
x
6
4
2
-4
-2
2
4
6
-2
-4
-6


y
8
4
2
-2
-4
-6
-8
-4
-2
2
4
6
x

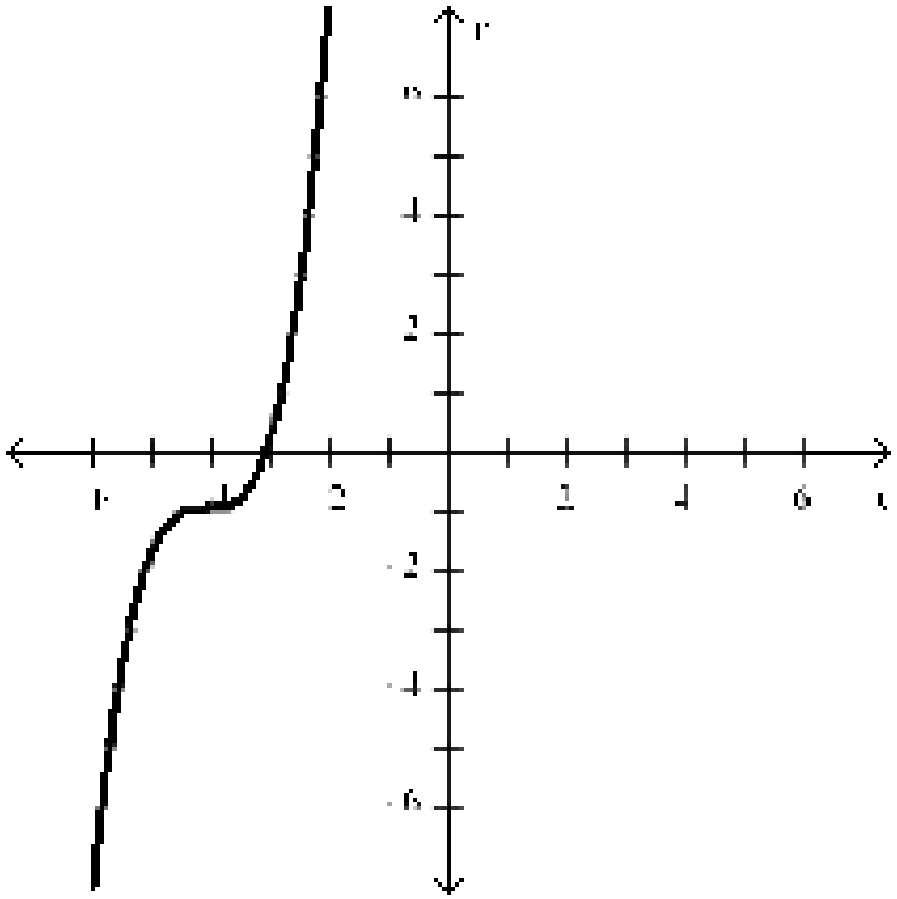

y
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4
2
-2
-4
-6
-8
-4
-2
2
4
6
x

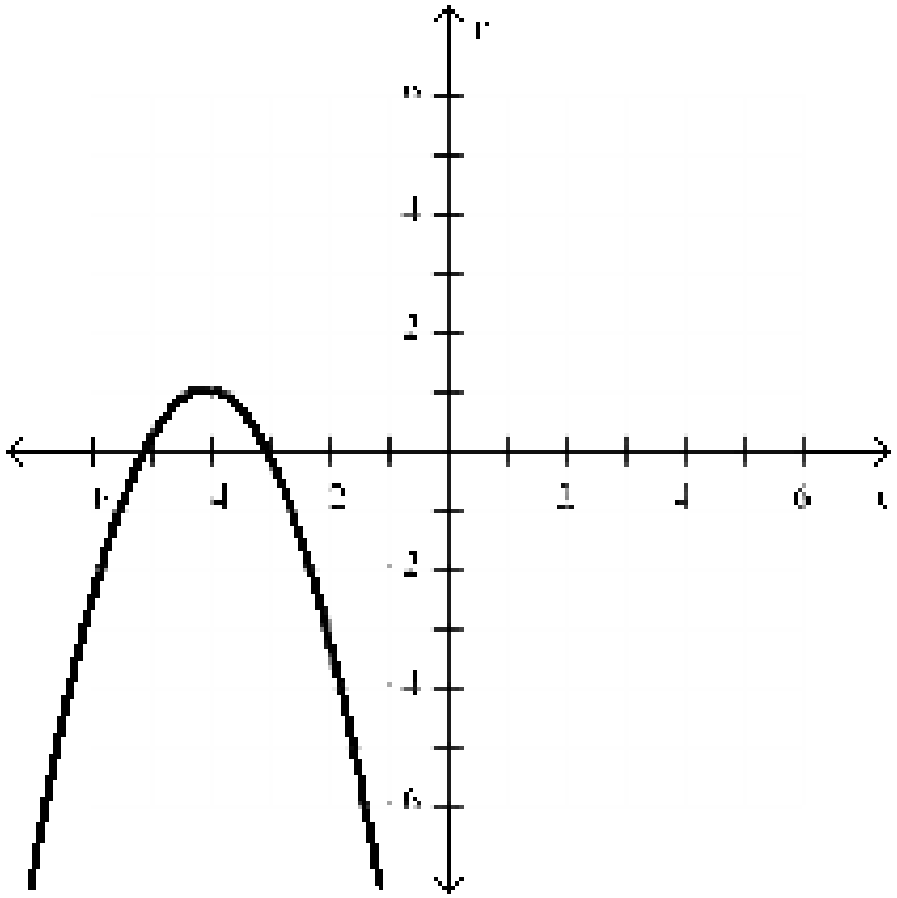


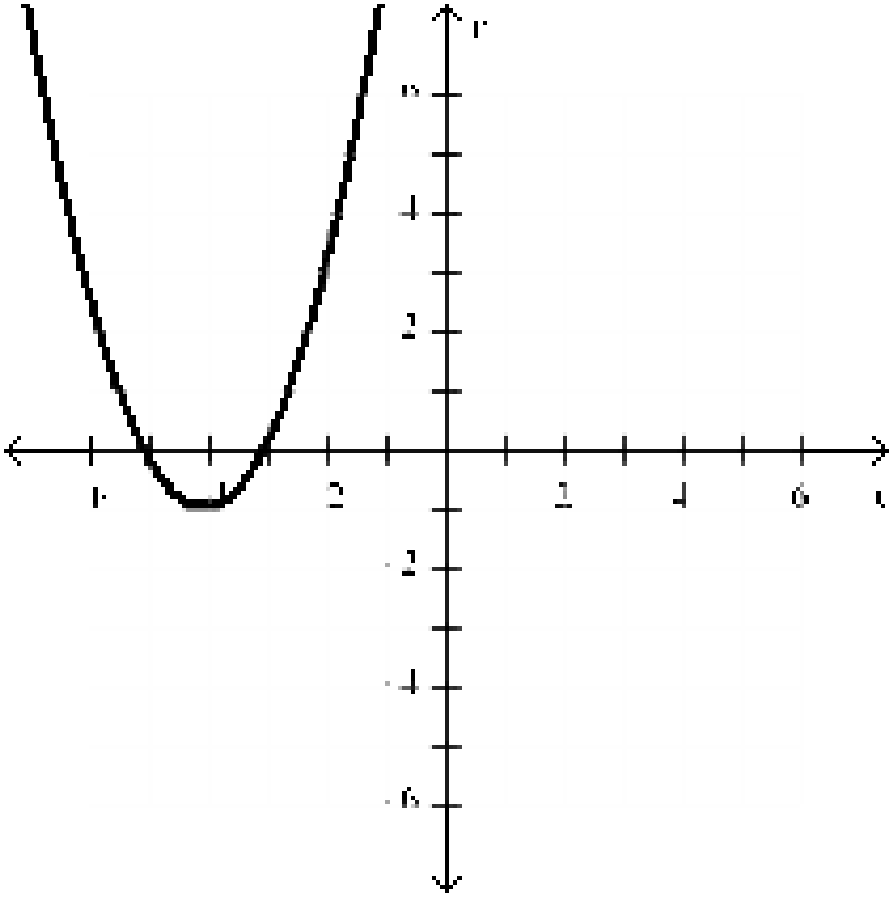
(A) graph of $x-1$ (B) graph of $-x-1$ (C) graph of $sqrt(-x-1)$ (D) graph of $sqrt(x-1)$

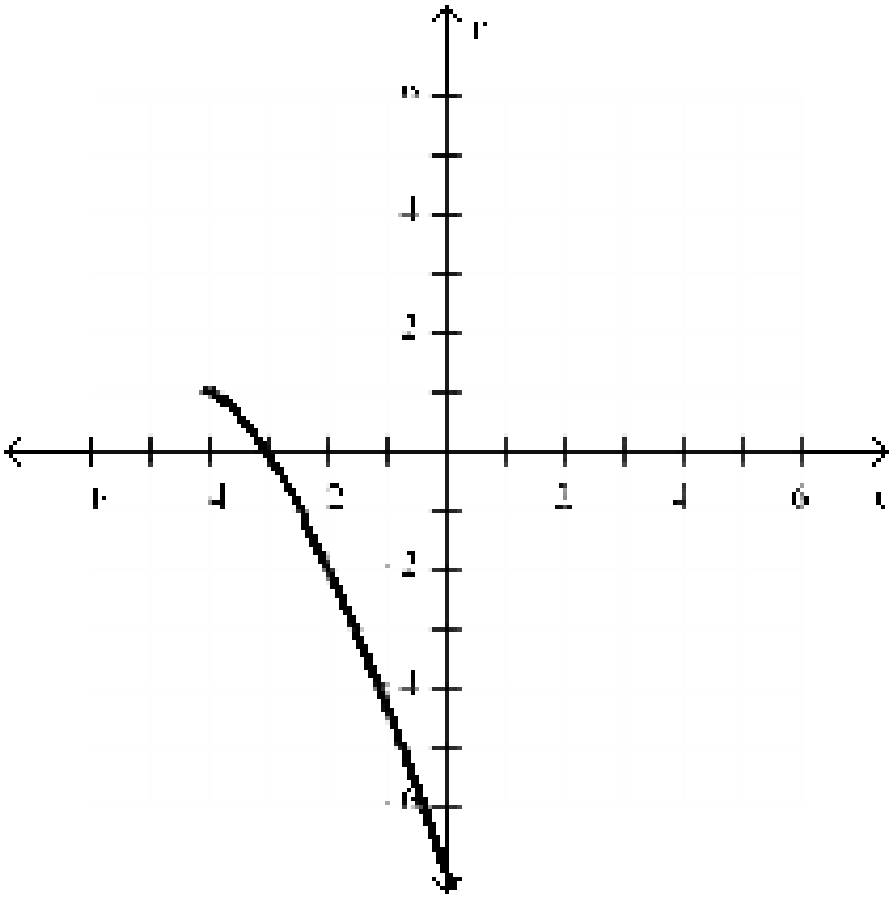
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

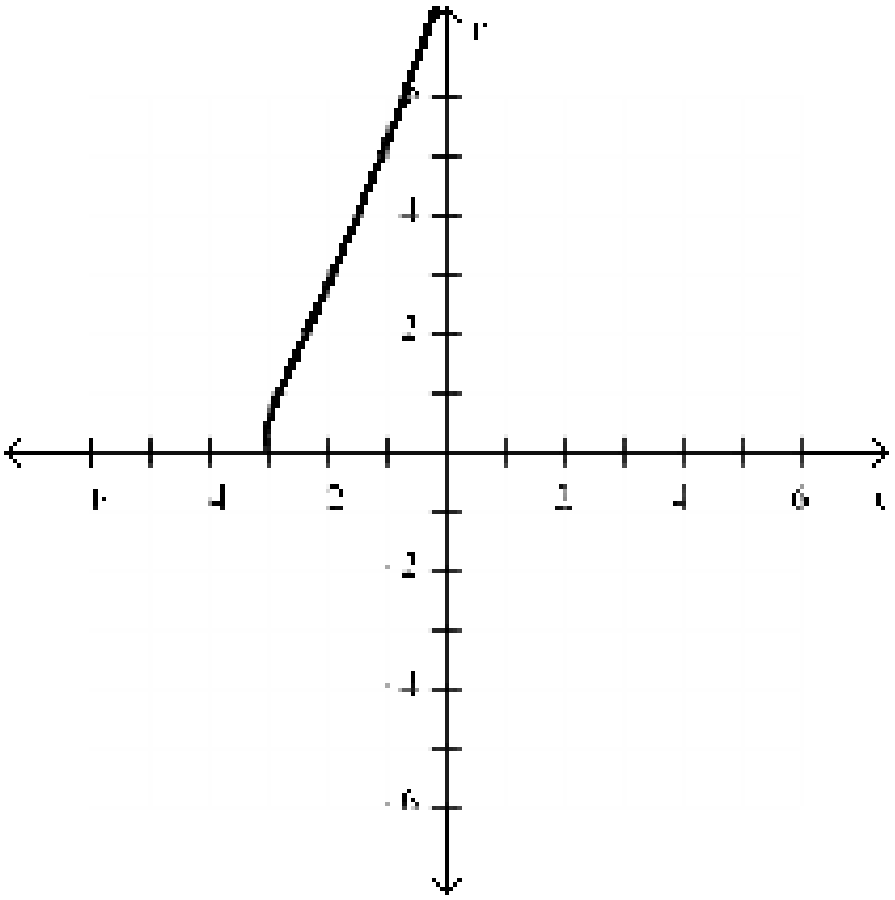
94. Which graph represents the square root of the graph shown? $−1*((x-(−4))^2+(−1))$ Recovered graph functions: $ −1*((x-(−4))^2+(−1)); ((x-(−4))^2)+(−1); −1*((sqrt(x-(−4))^3)+(−1))$. Rendered ExamView reference:


y
6
4
2
-2
-4
-6
-4
-2
2
4
6
x


y
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-2
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-6
-4
-2
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x


y
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-4
-2
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4
6


y
x
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-4
-2
2
4
6



(A) graph of −1*((x-(−4))^2+(−1))

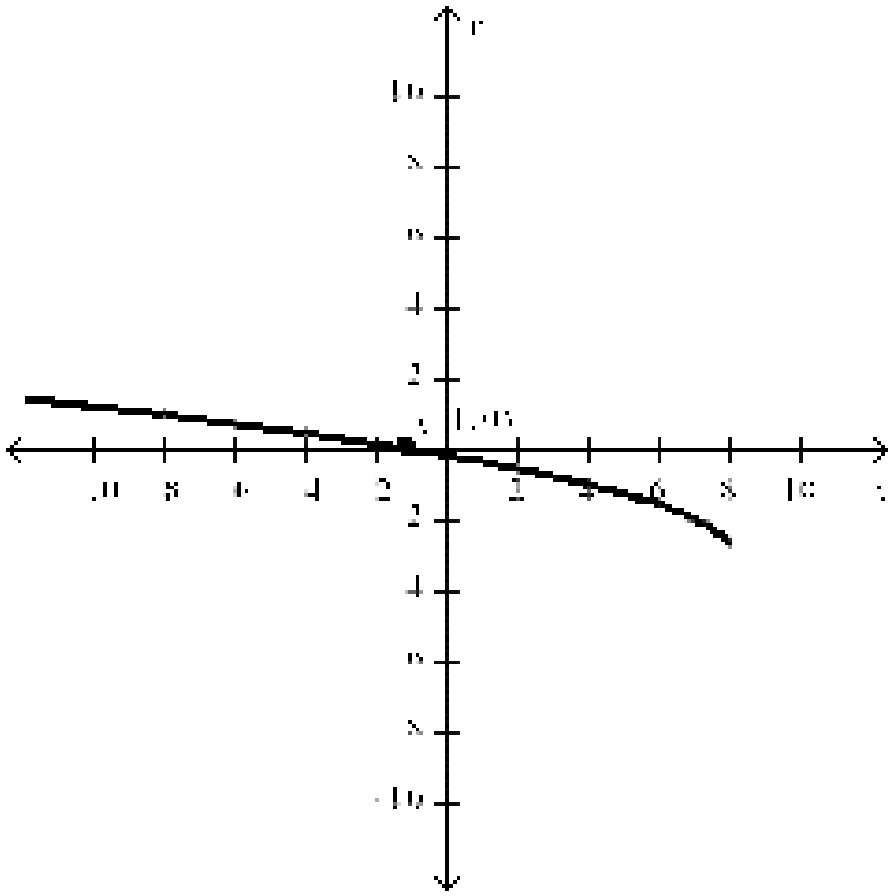
(B) graph of ((x-(−4))^2)+(−1)

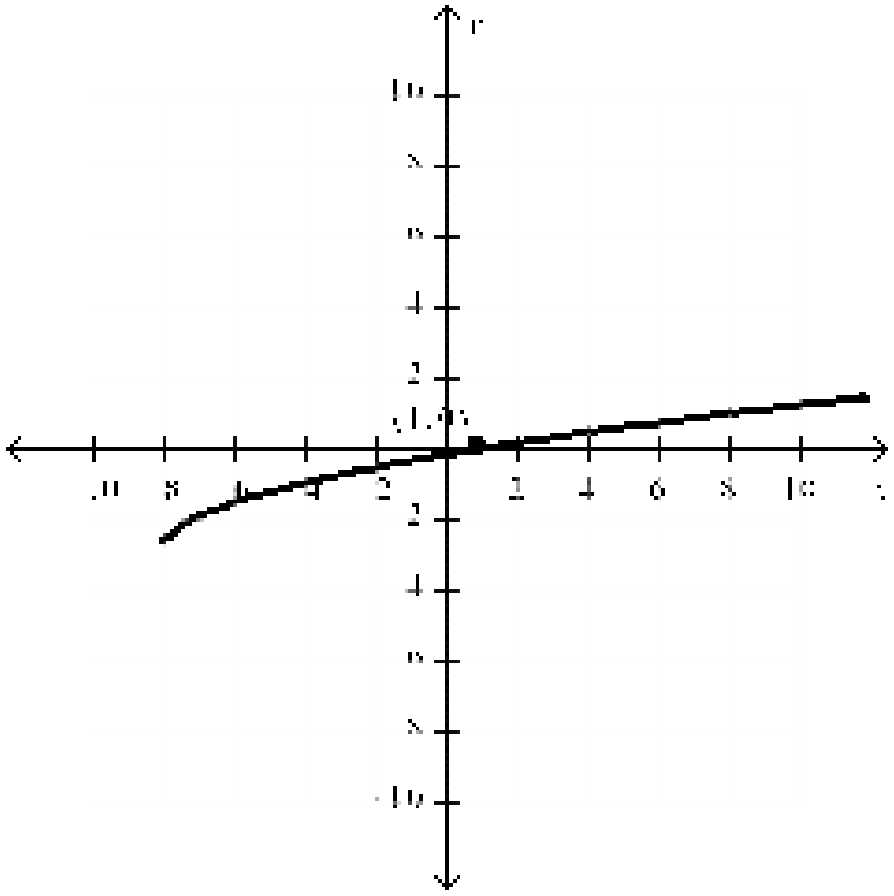
(C) graph of −1*((sqrt(x-(−4))^3)+(−1))

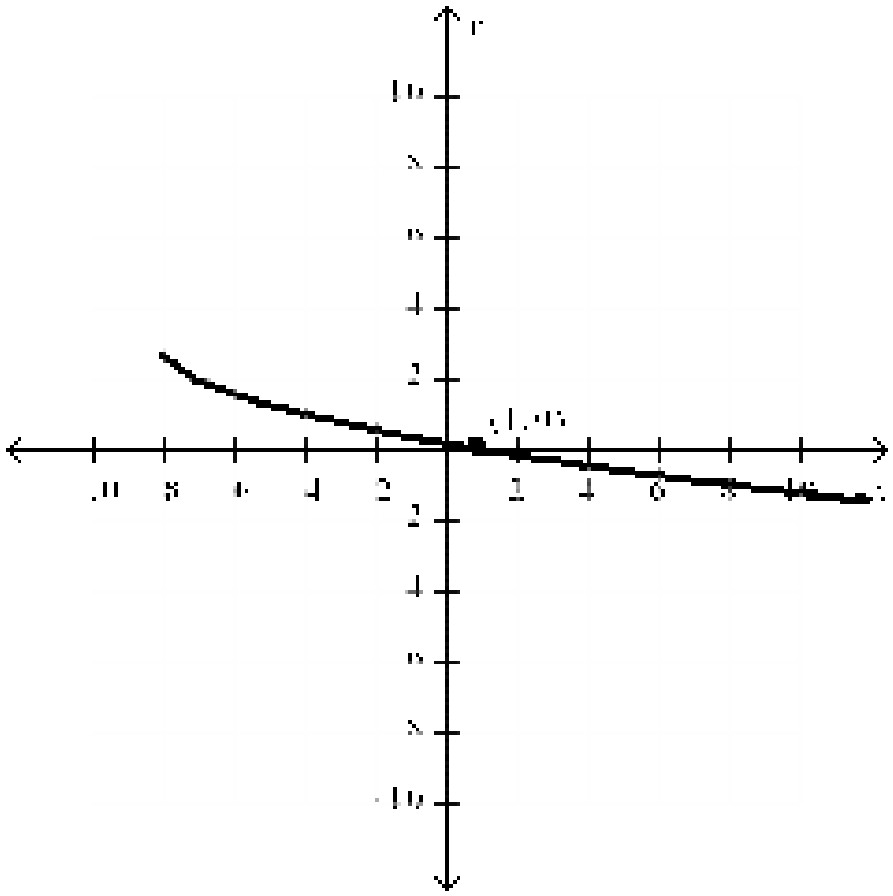
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

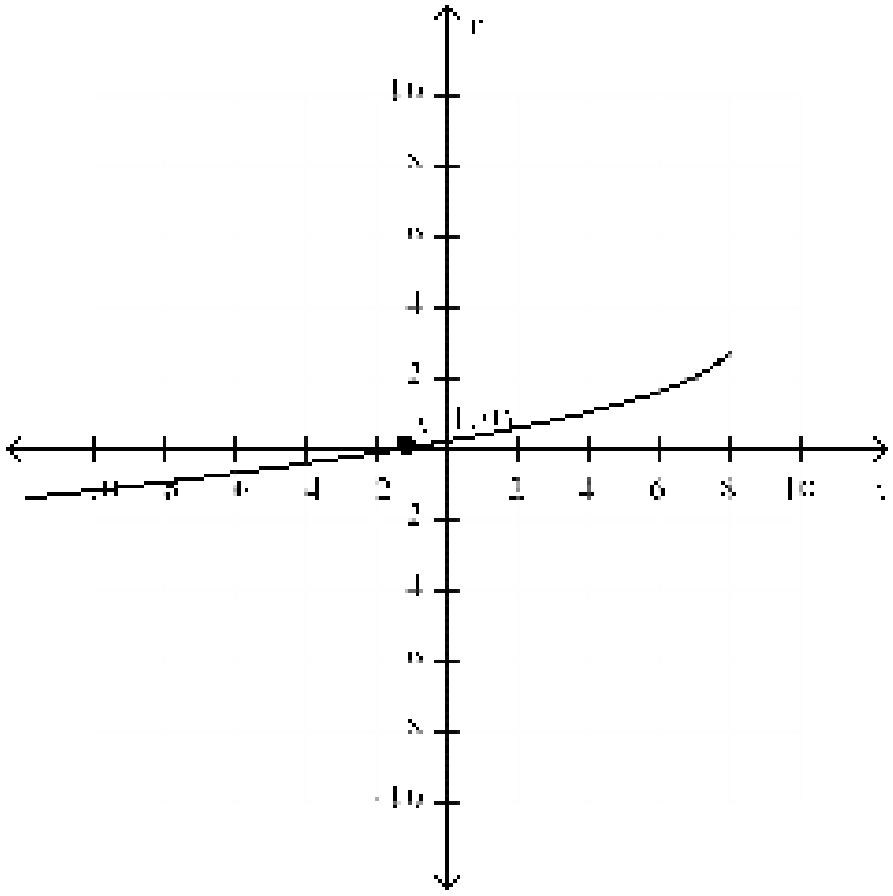
95. Which graph shows the graphical solution to the radical equation $+ 8$? Recovered graph functions: pow(($−3),2)-(8); −1*($pow(($−3),2)-(8)); sqrt(-x+(8))+(−3); sqrt(x+(8))+(−3); −1*(sqrt(x+(8))+(−3)); −1*(sqrt(-x+(8))+(−3))$. Rendered ExamView reference:

$$0 = \sqrt{x+8} - 3$$









(A) graph of pow((−3),2)-(8)

(B) graph of −1*(pow((−3),2)-(8))

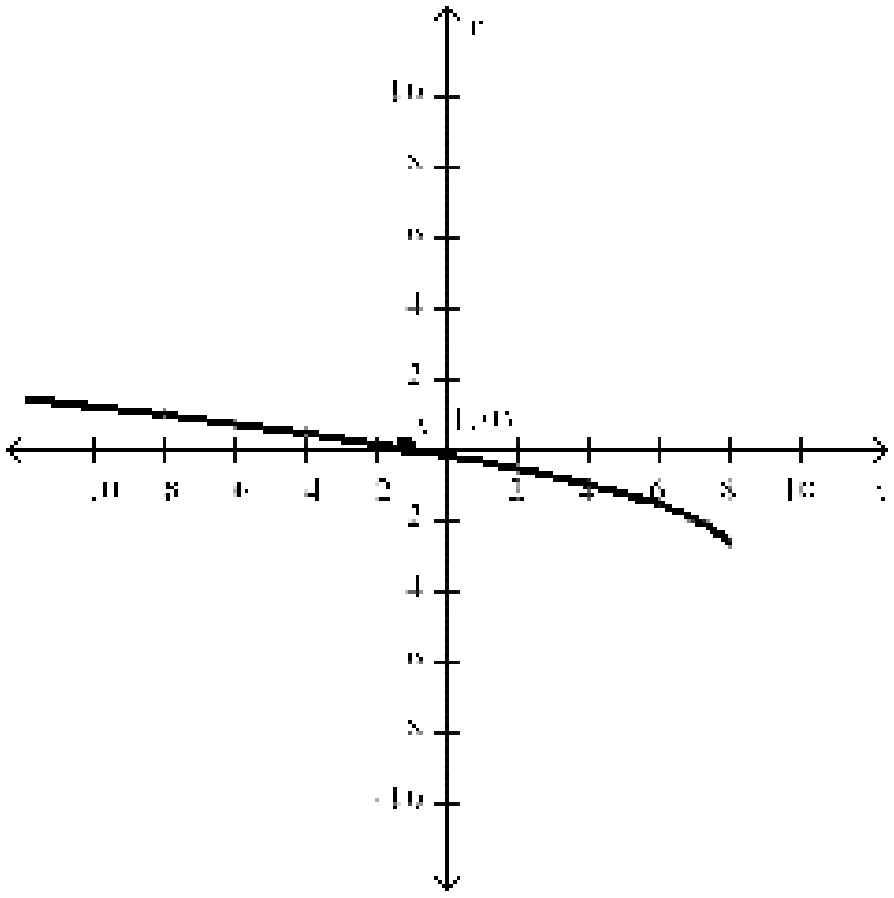
(C) graph of sqrt(-x+(8))+(−3)

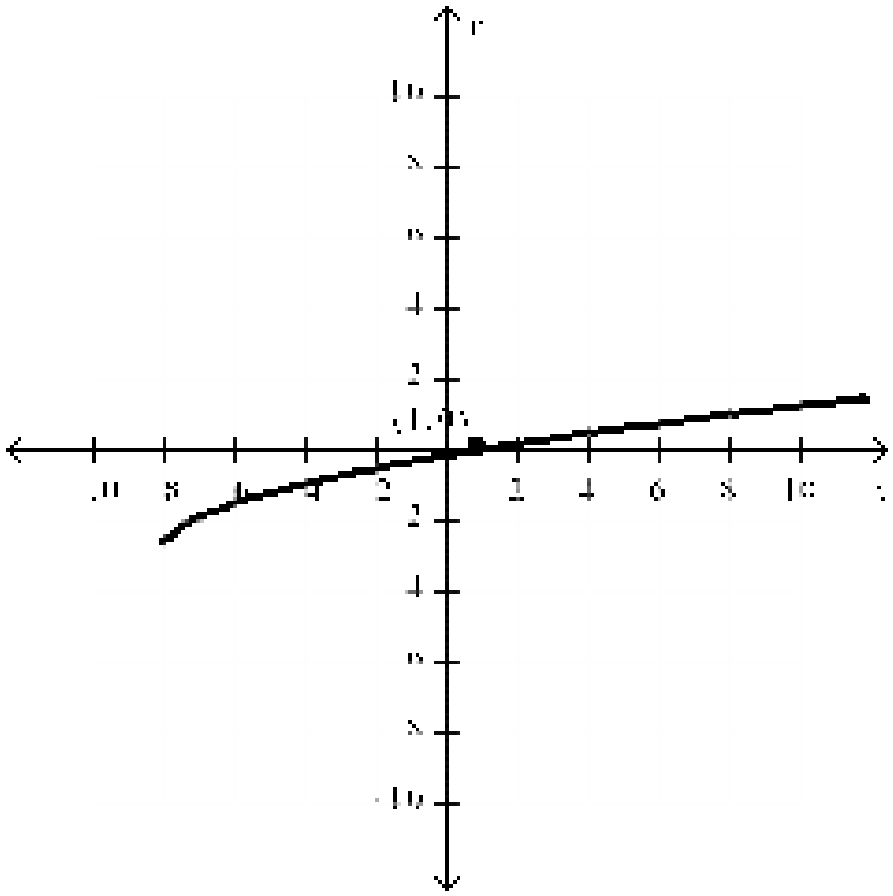
(D) graph of sqrt(x+(8))+(−3)

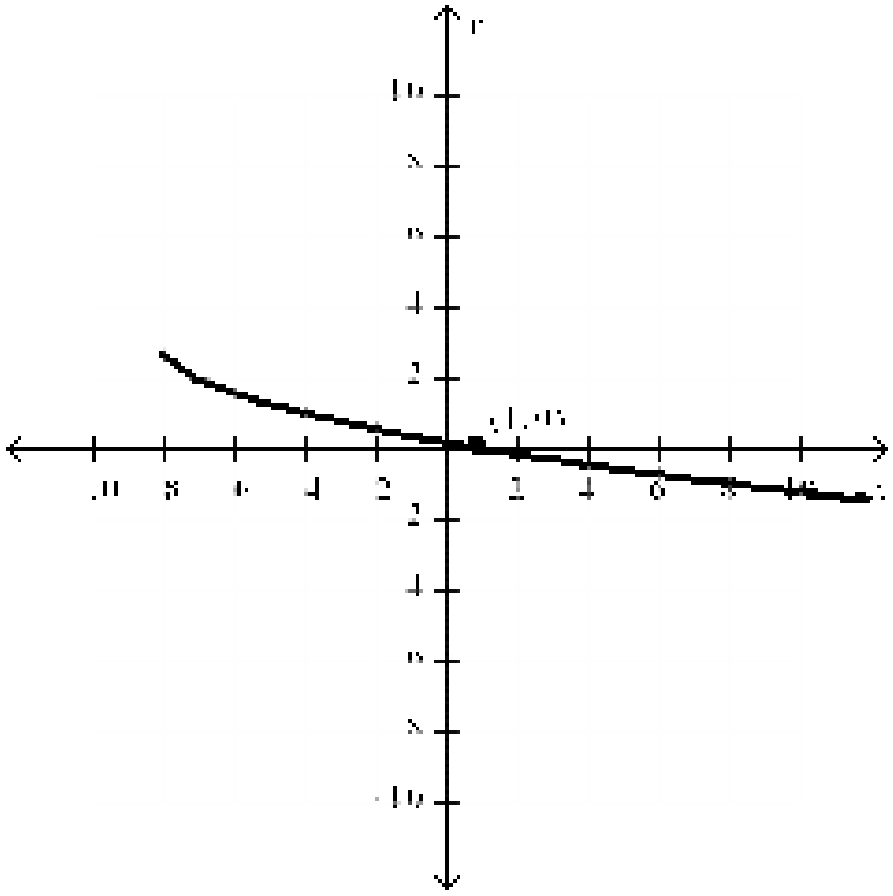
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

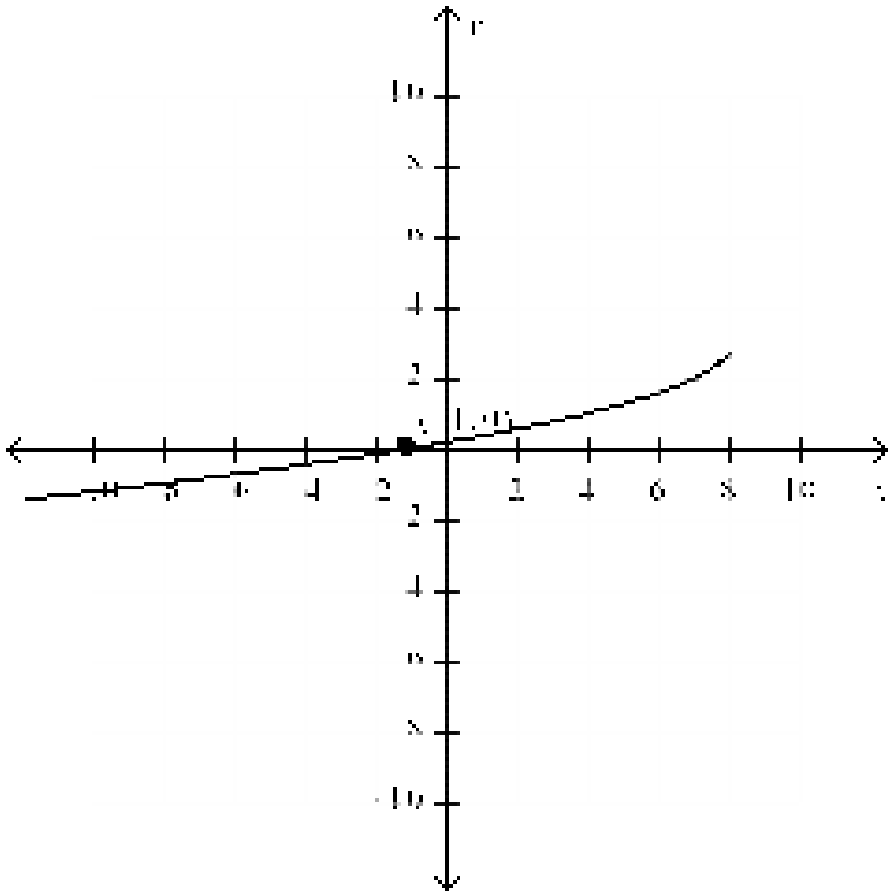
96. Which graph shows the graphical solution to the radical equation 2? Recovered graph functions: ($2)*(sqrt(b*(x+(−5))))+(−2); b*(sqrt((2)*(-x+(−5))))+(−2); −1*((2)*(sqrt(b*(-x+(−5))))+(−2)); −1*((2)*(sqrt(b*(x+(−5))))+(−2))$. Rendered ExamView reference:

$$0 = \sqrt{x+8} - 3$$







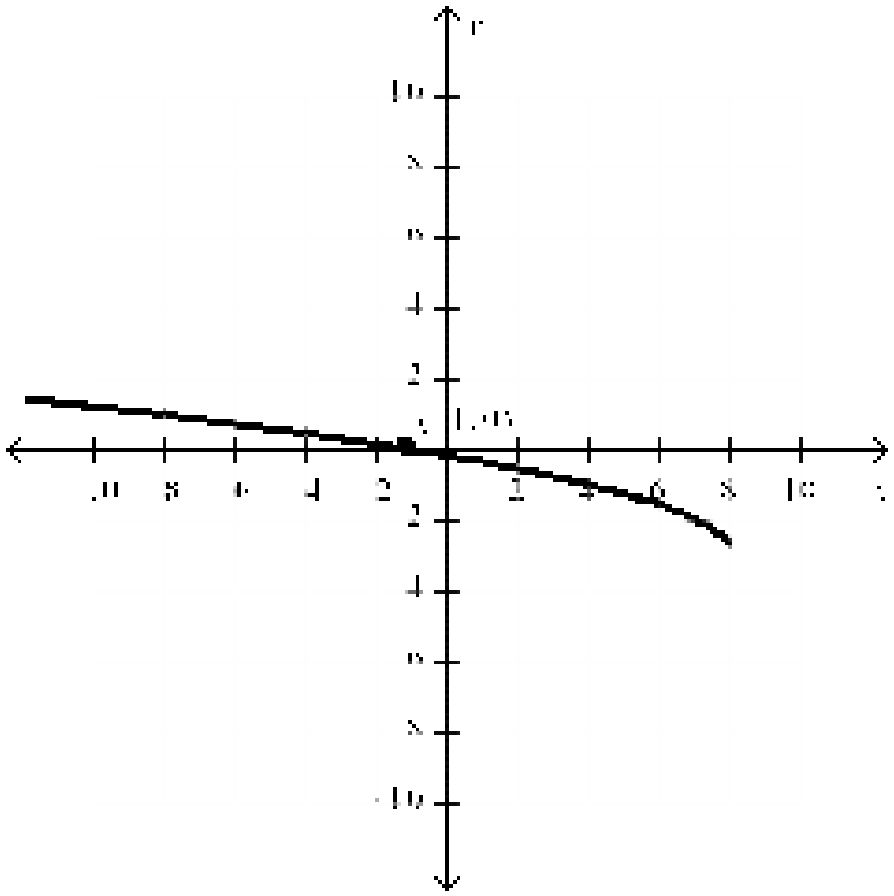


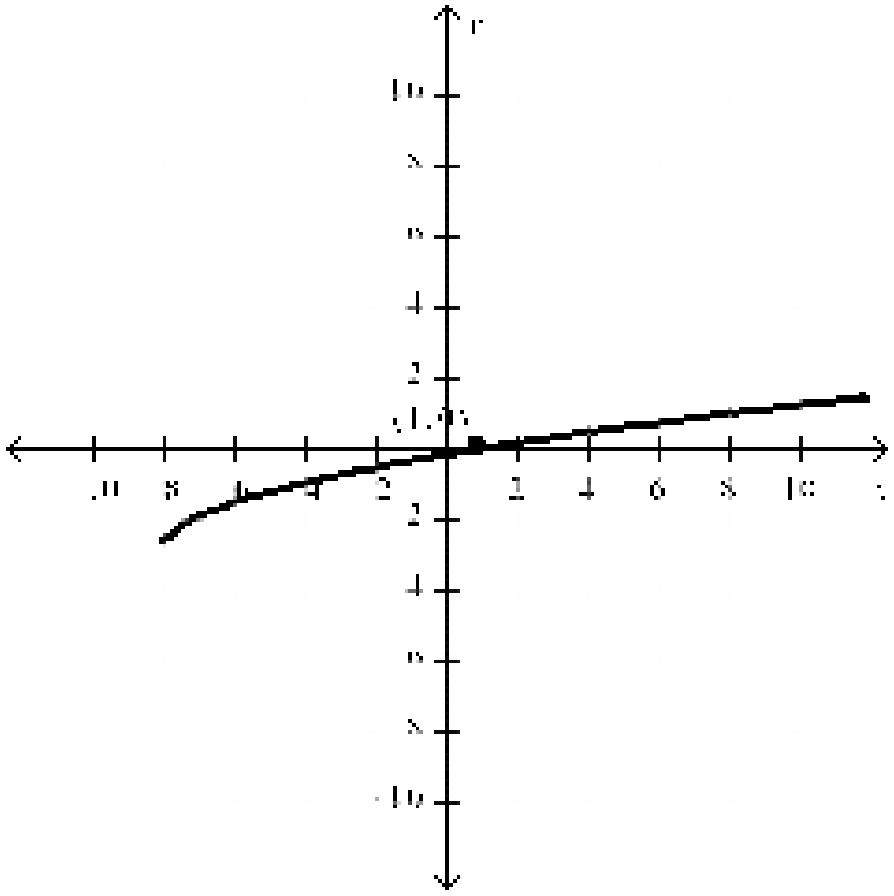
(A) $x >= – 2$ (B) $x >= – 5$ (C) $x >= range(1,5,1)$ (D) $x <= – 2$

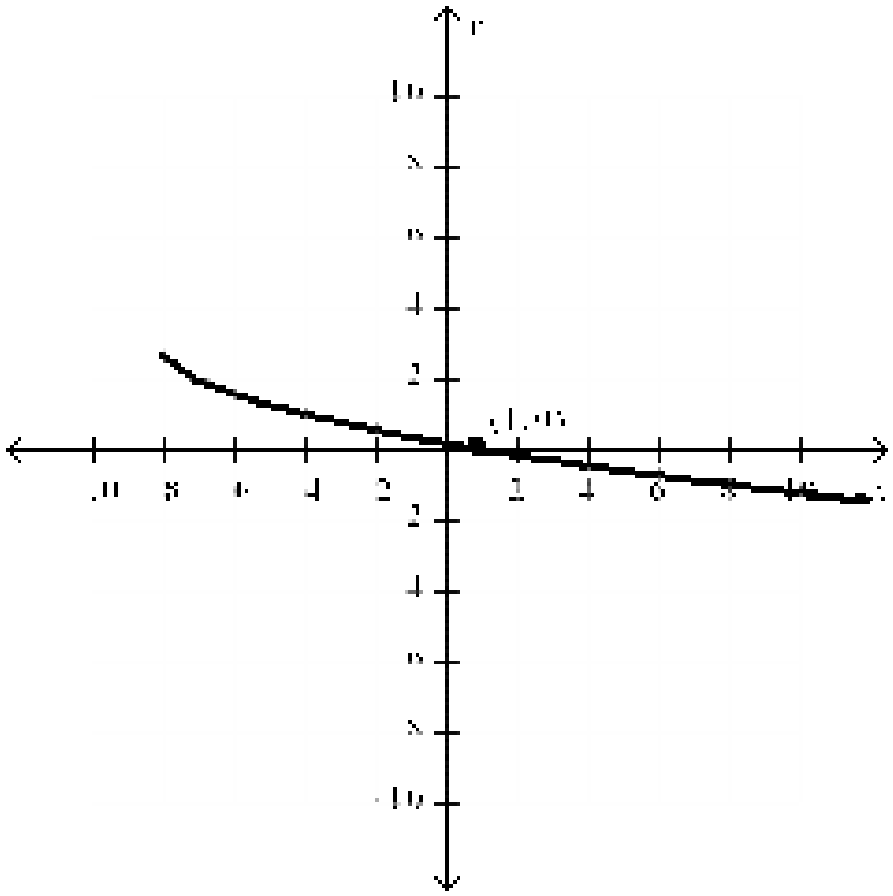
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

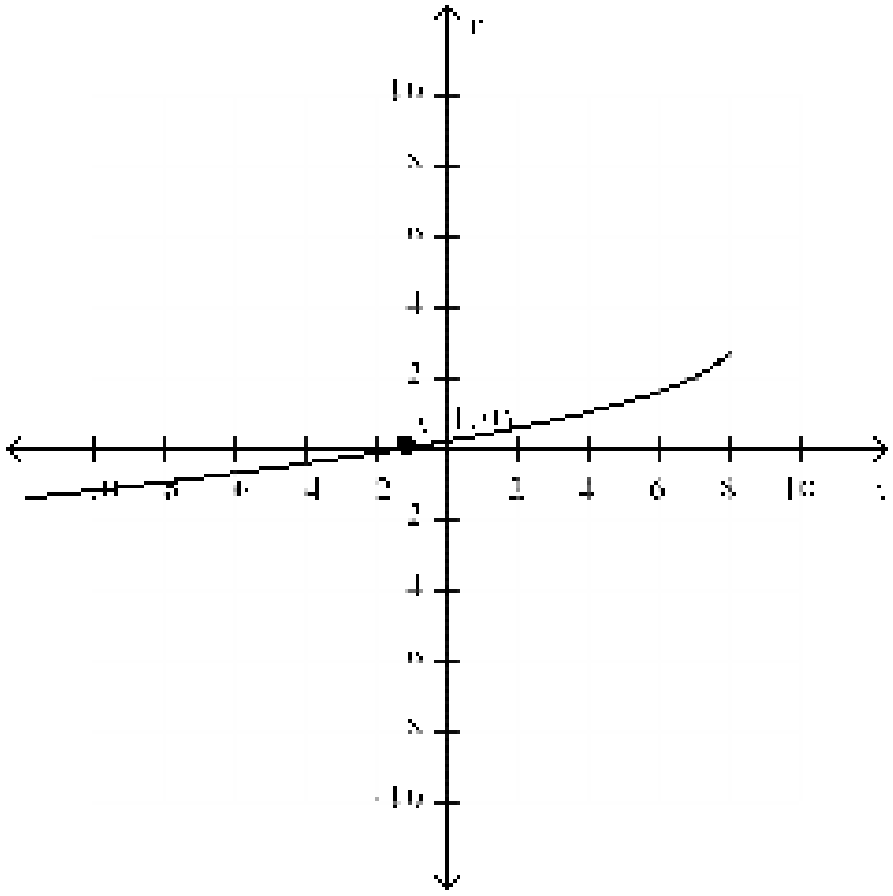
97. Which graph shows the solution to the radical equation 4? Recovered graph functions: $sqrt((8)-x); x+(4)$. Rendered ExamView reference:

$$0 = \sqrt{x + 8} - 3$$







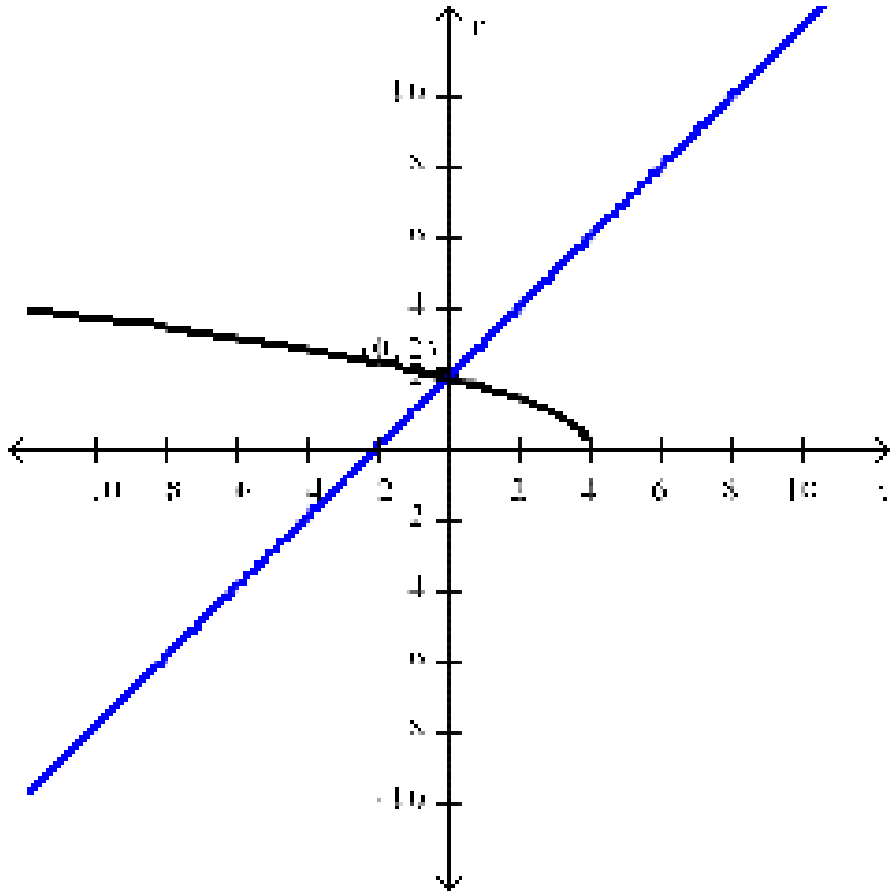


(A) graph of sqrt((8)-x)

(B) graph of x+(4)

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

98. Which radical equation can be solved using the graph shown below? Recovered graph functions: $sqrt((4)-x); x+(2)$.? 0 Rendered ExamView reference:



$$-\sqrt{4-x} = x+2$$

$$x+2 = -\sqrt{4+x}$$

$$\sqrt{4-x} = x+2$$

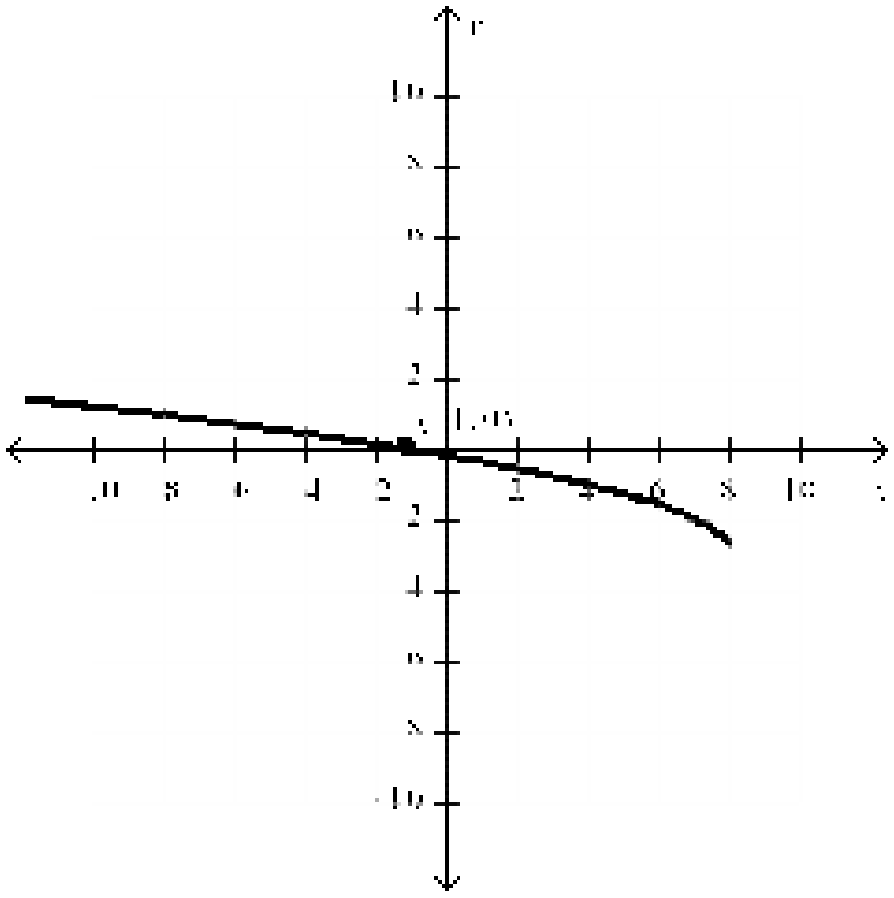
$$\sqrt{4+x} = x+2$$

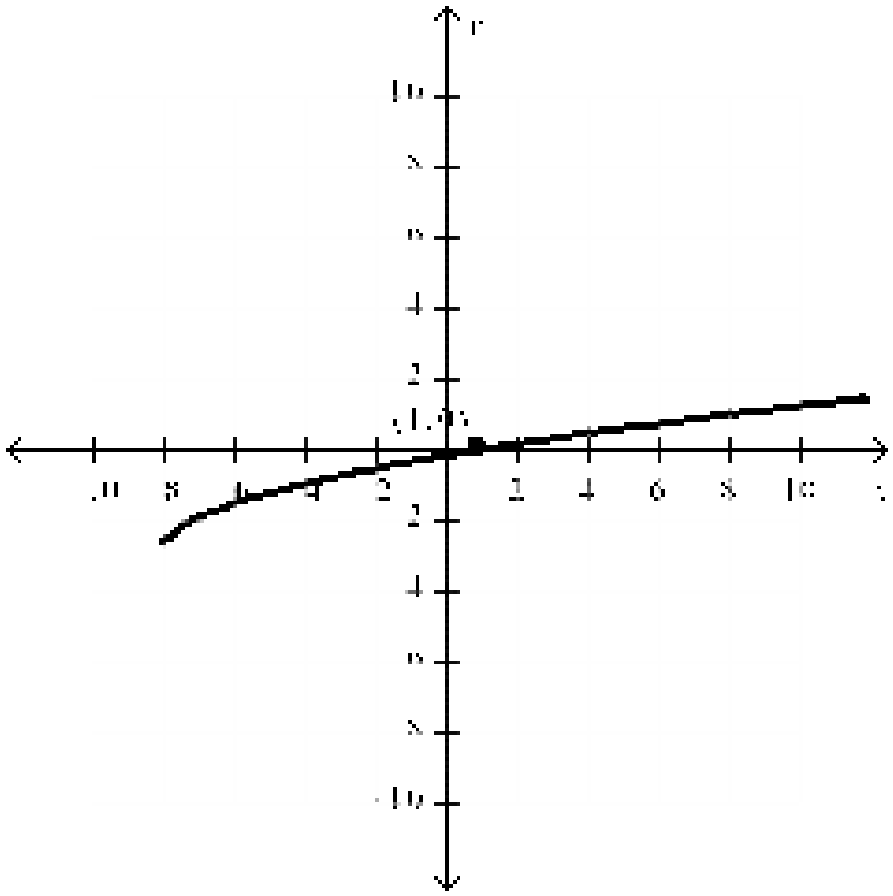
(A) 2 (B) 2 (C) 4 (D) 2

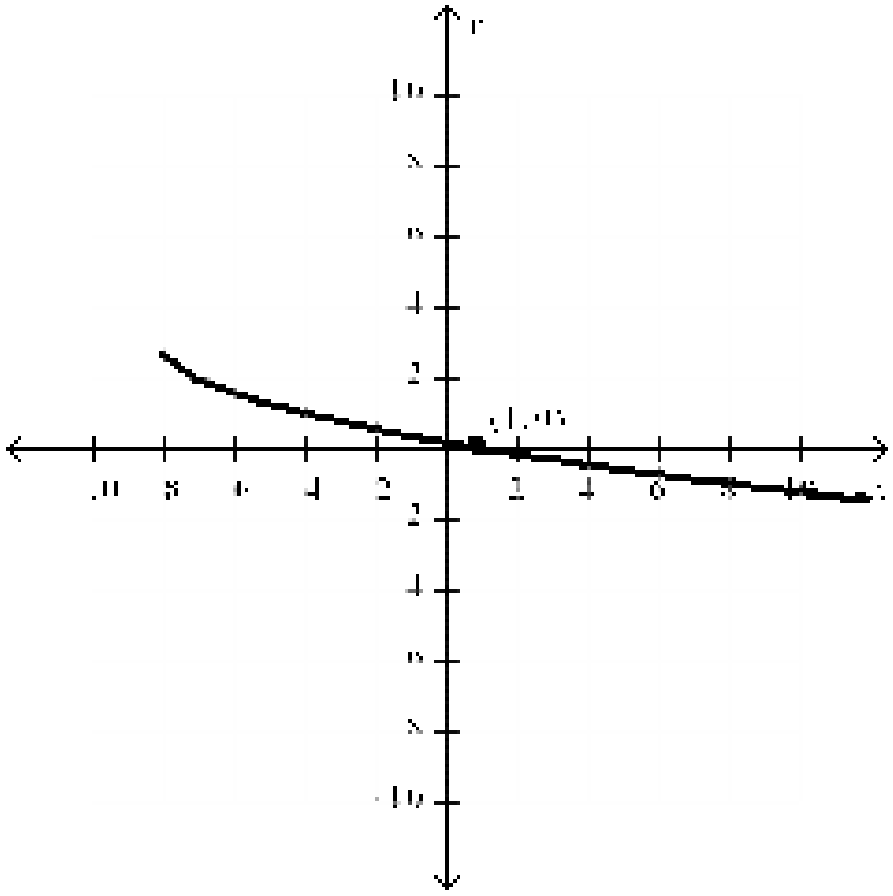
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

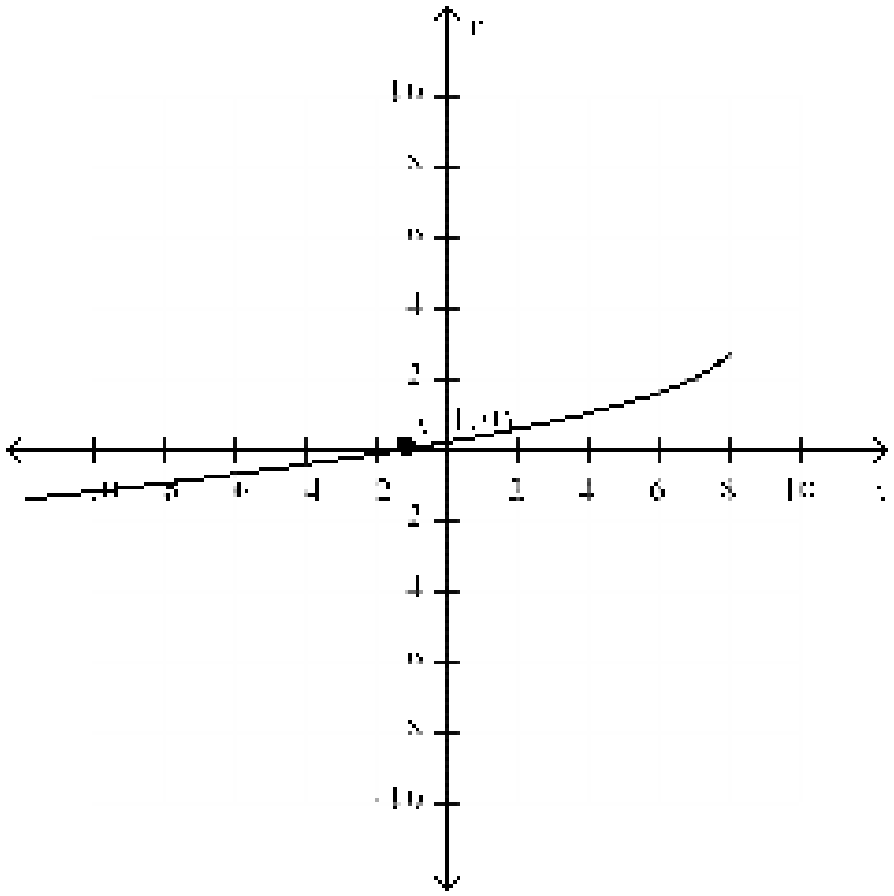
99. Which graph shows the solution to the radical equation 7? Recovered graph functions: $sqrt((7)-x); x-(−5); sqrt((7)+x); x+(−5); -sqrt((7)-x); -x+(−5); -x-(−5)$. Rendered ExamView reference:

$$0 = \sqrt{x+8} - 3$$









(A) graph of sqrt((7)-x) (B) graph of x-(−5) (C) graph of sqrt((7)+x) (D) graph of x+(−5)

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

100. What is the solution to the radical equation $+ 9$?

(A) 18 (B) $−18$ (C) 36 (D) 0

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

101. What is the solution to the radical equation 2?

(A) $x >= – 8$ (B) $x >= + 4$ (C) $x >= 2$ (D) $x <= – 8$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

102. What is the solution to the radical equation –10?

(A) $x >= –10$ (B) $x >= + 7$ (C) $x >= 1$ (D) $x <= –10$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

103. When solving the equation $+ 8,$ which values must be checked for extraneous roots?

(A) + 8 and + 8 (B) + 6 and –4 (C) + 8 and + 8 (D) 7 and 7

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

104. Which equation of a radical function would have the following domain and range? –$6;$ – 10

(A) + 6 (B) + 10 (C) + 10 (D) + 6

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

105. The variable

$$f(x) = a\sqrt{b(x-h)} + k$$

in the function $f(x) = a b(x - h)$ represents a _____.

TOP: bnk | short_answer **TYPE:** short_answer

106. The graph is translated 8 units to the left. Rendered ExamView reference:

$$f(x) = \sqrt{x}$$

$$g(x) = \sqrt{x+8} - 4$$

TOP: bnk | true_false **TYPE:** true_false

107. If the value of $f(x)$ satisfies $f(x) > 1$, then the graph of $y = sqrt(f(x))$ is below the graph of $y = f(x)$. Rendered ExamView reference:

$$y = \sqrt{f(x)}$$

TOP: bnk | short_answer **TYPE:** short_answer

108. range ($2,10,1) + 6 a<>b (a =$ –9)

TOP: bnk | completion **TYPE:** completion

109. When solving the equation –10 algebraically, the restriction on the variable is _____.

ANS: 2 **TOP:** bnk | completion **TYPE:** completion

110. The solution to the equation –4 is _____.

ANS: 7 **TOP:** bnk | completion **TYPE:** completion

111. $a*(sqrt(b*(x+k)))+h$

TOP: bnk | short_answer **TYPE:** short_answer

112. $b*(sqrt(a*(x+k)))+d$

TOP: bnk | short_answer **TYPE:** short_answer

113. $b*(sqrt(a*(x+h)))+k$

TOP: bnk | short_answer **TYPE:** short_answer

114. $b*(sqrt(a*(x+k)))+h$

TOP: bnk | short_answer **TYPE:** short_answer

115. $a*(sqrt(b*(x+h)))+k$

TOP: bnk | short_answer **TYPE:** short_answer

116. Create a graph of $sqrt(x)$ for each base function given, using transformations. a) [Graph diagram: Graph: $sqrt(x)$ | x: [$0.0, 10.0$], y: [$0.0, 3.2$]] b) [Graph diagram: Graph: $abs(sqrt(x))$ | x: [$0.0, 10.0$], y: [$0.0, 3.2$]] Rendered ExamView reference:

$$g(x) = f(x - 1) + 2$$

$$f(x) = \sqrt{x}$$

$$f(x) = \left| \sqrt{x} \right|$$

TOP: bnk | problem **TYPE:** problem

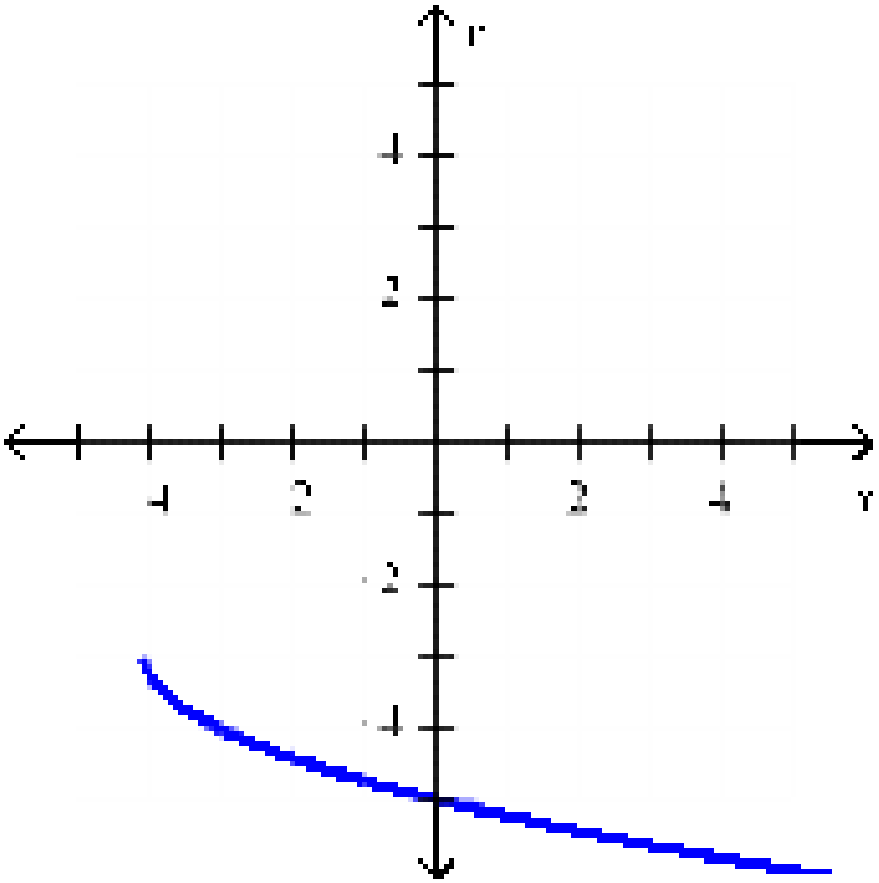
117. Determine the equation of each radical function, which has been transformed from $f(x) = x$ by the given translations.

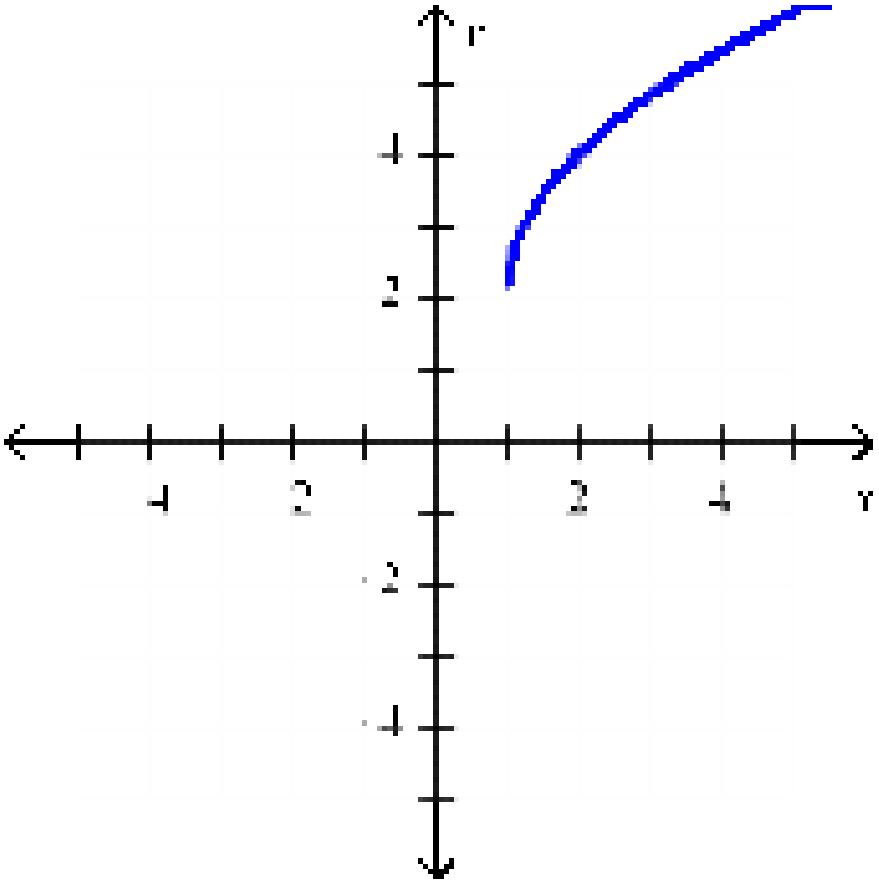
(a) vertical stretch by a factor of $5,$ then a horizontal translation of 6 units right
(b) horizontal stretch by a factor of $g(x) = 6x - 4$, then a vertical translation of 4 units down
(c) horizontal reflection in the $y-$axis, then a vertical translation of 9 units up and horizontal translation of 2 units right
(d) horizontal stretch by a factor of $g(x) = - (x - 2) + 9 or g(x) = - x + 2 + 9$, vertical reflection in the $x-$axis, and vertical stretch by a factor of $g(x) = - 2 3 1 1 / 3 x or g(x) = - 2 3 3x$

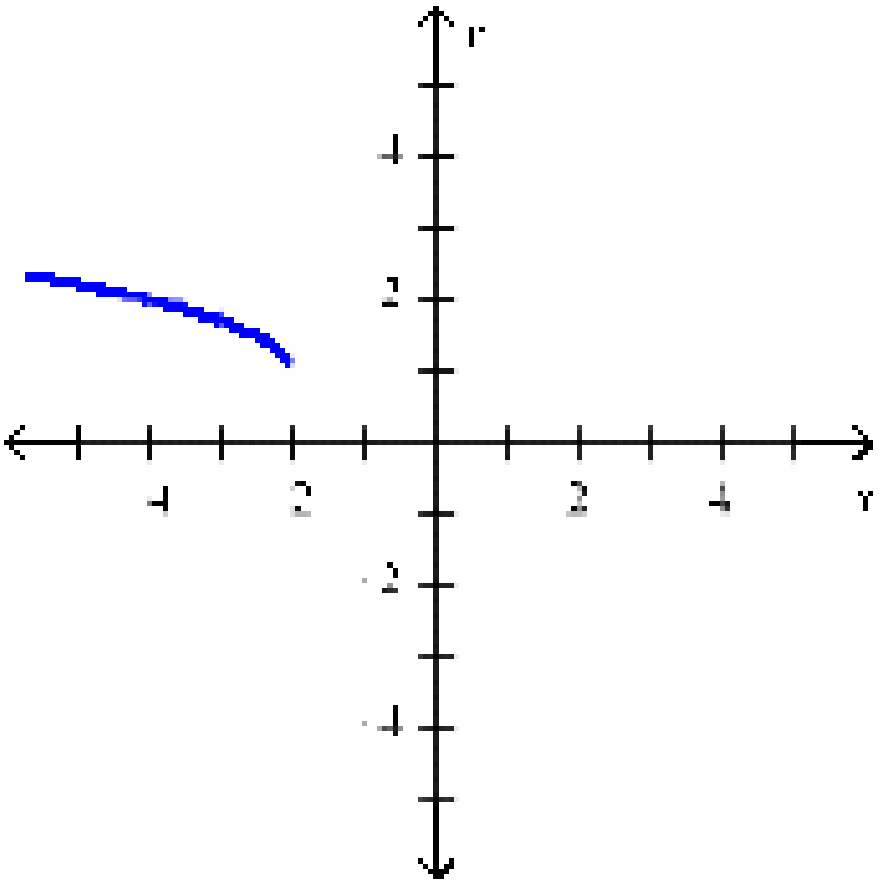
TOP: bnk | problem **TYPE:** problem

118. For each graph, write the equation of a radical function of the form $-sqrt(x+4)−3$. a) [Graph diagram: Graph: $-sqrt(x+4)−3$ | x: [$−4.0, 10.0$], y: [$−6.7, −3.0$]] b) [Graph diagram: Graph: $1/2*(sqrt(−2*(x+2)))+1$ | x: [$−10.0, −2.0$], y: [$1.0, 3.0$]] c) [Graph diagram: Graph: $1/2*(sqrt(−2*(x+2)))+1$ | x: [$−10.0, −2.0$], y: [$1.0, 3.0$]] Rendered ExamView reference:

$$y = a\sqrt{b(x-h)} + k$$







TOP: bnk | problem **TYPE:** problem

119. Determine the domain and range of each function. a) $f(x) = sqrt(-x) + 1/2$ b) $f(x) = 2sqrt((2/3)(x - 2)) + 1$ c) $y + 2 = 2sqrt(2(x - 1))$

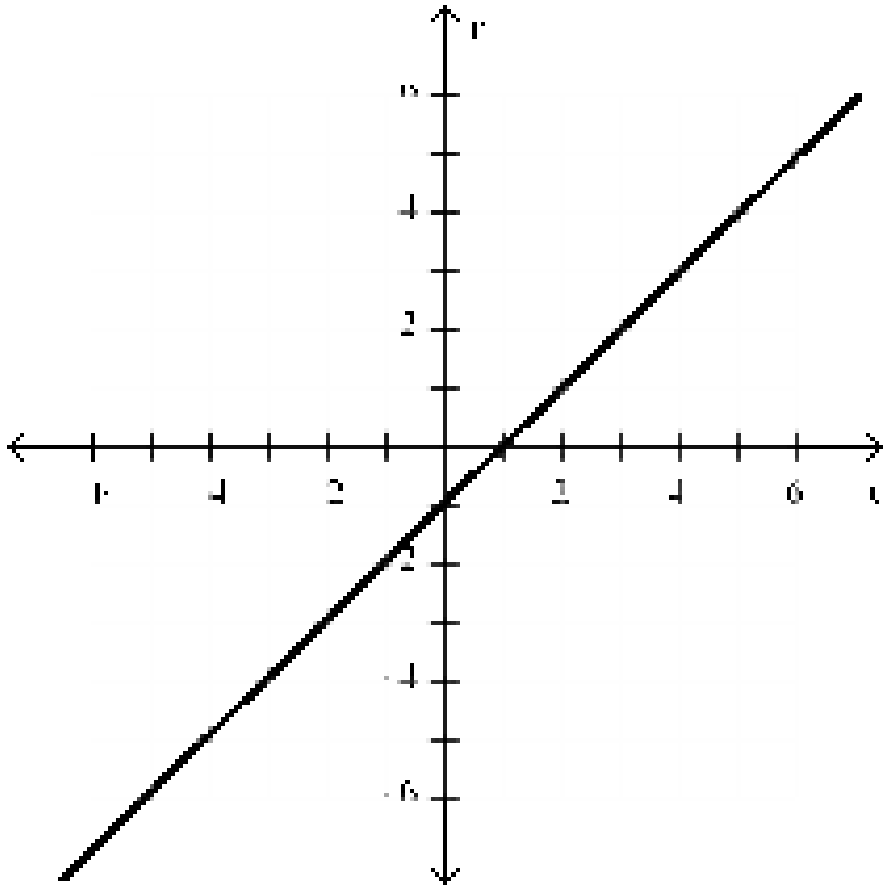
TOP: bnk | problem **TYPE:** problem

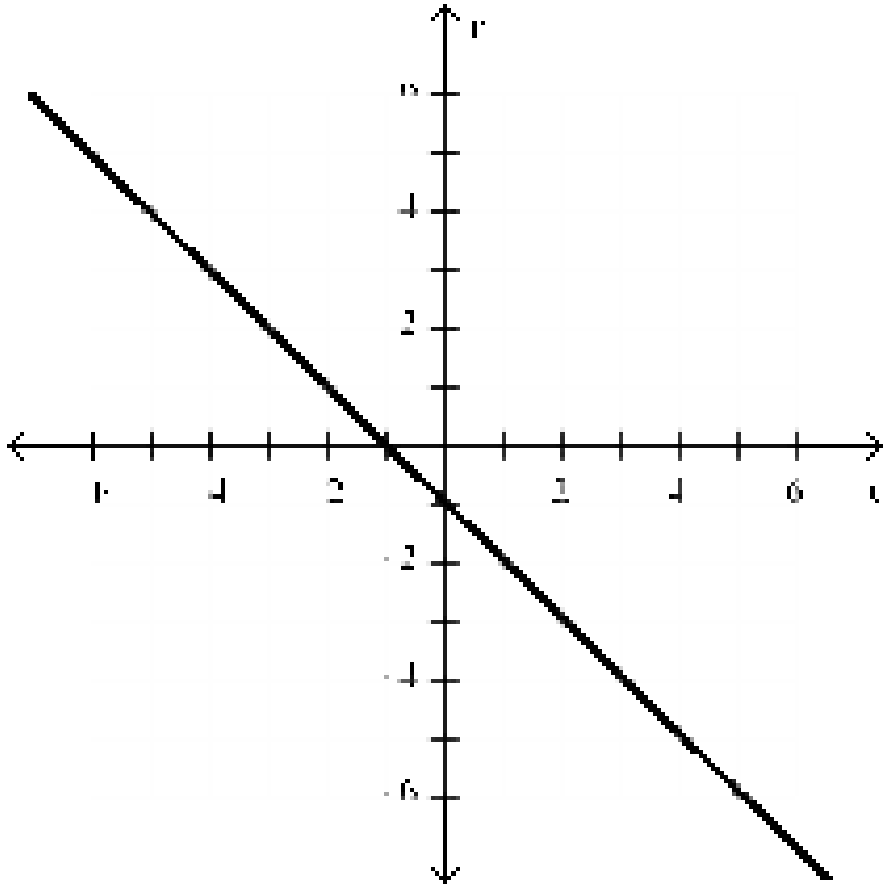
120. (a) Sketch the graph of each function $sqrt(x),$ and then sketch the graph of [Graph diagram: Graph: $sqrt(x)$ | x: [$0.0, 10.0$], y: [$0.0, 3.2$]].

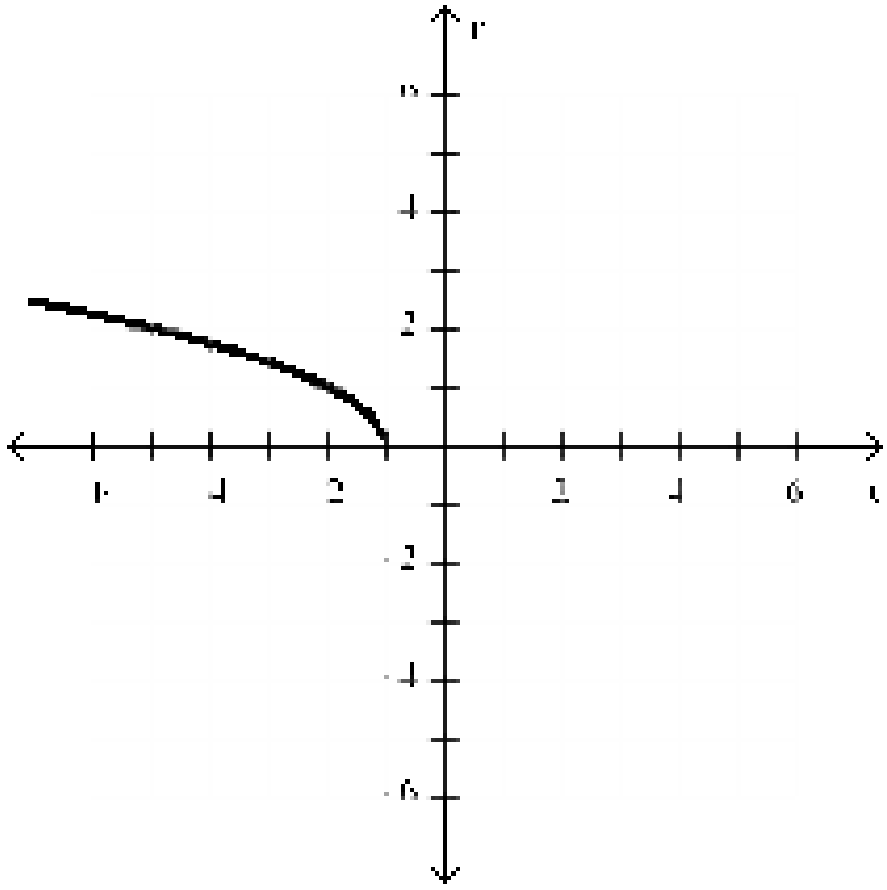
(b) [Graph diagram: Graph: $sqrt(x^2-4)$ | x: [$−10.0, 10.0$], y: [$0.0, 9.8$]] ii) [Graph diagram: Graph: x^2-4 | x: [$−10.0, 10.0$], y: [$−4.0, 96.0$]] iii) [Graph diagram: Graph: $sqrt(−0.5*x+2)$ | x: [$−10.0, 4.0$], y: [$0.0, 2.6$]] b) State the domain and range of each $f(x)$ and $g(x)$.

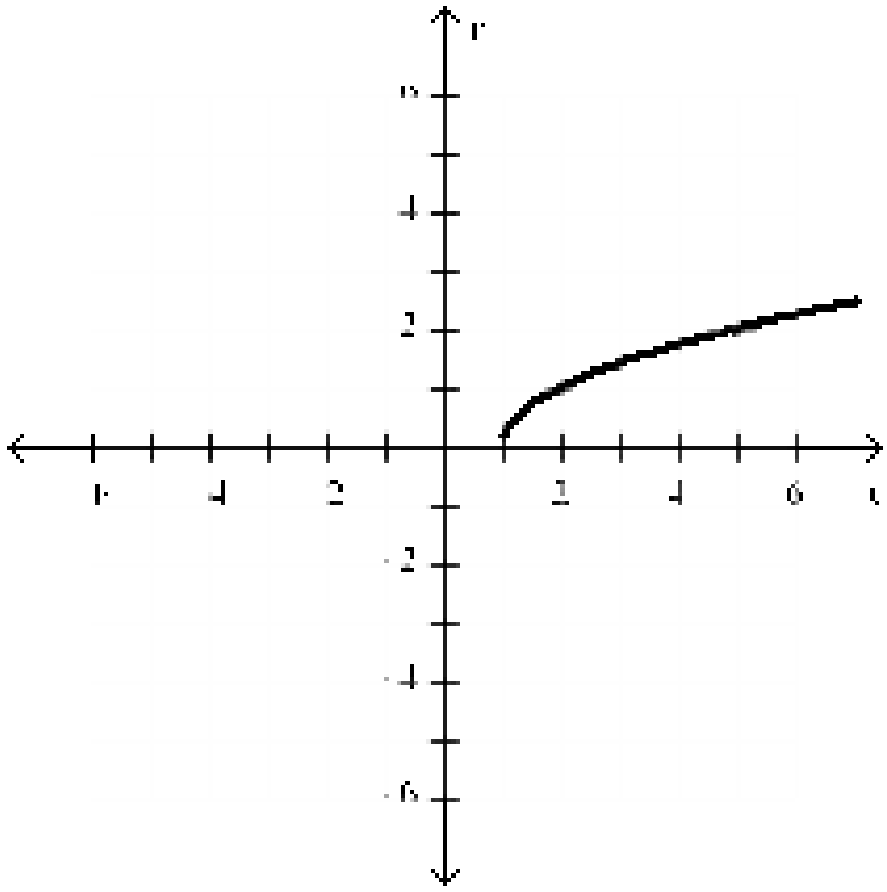
TOP: bnk | problem **TYPE:** problem

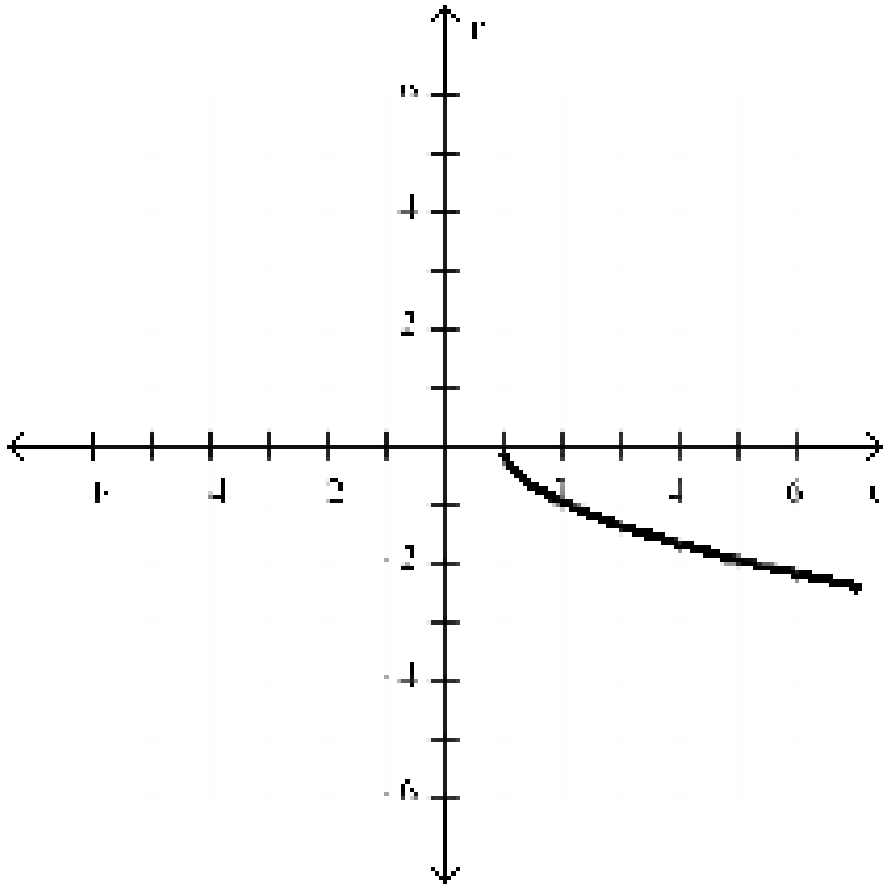
121. Which graph represents the square root of the graph shown? Original graph: $x+3$. Comparison graph: $-x-2$. Rendered ExamView reference:




y
6
4
2
-6
-4
-2
2
4
6
x

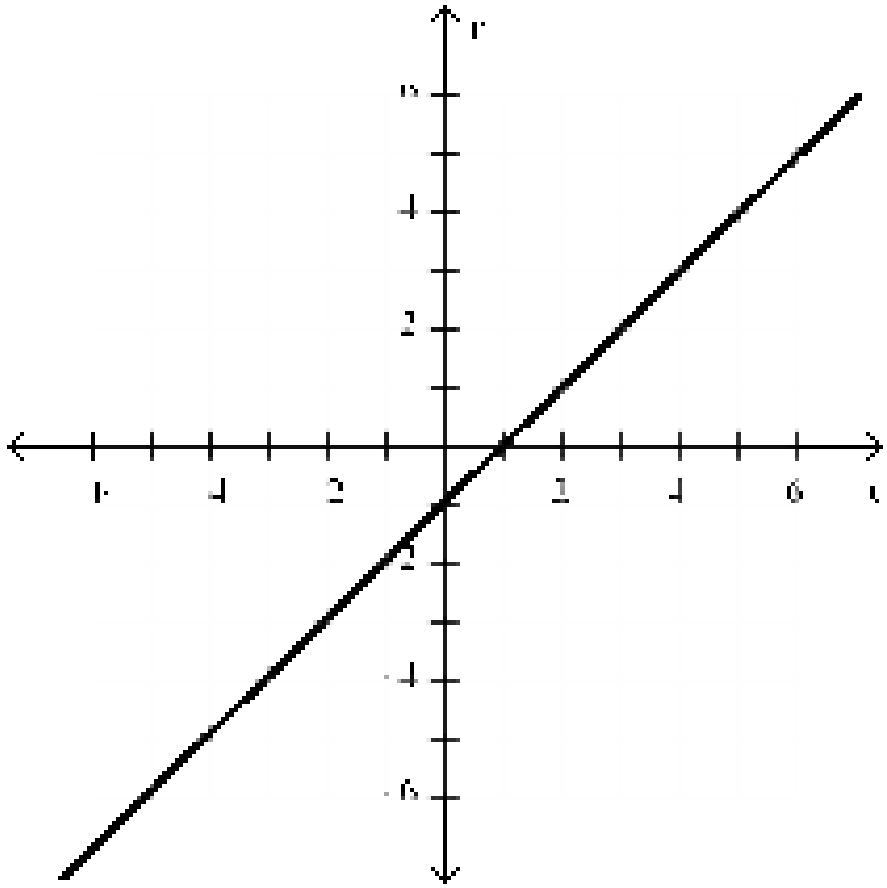

y
6
4
2
-6
-4
-2
2
4
6
x

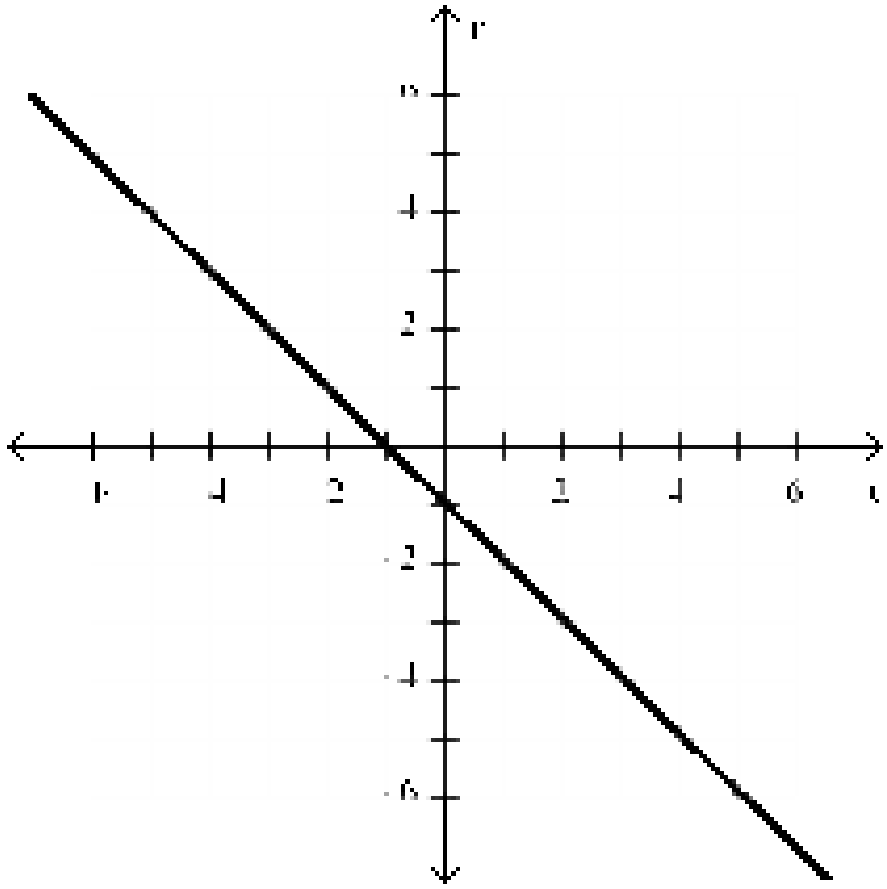


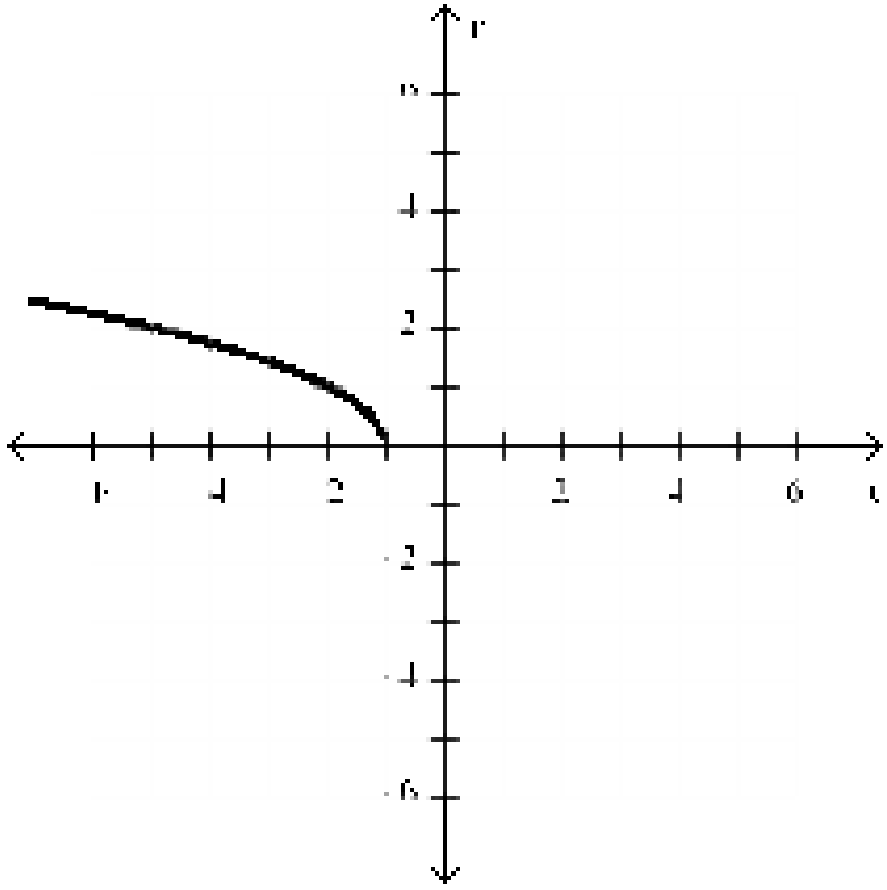


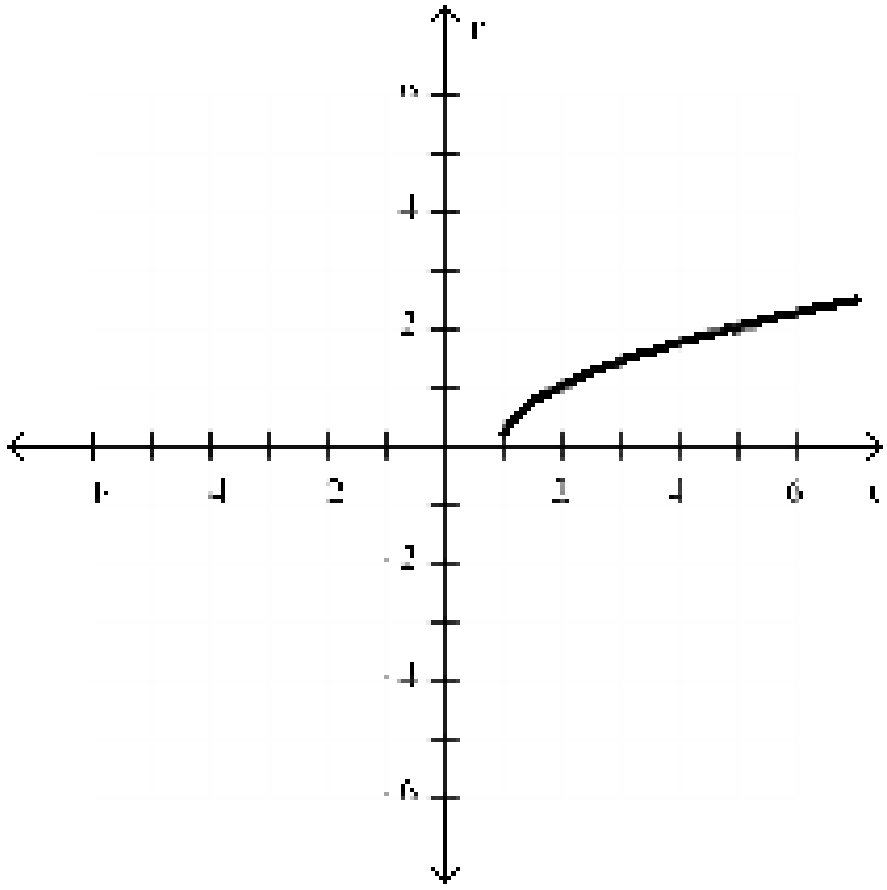
TOP: bnk | problem **TYPE:** problem

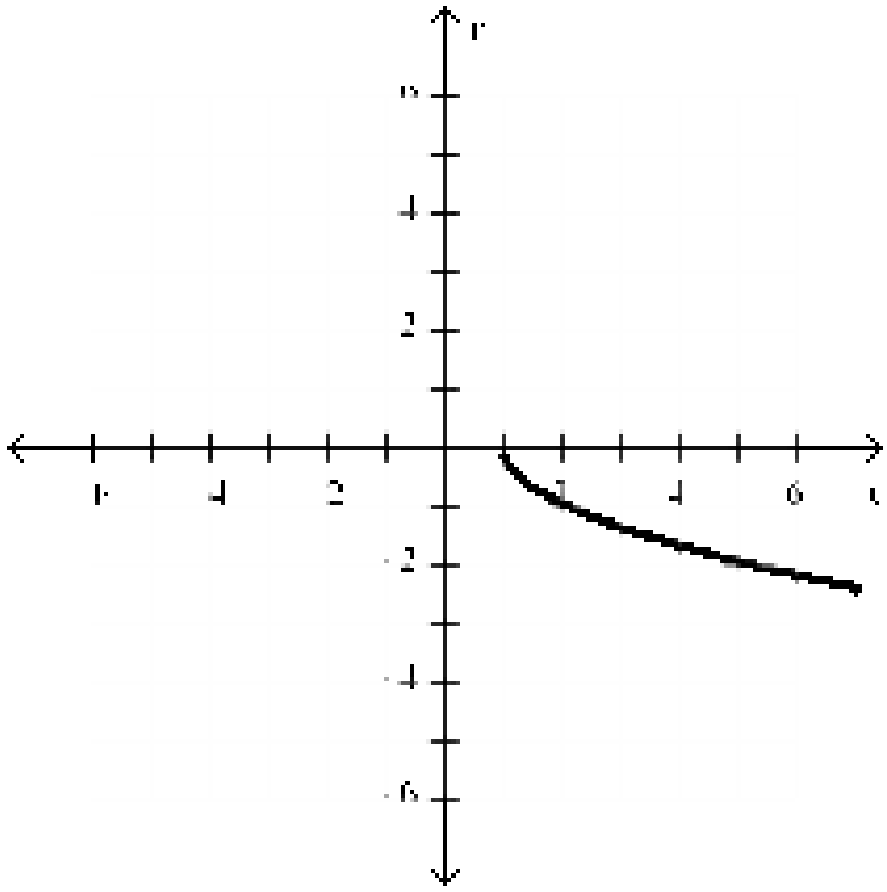
122. Which graph represents the square root of the graph shown? Original graph: $(−2*x+3)/6$. Comparison graph: $sqrt((−2*x+3)/6)$. Rendered ExamView reference:






y
8
4
2
-2
-4
-6
-8
-4
-2
2
4
6
x


y
8
4
2
-2
-4
-6
-8
-4
-2
2
4
6
x



TOP: bnk | short_answer **TYPE:** short_answer

123. Solve the equation $2x + 4 + 9 = 12$, $x - 2$, algebraically.

TOP: bnk | short_answer **TYPE:** short_answer

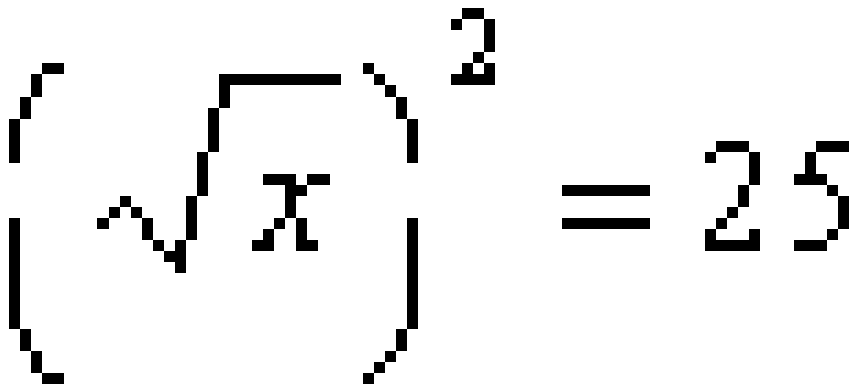
124. Solve the equation $3x - 6 = 12$ graphically.

TOP: bnk | short_answer **TYPE:** short_answer

125. Jim states that the equations

$$\sqrt{x^2} = 25$$

and



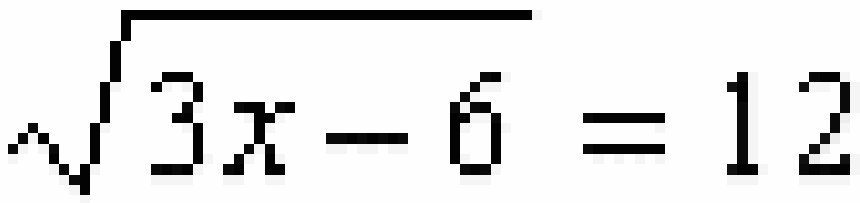
have the same solution. Is he correct? Justify your reasoning.

TOP: bnk | short_answer **TYPE:** short_answer

126. A student designs a special container as part of an egg drop experiment. She believes that the container can withstand a fall as long as the speed of the container does not exceed 80 ft$/s$. She uses the equation $v = (v_0) 2 + 2ad *$ to model the velocity, v, in feet per second, as a function of constant acceleration, a, in feet per second squared and the drop distance, d, in feet. Assuming the student s specifications are correct, will the egg break if the student drops the egg from shoulder height (5 ft) off a building 80 ft high? What is the maximum height the egg can be dropped from? (Note: The acceleration due to gravity is 32 ft$/s_2$.)

TOP: bnk | short_answer **TYPE:** short_answer

127. Solve the equation



graphically.

TOP: bnk | short_answer **TYPE:** short_answer

128. Solve the equation $1 x^3 - 7 = 1 x^3 = 8 x = 2$ algebraically.

TOP: bnk | short_answer **TYPE:** short_answer

129. The speed, s, in metres per second, of sound in dry air is can be described by the function –$8,$ where T is temperature, in degrees Celsius.

(a) Determine the domain and range of the function.
(b) Determine the speed of sound, to the nearest tenth of a metre per second, for each of the following temperatures.
(c) $12 C$

(i) $32 C$
(ii) –$8 C

(a) $ Graph the function and determine the $x-$intercept using the graph.
(b) What is the meaning of the $x-$intercept in this context?

TOP: bnk | problem **TYPE:** problem

130. The manufacturer of a new Global Positioning Satellite (GPS) system wants to predict the consumer interest in its new device. The company uses the function $−3*sqrt(-x-1)+15$ to model the number, I, in thousands, of pre$-$orders for the GPS as a function of the number, w, of weeks before the GPS release date.

(a) What are the domain and range and what do they mean in this situation?

(b) Identify the transformations represented by the function compared to [Graph diagram: Graph: $−3*sqrt(-x-1)+15$ | x: [$−10.0, −1.0$], y: [$6.0, 15.0$]].
(c) Graph the function and explain what the shape of the graph indicates about the situation.
(d) Determine the number of pre$-$orders the manufacturer can expect to have 10 weeks before the release date.

TOP: bnk | problem **TYPE:** problem

131. Which graph represents the square root of the graph shown? Original graph: $2*((x-1)^2)−3$. Comparison graph: $sqrt(2*((x-1)^2)−3)$. Rendered ExamView reference:

$$f(x) = 2(x-1)^2 - 3$$

TOP: bnk | problem **TYPE:** problem

132. For $5*(sqrt(-x))$ and [Graph diagram: Graph: $5*(sqrt(-x))$ | x: [$−10.0, 0.0$], y: [$0.0, 15.8$]], do the following.

(a) Graph $f(x)$ and $g(x)$ on the same set of axes.
(b) Determine the domain and range of each function.
(c) Explain which transformations would need to be applied to the graph of $f(x)$ to obtain the graph of $g(x)$.

TOP: bnk | problem **TYPE:** problem

133. The curves of best fit from the experiment are shown on the graph. $sqrt(3*(x-1))+1.6 a)$ When asked the type of function that could be used to model their findings, both groups argue that a radical function can be used. Do you agree with each group?

(a) How do these graphs differ from the graph of [Graph diagram: Graph: $sqrt(3*(x-1))+1.6$ | x: [$1.0, 10.0$], y: [$1.6, 6.8$]]?
(b) Write a function to approximate the graph for each group.
(c) What may have caused the differences in the data between the two groups? Justify your answer in terms of transformations.

TOP: bnk | problem **TYPE:** problem

134. The kinetic energy (energy of motion), E, in joules, of an object is given by the equation $sqrt((1/5)*x),$ where m represents the mass of the object, in kilograms, and v represents its speed, in metres per second.

(a) Determine the general equation for the velocity of a mass as a function of its kinetic energy.
(b) Find the speed of an object of mass 12 kg moving with a kinetic energy of
(c) $200 J$
(d) $ Graph the function if the mass is 12 kg.
(e) John conducts an experiment and graphs the data, resulting in the graph below. What is the mass of the object? [Graph diagram: Graph: $sqrt((1/5)*x)$ | x: [$0.0, 10.0$], y: [$0.0, 1.4$]]

TOP: bnk | problem **TYPE:** problem

135. Graph the functions below, and then identify and compare their domains and ranges. $3+1.2*(x^2)$ and [Graph diagram: Graph: $3+1.2*(x^2)$ | x: [$−10.0, 10.0$], y: [$3.0, 123.0$]] Rendered ExamView reference:

$$y_1 = 3 + 1.2x^2$$

$$y_2 = \sqrt{3 + 1.2x^2}$$

TOP: bnk | problem **TYPE:** problem

136. For the function, N represents the number, in thousands, of flowers pollinated and t represents the time, in days.

(a) If there are $10 000$ flowers, how long will it take to pollinate all of the flowers?
(b) Sun$-$Eui believes that the second generation of bees can reduce the time to pollinate the flowers by half by changing the function to [DIAGRAM NEEDED]. Is she correct? Justify your reasoning.
(c) Should Sun$-$Eui continue with the project?

TOP: bnk | problem **TYPE:** problem

137. Match the correct term with the correct part of the statement.

(A) quotient (B) remainder (C) divisor (D) dividend

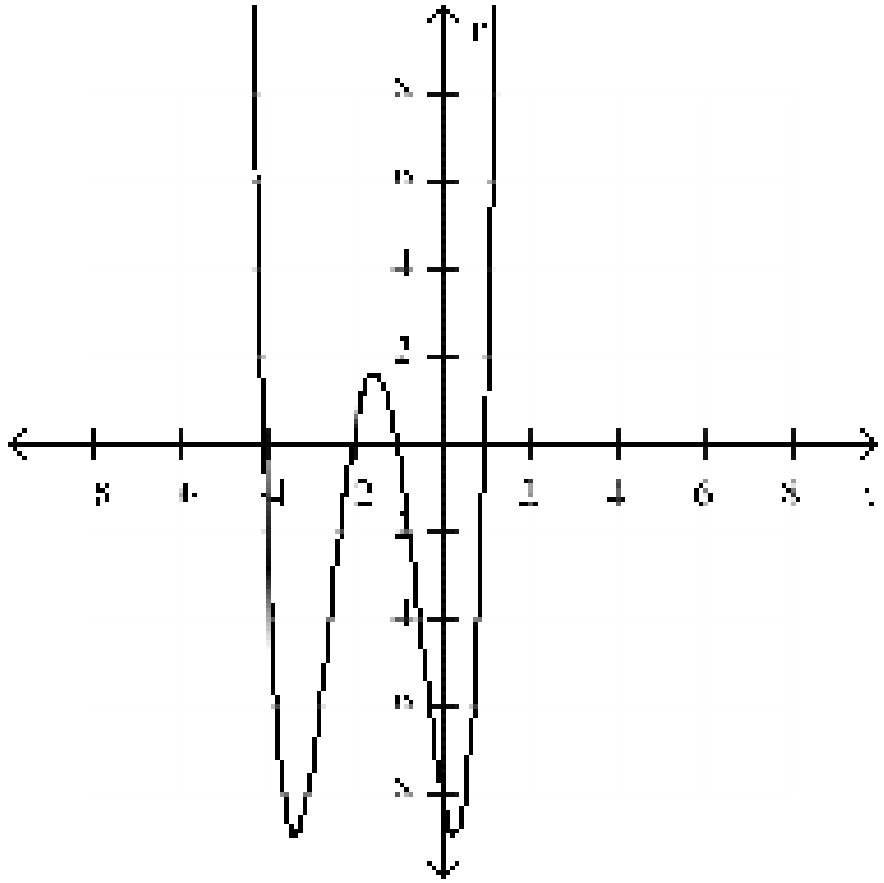
TOP: bnk | matching **TYPE:** matching

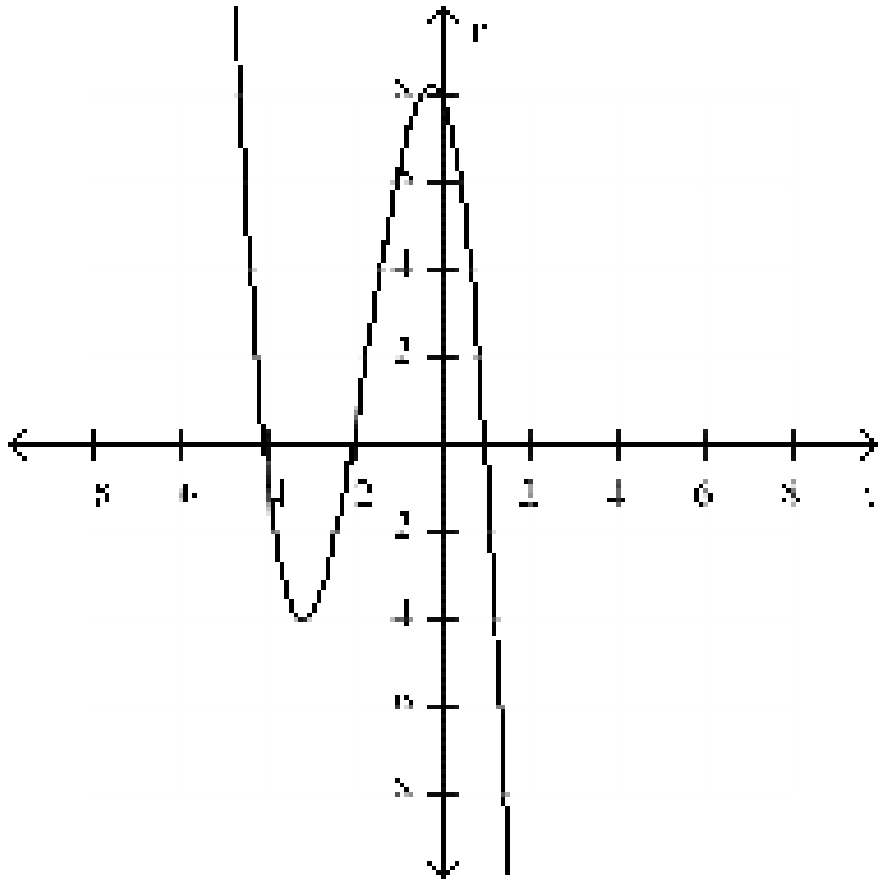
138. Which of the following is a polynomial function?

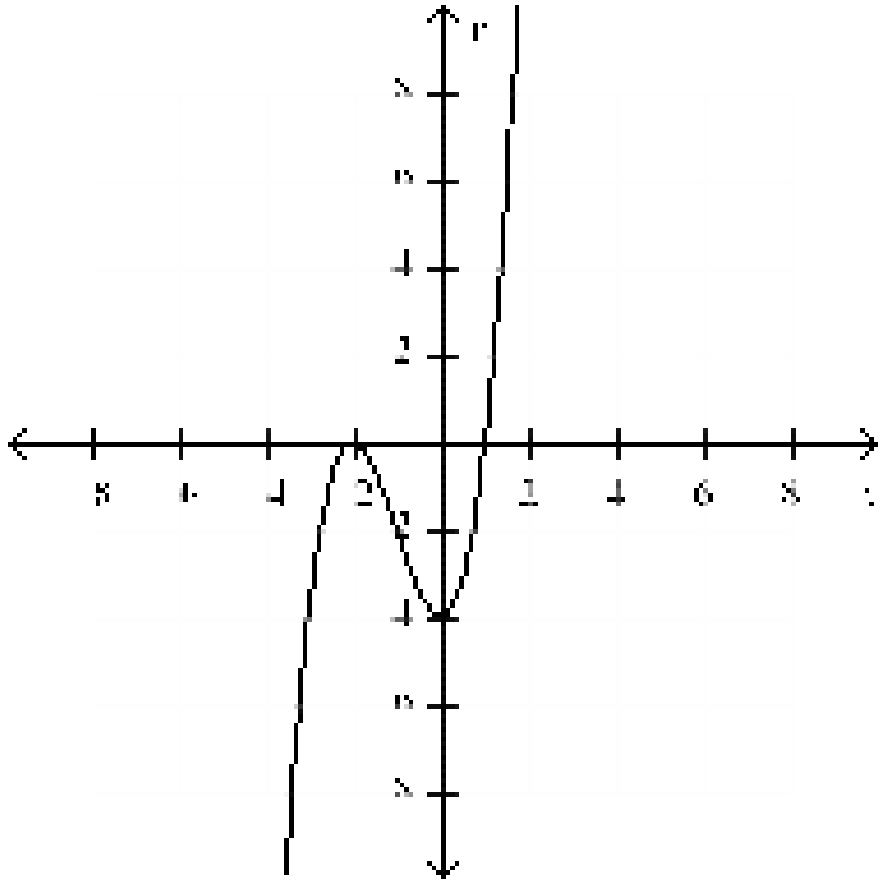
(A) $f(x) = −4x^3 + 4x^2 - 7x + 9$ (B) $f(x) = −4/x + 4$ (C) $f(x) = sqrt(x - 7) + 9$ (D) $f(x) = (−4x + 4)/(x + 9)$

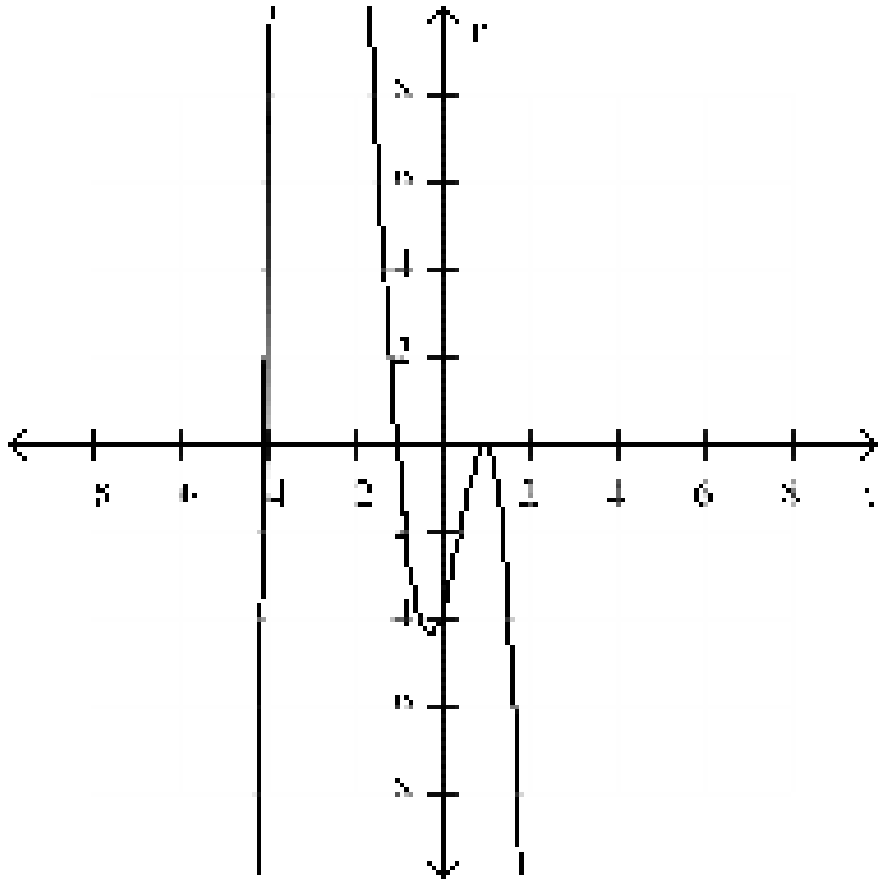
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

139. Which graph represents an odd$-$degree polynomial function with two $x-$intercepts? Recovered graph functions: ($x+a)*(x+(2))*(x+(4))*(x+d); −1*(x+a)*(x+(2))*(x+(4)); (x+a)*(x+(2))*(x+(2)); −1*(x+a)*(x+a)*(x+(4))*(x+d)$. Rendered ExamView reference:









(A) graph of (x+a)*(x+(2))*(x+(4))*(x+d)

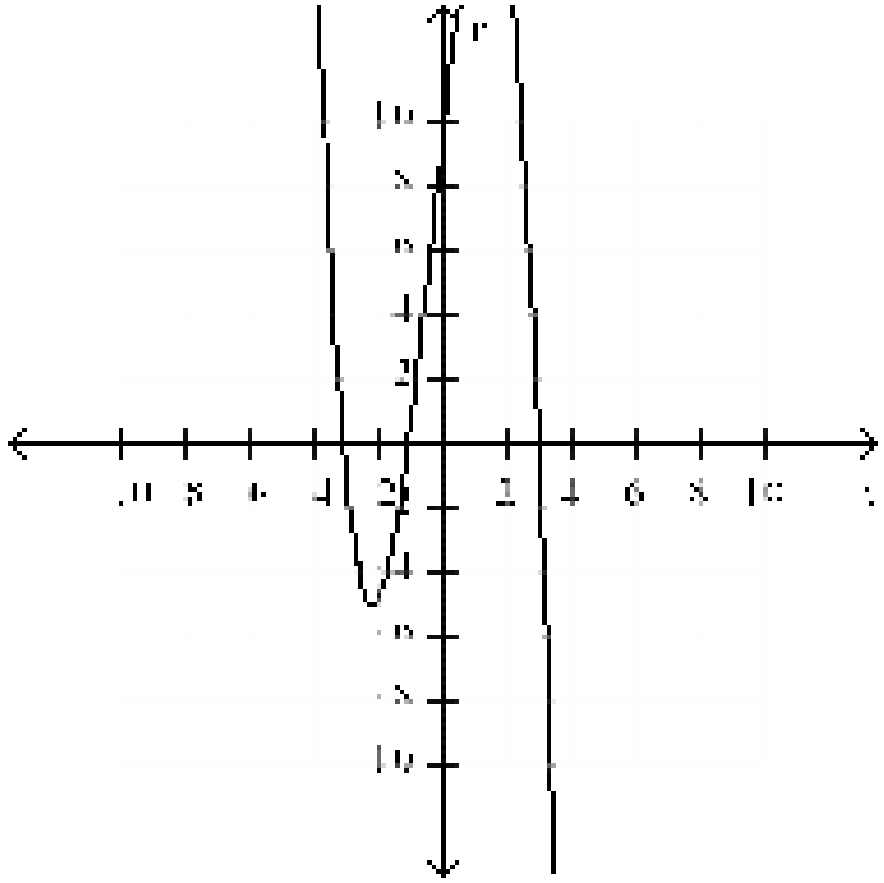
(B) graph of −1*(x+a)*(x+(2))*(x+(4))

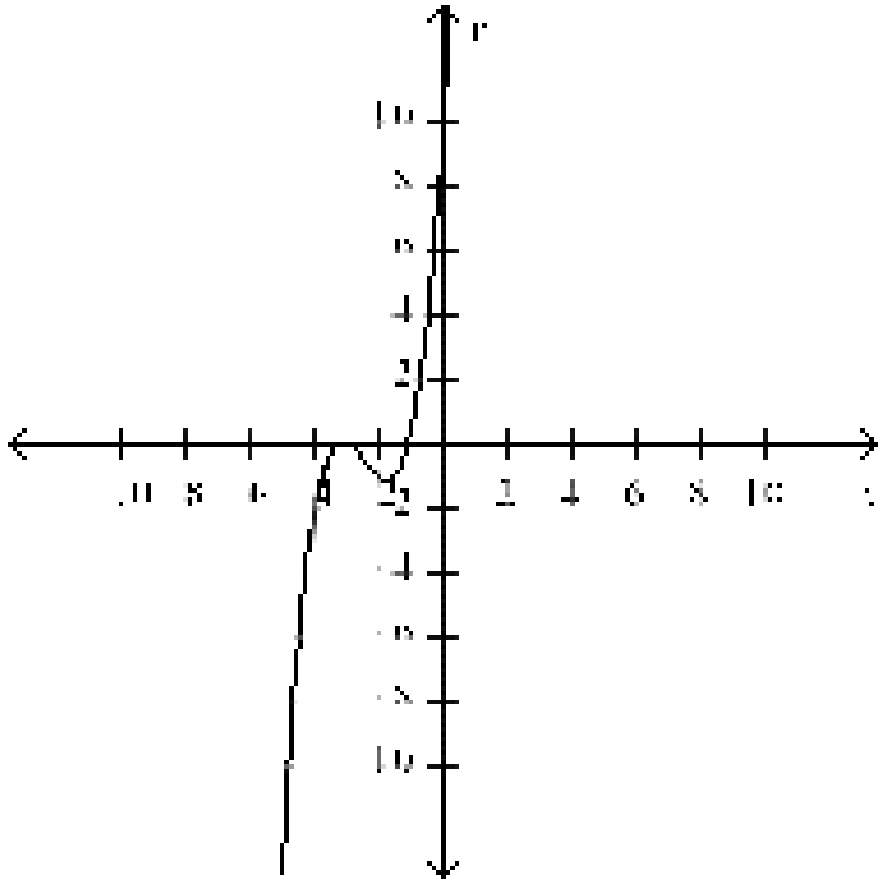
(C) graph of (x+a)*(x+(2))*(x+(2))

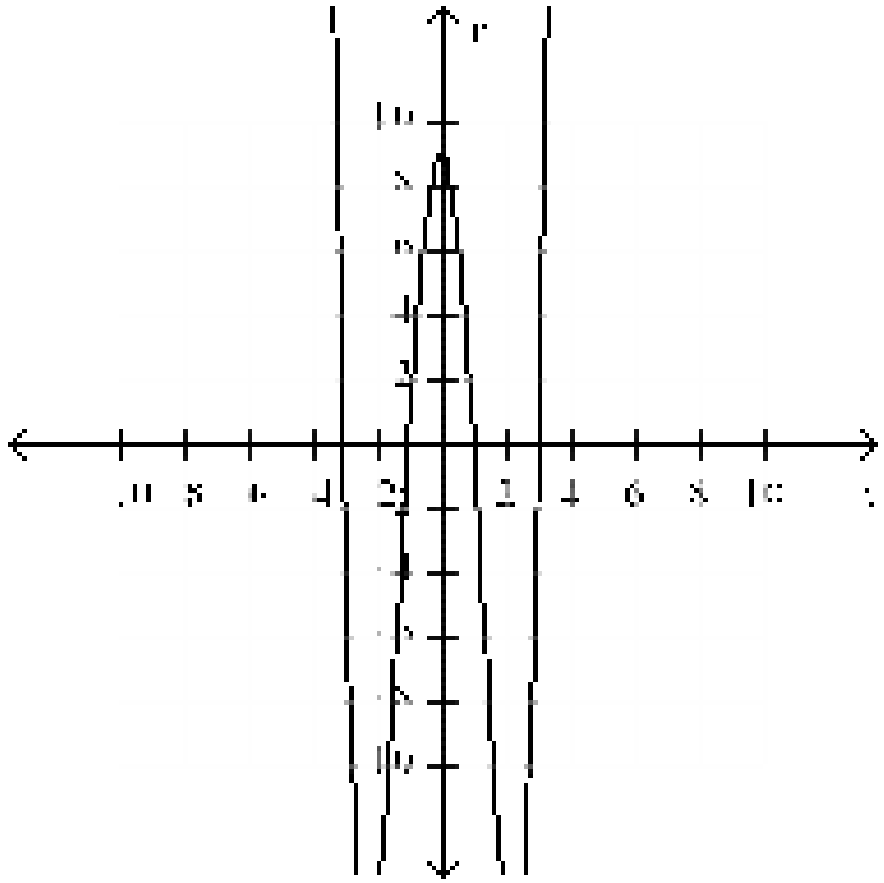
(D) graph of −1*(x+a)*(x+a)*(x+(4))*(x+d)

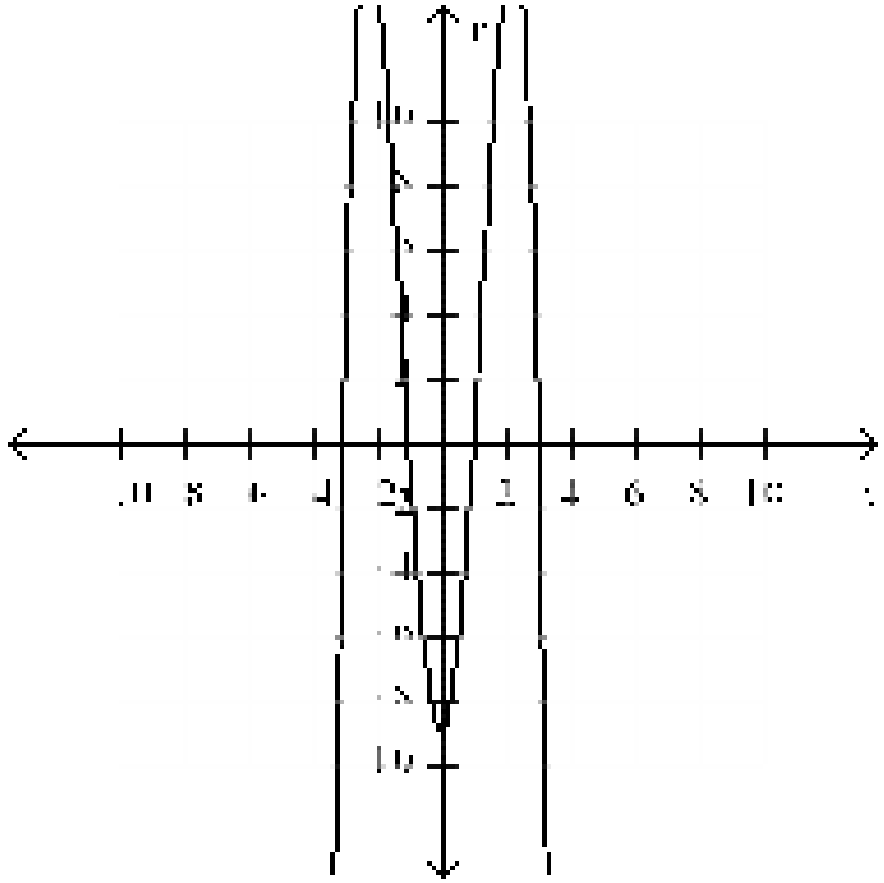
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

140. Which graph represents an even$-$degree polynomial function with a $y-$intercept of $+ 3$? Recovered graph functions: $−1*(x+a)*(x+(3))*(x+(−3)); (x+a)*(x+(3))*(x+(3)); (x+a)*(x+(3))*(x+(−3))*(x+d); −1*(x+a)*(x+(3))*(x+(−3))*(x+d)$. Rendered ExamView reference:









(A) graph of $-1*(x+a)*(x+(3))*(x+(-3))$

(B) graph of $(x+a)*(x+(3))*(x+(3))$

(C) graph of $(x+a)*(x+(3))*(x+(-3))*(x+d)$

(D) graph of $-1*(x+a)*(x+(3))*(x+(-3))*(x+d)$

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

141. How many $x-$intercepts are possible for the polynomial function 5?

(A) 5 (B) 5 (C) 4 (D) 5

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

142. What is the restriction on x if – 8 is divided by $+$?

(A) 8 (B) $−1$ (C) $−5$ (D) 2

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

143. If –9 is divided by $+ 9,$ then the restriction on x is

(A) $6/5$ (B) $−5/6$ (C) $5/6$ (D) $−6/5$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

144. What is the remainder when – 8 is divided by $+ 9$?

(A) 1070 (B) 962 (C) $−1070$ (D) $−962$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

145. If – 135 is divided by –2 to give a quotient of – 6 and a remainder of $+ 5,$ then which of the following is true?

(A) $−952$ (B) 952 (C) + 952

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

146. If –8 is divided by – $8,$ what is the remainder?

(A) $−160$ (B) 160 (C) $−320$ (D) 320

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

147. If –4 is divided by $+ 8,$ what is the remainder?

(A) $−296$ (B) 152 (C) 296 (D) $−152$

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

148. When 125 is divided by $125,$ the remainder is

(A) $x >= 1$ (B) $x >= 125$ (C) $x >= – 2$ (D) $x <= 1$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

149. For a polynomial $P(x),$ if $6 = 0,$ then which of the following must be a factor of $P(x)$?

(A) 6 (B) – 6 (C) – 6

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

150. A factor of $+ 16$ is

(A) – 3 (B) – 5 (C) – 8

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

151. Which of the following binomials is a factor of $+ 12$?

(A) – 2 (B) $−1$ (C) $−9$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

152. Which set of values for x should be tested to determine the possible zeros of $+$?

(A) 35 (B) 5 (C) 7 (D) 35

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

153. Determine the value of k so that $+ 2$ is a factor of $+ 10$.

(A) + 10 (B) + 10 (C) 14 (D) $−14$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

154. Which of the following is the fully factored form of $+ 2$?

(A) – 4 (B) + 5 (C) – 3

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

155. The fully factored form of $+ 10$ is

(A) – 3 (B) – 6 (C) – 5 (D) + 24

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

156. Which of the following is the fully factored form of $+ 9$?

(A) – 2 (B) – 9 (C) + 2

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

157. One root of the equation $+ 7$ is

(A) + 7 (B) + 7 (C) + 7 (D) $−3$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

158. What is the maximum number of real distinct roots that a cubic equation can have?

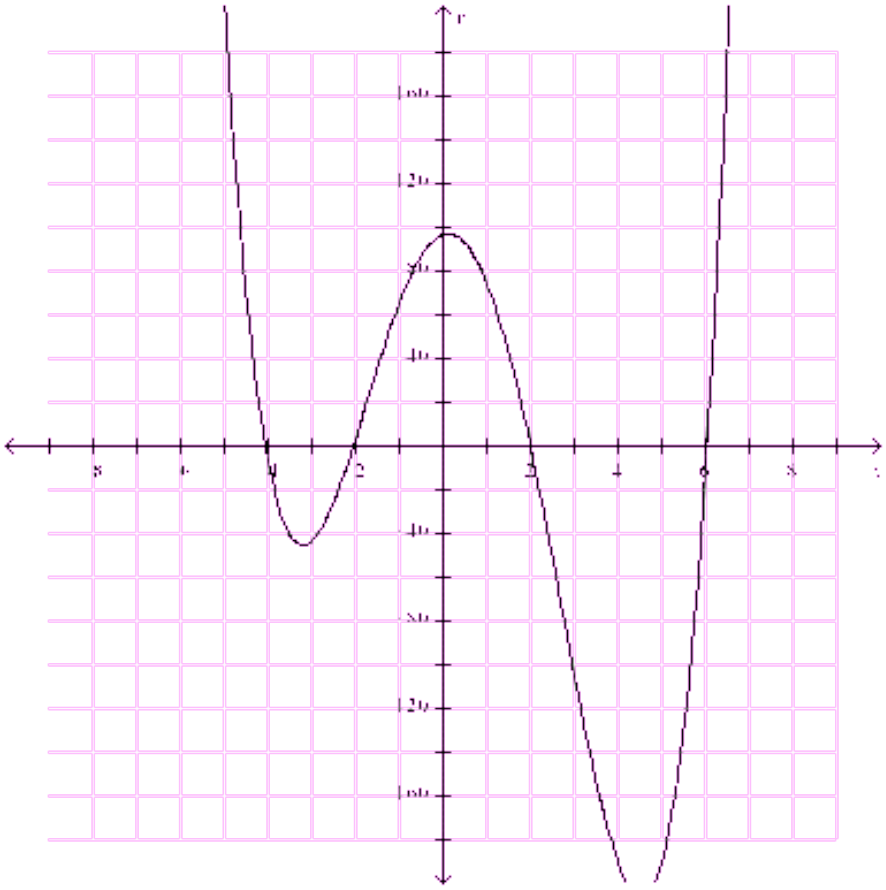
(A) cubic (B) 3 (C) 4 (D) 2

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

159. Based on the graph of – $2,$ what are the real roots of $+ 2$? – 2 Rendered ExamView reference:

$$f(x) = x^4 - 2x^3 - 24x^2 + 8x + 96$$

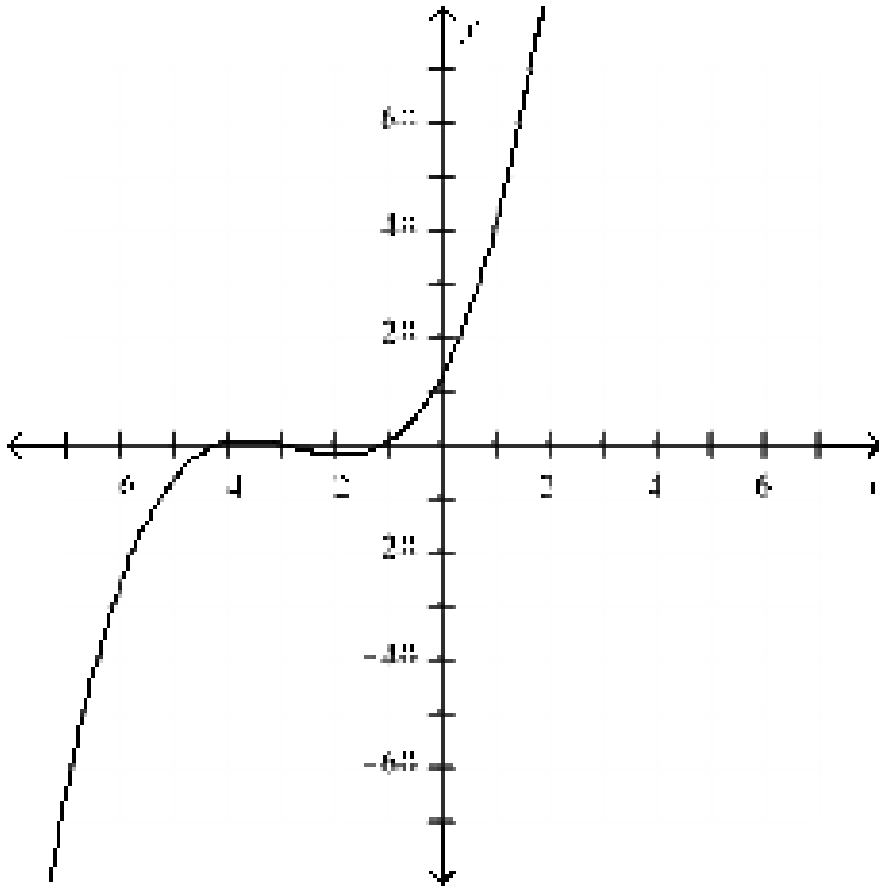
$$x^4 - 2x^3 - 24x^2 + 8x + 96 = 0$$

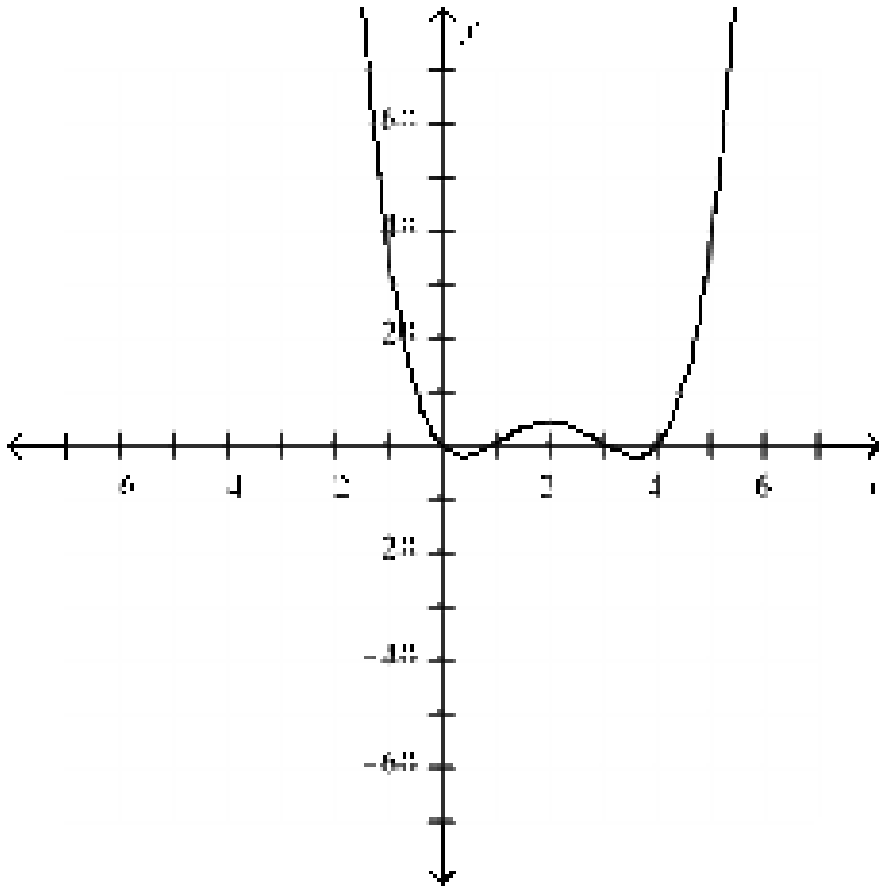


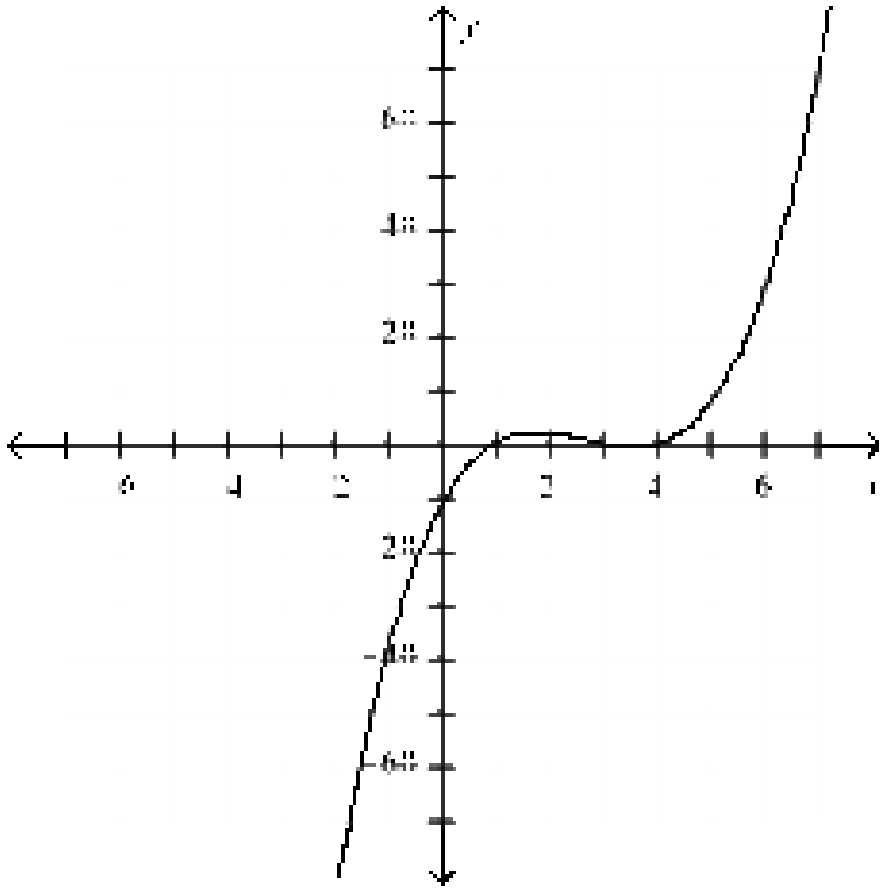
(A) 6 (B) 2 (C) $−2$ (D) $−4$

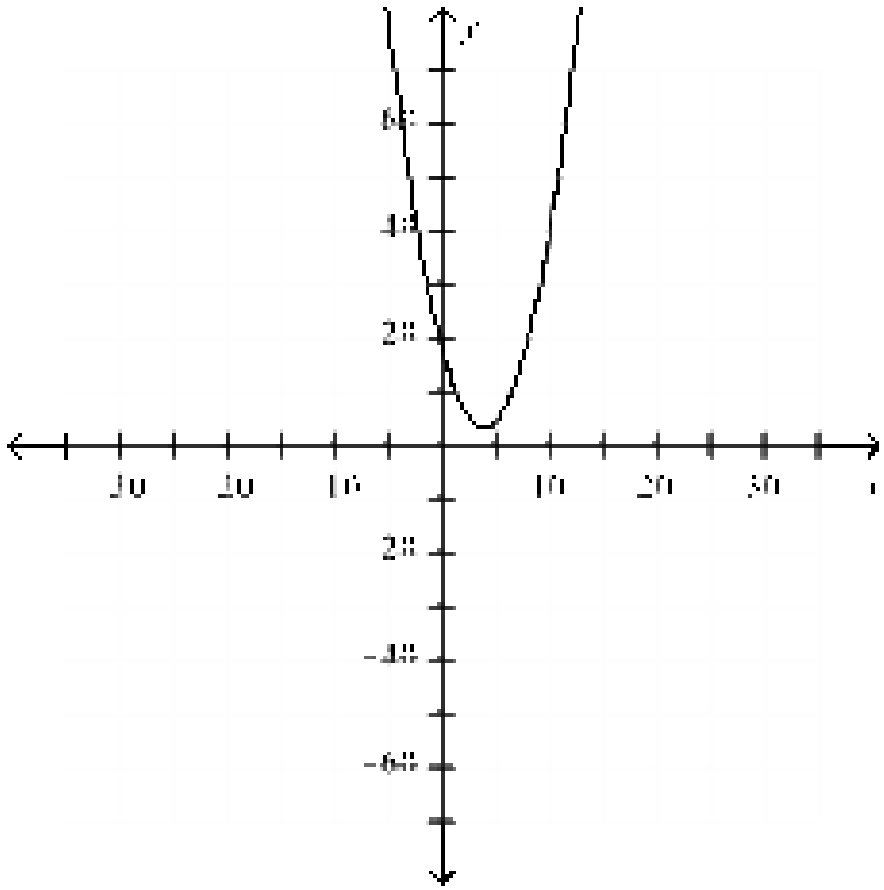
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

160. Which of the following graphs of polynomial functions corresponds to a cubic polynomial equation with roots $4, 1,$ and 3? Recovered graph functions: ($x-(−4))*(x-b)*(x-(−3)); x^4+(−8)*x^3+(19)*x^2+(−12)*x; x^3+(−8)*x^2+(19)*x+(−12); x^2+(−8)*x+(19)$. Rendered ExamView reference:







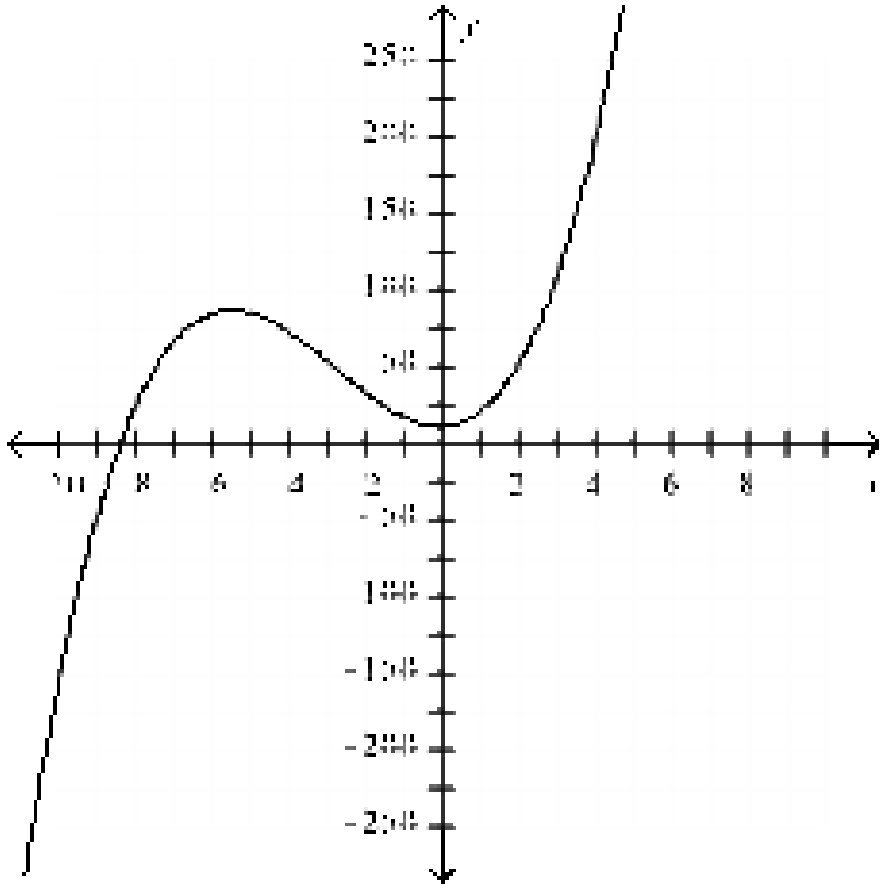


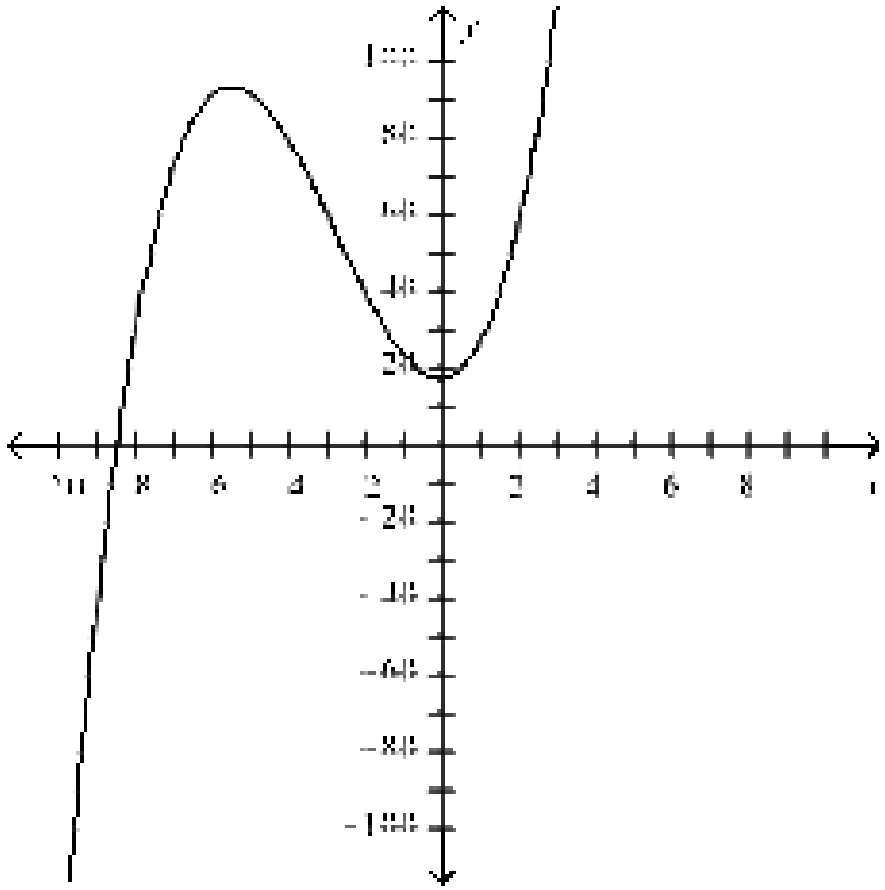
(A) 4 (B) 1 (C) 3

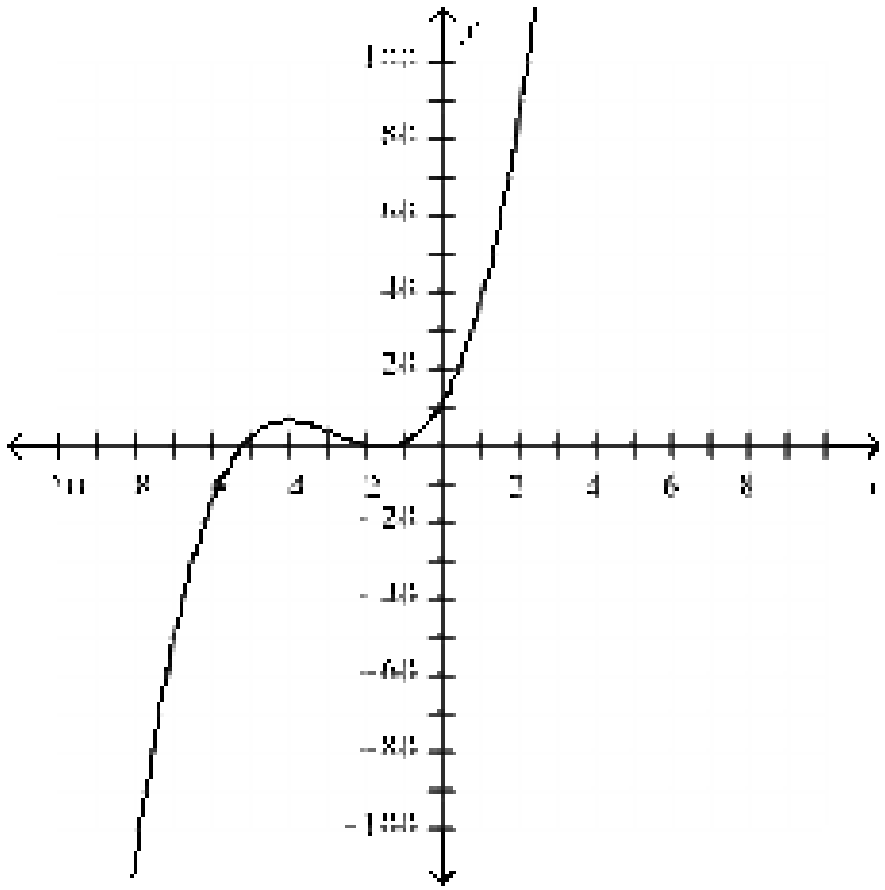
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

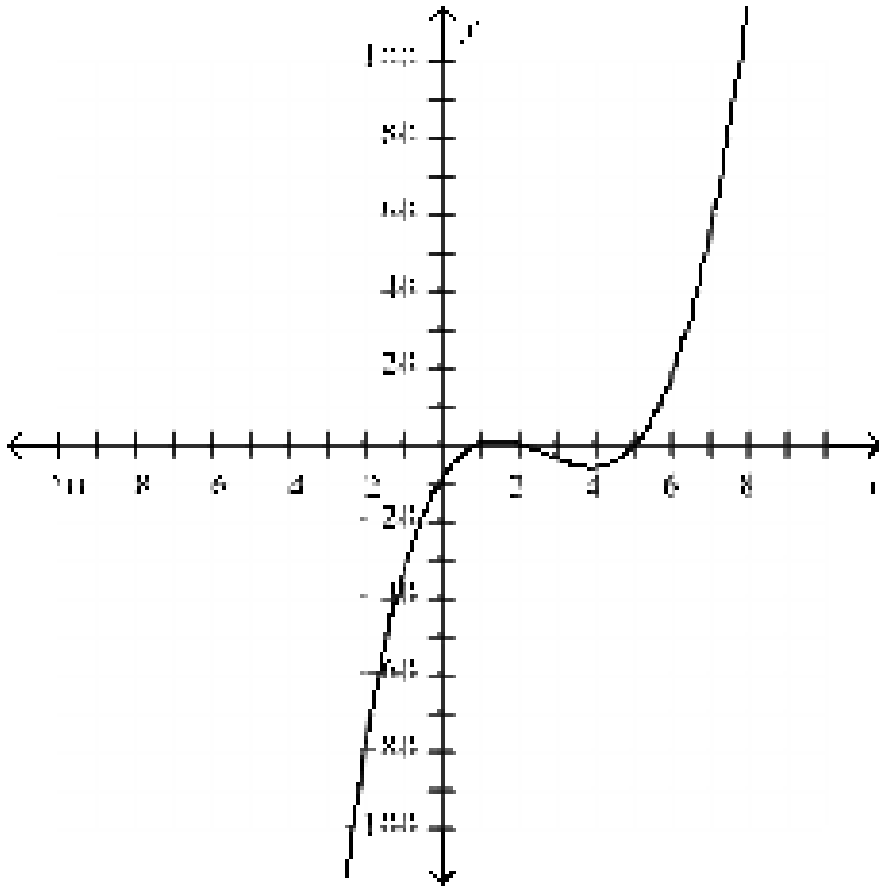
161. The graph that corresponds to the function $+ 8$ is Recovered graph functions: $x^3+(8)*x^2+(10); x^3+(8)*x^2+(17); x^3+(8)*x^2+(17)*x+(10); (x-a)*(x-(2))*(x-(5))$. Rendered ExamView reference:

$$f(x) = x^3 + 8x^2 + 17x + 10$$









(A) graph of x^3+(8)*x^2+(10)

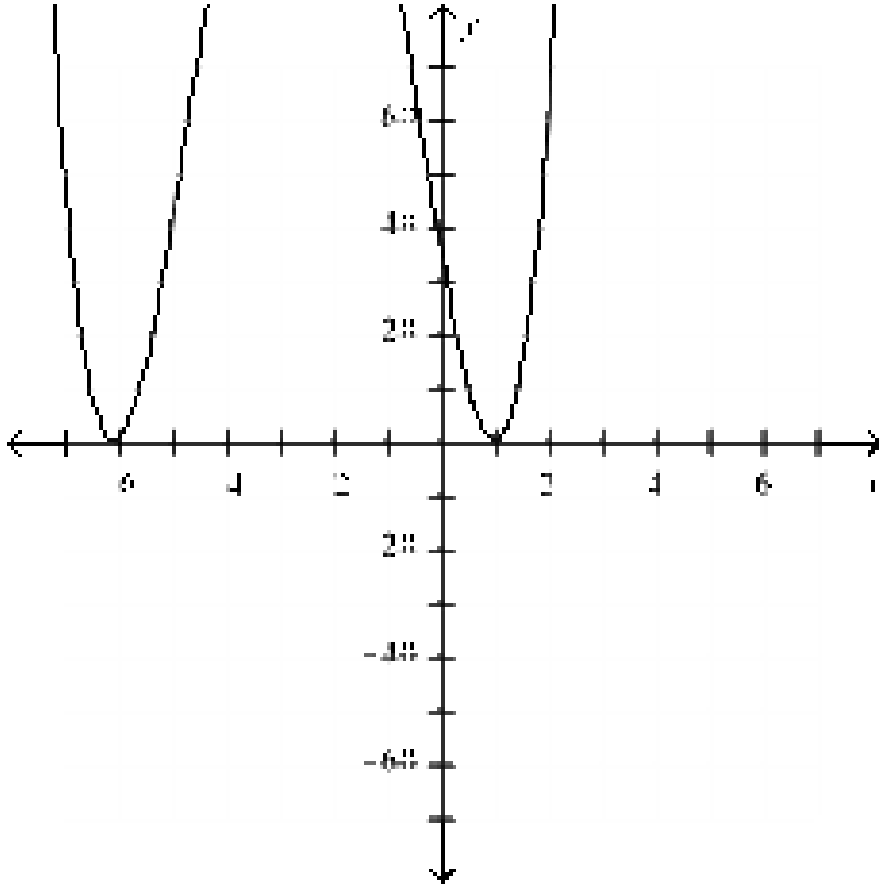
(B) graph of x^3+(8)*x^2+(17)

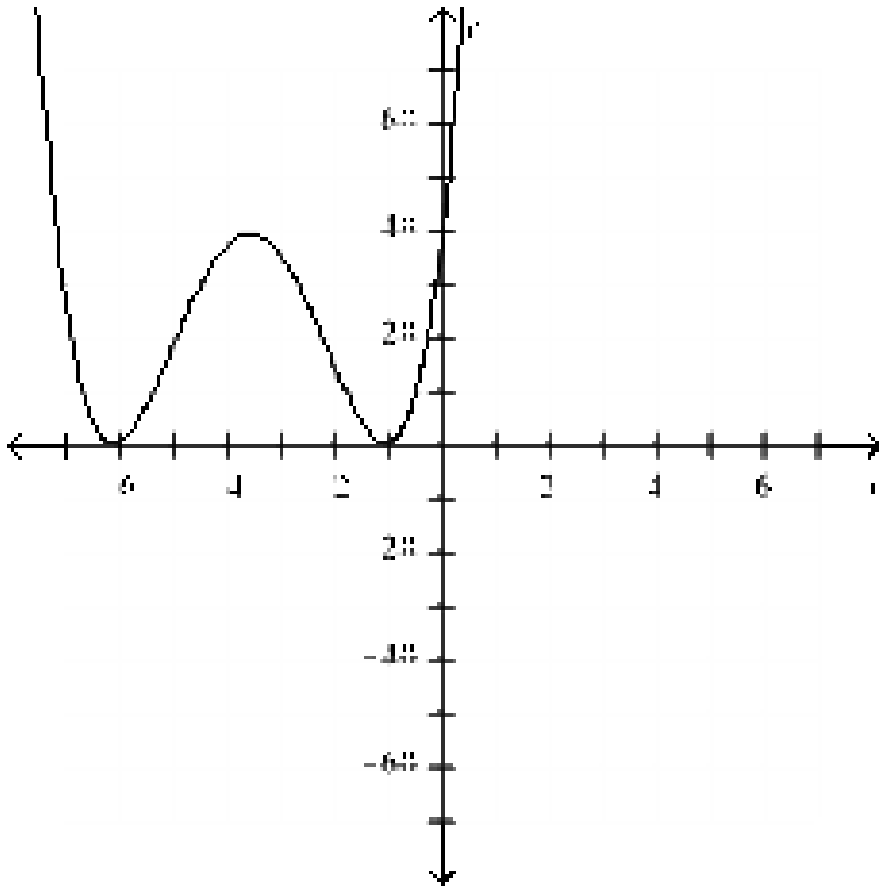
(C) graph of x^3+(8)*x^2+(17)*x+(10)

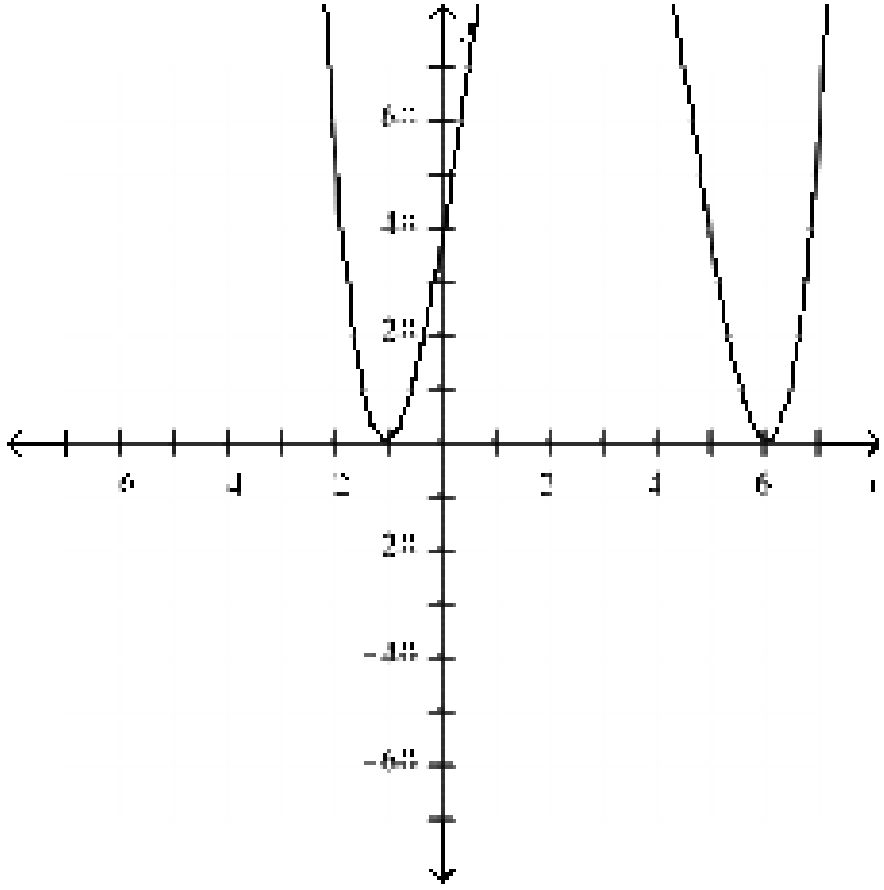
(D) graph of (x-a)*(x-(2))*(x-(5))

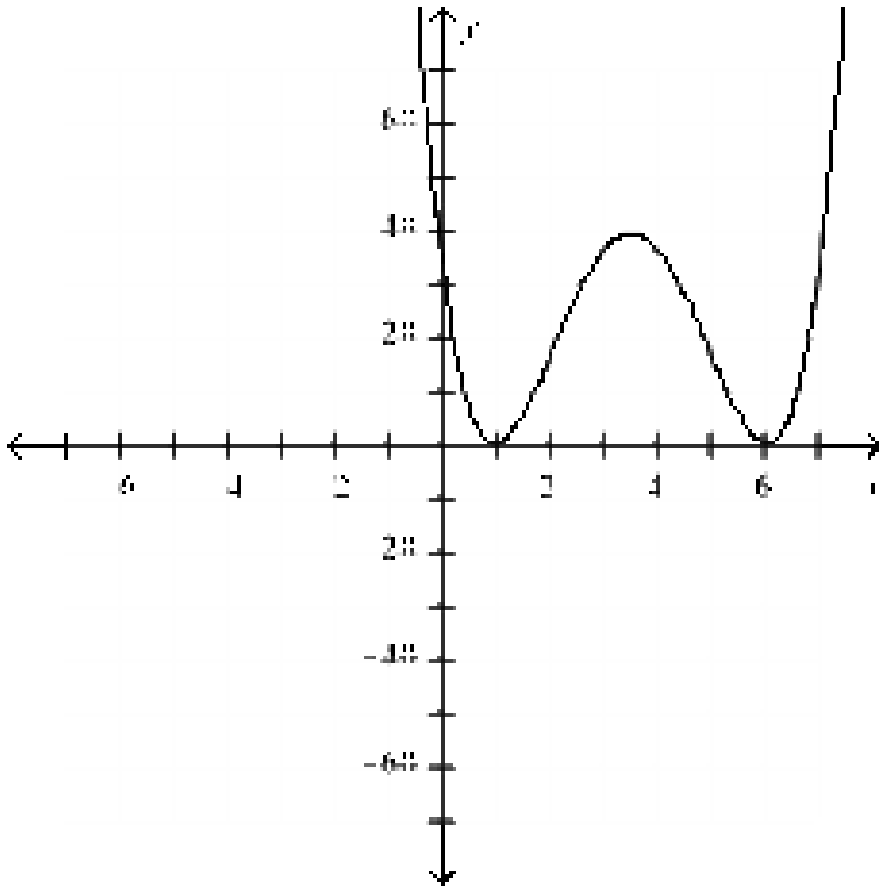
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

162. Which of the following graphs of polynomial functions corresponds to a polynomial equation with zeros –$6 ($multiplicity of $2)$ and –$1 ($multiplicity of $2)$? Recovered graph functions: ($x+(6))^2*(x-(1))^2; (x+(6))^2*(x+(1))^2; (x-(6))^2*(x+(1))^2; (x-(6))^2*(x-(1))^2$. Rendered ExamView reference:









(A) graph of (x+(6))^2*(x-(1))^2

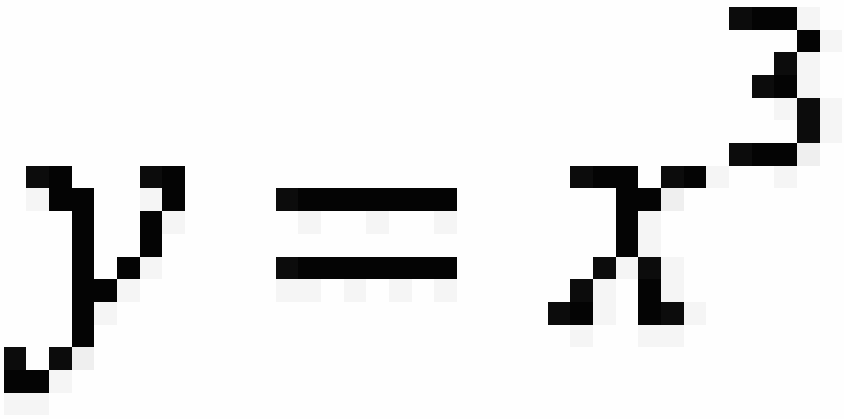
(B) graph of (x+(6))^2*(x+(1))^2

(C) graph of (x-(6))^2*(x+(1))^2

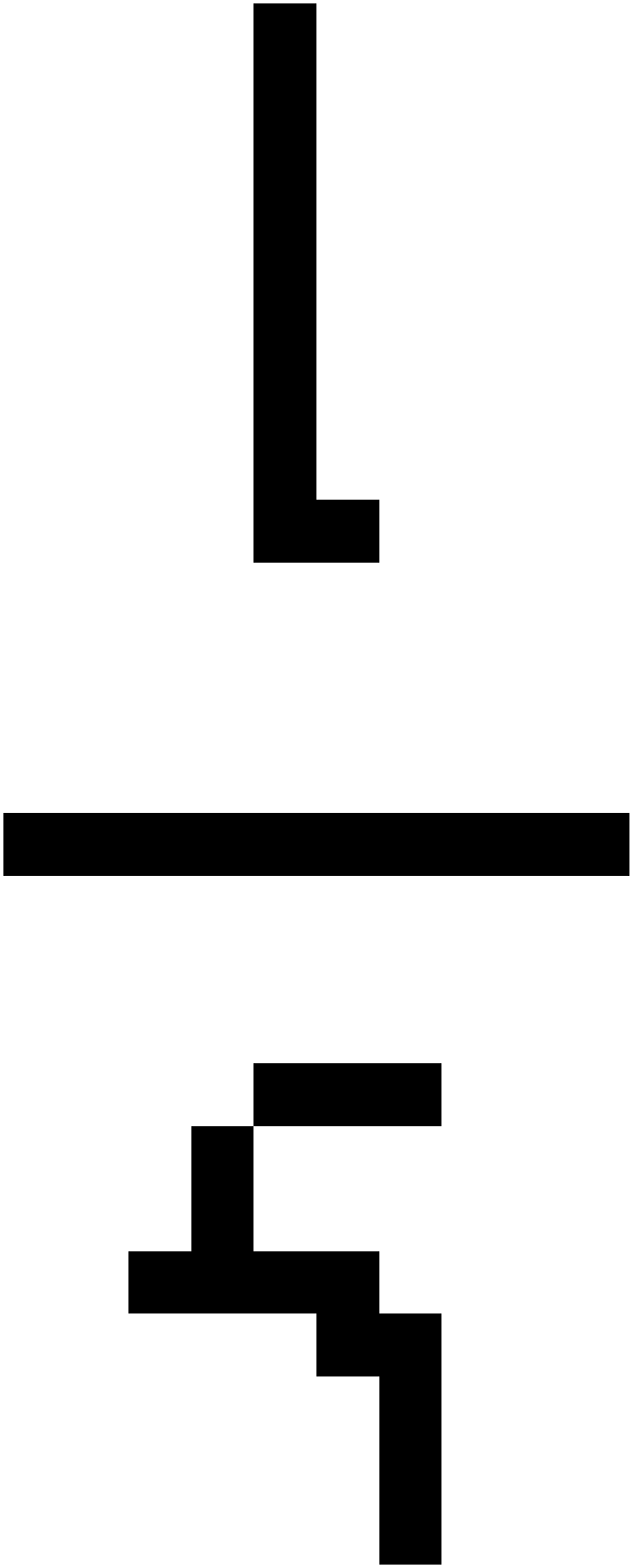
(D) graph of (x-(6))^2*(x-(1))^2

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

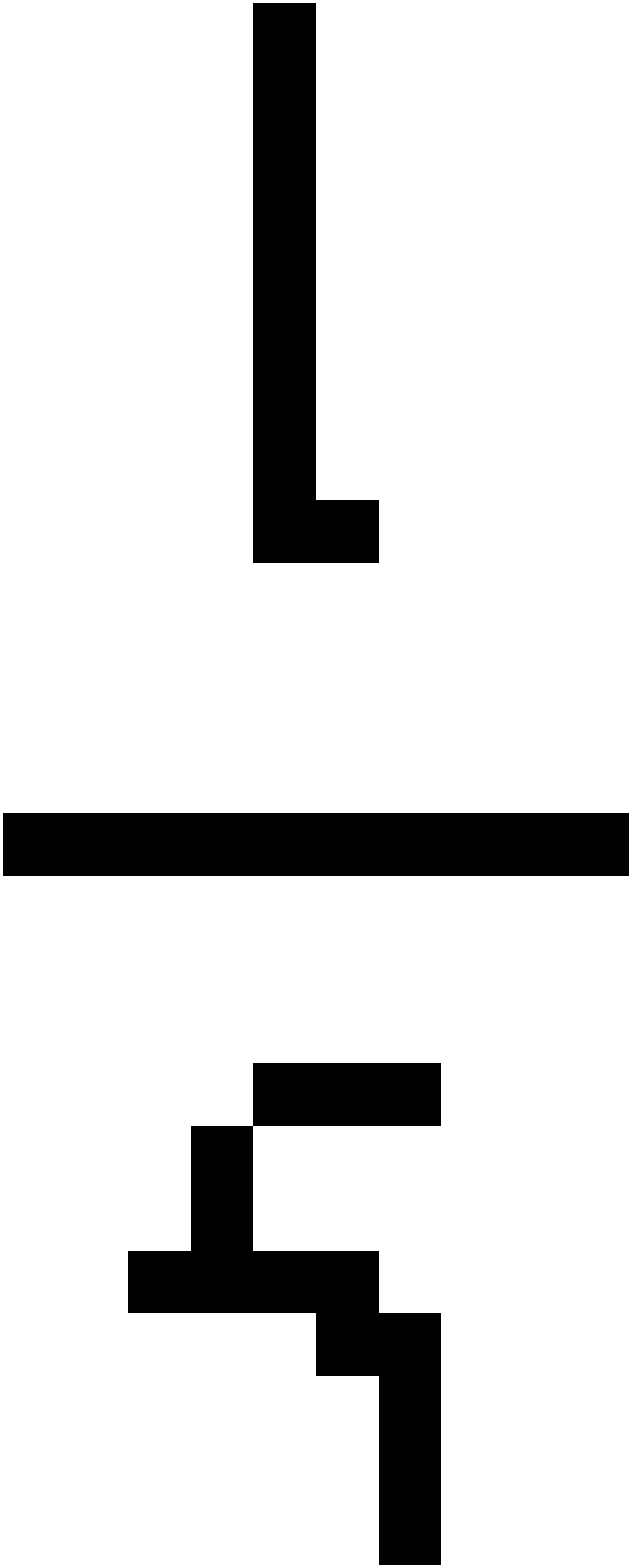
163. Given the function $y = x^3$, what are the parameters of the transformed function y $=$ $g (x h) 3 k$

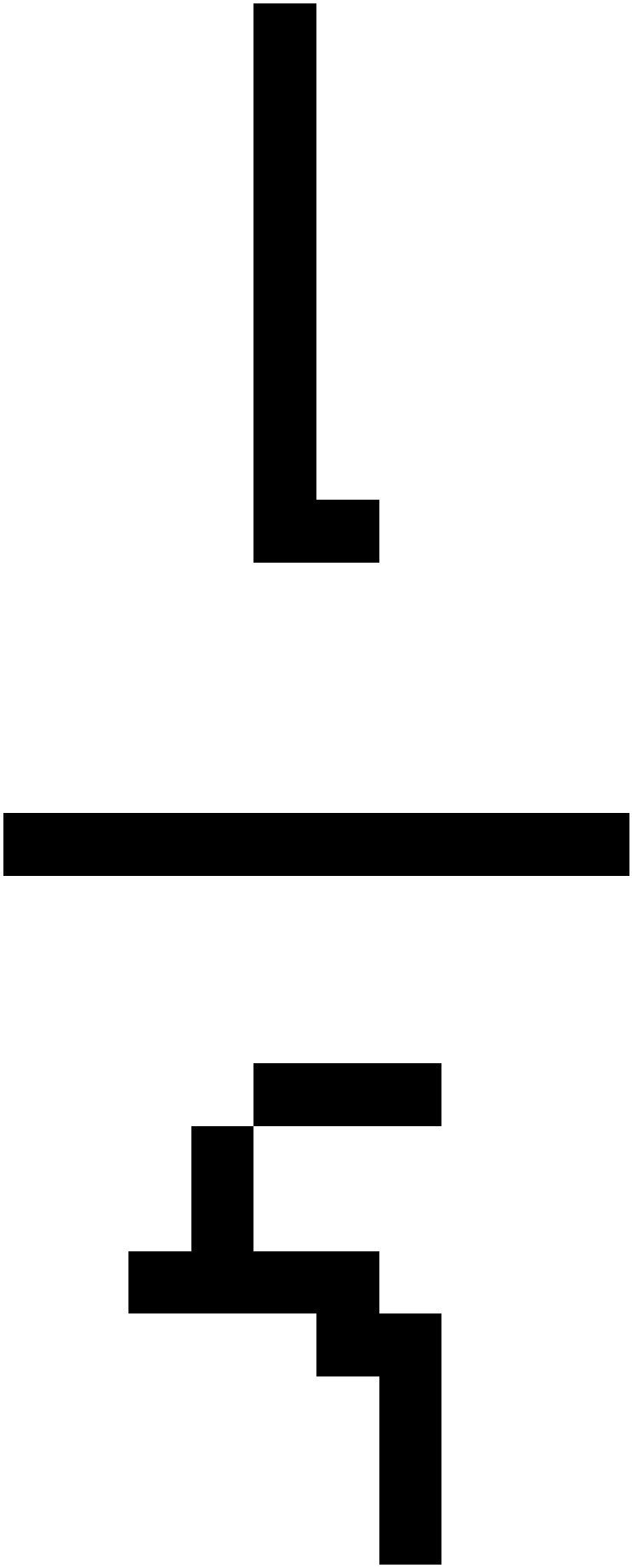


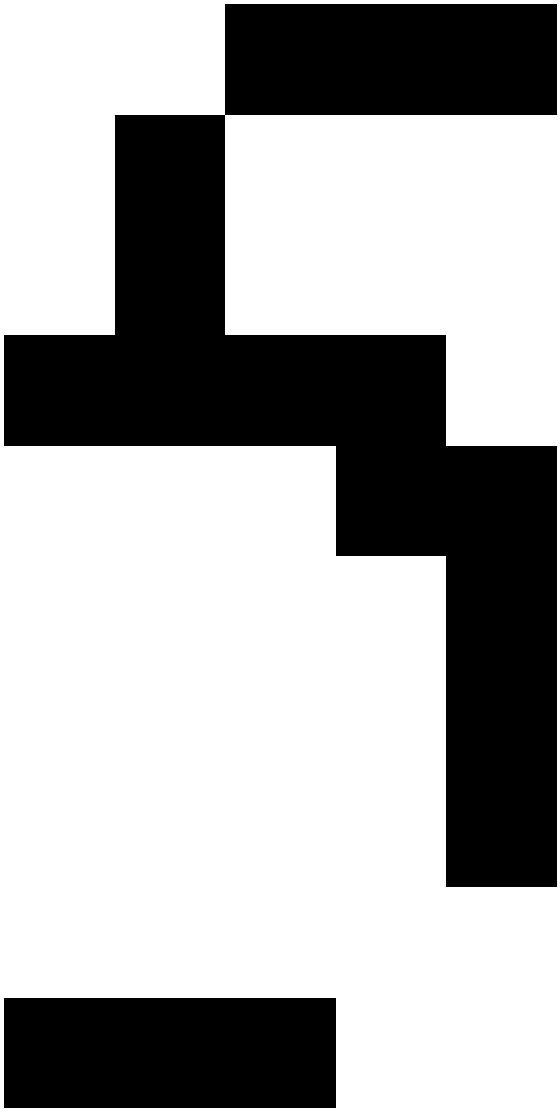
and what is the effect of each parameter on the graph of the original function? Rendered ExamView reference:

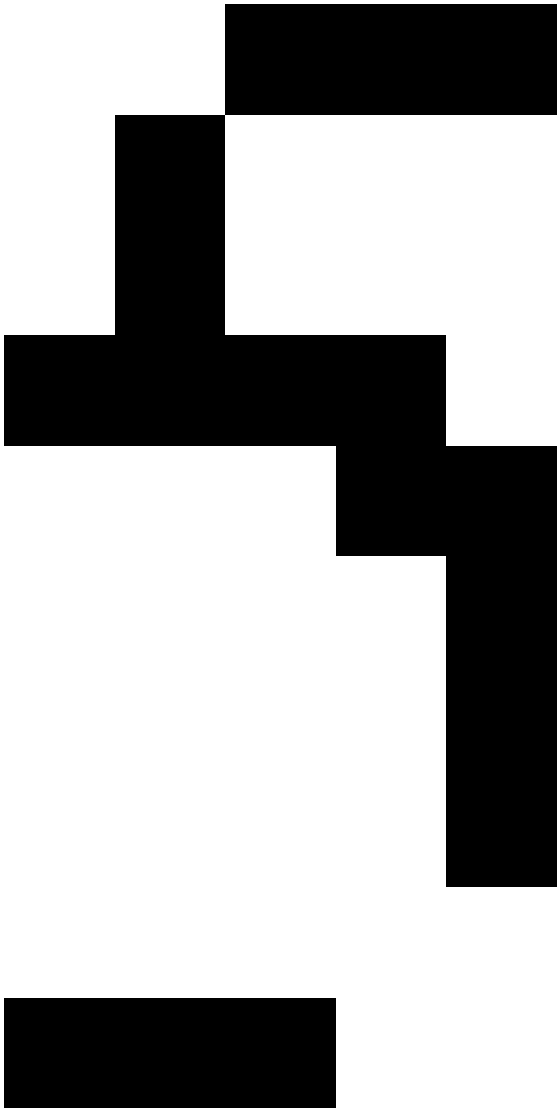


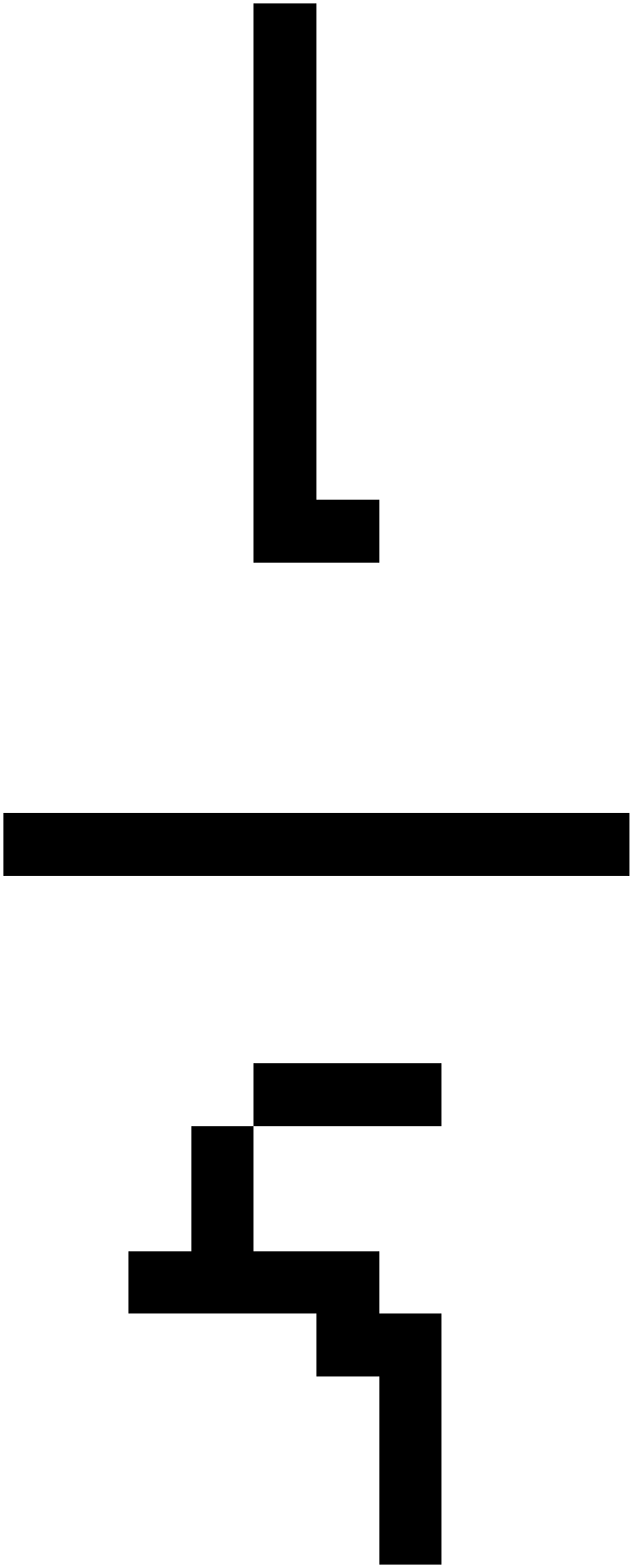
$$(x-2)^3-8$$

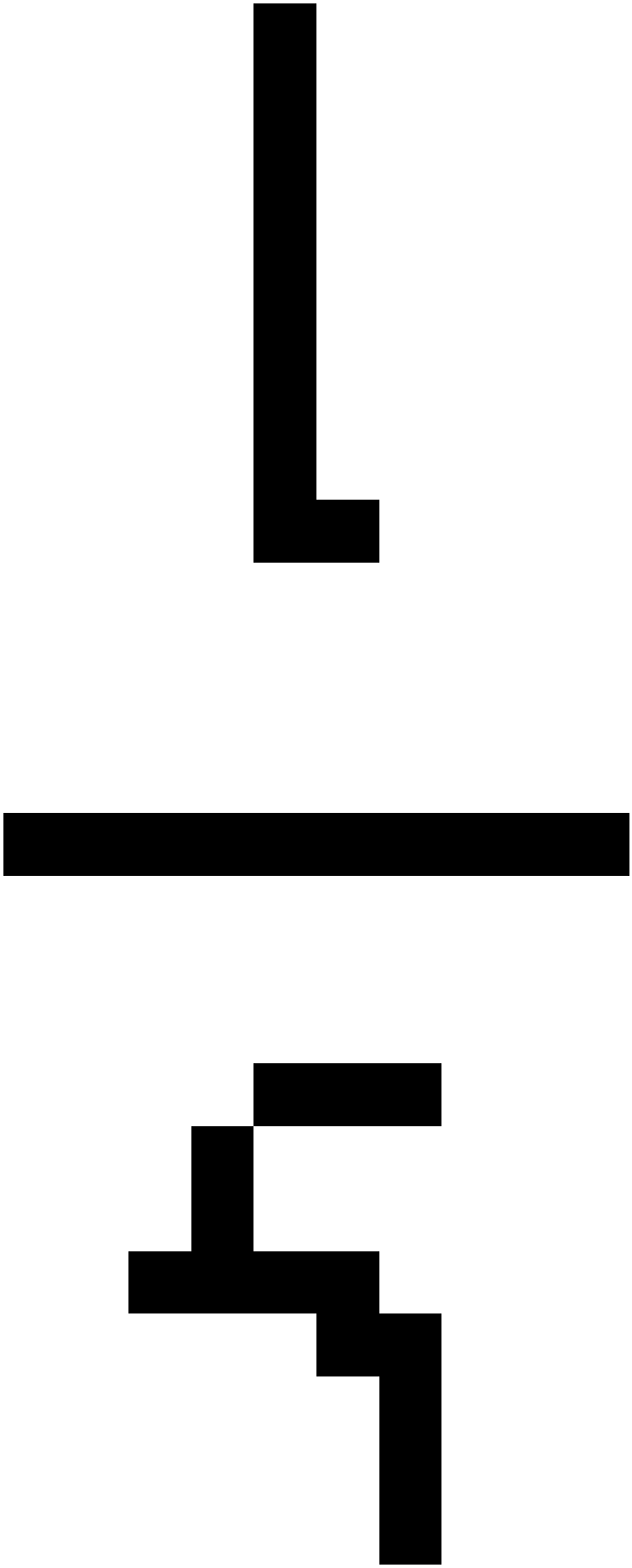


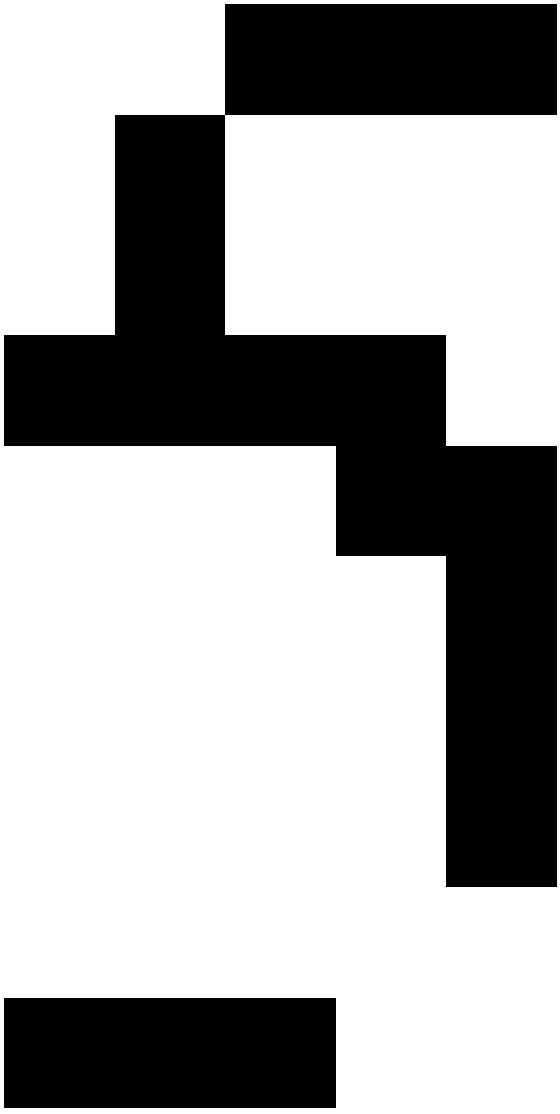


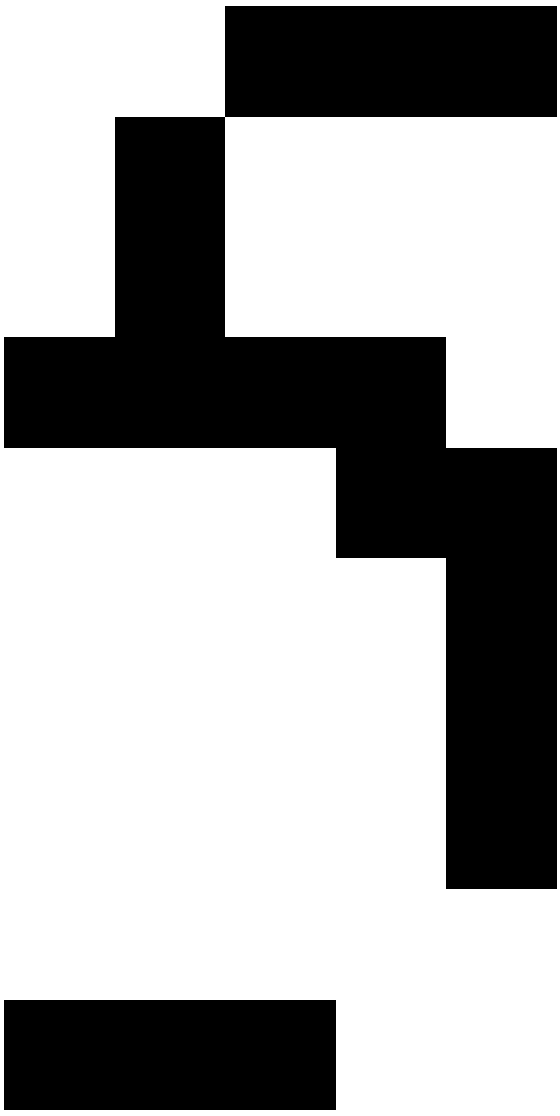












(A) a = – 2, vertical stretch about the x-axis by a factor of – 2 h = – 2, horizontal translation – 2 units – 2 k = – 2, vertical translation – 2 units – 2

(B) a = – 2, vertical stretch about the x-axis by a factor of – 2 h = – 2, horizontal translation – 2 units – 2 k = 1/5, vertical translation – 2 units – 8

(C) a = 1/5, vertical stretch about the x-axis by a factor of 1/5 h = 8, horizontal translation right units 2 k = 2, vertical translation down units 5

(D) a = 5, vertical stretch about the x-axis by a factor of 2 h = 2, horizontal translation left units 8 k = right, vertical translation 2 units 2

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

164. The function – is positive over the interval(s) _____.

TOP: bnk | problem **TYPE:** problem

165. The zeros of the function – are _____.

TOP: bnk | problem **TYPE:** problem

166. If $P(19/18) = 0$ for a polynomial $P(x),$ then _____ is a factor of $P(x)$.

TOP: bnk | problem **TYPE:** problem

167. $abs(a)>1$ and $abs(b)>1$ and $abs(c)>1$ and $abs(d)>1$ and $abs(e)>1 a*(−1*e)^3+b*e^2-c*e+d$ –657 If –9 is divided by – $3,$ the remainder is _____.

TOP: bnk | problem **TYPE:** problem

168. The minimum number of $x-$intercepts that the graph of an odd $-$degree polynomial function can have is _____. Rendered ExamView reference:

$$\frac{x^4-16x^3+99x^2-234x+216}{x+8}=x^3-24x^2+291x-2562+\frac{20\,712}{x+8}$$

ANS: 1 **TOP:** bnk | problem **TYPE:** problem

169.

$$x^4-16x^3+99x^2-234x+216$$

TOP: bnk | short_answer **TYPE:** short_answer

170.

$$x^4-16x^3+99x^2-234x+216$$

TOP: bnk | short_answer **TYPE:** short_answer

171.

$$-2x^3-6x^2+5x-7$$

Rendered ExamView reference:



$$-2x^2-20x-135$$

$$(x-7)$$

$$(-2x^2-20x-135)$$

$$-2x^3-6x^2+5x-7$$

$$(x-7)$$

$$(-2x^2-20x-135)$$

$$(x-7)$$

$$(-2x^2-20x-135)$$

$$-2x^3-6x^2+5x-7$$

$$(x-7)$$

$$(-2x^2-20x-135)$$

TOP: bnk | short_answer **TYPE:** short_answer

172.

$$P(x)=-8x^3-8x^2-8x-8$$

Rendered ExamView reference:



TOP: bnk | short_answer **TYPE:** short_answer

173. (a) Use long division to divide $x_3 + 3x2$ – 7 by x $+ 2$. Express the result in quotient form.
(b) Identify any restrictions on the variable.
(c) Write the corresponding statement that can be used to check the division.
(d) Verify your answer.

TOP: bnk | problem **TYPE:** problem

174. Show that x $+ 1$ is a factor of $P(x) = x_3 + 2x2 + 3x + 2$.

TOP: bnk | short_answer **TYPE:** short_answer

175. A variety of factoring techniques is required.

TOP: bnk | problem **TYPE:** problem

176. Factor fully.

(a) –n_3 – $4n2$ – $4n
(b) 12x3 + 16x2$ – $5x$ – $3
(c) 2x4 + x_3$ – $10x$ – 5

TOP: bnk | problem **TYPE:** problem

177. A variety of factoring techniques is required. Factor fully. a) $x_2(x$ – $2)(x + 2) + 3x + 6 b) 16x4$ – (x $+ 1)2$

$$x^3 + 2x^2 - 23x - 60$$

Rendered ExamView reference:

$$(x+3)(x-4)(x+5)$$

$$(x-3)(x+4)(x+5)$$

$$(x+3)(x+4)(x-5)$$

$(x-3)(x-4)(x-5)$

TOP: bnk | problem **TYPE:** problem

178. A variety of factoring techniques is required. Factor $2x^3 + 5x^2 - 14x - 8$ fully.

TOP: bnk | short_answer **TYPE:** short_answer

179. Solve by factoring.

(a) x_3 – $2x2$ – $5x + 6 = 0$
(b) x_3$ – $3x2$ – $4x + 12 = 0$
(c) x_3 = 4x2$

TOP: bnk | problem **TYPE:** problem

180. A variety of factoring techniques is required. Solve. a) $3x3 + 2x2$ – $8x + 3 = 0$

(a) 2x3 + x_2$ – $10x$ – $5 = 0$
(b) 5x4 = 7x2$ – 2

TOP: bnk | problem **TYPE:** problem

181. A variety of factoring techniques is required. Solve by factoring.

(a) $3x4$ – $48 = 0$
(b) $ –$x_3 + 12x2$ – $47x + 60 = 0$
(c) 21x4$ – $7x3$ – $6x2 + 2x = 0$

TOP: bnk | problem **TYPE:** problem

182. A variety of factoring techniques is required. Solve by factoring.

(a) $x_4 + 3x2$ – $28 = 0$
(b) 2x4$ – $54x = 0$

TOP: bnk | problem **TYPE:** problem

183. Solve by graphing using technology. Round the root(s) to one decimal place. a) $x_4 + x_3 + x_2 + x = 1$
b) 0.1x4 = x_3 + x_2$

TOP: bnk | problem **TYPE:** problem

184. Solve by graphing using technology. Round answers to one decimal place. a) x_3 – $7 > 0$
b) x^3-7$

TOP: bnk | problem **TYPE:** problem

185. Solve the following inequalities using an algebraic method. a) $x_2 + x$ – $2 > 0$
b)$ $0 x x x^3 9x - x^3 + 2x 2 + 4x - 8 0 * 7$ c) $x +$ d) $x -$

TOP: bnk | problem **TYPE:** problem

186. (a) If x $+ 1$ is a factor of ax$4 +$ bx$2 + c,$ what is the value of a $+ b + c$?
(b) Decide if x $+ 1$ is a factor of each polynomial. Explain your reasoning. i) $3x4 + 3x2$ – 6 ii) $17x4$ – $10x2$ – 7 iii) $x_4 + x_2 + 1$

TOP: bnk | problem **TYPE:** problem

187. Show that $x + a$ is a factor of the polynomial $P(x) = (x + a)^4 + (x + c)^4 - (a - c)^4$.

TOP: bnk | problem **TYPE:** problem

188. Given that –2 is a root of $x_3 + x =$ –$4x2 + 6,$ find the other root(s).

TOP: bnk | short_answer **TYPE:** short_answer

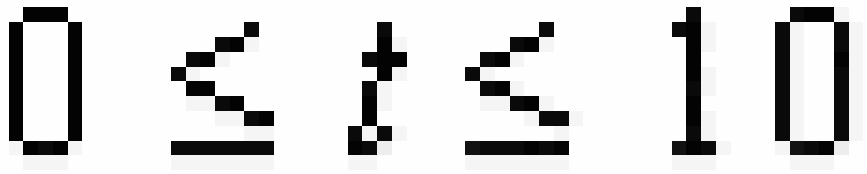
189. Solve $2x4$ – $x_3 = 7x2$ – $9x + 3$.

TOP: bnk | short_answer **TYPE:** short_answer

190. The height of a square$-$based box is 4 cm more than the side length of its square base. If the volume of the box is 225 cm$3,$ what are its dimensions?

TOP: bnk | short_answer **TYPE:** short_answer

191. The height, h, in metres, of a weather balloon above the ground after t seconds can be modelled by the function $h(t) =$ –$2t3 + 3t2 + 149t + 410,$ for



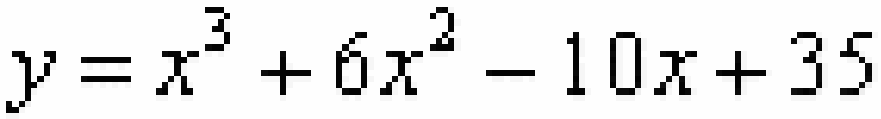
. When is the balloon exactly $980 m$ above the ground?

TOP: bnk | short_answer **TYPE:** short_answer

192. Amit has designed a rectangular storage unit to hold large factory equipment. His model has dimensions $1 m$ by $2 m$ by $4 m$. By what amount should he increase each dimension to produce an actual storage unit that is 9 times the volume of his model?

TOP: bnk | short_answer **TYPE:** short_answer

193. Let x represent the side length of the square to be cut from each corner. Then $V(x) = x(15$ – $2x)(25$ – $2x)$. Solve $x(15$ – $2x)(25$ – $2x) = 476$. $x = 1 3 .$ Let $P(x) = 4x 3 - 80x 2 + 375x - 476$, and try x $= 4$ in the factor theorem. $P(x) = 4x 3 - 80x 2 + 375x - 476 P(4) = 4(4) 3 - 80(4 2) + 375(4) - 476 = 256 - 1280 + 1500 - 476 P(4) = 0$ Thus, x 4 is a factor of $P(x)$. Divide to find another factor. $4x 3 - 80x 2 + 375x - 476 = 0 (x - 4)(4x 2 - 64x + 119) = 0 x = 4$ Thus, $x = 64 (- 64) 2 - 4(4)(119) 2(4) = 16 137 2 13.9$ Use the quadratic formula to solve the quadratic equation.



x $= 13.9$ cm is not possible because of the starting dimensions of the sheet of metal. The side length of the square cut from each corner is either 4 cm or approximately 2.15 cm.

TOP: bnk | short_answer **TYPE:** short_answer

194. Solve $5x 2 - 8x + 4$ algebraically and graphically.

TOP: bnk | short_answer **TYPE:** short_answer

195. Solve $3x 2 (x^2 - 8) + 6x + 5$ algebraically. Display your result on a number line.

TOP: bnk | short_answer **TYPE:** short_answer

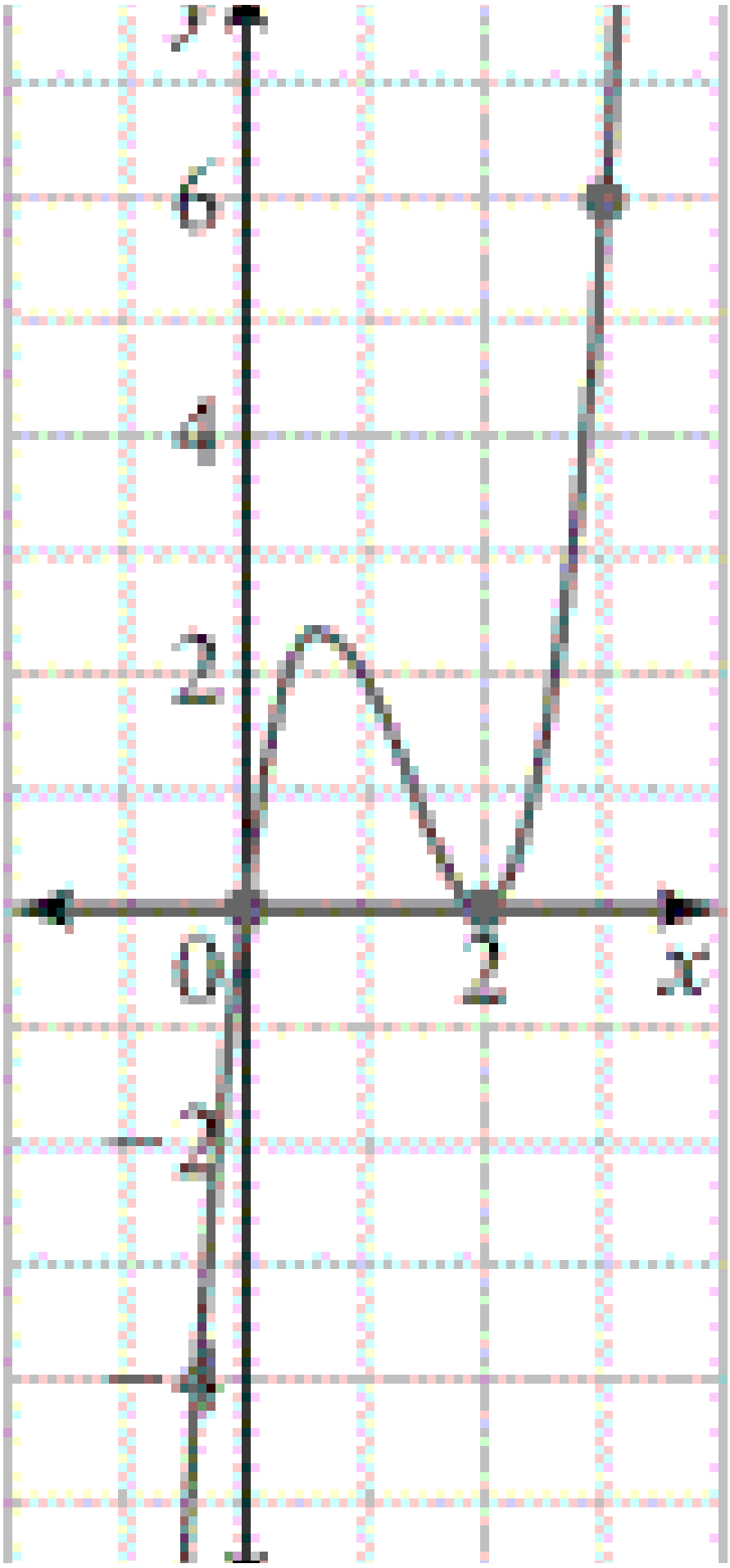
196. Create a cubic polynomial inequality for which x $= 3$ or ($x+2)*(x-3)^2$ is the solution. Explain your reasoning.

TOP: bnk | short_answer **TYPE:** short_answer

197. Determine an equation for the polynomial function represented by the table of values. –3 –30 –2 –10 –$1 10 30$

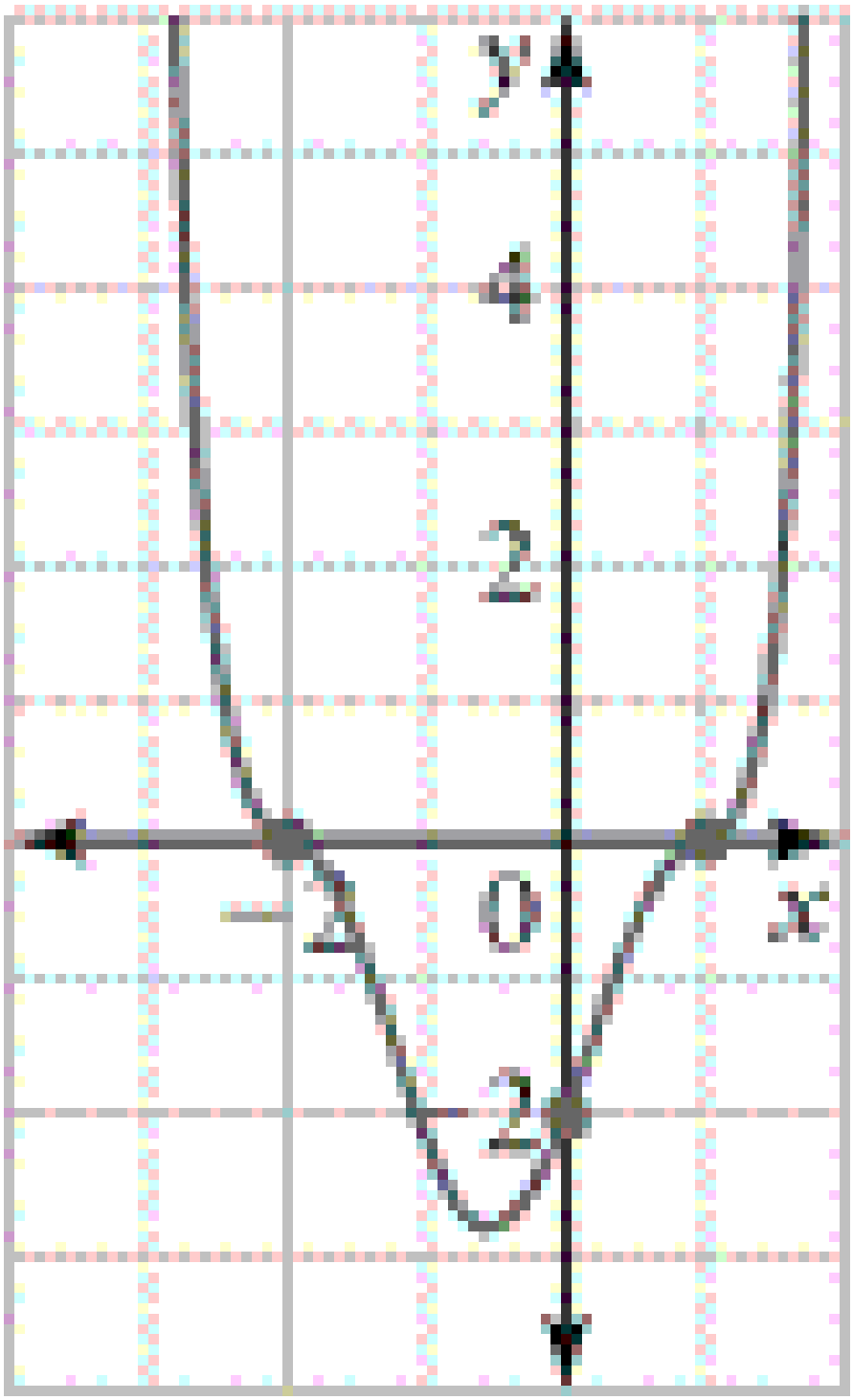
TOP: bnk | short_answer **TYPE:** short_answer

198. Determine an equation in expanded form for the polynomial function represented by the graph. $y = ax(x - 2) 2 6 = a(3) (3 - 2) 2 6 = 3a a = 2$ [image omitted: asset-1] Rendered ExamView reference:


y
6
4
2
0
2
x
-2
-4

TOP: bnk | short_answer **TYPE:** short_answer

199. Determine an equation in factored form for the polynomial function represented by the graph $7 y = a(x + 2) 3 (x - 1) 3 - 2 = a(0 + 2) 3 (0 - 1) 3 - 2 = - 8a a = 1 / 4 y = 1 / 4 (x + 2) 3 (x - 1) 3 D z0 z0 ,$ [image omitted: asset-2] Rendered ExamView reference:



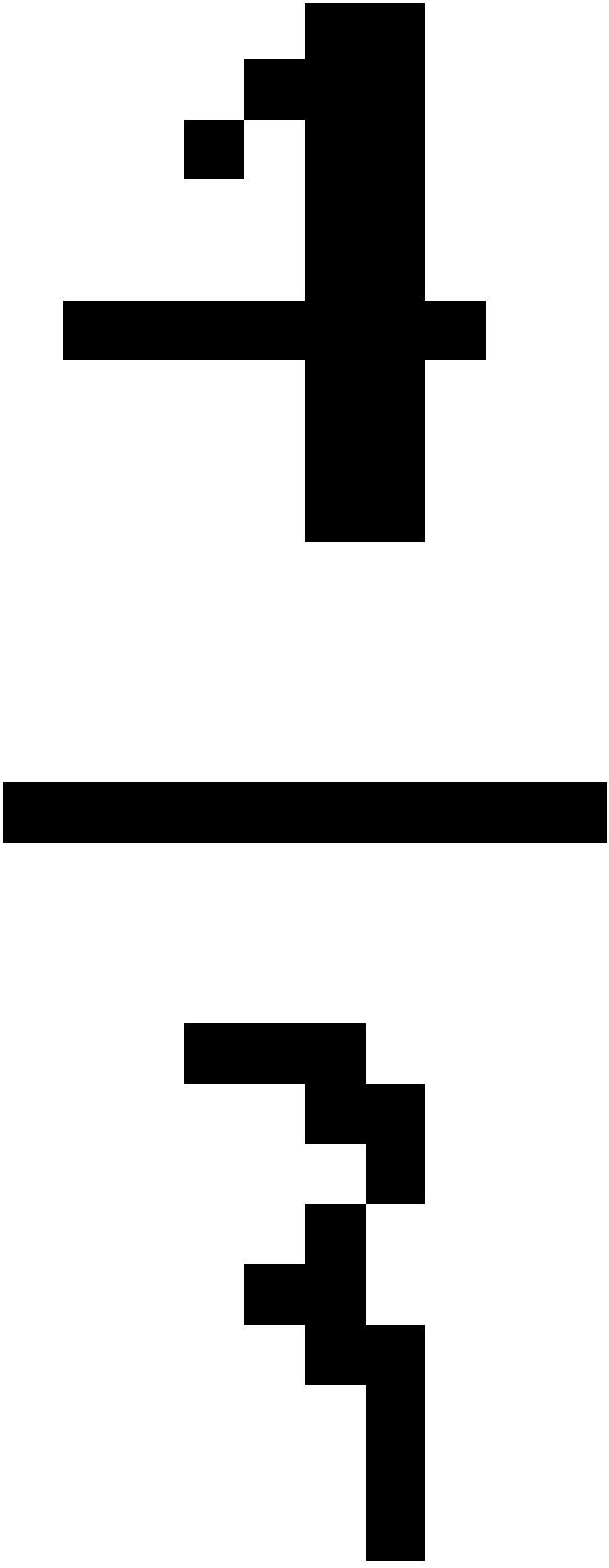
TOP: bnk | short_answer **TYPE:** short_answer

200. Match the correct term with the correct part of the statement. – 16
[image omitted: asset-3]

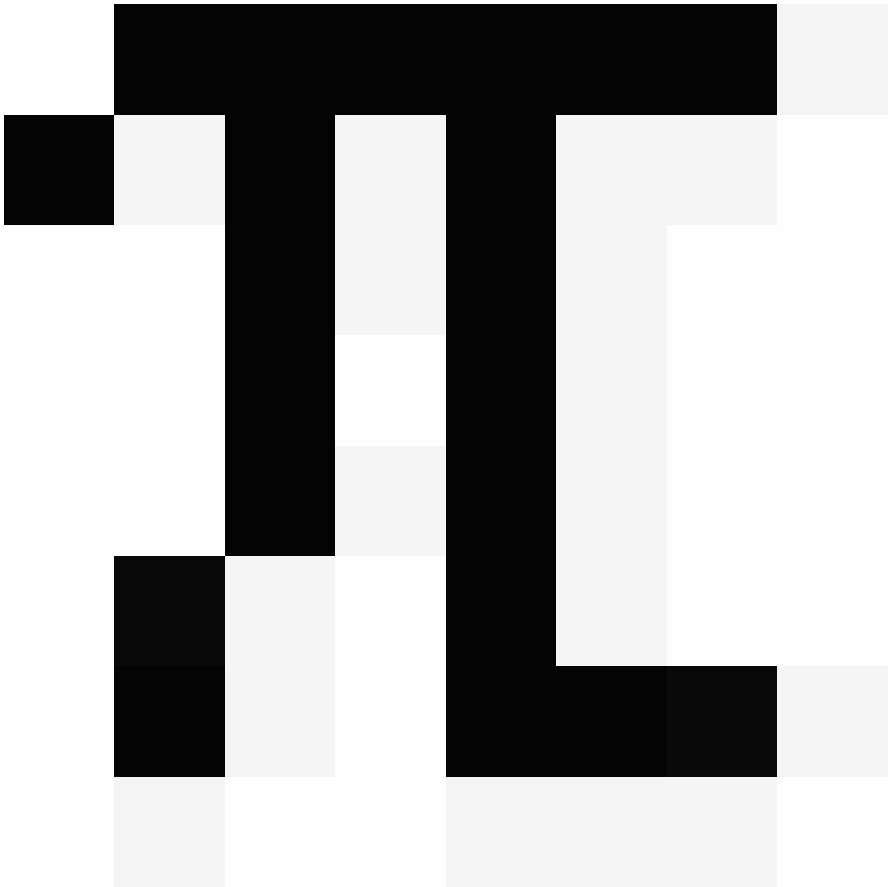
(A) $x >= + 8$ (B) $x >= + 216$ (C) $x >= – 3$ (D) $x <= + 8$

ANS: – 2562 **TOP:** bnk | mcq **TYPE:** mcq

201. $4/3$



radians is equal to how many degrees? Rendered ExamView reference:



(A) 240 (B) 420 (C) 150 (D) 330

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

202. The exact radian measure for an angle of 255

(A) $17/6$ (B) $12/17$ (C) $6/17$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

203. Which of the following angles, in degrees, is coterminal with, but not equal to, $1.2 6/5$ radians?

(A) 396 (B) 486 (C) 576

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

204. Determine the arc length of a circle with radius 5.5 cm if it is subtended by a central angle of $5/2 1.4$ radians. Round your answer to one decimal place.

(A) 1.4 (B) 4.4 (C) 43.2 (D) 6.9

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

205. Determine the equation of a circle with centre at the origin and radius 8.

(A) 16 (B) 8 (C) 64 (D) 8

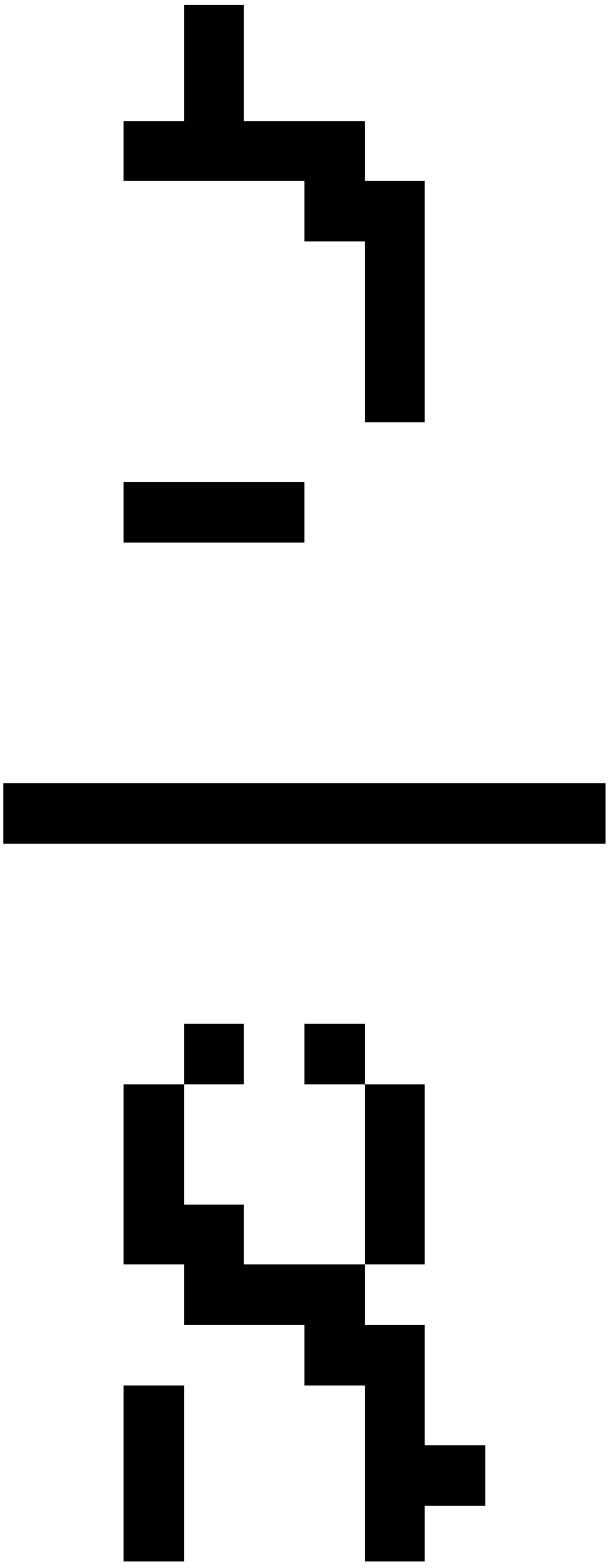
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

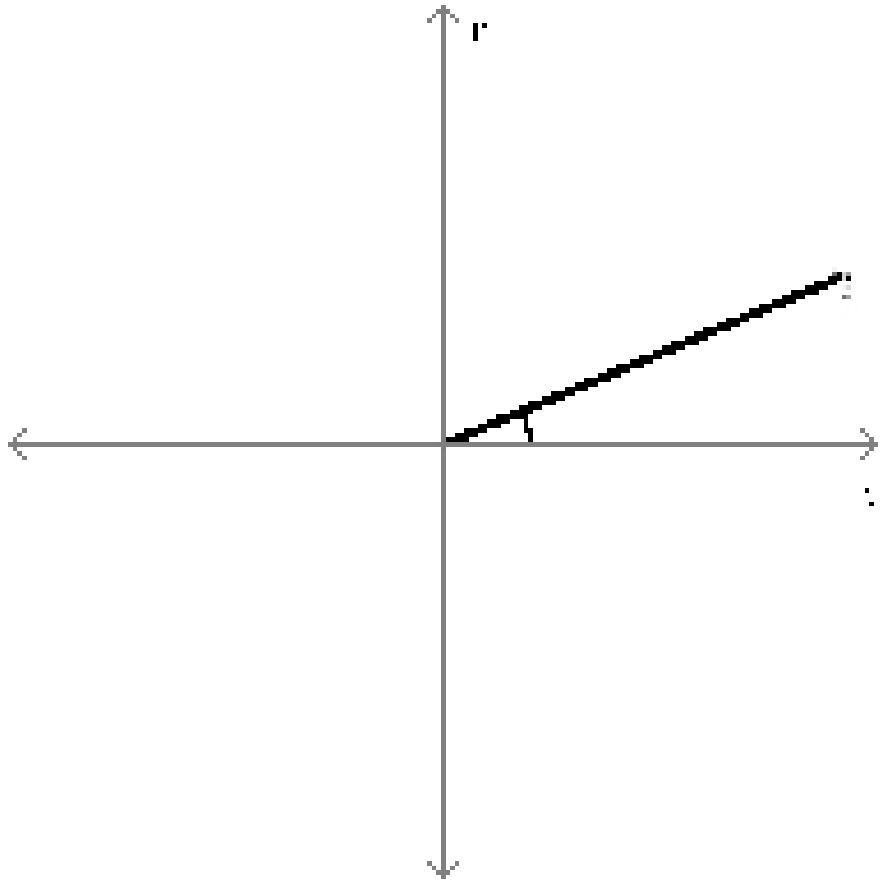
206. Determine the equation of a circle with centre at ($3,$ –$3)$ and radius 10.

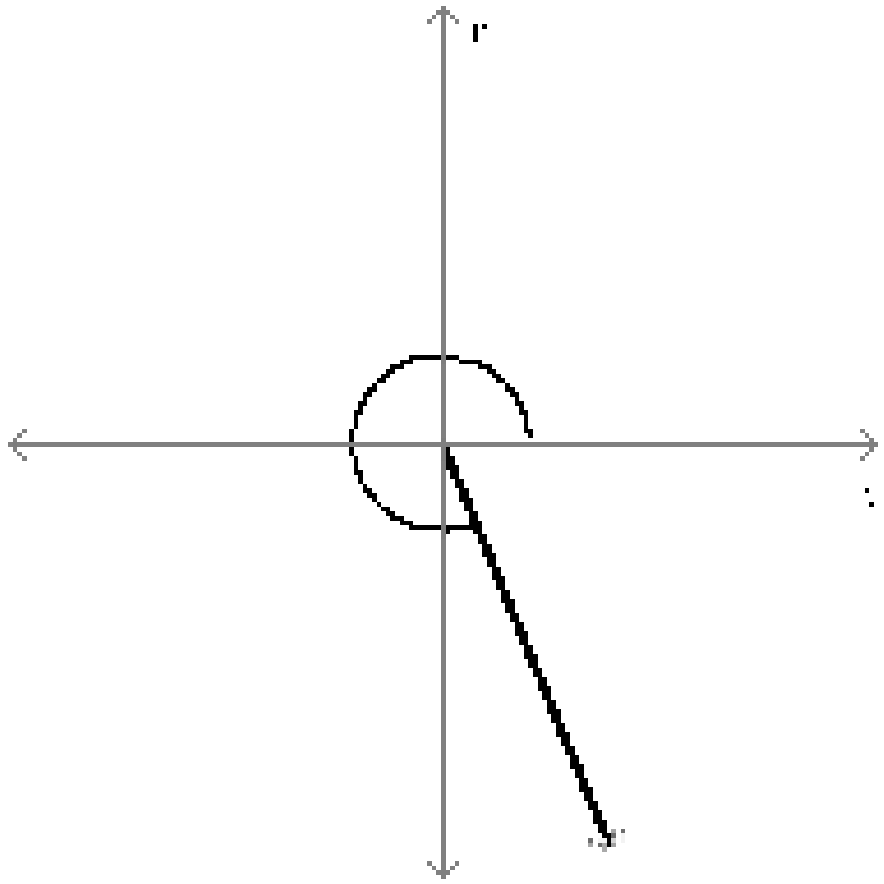
(A) 20 (B) – 3 (C) + 3 (D) 10

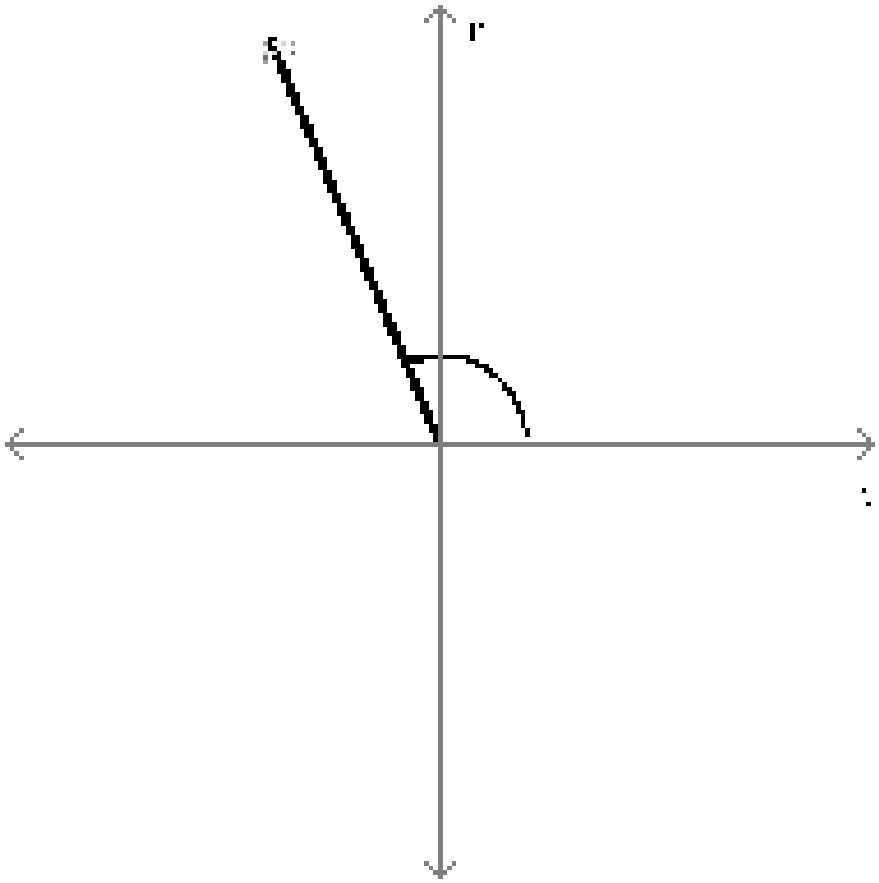
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

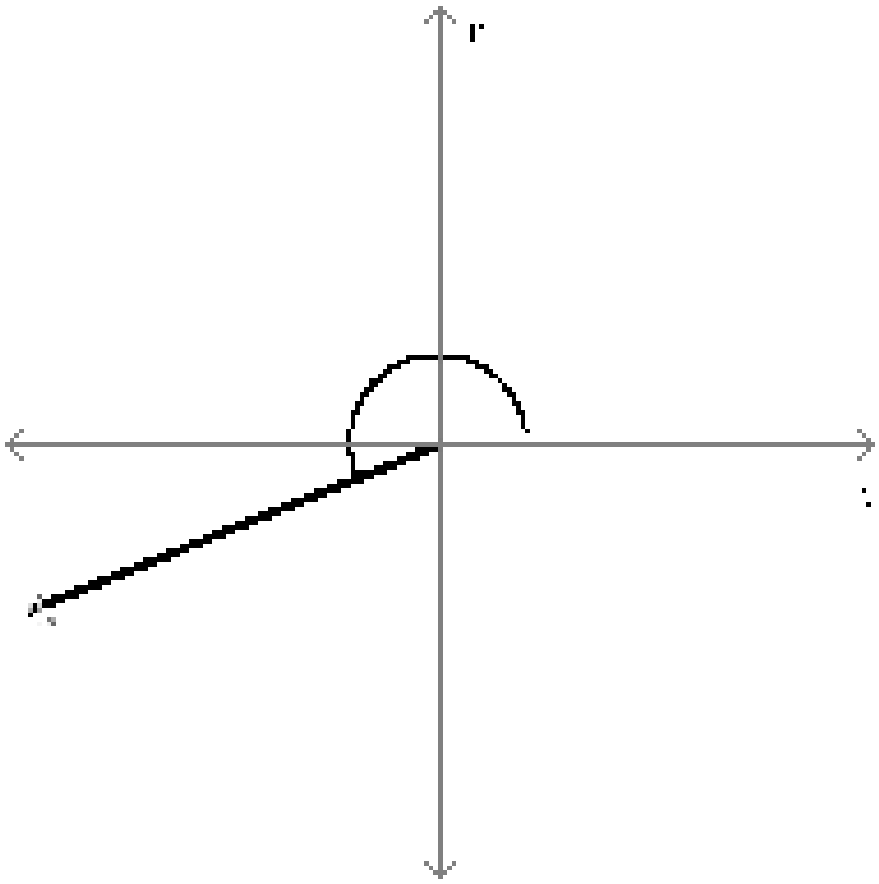
207. Which graph represents an angle in standard position with a measure of 4.619397662556 rad? Recovered graph functions: $5*cos((1.963495408494)); 5*sin((1.963495408494)); 5*cos((3.534291735289)); 5*sin((3.534291735289)); 5*cos((0.392699081699)); 5*sin((0.392699081699)); 5*cos(A_dis3); 5*sin(A_dis3)$. Rendered ExamView reference:











(A) graph of 5*cos((1.963495408494))

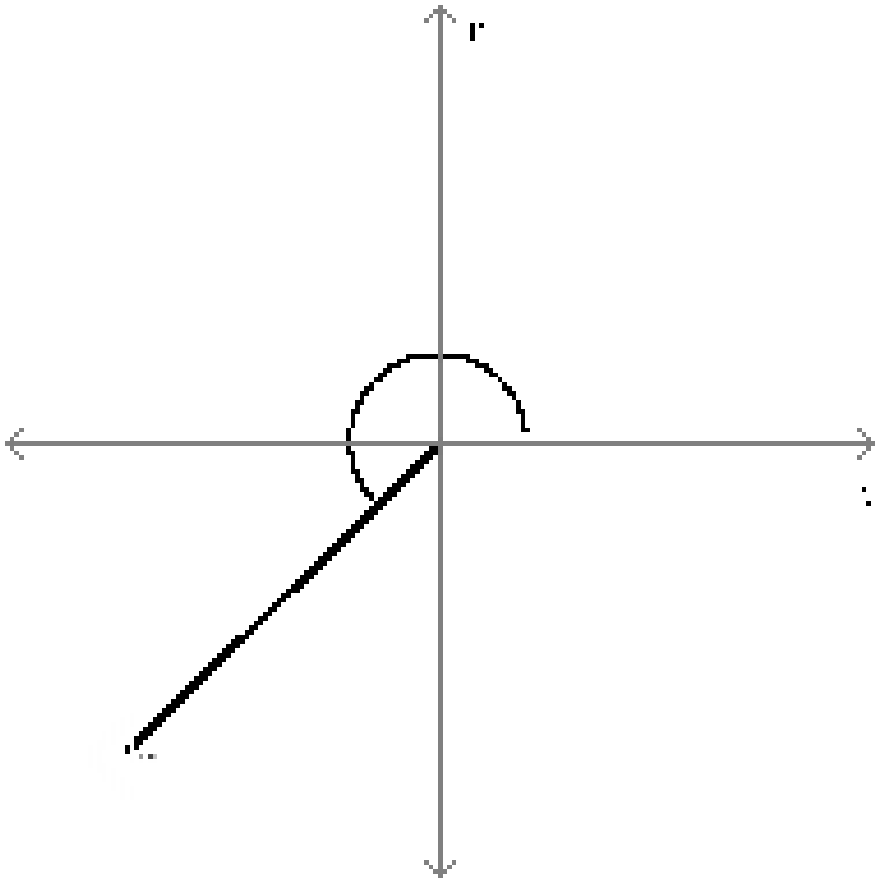
(B) graph of 5*sin((1.963495408494))

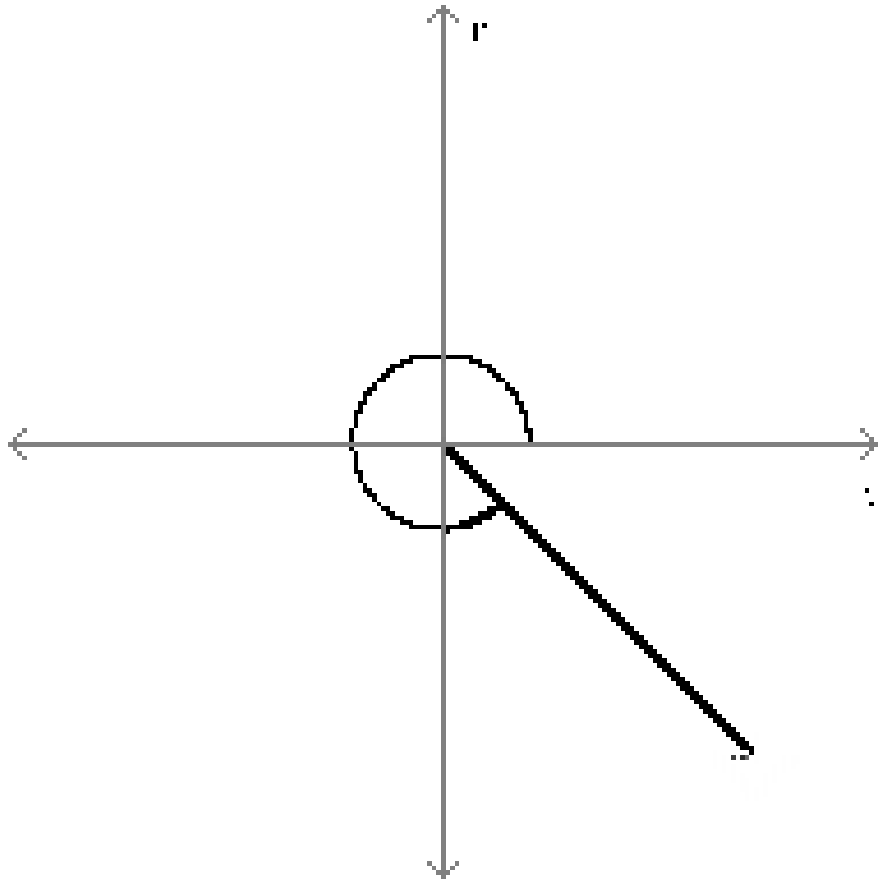
(C) graph of 5*cos((3.534291735289))

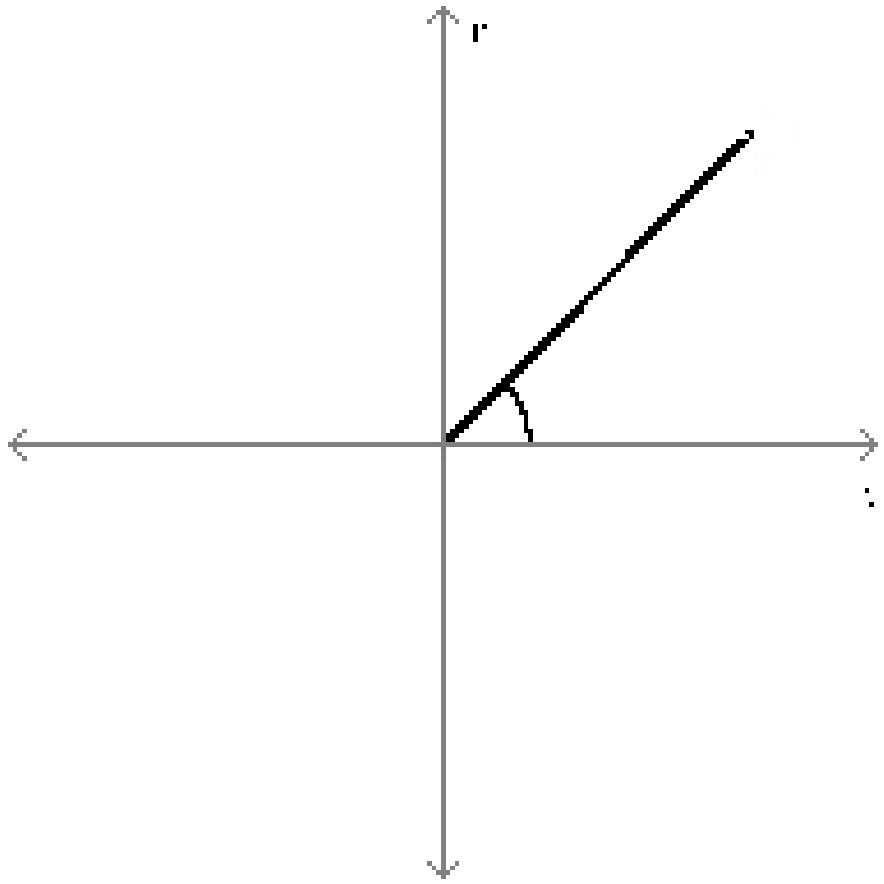
(D) graph of 5*sin((3.534291735289))

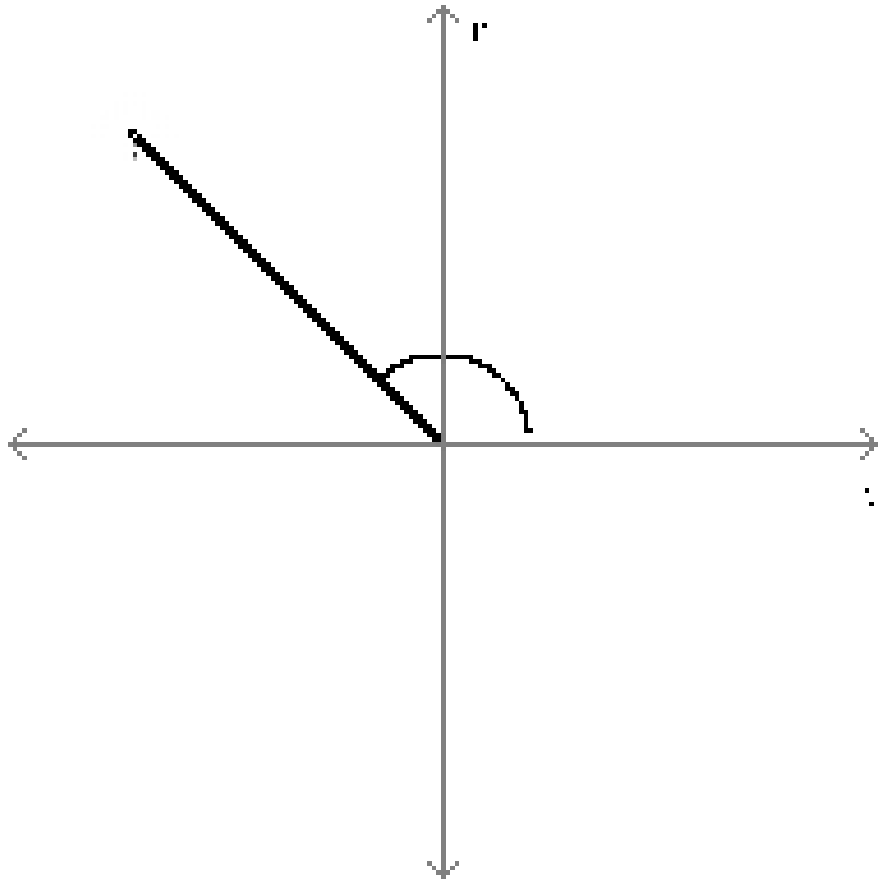
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

208. Which graph represents an angle in standard position with a measure of –3.535533905933 Recovered graph functions: $5*cos((2.356194490192)); 5*sin((2.356194490192)); 5*cos((3.926990816987)); 5*sin((3.926990816987)); 5*cos((0.785398163397)); 5*sin((0.785398163397)); 5*cos(A_dis3); 5*sin(A_dis3)$. Rendered ExamView reference:









(A) graph of 5*cos((2.356194490192))

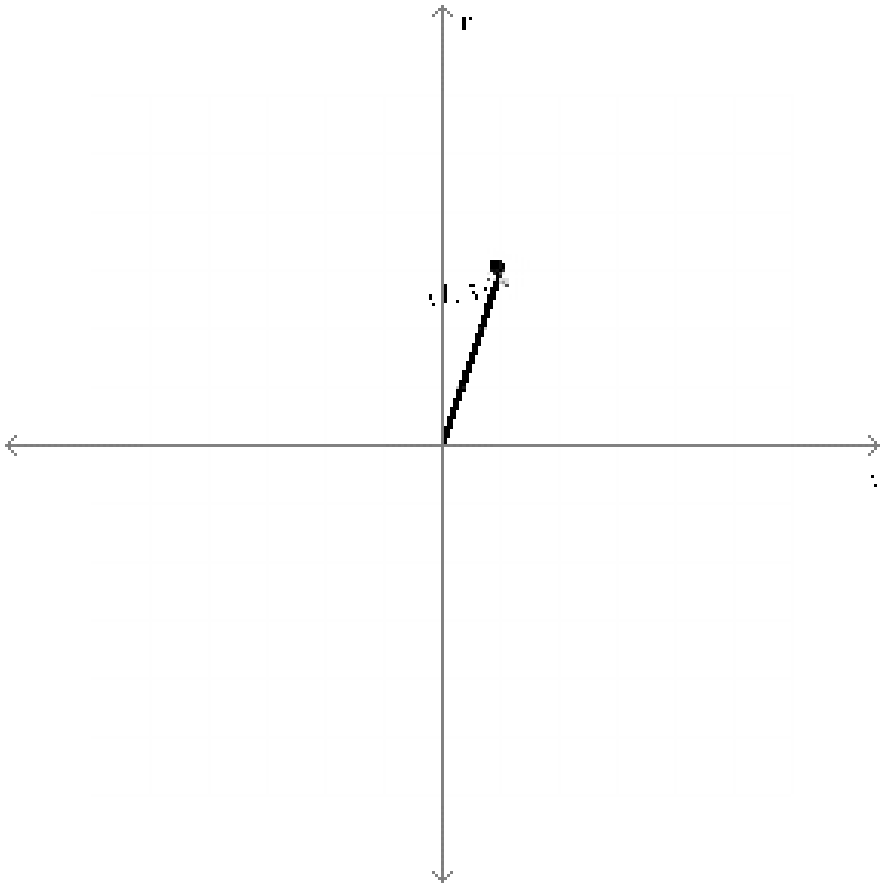
(B) graph of 5*sin((2.356194490192))

(C) graph of 5*cos((3.926990816987))

(D) graph of 5*sin((3.926990816987))

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

209. Determine the measure of the angle in standard position shown on the graph below. Round your answer to the nearest tenth of a degree. 1 Recovered graph functions: $atan((3)/(1)); cos((1.249045772398))$. Rendered ExamView reference:



(A) 3 (B) 1 (C) 3 (D) 71.6

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

210. What is the measure of the central angle of the slice, in radians? Round your answer to two decimal places, if necessary.

(A) 1.71 (B) 0.29 (C) 3.43 (D) 0.58

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

211. The coordinates of the point that lies at the intersection of the terminal arm and the unit circle at an angle of $cos((1.919862177194))$

(A) (0.94, –0.34) (B) (–0.34, –2.75) (C) (–0.34, 0.94) (D) (–2.75, 0.94)

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

212. Identify the point on the unit circle corresponding to an angle of 300° in standard position.

(A) (sqrt(3)/2, −1/2) (B) (−1/2, -sqrt(3)/2) (C) (1/2, -sqrt(3)/2) (D) (1, 0)

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

213. Identify the point on the unit circle corresponding to an angle of $4pi/3$ radians in standard position.

(A) (1/2, -sqrt(3)/2) (B) (-sqrt(3)/2, −1/2) (C) (−1/2, -sqrt(3)/2) (D) (sqrt(3)/2, −1/2)

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

214. Which point on the unit circle corresponds to $tan q =$ $sqrt(3)$?

(A) (1/2, -sqrt(3)/2) (B) (-sqrt(3)/2, −1/2) (C) (sqrt(3)/2, −1/2) (D) (−1/2, -sqrt(3)/2)

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

215. Identify a measure for the central angle q in the interval 0 such that point ($0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0, 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)$ is on the terminal arm.

(A) (−1, 0) (B) (0, 1) (C) (-sqrt(2)/2, sqrt(2)/2) (D) (-sqrt(2)/2, -sqrt(2)/2)

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

216. Which is a possible value of q, to the nearest hundredth of a radian, when $cos q =$ –0.58?

(A) $−2.19$ (B) 2.19 (C) $−0.62$ (D) 0.84

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

217. If the angle q is –5000 in standard position, it can be described as having made

(A) 13/8/9 (B) 27/7/9 (C) −27/7/9

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

218. How many degrees has the figure skater turned?

(A) $−220$ (B) 400 (C) 580

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

219. If the angle q is 1600 in standard position, in which quadrant does it terminate?

(A) quadrant III (B) quadrant IV (C) quadrant I

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

220. A ball is riding the waves at a beach. The ball s up and down motion with the waves can be described using the formula $2.3,$ where h is the height, in metres, above the flat surface of the water and t is the time, in seconds. What is the height of the ball, to the nearest hundredth of a metre, after t $= 17 s$?

(A) $−0.87$ (B) $−1.99$ (C) $−2.66$ (D) 1.99

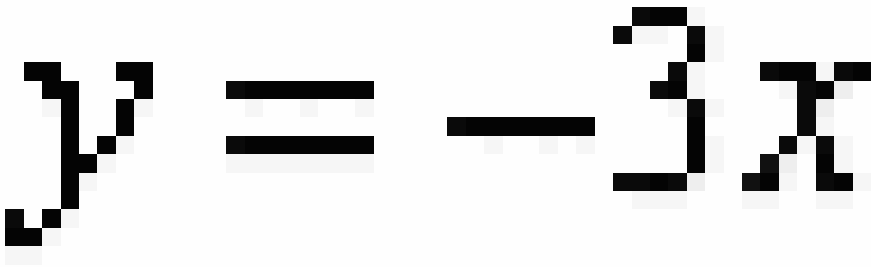
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

221. A tricycle has a front wheel that is 30 cm in diameter and two rear wheels that are each 12 cm in diameter. If the front wheel rotates through a angle of $32,$ through how many degrees does each rear wheel rotate, to the nearest tenth of a degree?

(A) 32.0 (B) 80.0 (C) 40.0 (D) 160.0

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

222. Determine the point in quadrant x^2+1 where the line represented by



intersects the unit circle.

(A) (–3, 0.95) (B) (0.94, –0.32) (C) (–0.32, –0.35) (D) (0.95, –0.32)

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

223. The point $P(0.391, 0.921)$ is the point of intersection of a unit circle and the terminal arm of an angle q in standard position. What is the equation of the line passing through the centre of the circle and the point P? Round the slope to two decimal places. Recovered graph functions: $cos((1.169370598836)); sin((1.169370598836))$. Rendered ExamView reference:

$$y = 2.36x$$

$$y = 2.36x + 0.92$$

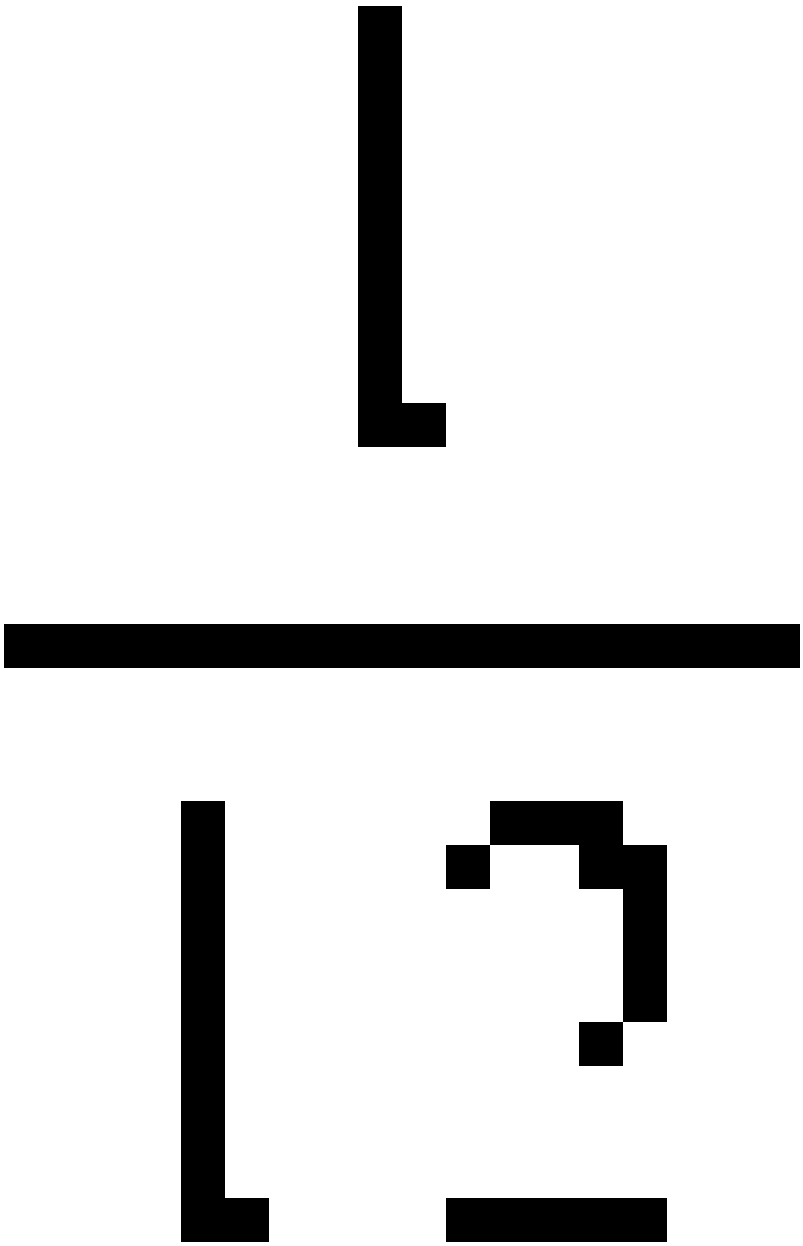
$$y = 0.42x$$

$$y = 2.36x + 0.39$$

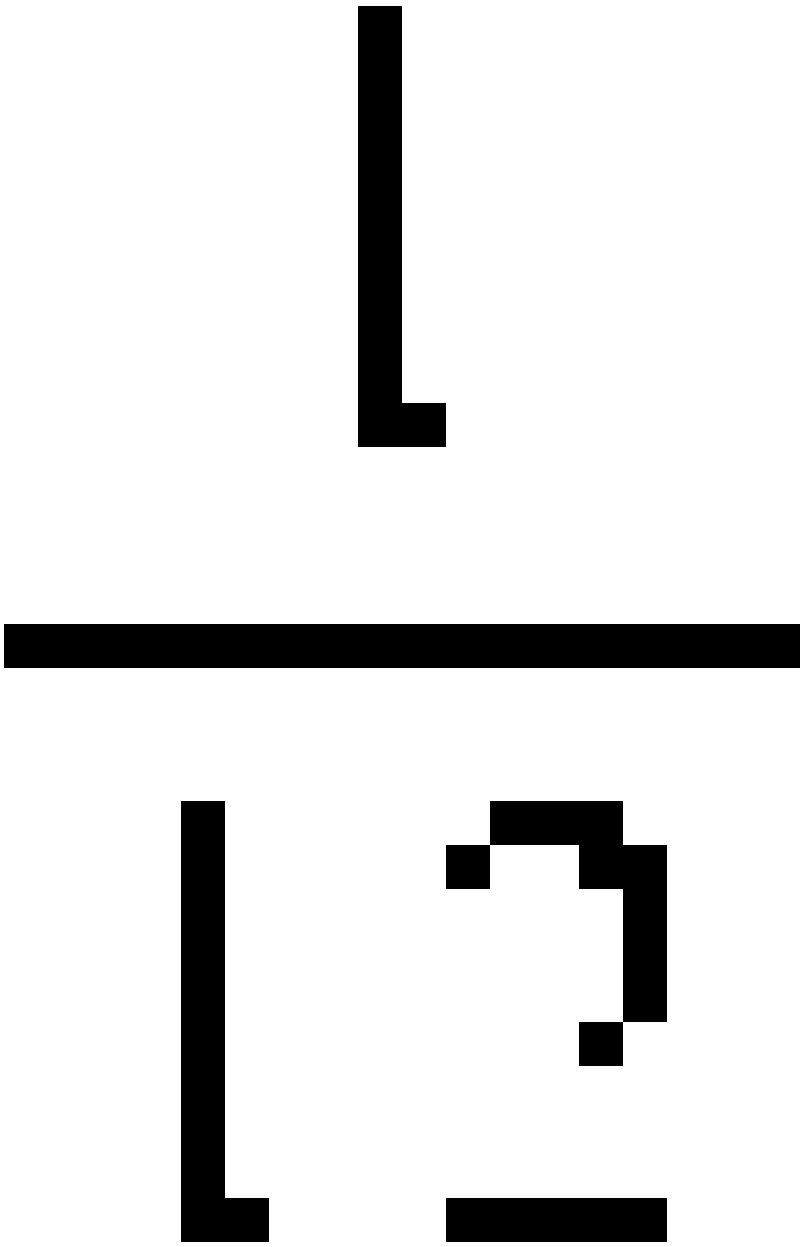
(A) 2.36 (B) 2.36 (C) 0.42 (D) 2.36

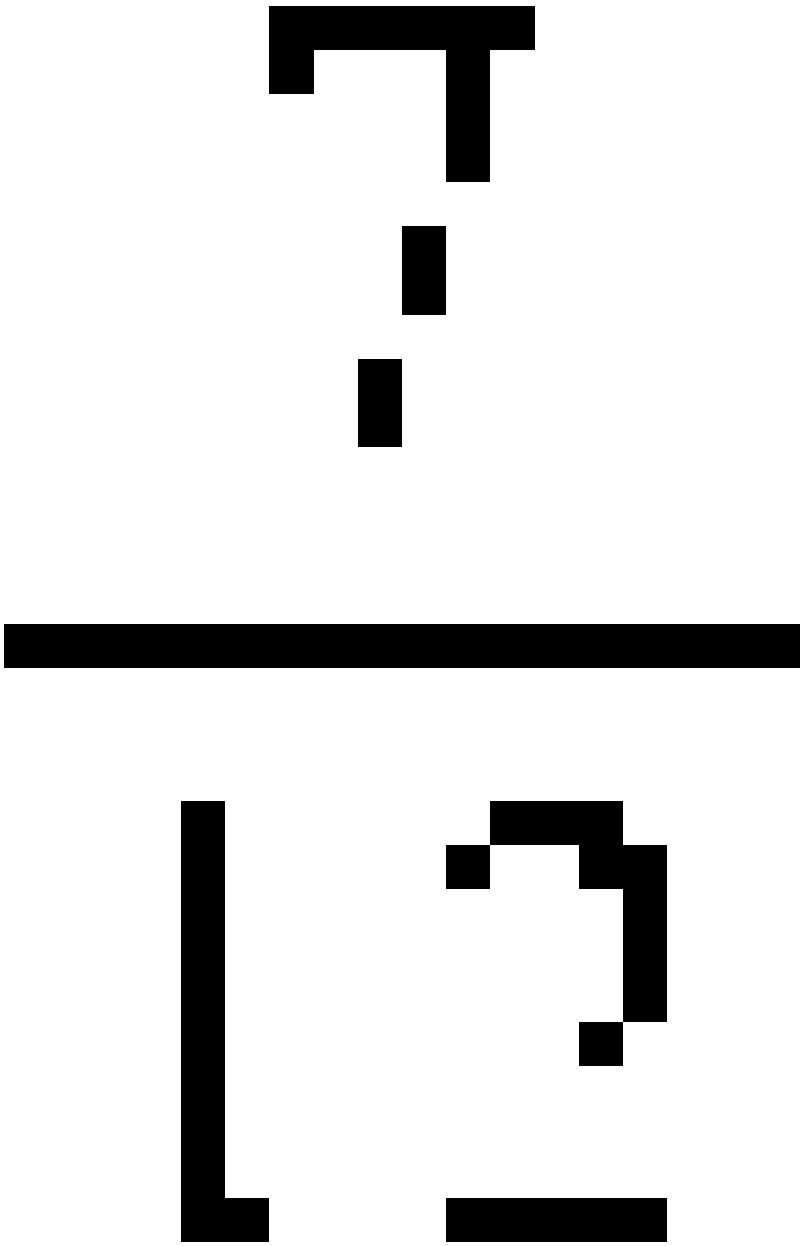
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

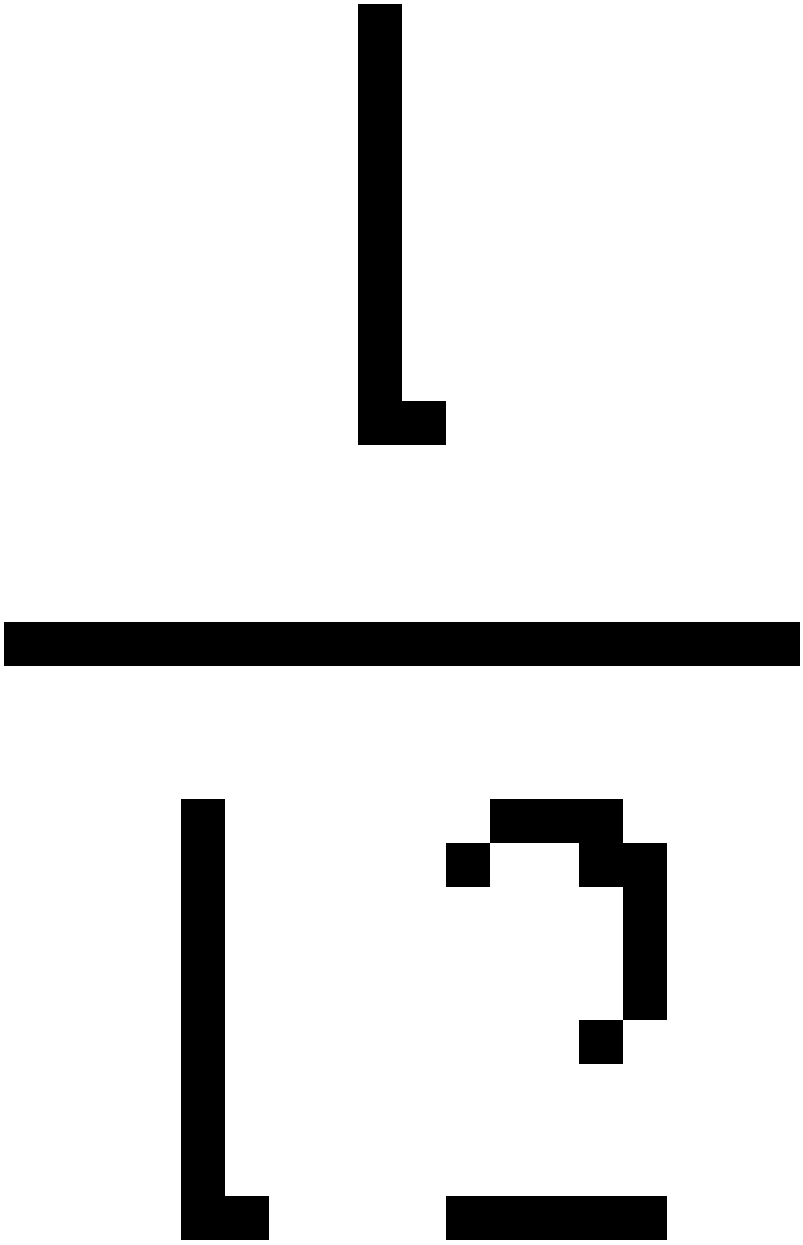
224. degree$<>0$ and degree$<>90$ and degree$<>180$ and degree$<>270$ and degree$<>360$ Giai got an answer of



when she was calculating the value of a trigonometric function. Assuming Giai did her calculation correctly, which of the following was she calculating? Rendered ExamView reference:







(A) \$1/12\$ (B) \$1/12\$ (C) \$7/12\$ (D) \$1/12\$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

225. A bottle is riding the waves at a beach. The bottle s up and down motion with the waves can be described using the formula \$1.4,\$ where h is the height, in metres, above the flat water surface and t is the time, in seconds. When is the first time, to the nearest tenth of a second, that the height of the bottle will be \$1.4 m\$?

(A) 14.8 (B) 0.9 (C) 1.1 (D) 1.5

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

226. The cosine of an angle is _____ in quadrant III.

TOP: bnk | problem **TYPE:** problem

227. The sine of an angle is _____ in quadrant I.

TOP: bnk | problem **TYPE:** problem

228. The reciprocal of tangent is _____.

TOP: bnk | completion **TYPE:** completion

229. For a point on the unit circle, the $y-$coordinate can be determined using the _____.

TOP: bnk | completion **TYPE:** completion

230. The

$$\sin\left(\frac{9\pi}{2} - A\right)$$

occurs when the angle in standard position is _____. Rendered ExamView reference:

$$\tan\left(A + \frac{9\pi}{2}\right)$$

$$\cos\left(\frac{13\pi}{2} - \theta\right)$$

$$\tan\left(\frac{5\pi}{2} - \theta\right)$$

$$\cos\left(\theta + \frac{5\pi}{2}\right)$$

$$\sin\left(\theta + \frac{11\pi}{2}\right)$$

TOP: bnk | completion **TYPE:** completion

231. $sin(theta)$

TOP: bnk | short_answer **TYPE:** short_answer

232. $cot(theta)$

TOP: bnk | short_answer **TYPE:** short_answer

233. $cos(theta)$

TOP: bnk | short_answer **TYPE:** short_answer

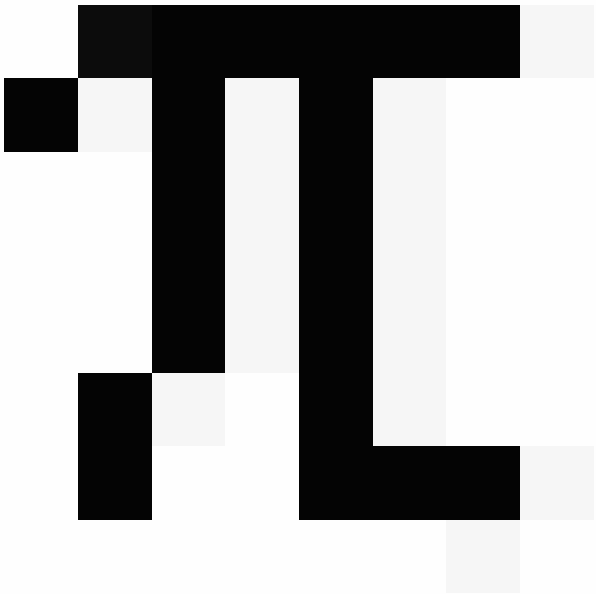
234. $cot(theta)$

TOP: bnk | short_answer **TYPE:** short_answer

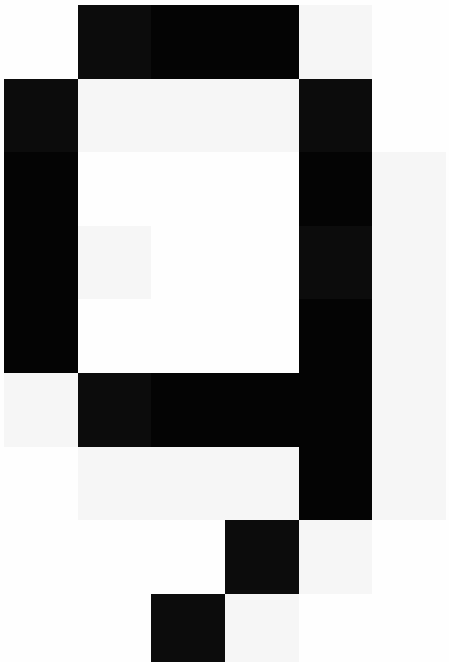
235. $sin(theta)$

TOP: bnk | short_answer **TYPE:** short_answer

236. A child swings on a playground swing set. If the length of the swing s chain is $3 m$ and the child swings through an angle of







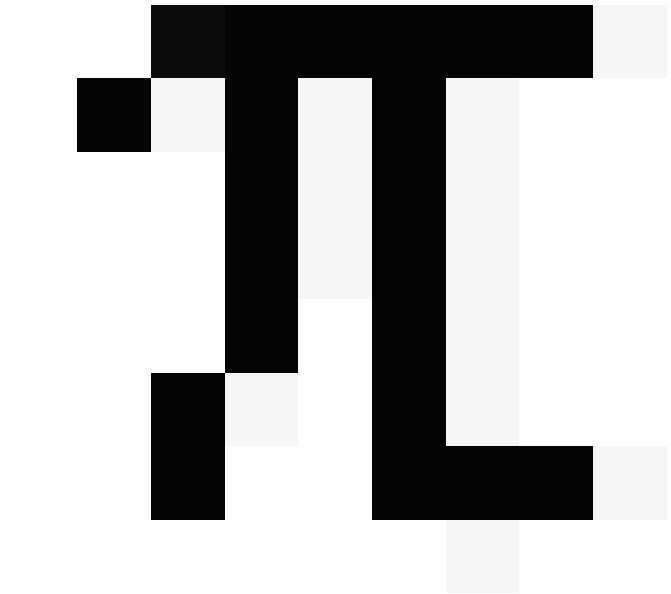
, what is the exact arc length through which the child travels?

TOP: bnk | short_answer **TYPE:** short_answer

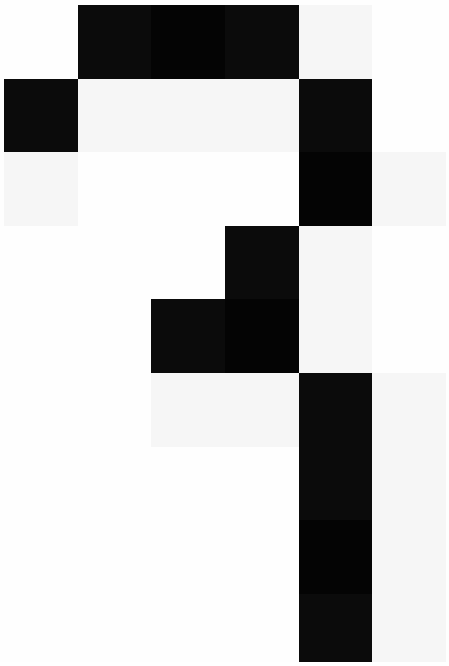
237. Determine the exact value of $cot pi / 3 + csc pi / 3 cos pi$.

TOP: bnk | short_answer **TYPE:** short_answer

238. A $3-m$ ladder is leaning against a vertical wall such that the angle between the ground and the ladder is







. What is the exact height that the ladder reaches up the wall?

TOP: bnk | short_answer **TYPE:** short_answer

239. Find the exact value of $sin 11 pi$.
[image omitted: asset-1]

TOP: bnk | short_answer **TYPE:** short_answer

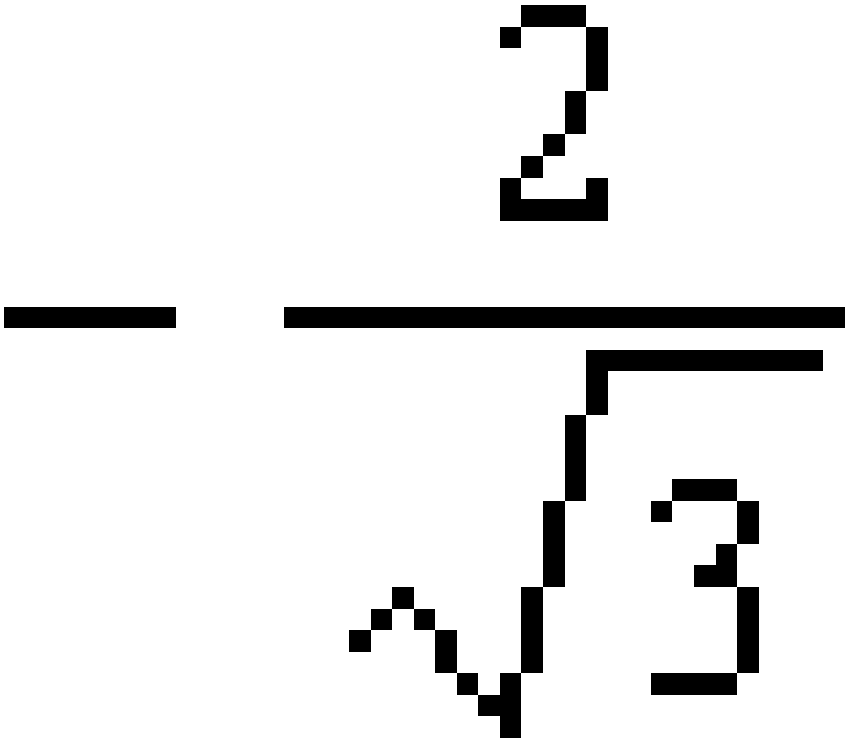
240. Find the exact value of $cos 5 pi$.
[image omitted: asset-2]

TOP: bnk | short_answer **TYPE:** short_answer

241. Given that $x = cos pi$ and that x lies in the first quadrant, determine the exact measure of angle x.

TOP: bnk | short_answer **TYPE:** short_answer

242. Without using a calculator, determine two angles between 0 and 360 that have a cosecant of



. Include an explanation of how you determined the two angles.

TOP: bnk | short_answer **TYPE:** short_answer

243. Given a circle of diameter 21 cm, determine the arc length subtended by a central angle of 1.2 radians.

TOP: bnk | short_answer **TYPE:** short_answer

244. Find the exact value of $cos 5 pi 6 2 - sin 5 pi 6 2$.

TOP: bnk | short_answer **TYPE:** short_answer

245. A and B are located in the first quadrant. If $sin(A) = 2$ and $cos(B) = 3$, determine the exact value of $sec A + sec B$.

TOP: bnk | short_answer **TYPE:** short_answer

246. Determine the exact measures for all angles where $tan(theta) = - 3$ in the domain

$$\tan\theta = -\sqrt{3}$$

. Rendered ExamView reference:

$$-180^\circ \leq \theta \leq 180^\circ$$

TOP: bnk | short_answer **TYPE:** short_answer

247. A grandfather clock shows a time of $7 o$ clock. What is the exact radian measure of the angle between the hour hand and the minute hand?

TOP: bnk | short_answer **TYPE:** short_answer

248. A $150-$cm$-$tall person stands on the bank of a narrow river and observes a flagpole on the opposite bank. If the angle of elevation to the top of the flagpole is 0.25 and the angle of depression to the bottom of the flagpole is $0.09,$ determine the flagpole s height, in metres, to two decimal places.
[image omitted: asset-3]

TOP: bnk | short_answer **TYPE:** short_answer

249. The volume of a drinking cup can be approximated by the formula $V = A top + A bottom 2 h$, where the top and bottom of the cup are circular and h is the height. If the cup has a bottom diameter of 6 cm, a height of 10 cm, and sides that slope outward at an angle of $0.09,$ determine the cup s volume.
[image omitted: asset-4]

TOP: bnk | short_answer **TYPE:** short_answer

250. The angle between the balls is 1.25 radians. After $2 s,$ the distance between the balls is $1 m$. How fast are the balls moving, to the nearest hundredth of a metre per second?
[image omitted: asset-5]

TOP: bnk | short_answer **TYPE:** short_answer

251. The piece of wood is nailed to the ground such that it makes a 30 angle with the floor.

(a) Represent this situation with a diagram.
(b) Which trigonometric ratio can be used to determine the length of the piece of wood?
(c) Determine the length of the piece of wood.

TOP: bnk | problem **TYPE:** problem

252. Two wires are connected to a tower at the same point on the tower. Wire #1 makes an angle of 45 with the ground and wire #2 makes an angle of 60 with the ground.

(a) Represent this situation with a diagram.
(b) Which wire is longer? Explain.
(c) If the point where the two wires connect to the tower is $35 m$ above the ground, determine exact expressions for the lengths of the two wires.
(d) Determine the length of each wire, to the nearest tenth of a metre.
(e) How do your answers to parts b) and d) compare?

[image omitted: asset-6]

TOP: bnk | problem **TYPE:** problem

253. Sarah and Simone are walking in a walk$-a-$thon down a straight street that leads to the finish line in the park. At the same time, they both notice a hot$-$air balloon directly over the finish line. Sarah sees the angle from the ground to the balloon as 30 and Simone, who is 0.25 km closer to the finish line than Sarah, sees the angle from the ground to the balloon as 45.

(a) Draw a diagram to represent this situation.
(b) If x represents the distance that Simone is from the finish line, write an expression for the distance from Sarah to the finish line.
(c) Write a trigonometric ratio for each girl s position that involves the height of the balloon, h; the distance, x, each girl is away from the finish line; and the angle from the girl to the balloon.
(d) Rearrange each equation from part c) to isolate h.
(e) Set the two expressions for h equal to each other and solve for x, to the nearest hundredth of a kilometre.
(f) Determine the height of the balloon, to the nearest hundredth of a kilometre.

[image omitted: asset-7]

TOP: bnk | problem **TYPE:** problem

254. A represent the angle that Gursant would measure. Let d represent the distance that Gursant is from the monument. Thus, $tan 41 = 6.0 x x = 6.0 tan$. $d = 3.5 + 6 tan$ Gursant would measure an angle of 30 from the ground to the top of the monument.
[image omitted: asset-8]

TOP: bnk | problem **TYPE:** problem

255. The point (–$5, 7)$ is located on the terminal arm of A in standard position.

(a) Determine the primary trigonometric ratios for A.
(b) Determine the primary trigonometric ratios for B with the same sine as A, but different signs for the other two primary trigonometric ratios.
(c) Use a calculator to determine the measures of A and B, to the nearest degree.

TOP: bnk | problem **TYPE:** problem

256. (a) Without using a calculator, determine two angles between 0 and 360 that have a sine ratio of [image omitted: asset-9].
(b) Use a calculator and a diagram to verify your answers to part a).

TOP: bnk | problem **TYPE:** problem

257. A such that $cos A =$ [image omitted: asset-10].

(a) In which quadrant(s) is this angle? Explain.
(b) If the sine of the angle is negative, in which quadrant is the angle? Explain.
(c) Sketch a diagram to represent the angle in standard position, given that the condition in part b) is true.
(d) Find the coordinates of a point on the terminal arm of the angle.
(e) Write exact expressions for the other two primary trigonometric ratios for the angle.

TOP: bnk | true_false **TYPE:** true_false

258. A in each diagram.
d) Calculate the length of BC in each diagram.
[image omitted: asset-11]
[image omitted: asset-12]

TOP: bnk | problem **TYPE:** problem

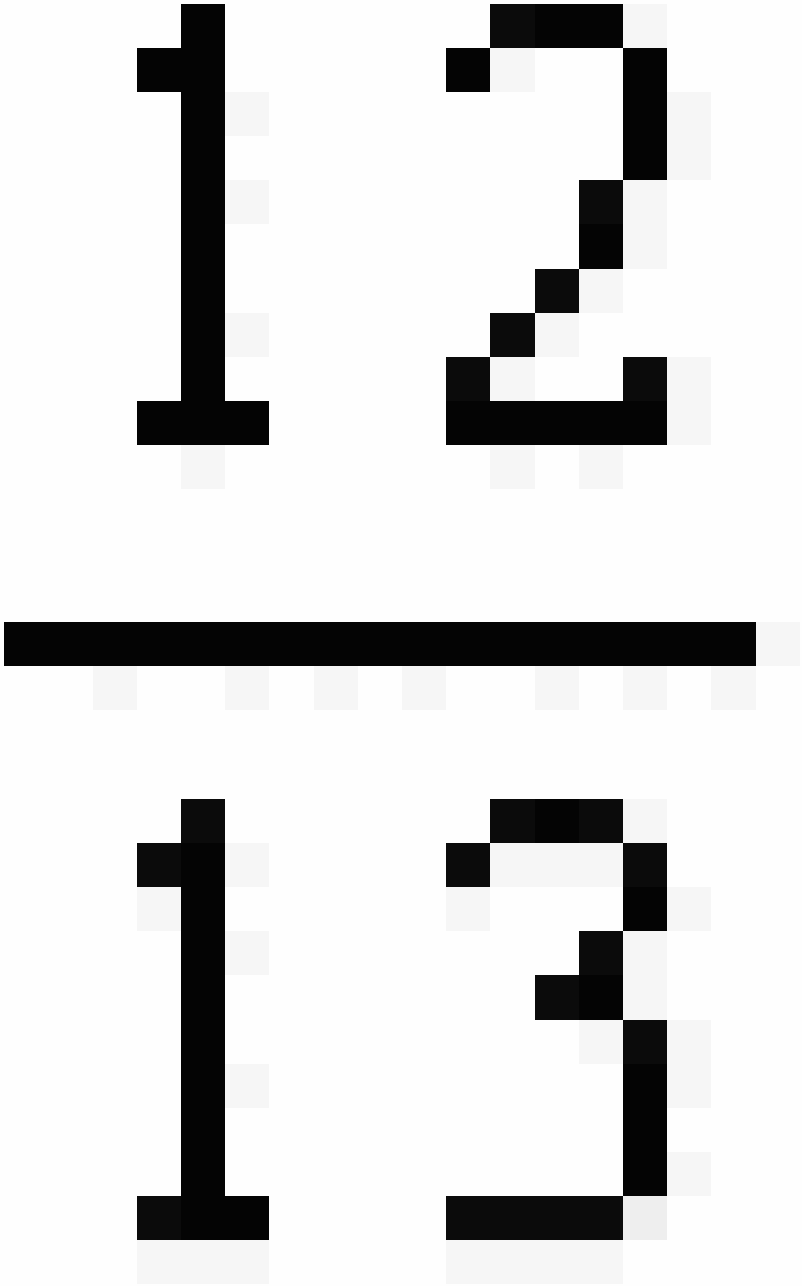
259. A new hotel has just been built on the road connecting town X and town Y, exactly halfway between the two towns. From where Jason is standing, he sees that the angle of elevation to the top of the hotel is 1. Determine the height of the hotel, to the nearest tenth of a metre. Include a diagram with your solution.
[image omitted: asset-13]

TOP: bnk | short_answer **TYPE:** short_answer

260. A rectangle has a diagonal of 8 cm. The diagonal creates a 60 angle at the base of the rectangle. a) Write an exact expression for the base and the height of the rectangle.
b) Use your expressions to find the exact area of the rectangle.

TOP: bnk | problem **TYPE:** problem

261. Let x represent the width of the canyon. Use the cosine law.



The width of the canyon is approximately $173.3 m$.

TOP: bnk | short_answer **TYPE:** short_answer

262. Determine the height of the totem pole, to the nearest tenth of a metre.
[image omitted: asset-14]
[image omitted: asset-15]

TOP: bnk | short_answer **TYPE:** short_answer

263. (a) For the given trigonometric ratio, determine two other angles that have the same value.
(b) $sin 45$ ii) $tan 300$ iii) $cos 120 b)$ Explain how you determined the angles in part a).

TOP: bnk | problem **TYPE:** problem

264. The point $P($–$3,$ –$6)$ lies on the terminal arm of an angle in standard position.

(a) Which primary trigonometric ratios are positive and which are negative?
(b) Which reciprocal trigonometric ratios are positive and which are negative?
(c) Determine exact values for the primary trigonometric ratios.
(d) Determine exact values for the reciprocal trigonometric ratios.

TOP: bnk | problem **TYPE:** problem

265. The slice is in the shape of a sector with arc length 8 cm. What is the measure of the central angle of the slice, in radians, rounded to two decimal places?

TOP: bnk | short_answer **TYPE:** short_answer

266. A bicycle tire revolves at 150 rpm (revolutions per minute). What is its angular velocity, in radians per second, rounded to two decimal places?

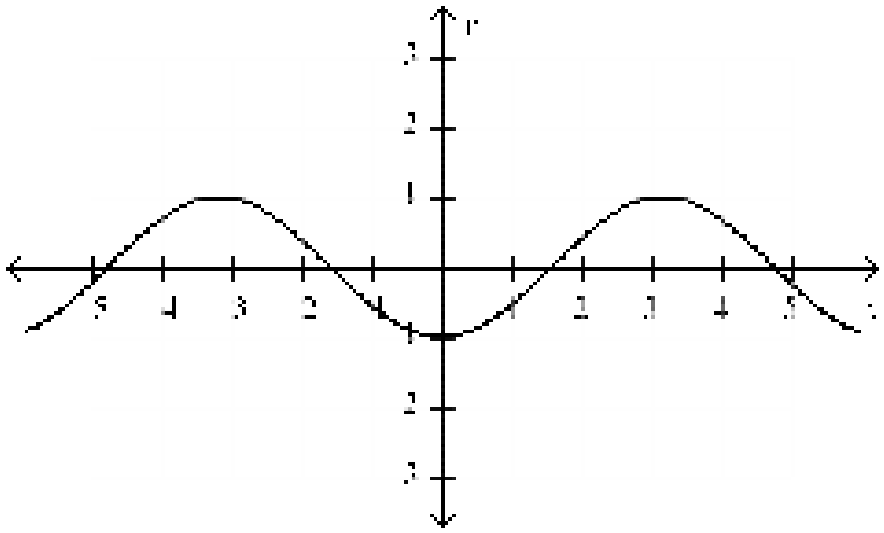
TOP: bnk | short_answer **TYPE:** short_answer

267. A group of marketing students are creating a strategy to sell ceiling fans. They determine that the number of sales will vary according to the month of the year. The formula $s = 9.2 + 2.4cos pi / 6 (t - 6)$ gives the expected sales, in thousands, according to the month, t, where t $= 1$ represents January, t $= 2$ is February, and so on.

(a) In which month are sales of fans expected to reach $11 600$?
(b) In which month are sales expected to be the least?
(c) Does this formula seem reasonable? Explain.

TOP: bnk | problem **TYPE:** problem

268. Which function, where x is in radians, is represented by the graph shown below? Recovered graph functions: $sin(x); cos(x); -sin(x); -cos(x)$.? $sin(x)$ Rendered ExamView reference:



$$y = -\cos x$$

$$y = \cos x$$

$$y = \dot{\sin} x$$

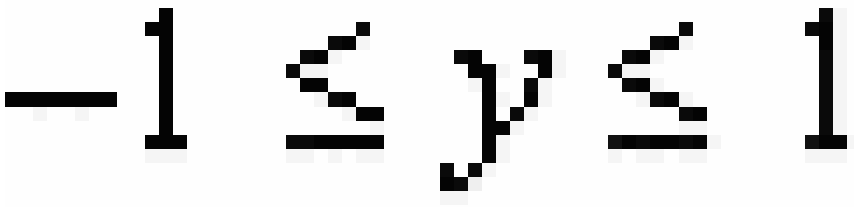
$$y = -\dot{\sin} x$$

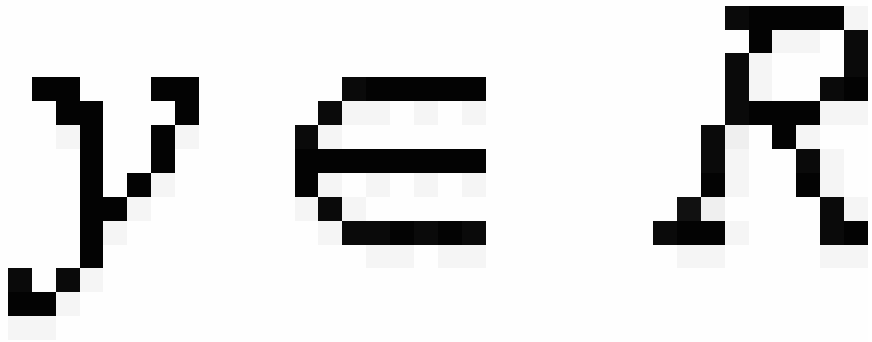
(A) -cos (B) cos (C) sin (D) -sin

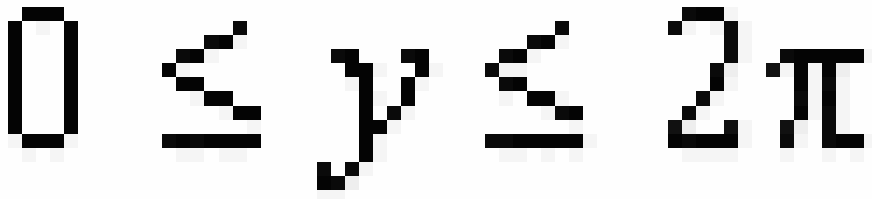
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

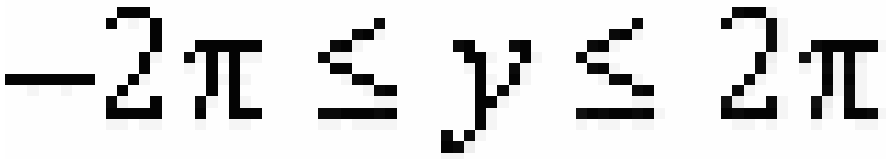
269. The range (in radians) of the graph of cos is Rendered ExamView reference:

$$y = \cos x$$





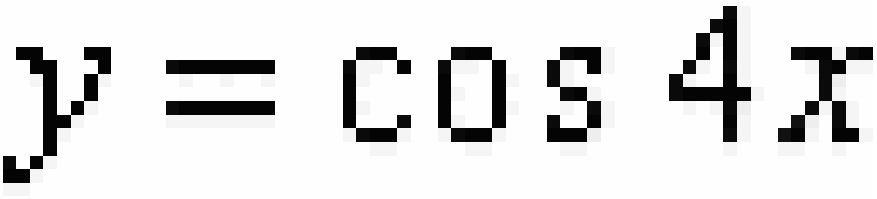




(A) $−1 <= y <= 1$ (B) $0 <= y <= 2 pi$ (C) all real numbers (D) $-pi <= y <= pi$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

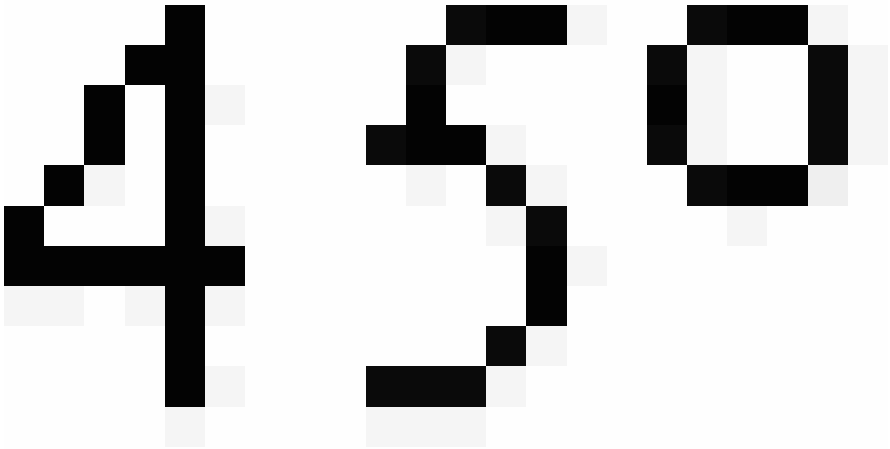
270. The period (in degrees) of the graph of cos is Rendered ExamView reference:



270°

90°

180°



(A) 270 (B) 180 (C) 45

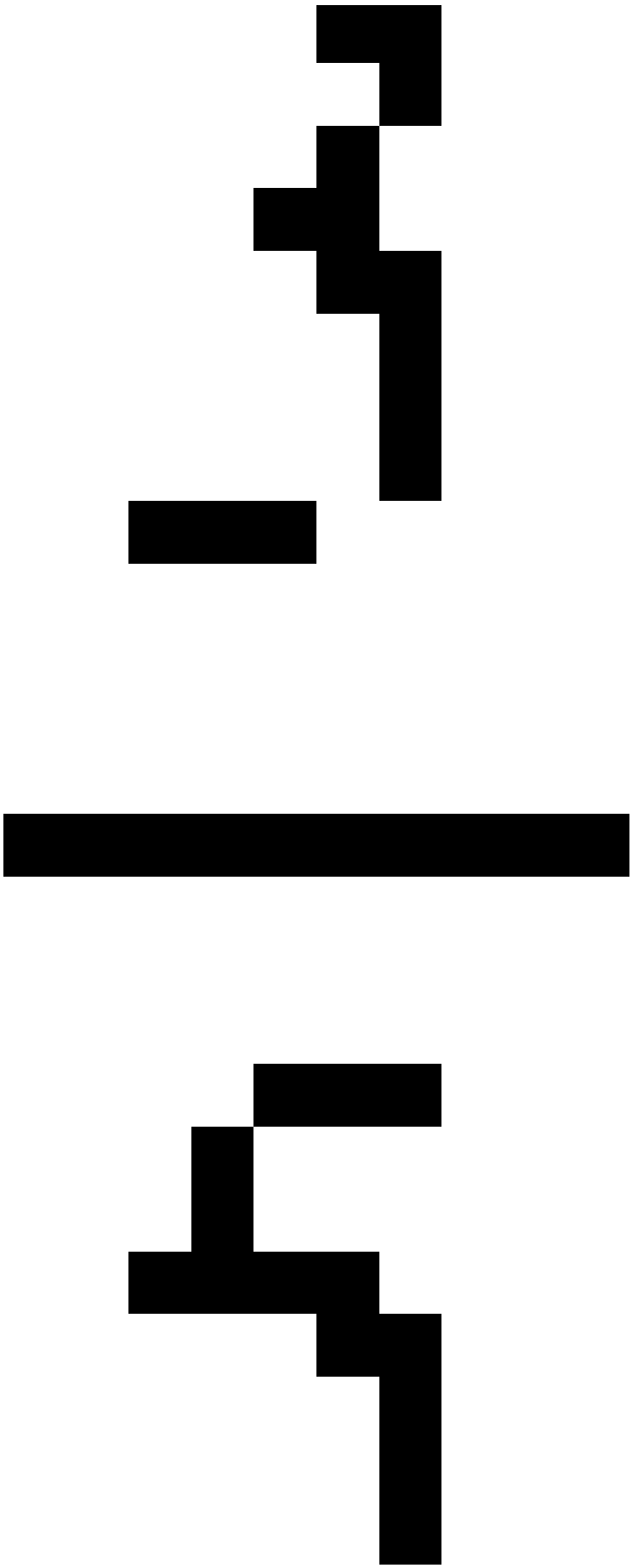
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

271. The amplitude and period (in –$2)$ of degrees are

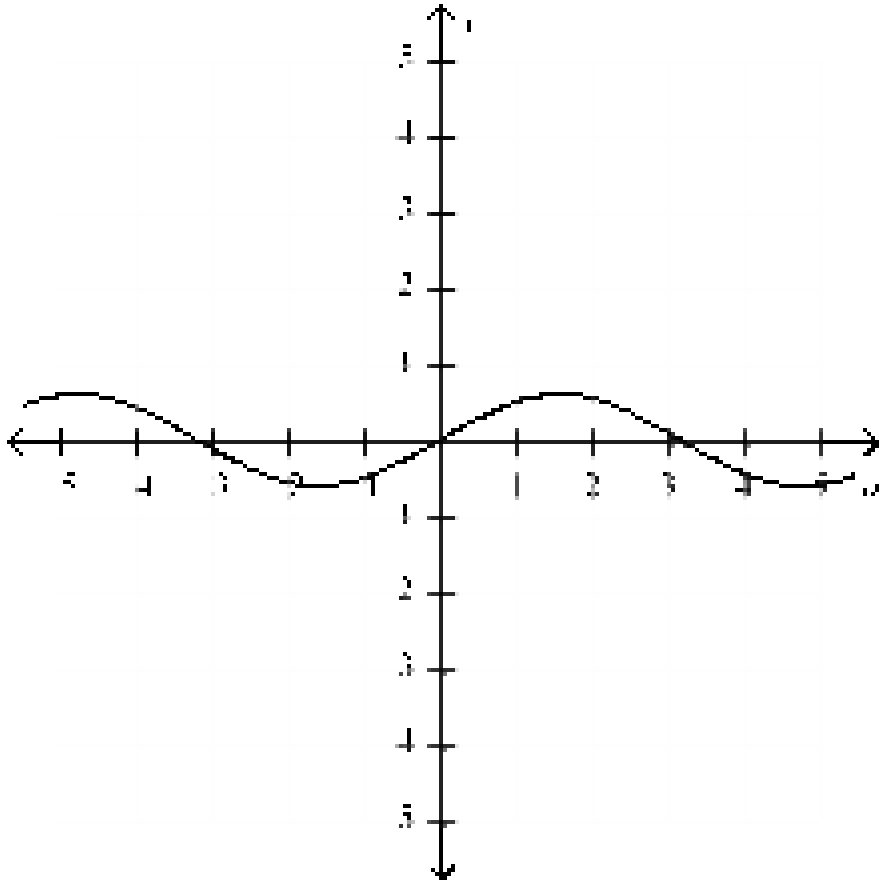
(A) $−2$ (B) $4/5$ (C) 90 (D) 1

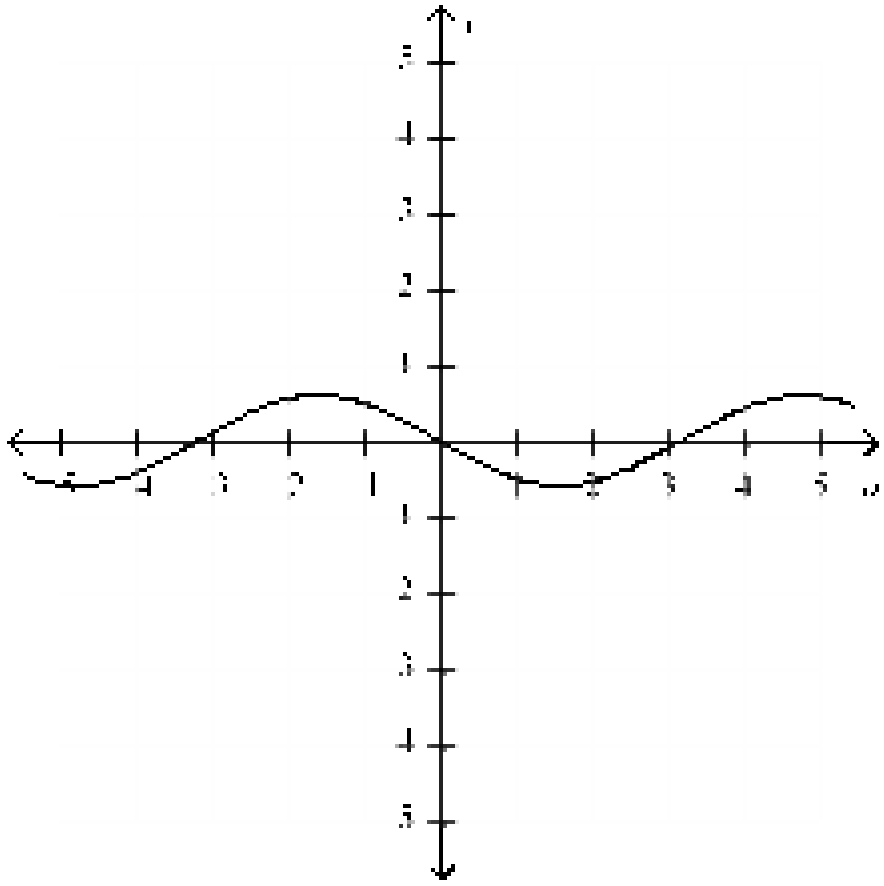
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

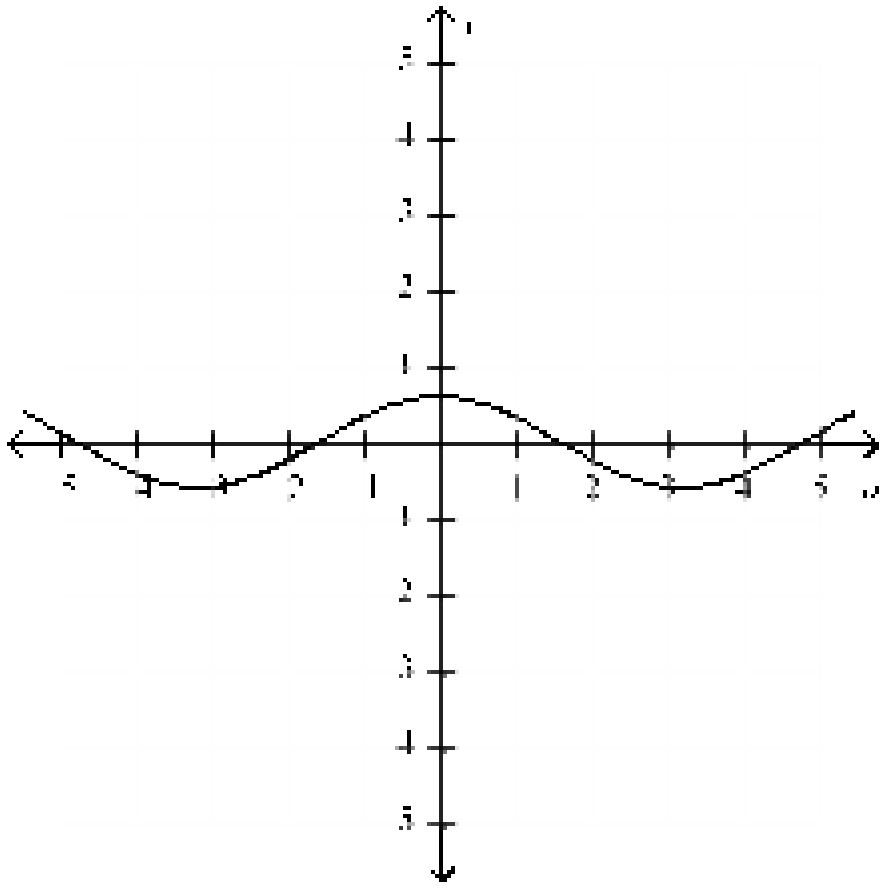
272. Which graph represents the function y $= 3/5 sin (−1,$ where is in radians? Recovered graph functions: $ −1*(0.6)*sin((−1)*x); (0.6)*sin((−1)*x); (0.6)*cos((−1)*x); −1*(0.6)*cos((−1)*x)$. Rendered ExamView reference:

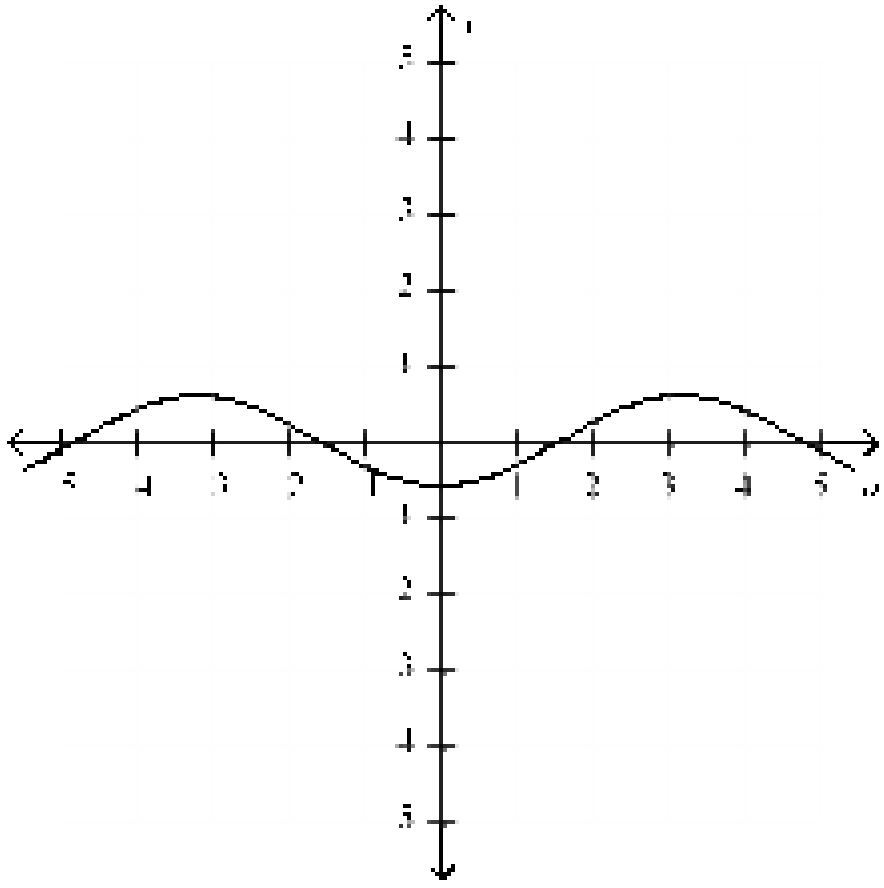












(A) graph of $-1*(0.6)*\sin((-1)*x)$

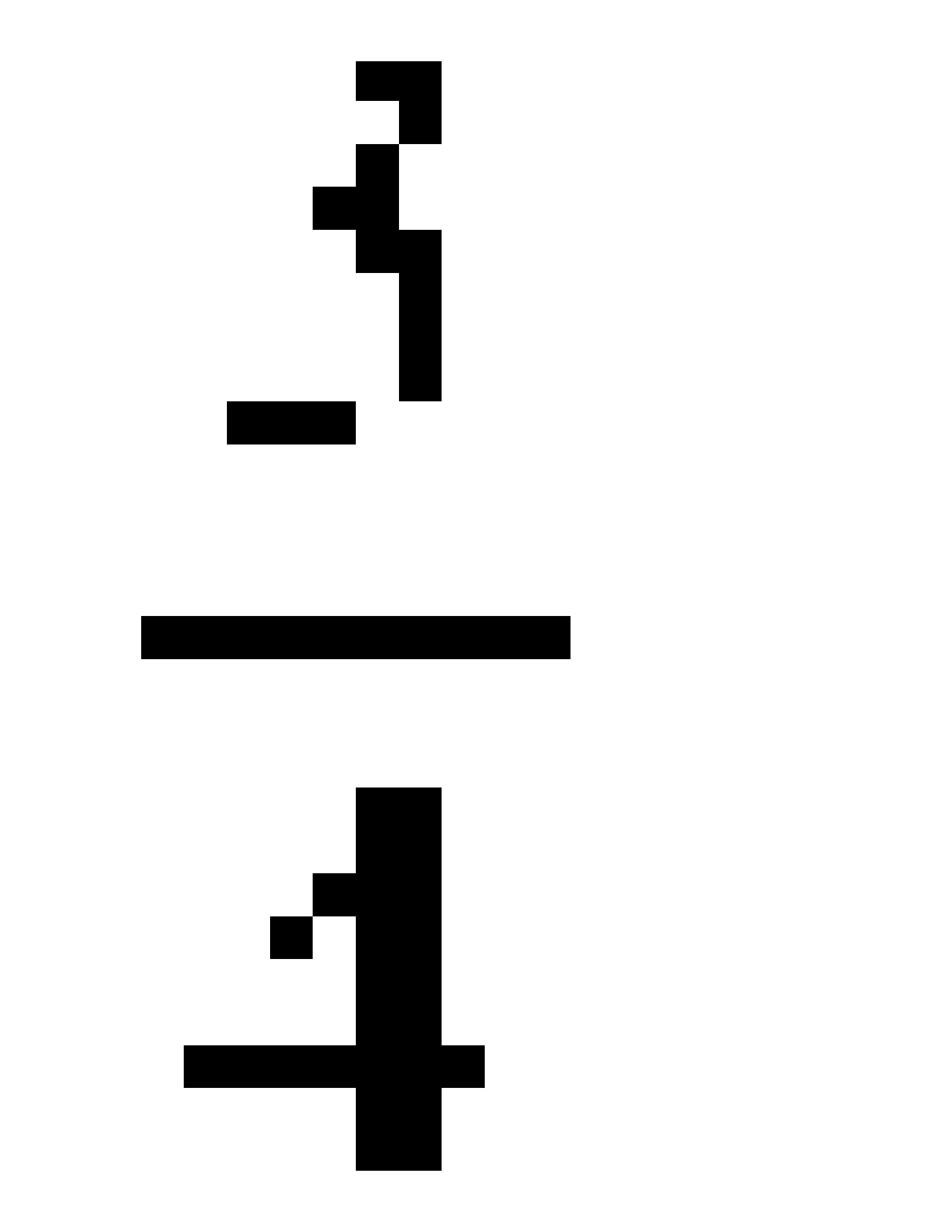
(B) graph of $(0.6)*\sin((-1)*x)$

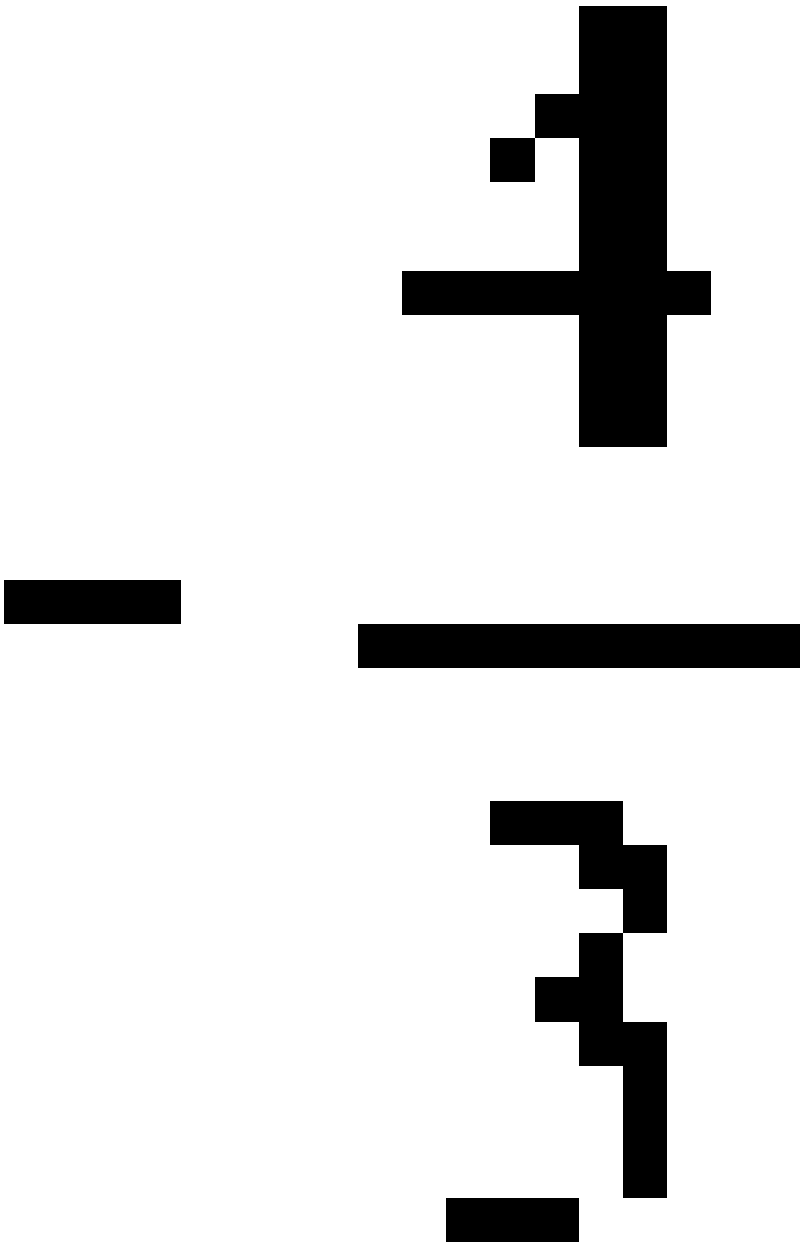
(C) graph of $(0.6)*\cos((-1)*x)$

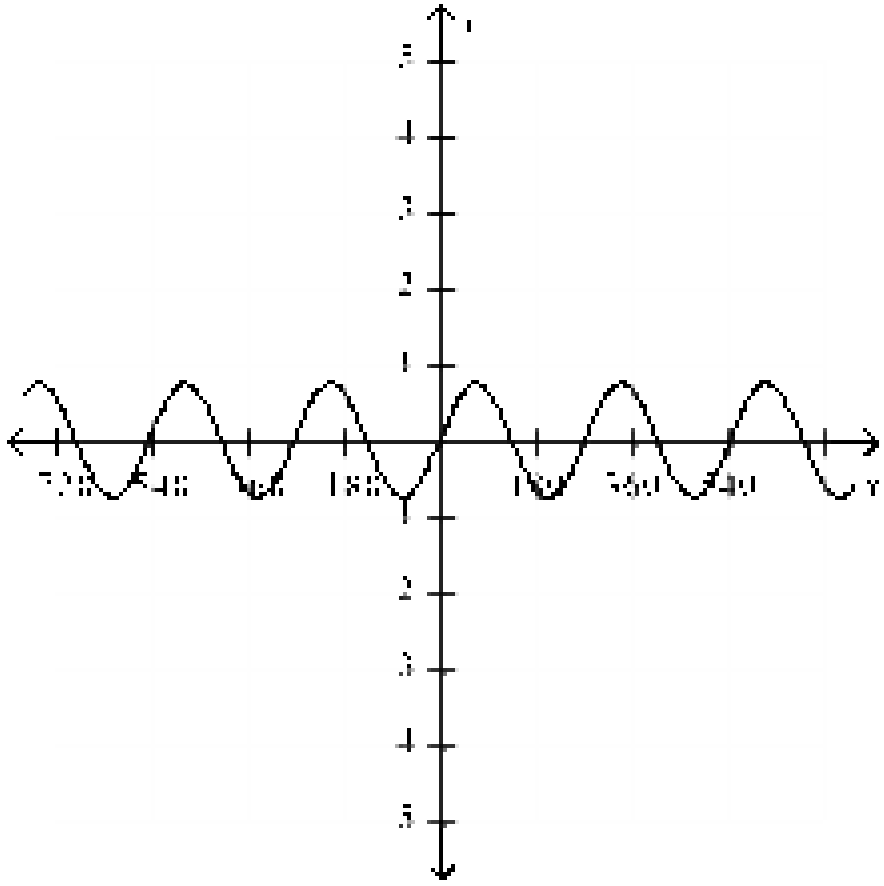
(D) graph of $-1*(0.6)*\cos((-1)*x)$

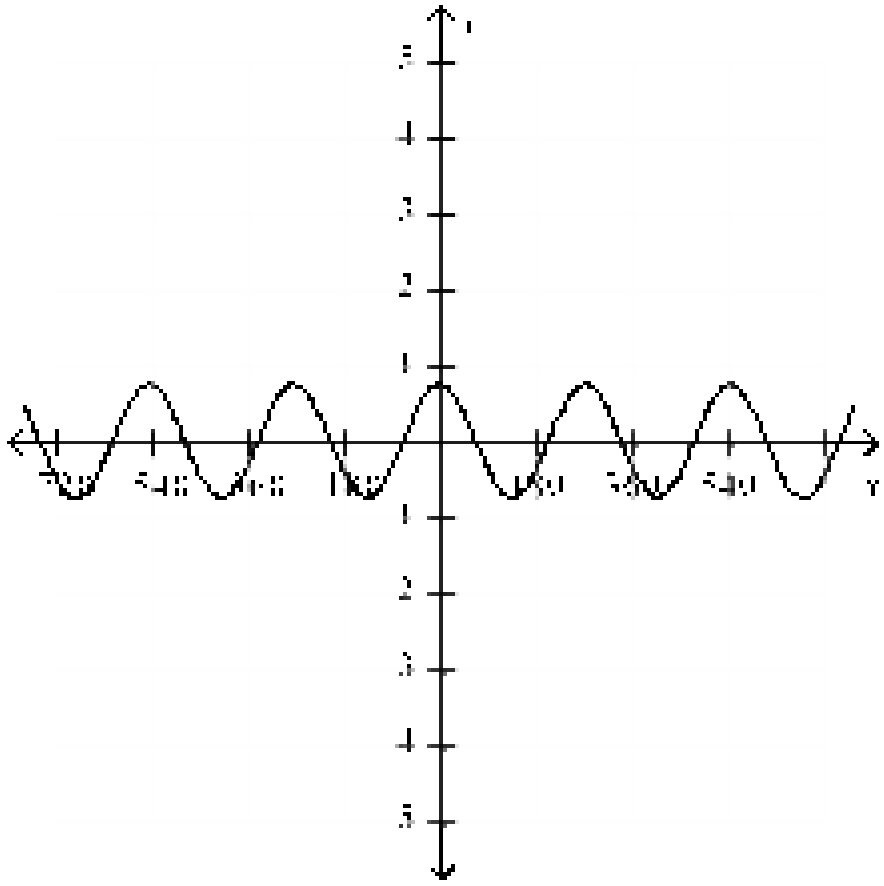
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

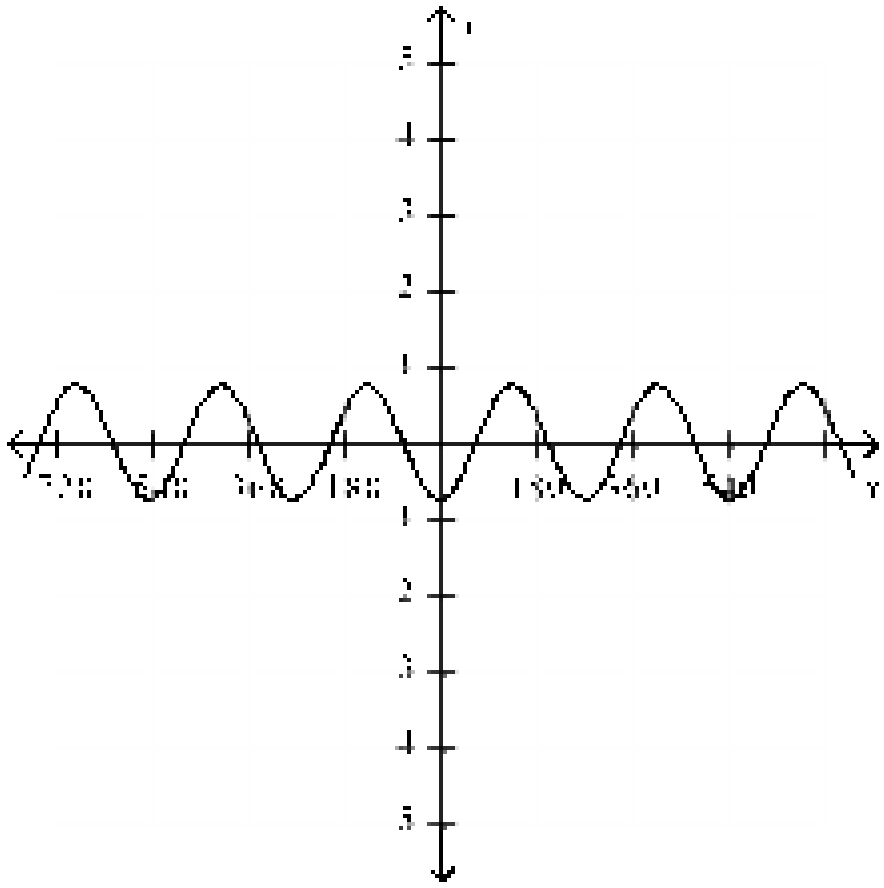
273. Which graph represents the function y $= 3/4 cos(3/4 x),$ where x is in degrees? Recovered graph functions: $ −1*(0.75)*sin((−1.333333333333)*(pi*x/180)); (0.75)*cos((−1.333333333333)*(pi*x/180)); −1*(0.75)*cos((−1.333333333333)*(pi*x/180)); (0.75)*sin((−1.333333333333)*(pi*x/180))$. Rendered ExamView reference:

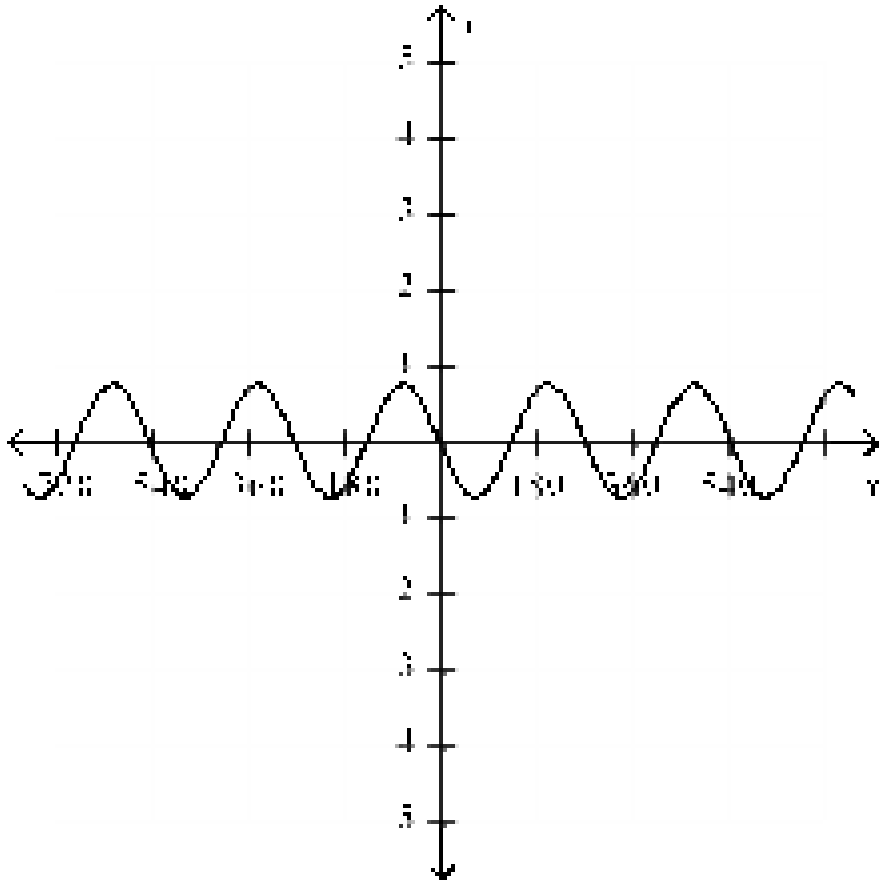












(A) graph of $-1*(0.75)*\sin((-1.333333333333)*(pi*x/180))$

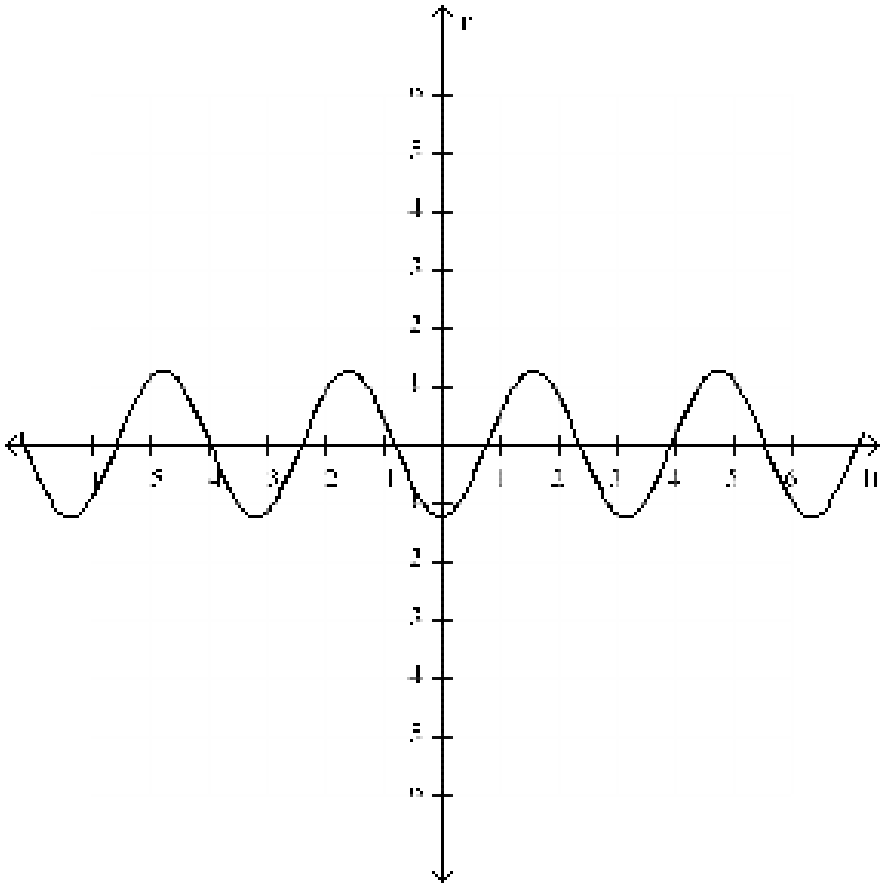
(B) graph of $(0.75)*\cos((1.333333333333)*(pi*x/180))$

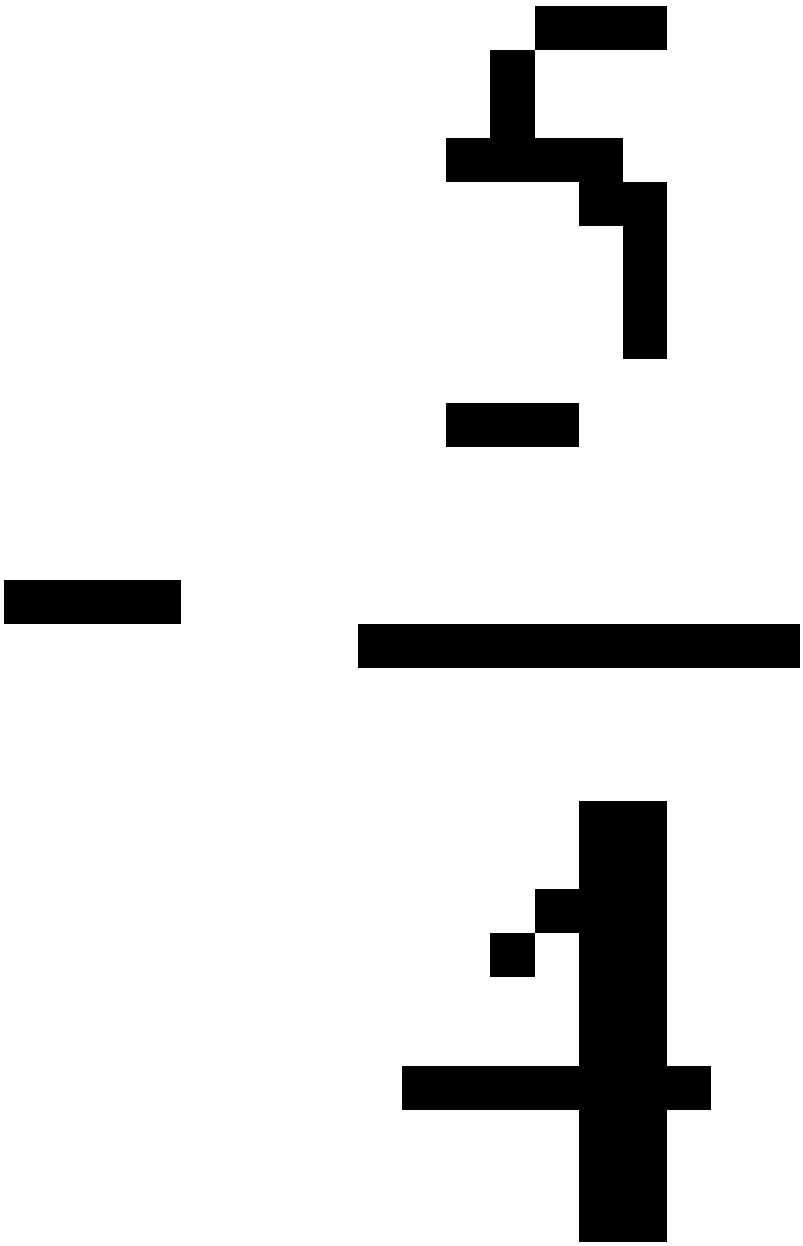
(C) graph of $-1*(0.75)*\cos((-1.333333333333)*(pi*x/180))$

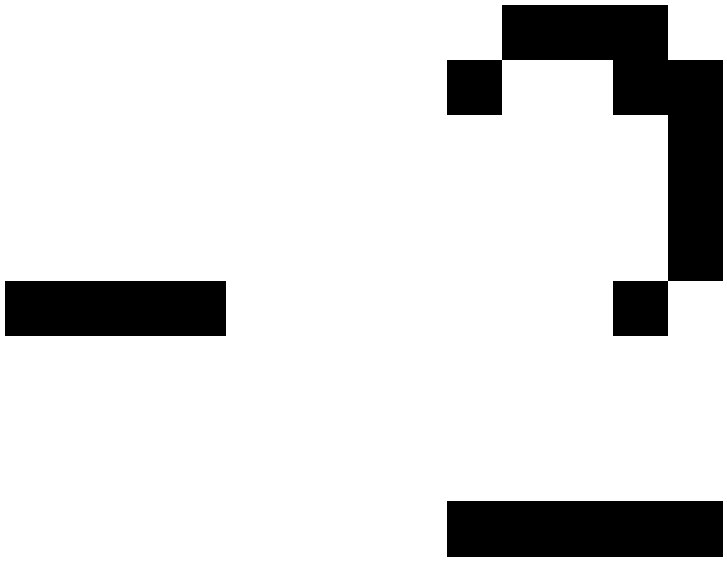
(D) graph of $(0.75)*\sin((1.333333333333)*(pi*x/1$

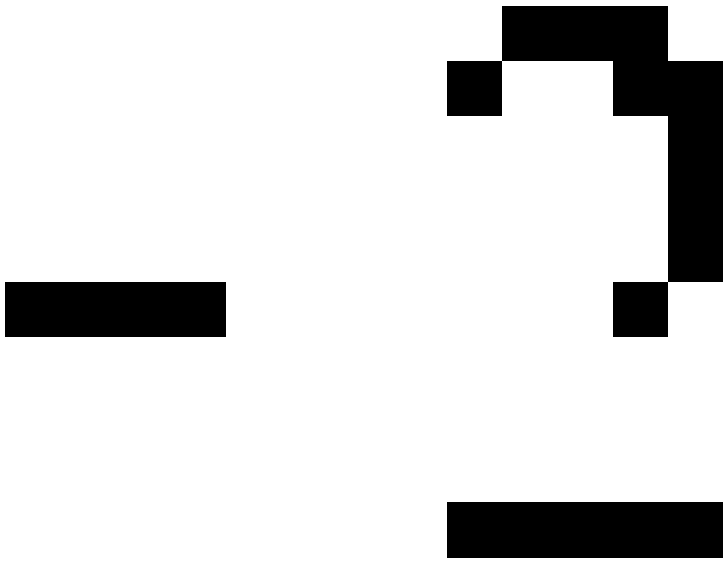
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

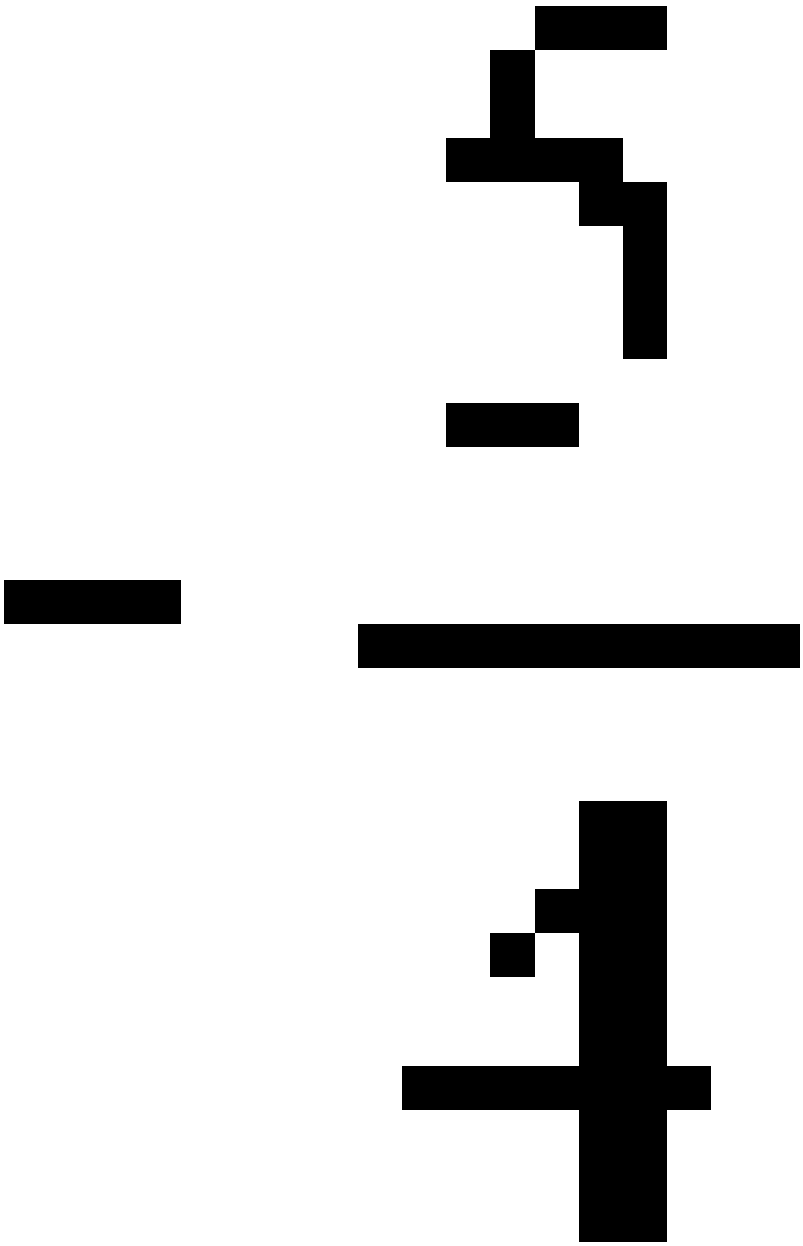
274. Which function is represented by the graph shown below? Recovered graph function: ($−1.25)*cos((−2)*x),$ where is in radians? ($−1.25)*cos((−2)*x)$ Rendered ExamView reference:

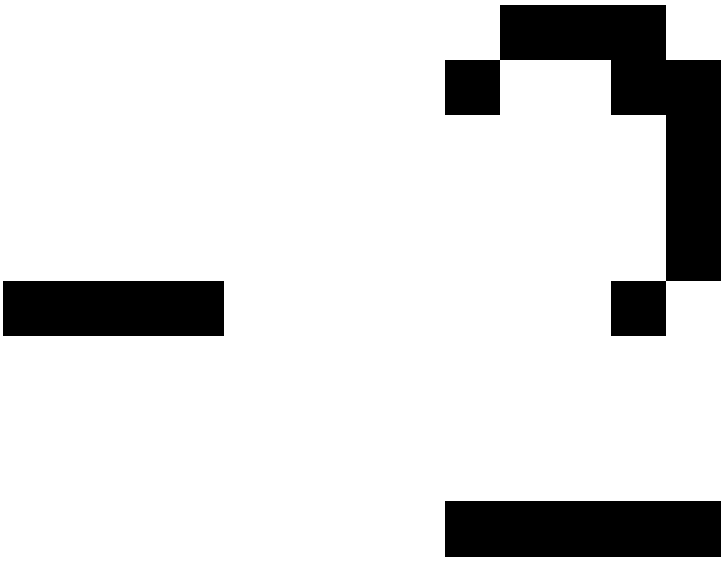


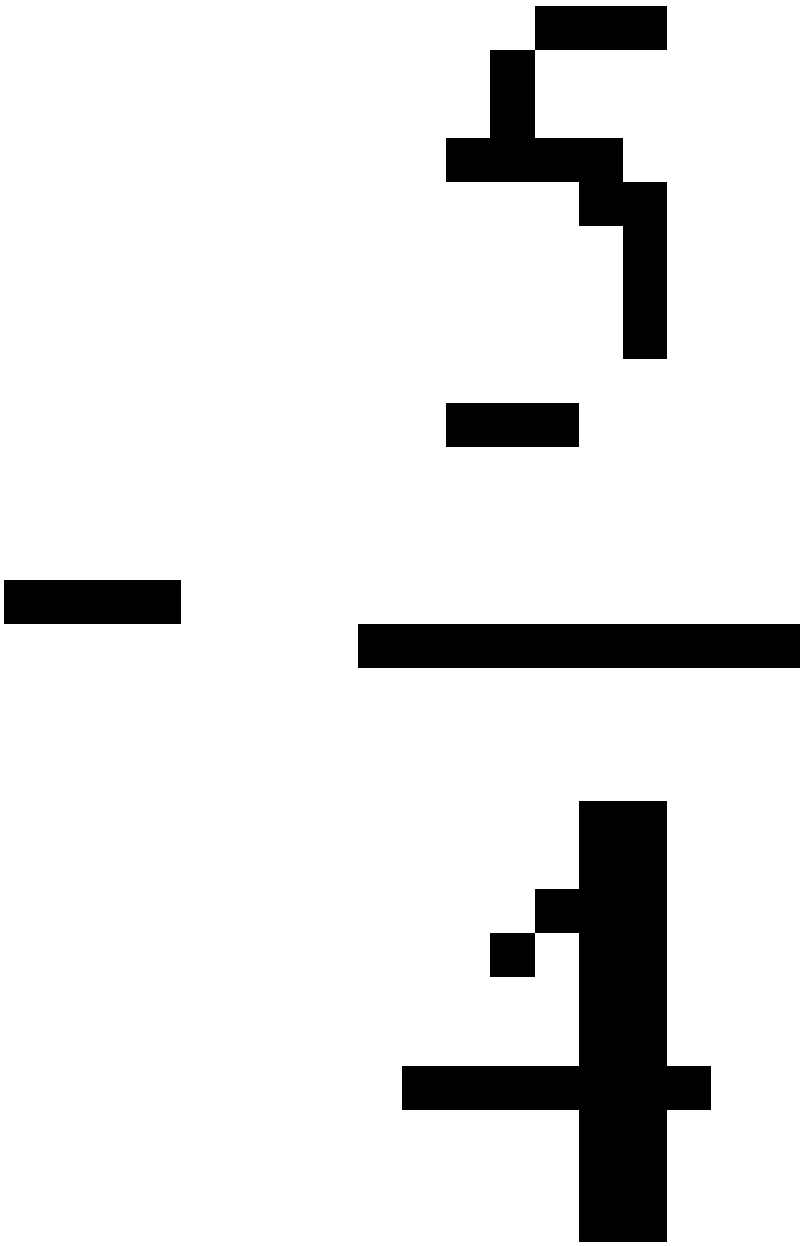


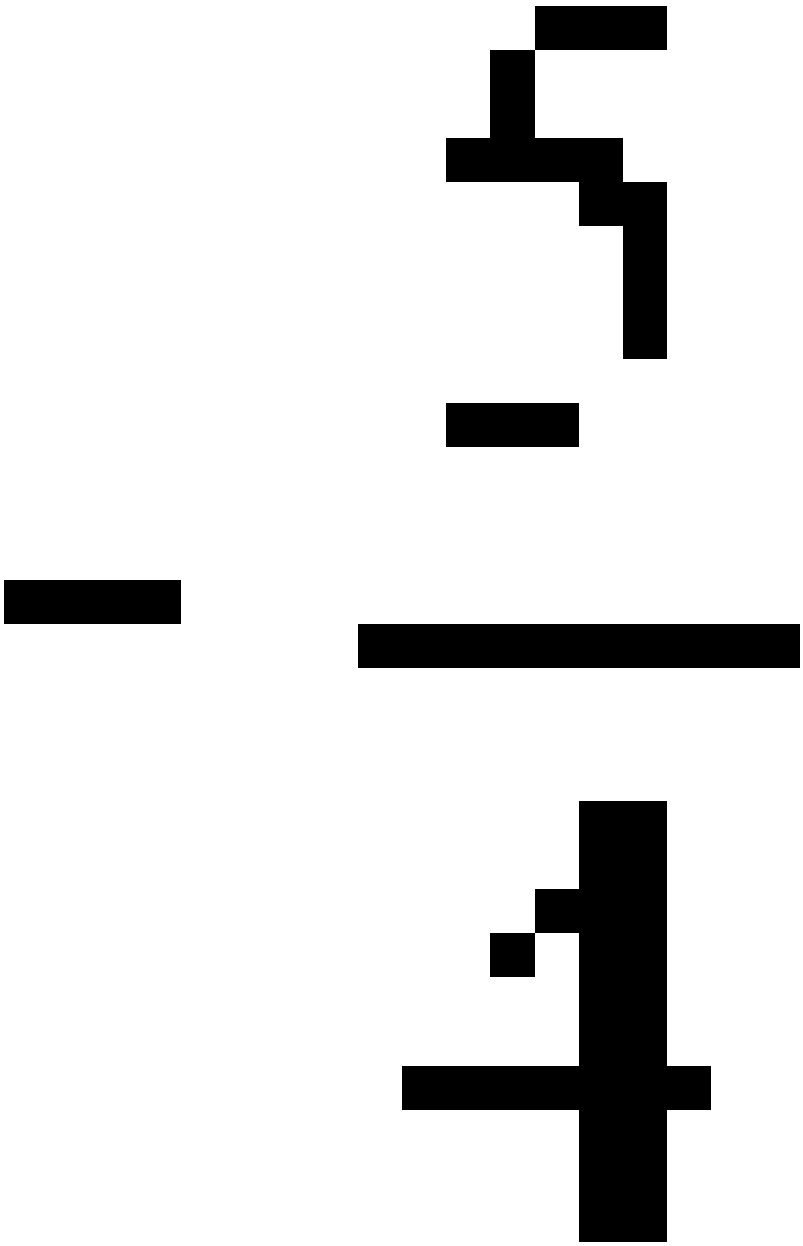


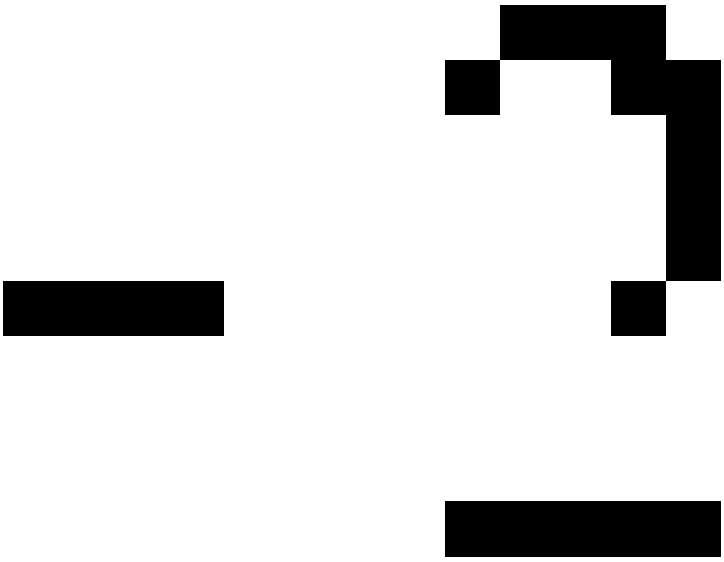








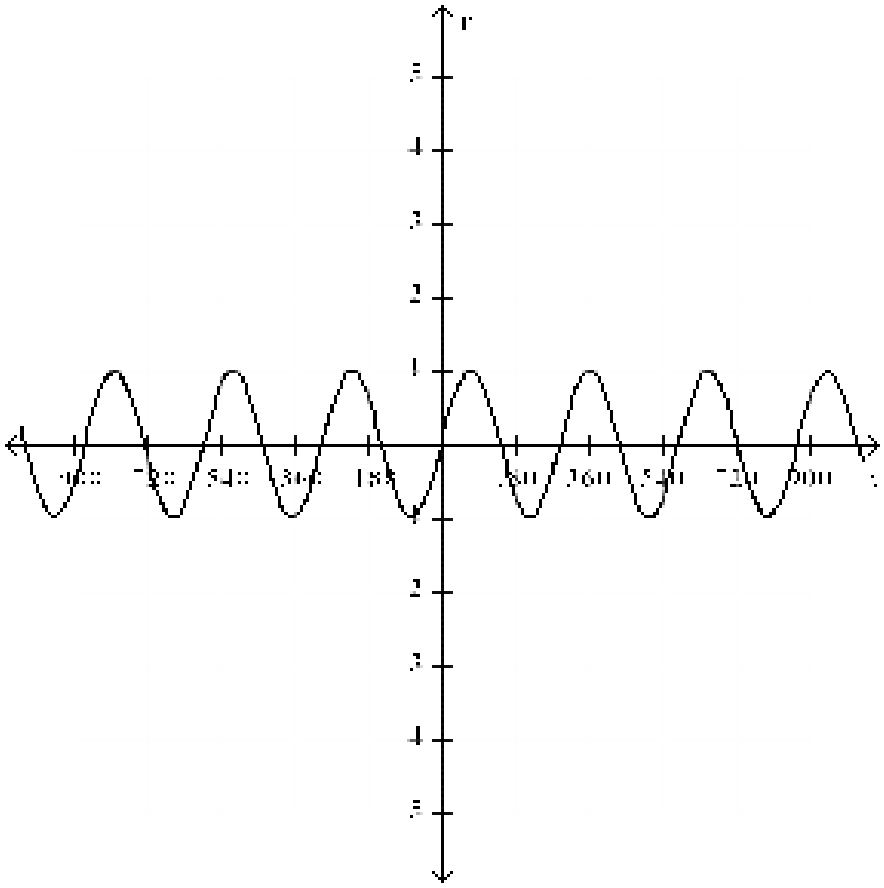


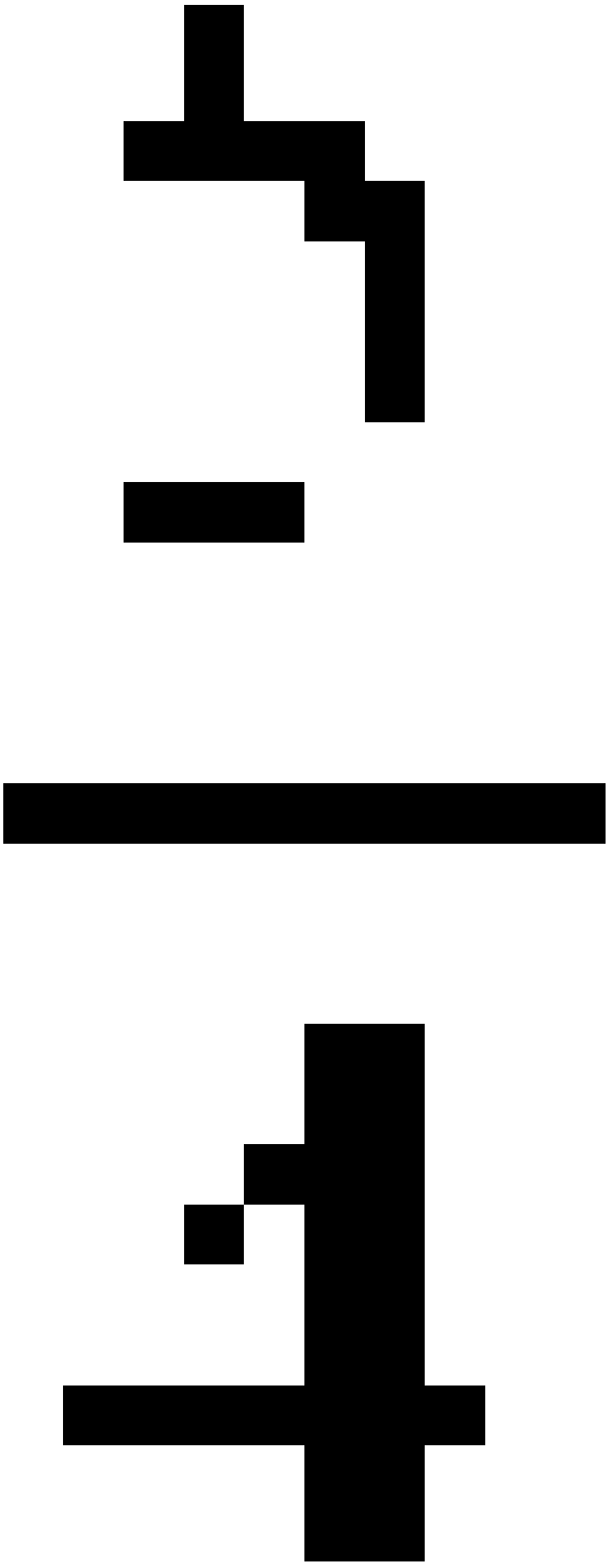


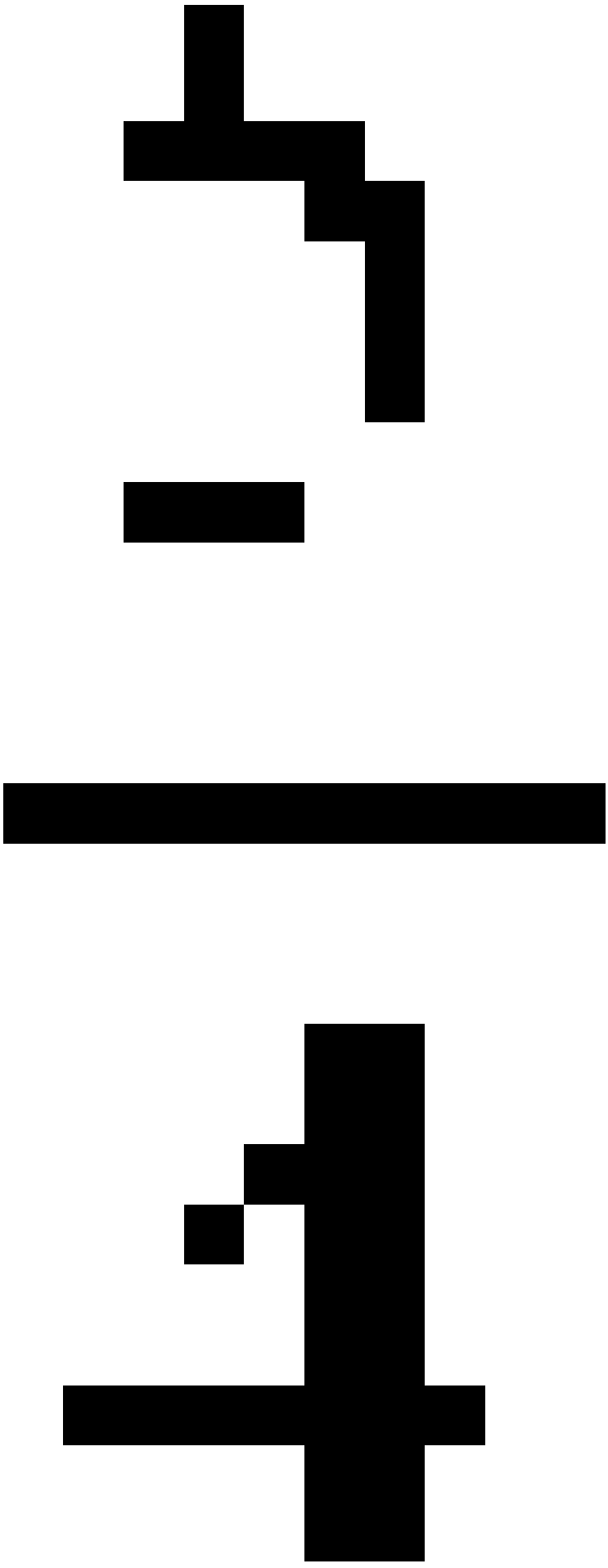
(A) y = −5/4 sin(−2 x) (B) y = −2 sin(−5/4 x) (C) y = −2 cos(−5/4 x) (D) y = −5/4 cos(−2 x)

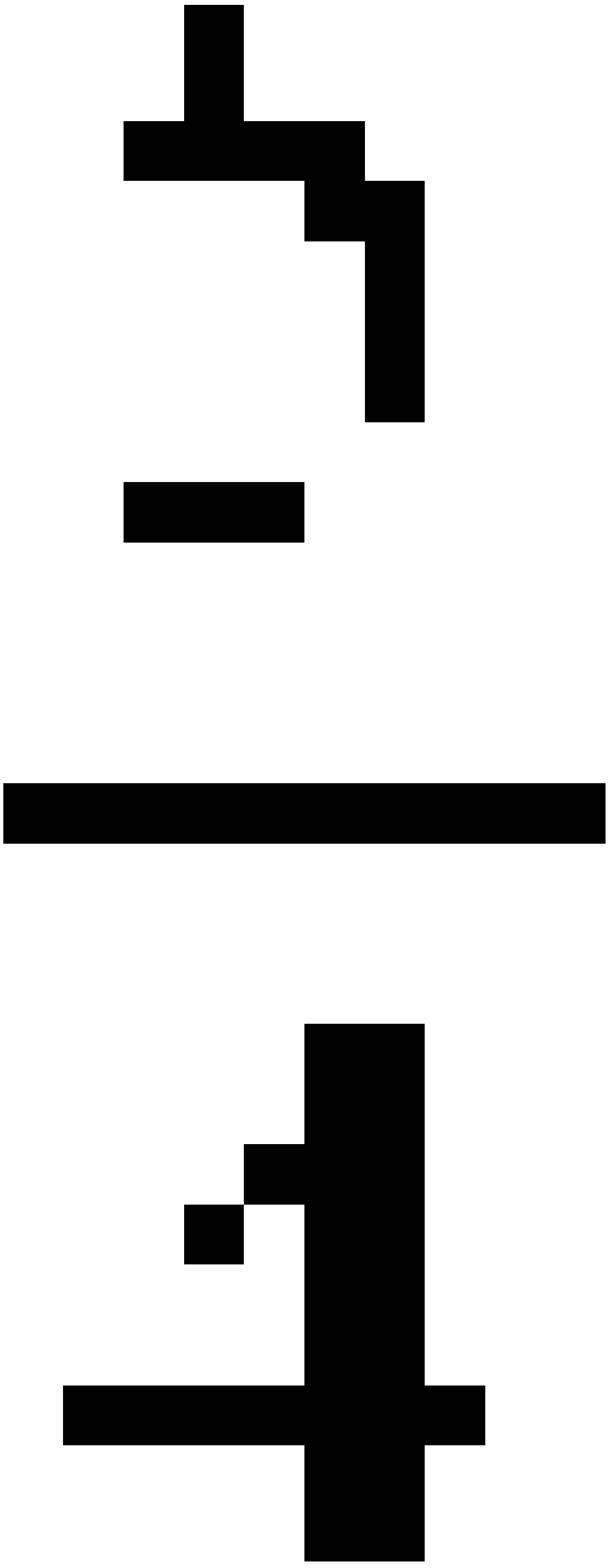
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

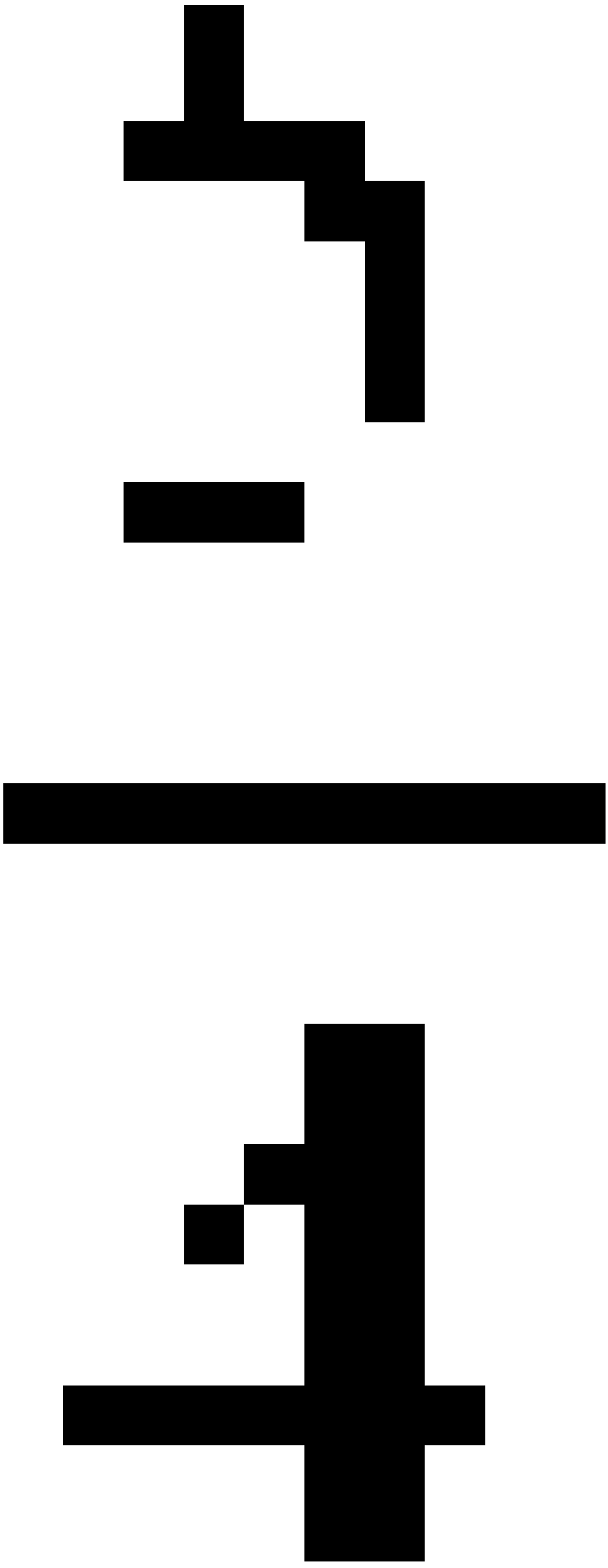
275. Which function is represented by the graph shown below? Recovered graph function: ($1)*sin((1.25)*($pi$*x/180)),$ where x is in degrees? $5/4 a$. y $= 1 sin(1 x) c$. y $= 1 cos(1 x) b$. y $= 1 cos(5/4 x) d$. y $= 1 sin(5/4 x)$ Rendered ExamView reference:







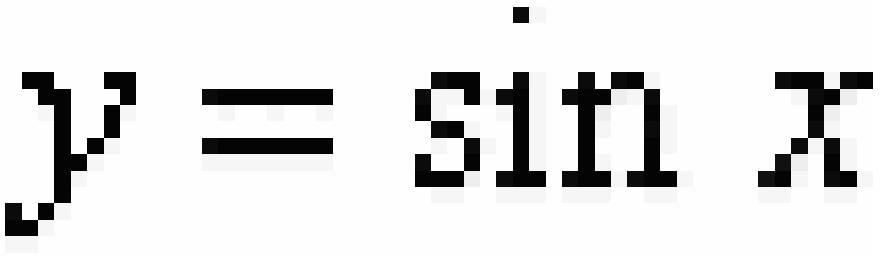


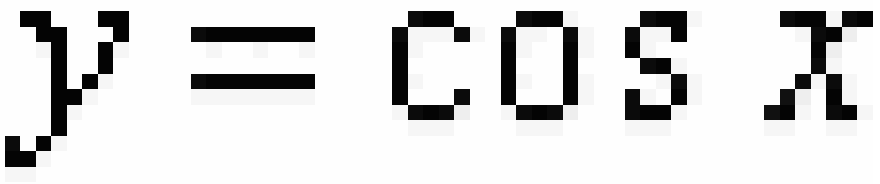


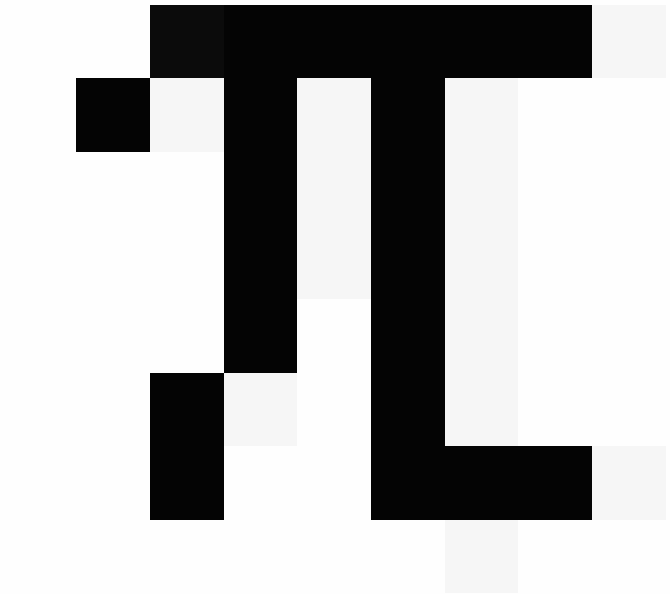
(A) $5/4$ (B) 1 (C) $5/4$ (D) 1

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

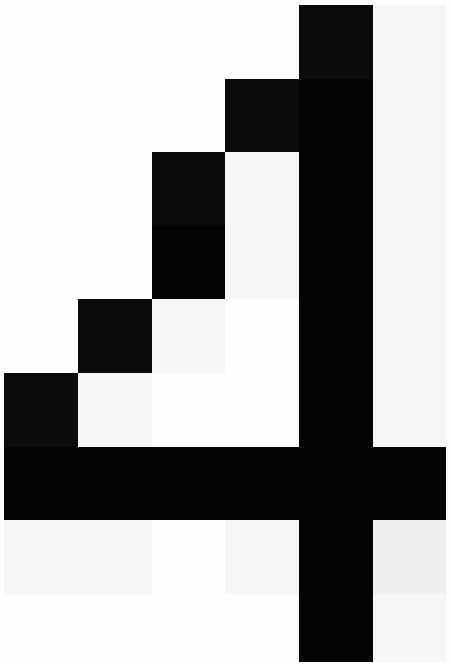
276. The graph of sin can be obtained by translating the graph of cos Rendered ExamView reference:

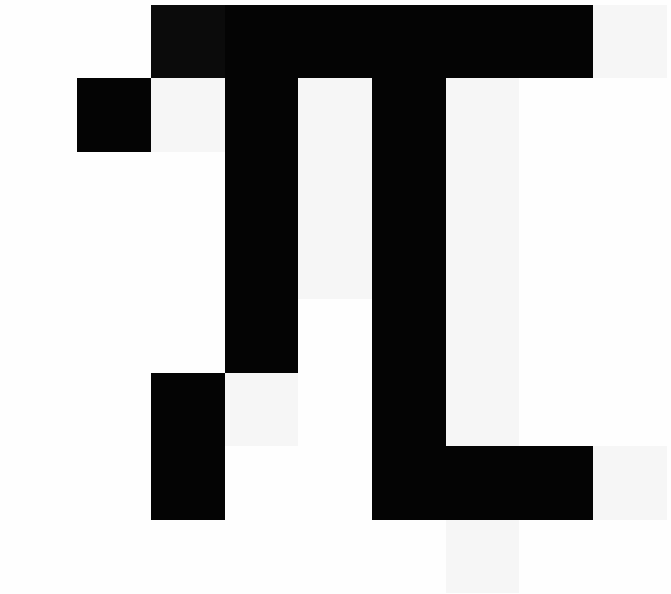




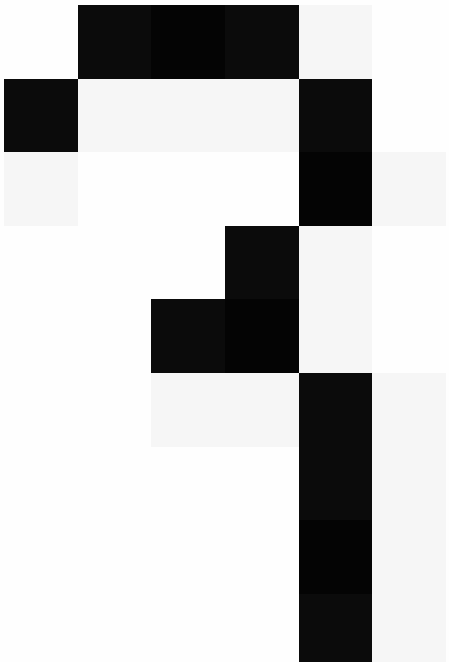


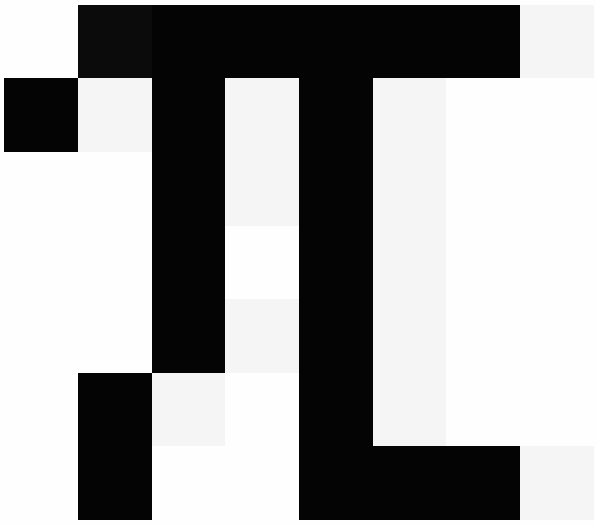




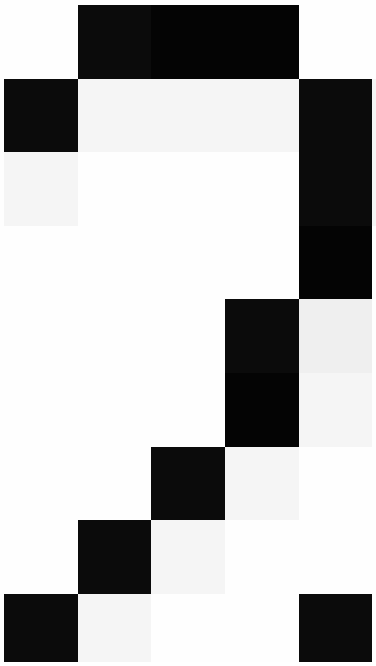


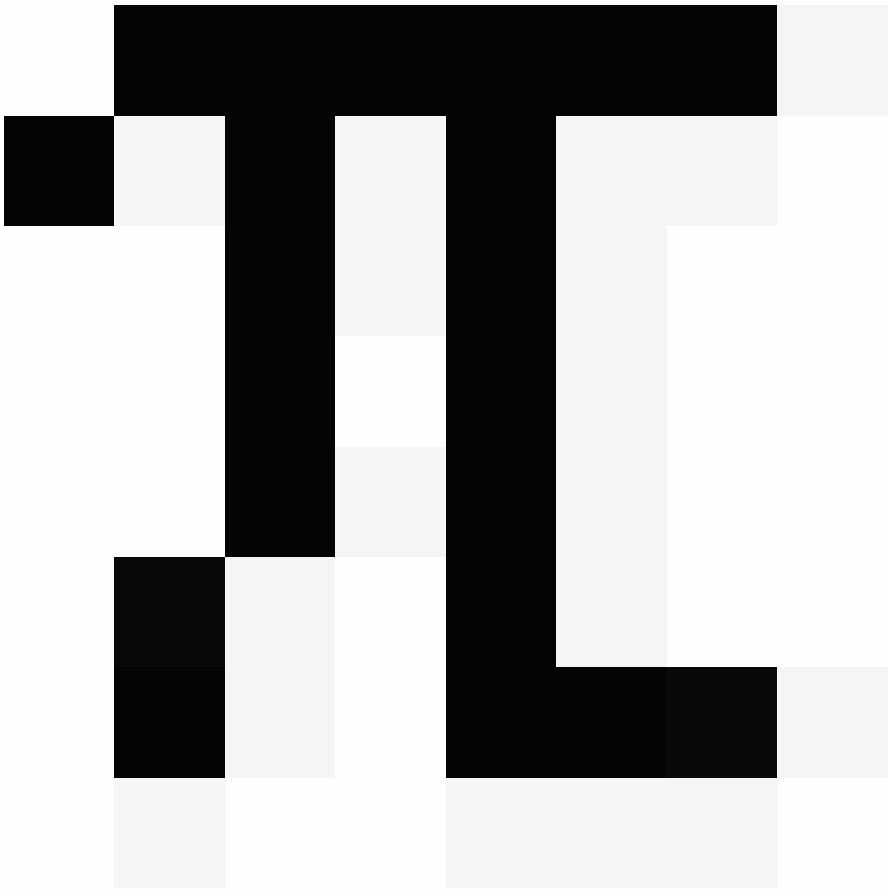












(A) $pi/2$ units to the left

(B) $pi/2$ units to the right

(C) pi units to the right

(D) pi units to the left

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

277. What is the amplitude of the sinusoidal function 7?

(A) – 5 (B) – 5 (C) $–8$ (D) 7

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

278. What is the period of the sinusoidal function –?

(A) $1/8$ (B) $1/4$ (C) 4 (D) $1/2$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

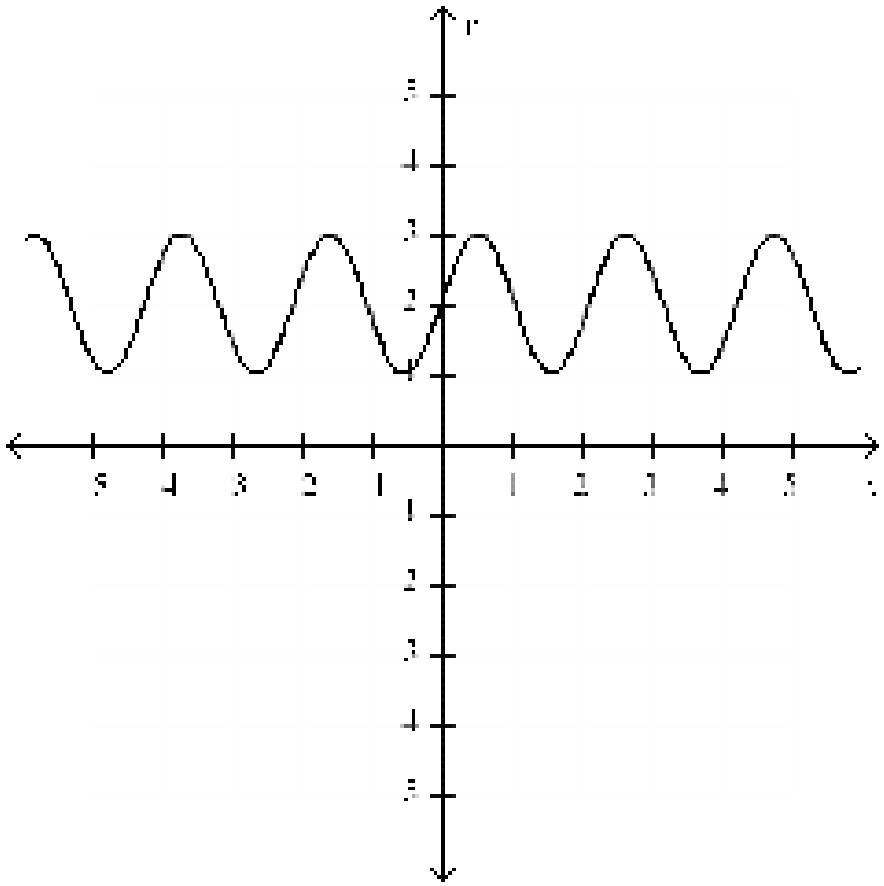
279. Determine the phase shift of the sinusoidal function –6.

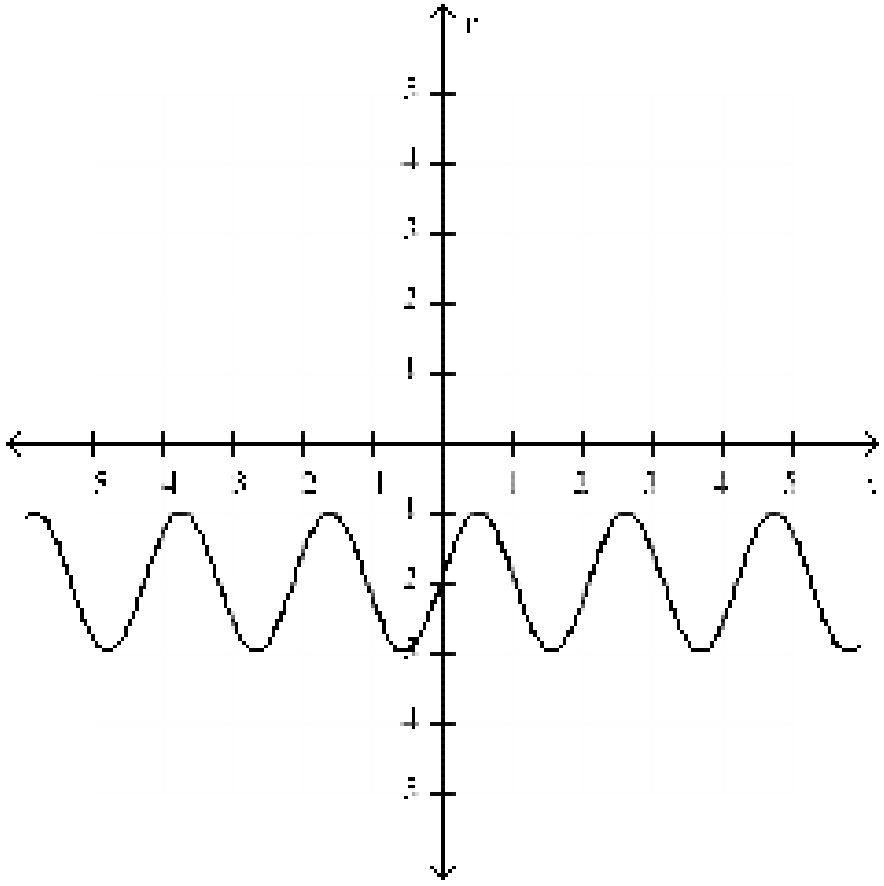
(A) $2/5$ (B) $5/2$ (C) $1/3$ (D) $–6/5$

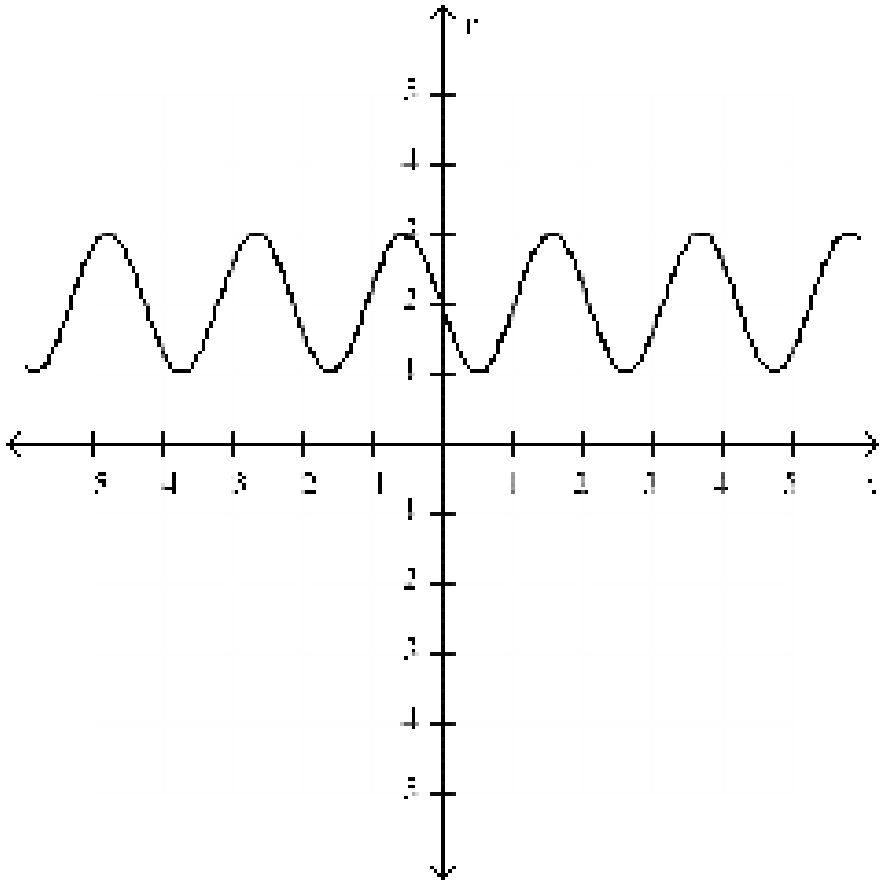
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

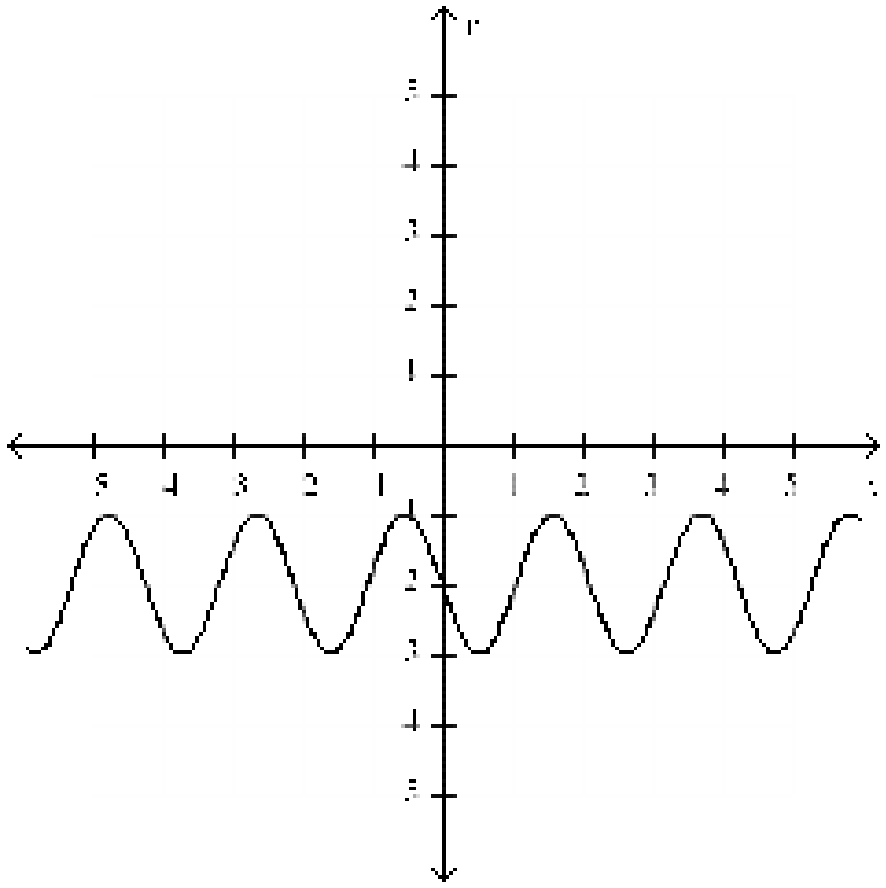
280. Which graph represents the sinusoidal function –? Recovered graph functions: $-a*cos((3)*(x+$pi$/(6)))-(−2); -a*cos((3)*(x+pi/(6)))+(−2); a*cos((3)*(x+pi/(6)))-(−2); a*cos((3)*(x+pi/(6)))+(−2)$. Rendered ExamView reference:

$$y = \cos\left[3\left(x + \frac{\pi}{6}\right)\right] - 2$$









(A) graph of -a*cos((3)*(x+pi/(6)))-(−2)

(B) graph of -a*cos((3)*(x+pi/(6)))+(−2)

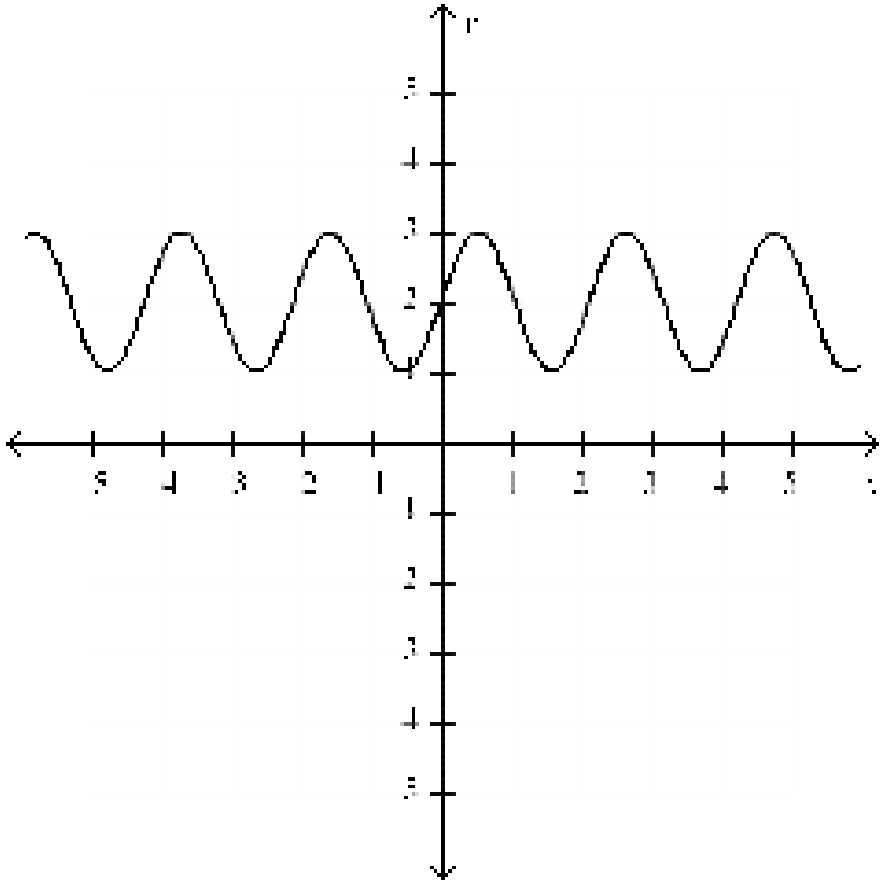
(C) graph of a*cos((3)*(x+pi/(6)))-(−2)

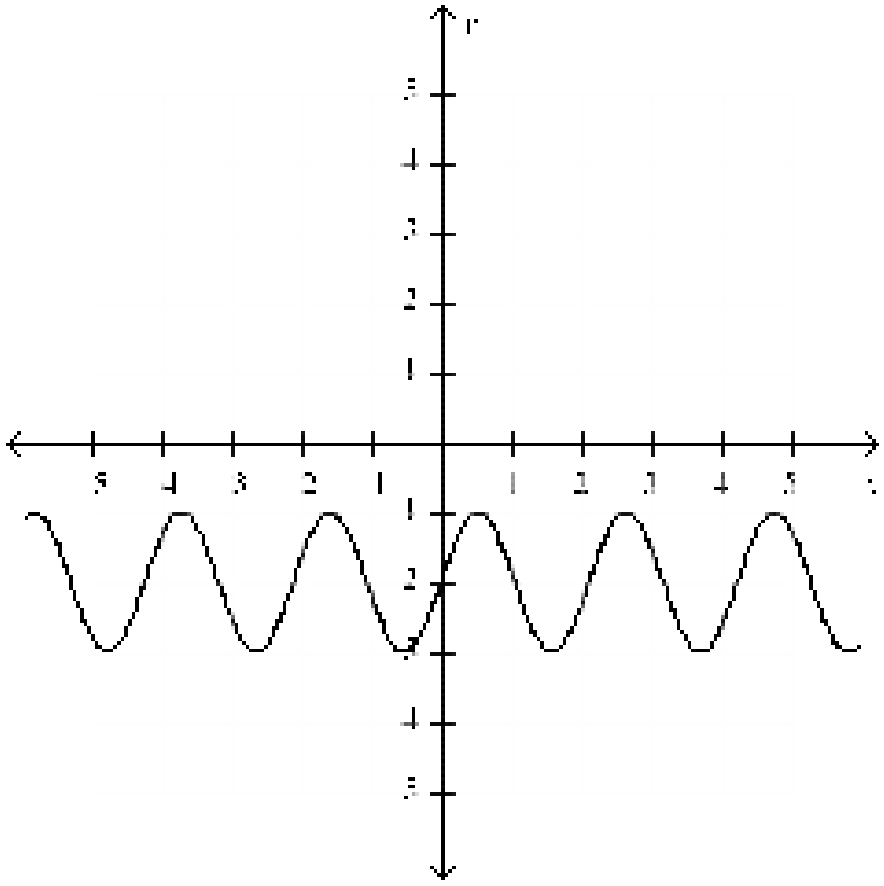
(D) graph of a*cos((3)*(x+pi/(6)))+(−2)

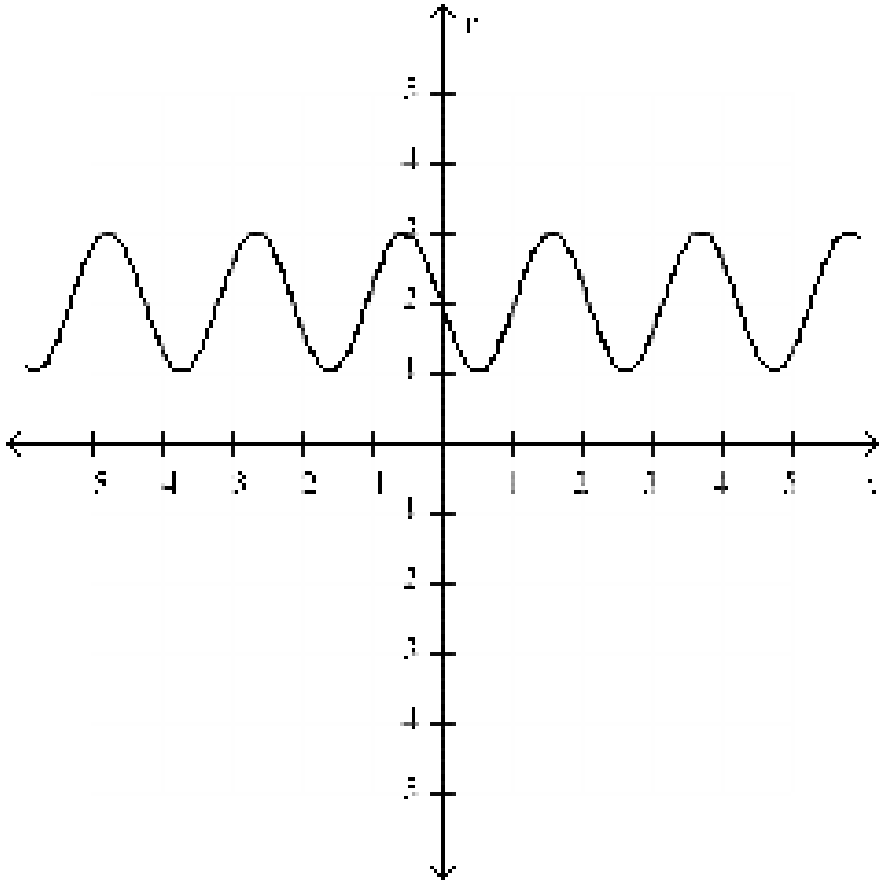
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

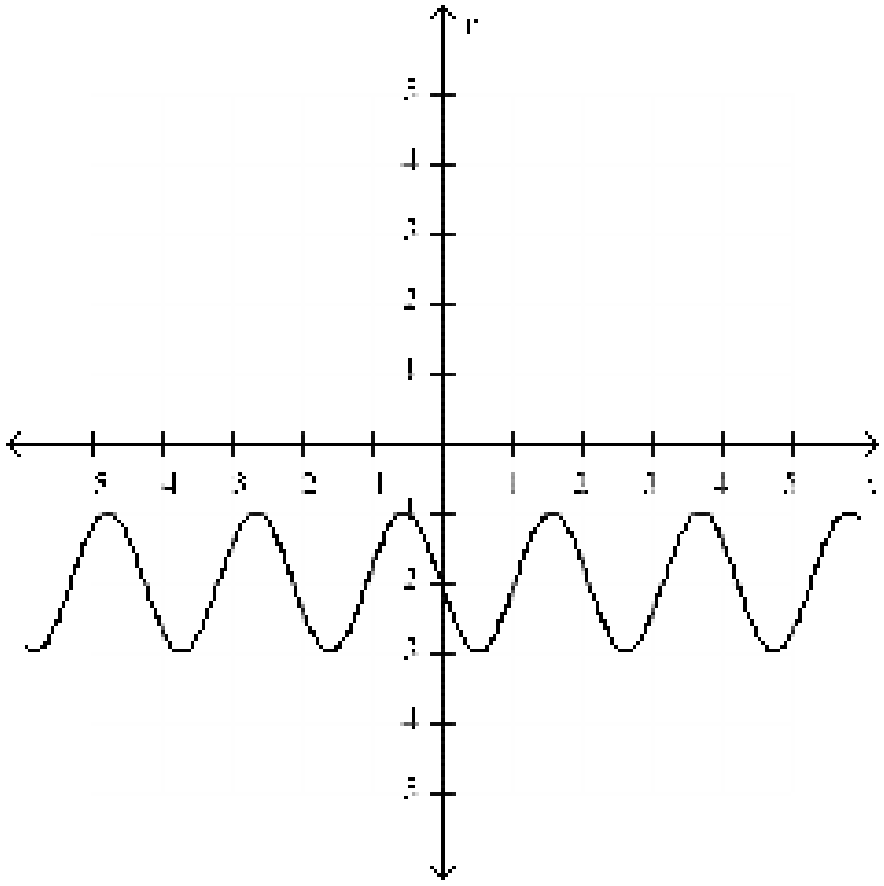
281. Which graph represents the sinusoidal function –? Recovered graph functions: $a*sin((−3)*(x-$pi$/(4)))-(2); a*sin((−3)*(x+pi/(4)))+(2); a*sin((−3)*(x+pi/(2)))+(2); -a*sin((−3)*(x+pi/(4)))+(2)$. Rendered ExamView reference:

$$y = \cos\left(3\left(x + \frac{\pi}{6}\right)\right) - 2$$









(A) graph of a*sin((−3)*(x-pi/(4)))-(2)

(B) graph of a*sin((−3)*(x+pi/(4)))+(2)

(C) graph of a*sin((−3)*(x+pi/(2)))+(2)

(D) graph of -a*sin((−3)*(x+pi/(4)))+(2)

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

282. Give an equation for a transformed sine function with an amplitude of $5/8, a$ period of $4/5, a$ phase shift of $3/2$ rad to the right, and a vertical translation of 9 units down.

(A) – 9

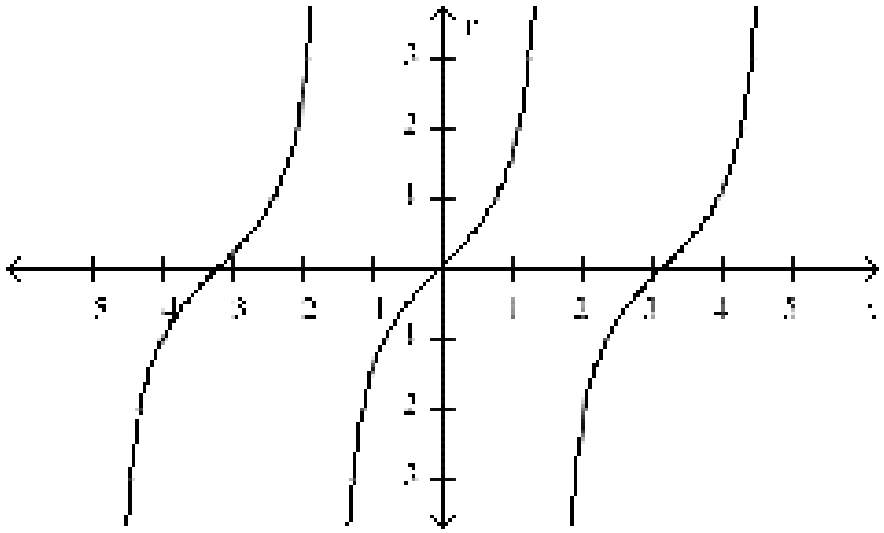
(B) $5/8$

(C) $5/2$

(D) – 9

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

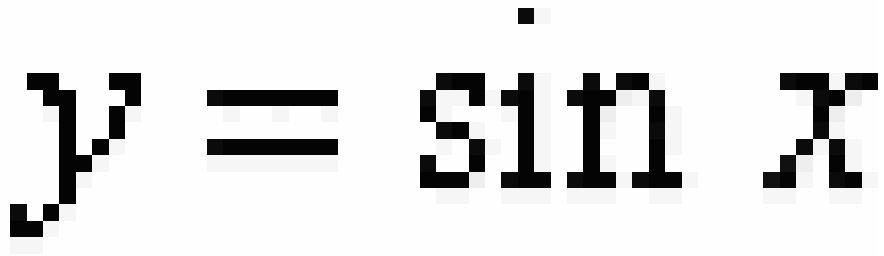
283. Which function represents the graph shown, where x is in radians? $tan(x)$ Rendered ExamView reference:



$y = \tan x$

$y = -\tan x$

$y = \cos x$



(A) graph of tan(x)

(B) graph of -tan(x)

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

284. Which of the following is not an asymptote of the function $−7/2$?

(A) $−7/2$ (B) $−5/2$ (C) $−9/2$ (D) –

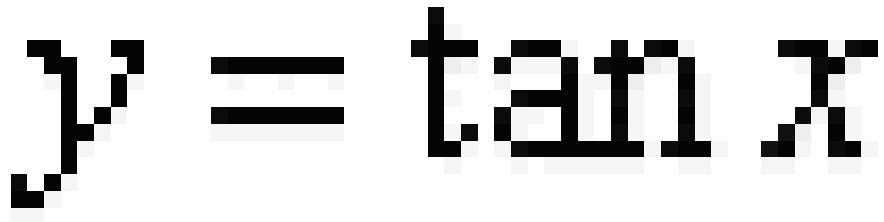
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

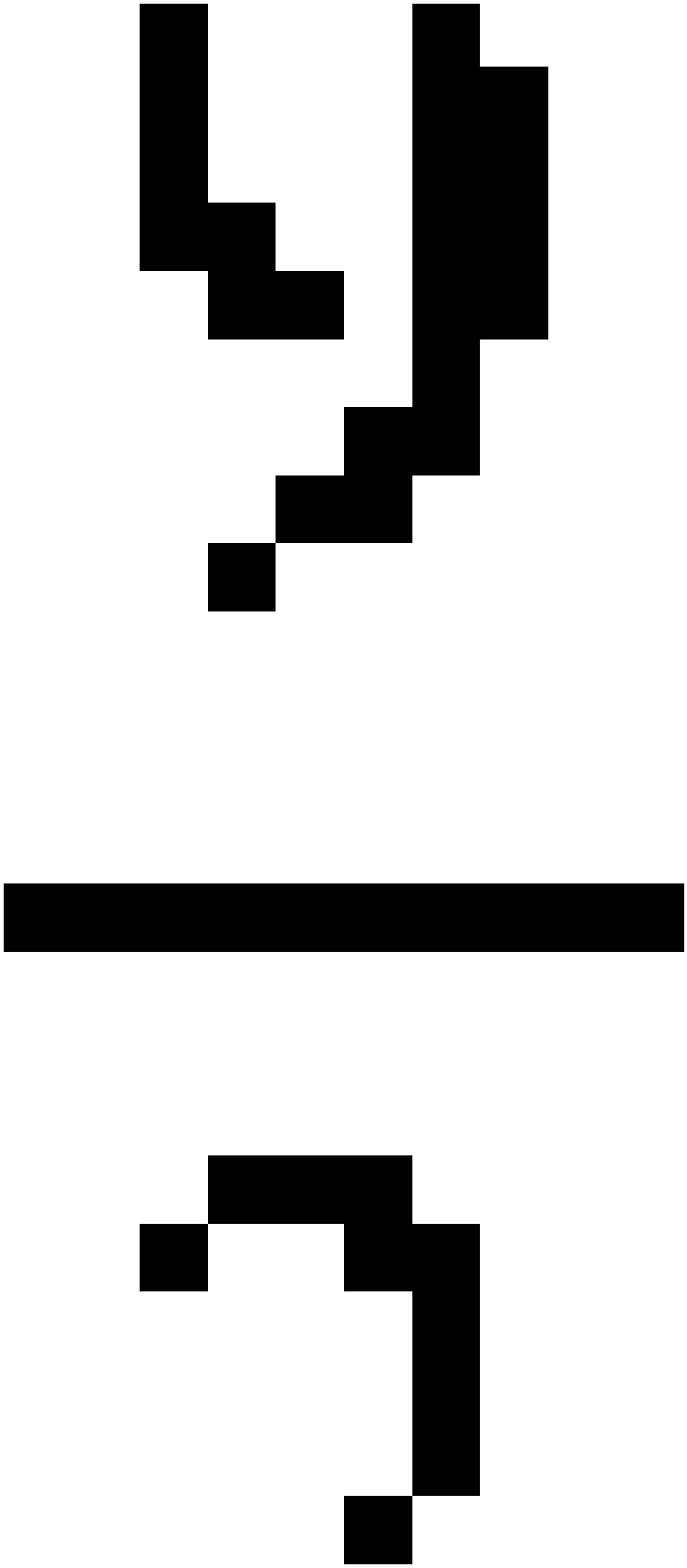
285. Which function has zeros only at $2/3$?

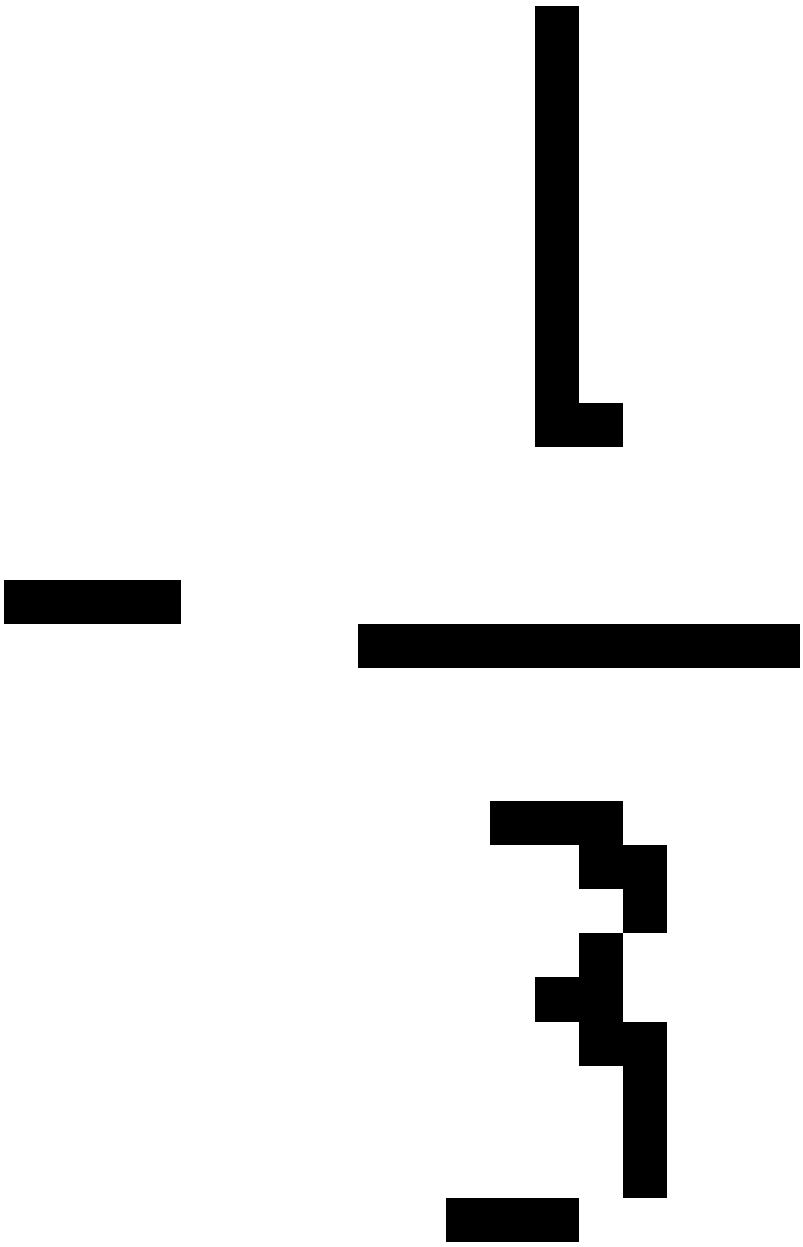
(A) $2/3$ (B) $−7/6$ (C) – (D) $5/4$

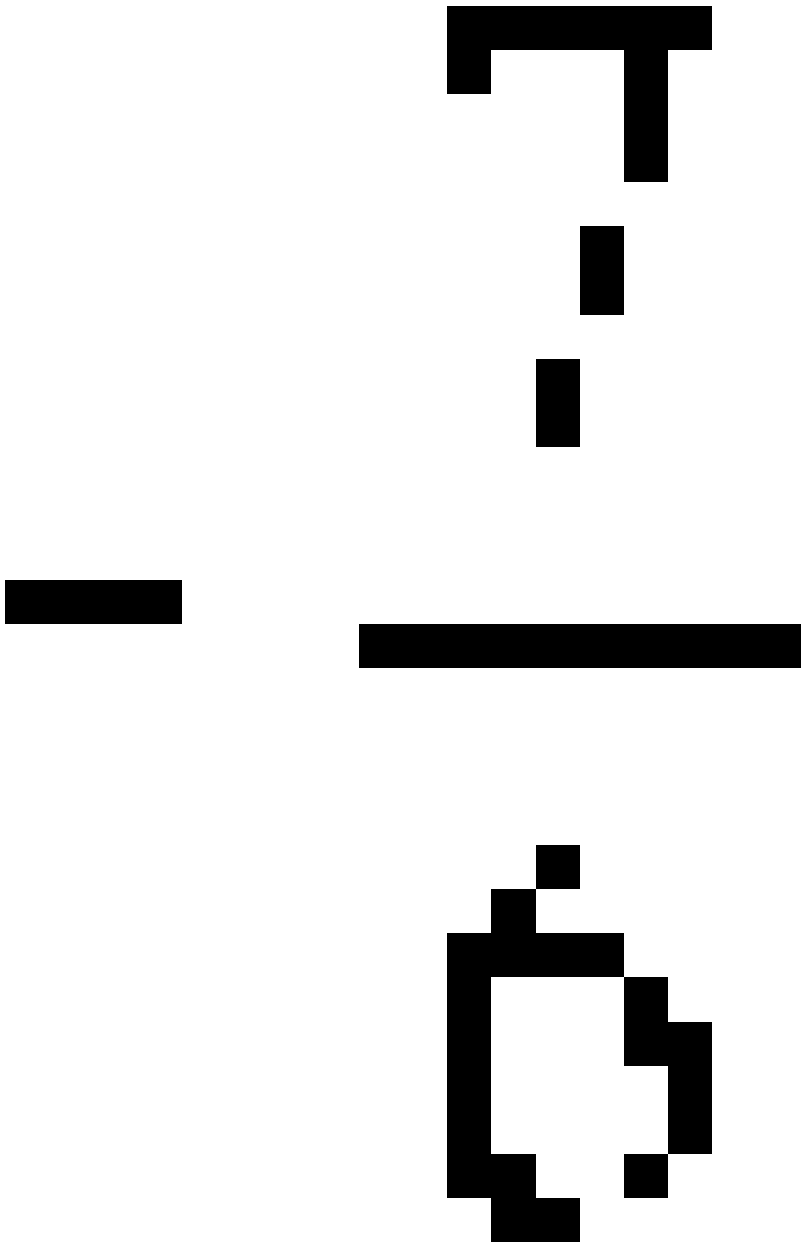
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

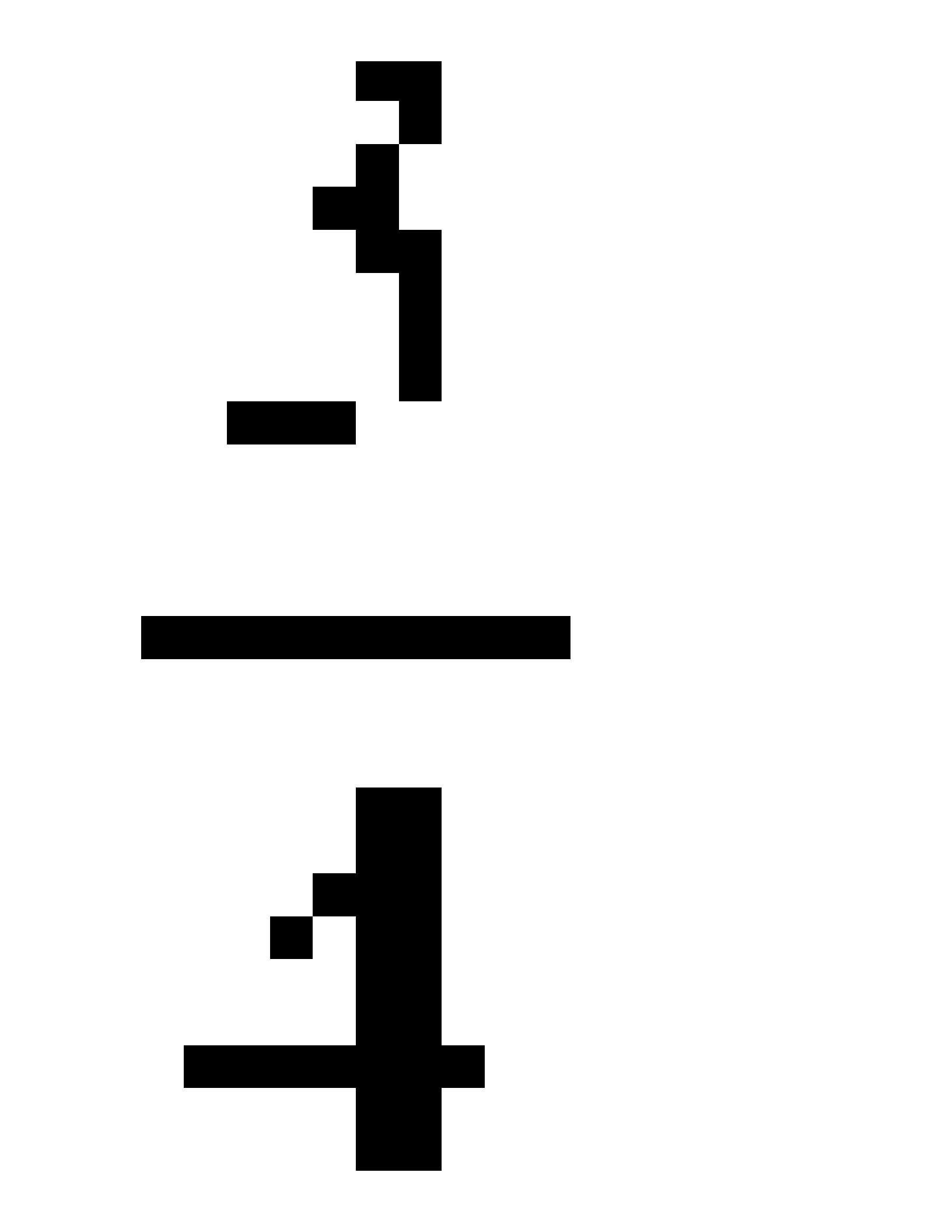
286. Given the tangent function expression shown for parameter $n = 9$, which is the $x-$coordinate at which the function is undefined? Rendered ExamView reference:











(A) $9/2$ (B) $−1/3$ (C) $−7/6$ (D) $3/4$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

287. Given the trigonometric function –$7,$ find the value of the $y-$coordinate of the point with $x-$coordinate $y = tan(x)$

(A) $−1125$ (B) $−1215$ (C) $−1170$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

288. What are the solutions for $1/2 45 = 0$ in the interval 225?

(A) 1 (B) 60 (C) 240 (D) 45

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

289. What are the solutions for $1/2 90 = 0$ in the interval 270?

(A) 30 (B) 210 (C) 315 (D) 1

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

290. What is the radius of the Ferris wheel?

(A) 9 m (B) 18 m (C) 36 m (D) 27 m

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

291. How long does it take for the wheel to revolve once?

(A) 80 s (B) 320 s (C) 160 s (D) 40 s

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

292. What is the minimum height of a car?

(A) 0 m (B) 9 m (C) 18 m (D) 31 m

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

293. What is the maximum height of a car?

(A) 9 m (B) 15.5 m (C) 40 m (D) 31 m

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

294. What is the piston s minimum height?

(A) 7 cm (B) −7 cm (C) 0 cm (D) 14 cm

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

295. What is the period?

(A) $7/40$ (B) 8 (C) $1/14$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

296. Solve 6 to three decimal places on the interval $+ 2$.

(A) 0.340 (B) 5.943 (C) 1.231
(D) 1.231 (E) 5.052

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

297. Solve $+ 3,$ to the nearest tenth of a degree, if necessary, on the interval –5.

(A) x = + 3, x = 53.1 (B) x = 323.1, x = 126.9 (C) x = 126.9, x = 36.9 (D) x = 233.1, x = 53.1

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

298. For the function cos, d affects the _____ of the graph. Rendered ExamView reference:

$$y = a\cos b(x - c) + d$$

ANS: vertical displacement **TOP:** bnk | problem **TYPE:** problem

299. The period of the basic sine function is _____.

ANS: 2p **TOP:** bnk | problem **TYPE:** problem

300. The maximum value of the function 3 is _____.

ANS: –6 **TOP:** bnk | problem **TYPE:** problem

301. The period of the function –8 is _____.

ANS: 6 **TOP:** bnk | problem **TYPE:** problem

302. The phase shift of the function –8 is _____ unit(s) to the _____.

ANS: 7 **TOP:** bnk | problem **TYPE:** problem

303. $y = a 1 sin b 1^(x) + pi c 1 d$

TOP: bnk | short_answer **TYPE:** short_answer

304. $y = a 2 sin b 2^(x) - pi c 2 d$

TOP: bnk | short_answer **TYPE:** short_answer

305. $y = a 3 cos b 3^(x) - pi c 3 d$

TOP: bnk | short_answer **TYPE:** short_answer

306. $y = a 4 cos b 4^(x) + pi c 4 d$

TOP: bnk | short_answer **TYPE:** short_answer

307. Explain how you could graph the function $y = cos(x)$ given a table of values containing ordered pairs for the function $y = sin(x)$. Rendered ExamView reference:

$$y = \cos x$$

$$y = \sin x$$

TOP: bnk | short_answer **TYPE:** short_answer

308. When a ball is thrown, the horizontal distance, d, in metres, travelled can be modelled by the equation $d = v^2 g sin(2 deg) theta$, where $g = 9.8 m / s^2$; v is the velocity with which the ball is thrown, in metres per second; and is the angle of elevation, in radians. If a ball is thrown at $30 m/s,$ what angle of elevation is required for the ball to travel $30 m$?

TOP: bnk | short_answer **TYPE:** short_answer

309. Solve $sin^2(x) - 5cos(x) - 2 = 0$ on the interval

$$6\sin^2 x - 5\cos x - 2 = 0$$

. Rendered ExamView reference:

$$x \in [0, 2\pi]$$

TOP: bnk | short_answer **TYPE:** short_answer

310. A Ferris wheel of diameter $18.5 m$ rotates at a rate of 0.2 rad$/s$. If passengers board the lowest car at a height of $3 m$ above the ground, determine a sinusoidal function that models the height, h, in metres, of the car relative to the ground as a function of the time, t, in seconds.

TOP: bnk | short_answer **TYPE:** short_answer

311. The water level at an ocean inlet has a depth, d, in metres, that varies with the time, t, in hours after midnight, according to the equation $d = 5sin^2(pi) t - 4 12.4 + 6$. What is the water depth at $2:30 a$.m., to the nearest hundredth of a metre?

TOP: bnk | short_answer **TYPE:** short_answer

312. Describe the transformations that, when applied to the graph of y $= cos x,$ result in the graph of

$$y = -2\cos\left[\frac{1}{8}\left(x - \frac{\pi}{3}\right)\right] + 1$$

.

TOP: bnk | short_answer **TYPE:** short_answer

313. Sketch the graph of $tan(x)$ for two cycles and state the domain, range, period, and equations of the asymptotes. x is measured in radians. Rendered ExamView reference:

$$y = \tan x$$

TOP: bnk | problem **TYPE:** problem

314. A pebble is embedded in the tread of a rotating bicycle wheel of diameter 60 cm. If the wheel rotates at 4 revolutions per second, determine a relationship between the height, h, in centimetres, of the pebble above the ground as a function of time, t, in seconds.

TOP: bnk | short_answer **TYPE:** short_answer

315. A population, p, of bears varies according to $p(t) = 250 + 30 cos(t)$, where t is the time, in years, and angles are measured in radians.

(a) What are the maximum and minimum populations?
(b) What is the first interval, in years and months, over which the population is increasing? [DIAGRAM NEEDED]

TOP: bnk | problem **TYPE:** problem

316. A girl jumps rope such that the height, h, in metres, of the middle of the rope can be approximated by the equation $h = 0.7sin 72t + 9 + 0.75$, where t is the time, in seconds.

(a) What is the amplitude of this function?
(b) How many revolutions of the rope does the girl make in $1 min$?

TOP: bnk | problem **TYPE:** problem

317. Determine the number of solutions for the trigonometric equation $x = 0$ on the interval $x = x$.

TOP: bnk | short_answer **TYPE:** short_answer

318. When a pendulum that is $0.5 m$ long swings back and forth, its angular displacement„ in radians, from rest position is given by $theta = 1 / 4 sin pi / 2 t$, where t is the time, in seconds. At what time(s) during the first $4 s$ is the pendulum displaced 1 cm vertically above its rest position? (Assume the pendulum is at its rest position at t $= 0$.) [image omitted: asset-1]

TOP: bnk | short_answer **TYPE:** short_answer

319. Sketch the graph of .$5*sin($pi$*(x-1))+3$ for two cycles, where angles are in radians. Rendered ExamView reference:

$$y = \frac{1}{2} \sin\left[\pi(x-1)\right] + 3$$

TOP: bnk | problem **TYPE:** problem

320. Sketch the graph of $3*cos($.$5*x-2*$pi)$−1$ for [Graph diagram: Graph: $3*cos(.5*x-2*pi)−1$ | x: [$−10.0, 10.0$], y: [$−4.0, 2.0$]]. Rendered ExamView reference:

$$y = 3\cos\left(\frac{1}{2}x - 2\pi\right) - 1$$

$$-2\pi \leq x \leq 2\pi$$

TOP: bnk | problem **TYPE:** problem

321. The period is $12 h 24 min,$ or $12.4 h$. $b = pi / 6.2$ An equation that models the depth of the water is $d = 4cos pi / 6.2 t$.
At $10:00 a$.m., t $= 2$. Substitute t $= 2$ into the equation. $d = 4cos pi / 6.2 t = 4cos pi / 6.2 (2) 2.1$ The depth of the water at $10 a$.m. is $2.1 m$.

TOP: bnk | short_answer **TYPE:** short_answer

322. A mass is suspended by a spring such that it hangs at rest $0.5 m$ above the ground. The mass is raised 40 cm and released at time t $= 0 s,$ causing it to oscillate sinusoidally. If the mass returns to the high position every $1.2 s,$ determine the height of the mass above the ground at t $= 0.7 s$.

TOP: bnk | short_answer **TYPE:** short_answer

323. A windmill has blades that are $20 m$ in length, and the centre of their circular motion is a point $23 m$ above the ground. The blades have a frequency of 4 revolutions per minute when in operation.

(a) Use a sinusoidal function to model the height above the ground of the tip of one blade as a function of time.
(b) Graph the function over three complete cycles.
(c) How far above the ground is the tip of the blade after $10 s$?

TOP: bnk | problem **TYPE:** problem

324. The table shows the hours of daylight measured on the first day of each month, over a $1-$year period in a northern Ontario city. Hours of Daylight (h:$min) 8:25 9:55 11:35 13:30 15:48 16:15 15:25 14:26 12:35 10 10:39 11 9:01 12 8:00 a)$ Graph the table data.

(a) Use the graph and the table to develop a sinusoidal model to represent the information.
(b) Graph the model on the same set of axes as the data. Comment on the fit.
(c) Use your model to estimate the number of hours of daylight, to the nearest tenth of an hour, on January $15,$ and verify the solution using the graph.

TOP: bnk | problem **TYPE:** problem

325. The table shows the fraction of the Moon that can be seen at midnight from Simone s town. Day 1 represents January 1. $10 14 19 21$ Fraction Visible $0.25 0.17 0.12 0.06 0.02 0.00 0.10 0.56 0.98 1.00 24 30 35 41 45 51 56 60 65 66 0.78 0.33 0.15 0.65 0.30 0.01 a)$ What is the period of the sine function that could be used to model the data?

(a) What is the amplitude of the function?
(b) What is the phase shift of the function?
(c) What is the vertical shift?
(d) Use your answers to parts a) to d) to write an equation for the function.
(e) Use your function to determine the fraction of the moon visible to Simone on day i) 100

(i) 150
(ii) 200

TOP: bnk | problem **TYPE:** problem

326. The air quality index, I, in a large city can be modelled by the equation $I = 32 sin$, where t represents the time, in hours, after midnight. New legislation has been introduced that is expected to decrease the pollution levels in the city so that the index values will decrease to 90% of current values in 10 years.

(a) What are the current minimum and maximum values of the index in the city?
(b) At what time of the day is the air quality index at a maximum?
(c) If an air quality alert is issued for times when the index is above $48,$ during what time period will an air quality alert be issued?
(d) What factors in the equation for air quality index will be affected by the legislation? Explain how they will be affected.
(e) What will the new air quality index equation be based on your answers in part d)?
(f) How will these changes affect the times when an air quality alert will be issued?

TOP: bnk | problem **TYPE:** problem

327. The table shows the number of cars sold by George each month for a year. Note that month 1 corresponds to January. Number of New Cars Sold $20 30 37 39 36 19 10 11 12 a)$ Use the table to determine a function that can be used to model the information.

(a) Over what domain and range is this model valid?
(b) The value for the second month is missing. Use your model to determine how many cars George sold in that month.
(c) Do you expect the data to continue to be sinusoidal? Explain.

TOP: bnk | problem **TYPE:** problem

328. The flapping of a bird's wing can be modelled by the function $y = 5 sin(7200t) + 5$, where y represents the distance the tip of the wing travels, in centimetres, and t represents the time, in seconds. a) Determine the period of the motion of the wing. b) Determine the amplitude, the minimum value, and the maximum value. c) What are the first times after $t = 0$ that the tip of the wing reaches the minimum and maximum values? d) Determine the position of the wing tip at i) $1/60$ s ii) $1/20$ s iii) $1/10$ s.

TOP: bnk | problem **TYPE:** problem

329. Wilson places a measuring tape on a pillar of a dock to record the water level in his local coastal community. He finds that a high tide of $1.77 m$ occurs at $5:17 a$.m., and a low tide of $0.21 m$ occurs at $11:38 a$.m.

(a) Estimate the period of the fluctuation of the water level.
(b) Estimate the amplitude of the pattern.
(c) Predict when the next two high tides will occur.
(d) Predict when the next two low tides will occur.

TOP: bnk | problem **TYPE:** problem

330. Consider the function y $= sin x$ – $cos x$.

(a) Predict the $y-$intercept of the function.
(b) Predict the $x-$intercepts from 0 to 360.
(c) Explain why your answers in part b) can be found by solving $tan x = 1$.
(d) Graph the function for the interval $sin($pi$*x/180)-cos($pi$*x/180)$.

TOP: bnk | problem **TYPE:** problem

331. Consider the graph of $tan(x*$pi$/180)$. a) Describe the transformations from the function $2*tan(pi*(x-45)/180)+3$.

(a) Graph the two functions on the same set of axes over the interval [Graph diagram: Graph: $tan(x*pi/180)$ | x: [$ −10.0, 10.0$], y: [$−0.2, 0.2$]].
(b) Given that [Graph diagram: Graph: $2*tan(pi*(x-45)/180)+3$ | x: [$−10.0, 10.0$], y: [$0.1, 1.6$]], explain where the function $2*tan(pi*(x-45)/180)+3$ has vertical asymptotes.

TOP: bnk | problem **TYPE:** problem

332. The graph of $2*cos($pi$*(x+60)/180)−2$ is transformed so that the amplitude becomes 2 and the $x-$intercepts coincide with the maximum values. a) What is the equation of the transformed function?

(a) What phase shift of the transformed function will produce a $y-$intercept of –1?
(b) What is the equation of the function after the transformation in part b)?
(c) Verify your solution to part c) by graphing.

TOP: bnk | problem **TYPE:** problem

333. A cosine function has half the period of the function $cos(2*$pi$*x/180)$. All other properties of the two functions are identical.

(a) Determine the number of points of intersection if these two functions are graphed on the interval [Graph diagram: Graph: $cos(2*pi*x/180)$ | x: [$−10.0, 10.0$], y: [$0.9, 1.0$]]. Explain.
(b) Determine the values of x of the first two points of intersection to the right of x $= 0$.
(c) If the graph of the two functions were extended to $720,$ how many points of intersection would there be? Explain.
(d) Use the concept of coterminal angles to determine the values of x where the points of intersection would occur.
(e) Graph the two functions over the interval [Graph diagram: Graph: $cos(pi*x/180)$ | x: [$−10.0, 10.0$], y: [$1.0, 1.0$]] to verify your answer for parts b), c), and d).

TOP: bnk | problem **TYPE:** problem

334. A sinusoidal function has an amplitude of $2, a$ period of $180,$ and a maximum at ($0, 4)$.

(a) Represent this function with an equation using a sine function.
(b) Represent this function with an equation using a cosine function.
(c) Explain how these two functions are related.

TOP: bnk | problem **TYPE:** problem

335. Consider the function $f(x) = 1/2 sin(3(x - 30 deg)) + 4$. a) Determine the amplitude, the period, the phase shift, and the vertical shift of the function with respect to $y = sin x$. b) What are the minimum and maximum values of the function? c) Determine the $x-$intercepts of the function in the interval $0 deg <= x <= 360 deg$. d) Determine the $y-$intercept of the function.

TOP: bnk | problem **TYPE:** problem

336. Use the following information to answer the questions. The height, h, in metres, above the ground of a car as a Ferris wheel rotates can be modelled by the function $18,$ where t is the time, in seconds.

TOP: bnk | problem **TYPE:** problem

337. Which equation is a reciprocal identity? (fn$1 = sec)$

(A) tan (B) csc (C) sin

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

338. Which equation is a quotient identity? (fn$1 = cot)$

(A) sin (B) cos (C) cot (D) sin

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

339. Which equation is a Pythagorean identity?

(A) $a = 5 m$ (B) $sin^2(theta) + cos^2(theta) = 1$ (C) $tan^2(theta) = sec^2(theta)$ (D) $cot^2(theta) + 1 = csc^2(theta)$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

340. What does the expression $cot(theta)$ $cot^2(theta)$ $tan^2(theta)$ $tan(theta)$ $sin(theta)$ $cos(theta)$ simplify to?

(A) $csc^2(theta)$ (B) $sec^2(theta)$ (C) $tan^2(theta)$ (D) $sin(theta)$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

341. Which of the following sets is the set of reciprocal trigonometric ratios?

(A) csc x, sec x, and cot x (B) sec x, cos x, and cot x (C) sin x, csc x, and cot x (D) csc x, sec x, and tan x

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

342. What is an equivalent expression for $0 1 0$?

(A) $cos(A)$ (B) $sin(A)$ (C) $tan(A)$ (D) $csc(A)$

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

343. Which expression is equivalent to $1 + tan^2 theta$?

(A) $csc^2 theta$ (B) $tan^2 theta$ (C) $sec^2 theta$ (D) $1 - tan^2 theta$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

344. Which expression is equivalent to $1 + tan^2 theta$?

(A) $csc^2 theta$ (B) $tan^2 theta$ (C) $sec^2 theta$ (D) $1 - tan^2 theta$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

345. Which expression is equivalent to sec?

(A) sin (B) tan (C) cos (D) sin

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

346. Which expression is equivalent to tan?

(A) sin (B) tan (C) sin (D) cos

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

347. Which expression is equivalent to $1 + tan^2 theta$?

(A) $sec^2 theta$ (B) $csc^2 theta$ (C) $tan^2 theta$ (D) $1 - tan^2 theta$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

348. Which expression is equivalent to sin?

(A) cos (B) cos (C) cos (D) sin

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

349. Express $cos(a) - sec(a) sec(a) + cos^2(a) tan^2(a) tan(a) - sin(a) tan(a)$ in terms of cosine and sine only, and simplify.

(A) 0 (B) $cos^2(a)$ (C) 1 (D) $sin^2(a)$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

350. Which expression is equivalent to cos?

(A) if(num=1,chr(43),chr(45)) (B) sin (C) sin (D) cos

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

351. Which expression is equivalent to if(num$=1,$chr($45),$chr(43))?

(A) if(num=1,chr(43),chr(45)) (B) if(num=1,chr(45),chr(43)) (C) if(num=1,chr(43),chr(45)) (D) if(num=1,chr(45),chr(43))

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

352. Determine the exact value of $cos(15 deg)$.

(A) $frac(sqrt(6) - sqrt(2), 4)$ (B) $frac(sqrt(2) - sqrt(6), 4)$ (C) $frac(-sqrt(6) - sqrt(2), 4)$ (D) $frac(sqrt(6) + sqrt(2), 4)$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

353. Simplify 19.

(A) 19 (B) 161 (C) 19 (D) 161

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

354. Simplify 168. Round your answer to the nearest hundredth.

(A) 0.47 (B) $−0.75$ (C) $−1.15$ (D) 0.42

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

355. Simplify 116.

(A) 116 (B) $−26$ (C) 116 (D) $−26$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

356. Which expression is equivalent to $cos(x - y)$?

(A) $cos x cos y - sin x sin y$ (B) $cos x cos y + sin x sin y$ (C) $sin x cos y + cos x sin y$ (D) $sin x cos y - cos x sin y$

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

357. Which expression is equivalent to $cos(2x)$?

(A) $2 sin x cos x$ (B) $sin^2 x + cos^2 x$ (C) $1 + tan^2 x$ (D) $cos^2 x - sin^2 x$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

358. $sin(2x)$ is equivalent to

(A) $2 sin x cos x$ (B) $cos^2 x - sin^2 x$ (C) $1 - 2 sin^2 x$ (D) $2 cos^2 x - 1$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

359. $sin(x + y)$ can be rewritten as

(A) $sin x cos y + cos x sin y$ (B) $sin x cos y - cos x sin y$ (C) $cos x cos y - sin x sin y$ (D) $cos x cos y + sin x sin y$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

360. $sin^2(x / 2)$ can be rewritten as

(A) $frac(1 + cos x, 2)$ (B) $frac(1 - sin x, 2)$ (C) $frac(1 + sin x, 2)$ (D) $frac(1 - cos x, 2)$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

361. What is the solution to the equation $sin(x) + 4csc(x) + 5 = 0$ over the domain $0 <= x <= 2 * pi$?

(A) $pi / 2$ (B) pi (C) 0 (D) $3 pi / 2$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

362. What is the solution to the equation $tan^2(x) + 3sec(x) + 3 = 0$ over the domain $0 x$?

(A) 60 and 240 (B) 120, 180, and 240 (C) no solution (D) 120

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

363. What is the general solution, in degress, to the equation $2x - 2sin(x)sin(2x) = - 1$?

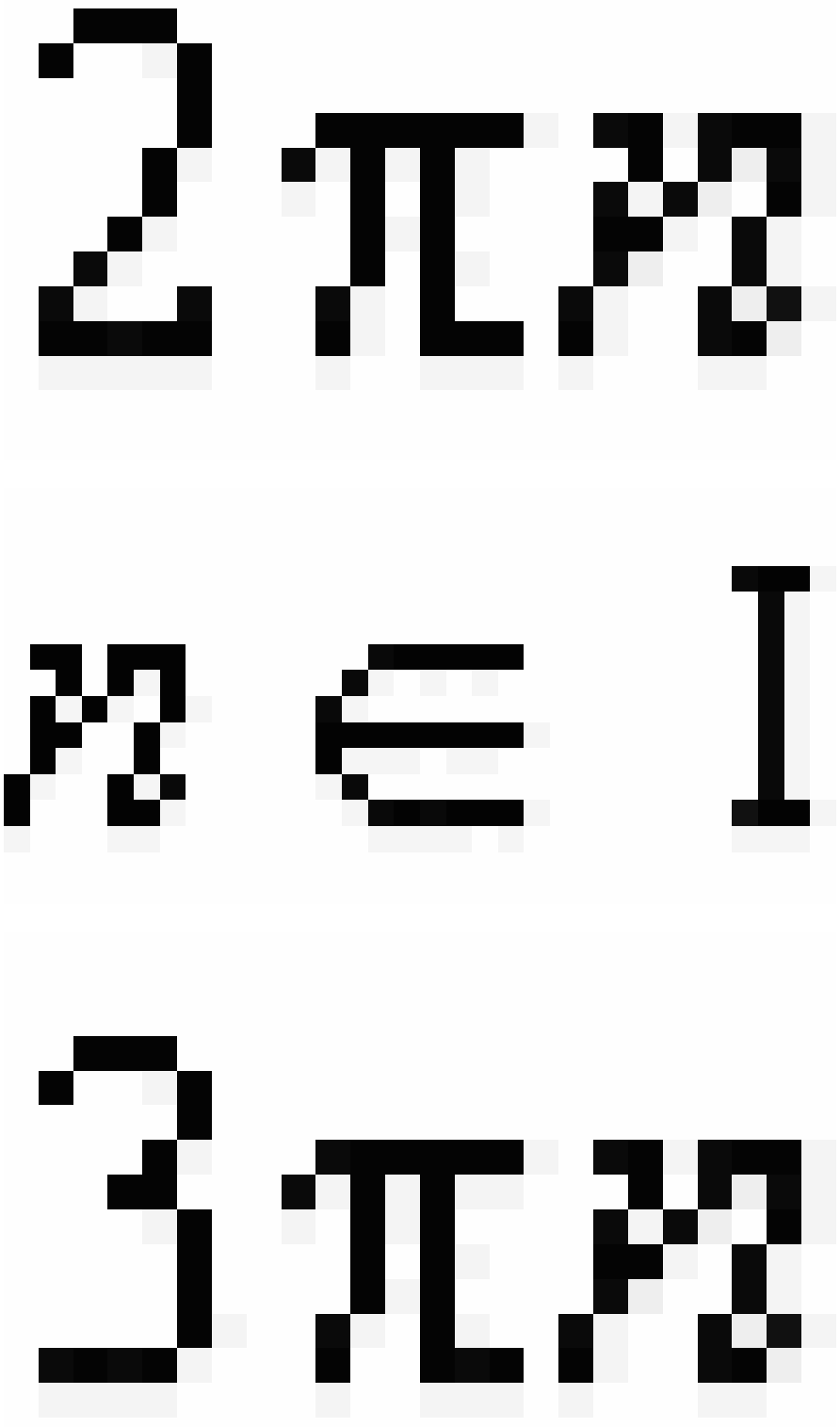
(A) $2 sin x cos x$ (B) $sin^2 x + cos^2 x$ (C) $cos^2 x - sin^2 x$ (D) $1 + tan^2 x$

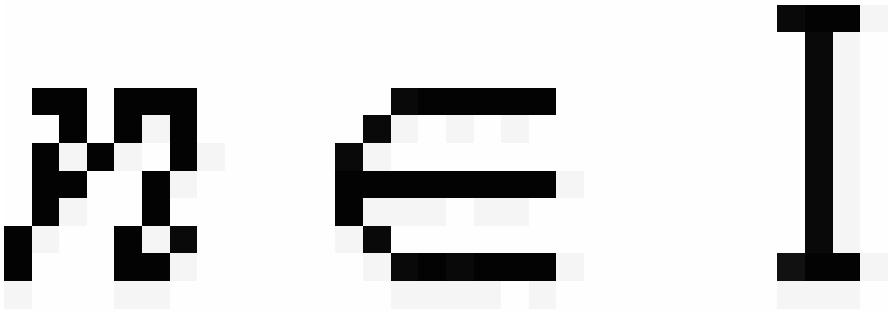
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

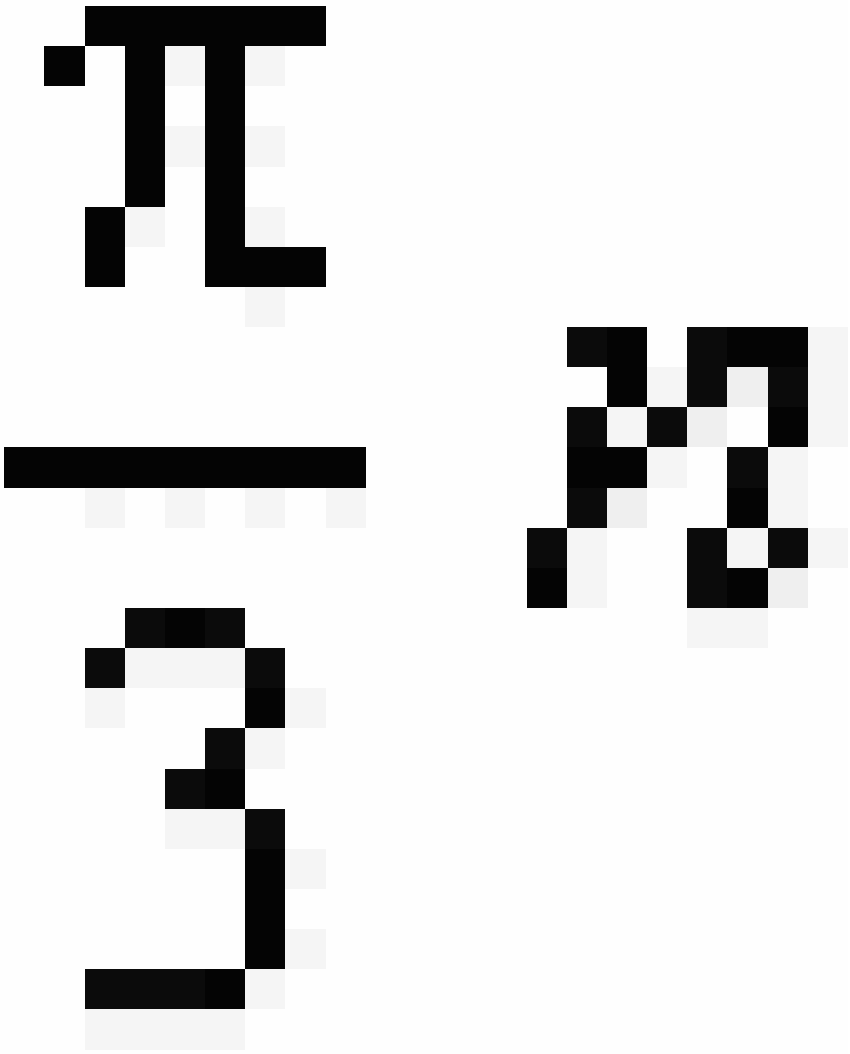
364. What is the general solution, in radians, to the equation

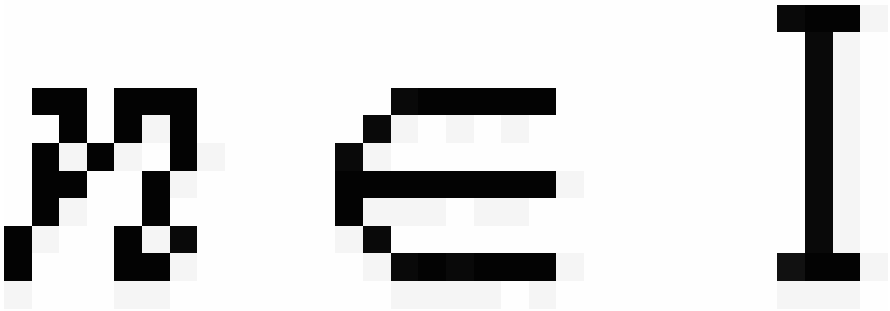
$$(4\cos^2 2\theta + 1)\sin\frac{1}{3}\theta = 0$$

? Rendered ExamView reference:









(A) $2 sin x cos x$ (B) $sin^2 x + cos^2 x$ (C) $cos^2 x - sin^2 x$ (D) $1 + tan^2 x$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

365. What is the general solution, in radians, to the equation $sin(2x) + 2cos(x) = 0$?

(A) $2 sin x cos x$ (B) $sin^2 x + cos^2 x$ (C) $1 + tan^2 x$ (D) $cos^2 x - sin^2 x$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

366. $0 1$

$$\tan x = \frac{\sin x}{\cos x} \text{ and } \cot x = \frac{\cos x}{\sin x}$$

are known as the _____ identities.

TOP: bnk | true_false **TYPE:** true_false

367. An equation with trigonometric expressions that is true for all values of the variable for which the expressions are defined is called a trigonometric _____.

TOP: bnk | true_false **TYPE:** true_false

368. The general solution, in degrees, for the equation $0 0 1$ is _____.

TOP: bnk | true_false **TYPE:** true_false

369. The $1 0 1$ identities are defined as 0 $cos 2 + sin^2(x), 1 + tan^2(x), 1 + cot^2(x)$

reciprocal

Rendered ExamView reference:

$\csc x =$ _____,

$\sec x =$ _____, and

$\cot x =$ _____

TOP: bnk | true_false **TYPE:** true_false

370. A trigonometric identity is a relation among trigonometric ratios that is _____ for all values of the variables for which both sides are _____.

TOP: bnk | true_false **TYPE:** true_false

371. $sin(a) - 1 cos(a)$

TOP: bnk | short_answer **TYPE:** short_answer

372. $csc(a) - 1 1 + csc(a)$

TOP: bnk | short_answer **TYPE:** short_answer

373. $sin(a) + cos(a) tan(b) cos(a) - sin(a) tan(b)$

TOP: bnk | short_answer **TYPE:** short_answer

374. $T = sin(4 deg) a - sin(2 deg) a cos(2 deg) a + cos(4 deg) a$

TOP: bnk | short_answer **TYPE:** short_answer

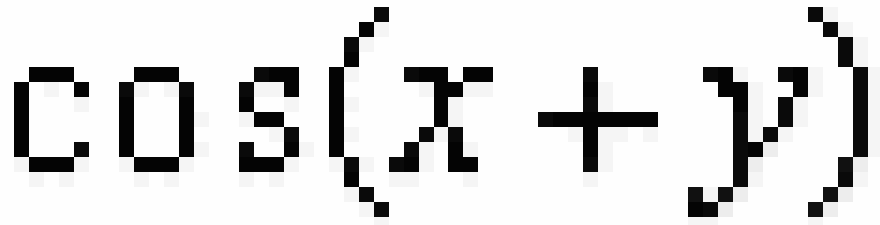
375.

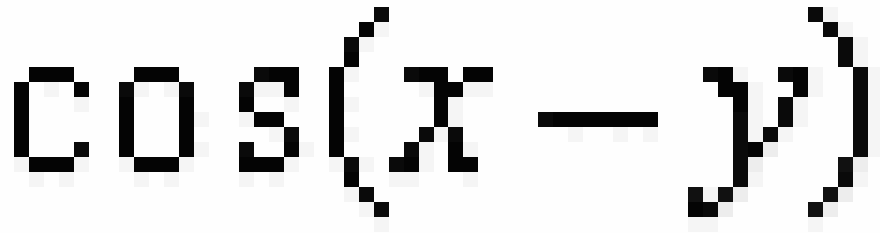
$$\cos x \cos y - \sin x \sin y$$

Rendered ExamView reference:

$$\sin(x + y)$$

$$\sin(x - y)$$





TOP: bnk | short_answer **TYPE:** short_answer

376. Find the exact value of $cos^2(pi) 9 cos pi / 18 + sin^2(pi) 9 sin pi$.

TOP: bnk | short_answer **TYPE:** short_answer

377. Find the exact value of $tan pi / 10 + tan pi / 15 1 - tan pi / 10 tan pi$.

TOP: bnk | short_answer **TYPE:** short_answer

378. Determine the value of $cos(theta) sin(theta) sec(theta) + tan(theta) sec(theta) tan(theta)$ if $sin(theta) = 13$ and $cos(theta) = 84$.

TOP: bnk | short_answer **TYPE:** short_answer

379. Use a counterexample to show that $cos(x + y) = cos(x) + cosy$ is not an identity.

TOP: bnk | problem **TYPE:** problem

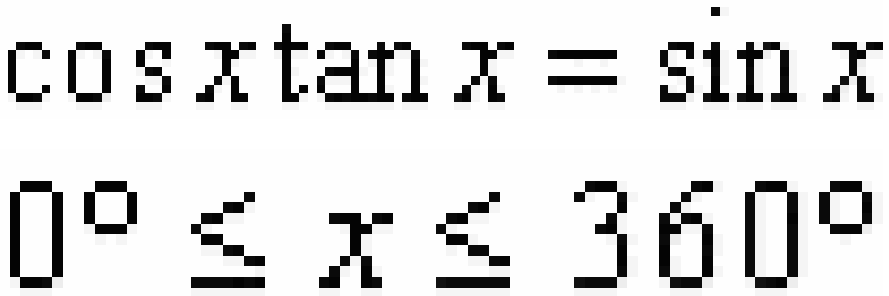
380. Prove the identity $tan^2(theta) - sin^2(theta) = sin^2(theta) tan^2(theta)$.

TOP: bnk | short_answer **TYPE:** short_answer

381. Is the equation $csc^2(theta) = 1 + cot^2(theta)$ true for all values of

TOP: bnk | true_false **TYPE:** true_false

382. Graph P P Graph P P The graphs are the same for the range P. Recovered graph functions: $cos(x*$pi$/180)*tan(x*$pi$/180); sin(x*$pi$/180)$. Rendered ExamView reference:



TOP: bnk | problem **TYPE:** problem

383. What is the solution for $cos(x) - 3 = 0$ for

$$2\cos x - \sqrt{3} = 0$$

? Rendered ExamView reference:

$$0 \leq x \leq 2\pi$$

TOP: bnk | short_answer **TYPE:** short_answer

384. Solve $cot^2(theta) + cot(theta) = 0$. State the solution in general form.

TOP: bnk | short_answer **TYPE:** short_answer

385. Solve $sec^2(theta) - 2tan theta - 3 = 0$. State the general solution to the nearest degree.

TOP: bnk | short_answer **TYPE:** short_answer

386. A skateboard ramp is built with an incline angle of [image omitted: asset-1]. If the base of the ramp is $1 m$ in length, determine the exact height of the ramp.

TOP: bnk | short_answer **TYPE:** short_answer

387. Prove the identity $cos(theta) = sin^2(theta)/(1 - cos(theta))$.

TOP: bnk | short_answer **TYPE:** short_answer

388. Prove the identity $sin pi / 2 - x cot pi / 2 + x = - sin(x)$.

TOP: bnk | short_answer **TYPE:** short_answer

389. Prove the identity $cos(theta) - sin(theta) cos(theta) + sin(theta) = sec(2 deg) theta - tan(2 deg) theta$.

TOP: bnk | short_answer **TYPE:** short_answer

390. Express $sin(x + y + z)$ in terms of cosines and sines of x, y, and z.

TOP: bnk | short_answer **TYPE:** short_answer

391. If $x = k$, for what value(s) of k does

$$\sin x + \cos x = k$$

? Rendered ExamView reference:

$$\sin x \cos x = 1$$

TOP: bnk | short_answer **TYPE:** short_answer

392. Prove the identity $cos(2 deg) theta + sin(2 deg) theta 1 + cos(2 deg) theta + sin(2 deg) theta = tan(theta)$.

TOP: bnk | short_answer **TYPE:** short_answer

393. Prove the identity $cos^2(theta) - sin^2(theta) cos^2(theta) + sin(theta) cos(theta) = 1 - tan(theta)$.

TOP: bnk | short_answer **TYPE:** short_answer

394. Prove the identity $cos(theta) + 1 / 1 - cos(theta) = 2 sin^2(theta)$.

TOP: bnk | short_answer **TYPE:** short_answer

395. Prove the identity

$$\frac{1+\tan\theta}{1+\cot\theta} = \frac{1-\tan\theta}{\cot\theta-1}$$

.

TOP: bnk | short_answer **TYPE:** short_answer

396. An angle satisfies the relation $sqrt($.$2-x^2)$.

(a) Use the definition of the reciprocal trigonometric ratios to express the left side of the relation in terms of the sine and$/$or cosine ratios.
(b) Determine the value(s) for the angle. Do not use a calculator.
(c) Verify your answer to part b) using a calculator.
(d) Show your answer to part b) using a unit circle.

TOP: bnk | problem **TYPE:** problem

397. Use the expressions for the reciprocal trigonometric ratios in terms of x, y, and r to show that $csc^2(A) = 1 + cot^2(A)$ for all values of A.

TOP: bnk | short_answer **TYPE:** short_answer

398. Prove that the equation $csc^2(x) = 1 / 1 + cos(x) + 1 / 1 - cos(x)$ is true for all values of x.

TOP: bnk | true_false **TYPE:** true_false

399. Prove the identity $sin(3 deg) theta + sin(theta) cos(3 deg) theta + cos(theta) = tan(2 deg) theta$.

TOP: bnk | short_answer **TYPE:** short_answer

400. What is the solution to the equation $cos(2x) + 2 = sin(x)$ for $0 x$?

TOP: bnk | short_answer **TYPE:** short_answer

401. Solve $cos(x / 2) + sin(x / 2) = 1$.

TOP: bnk | short_answer **TYPE:** short_answer

402. Solve $sin(3x) + sin(x) = cos(x)$ for $0 <= x <= 2 * pi$.

TOP: bnk | short_answer **TYPE:** short_answer

403. What is the general solution to $tan(x) csc(x) + 2 = 0$?

TOP: bnk | short_answer **TYPE:** short_answer

404. Solve $sin^4(x) + 3sin^2(x) - 1 = 0$ over the domain $0 x$.

TOP: bnk | short_answer **TYPE:** short_answer

405. Solve $sin^2(x) = 0$ over the domain $0 x$.

TOP: bnk | short_answer **TYPE:** short_answer

406. Solve $sin^2(x) = 0$ over the domain

$$4\sin^4 x + 3\sin^2 x - 1 = 0$$

. Rendered ExamView reference:

$$0^\circ \leq x \leq 360^\circ$$

TOP: bnk | short_answer **TYPE:** short_answer

407. Which set of properties does the function 2 have?

(A) no x-intercept, no y-intercept (B) x-intercept is 1, no y-intercept (C) no x-intercept, y-intercept is 1 (D) x-intercept is 0, y-intercept is 0 2 4 7 8

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

408. Which choice best describes the function $6 0$?

(A) 6 (B) 1 (C) increasing (D) decreasing

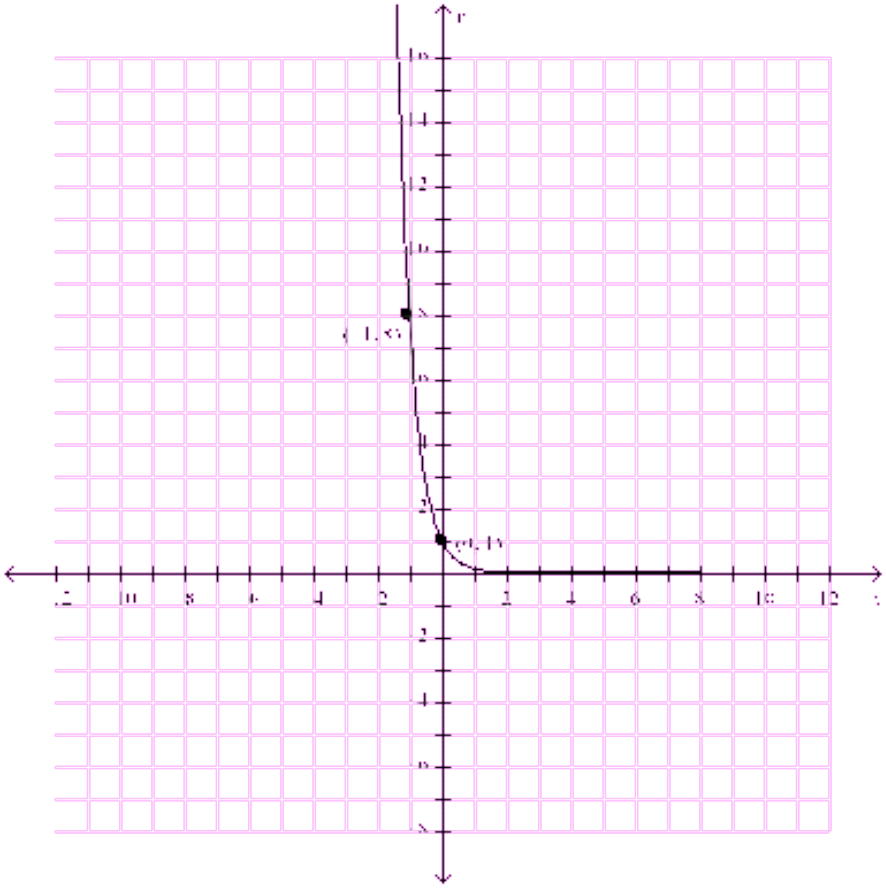
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

409. Which set of properties is correct for the function $9 1$?

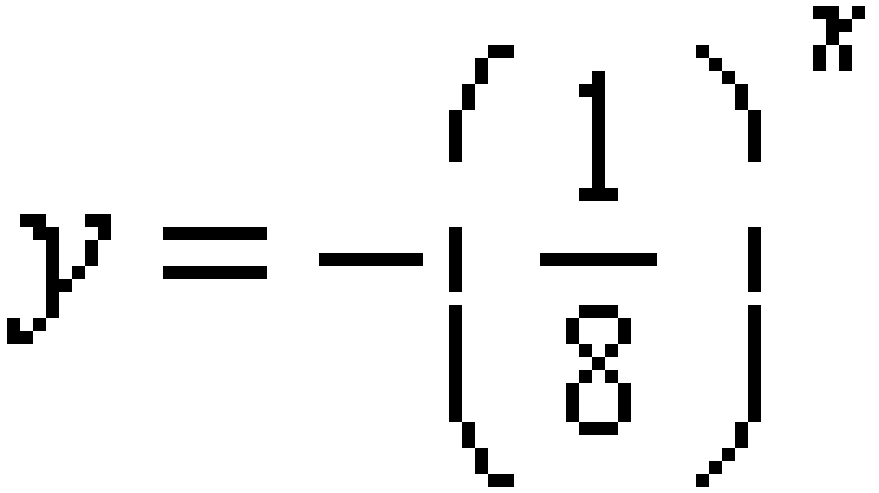
(A) 9 (B) 1 (C) 9 (D) 0

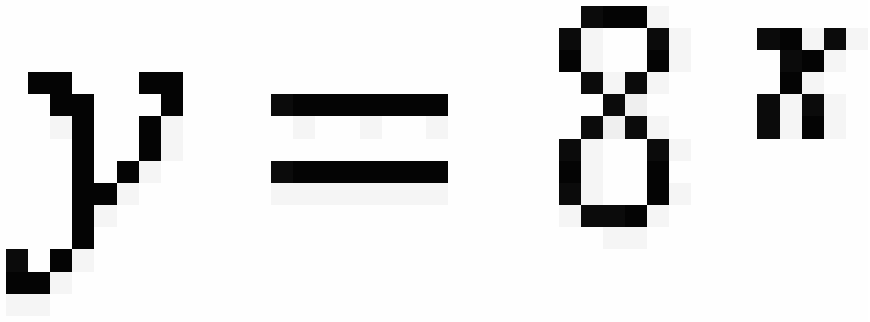
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

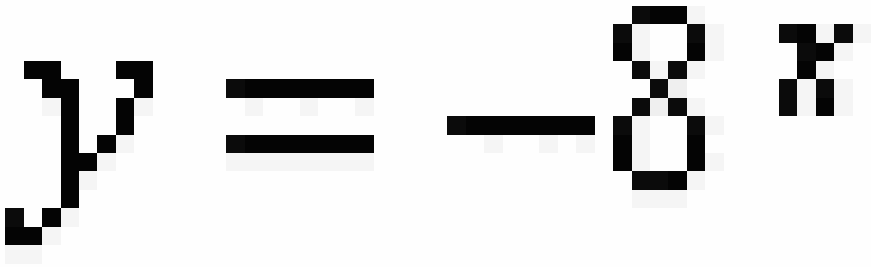
410. Which exponential equation matches the graph shown? 1 Recovered graph functions: (($1)/(8))^x; (4)^x$. Rendered ExamView reference:



$$y = \left(\frac{1}{8}\right)^x$$







(A) 8 (B) 1 (C) 4 (D) 0

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

411. A radioactive sample with an initial mass of 1 mg has a half$-$life of 9 days. What is the equation that models the exponential decay, A, for time, t, in $9 -$day intervals?

(A) $A = 1(1/2)^t$ (B) $A = 1(2)^t$ (C) $A = 9(1/2)^t$ (D) $A = 1(1/2)^(t/9)$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

412. A colony of ants has an initial population of 750 and triples every day. Which function can be used to model the ant population, p, after t days?

(A) 3 (B) 750 (C) 750 (D) 3

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

413. A bacteria colony initially has 1500 cells and doubles every week. Which function can be used to model the population, p, of the colony after t days?

(A) 1500 (B) 2 (C) 1500 (D) 3

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

414. An investment of $ 150 is placed into an account that earns interest, compounded annually, at a rate of 5 % for 12 years. The amount, A, in the account can be modelled by the function $150,$ where t is the time, in years. What is the 5 of this function?

(A) domain 12 (B) 12 150 (C) 150 12 (D) 12 150

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

415. A rumour is being spread in a community that the mayor is going to retire before the next election. Initially, 6 people were told the rumour and each of these people told 6 others the following day. Each of the 6 people told by the initial people told 6 others the day after they were told. If this process keeps up, how many people will be told the rumour on the eighth day?

(A) 1 679 616 (B) 10 077 696 (C) 279 936 (D) 48

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

416. To the nearest year, how long would an investment need to be left in the bank at 5 %, compounded annually, for the investment to triple?

(A) 15 (B) 28 (C) 26 (D) 23

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

417. How much was in the account 10 years ago, to the nearest cent?
[Hint: Use $P = A 1 + i - n$.]

(A) 163.18 (B) 42.86 (C) 30.00

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

418. How much will the car be worth 8 years after it is purchased?

(A) 80 000 (B) 2684.35 (C) 2000 (D) 68 797.07

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

419. The equation 90 can also be written as

(A) 45 (B) 5 (C) 90 (D) 5

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

420. For the exponential function $100,$ which of the following statements is not true?

(A) 9 (B) increasing (C) decreasing (D) greater

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

421. Which of the following transformations maps the function $+ 5$ onto the function 8?

(A) a horizontal shift of 8 units to the 8 and a vertical shift of + 7 units + 7

(B) a horizontal shift of 8 units to the + 5 and a vertical shift of + 7 units 5

(C) a horizontal shift of left units to the 7 and a vertical shift of up units 5

(D) a horizontal shift of right units to the 7 and a vertical shift of down units 5

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

422. Which function results when the graph of 6 is translated 2 units down? Rendered ExamView reference:

$$y = 6^{x}$$

$$y = 6^{x-2}$$

$$y = 6^{x} - 2$$

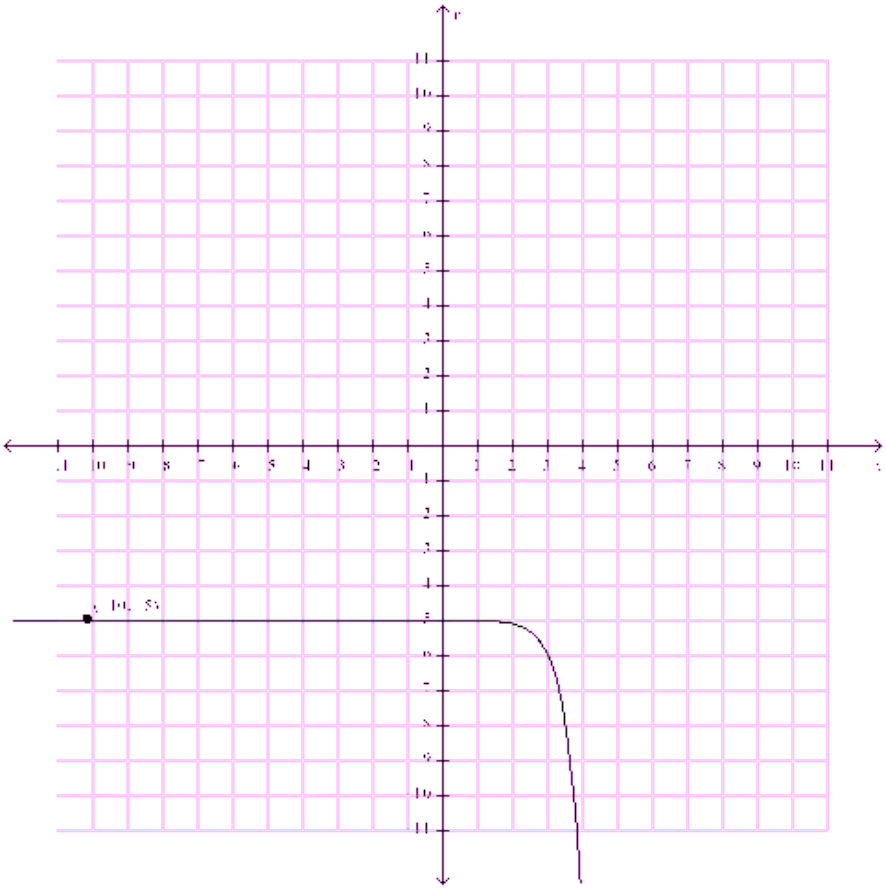
$$y = 6^{x+2}$$

$$y = 6^{x} + 2$$

(A) 6 (B) + 2 (C) 6 (D) + 2

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

423. Which function is represented by the following graph? –10 Rendered ExamView reference:



$$y = -9(9)^{(x+4)} + 5$$

$$y = -9(9)^{(x-4)} - 5$$

$$y = 9(9)^{-(x+4)} + 5$$

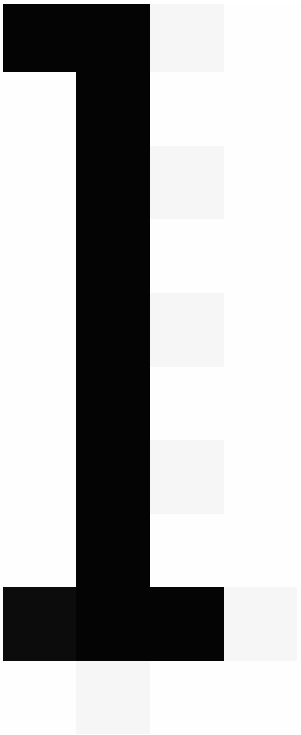
$$y = 9(9)^{-(x-4)} - 5$$

(A) 9 (B) – (C) – 4 (D) – 5

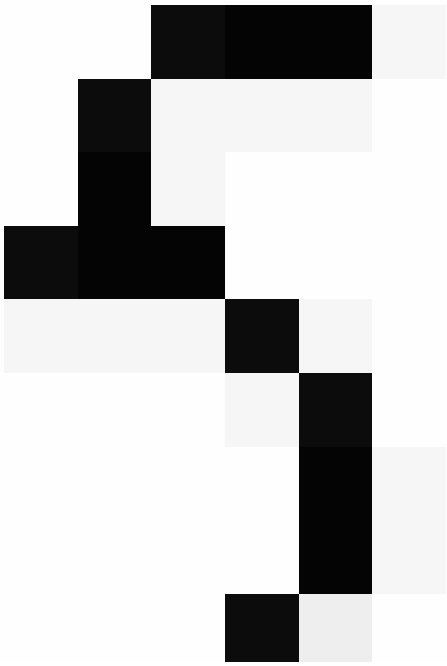
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

424. Which function results when the graph of the function 5 is reflected in the $y-$axis, compressed vertically by a factor of $5,$ and shifted 4 units down? Rendered ExamView reference:

$$y = 4^x$$







$$y = \frac{1}{5}(4)^{-x} - 2$$

$$y = \frac{1}{5}(4)^{x} + 2$$

$$y = \frac{1}{5}(4)^{x} - 2$$

$$y = \frac{1}{5}(4)^{-x} + 2$$

(A) – 2 (B) 5 (C) 4 (D) + 2

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

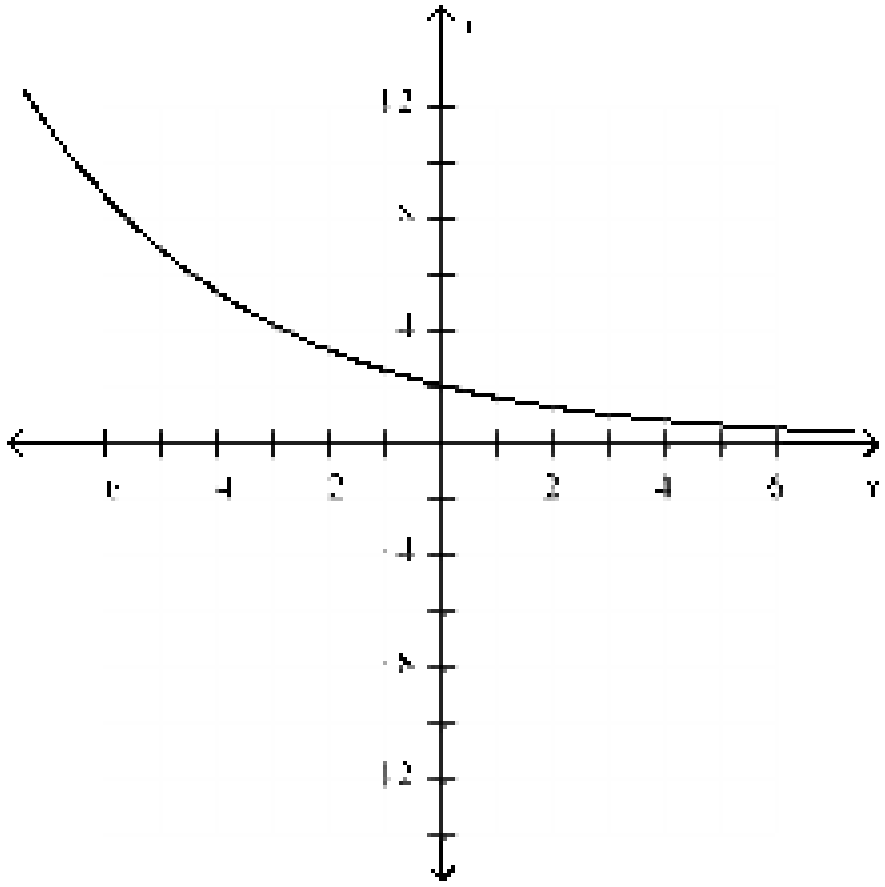
425. What is the exponential equation for the function that results from the transformations listed being applied to the base function 6? a reflection in the $6 -$axis a vertical stretch by a factor of $9 a$ horizontal stretch by a factor of y

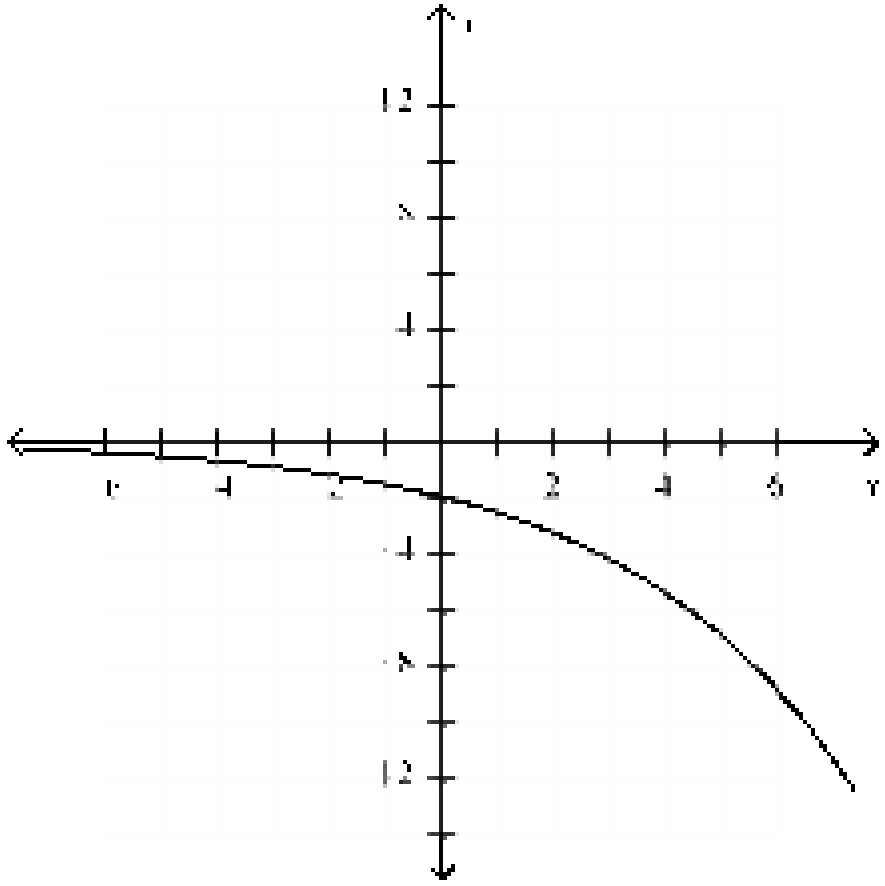
(A) $−6$ (B) 9 (C) if(a>0,1,-1) (D) 7

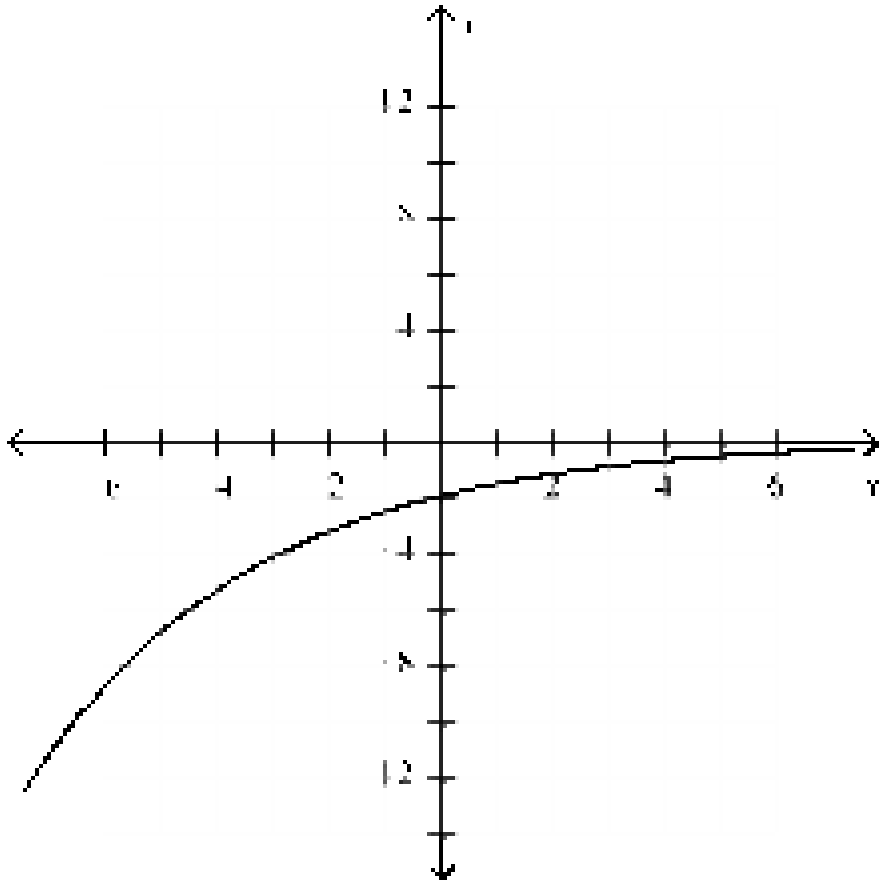
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

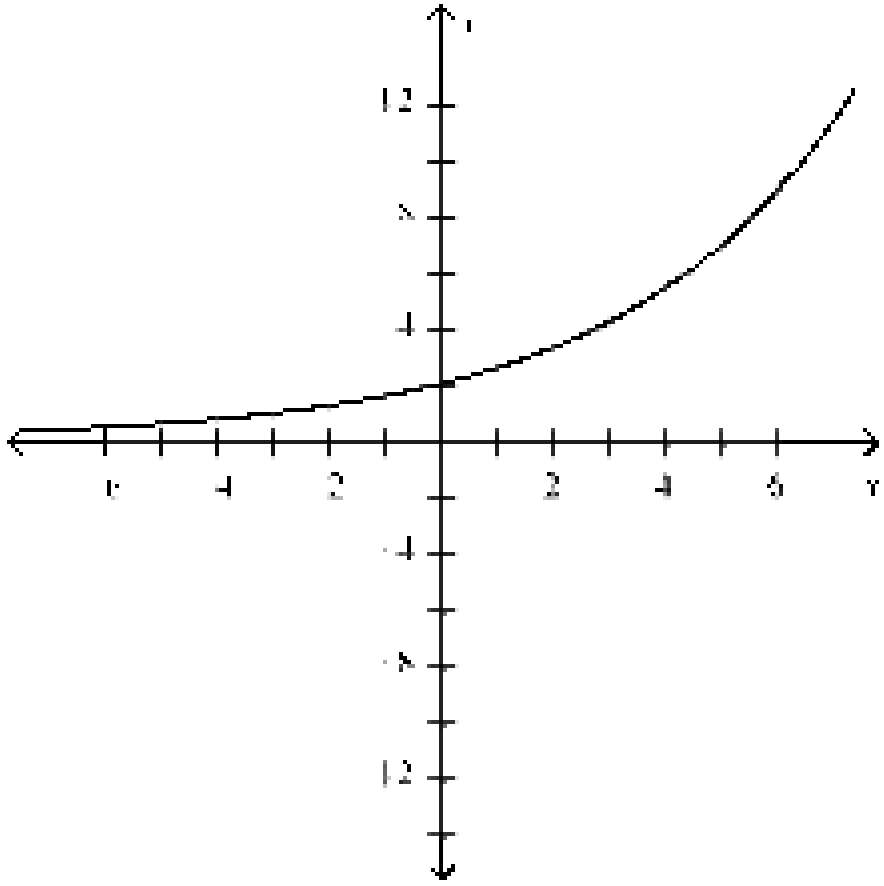
426. Which graph represents the function 2? Recovered graph functions: ($2)*((7)/(9))^x; -(2)*((9)/(7))^x; -(2)*((7)/(9))^x; (2)*((9)/(7))^x$. Rendered ExamView reference:

$$y = 2\left(\frac{7}{9}\right)^x$$









(A) graph of (2)*((7)/(9))^x

(B) graph of -(2)*((9)/(7))^x

(C) graph of -(2)*((7)/(9))^x

(D) graph of (2)*((9)/(7))^x

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

427. Which equation can be used to model the given information, where the population has been rounded to the nearest whole number? Year (x) Population (y) 100

(A) 100

(B) 1.4

(C) 100

(D) 1.4

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

428. The population of a bacterial culture triples every hour. When the scientist observed the culture, it had already been growing for some time. She developed the equation for the population, P, after t hours as $700,$ based on t $= 0$ representing the time she started her measurements. How many bacterial cells were there $3 h$ before she started measuring?

(A) 78

(B) 350

(C) 26

(D) 233

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

429. Solve for x. $1 562 500$

(A) 9

(B) 7

(C) 11

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

430. Solve for x, to one decimal place. 7333

(A) 1466.6

(B) 36 667.0

(C) 11.1

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

431. Solve for x. 36

(A) 36

(B) 0.3

(C) 7

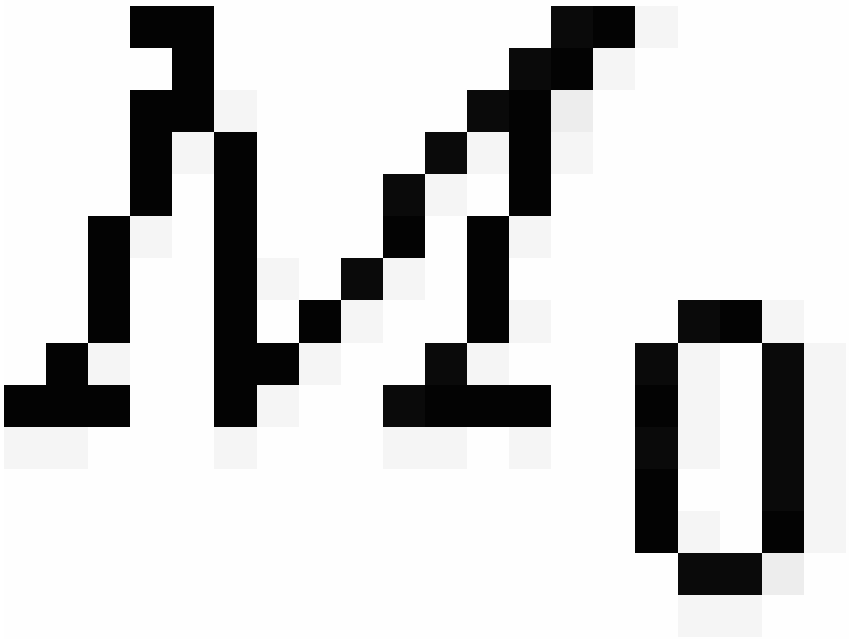
(D) 3.0

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

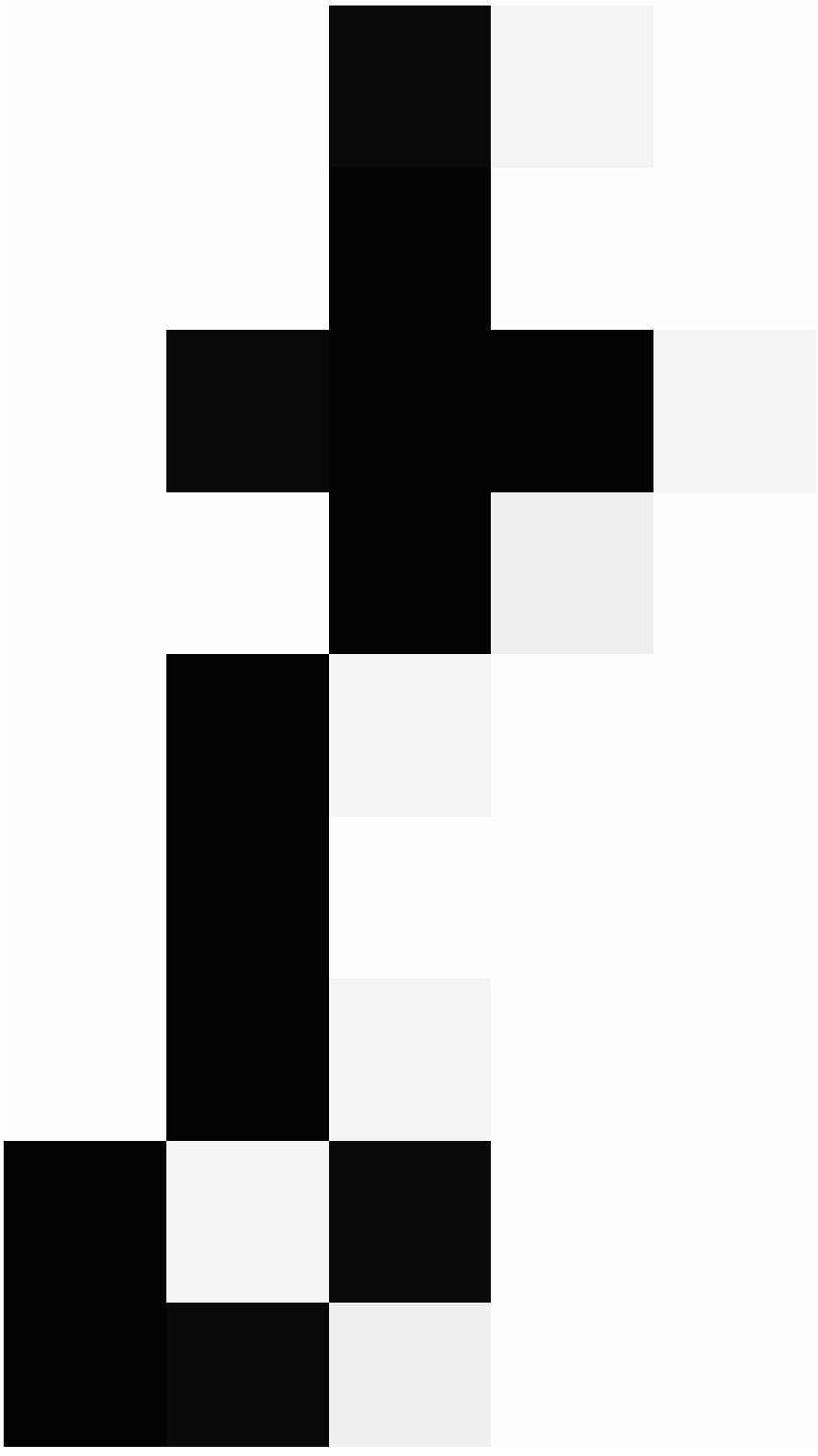
432. The half$-$life of a radioactive element can be modelled by $32,$ where 45 is the initial mass of the element; $M = M_0 1 c t b$ is the elapsed time, in hours; and

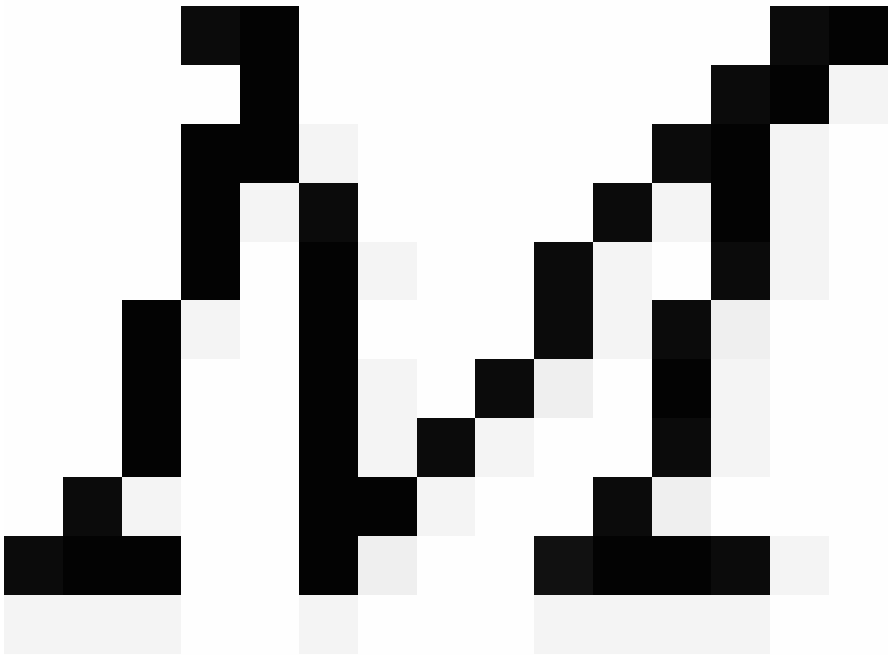
$$M = M_0 \left(\frac{1}{32} \right)^{\frac{t}{45}}$$

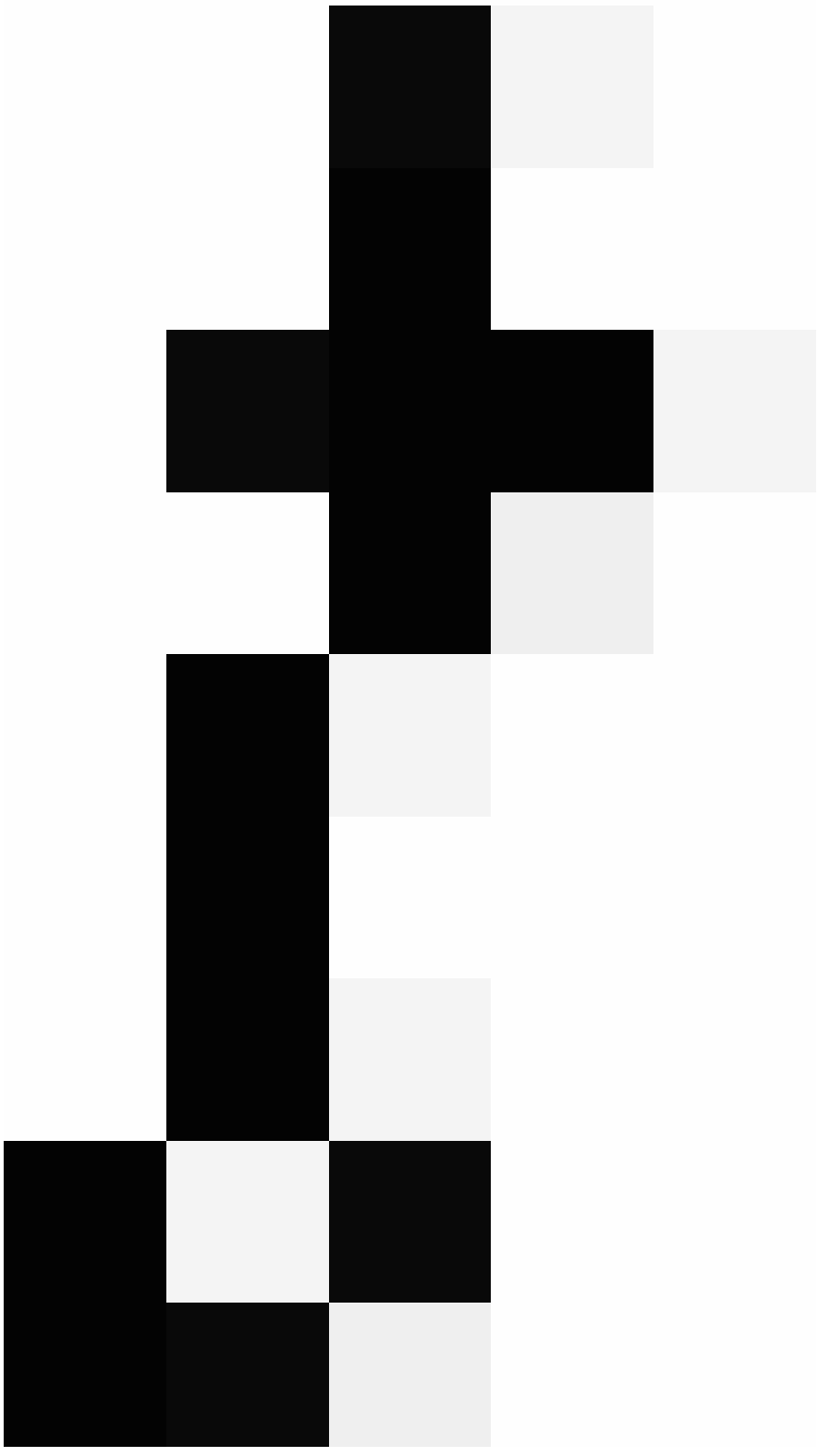
is the mass that remains after time



. The half$-$life of the element is Rendered ExamView reference:



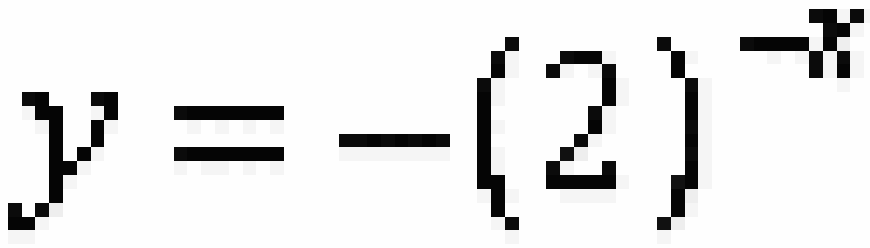




(A) 11 (B) 18 (C) 10 (D) 9

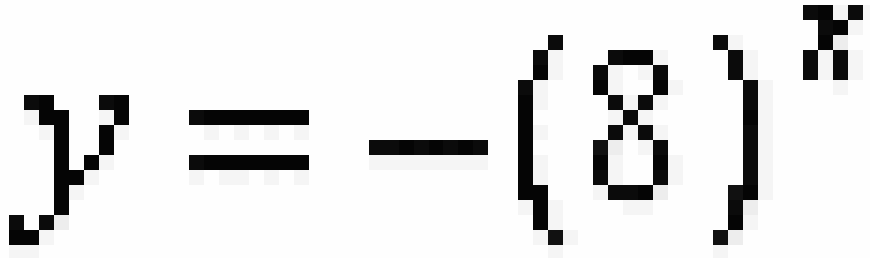
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

433. The graph of – has domain _____ and range _____. Rendered ExamView reference:



TOP: bnk | completion **TYPE:** completion

434. The graph of – has a $y-$intercept of _____ and an $x-$intercept of _____. Rendered ExamView reference:



TOP: bnk | completion **TYPE:** completion

435. The equation of the horizontal asymptote of the function $4 8$ is _____.

TOP: bnk | true_false **TYPE:** true_false

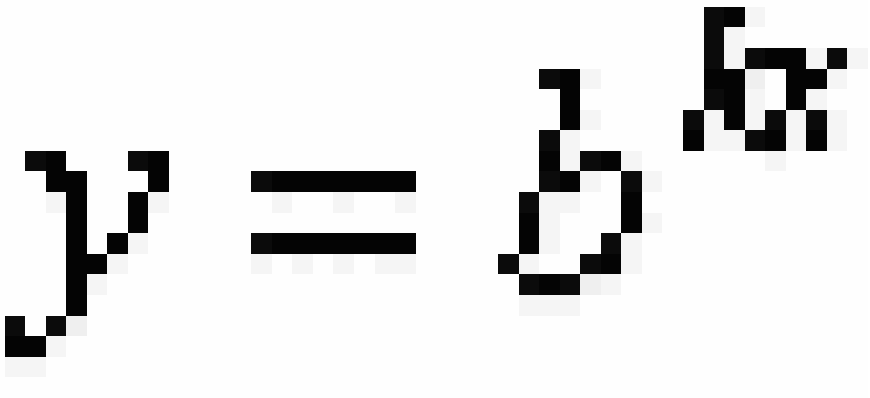
436. The expression 2 written as a single power of 6 is _____.

ANS: 6 **TOP:** bnk | completion **TYPE:** completion

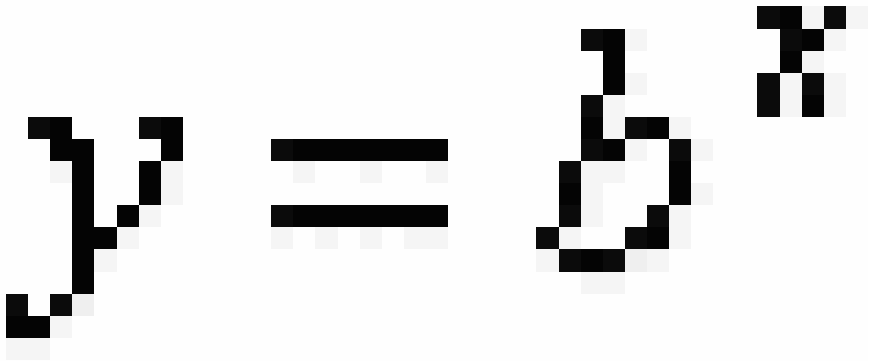
437. The value of t that makes the equation 64 true is _____.

ANS: 2/13 **TOP:** bnk | true_false **TYPE:** true_false

438. For the function $y = b kx$, if k $< 0,$ the graph of the function is _____ in the _____$-$axis, when compared to the graph of



. Rendered ExamView reference:

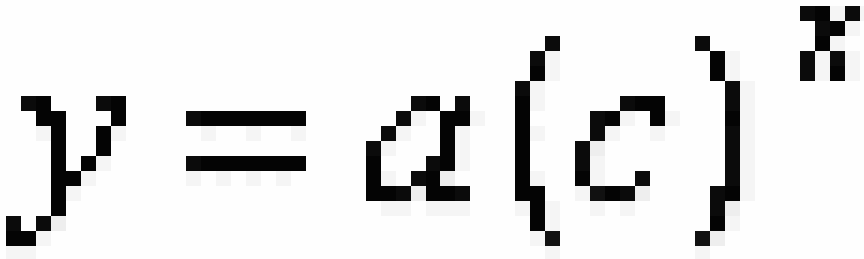


TOP: bnk | completion **TYPE:** completion

439. a pattern of growth in which each term is multiplied by a constant amount (between 0 and $1)$ to produce the next term

TOP: bnk | short_answer **TYPE:** short_answer

440. an exponential function of the form $y = a c x$, where a > 0 and



c [rendered RTF object missing]

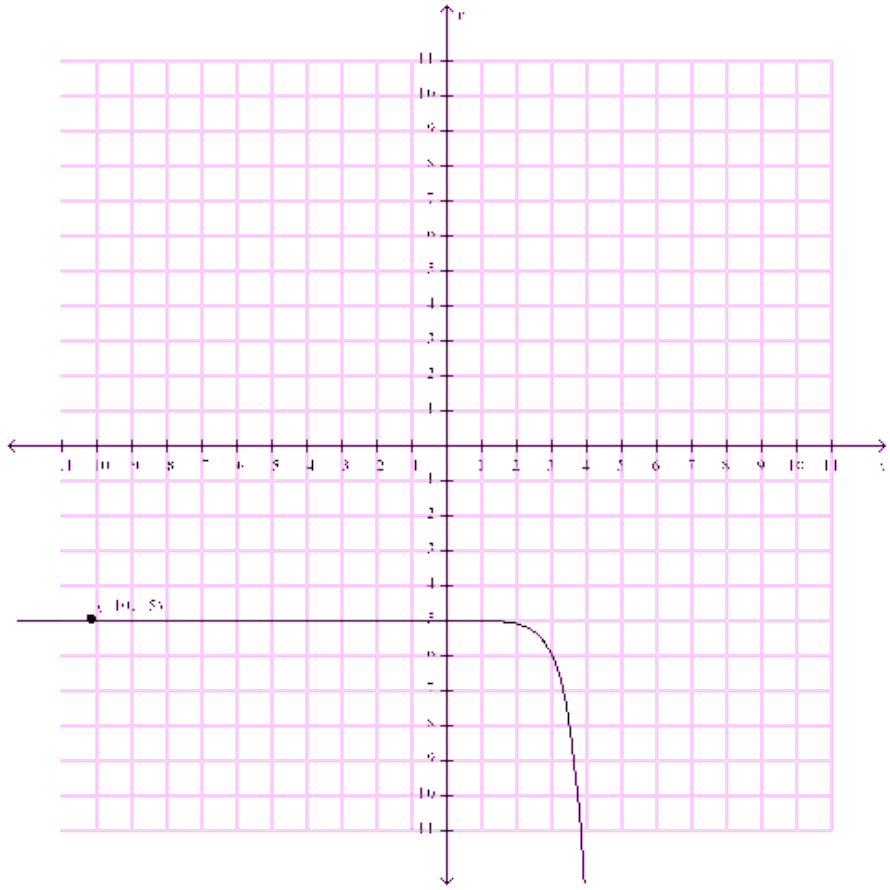
TOP: bnk | short_answer **TYPE:** short_answer

441. the horizontal asymptote of an exponential function of the form

$$y = 4(8)^{-6(x-7)} - 9$$

TOP: bnk | short_answer **TYPE:** short_answer

442. occurs for the graph of $y = a c b(x - h) + k$ when



$$y = -9(9)^{(x+4)} + 5$$

Rendered ExamView reference:

$$y = -9(9)^{(x-4)} - 5$$

$$y = 9(9)^{-(x+4)} + 5$$

$$y = 9(9)^{-(x-4)} - 5$$

TOP: bnk | short_answer **TYPE:** short_answer

443. occurs for the graph of $0.5*x^2$ when [Graph diagram: Graph: $0.5*x^2$ | x: [$−10.0, 10.0$], y: [$0.0, 50.0$]] < 0 Rendered ExamView reference:

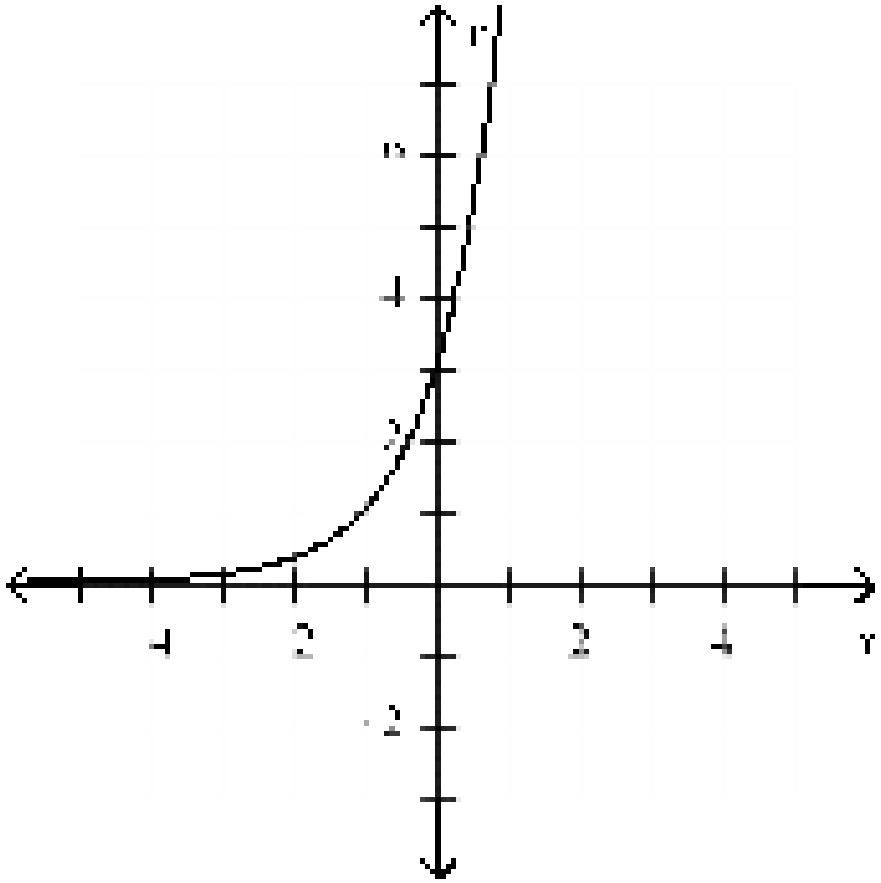
$$y = a(c)^{bx}$$

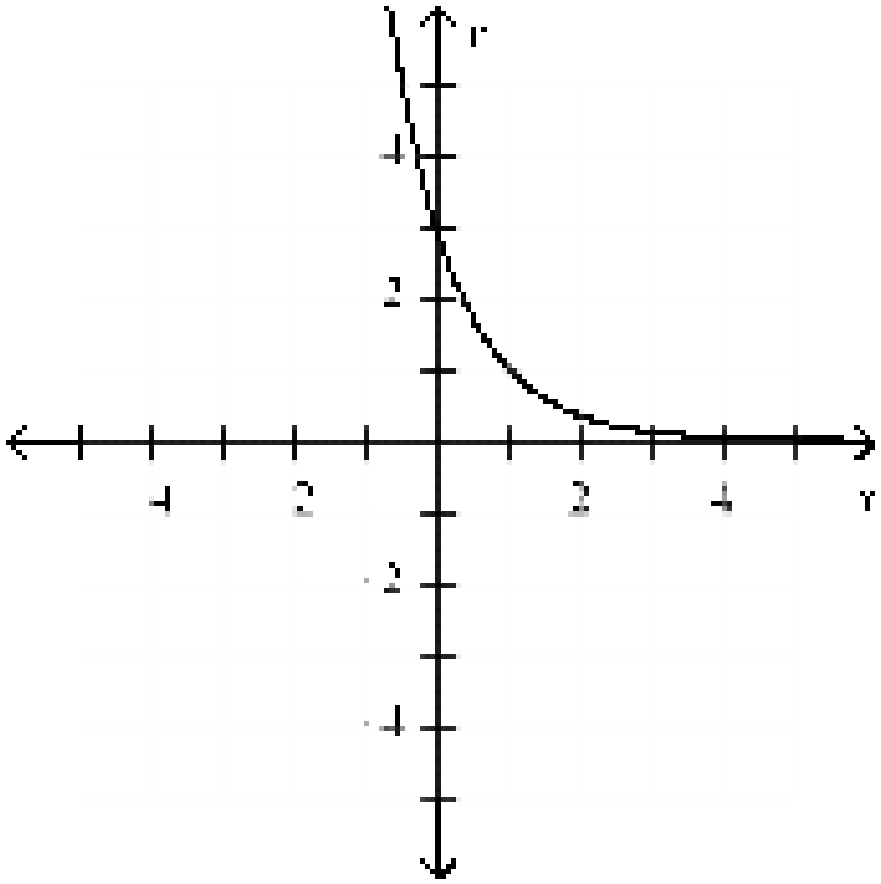
TOP: bnk | short_answer **TYPE:** short_answer

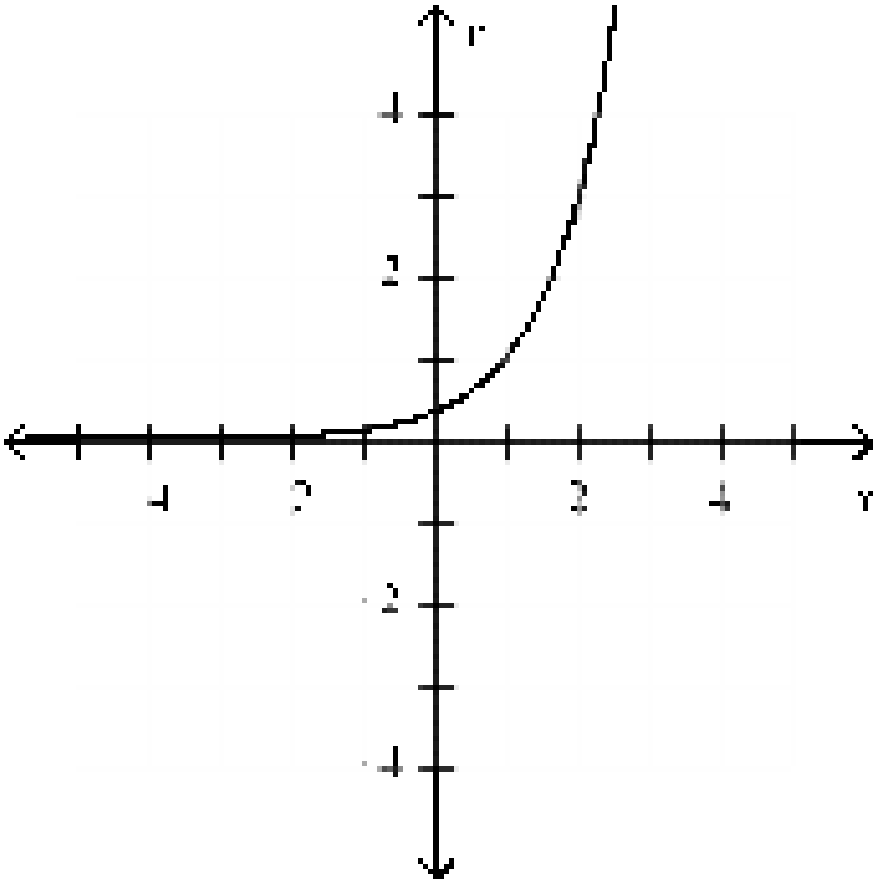
444. (a) Determine the type of function shown in each graph.
(b) $0.5*x^2$ ii) [Graph diagram: Graph: $0.5*x^2$ | x: [$−10.0, 10.0$], y: [$0.0, 50.0$]] iii) [Graph diagram: Graph: $0.5*2^x$ | x: [$−10.0, 10.0$], y: [$0.0, 512.0$]] b) Describe what you would expect to see in the first differences column of a table of values for each graph in part a).

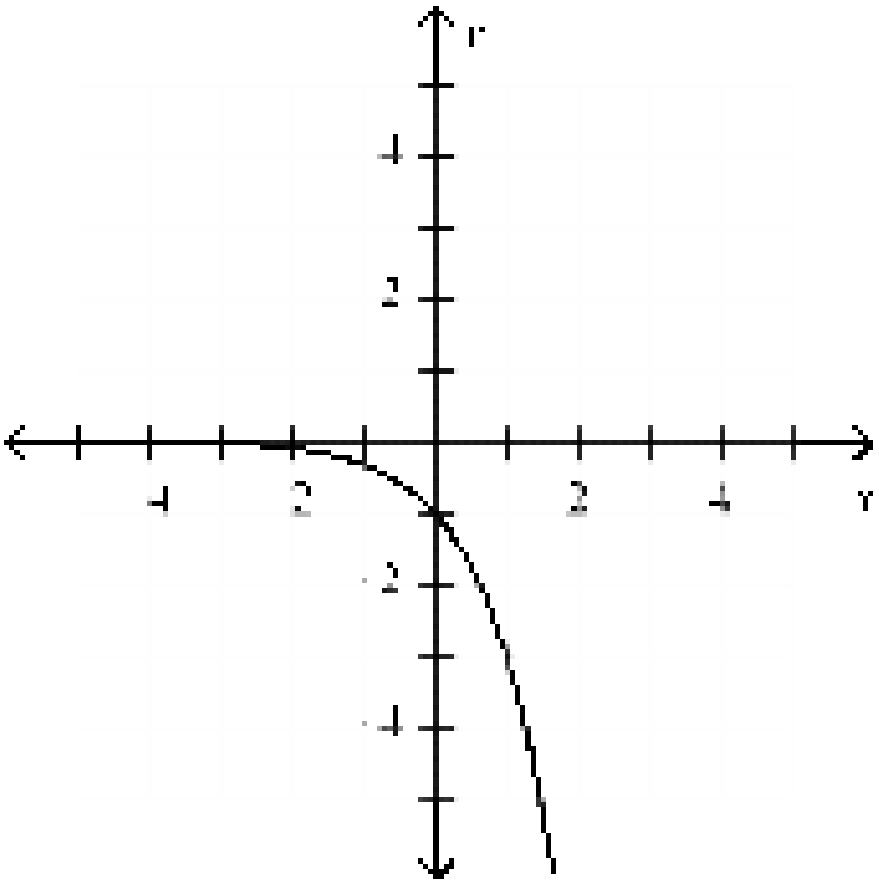
TOP: bnk | problem **TYPE:** problem

445. Match each graph with the correct corresponding equation. a) $3*3^x b)$ [Graph diagram: Graph: $3*3^x$ | x: [$−10.0, 10.0$], y: [$0.0, 177147.0$]] c) [Graph diagram: Graph: $3*(1/3)^x$ | x: [$−10.0, 10.0$], y: [$0.0, 177147.0$]] d) [Graph diagram: Graph: $(1/3)*3^x$ | x: [$−10.0, 10.0$], y: [$0.0, 19683.0$]] i) [Graph diagram: Graph: $−3^x$ | x: [$−10.0, 10.0$], y: [$−59049.0, −0.0$]] ii) [Graph diagram: Graph: $−3^x$ | x: [$−10.0, 10.0$], y: [$−59049.0, −0.0$]] iii) [Graph diagram: Graph: $−3^x$ | x: [$−10.0, 10.0$], y: [$−59049.0, −0.0$]] iv) [Graph diagram: Graph: $−3^x$ | x: [$−10.0, 10.0$], y: [$−59049.0, −0.0$]] Rendered ExamView reference:


y
6
4
2
-4
-2
2
4
x
-2


y
4
2
-4
-2
2
4
x
-2
-4


y
4
2
-4
-2
2
4
x
-2
-4


y
4
2
-4
-2
2
4
x
-2
-4

$$y = \frac{1}{3}(3)^x$$

$$y = 3(3)^x$$

$$y = -3^x$$

$$y = 3\left(\frac{1}{3}\right)^x$$

446. Sketch the graph of an exponential function with all of the following characteristics: domain $3*(1/3)^x$ range [Graph diagram: Graph: $3*(1/3)^x$ | x: [$−10.0, 10.0$], y: [$0.0, 177147.0$]] $y-$intercept of 3 no $x-$intercept the function is always decreasing Rendered ExamView reference:

$$\{x | x \in R\}$$

$$\{y | y > 0, y \in R\}$$

TOP: bnk | problem **TYPE:** problem

447. Sketch an exponential function with all of the given conditions: domain $y = - 4 1 / 2^(x)$ range

$$\{x | x \in R\}$$

$y-$intercept of –4 no $x-$intercept the function is always increasing Rendered ExamView reference:

$$\{y | y < 0, y \in R\}$$

TOP: bnk | problem **TYPE:** problem

448. For the function $3^x,

(a) $ describe the transformations of the function when compared to the function $0.5*3^(x-2)$ b) sketch the graph of the given function and [Graph diagram: Graph: 3^x | x: [$−10.0, 10.0$], y: [$0.0, 59049.0$]] on the same set of axes
(b) state the domain, the range, and the equation of the asymptote

TOP: bnk | problem **TYPE:** problem

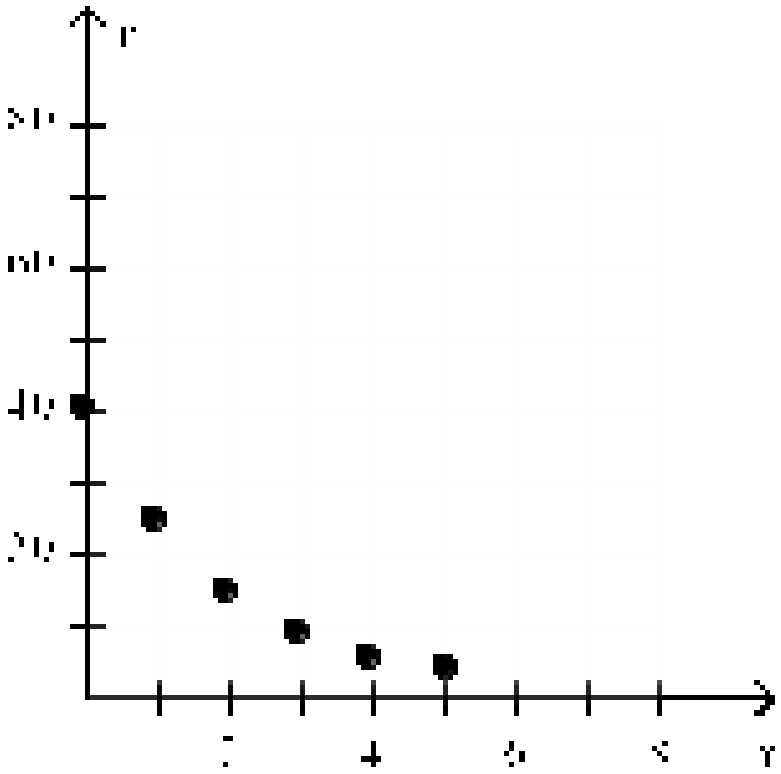
449. Write the equation for the function that results from each transformation or set of transformations applied to the base function $y = 5 - x$.

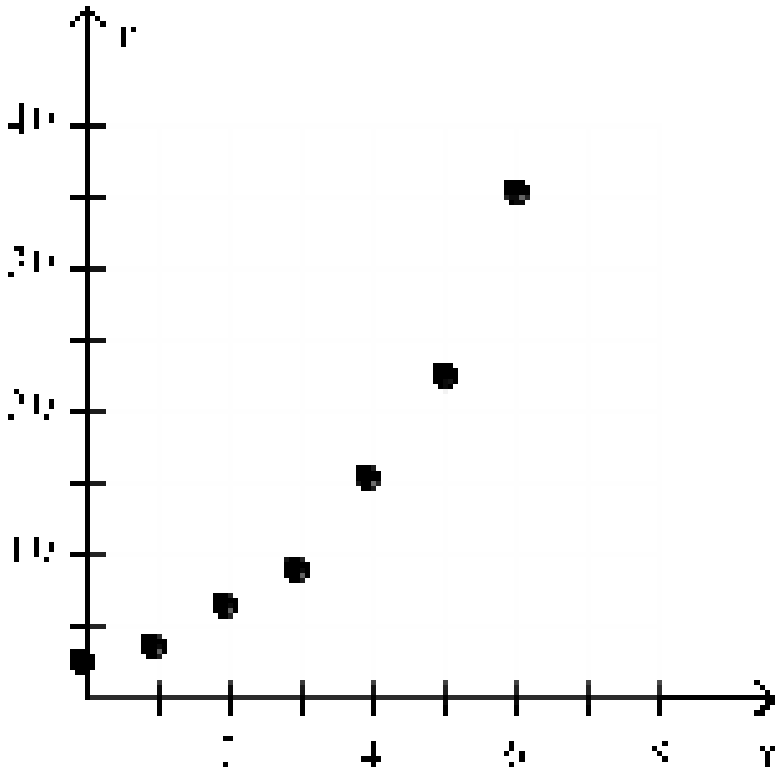
(a) reflect in the $y-$axis
(b) shift 3 units to the right
(c) shift 1 unit down and 4 units to the left
(d) reflect in the $x-$axis and shift 2 units down

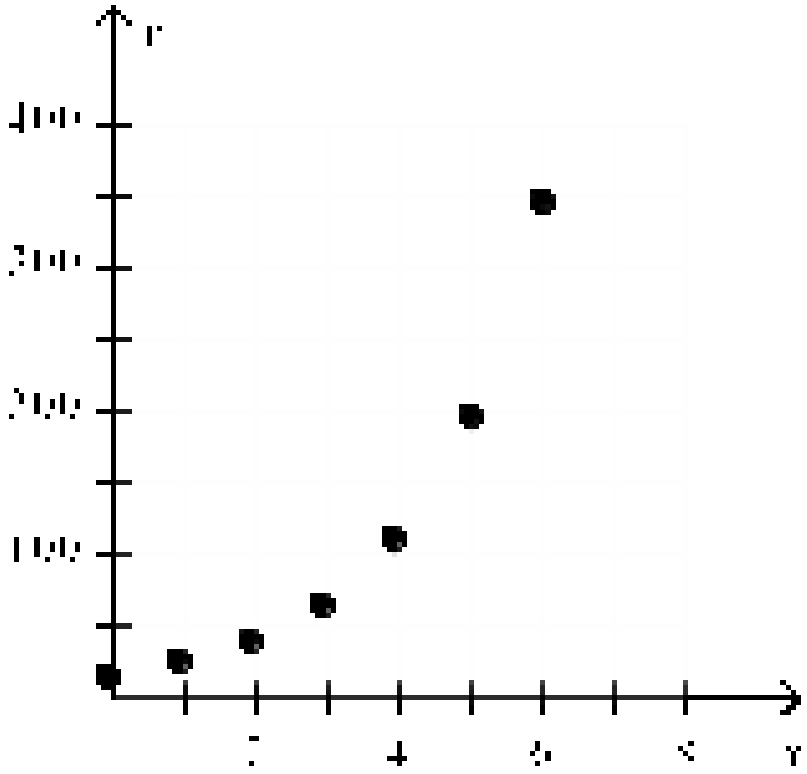
TOP: bnk | problem **TYPE:** problem

450. Match each exponential scatter plot with the corresponding equation of its curve of best fit. a) [Scatter plot: decreasing exponential, starts near $(0, 40)$ and approaches 0 as x increases] b) [Scatter plot: increasing exponential, starts near

$(0, 2)$ and rises to about 20 by $x = 5$] c) [Scatter plot: increasing exponential, starts near $(0, 10)$ and rises steeply past 150 by $x = 5$] i) $y = 2(1.6)^x$ ii) $y = 40(0.6)^x$ iii) $y = 10(1.8)^x$. Rendered ExamView reference:







$$y = 2(1.6)^x$$

$$y = 40(0.6)^x$$

$$y = 10(1.8)^x$$

TOP: bnk | problem **TYPE:** problem

451. The scatter plot shows how his money has grown over time. [DIAGRAM NEEDED] a) Do the data show an exponential trend? Explain your reasoning.

(a) Estimate values of a and b to develop a model for the information of the form $V(t) = a b^t$, for t in years.
(b) Use your model to predict the expected value of Michael s investment in 30 years.
(c) How does the result in part c) illustrate the power of exponential growth?

TOP: bnk | problem **TYPE:** problem

452. Solve for n: $n = 3 n = 1$

TOP: bnk | short_answer **TYPE:** short_answer

453. Solve for x: $x = 9^(x)^2 - 1$

TOP: bnk | short_answer **TYPE:** short_answer

454. Graphically solve the system $y = 3^(x) , 3x - y = 0$. $y = 3 x$

TOP: bnk | short_answer **TYPE:** short_answer

455. A colony of ants starts with an initial population of 50 and doubles every week for 8 weeks.

(a) Create a table of values for weeks 0 to 8 for the population of the colony.
(b) Graph the data from your table of values.
(c) Is the relationship between the ant population and the number of weeks exponential? Explain.
(d) Model the information using an equation.

TOP: bnk | problem **TYPE:** problem

456. An initial investment of $$4000$ earns interest at 4% per year, compounded annually. a) Create a table of values for the first 10 years of the investment. Round the amounts to the nearest cent.

(a) Graph the data from your table of values.
(b) Is the relationship between the time and the amount of the investment exponential? Explain.
(c) Write an equation to represent the data.

TOP: bnk | problem **TYPE:** problem

457. Jeff buys a new vehicle for $35 000. It is known that the vehicle will depreciate by 20% of its current value every year. a) Write an equation to relate the depreciated value, V, of the vehicle to the age, t, in years, of the vehicle.

(a) Use the equation to determine the value of the vehicle 2 years after Jeff buys it.
(b) Approximately how long will it take the vehicle to depreciate to $3000?

TOP: bnk | problem **TYPE:** problem

458. A sample of 60 mg of the material is present today.

(a) Write an equation to relate the amount of cobalt$−60$ remaining and the number of half$-$life periods.
(b) What amount will be present in 10.6 years?
(c) How many years will it take for the amount of cobalt$−60$ to decay to 12.5% of its initial amount?

TOP: bnk | problem **TYPE:** problem

459. Buket wants to invest some money in an account that pays 3.5% interest, compounded yearly. She wants to have an accumulated amount of $6000 at the end of 4 years.

(a) Identify the values of A, i, and n to be used in the formula $P $= A 1$$.
(b) Determine how much Buket needs to invest today in order for her to reach her goal in 4 years.
(c) How much less would she need to invest if the financial institution were to offer her 4% annual interest? Show all your work.

TOP: bnk | problem **TYPE:** problem

460. When interest is compounded semi$-$annually, the formula used to find the amount of an investment is $A = P(1 + i/2)^(2n)$, where A represents the amount; P represents the principal invested; i represents the annual interest rate, as a decimal; and n represents the number of years of the investment. a) Use the formula to determine the amount that each investment would be worth. i) $5000 at a rate of 4%, compounded semi-annually, for 10 years ii) $$4000$ at a rate of 5

%, compounded semi$-$annually, for 20 years b) If interest is compounded quarterly, the formula becomes $A = P(1 + i/4)^(4n)$. Use the formula to determine the amount that the investments from part a) would be worth if interest were compounded quarterly. c) Explain the difference in the answers for parts a) and b).

TOP: bnk | problem **TYPE:** problem

461. A radioactive sample with an initial mass of 72 mg has a half$-$life of 10 days.

(a) Write a function to relate the amount remaining, A, in milligrams, to the time, t, in days.
(b) What amount of the radioactive sample will remain after 20 days?
(c) What amount of the radioactive sample was there 30 days ago?
(d) How long, to the nearest day, will it take for there to be 0.07 mg of the initial sample remaining?

TOP: bnk | problem **TYPE:** problem

462. Graph each function. Identify the

(a) domain

(i) range
(ii) intercepts
(iii) intervals of increase$/$decrease

(a) equation of the asymptote
(b) ($1/3)^x b)$ [Graph diagram: Graph: $(1/3)^x$ | x: [$−10.0, 10.0$], y: [$0.0, 59049.0$]]

TOP: bnk | problem **TYPE:** problem

463. (a) Rewrite the function $−1*2^(−2*x+4)−6$ in the form [Graph diagram: Graph: $−2^(−2*x+4)−6$ | x: [$−7.5, 10.0$], y: [$−524294.0, −6.0$]].
(b) Describe the transformations that must be applied to the graph of [Graph diagram: Graph: $−2^(−2*x+4)−6$ | x: [$−7.5, 10.0$], y: [$−524294.0, −6.0$]] to obtain the graph of the given function.
(c) Graph the function.
(d) Determine the equation of the function that results after the graph in part c) is reflected in the $x-$axis.
(e) Graph the function from part d).

TOP: bnk | problem **TYPE:** problem

464. (a) Write the equation for a transformed exponential function with a base of 4 that passes through the point $y = 1 / 2 4 2x + 1$.
(b) Write two equations, different from the one in part a), that satisfy these criteria.
(c) Use algebraic and$/$or graphical reasoning to explain why each equation is a solution.

TOP: bnk | problem **TYPE:** problem

465. (a) Write the equation for a transformed exponential function whose horizontal asymptote is y $= 2,$ with a $y-$intercept of 4.
(b) Is this function unique? Explain.
(c) Illustrate your answer to part b) graphically, using examples where appropriate.

TOP: bnk | problem **TYPE:** problem

466. An antique table was worth $400 in 2000. Each subsequent year, the table increased in value until it was sold at an auction in 2004. The table shows the value of the antique table for the years 2000 to 2004. Value ($) $2000 400.00 2001 440.00 2002 484.00 2003 532.40 2004 585.64 a)$ Create a scatter plot of the data.

(a) Describe the shape of the curve.
(b) Determine the equation for the curve of best fit, using t $= 0$ for the year 2000.
(c) Use this equation to determine the value of the antique table in $2010,$ assuming that the value continues to increase according to the equation of the curve of best fit.
(d) Approximately how long will it take for the table to be worth $1250?

TOP: bnk | problem **TYPE:** problem

467. The table shows the value, V, of Rachel s car t years after it was purchased. Years After Purchase Value ($) 42 000 37 800 34 020 30 618 27 556 a) Make a scatter plot for the data.

(a) Describe the shape of the curve.
(b) Determine an equation to model the data.
(c) Use the equation to determine the value of the car in 8 years.
(d) Approximately how long will it take for the car to be worth less than $$10 000$?

TOP: bnk | problem **TYPE:** problem

468. The data are shown below. Price ($) 8.00 8.40 8.82 9.26 9.77 10.21 a) Make a scatter plot for the data.

(a) Describe the shape of the curve.
(b) Determine an equation to model the data.
(c) Use the equation to determine the value of the stock in 9 weeks.
(d) Approximately how long will it take for the share price of the stock to exceed $20?

TOP: bnk | problem **TYPE:** problem

469. Solve the equation $x = 64^(x) - 3$.

TOP: bnk | short_answer **TYPE:** short_answer

470. Solve the equation $3x = 4$.

TOP: bnk | short_answer **TYPE:** short_answer

471. A $200-g$ sample of a radioactive substance is placed in a chamber to be tested. After $3 h, 140 g$ of the sample remains.

(a) Determine the half$-$life of this substance.
(b) Determine the equation that represents the decay of the substance from your answer to part a).
(c) Graph the amount of the substance remaining as a function of time. Use your graph to determine the amount of time it will take for 85% of the substance to decay. $200*$.$5^(x/6)$

TOP: bnk | true_false **TYPE:** true_false

472. A $$21 500$ investment earns 5.25% interest, compounded quarterly.

(a) Determine the value of the investment in 5 years.
(b) How long will it take the original investment to double in value?

TOP: bnk | problem **TYPE:** problem

473. A chemist has a $20-$mg sample of polonium$−218$. He needs approximately 81.5% of it for an experiment. Given that the half$-$life of polonium$−218$ is approximately $3.1 min,$ how many seconds will it take for the sample to decay to the desired mass?

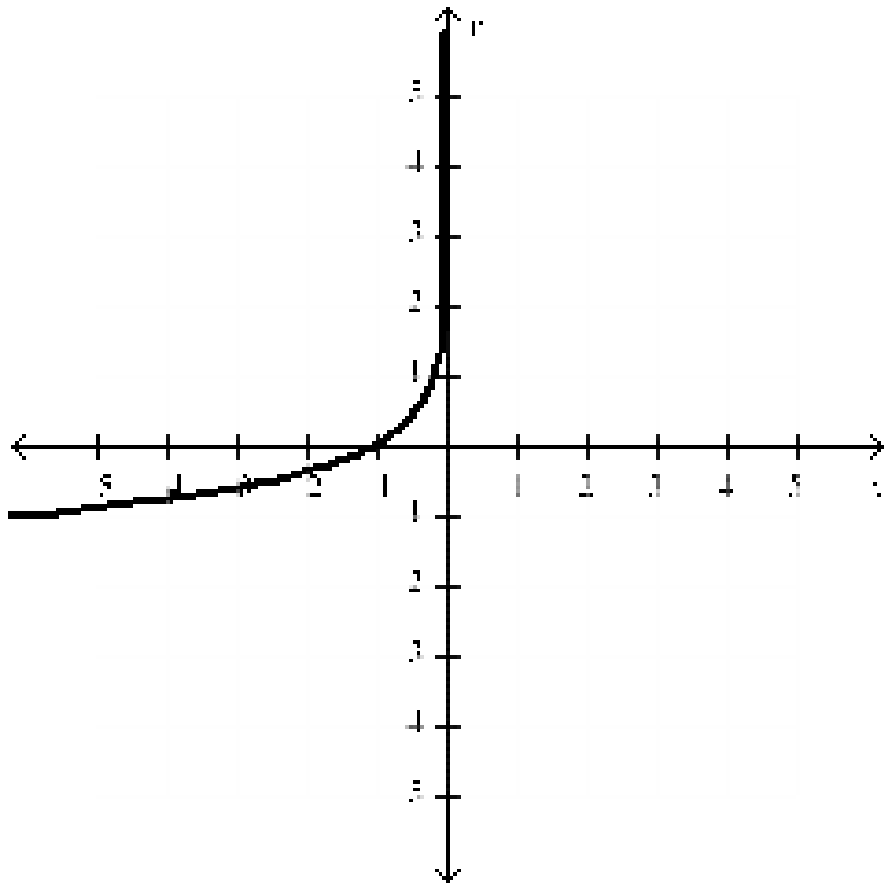
TOP: bnk | short_answer **TYPE:** short_answer

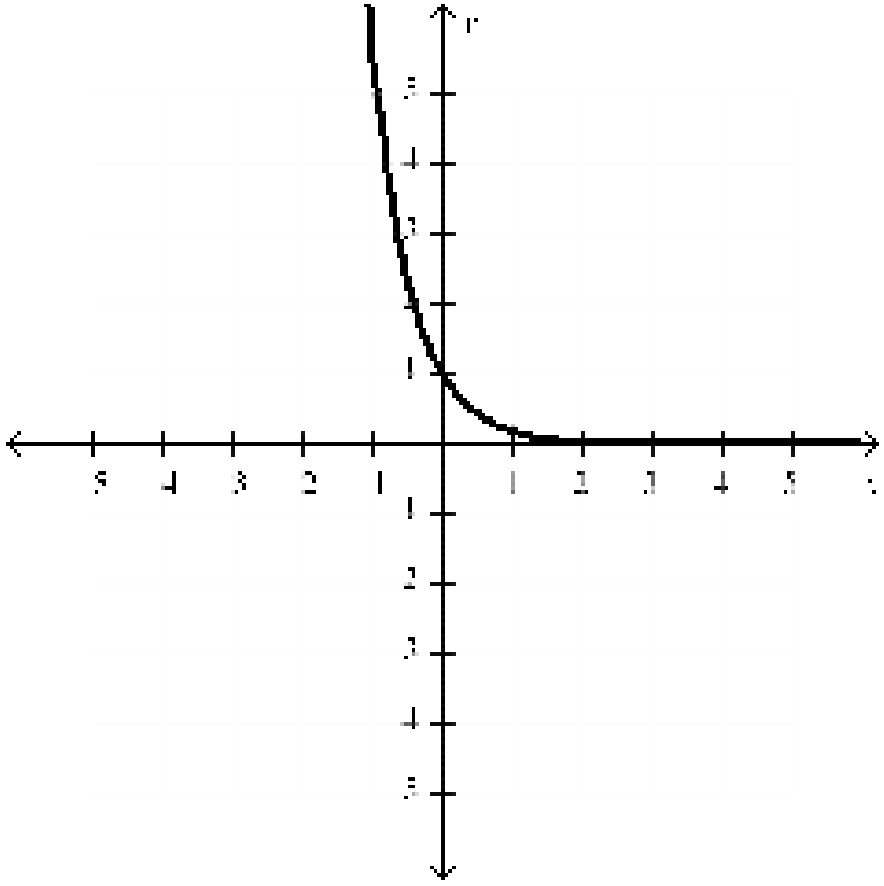
474. An investment offers a bonus of 2% of the principal after being invested for 5 years. If $$50 000$ is invested at 4.75 %, compounded annually, for 10 years, describe how the graph of the investment with the bonus differs from the graph of the investment without the bonus.

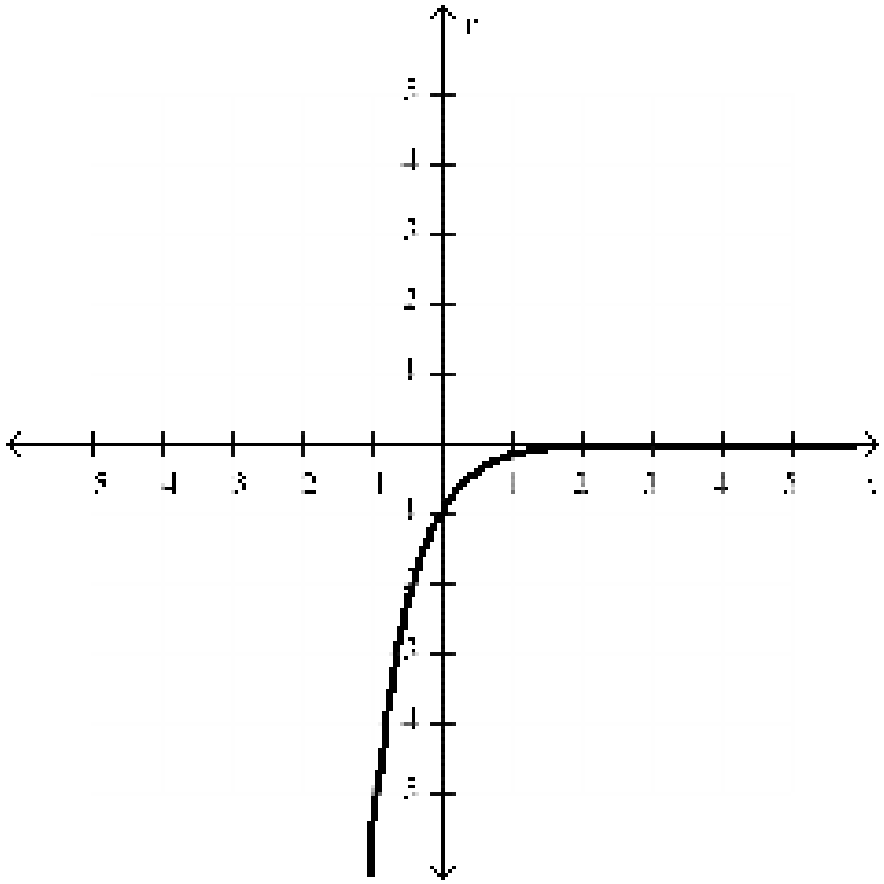
TOP: bnk | short_answer **TYPE:** short_answer

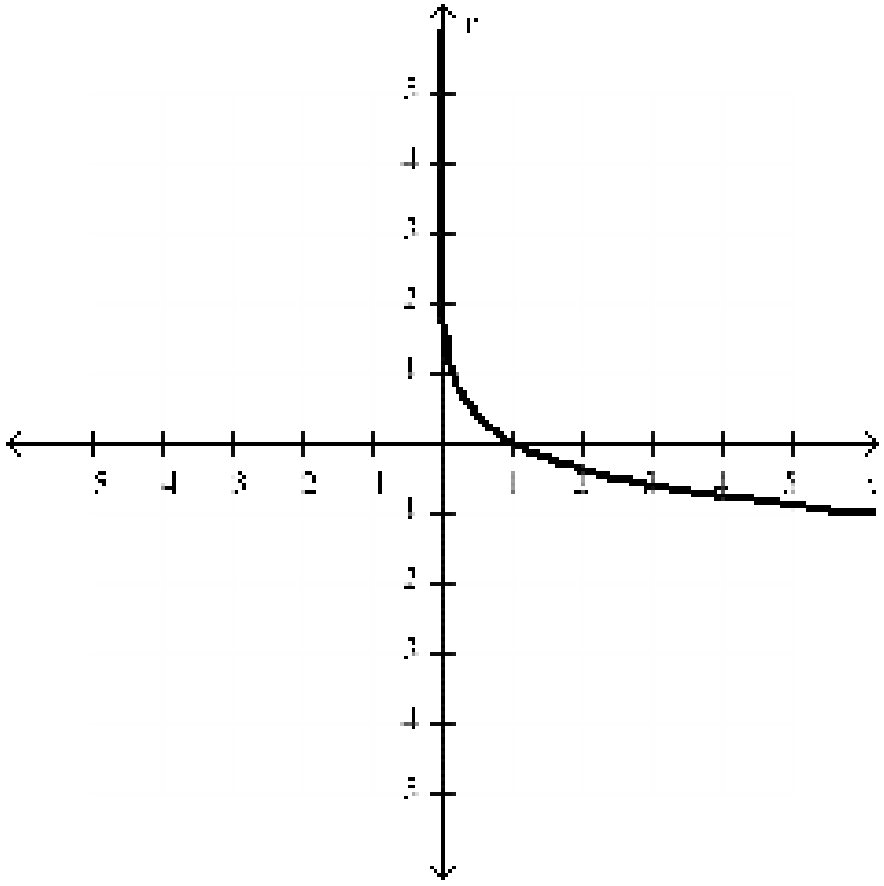
475. Which graph represents the inverse of 6? Recovered graph functions: ($1/(6))^x; −1*(1/(6))^x$. Rendered ExamView reference:

$$y = \left(\frac{1}{6}\right)^x$$









(A) graph of (1/(6))^x (B) graph of −1*(1/(6))^x

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

476. Another way of writing 5 is a. $5 c$. b. d.

(A) 5 (B) 5 (C) 5 (D) 3125

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

477. Which of the following represents a?

(A) 343 (B) a (C) 343 (D) 7

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

478. Another way of writing 7 is

(A) 343 (B) 7 (C) 343 (D) 3

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

479. Evaluate 4.

(A) 4096 (B) 0.13 (C) 5.33

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

480. The range of the function –8 is

(A) $x >= + 9$ (B) $x >= + 8$ (C) $x >= 10$ (D) $x <= + 9$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

481. The domain of the function 8 is

(A) $x >= + 7$ (B) $x >= + 8$ (C) $x >= 8$ (D) $x <= + 7$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

482. Compared to the graph of the base function, the graph of the function is translated 4 units down. Rendered ExamView reference:

$$y = \log_{10} x$$

$$y = \log_{10} x + 4$$

(A) 4 units down (B) 4 units up (C) 4 units left (D) 4 units right

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

483. Compared to the graph of the base function $f(x) = x^2$, the graph of $g(x) = (x + 6)^2 + 7$ is translated 6 units right and 7 units up. Rendered ExamView reference:

$$y = \log_{,} x$$

$$y = \log_{,}(x - 6) + 7$$

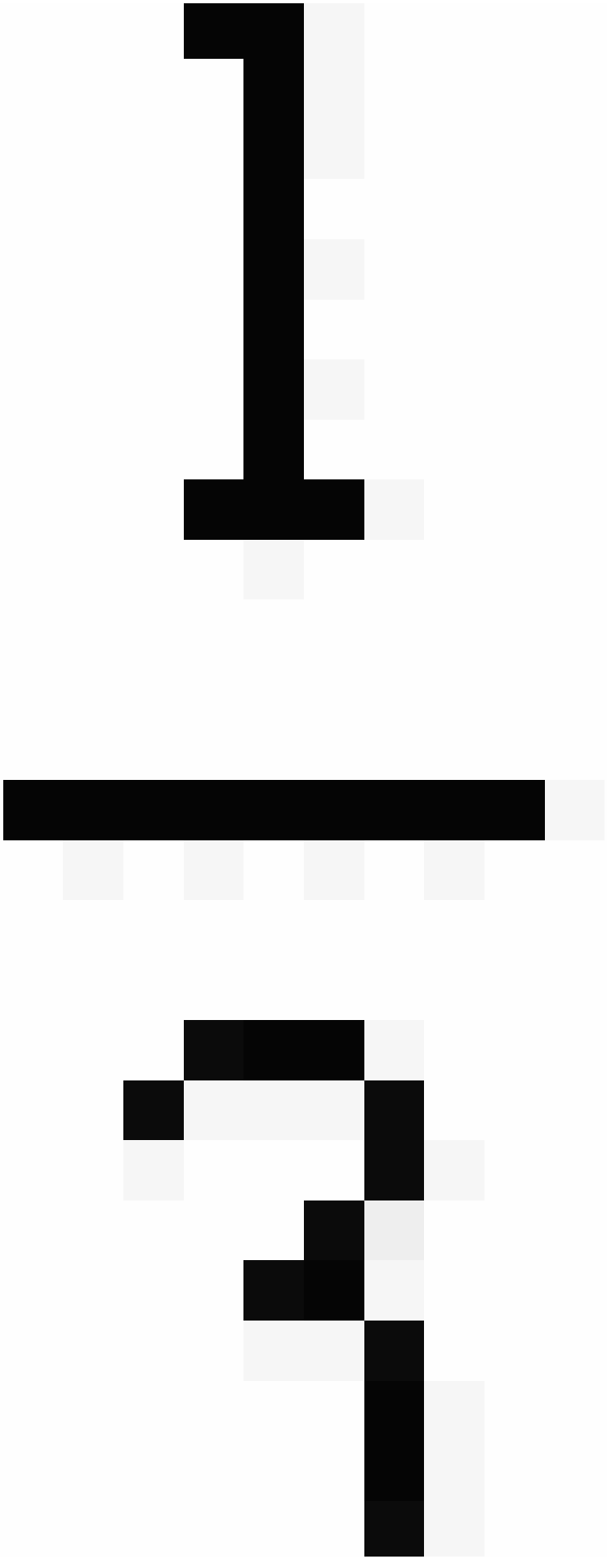
(A) 6 units right and 7 units up (B) 6 units left and 7 units up (C) 6 units right and 7 units down (D) 6 units left and 7 units down

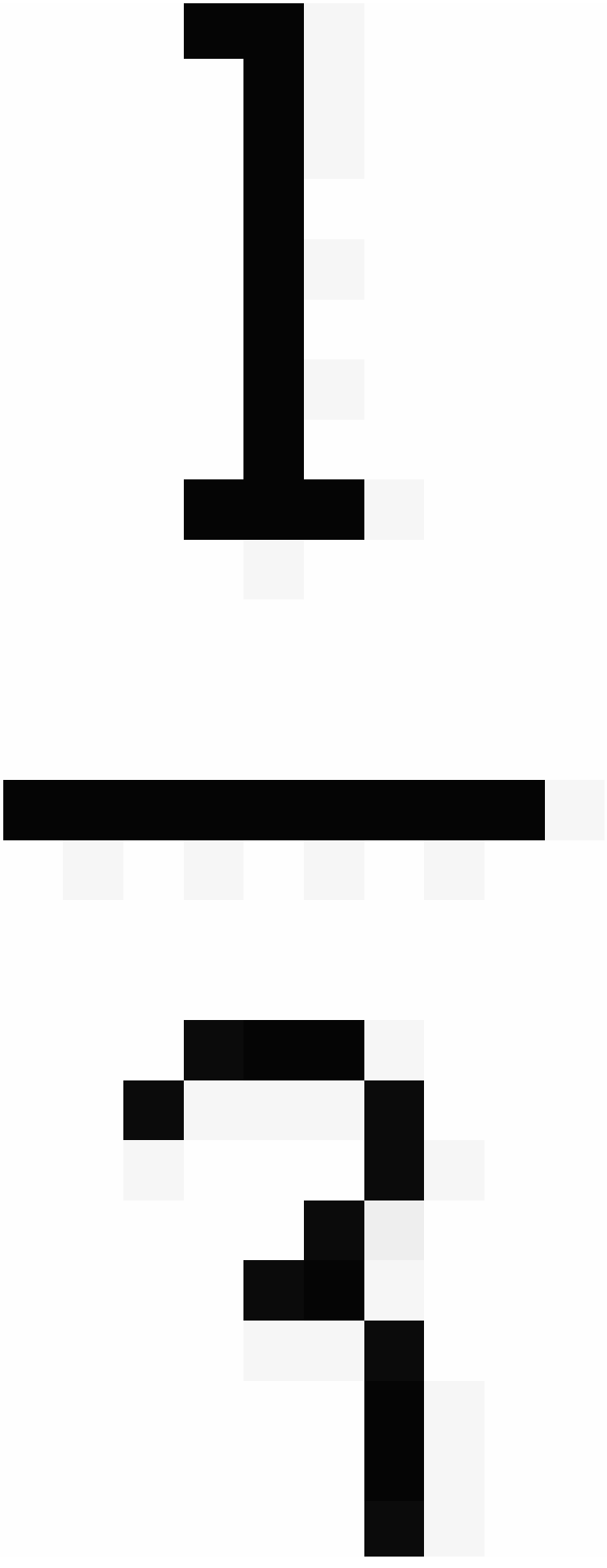
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

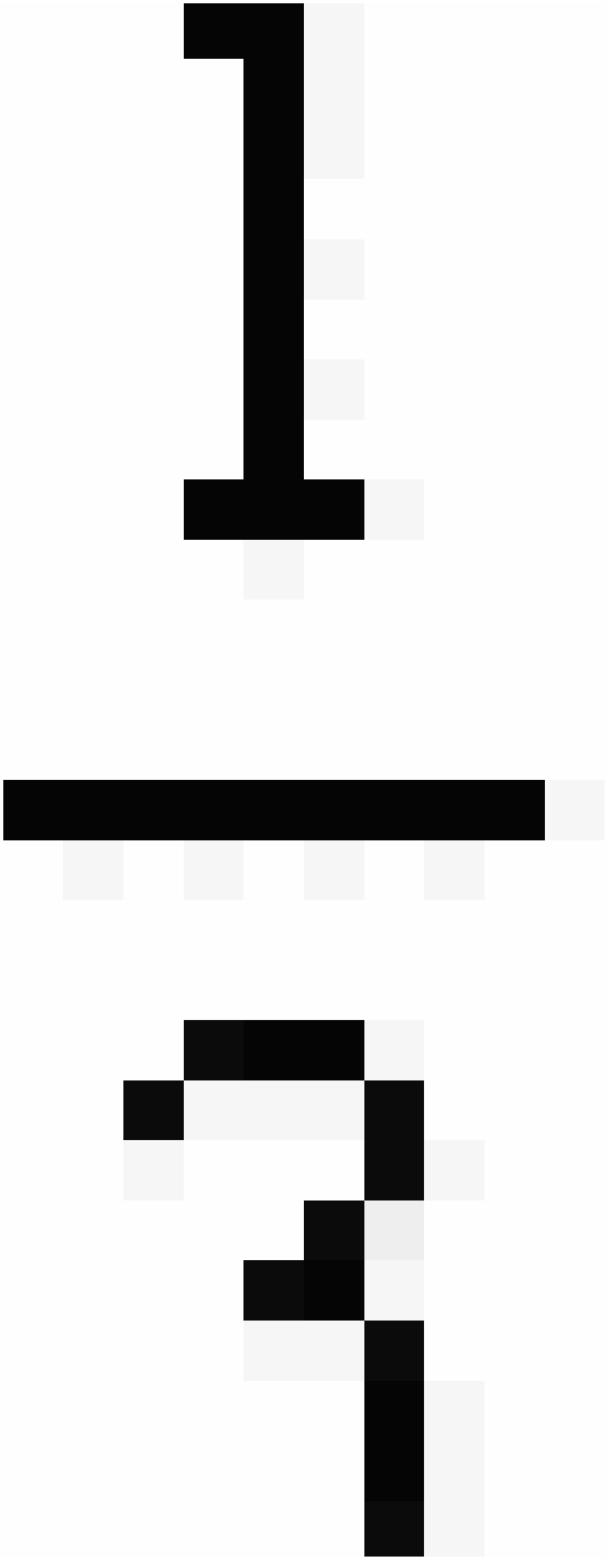
484. Compared to the graph of the base function – $10,$ the graph of the function 9 is Rendered ExamView reference:

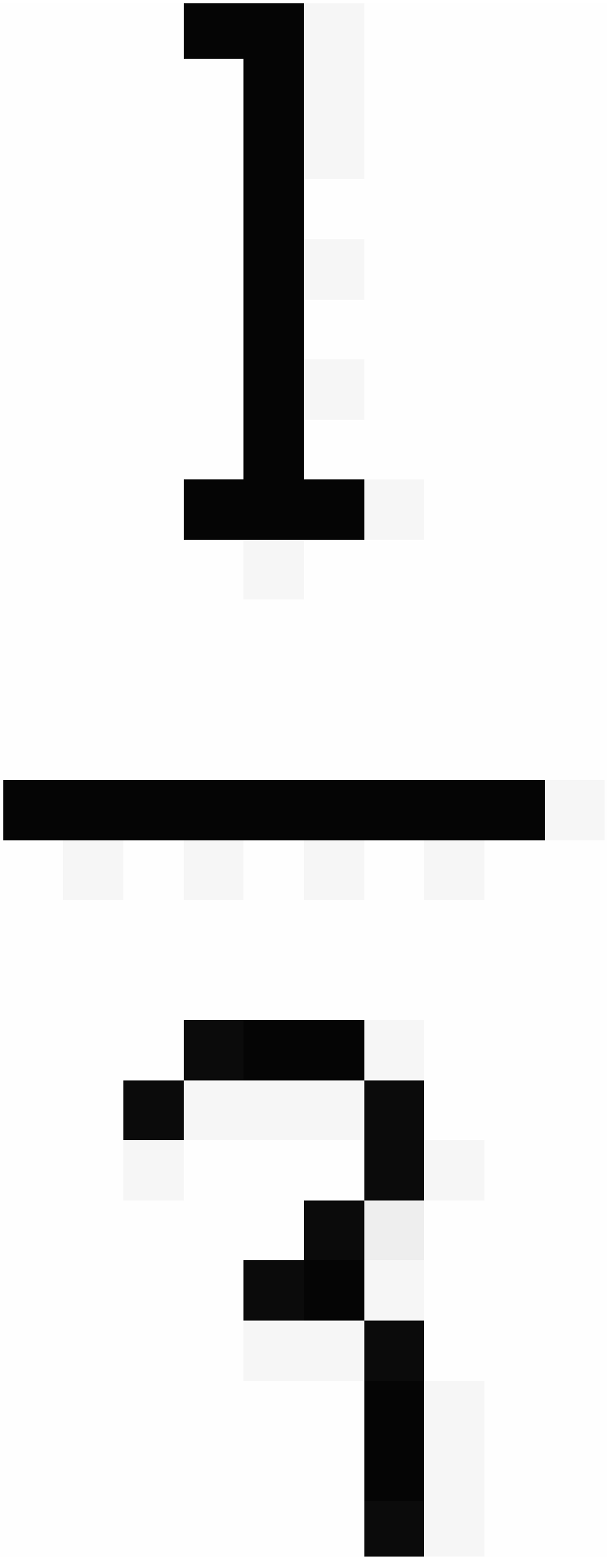
$$f(x) = \log_{g} x$$

$$g(x) = 3\log_9[3(x+6)] - 10$$









(A) $x >= – 10$ (B) $x >= + 6$ (C) $x >= 3$ (D) $x <= – 10$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

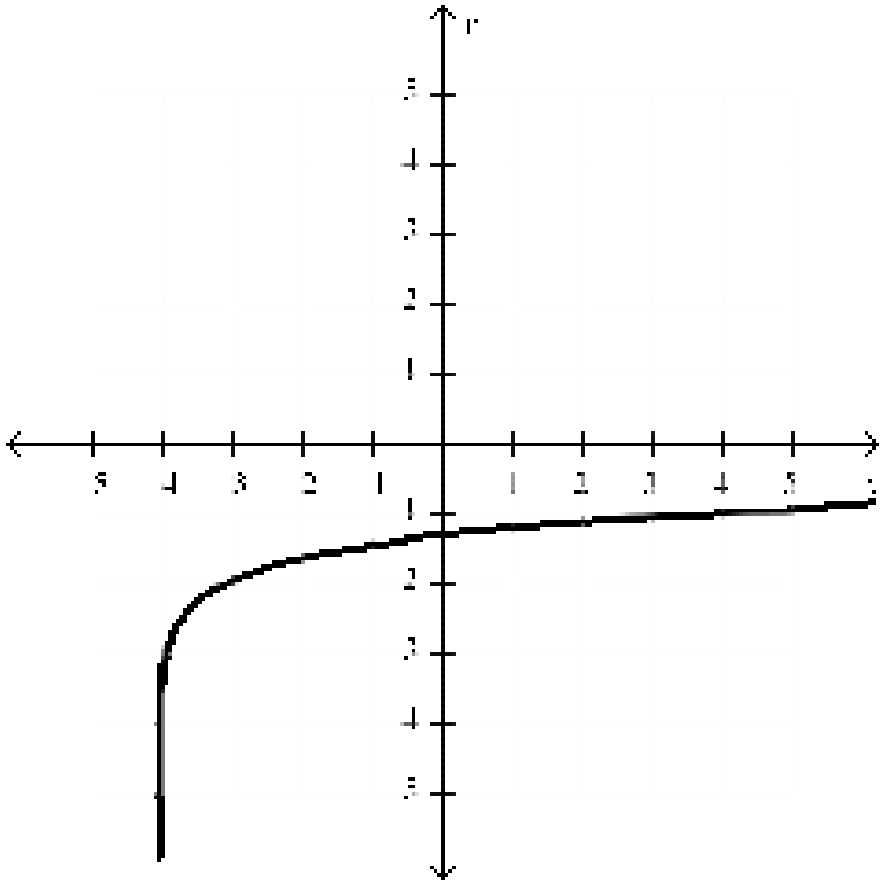
485. Which function represents a vertical translation of – 8 units – $8, a$ horizontal translation of 9 units – $8, a$ horizontal stretch by a factor of $7,$ down, a vertical stretch by a factor of $8,$ and right, when compared to the base function 6.

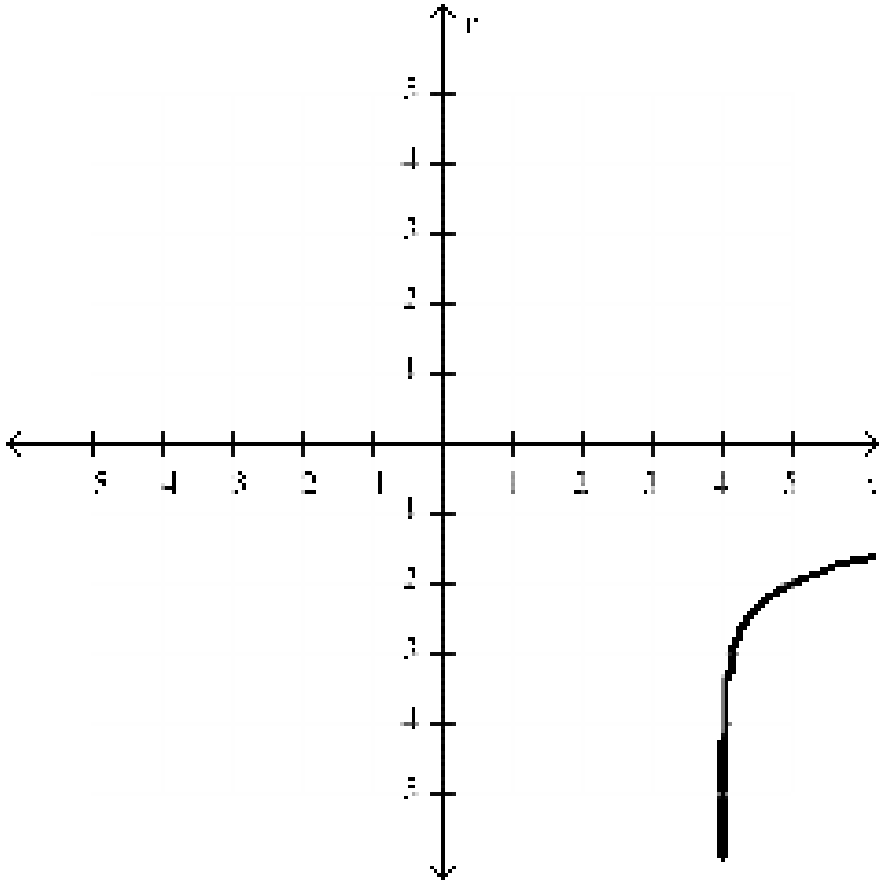
(A) $x >= – 7$ (B) $x >= – 8$ (C) $x >= 6$ (D) $x <= – 7$

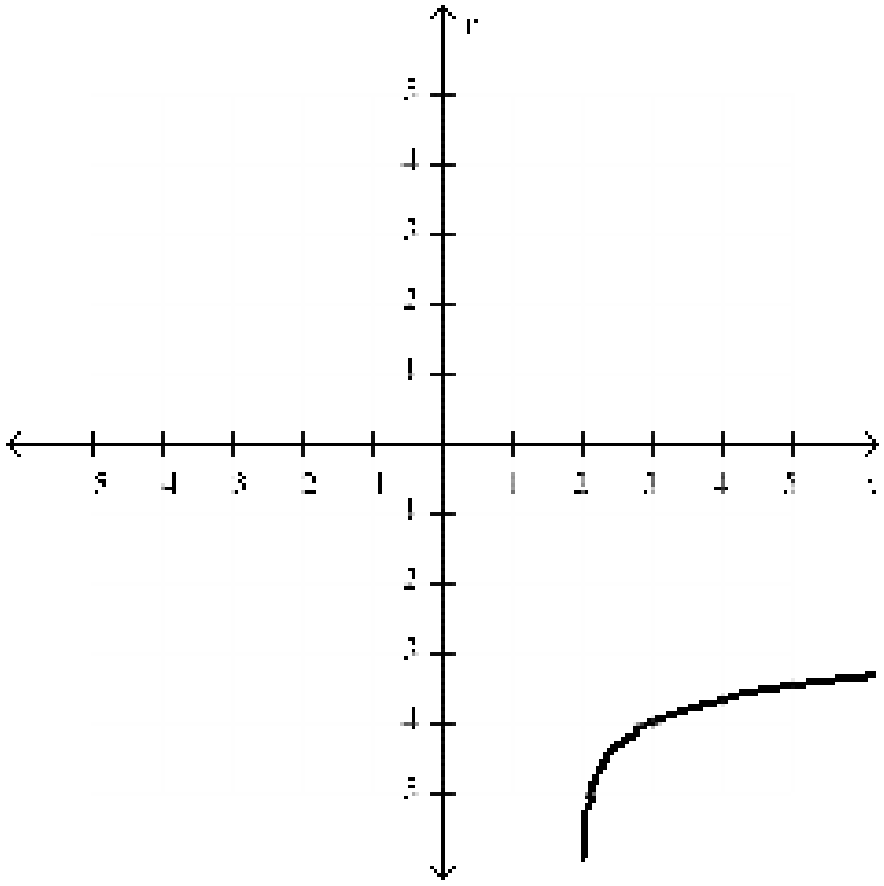
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

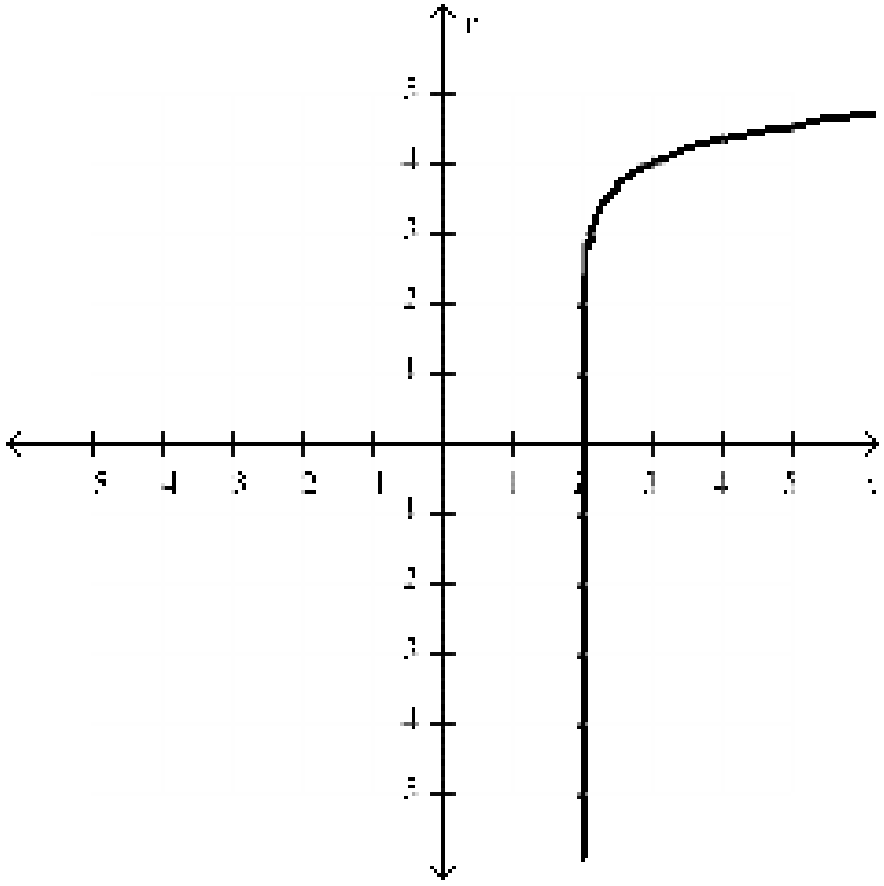
486. Which graph represents the function 8? Recovered graph functions: ($8)^(y-(−2))+(−4); (8)^(y-(−2))-(−4); (8)^(y-(−4))-(−2); (8)^(y+(−4))-(−2)$. Rendered ExamView reference:

$$y = \log_8(x - 2) - 4$$









(A) graph of (8)^(y-(−2))+(−4)

(B) graph of (8)^(y-(−2))-(−4)

(C) graph of (8)^(y-(−4))-(−2)

(D) graph of (8)^(y+(−4))-(−2)

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

487. What is the equation for the asymptote of the function $+ 2$?

(A) $x >= – 3$

(B) $x >= + 2$

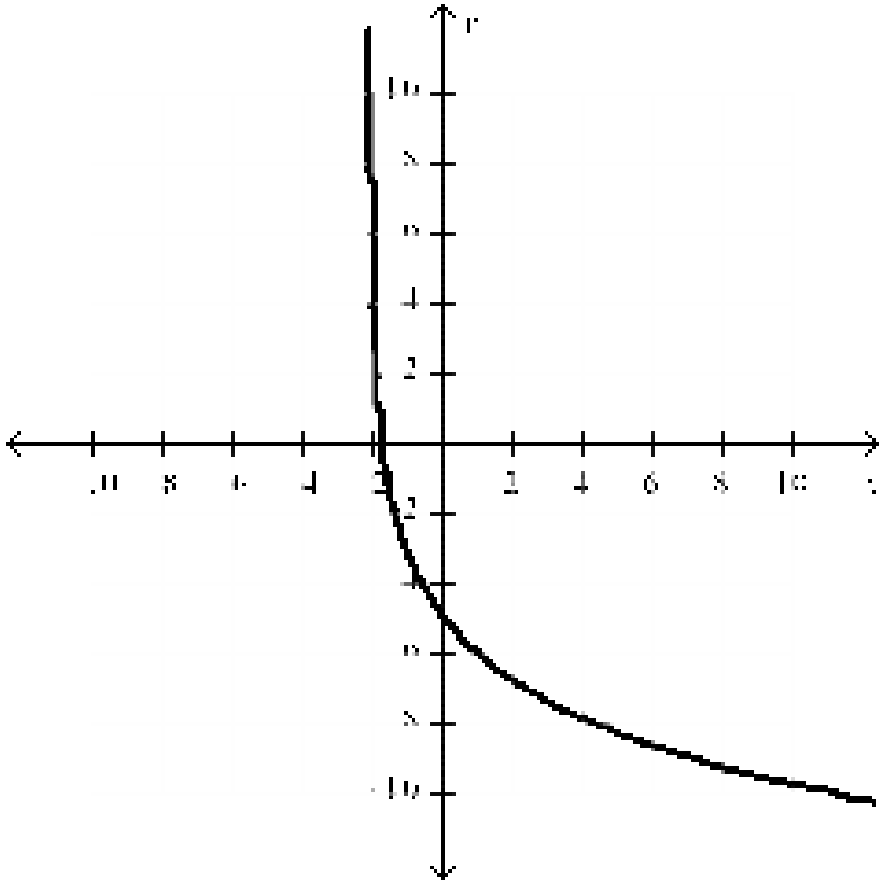
(C) $x >= –5$

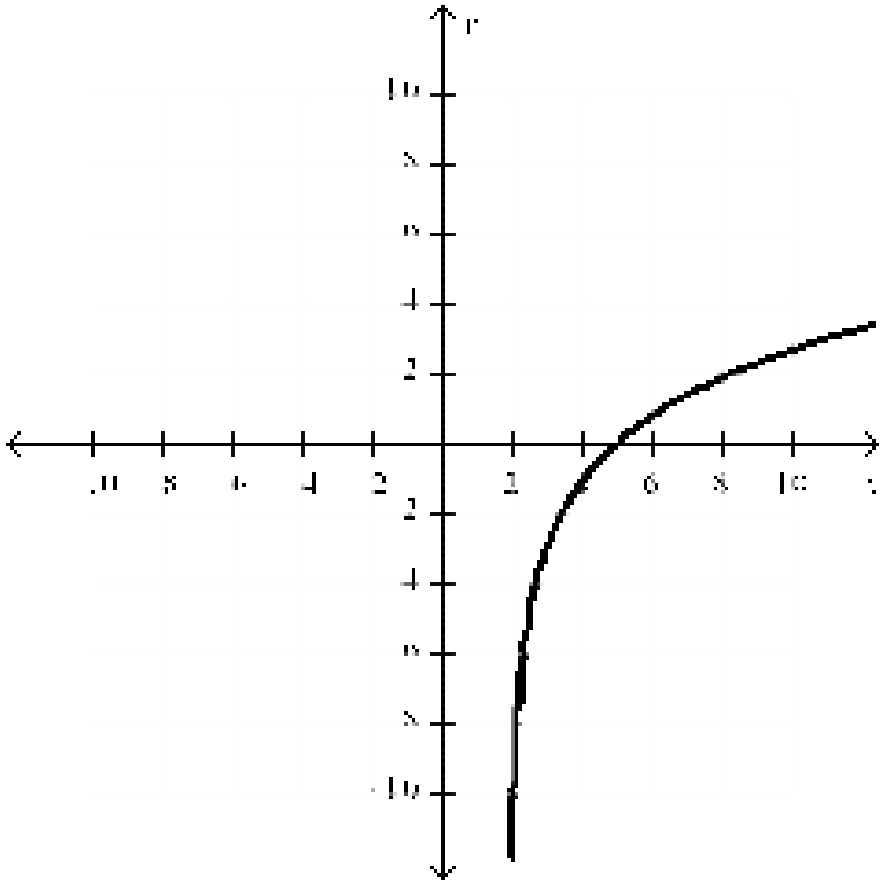
(D) $x <= – 3$

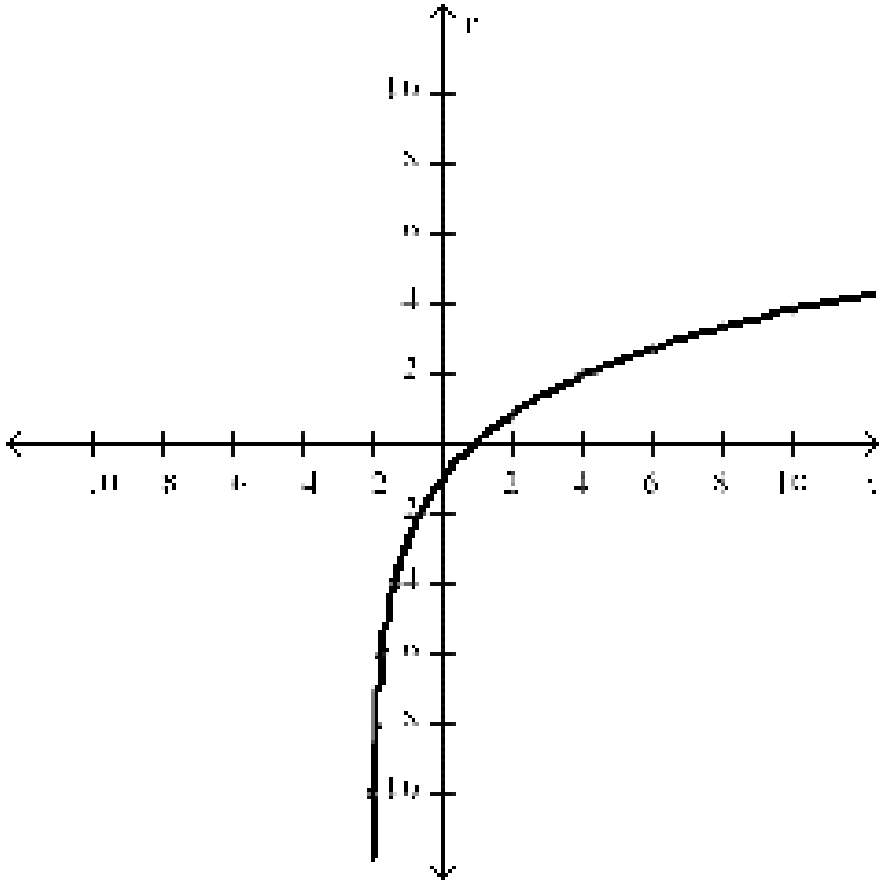
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

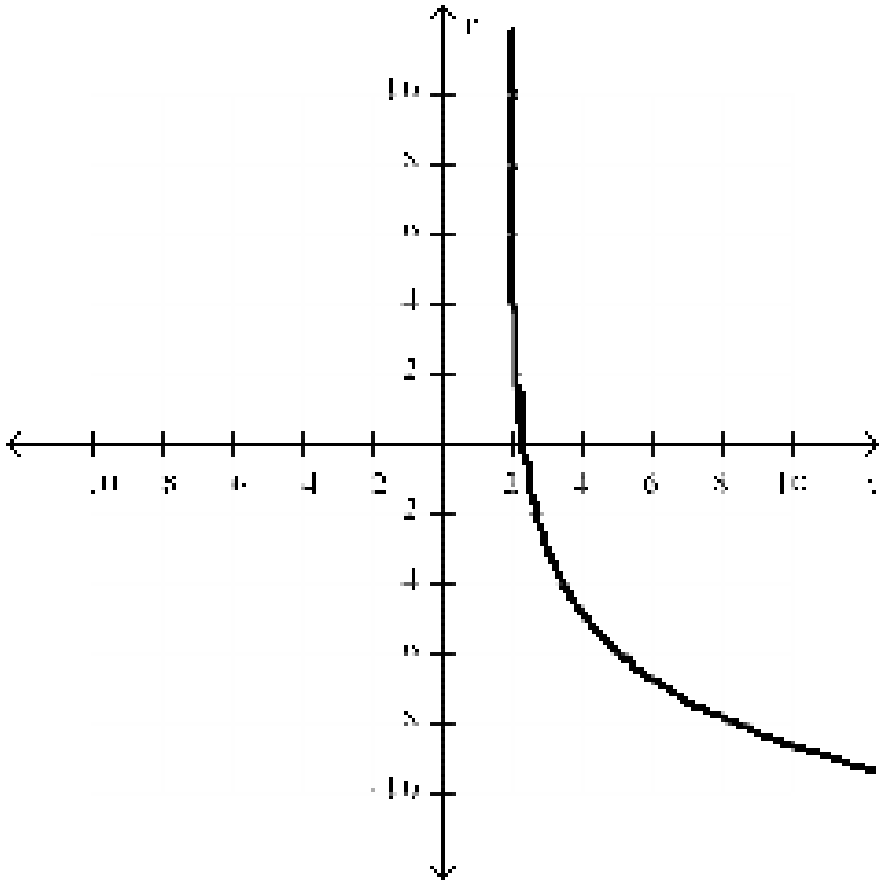
488. Which graph represents the function –3? Rendered ExamView reference:

$$y = -3\log_3[(x-2)] - 3$$







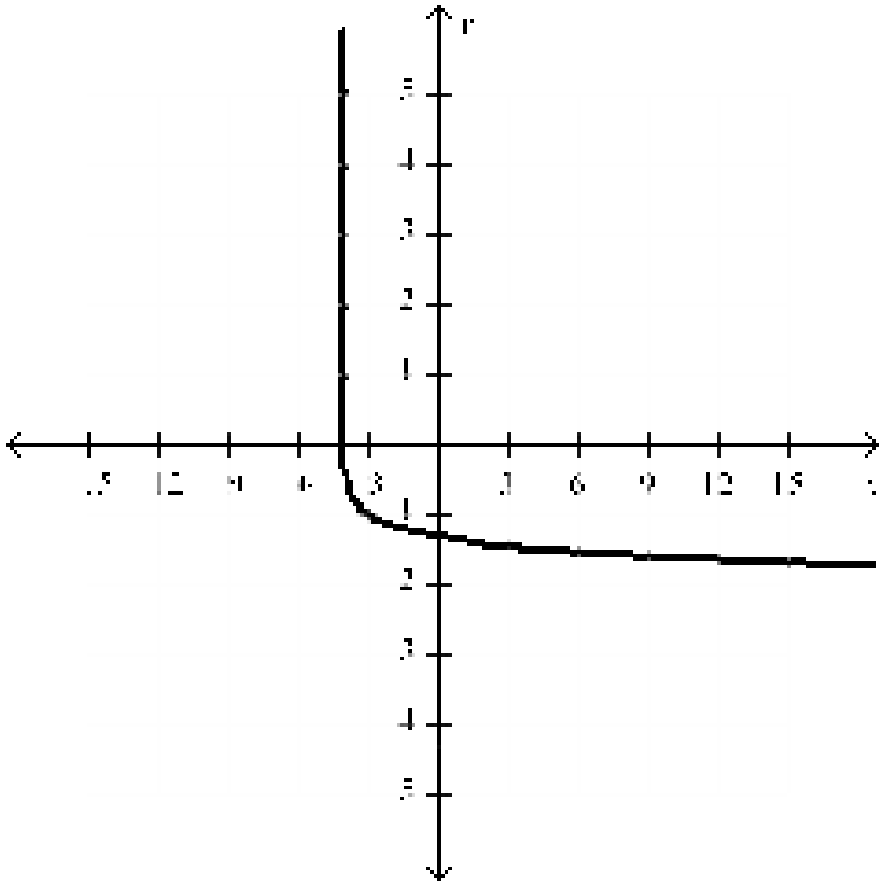


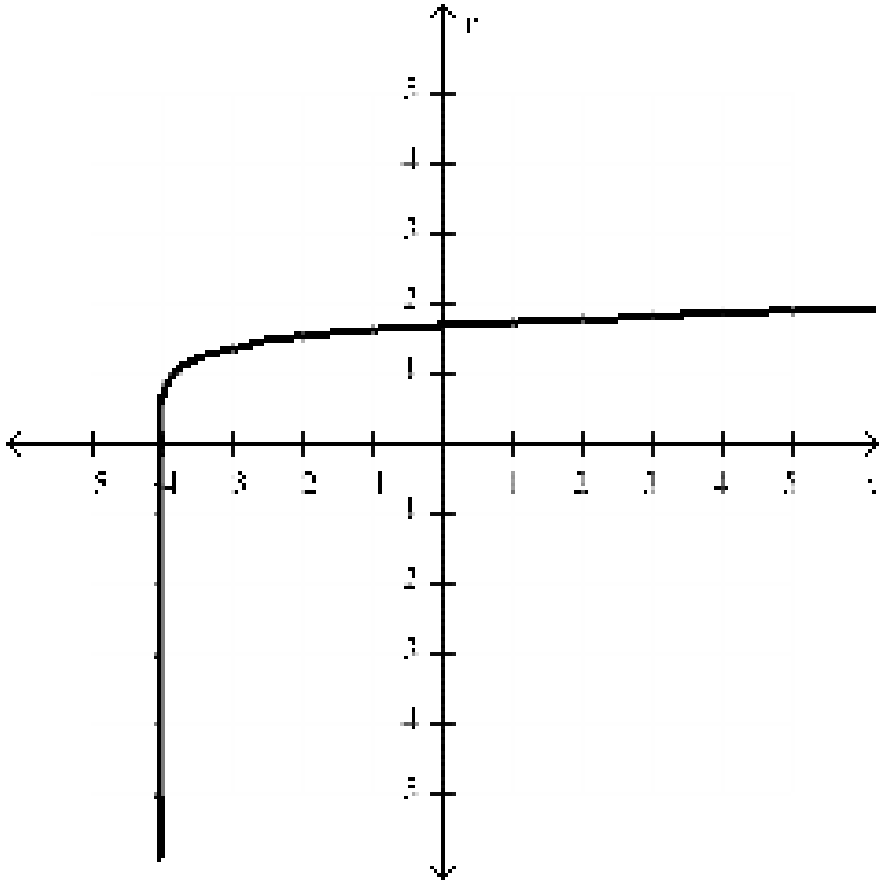
(A) $x >= – 3$ (B) $x >= – 2$ (C) $x >= range(−5,5,1)$ (D) $x <= – 3$

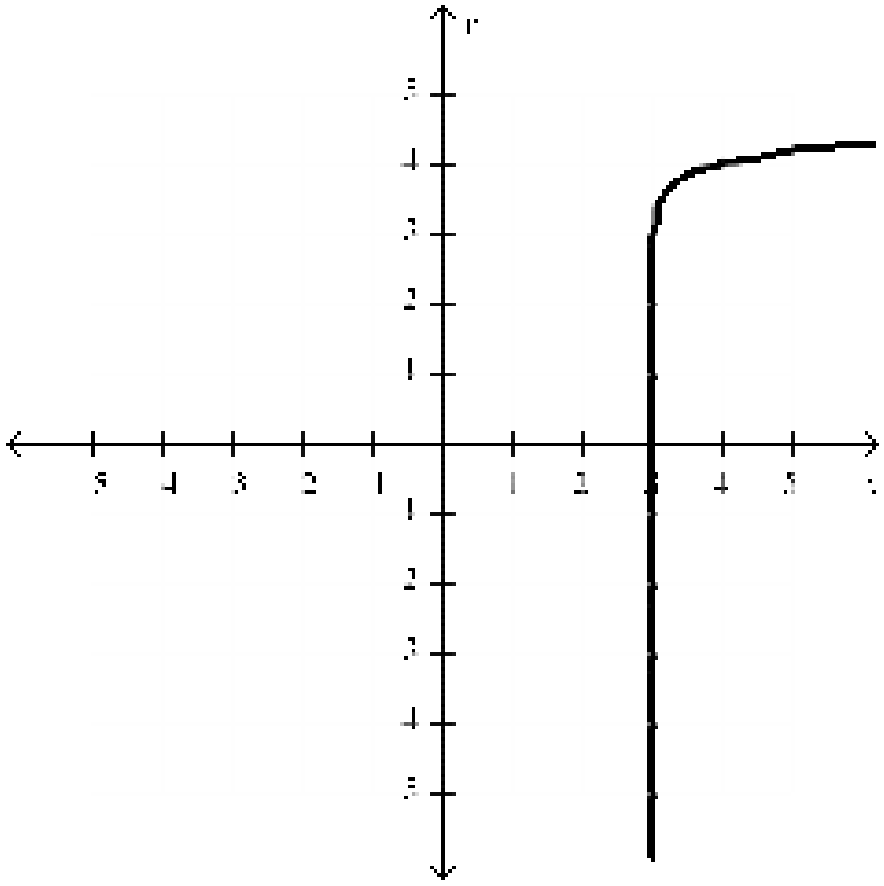
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

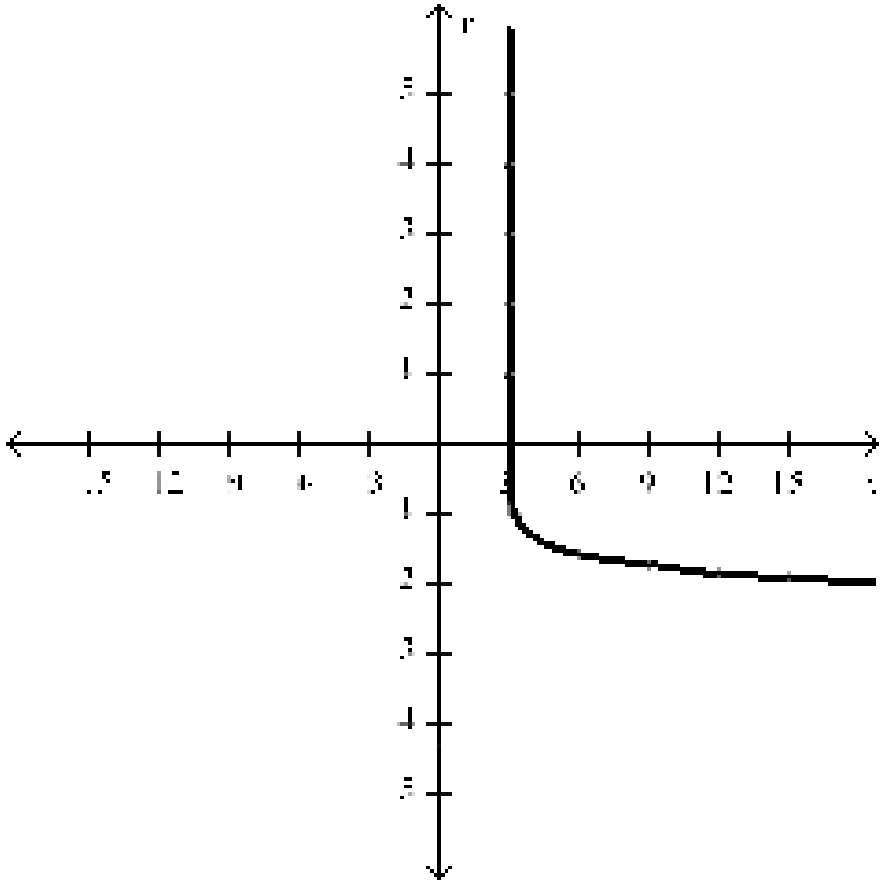
489. Which graph represents the function 3? Rendered ExamView reference:

$$y = \frac{1}{3}\log_4(x-3)+4$$









(A) 3 units right and 4 units up

(B) 3 units left and 4 units up

(C) 3 units right and 4 units down

(D) 3 units left and 4 units down

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

490. Which of the following is equivalent to the expression 4?

(A) s

(B) z

(C) 28

(D) v

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

491. If $3, s,$ and v, an algebraic expression in terms of z, s, and v for z is

(A) v 2 s + 2 z

(B) v 2(s + z)

(C) v 2(s z)

(D) v 2 s + z

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

492. Which if the following is equivalent to the expression 4?

(A) 4

(B) w

(C) 4

(D) y

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

493. Evaluate the expression 6 to the nearest hundredth.

(A) 7.76

(B) 10.82

(C) 9.76

(D) 7.01

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

494. The pH scale is used to measure the acidity or alkalinity of a solution. pH is defined as $4.3,$ where –6 is the concentration of hydronium ions, measured in moles per litre. Determine the pH of a solution with a concentration of 6.00. Round your answer to two decimal places.

(A) 6.00

(B) 0.78

(C) 5.37

(D) 3.52

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

495. Solve 3.

(A) 6 (B) $2/3$ (C) $3/2$ (D) 729

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

496. Solve $2x^2 + 5x + 4 = 0$.

(A) 3.06 (B) 2.95 (C) 7.26 (D) -1.40

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

497. A graduate student determines that the relationship between the length, s, in metres, of the skull and the body mass, m, in kilograms, of a particular species can be expressed using the equation 3.6045. Determine the body mass of an animal with a skull size of $3.4425 m$. Round your answer to the nearest kilogram.

(A) 547 (B) 343 (C) 376 (D) 313

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

498. Solve 3 to the nearest hundredth.

(A) -6.43 (B) 1.15 (C) 3.46 (D) -2.14

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

499. Solve 8. Round your answer to two decimal places. a. $486 c$. b. d.

(A) 3.59 (B) 1.78 (C) 2.97 (D) 0.34

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

500. The eye size of many vertebrates is related to body mass by the logarithmic equation $43,$ where E is the eye axial length, in millimetres, and m is the body mass, in kilograms. Predict the mass of a vertebrate with an eye axial length of 2.66 mm. Round your answer to the nearest hundredth of a kilogram.

(A) 2.66 (B) 1242.98 (C) 868.60 (D) 1.32

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

501. The function 9 written in exponential form is _____.

TOP: bnk | problem **TYPE:** problem

502. The variable

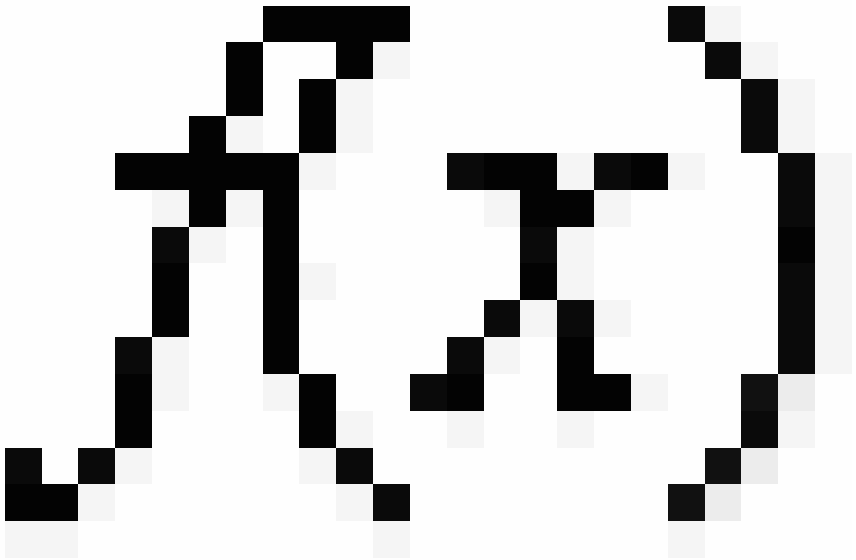
$$f(x) = a\log_c[b(x-h)] + k$$

in the function $f(x) = alog$ represents _____.

TOP: bnk | short_answer **TYPE:** short_answer

503. The graph of – 6 – 6 is the graph of – 6 translated _____. Rendered ExamView reference:

$$g(x) = \log_c(x-6)$$



TOP: bnk | true_false **TYPE:** true_false

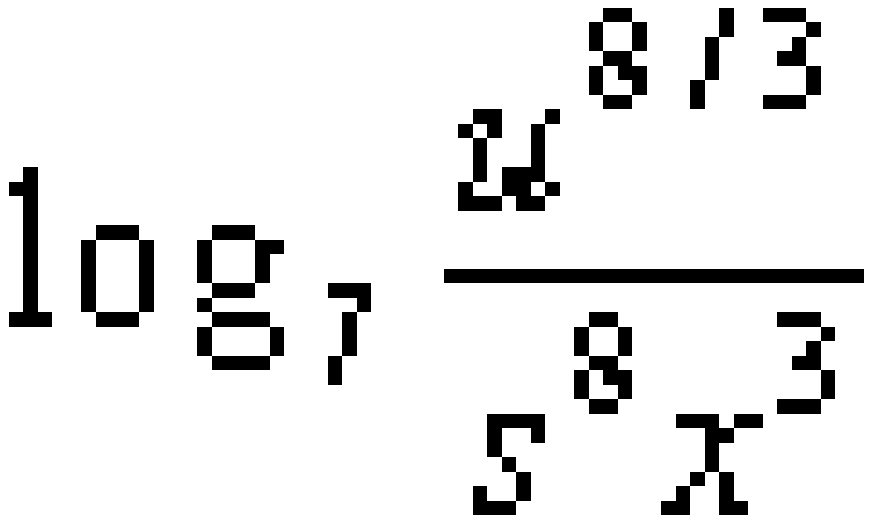
504. The domain of the function 5 is _____.

TOP: bnk | true_false **TYPE:** true_false

505. The equation power $log_c(MN) = log_c(M) + log_c(N)$ $log_c(M) N = log_c(M) - log_c(N)$ is an example of the _____ law of logarithms.

TOP: bnk | true_false **TYPE:** true_false

506.



TOP: bnk | short_answer **TYPE:** short_answer

507.

$$\log_7 \frac{w^{8/3}}{s^8 x^3}$$

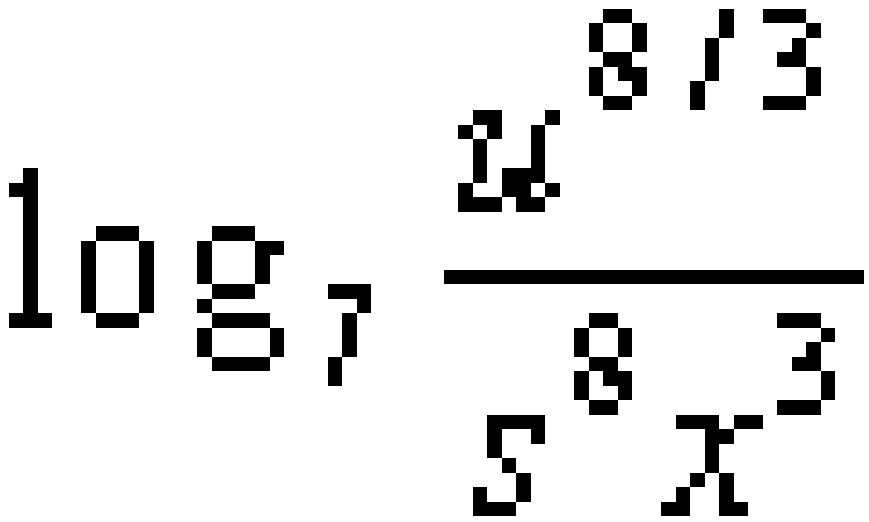
TOP: bnk | short_answer **TYPE:** short_answer

508.

$$\log_7 \frac{w^{8/3}}{s^8 x^3}$$

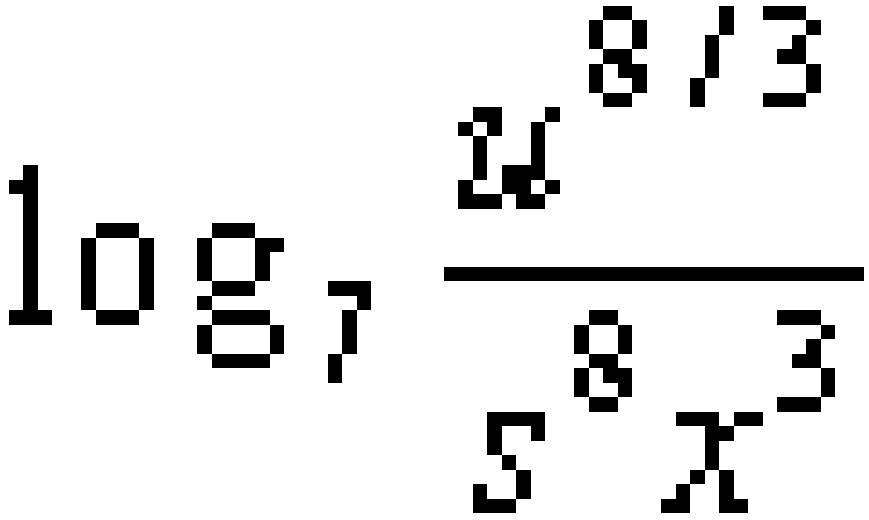
TOP: bnk | short_answer **TYPE:** short_answer

509.



TOP: bnk | short_answer **TYPE:** short_answer

510.



TOP: bnk | short_answer **TYPE:** short_answer

511. Evaluate $log_2(32)$.

TOP: bnk | short_answer **TYPE:** short_answer

512. Evaluate $log_2(64) + log_3(27) log_4(1)$.

TOP: bnk | short_answer **TYPE:** short_answer

513. Solve the system of equations: 0 Add these equations to find m: 0 Subtract the first equation from the second to find n: 0

TOP: bnk | problem **TYPE:** problem

514. Sketch the graph of the function $2*log(x-4)+1$. Rendered ExamView reference:

$$y = 2\log(x-4)+1$$

TOP: bnk | problem **TYPE:** problem

515. Sketch the graph of the function $log(−3*x+9)−2$. Rendered ExamView reference:

$$y = \log(-3x+9)-2$$

TOP: bnk | problem **TYPE:** problem

516. Graph the function $-log(x+2)−1$. Identify the domain, the range, and the equation of the vertical asymptote. Rendered ExamView reference:

$$f(x) = -\log(x+2)-1$$

TOP: bnk | problem **TYPE:** problem

517. A certain type of exponential growth can be described by the equation $N = N_0 10^(kt)$, where N_0 is the initial amount; k is the doubling time, in years; and N is the amount after time, t, in years, has passed. Suppose that the population of a small town doubles every 25 years. How long does it take to triple, to the nearest tenth of a year?

TOP: bnk | short_answer **TYPE:** short_answer

518. Given

$$\log_2 7 \approx 2.8074$$

, find the value of

$$\log_2 14$$

.

TOP: bnk | short_answer **TYPE:** short_answer

519. Solve the equation $3x + 1 = 2 2x - 3$. Leave your answer in exact form.

TOP: bnk | short_answer **TYPE:** short_answer

520. Solve for x. $log 4 (x + 4) - log 4 (x + 12) = 1$

TOP: bnk | short_answer **TYPE:** short_answer

521. What is the intensity of the mixture if 50 mL of dye was spilled? Round your answer to two decimal places.

TOP: bnk | short_answer **TYPE:** short_answer

522. The magnitude of an earthquake is defined as $M = log(A / A_0)$, where A is the amplitude of the ground motion and A_0 is the amplitude corrected for the distance from the actual earthquake that would be expected for a standard earthquake. On March $2, 2012,$ an earthquake with an amplitude $M = log 10 5.1 A_0 A_0 M = log10 5.1 M = 5.1$ times [DIAGRAM NEEDED] was recorded in Norman Wells, Northwest Territories.

(a) What was the earthquake s magnitude on the Richter scale?
(b) How does the earthquake in Norman Wells compare to the earthquake off Vancouver Island in 1946 that measured 7.3 on the Richter scale?

TOP: bnk | problem **TYPE:** problem

523. List the steps and explain the effect of each transformation to graph the function

$$y = -3\log\left[-2(x-1)\right] + 4$$

.

TOP: bnk | short_answer **TYPE:** short_answer

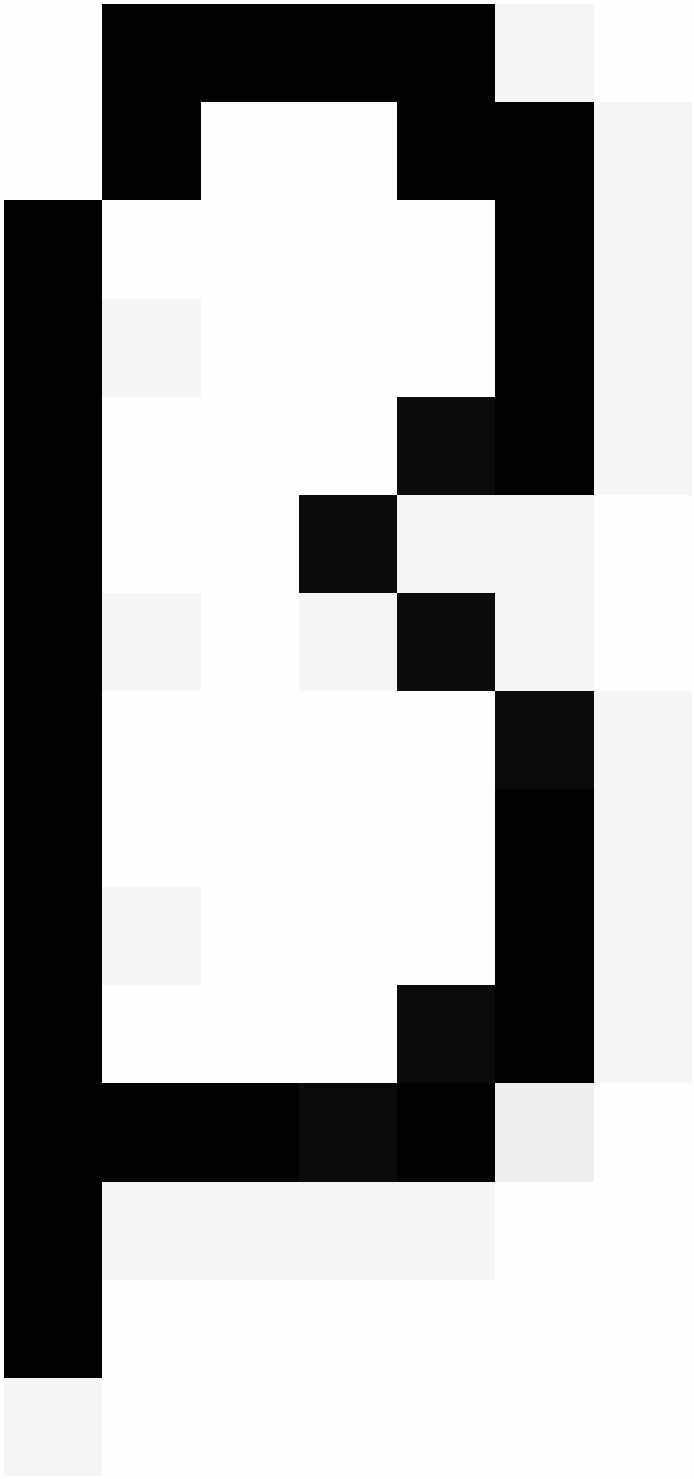
524. Given $log(7) = 0.8451$ and $log(2) = 0.3010$, find the value of $log(28)$.

TOP: bnk | short_answer **TYPE:** short_answer

525. (a) Sketch the graph of the function $3*log(2*(x-1))+4$. Identify the key features.
(b) Identify the domain and range.
(c) Identify the equation of the vertical asymptote. [Graph diagram: Graph: $3*log(2*(x-1))+4$ | x: [$1.5, 10.0$], y: [$4.0, 7.8$]]

TOP: bnk | problem **TYPE:** problem

526. The intensity level,



, in decibels, of sound is defined as

$$\beta = 10 \log\left(\frac{I}{I_0}\right)$$

, where I is the intensity of the sound in watts per square metre.The ratio of the intensity of sound pollution measured at a small airport runway versus that of the local highway is 6420.4. If the sound level on the local highway is 91 dB, determine the sound level on the runway, to the nearest decibel.

TOP: bnk | short_answer **TYPE:** short_answer

527. Find the difference in pH levels of a swimming pool with a hydronium ion concentration of

$$4.4 \times 10^{-8}$$

mol$/L$ and a swimming pool with a hydronium ion concentration of

$$5.7 \times 10^{-8}$$

mol$/L,$ to the nearest hundredth.

TOP: bnk | short_answer **TYPE:** short_answer

528. The stellar magnitude scale compares the brightness of stars using the equation $m_2 - m_1 = log(b_1 / b_2)$, where m_1 and m_2 are the apparent magnitudes (how bright the stars appear in the sky) of the two stars being compared, and b_1 and b_2 are their brightnesses (how much light they emit). a) The brightest appearing star in our sky, Sirius, has an apparent magnitude of $−1.5$. How much brighter does Sirius appear than Betelgeuse, whose apparent magnitude is 0.12? Round your answer to the nearest whole number. b) The Sun appears about $1.3 times 10^10$ times as bright in the sky as does Sirius. What is the apparent magnitude of the Sun, to the nearest tenth?

TOP: bnk | problem **TYPE:** problem

529. Prove that

$$\log a + \log a^2 + \log a^3 - \log a^6 = \log 1$$

.

TOP: bnk | short_answer **TYPE:** short_answer

530. Show that

$$\frac{1}{\log_a b} = \log_b a$$

.

TOP: bnk | short_answer **TYPE:** short_answer

531. Show that

$$3\log\sqrt{x} + 2\log x - \frac{1}{2}\log x = 3\log x$$

.

TOP: bnk | short_answer **TYPE:** short_answer

532. Sketch the graphs of $y = log_(1/2)(x)$, $y = log_(1/3)(x)$, and $y = log_(1/4)(x)$, and then describe the pattern in terms of n for $y = log_(1/n)(x)$. Rendered ExamView reference:

$$y = \log_{\frac{1}{2}} x$$

$$y = \log_{\frac{1}{3}} x$$

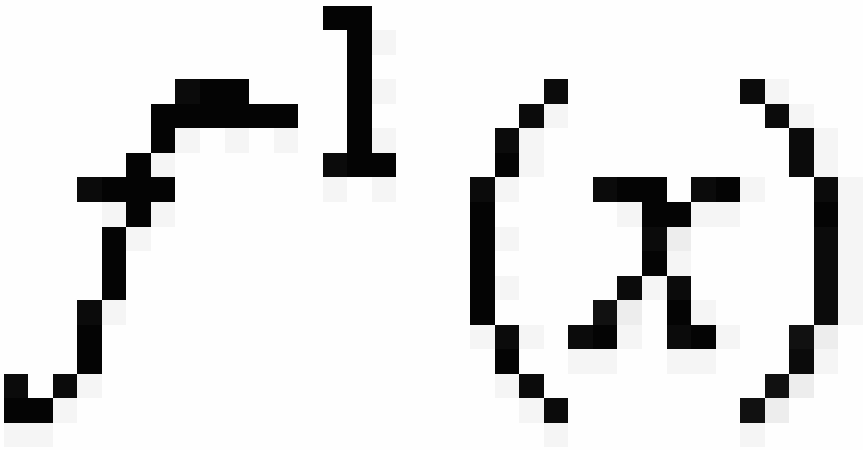
$$y = \log_{\frac{1}{4}} x$$

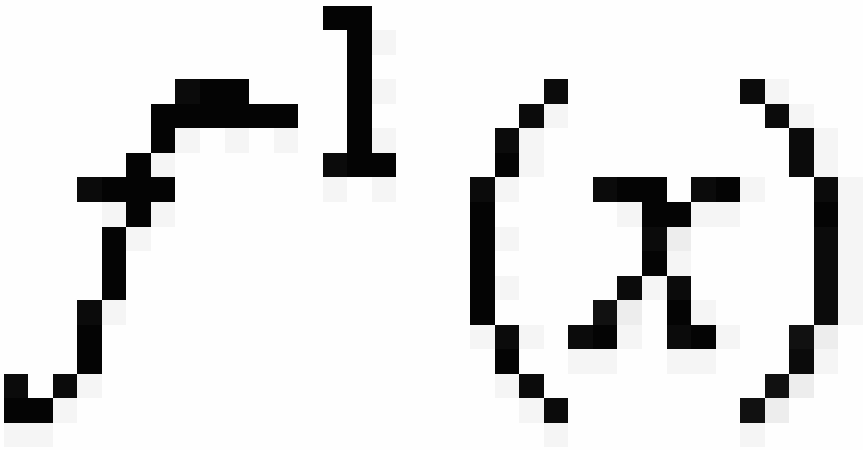
$$y = \log_{\frac{1}{x}} x$$

TOP: bnk | problem **TYPE:** problem

533. Sketch a graph of the function $2^(2*x)−1$ and its inverse [Graph diagram: Graph: $2^(2*x)−1$ | x: [$−10.0, 9.5$], y: [$−1.0, 524287.0$]]. Compare the graph of [Graph diagram: Graph: $2^(2*x)−1$ | x: [$−10.0, 9.5$], y: [$−1.0, 524287.0$]] to the graph of the function $2^(2*x)−1$. What conclusions can you make? Rendered ExamView reference:

$$f(x) = 2^{2x} - 1$$





$$g(x) = \frac{1}{2}\log_2(x+1)$$

TOP: bnk | problem **TYPE:** problem

534. A $200-g$ sample of a radioactive substance is placed in a chamber to be tested. After $3 h, 140 g$ of the sample remains.

(a) Determine the half$-$life of this substance, to the nearest hundredth of an hour.
(b) Graph the amount of the substance remaining as a function of time.

TOP: bnk | problem **TYPE:** problem

535. Prove that

$$\log_a M^p = p \log_a M$$

. Rendered ExamView reference:

$$\log_7 s - \log_7 w + 3\log_7 x$$

$$8 / 3\log_7 w - 8\log_7 s + 3\log_7 x$$

$$8\log_7 s - \log_7 w - 3\log_7 x$$

$$8\log_7 s - 8 / 3\log_7 w + 3\log_7 x$$

$$8 / 3\log_7 w - 8\log_7 s - 3\log_7 x$$

$$8\log_7 s + 8 / 3\log_7 w + 3\log_7 x$$

TOP: bnk | short_answer **TYPE:** short_answer

536. (a) Graph the function $log(x)/log(3)$.
(b) Graph the following functions on the same graph: [Graph diagram: Graph: $log(x)/log3$ | x: [$0.5, 10.0$], y: [$−0.6, 2.1$]] c) Explain the effect of the constant k in the function $log(3*x)/log3$.

TOP: bnk | problem **TYPE:** problem

537. Explain the steps used to solve the equation $log_2(x) 2 - 8x = log 2$.

TOP: bnk | short_answer **TYPE:** short_answer

538. Solve the equation $log_x(2) + 48x 3 = 2$.

TOP: bnk | short_answer **TYPE:** short_answer

539. For his dream car, Bruce invested $18 000 at 7.8% interest, compounded monthly, for 5 years. After the 5 years, he still did not have enough money. How much longer will he have to invest the money at 5% interest, compounded daily, to have a total of $$35 000$? Round to the nearest tenth of a year.

TOP: bnk | short_answer **TYPE:** short_answer

540. Sketch the graph of the function $2*(10^((y+1)/-2))−1-0.370$. Determine the $x-$intercept algebraically, to the nearest hundredth. Rendered ExamView reference:

$$f(x) = -2\log\left\lfloor \frac{1}{2}(x+1) \right\rfloor - 1$$

TOP: bnk | problem **TYPE:** problem

541. Which of the reciprocal functions has a vertical asymptote with equation 9?

(A) 2 (B) 9 (C) 9 (D) 2

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

542. What is true about the behaviour of the function \$4\$ as \$+ 5 −5/4\$ \$f(x)\$ (right to left)?

(A) + 5 (B) \$−5/4\$ (C) if(denom<0,chr(43),chr(45)) (D) if(denom>0,chr(43),chr(45))

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

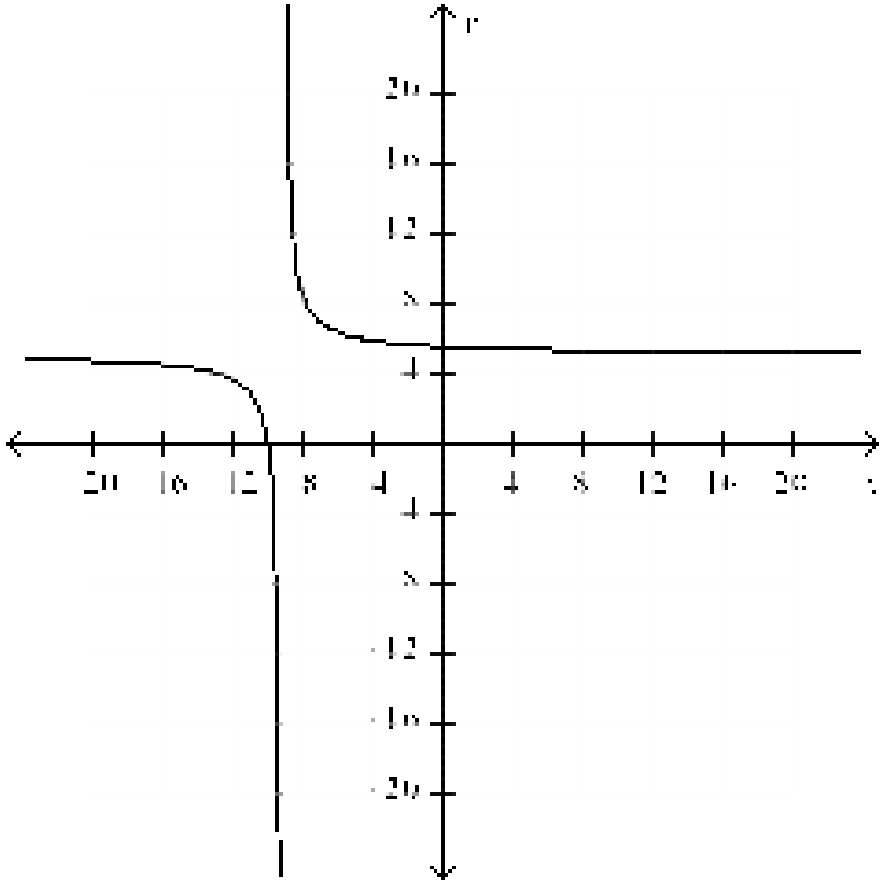
543. What is the \$x-\$intercept of 2?

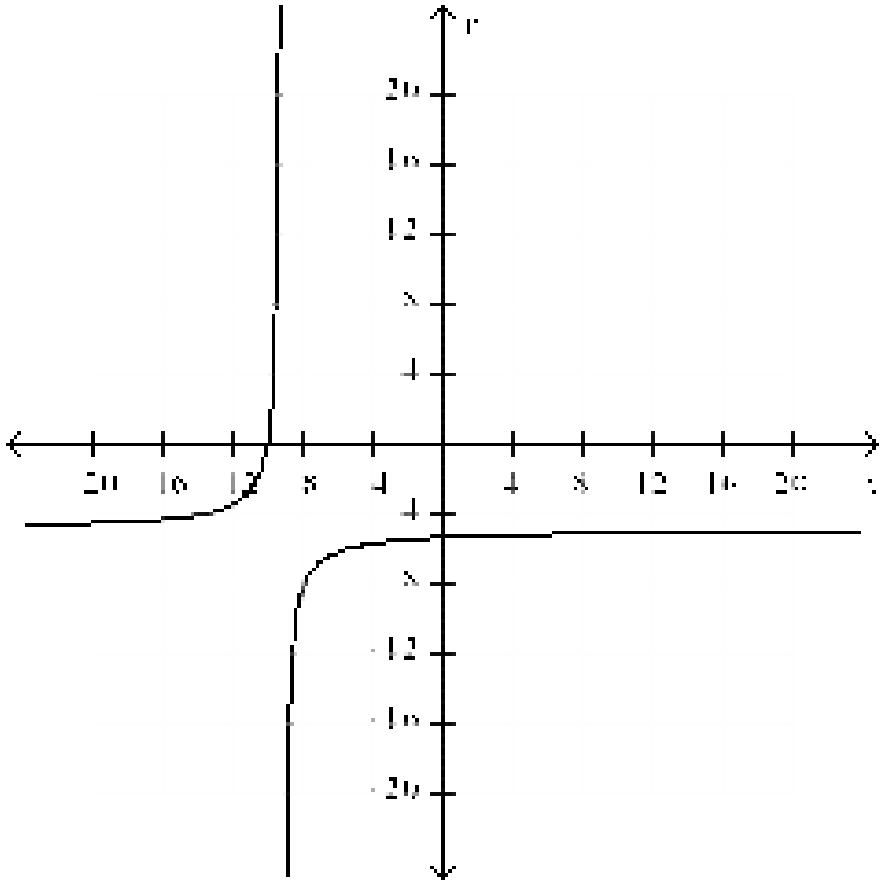
(A) 2 (B) + 4 (C) −2 (D) \$−1/2\$

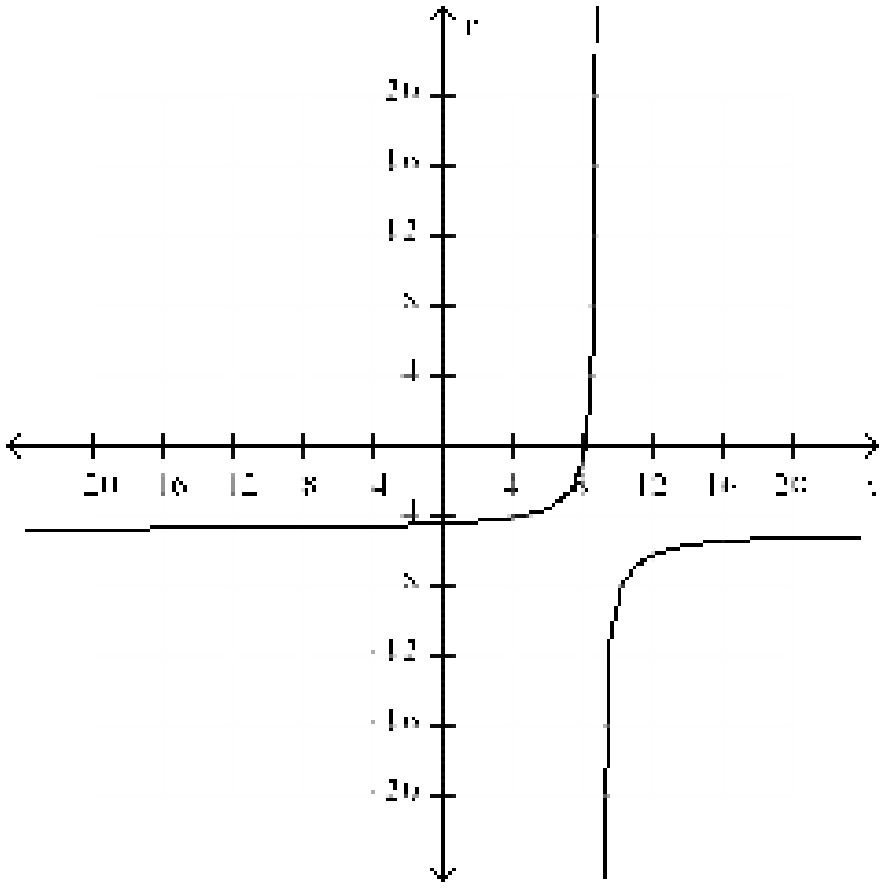
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

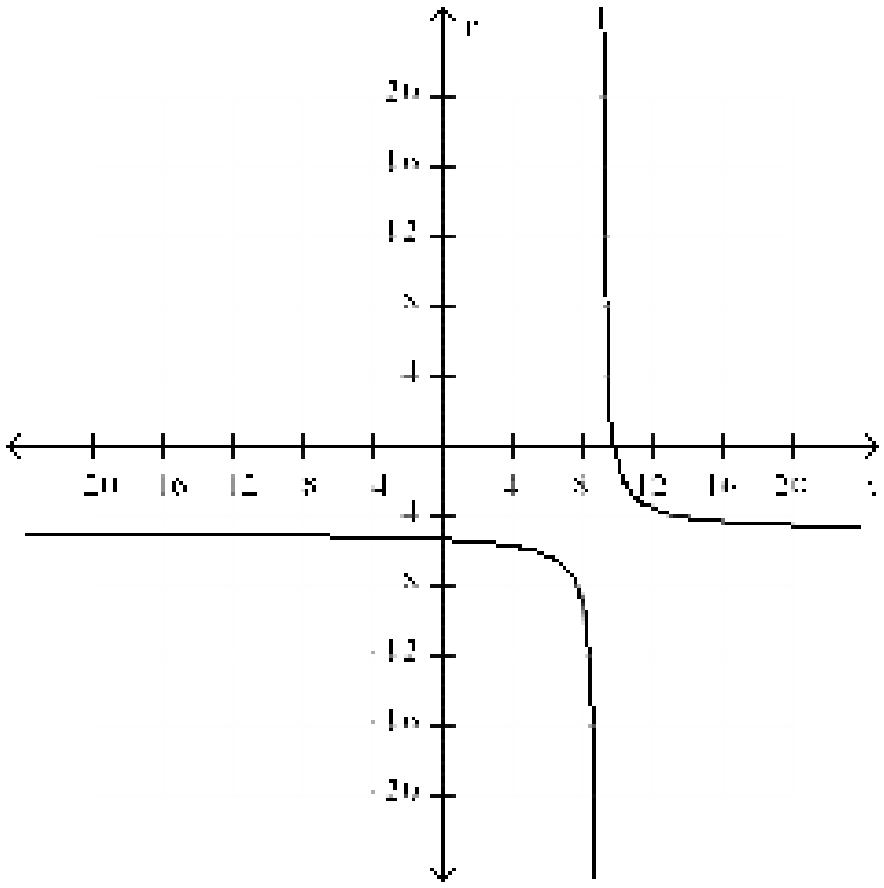
544. Which graph represents the function –\$4\$? Recovered graph functions: (\$-(−4)/(x-(−9)))-(−5); ((−4)/(x-(−9)))+(−5); ((−4)/(x+(−9)))+(−5); (-(−4)/(x+(−9)))+(−5)\$. Rendered ExamView reference:

$$f(x) = \frac{-4}{x-9} - 5$$


y
20
16
12
8
4
-4
-8
-12
-16
-20
-20
-16
-12
-8
-4
4
8
12
16
20
x


y
20
16
12
8
4
-4
-8
-12
-16
-20
-20
-16
-12
-8
-4
4
8
12
16
20
x





(A) graph of (-(−4)/(x-(−9)))-(−5)

(B) graph of ((−4)/(x-(−9)))+(−5)

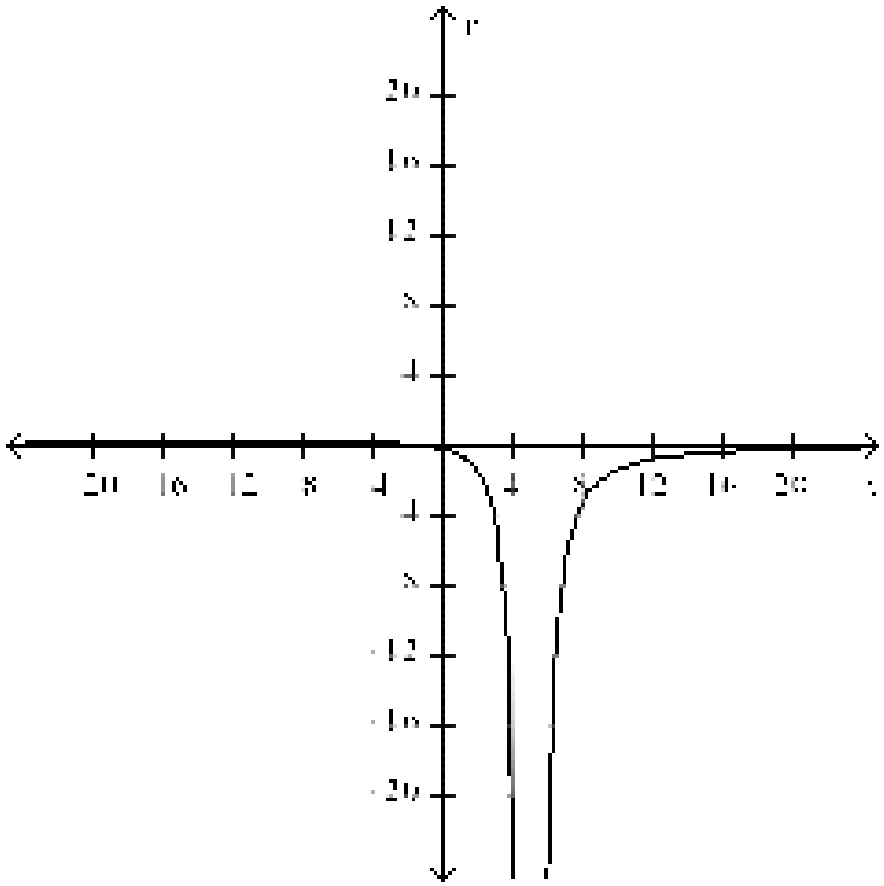
(C) graph of ((−4)/(x+(−9)))+(−5)

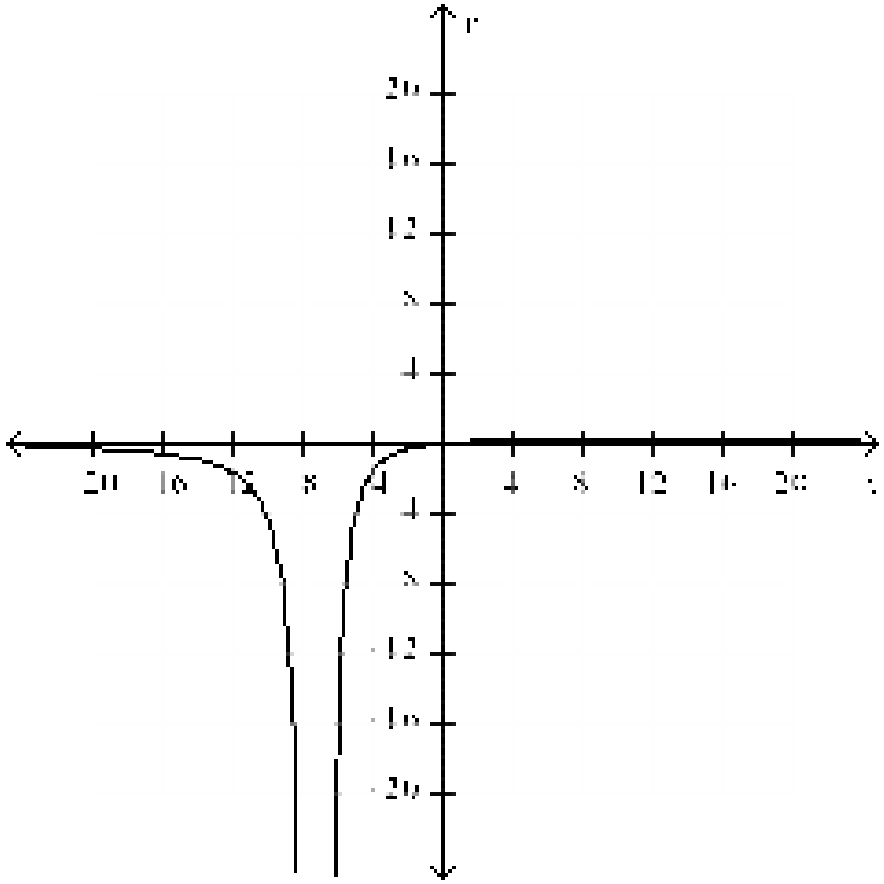
(D) graph of (-(−4)/(x+(−9)))+(−5)

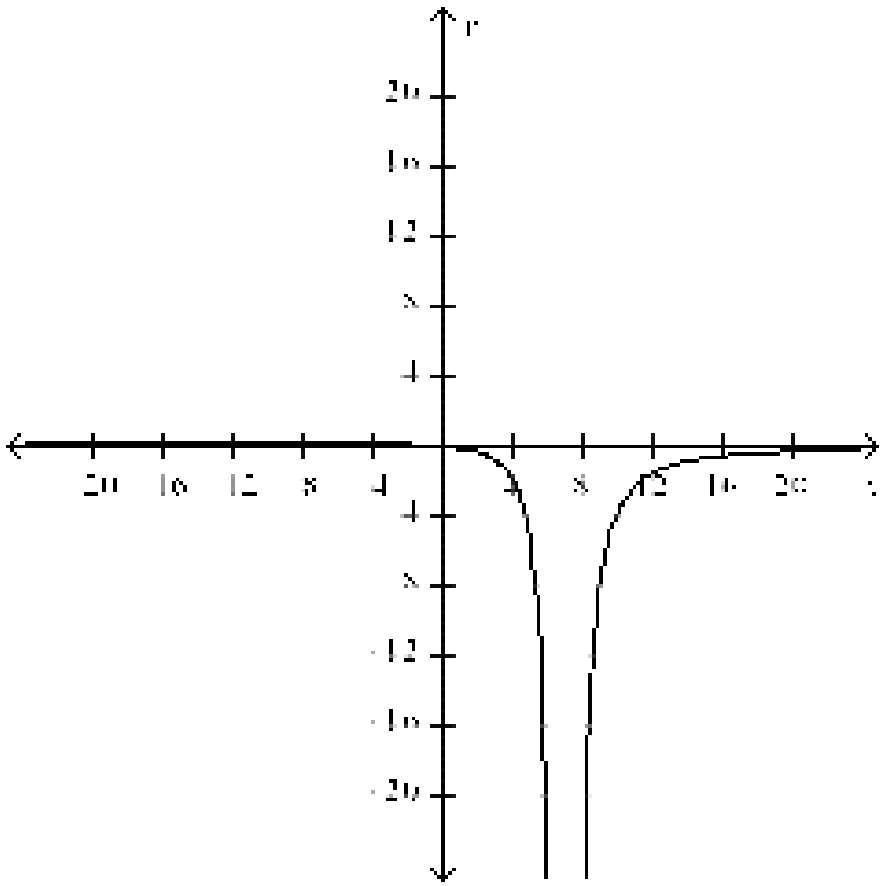
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

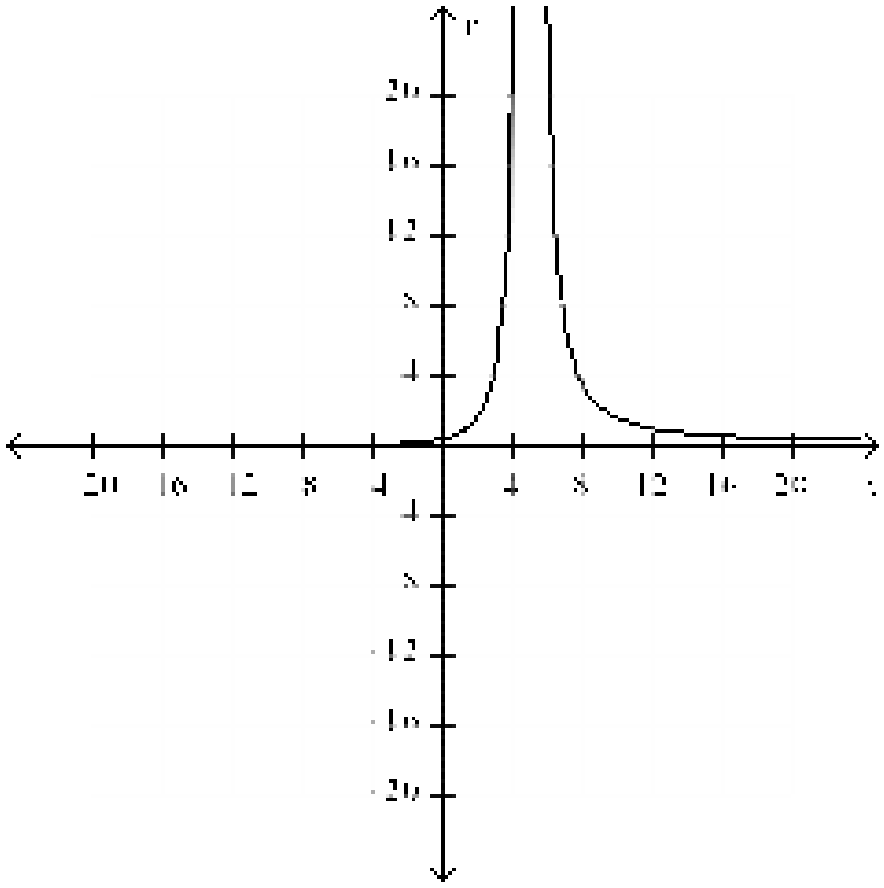
545. The rational function –3 is represented by which graph? Recovered graph functions: (($−3)*x+(−7))/(x+(−5))^2; -((−3)*x-(−5))/(x-(−7))^2; ((−3)*x+(−5))/(x+(−7))^2; -((−3)*x+(−7))/(x+(−5))^2$. Rendered ExamView reference:

$$f(x) = \frac{-3x - 7}{(x - 5)^2}$$









(A) graph of ((−3)*x+(−7))/(x+(−5))^2

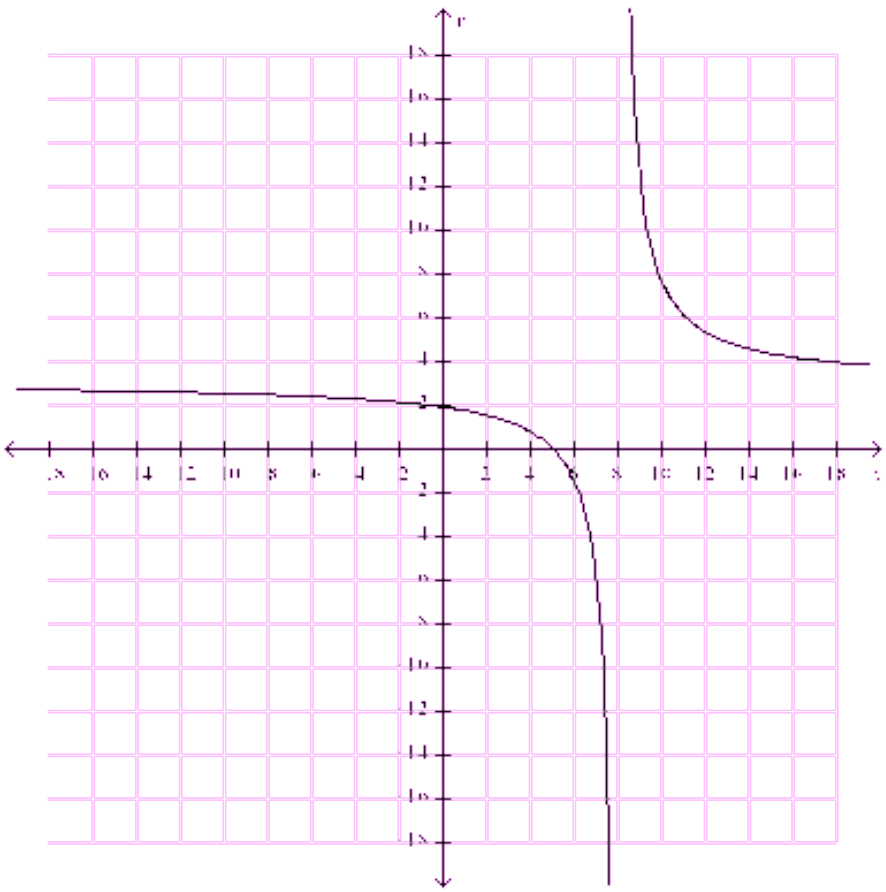
(B) graph of -((−3)*x-(−5))/(x-(−7))^2

(C) graph of ((−3)*x+(−5))/(x+(−7))^2

(D) graph of -((−3)*x+(−7))/(x+(−5))^2

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

546. Which function represents the graph shown below? Recovered graph function: (($9)/(x+(−8)))+(3)$. 9 Rendered ExamView reference:



$$f(x) = \frac{9}{x-8} + 3$$

$$f(x) = \frac{-9}{x+3} - 8$$

$$f(x) = \frac{-9}{x-8} + 3$$

$$f(x) = \frac{9}{x+3} - 8$$

(A) + 3 (B) 9 (C) + 3 (D) – 8

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

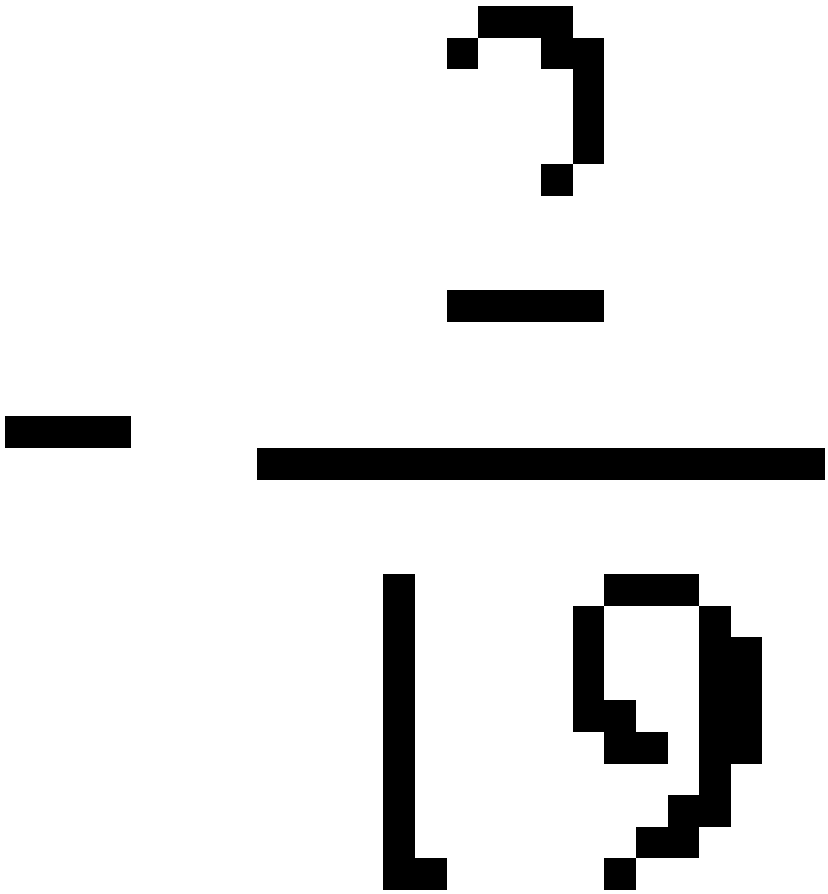
547. Which of the following functions has a slant asymptote when graphed? (e $=$ – $10, a = 5, e =$ – $10, f =$ – $15, g =$ – $3, h =$ – $4, d =$ – $4, d =$ – $4, d =$ – $4, d =$ – $4, a = 5, e =$ – $10, f =$ – $15, b =$ – $3, d =$ – $4, d =$ – $4, d =$ – $4, d =$ – $4, a = 5, e =$ – $10, f =$ – $15, g_dis =$ – $2, h_dis =$ – $3, d =$ – $4, d =$ – $4, d =$ – $4, d =$ – 4)

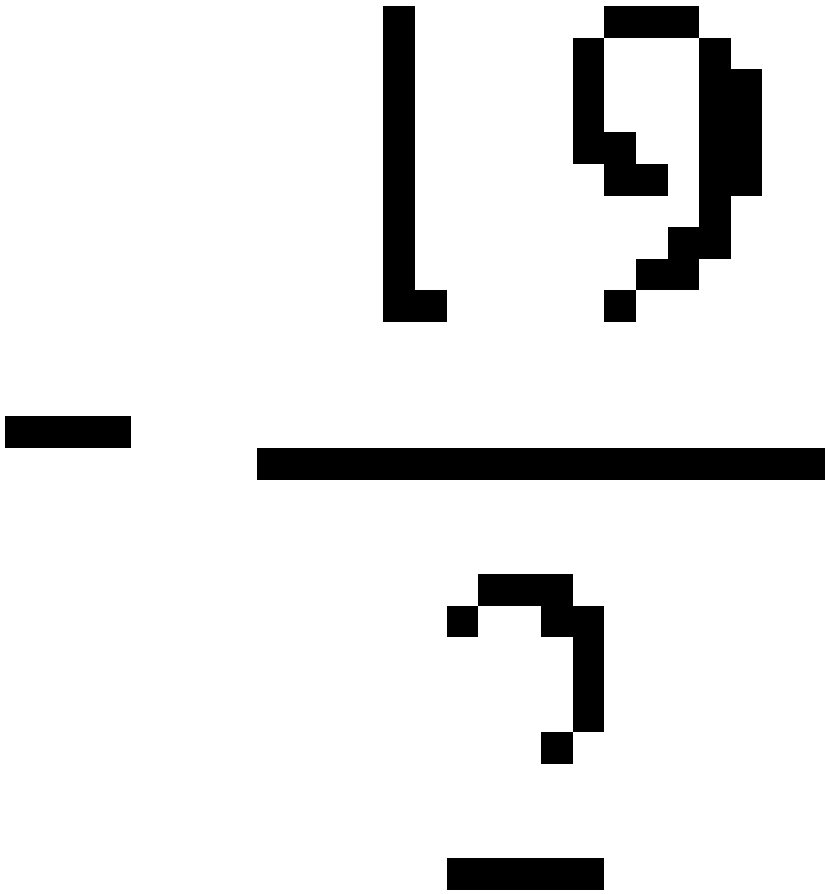
(A) $f(x) = (5x^2 - 10x - 15)/(x - 3)$ (B) $f(x) = (5x - 10)/(x - 3)$ (C) $f(x) = (−3x - 4)/(x - 3)$ (D) all of the above

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

548. What is the value of k in the function 5 if its graph passes through the point ($4, 4)$? Rendered ExamView reference:

$$f(x) = \frac{2-k}{5x+k}$$





(A) $−2/19$ (B) $−19/2$ (C) $−10$ (D) 4

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

549. Which function has vertical asymptotes with equations $x = −9$ and $x = −0.857$?

(A) $f(x) = 1/(7x^2 - 69x + 54)$ (B) $f(x) = 1/(7x^2 + 69x + 54)$ (C) $f(x) = 1/(7x^2 + 69x - 54)$ (D) $f(x) = (7x)/(x^2 + 69x + 54)$

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

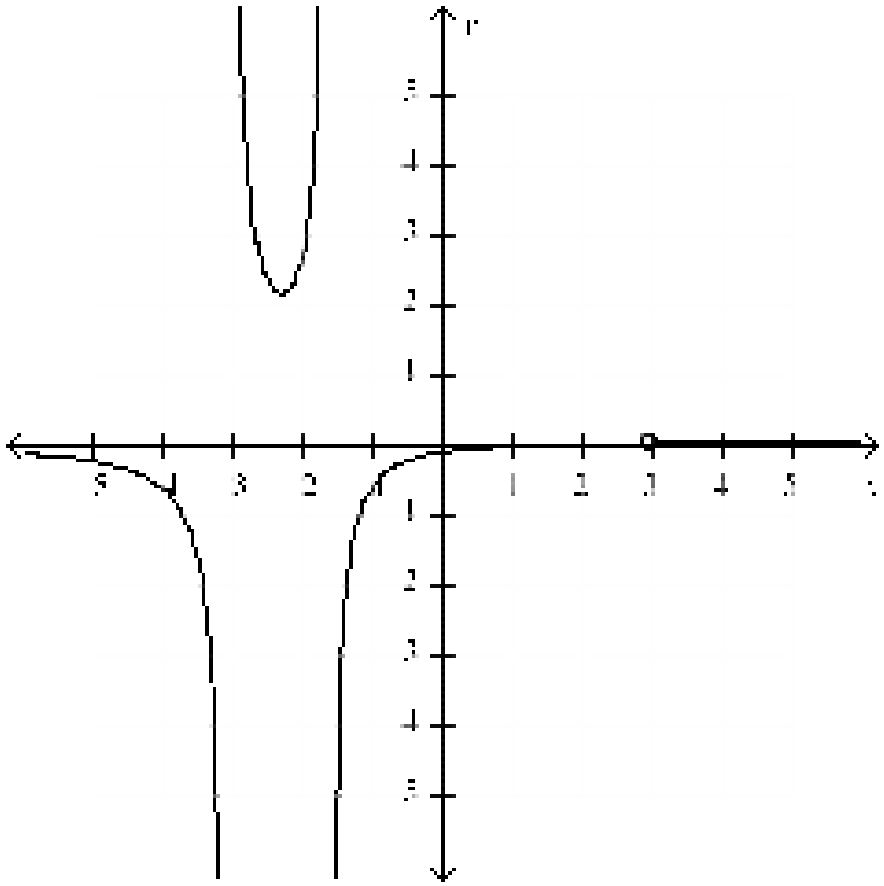
550. Which function has a point of discontinuity at $x = + 3$?

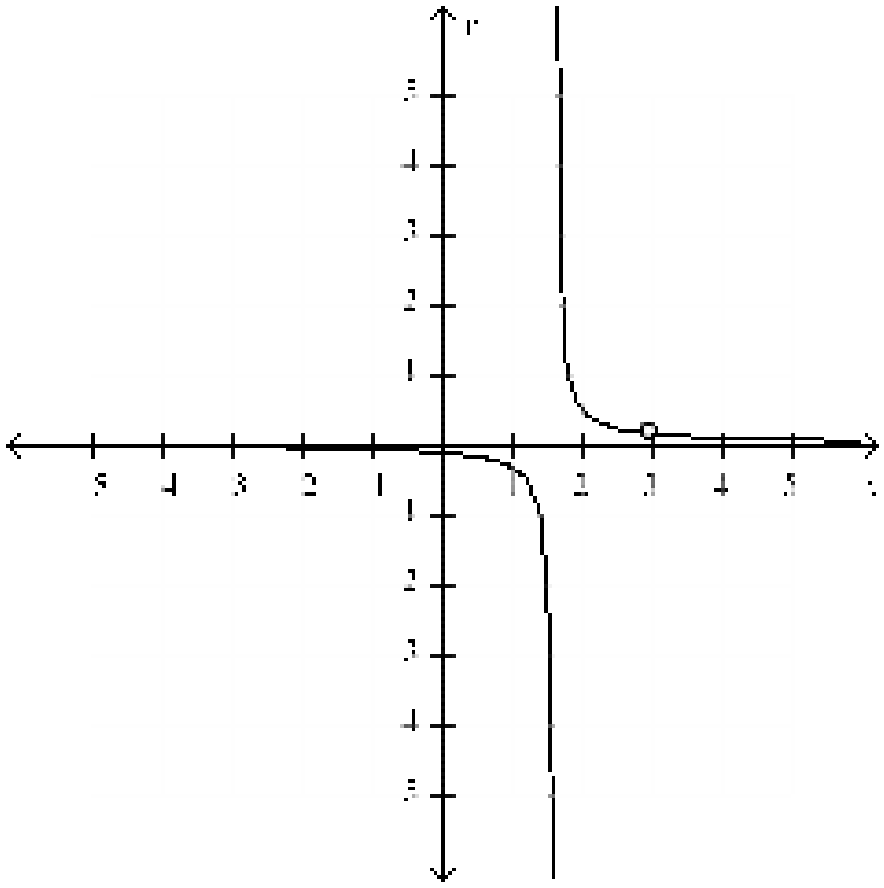
(A) $f(x) = ((x - 3)(x + 1))/((x - 3)(x + 2))$ (B) $f(x) = 1/(x - 3)$ (C) $f(x) = ((x + 3)(x + 1))/((x + 3)(x + 2))$ (D) $f(x) = ((x + 3)(x + 1))/((x - 3)(x + 2))$

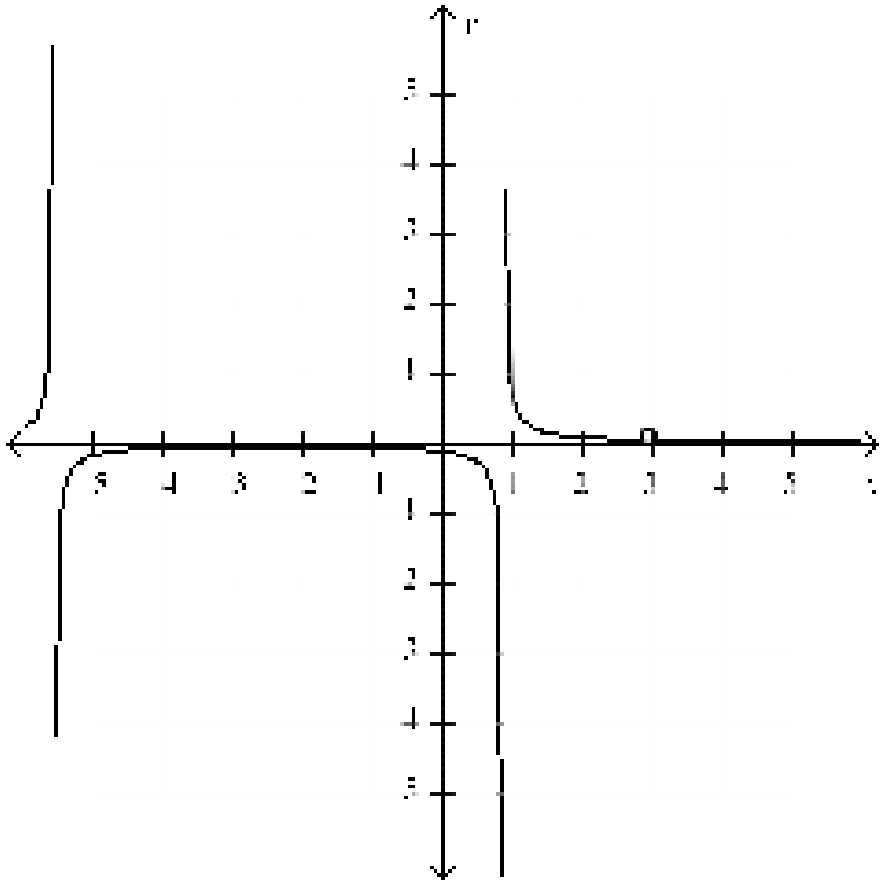
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

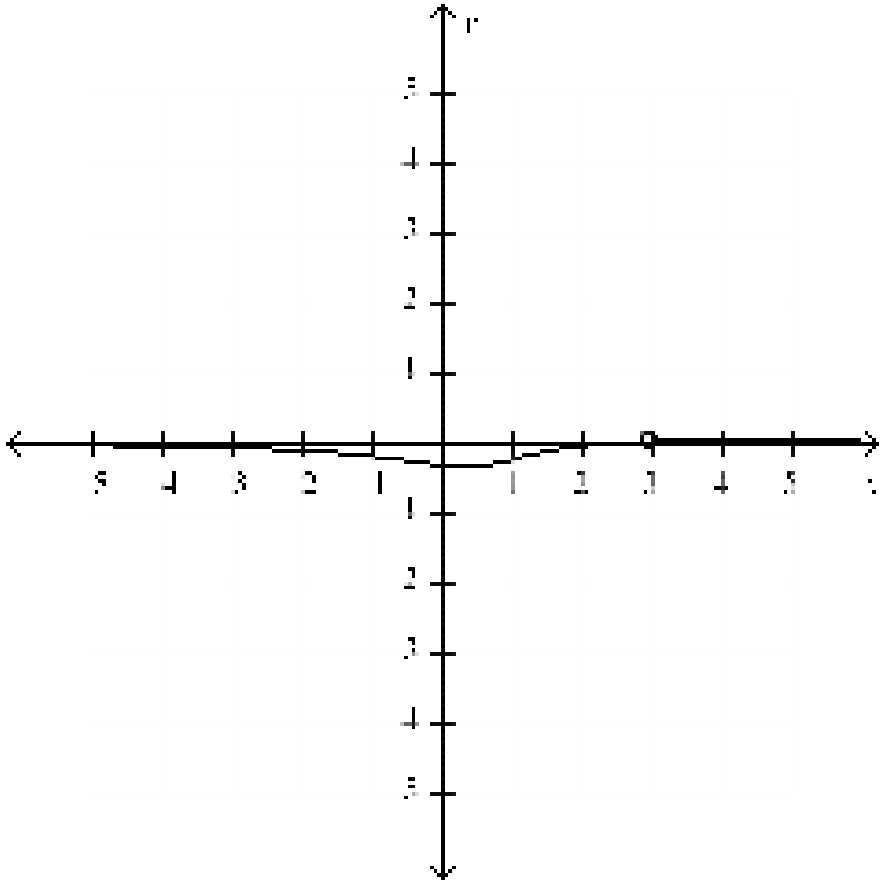
551. Which graph represents – 3? Recovered graph functions: ($x+(−3))/((5)*x^2-(−23)*x+(24)); (x+(−3))/((5)*x^2+(−23)*x+(24)); (3)/((5)*x^2-(−23)*x-(24)); (x+(−3))/((5)*x^2+(−6)*x+(9))$. Rendered ExamView reference:

$$f(x) = \frac{x-3}{5x^2 - 23x + 24}$$









(A) graph of (x+(−3))/((5)*x^2-(−23)*x+(24))

(B) graph of (x+(−3))/((5)*x^2+(−23)*x+(24))

(C) graph of (3)/((5)*x^2-(−23)*x-(24))

(D) graph of (x+(−3))/((5)*x^2+(−6)*x+(9))

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

552. Which function has a $y-$intercept of $−8/27$?

(A) $f(x) = 8/(x^2 - 12x + 27)$

(B) $f(x) = −8/(x^2 - 12x - 27)$

(C) $f(x) = −8/(x^2 - 12x + 27)$

(D) all of the above

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

553. Which function has a horizontal asymptote with equation $y = 2/7$?

(A) $f(x) = (−2x - 3)/(7x + 8)$

(B) $f(x) = (7x + 8)/(2x - 3)$

(C) $f(x) = (2x + 8)/(7x - 3)$

(D) $f(x) = (2x - 3)/(7x + 8)$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

554. Which function has an $x-$intercept of $1/3$?

(A) $f(x) = (−6x - 2)/(5x - 3)$

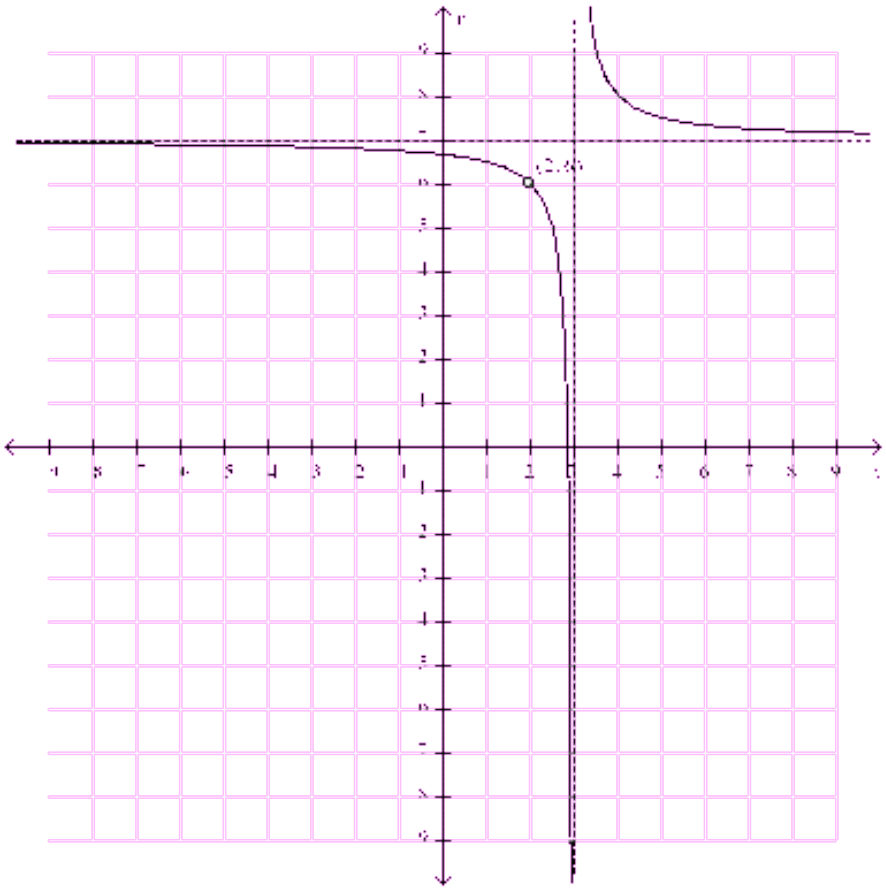
(B) $f(x) = (5x - 3)/(6x - 2)$

(C) $f(x) = (6x - 2)/(5x - 3)$

(D) $f(x) = (6x - 3)/(5x - 2)$

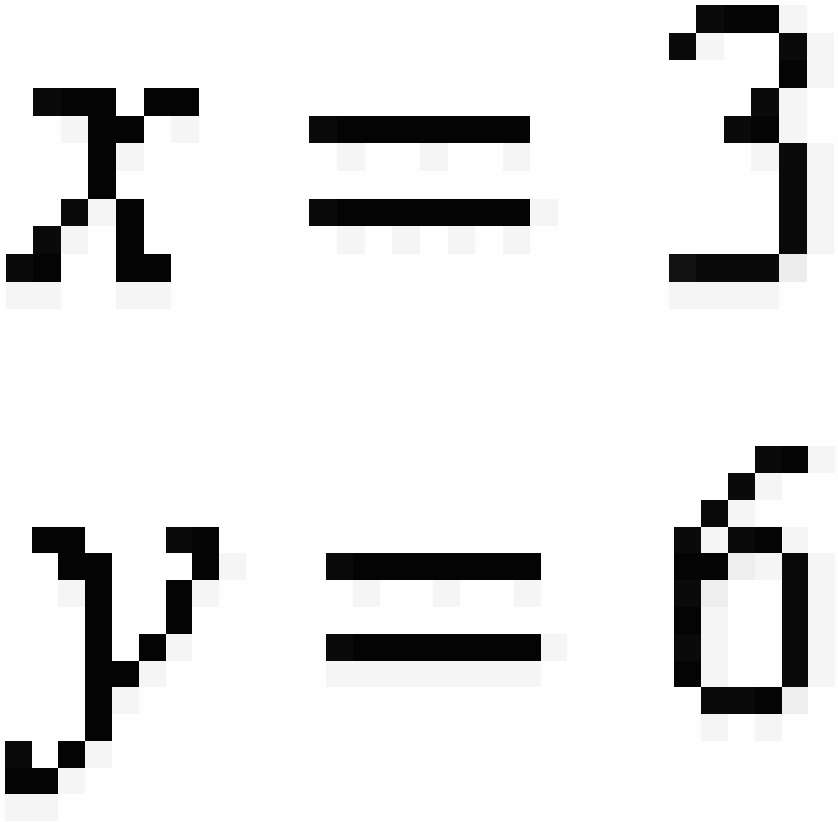
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

555. What is the equation for the vertical asymptote of the graph of the function shown? – 3 Rendered ExamView reference:



$$x = 2$$

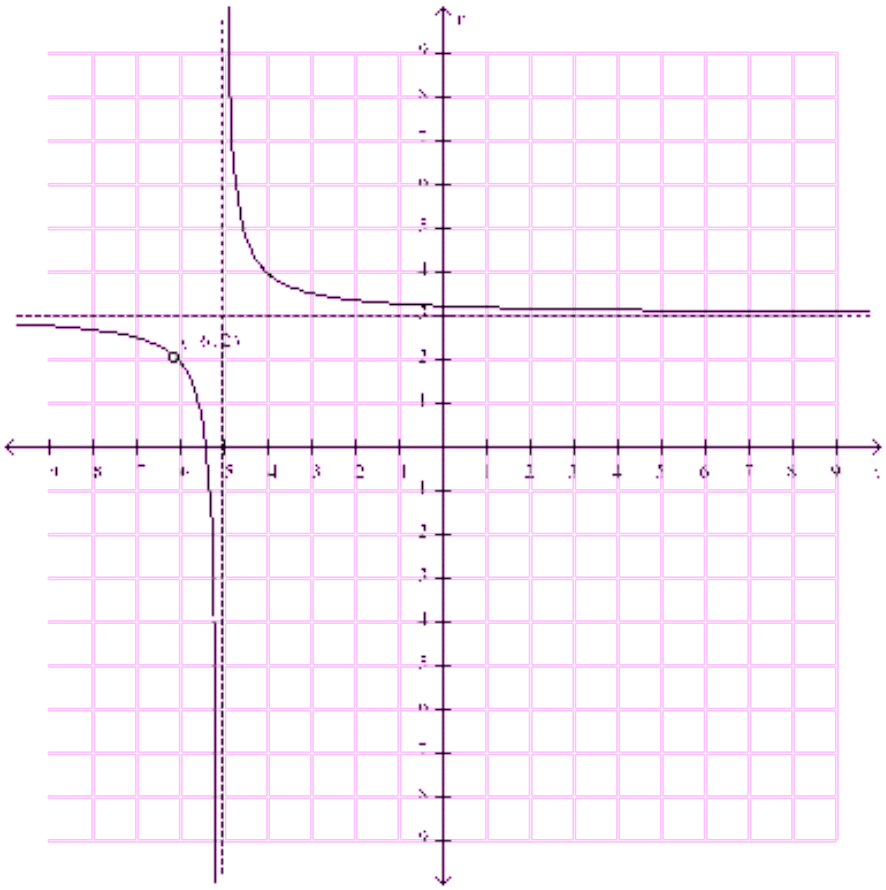
$$y = 7$$



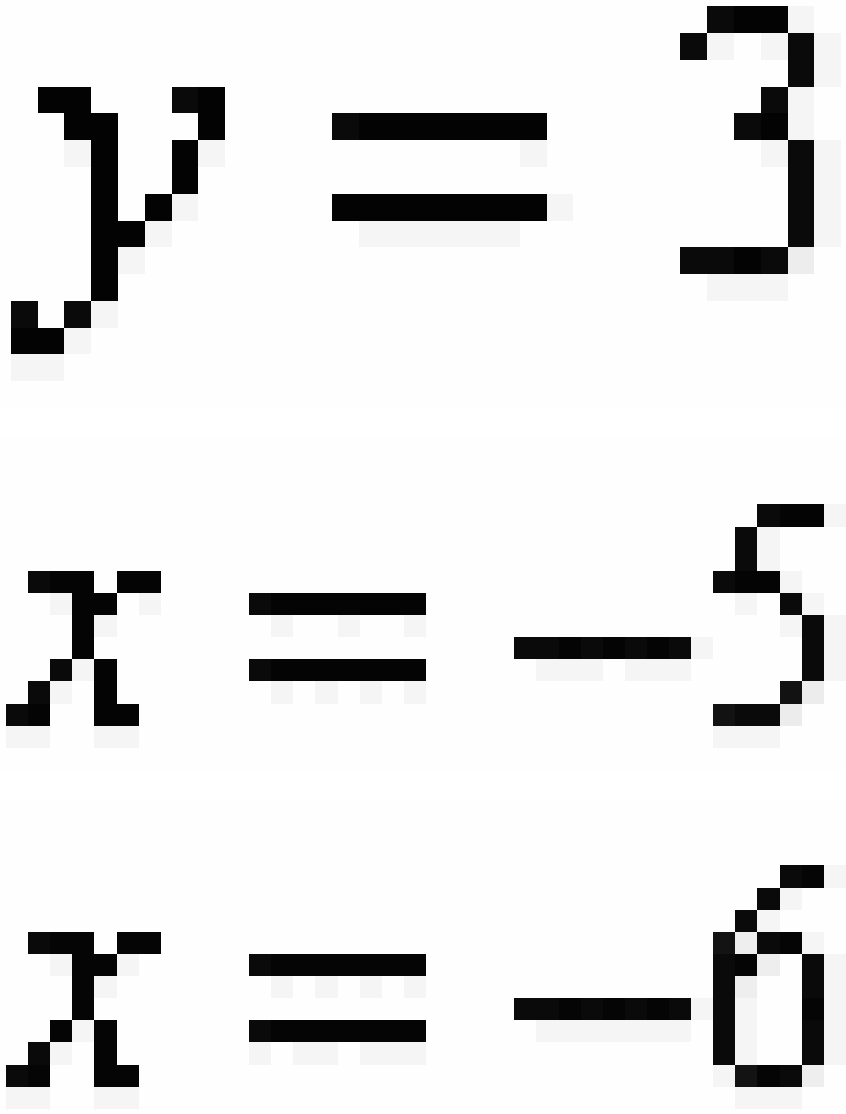
(A) 2 (B) 7 (C) 3 (D) 6

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

556. What is the equation for the horizontal asymptote of the graph of the function shown? $+ 6$ Rendered ExamView reference:



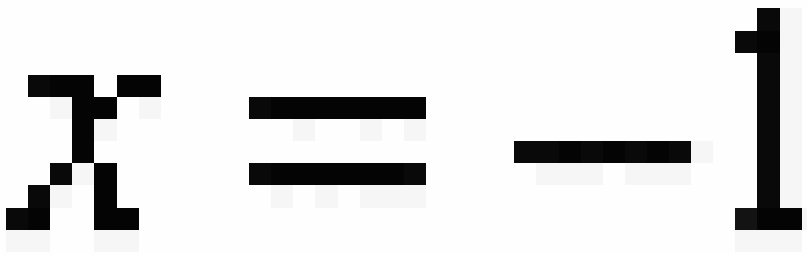
$$y = 2$$



(A) 2 (B) 3 (C) $–5$ (D) $–6$

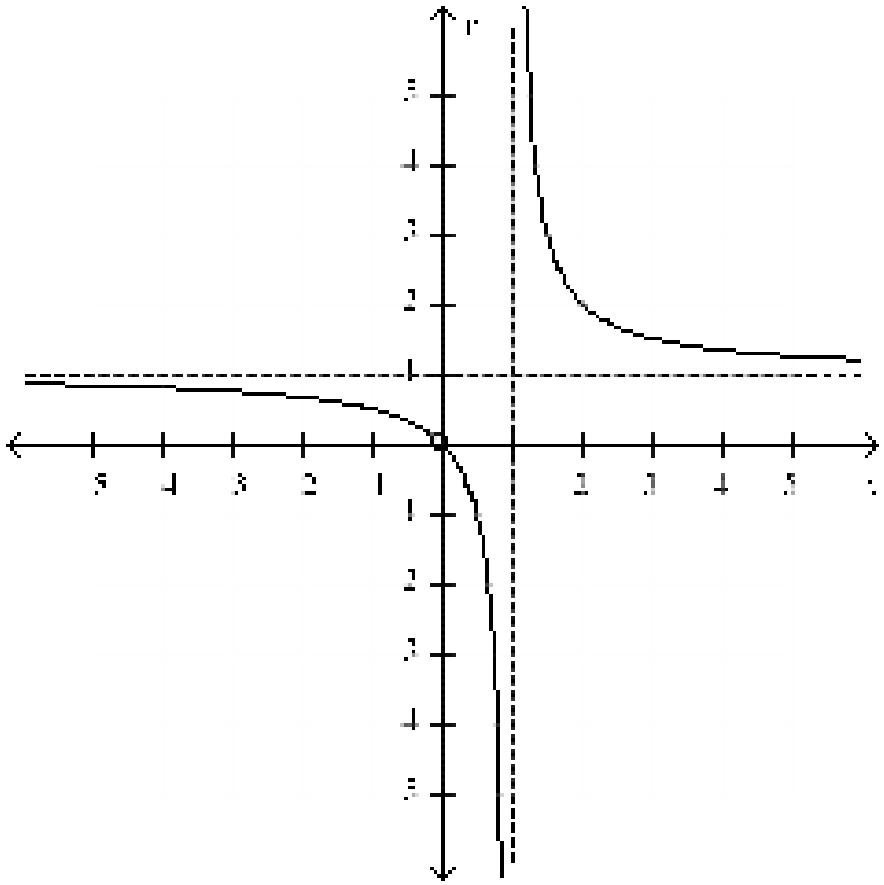
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

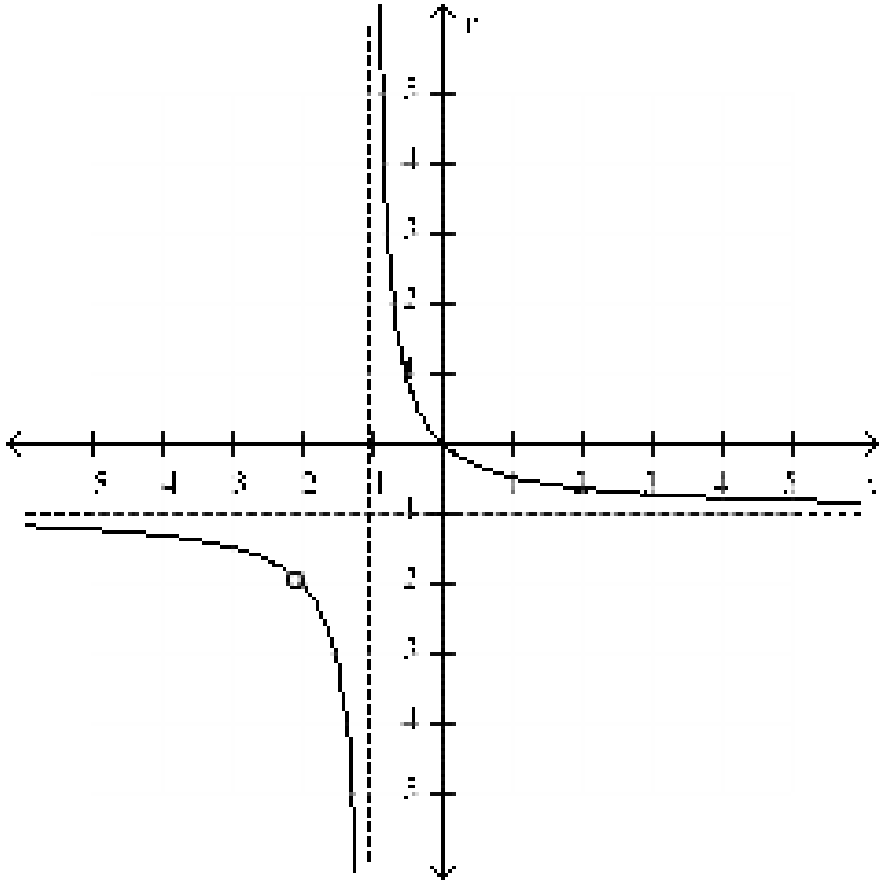
557. Which graph of a rational function has the following characteristics? a vertical asymptote with equation –$1 a$ horizontal asymptote with equation –$1 a$ point of discontinuity at –2 Recovered graph functions: ($x+(2))/((x+(2))*(x+(−1)))+b; (x+(2))/((x+(2))*(x+b))+(−1); −1*((x+(2))/((x+(2))*(x+(−1)))+b); −1*((x+(2))/((x+(2))*(x+(−1)))-b)$. Rendered ExamView reference:

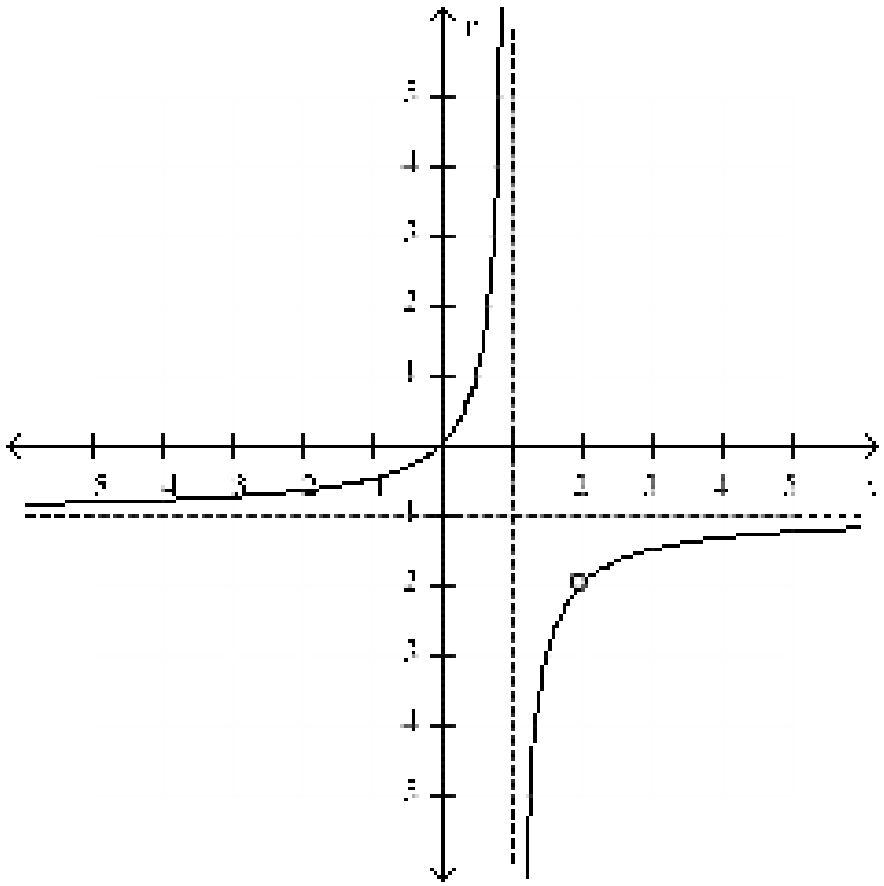


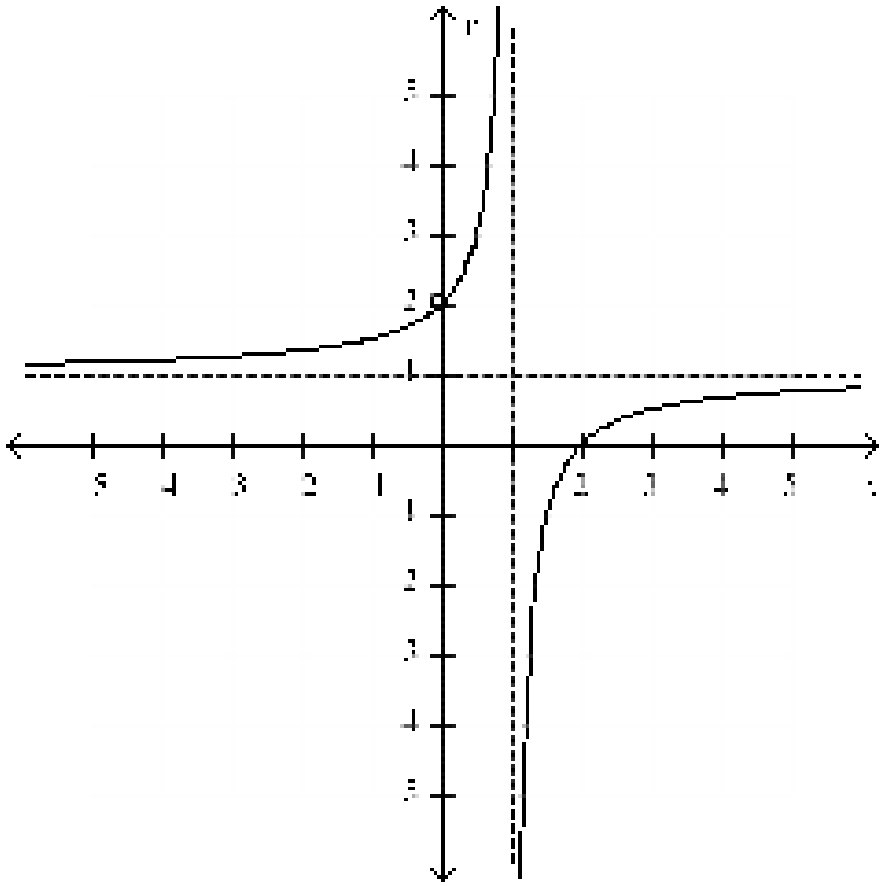
$$y = -1$$

$$(-2, -2)$$









(A) graph of (x+(2))/((x+(2))*(x+(−1)))+b

(B) graph of (x+(2))/((x+(2))*(x+b))+(−1)

(C) graph of −1*((x+(2))/((x+(2))*(x+(−1)))+b)

(D) graph of −1*((x+(2))/((x+(2))*(x+(−1)))-b)

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

558. Which rational function has the following characteristics? a vertical asymptote with equation $2 a$ horizontal asymptote with equation $3 a$ point of discontinuity at 9

(A) – 2 (B) – 3 (C) + 2 (D) + 8

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

559. Which function has a graph in the shape of a parabola? (a $=$ – $3, n = 2, b =$ – $7, a =$ – $3, b =$ – $7, a =$ – $3, a =$ – $3, m = 3, b =$ – $7, a =$ – $3, b =$ – $7, a =$ – 3) Rendered ExamView reference:

$$f(x) = \frac{(x-3)^2(x-7)}{(x-3)(x-7)}$$

$$f(x) = \frac{x-3}{(x-3)^3(x-7)}$$

$$f(x) = \frac{(x-3)^2(x-7)}{x-3}$$

(A) – 7 (B) – 3 (C) – 7 (D) – 3

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

560. Solve the equation –1.

(A) −7 (B) $−6$ (C) $−8$ (D) $−1/7$

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

561. Solve the equation –2.

(A) $18/5$ (B) $−18/5$ (C) 4

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

562. Solve the equation $+ 3$.

(A) $−3$ (B) $−91/12$ (C) $91/12$ (D) $−9$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

563. Solve the equation 2.

(A) 9 (B) 2 (C) 9 (D) $3/2$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

564. What are the $x-$intercepts of the graph of $+ 35$? Rendered ExamView reference:

$$f(x) = \frac{x^2+7x-18}{x^2+12x+35}$$

(A) 7 (B) 5 (C) 2 (D) $−9$

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

565. Solve the equation 6 graphically.

(A) 6 (B) – (C) – 8 (D) −8

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

566. The rational function –17 has a vertical asymptote with equation _____.

ANS: −1 **TOP:** bnk | problem **TYPE:** problem

567. The rational function 13 has a $y-$intercept of _____.

ANS: −13/11 **TOP:** bnk | problem **TYPE:** problem

568. The number of $x-$intercepts in the graph of $+ 3$ is _____. $+ 8$ Rendered ExamView reference:

$$f(x) = \frac{(x+3)(x+8)(x+4)}{x(x+8)}$$

ANS: 2 **TOP:** bnk | problem **TYPE:** problem

569. Given a rational function, if location of a point of discontinuity is a restriction, then equation of a vertical asymptote could be the location of a point of discontinuity or the equation of a vertical asymptote.

TOP: bnk | problem **TYPE:** problem

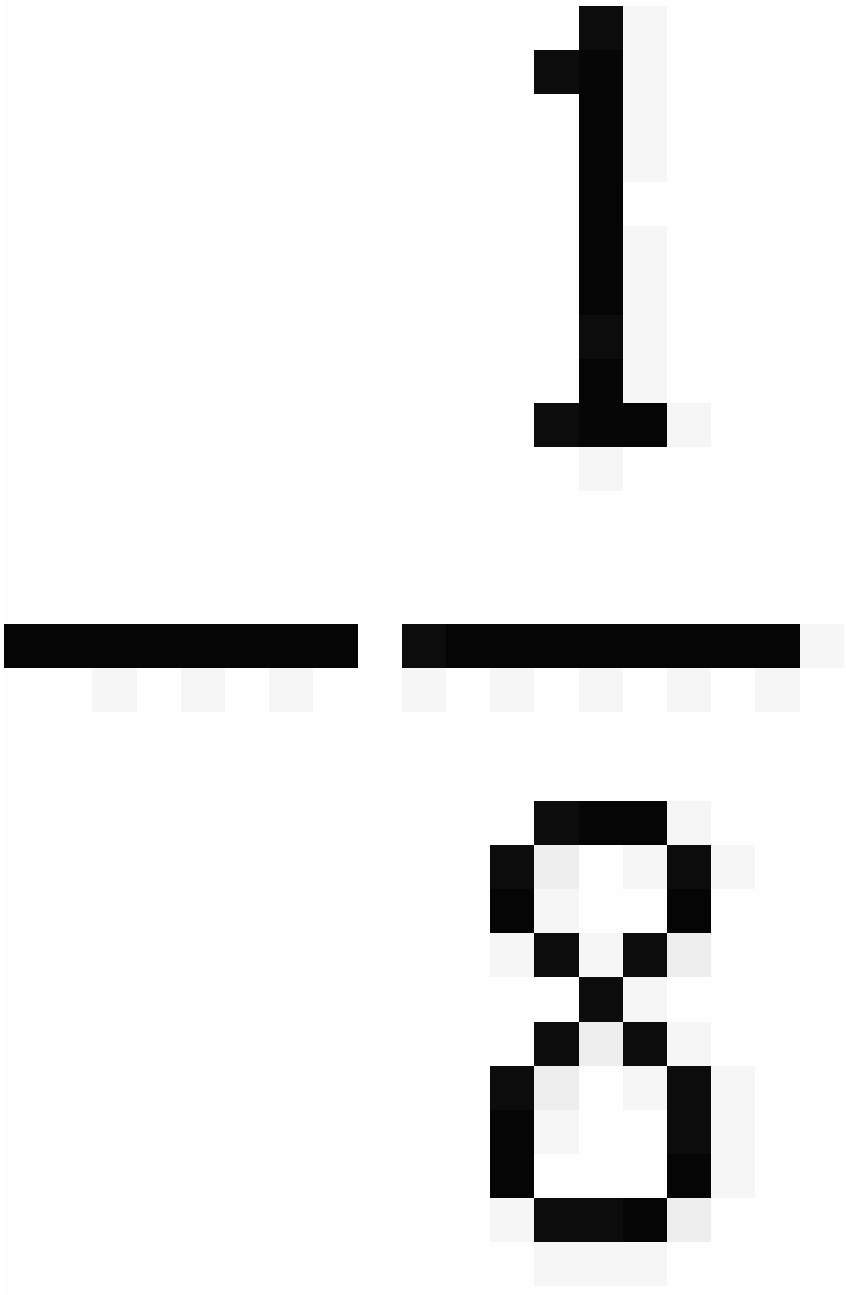
570. When solving the rational function – 8 graphically, the single function that could be graphed is _____.

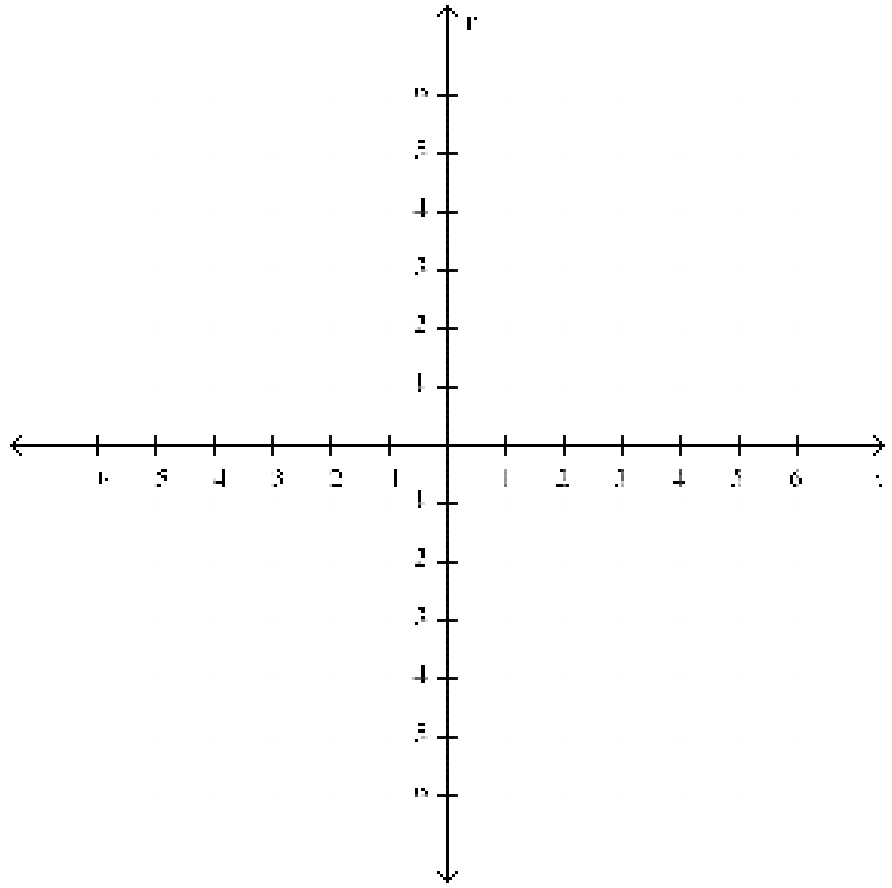
TOP: bnk | problem **TYPE:** problem

571. Graph the rational function $−1/(x+b1)$ and identify its key features. Rendered ExamView reference:

$$f(x) = \frac{1}{kx - c}$$

$$x = 2$$



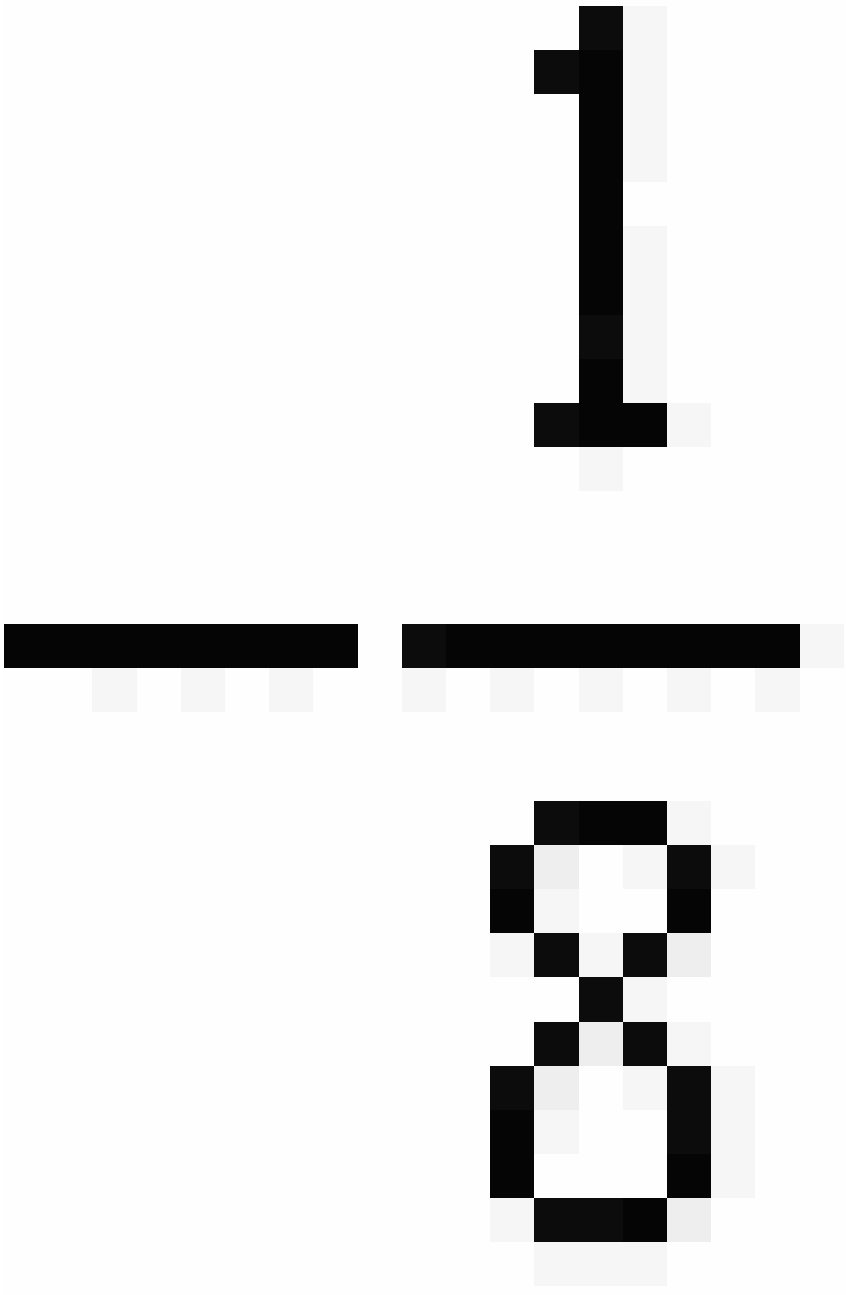


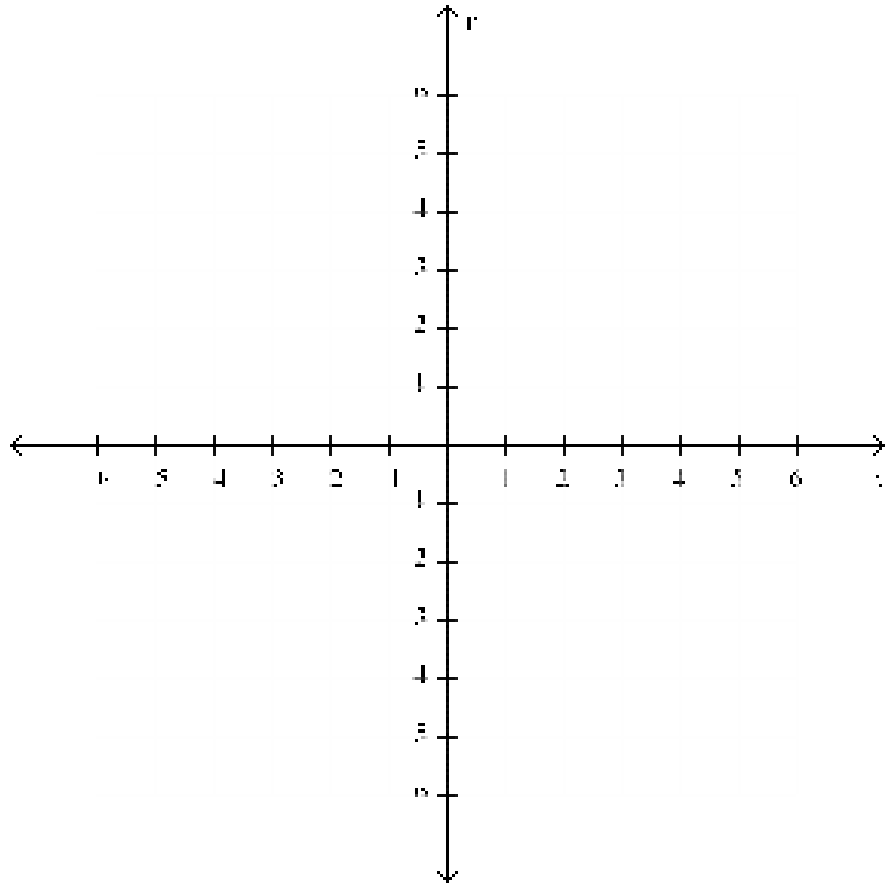
TOP: bnk | problem **TYPE:** problem

572. Graph the rational function $x/(f*(x+b_1))$ and identify its key features. Rendered ExamView reference:

$$f(x) = \frac{1}{kx - c}$$

$$x = 2$$



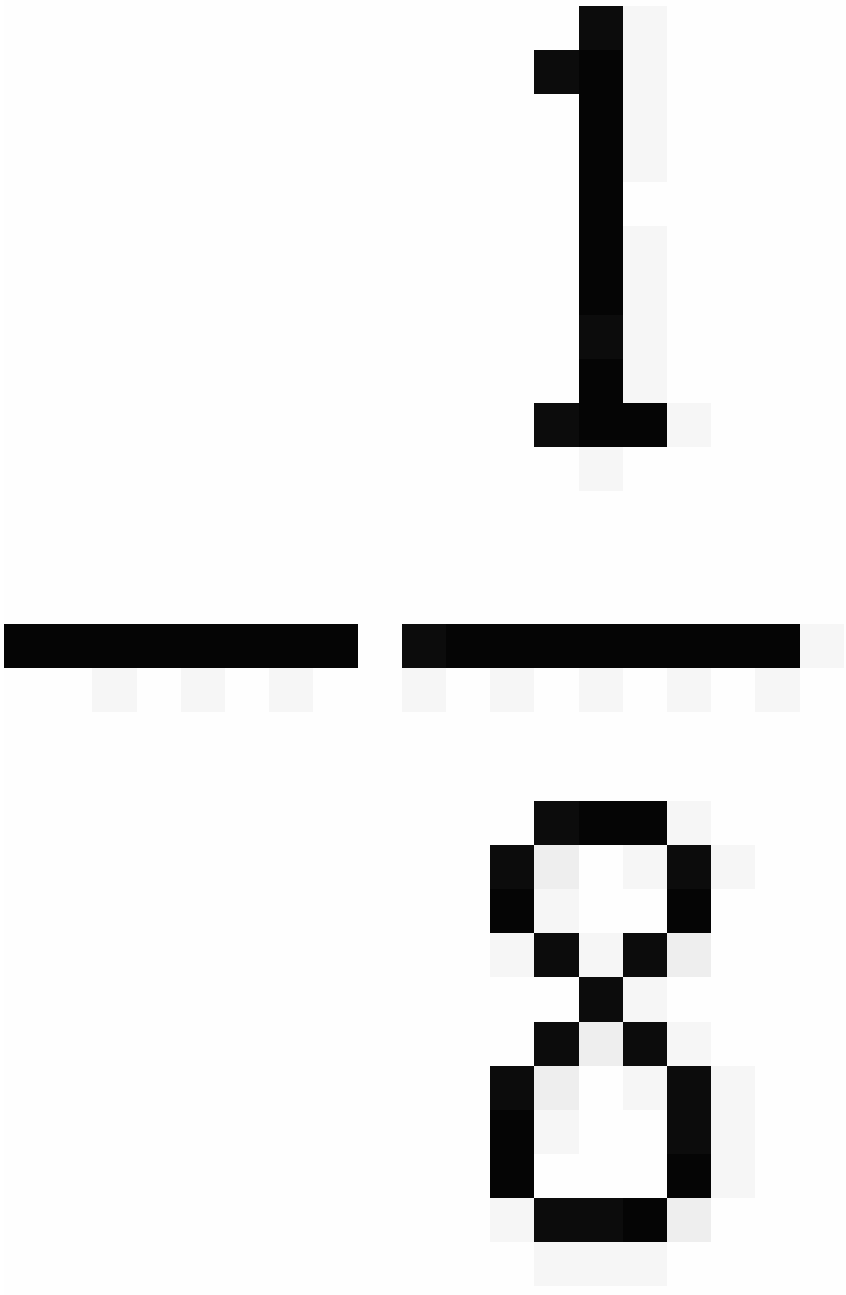


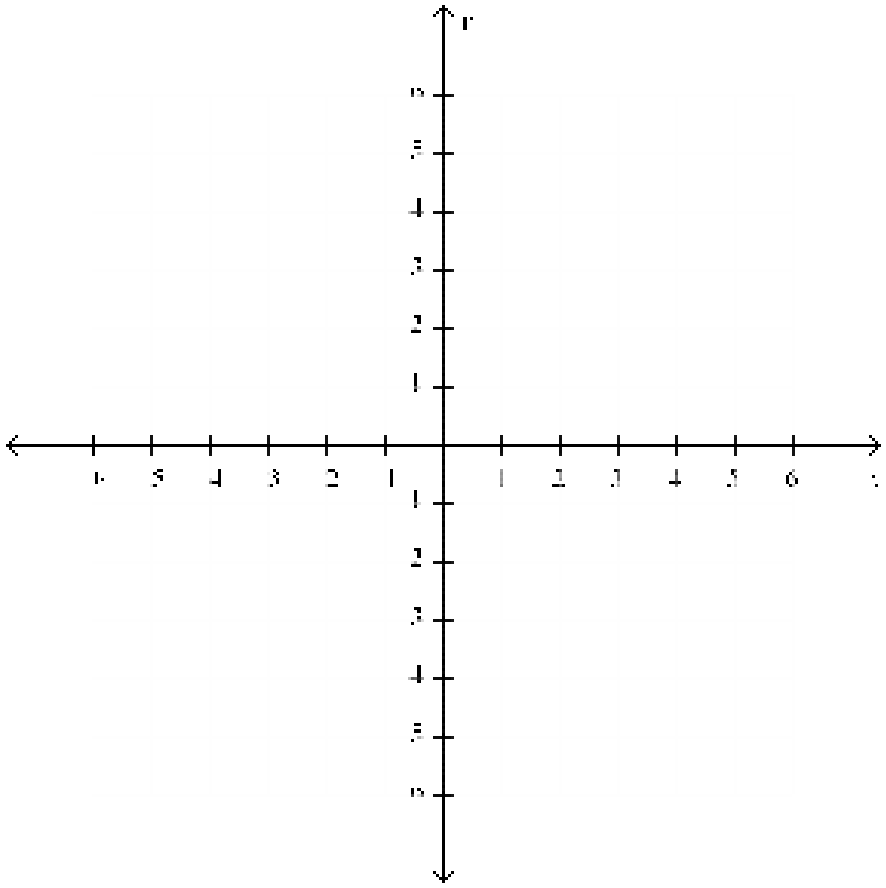
TOP: bnk | problem **TYPE:** problem

573. Graph the rational function $a/(x^2+d*x+e)$ and identify its key features. Rendered ExamView reference:

$$f(x) = \frac{1}{kx - c}$$

$$x = 2$$



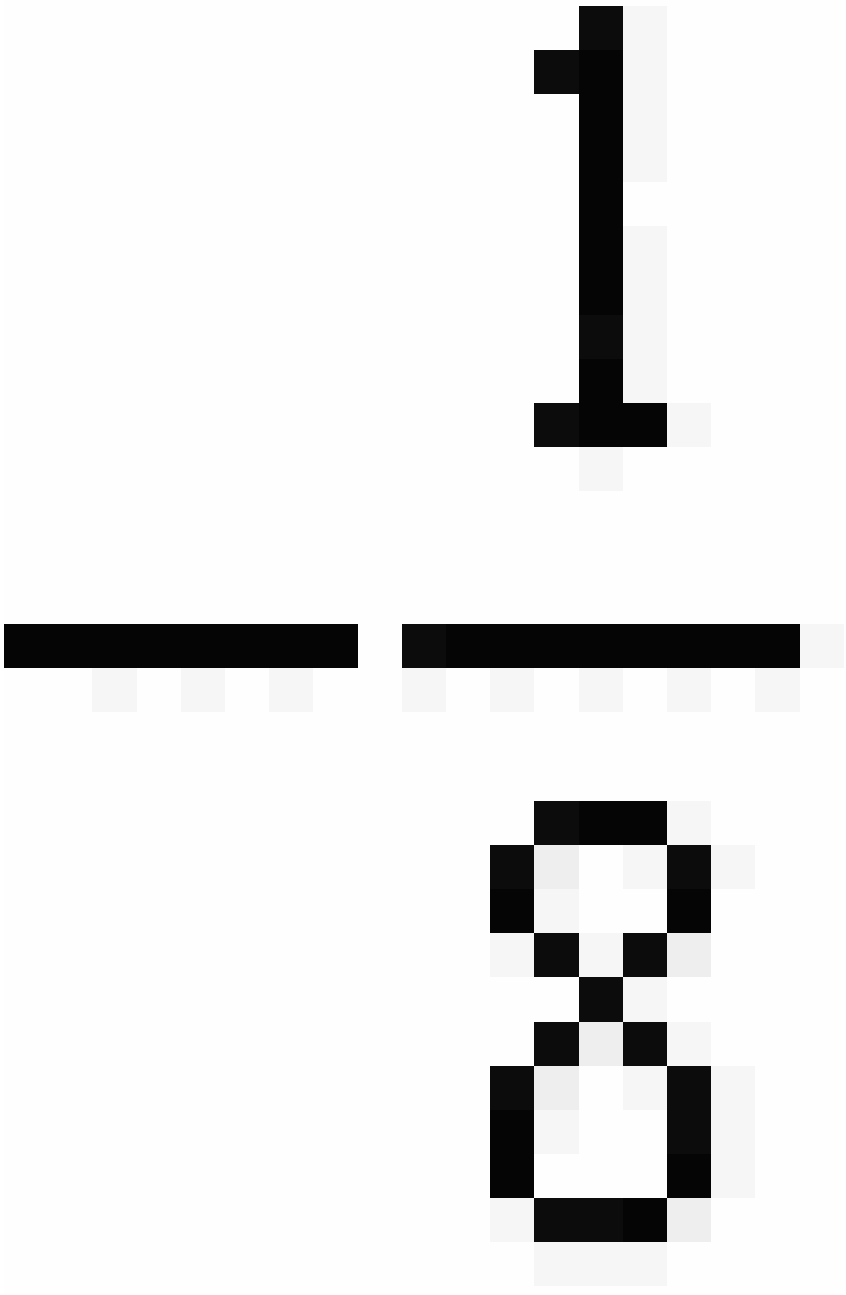


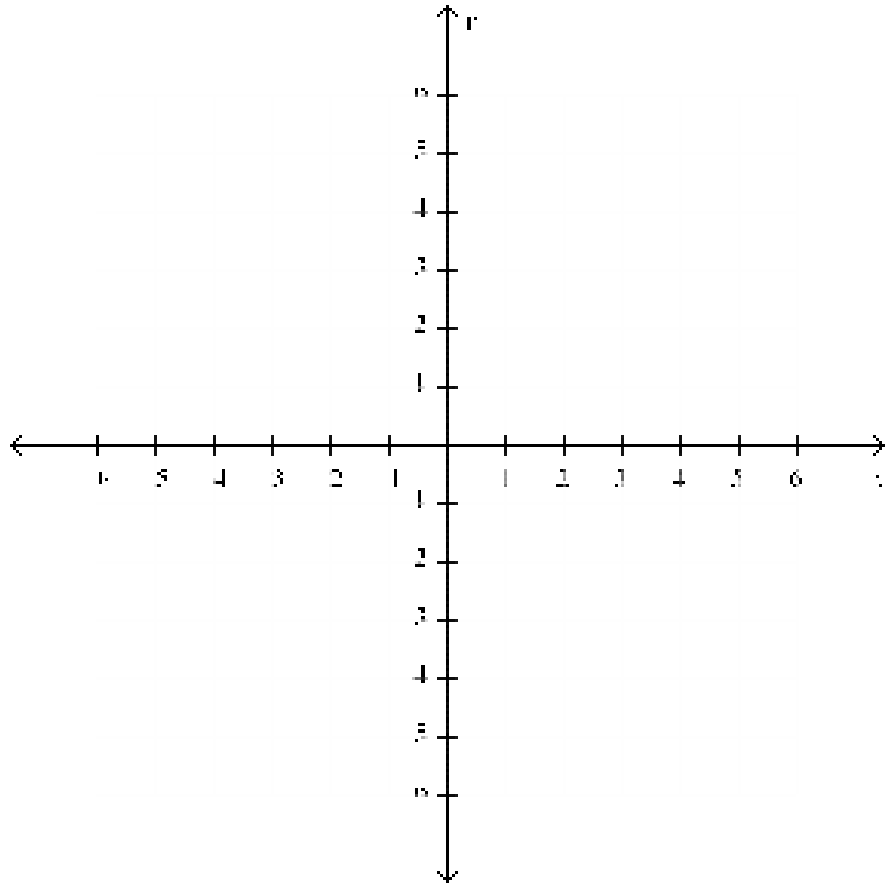
TOP: bnk | problem **TYPE:** problem

574. Graph the rational function $(f*x-a)/(x+b1)$ and identify its key features. Rendered ExamView reference:

$$f(x) = \frac{1}{kx - c}$$

$$x = 2$$



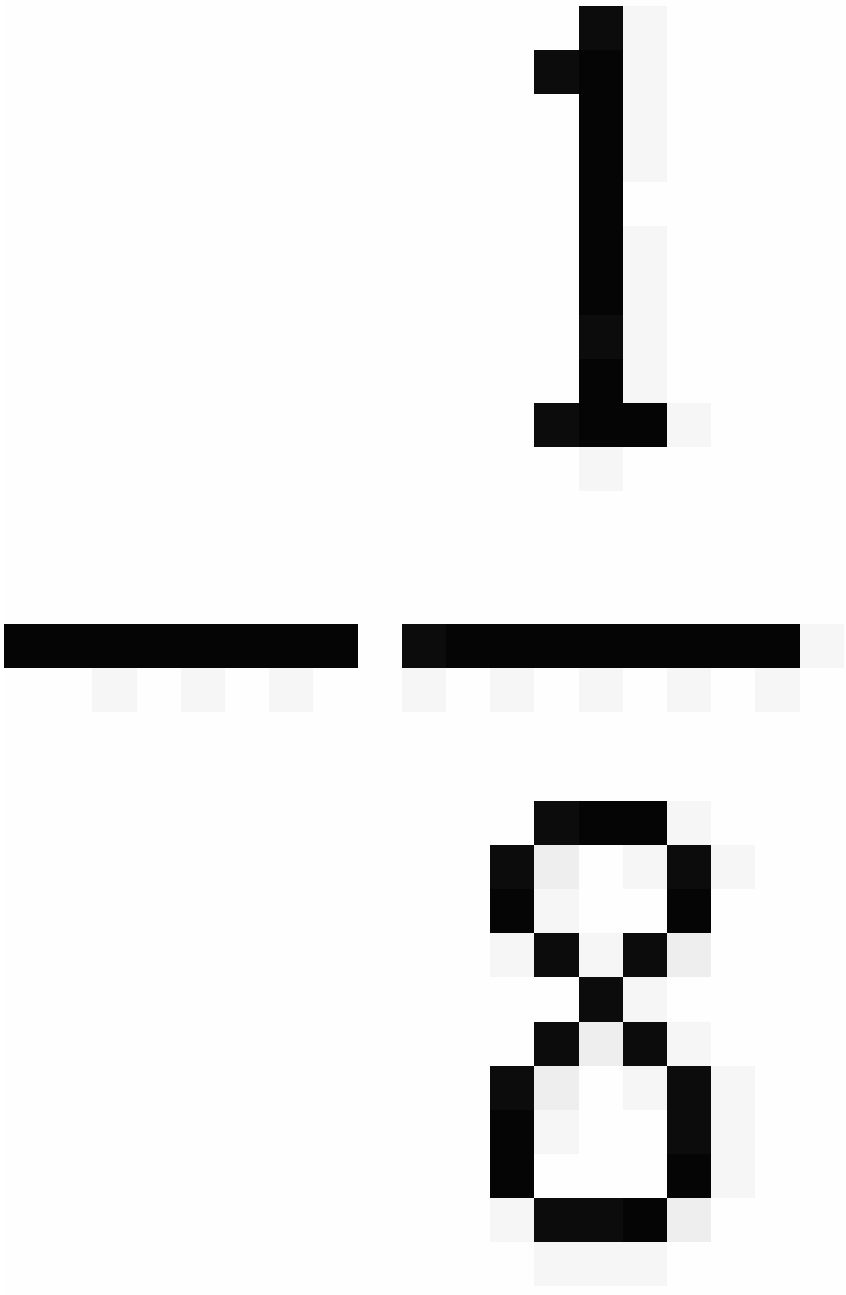


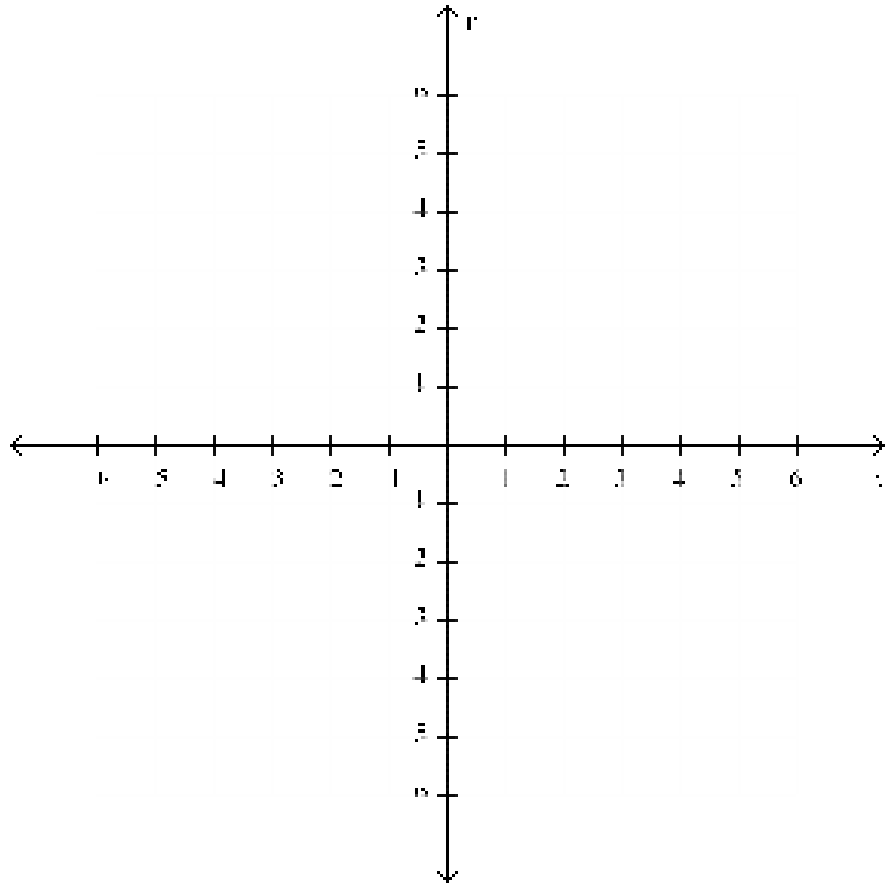
TOP: bnk | problem **TYPE:** problem

575. Graph the rational function $a/(x^2+b)$ and identify its key features. Rendered ExamView reference:

$$f(x) = \frac{1}{kx - c}$$

$$x = 2$$





TOP: bnk | problem **TYPE:** problem

576. (a) Determine an equation in the form $1/(4*x-8)$ for a function with a vertical asymptote at [Graph diagram: Graph: $1/(4*x-8)$ | x: [$−10.0, 10.0$], y: [$−0.5, 0.5$]] and a $y-$intercept of [Graph diagram: Graph: $1/(4*x-8)$ | x: [$−10.0, 10.0$], y: [$−0.5, 0.5$]].

(b) Sketch the graph of the function. [Graph diagram: Graph: $1/(4*x-8)$ | x: [$−10.0, 10.0$], y: [$−0.5, 0.5$]]

TOP: bnk | problem **TYPE:** problem

577. Sketch the graph of the function $5/(x+3)$ using transformations and identify

(a) the vertical asymptotes (if any)
(b) the horizontal asymptotes (if any)
(c) the domain and range
(d) the behaviour near any non$-$permissible values
(e) the end behaviour [Graph diagram: Graph: $5/(x+3)$ | x: [$−10.0, 10.0$], y: [$−10.0, 10.0$]]

TOP: bnk | problem **TYPE:** problem

578. Determine the approximate slope of the tangent to the curve $f(x) = 3x + 1 2x + 7$ at

$$f(x) = \frac{3x+1}{2x+7}$$

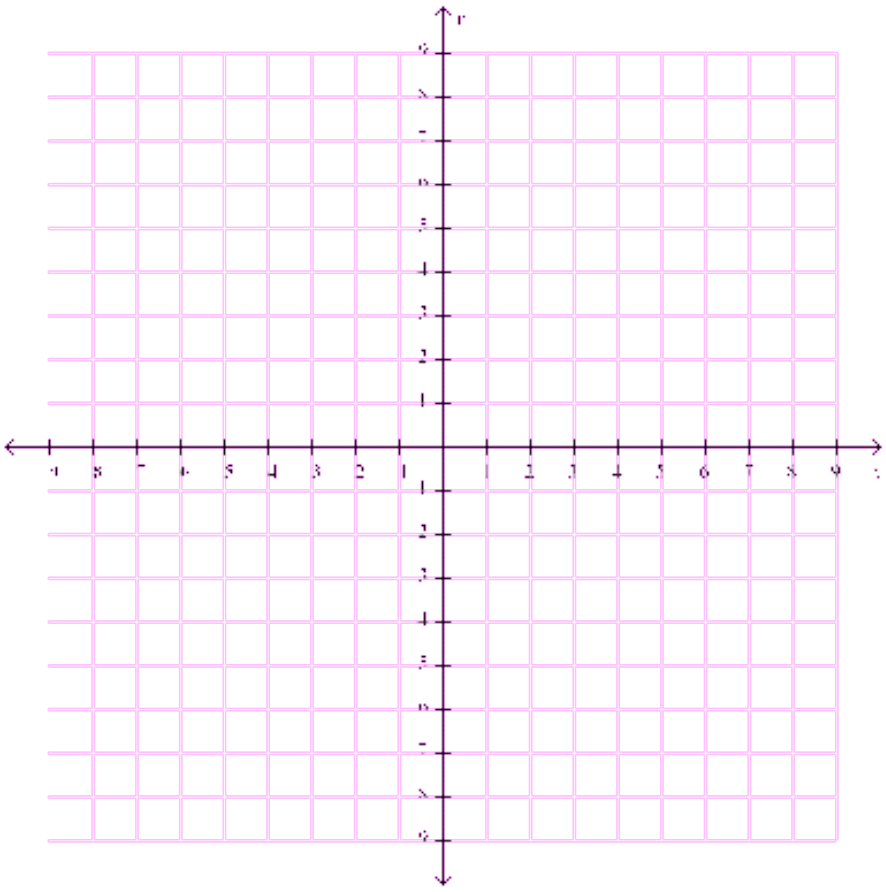
, to one decimal place. Rendered ExamView reference:

$$x = -2$$

TOP: bnk | short_answer **TYPE:** short_answer

579. Consider the function $−2/(x+4)$. a) Determine the key features of the function: i) domain and range ii) intercepts iii) equations of any asymptotes b) Sketch the graph of the function. [Graph diagram: Graph: $−2/(x+4)$ | x: [$−10.0, 10.0$], y: [$−4.0, 4.0$]] Rendered ExamView reference:

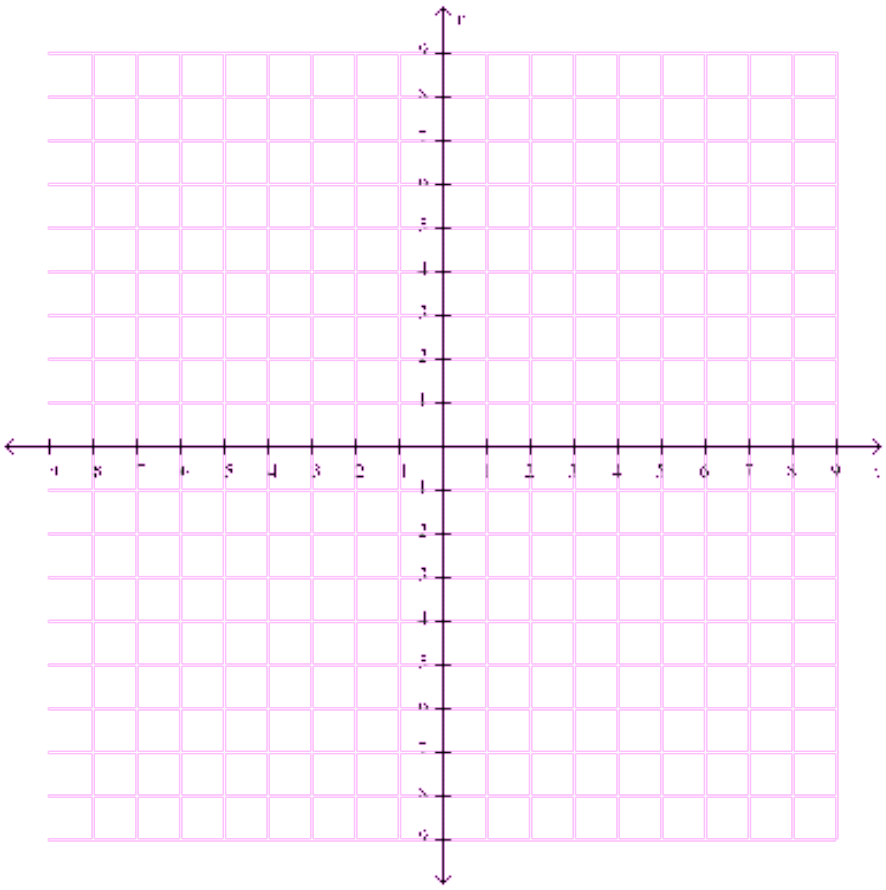
$$f(x) = -\frac{2}{x+4}$$



TOP: bnk | problem **TYPE:** problem

580. Consider the function \$3/(4*x-5)\$. a) Determine the key features of the function: i) domain and range ii) intercepts iii) equations of any asymptotes b) Sketch the graph of the function. [Graph diagram: Graph: \$3/(4*x-5)\$ | x: [\$−10.0, 10.0\$], y: [\$−3.0, 3.0\$]] Rendered ExamView reference:

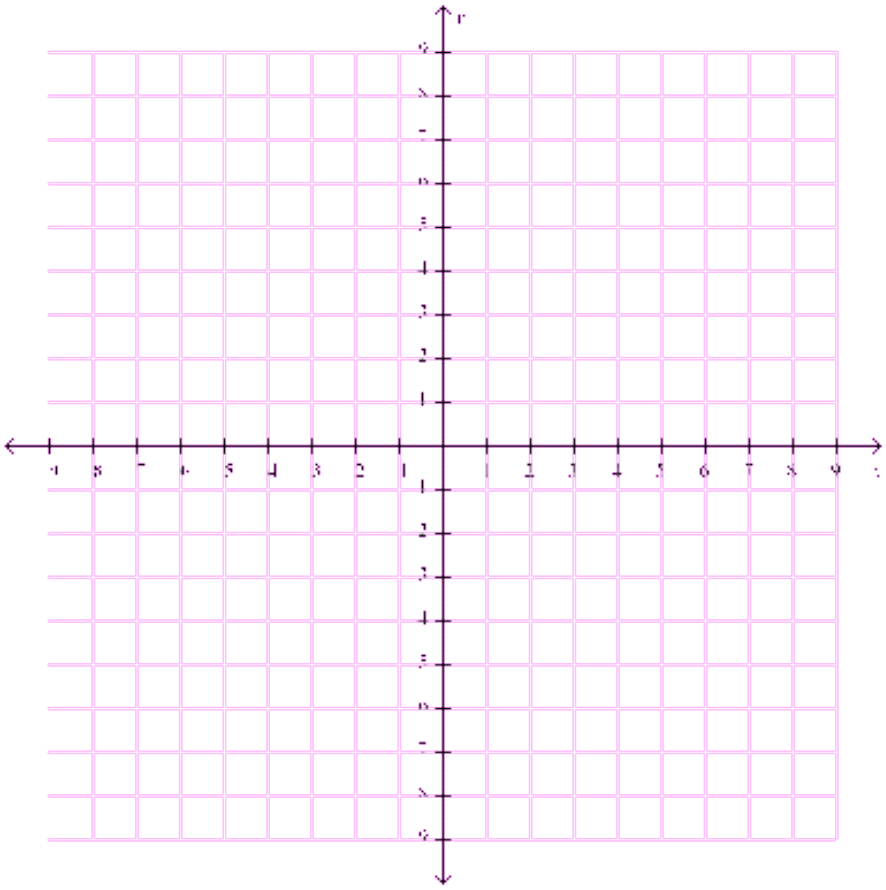
$$f(x) = -\frac{2}{x+4}$$



TOP: bnk | problem **TYPE:** problem

581. Consider the function \$1/(x^2+2*x-8)\$. a) Determine the key features of the function: i) domain and range ii) intercepts iii) equations of any asymptotes b) Sketch the graph of the function. [Graph diagram: Graph: \$1/(x^2+2*x-8)\$ | x: [\$−10.0, 10.0\$], y: [\$−0.4, 0.3\$]] Rendered ExamView reference:

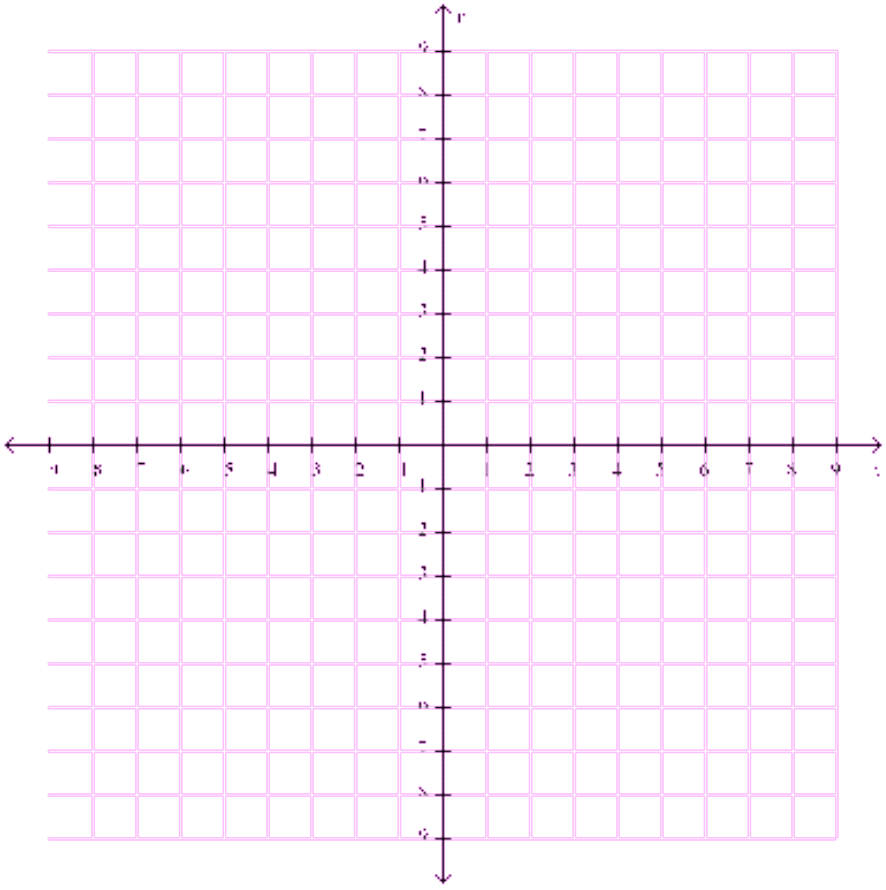
$$f(x) = \frac{1}{x^2 + 2x - 8}$$



TOP: bnk | problem **TYPE:** problem

582. Consider the function ($3*x+8)/(x-2)$. a) Determine the key features of the function: i) domain and range ii) intercepts iii) equations of any asymptotes b) Sketch the graph of the function. [Graph diagram: Graph: $(3*x+8)/(x-2)$ | x: [$ −10.0, 10.0$], y: [$−25.0, 31.0$]] Rendered ExamView reference:

$$f(x) = \frac{3x + 8}{x - 2}$$



TOP: bnk | problem **TYPE:** problem

583. Consider the function ($x+3)/(x^2-x-12)$. a) Determine the key features of the function: i) domain and range ii) intercepts iii) equations of any asymptotes b) Sketch the graph of the function. Rendered ExamView reference:

$$f(x) = \frac{x+3}{x^2 - x - 12}$$

TOP: bnk | problem **TYPE:** problem

584. Solve. a) $1 x - 4 = - 6 x x + 5 x - 3 = 2x + 7 x$ b) $1 x = 1 x - 2$ c) $1 x - 4 = - 6 x x = - 6(x - 4) x = - 6x + 24 7x = 24 x = 24$ d) $5 x - 3 = 2x + 7 x x(x + 5) = (x - 3)(2x + 7) x^2 + 5x = 2x 2 + x - 21 x^2 - 4x - 21 = 0 (x - 7)(x + 3) = 0 x = 7 or x = - 3$ $2x - 4 = 4 x - 2 3(x - 2) = 4(2x - 4) 3x - 6 = 8x - 16 5x = 10 x = 2$

TOP: bnk | problem **TYPE:** problem

585. Solve each equation by graphing, using technology. Express your answers to two decimal places. a) $3x x - 2 = - 1 x + 3 1 x + 1 x - 5 = - 1 * x^2 (x - 5) 3 = - 2 x - 5 3x - 1 x = x + 5 x - 3$ b) $x - 3.52$ c) $x - 1.19$ d)

$$\frac{3x}{x-2} = -\frac{1}{x+3}$$

Rendered ExamView reference:

$$\frac{1}{x} + \frac{1}{x-5} = -1$$

$$\frac{x^2}{(x-5)^3} = -\frac{2}{x-5}$$

$$\frac{3x-1}{x} = \frac{x+5}{x-3}$$

TOP: bnk | problem **TYPE:** problem

586. The time, t, in hours, that it takes Alistair to jog 5 km is inversely proportional to his average speed, v, in kilometres per hour. a) Write a function to represent the time as a function of the speed. b) Sketch the graph of this function. c) If Alistair jogs at 4.5 km$/h,$ how long does it take him to complete a $5-$km run, to the nearest minute?

TOP: bnk | problem **TYPE:** problem

587. The pressure exerted on the floor by the heel of someone s shoe is inversely proportional to the square of the width of the heel of the shoe. When Megumi wears $2-$cm$-$wide heels, she exerts a pressure of 400 kPa. a) Determine a function

to represent the pressure, p, exerted by Megumi if she wears heels of width w. b) Sketch the graph of this function. c) If she wears spike heels with a width of 0.5 cm, what pressure does she exert?

TOP: bnk | problem **TYPE:** problem

588. A photographer uses a light meter to measure the intensity of light from a flash bulb. The intensity, I, in lux, of the flash bulb is a function of the distance, d, in metres, from the light and can be represented by

$$I(d) = \frac{10}{d^2}, d > 0$$

. a) Determine the following, to two decimal places: i) the intensity of light $3 m$ from the flash bulb ii) the average rate of change in the intensity of light for the interval

$$1 < d < 3$$

b) What does the sign of your answer to part a)ii) indicate about the light intensity?

TOP: bnk | problem **TYPE:** problem

589. The population, P, of a small town is modelled by the function $P(t) = 20(4t + 3) 2t + 5$, where t is time, in years, since the start of 1990.

(a) Is the population increasing or decreasing? Justify your answer.
(b) The town will need its own transit system if the population exceeds $50 000$. Will the town s population ever exceed $50 000$? Explain.

TOP: bnk | problem **TYPE:** problem

590. The distance, d, in metres, between the boat and the dock as a function of time, t, in seconds, is given by ($−7*x+70)/(x+4)$.

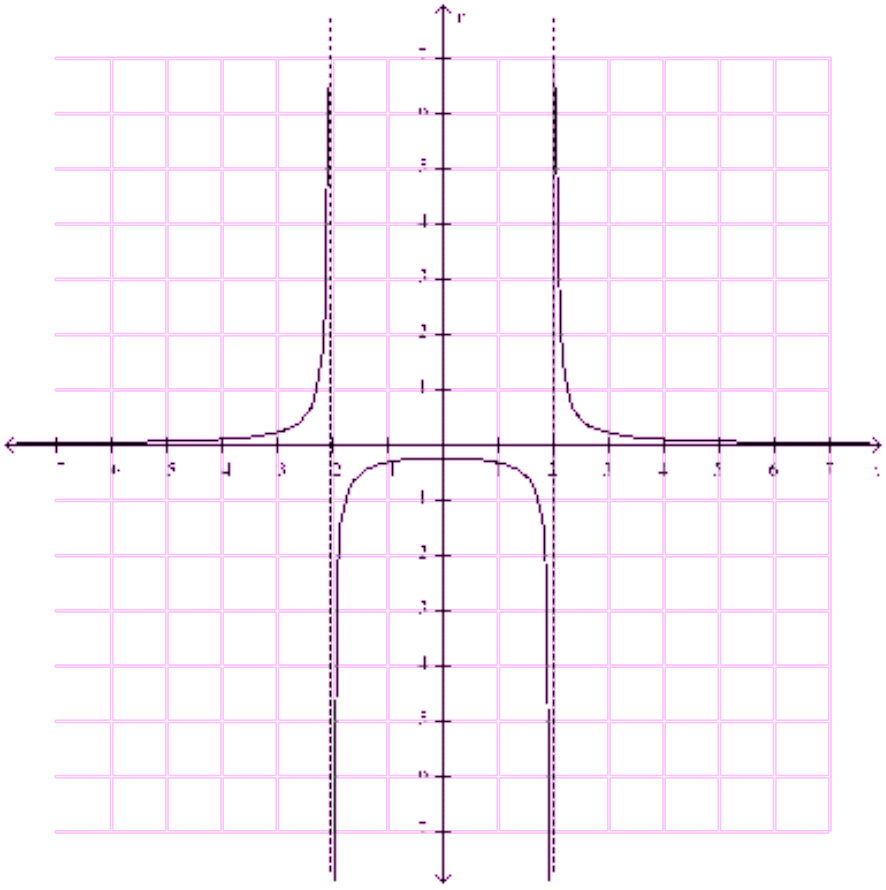
(a) What is the average velocity of the boat during the time interval from when Ingrid turns the boat off to when it meets the dock?
(b) Determine the velocity of the boat when Ingrid turns off the power, to one decimal place. c) Determine the velocity of the boat when it meets the dock, to one decimal place.
(c) Sketch the graph of the function. On the same set of axes, sketch the graph of the speed of the boat in relation to the time.

TOP: bnk | problem **TYPE:** problem

591. a) Write an equation to represent a rational function with the following conditions: reciprocal of a quadratic function; asymptotes with equations $x = −5$, $x = 4/3$, and $y = 0$; and $y < 0$ whenever $x < −5$ or $x > 4/3$. b) How many such equations are there? Explain your answer.

TOP: bnk | problem **TYPE:** problem

592. Write an equation for the graph of the rational function shown. Explain your reasoning. $1/(x^2-4)$ Rendered ExamView reference:



TOP: bnk | short_answer **TYPE:** short_answer

593. a) The graph of a function of the form ($2*x+6)/(x+2)$ is shown. Determine the key features: i) domain and range ii) intercepts iii) equations of any asymptotes [Graph diagram: Graph: $(2*x+6)/(x+2)$ | x: [$−10.0, 10.0$], y: [$−2.0, 6.0$]] b) State a possible equation for this function. Rendered ExamView reference:

$$f(x) = \frac{x+3}{x^2 - x - 12}$$

TOP: bnk | problem **TYPE:** problem

594. a) Determine an equation in the form ($3*x+8)/(4*x+4)$ for a function that has asymptotes with equations [Graph diagram: Graph: $(3*x+8)/(4*x+4)$ | x: [$−10.0, 10.0$], y: [$−1.8, 3.3$]] and [Graph diagram: Graph: $(3*x+8)/(4*x+4)$ | x: [$−10.0, 10.0$], y: [$−1.8, 3.3$]] and a $y-$intercept of 2. b) Sketch the graph of your function. Rendered ExamView reference:

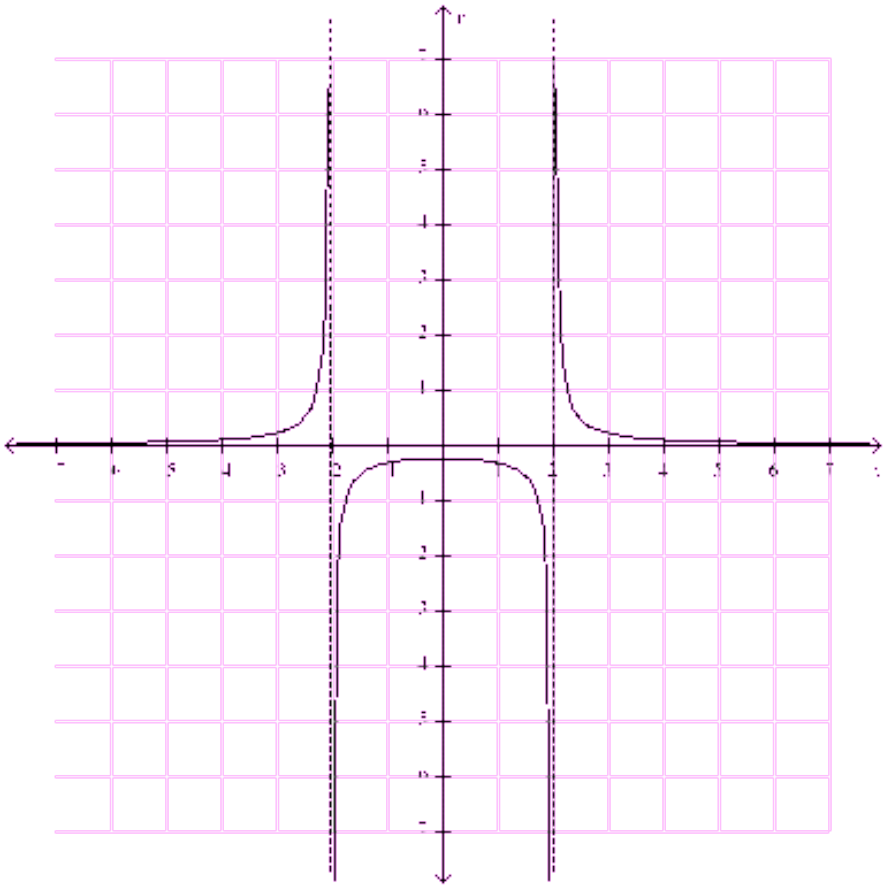
$$f(x) = \frac{x+3}{x^2 - x - 12}$$

TOP: bnk | problem **TYPE:** problem

595. The equation 3 has one solution, x $=$ –6. However, there are infinitely many solutions to $3,$ with x $=$ –6 being one of them, acting as a boundary on the interval of solutions for x.

TOP: bnk | problem **TYPE:** problem

596. a) Use long division to express the function $f(x) = x - 5 + 17 x + 2$ as the sum of a polynomial function and a rational function of the form



, where A is a constant. b) What does your result in part a) allow you to predict about the graph of $y = x - 5$, for large values of x? Why?

TOP: bnk | problem **TYPE:** problem

597. Write an equation for a rational function whose graph has all of the following features: vertical asymptote with equation x $= 3$ horizontal asymptote with equation y $= 2$ hole at x $= 1$ no $x-$intercepts

TOP: bnk | short_answer **TYPE:** short_answer

598. (a) Use the asymptotes and intercepts to make a quick sketch of the function ($x+1)/(x-5)$ and its reciprocal [Graph diagram: Graph: $(x-5)/(x+1)$ | x: [$−10.0, 10.0$], y: [$−11.0, 13.0$]] on the same set of axes.

(b) Describe the symmetry in the graphs in part a). c) Determine the equation of the mirror line in your graph from part a).
(c) Determine intervals where f is positive and where f is negative. Determine intervals where g is positive and where g is negative. How do the two sets of intervals compare to each other?
(d) Does the pattern from part d) occur for all pairs of functions [Graph diagram: Graph: $(x-5)/(x+1)$ | x: [$−10.0, 10.0$], y: [$−11.0, 13.0$]] and [Graph diagram: Graph: $(x+1)/(x-5)$ | x: [$−10.0, 10.0$], y: [$−11.0, 13.0$]], [Graph diagram: Graph: $(x-5)/(x+1)$ | x: [$−10.0, 10.0$], y: [$−11.0, 13.0$]]? Explain why or why not.

TOP: bnk | true_false **TYPE:** true_false

599. (a) Use the asymptotes and intercepts to make a quick sketch of the function ($x-2)/(x+4)$ and its reciprocal, [Graph diagram: Graph: $(x+4)/(x-2)$ | x: [$−10.0, 10.0$], y: [$−11.0, 13.0$]], on the same set of axes.
(b) Describe the symmetry in the graphs in part a). c) Determine the equation of the mirror line in your graph from part a).
(c) Determine intervals of increase and decrease for both f and g. How do the sets of intervals compare? e) Does the pattern from part d) occur for all pairs of functions [Graph diagram: Graph: $(x+4)/(x-2)$ | x: [$−10.0, 10.0$], y: [$−11.0, 13.0$]] and [Graph diagram: Graph: $(x-2)/(x+4)$ | x: [$−10.0, 10.0$], y: [$−11.0, 13.0$]], [Graph diagram: Graph: $(x+4)/(x-2)$ | x: [$−10.0, 10.0$], y: [$−11.0, 13.0$]]? Explain why or why not.

TOP: bnk | problem **TYPE:** problem

600. Solve each inequality by graphing, using technology. Express your answers to one decimal place. a) $2x + 1 x - 3 x$ b) $3x 2 - 4 x^2 - 9 - 1 x^2$ c) $2x x - 3$ d) $3x 2x + 4$

TOP: bnk | problem **TYPE:** problem

601. Solve.
a) $6 x + 3$ b) $2x + 4 x - 2 0^(x)^2 - 4x - 12 x^2 - 25$ c) $x + 4$ d) $x - 2 or x$

TOP: bnk | problem **TYPE:** problem

602. Solve. a) $2 x + 3 2x x - 2 = 2 - 1 x + 2 2x + 1 x + 2 = x + 3 x + 4$ b) $1 x + 1 x^2 = - 1 x^3 F -$ c) $x = - 6 5 l l x = - 2 6$ d)

$$\frac{1}{x-4} = -\frac{6}{x}$$

$$\frac{x+5}{x-3} = \frac{2x+7}{x}$$

Rendered ExamView reference:

$$7 + \frac{1}{x} = \frac{1}{x - 2}$$

$$\frac{3}{2x - 4} = \frac{4}{x - 2}$$

TOP: bnk | problem **TYPE:** problem

603. An airplane makes a $990-$mi flight with a tail wind and returns, flying into the wind. The total flying time is $3 h 20 min,$ and the plane s airspeed is 600 mph. What is the wind speed?

TOP: bnk | short_answer **TYPE:** short_answer

604. What is the solution to the rational equation $2 x + 4 + x + 1 x + 6 = 11x + 32 x^2 + 10x + 24$?

TOP: bnk | short_answer **TYPE:** short_answer

605. A ski club charters a bus for a ski trip at a cost of $480. In an attempt to lower the bus fare per skier, the club invites non-members to go along. After five non-members join the trip, the fare per skier decreases by $$4.80$. How many club members are going on the trip?

TOP: bnk | short_answer **TYPE:** short_answer

606. Given the functions – 12 and – $3,$ determine the equation for the combined function –9.

(A) 1 (B) – (C) + 6 (D) 0

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

607. Given the functions – 8 and – $8,$ determine the equation for the combined function –2.

(A) 1 (B) 16 (C) 4

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

608. Given the functions $f(x) = x^2 - 81$ and $g(x) = x + 9$, determine the simplified equation for the combined function $h(x) = f(x) / g(x)$.

(A) $h(x) = x + 9$ (B) $h(x) = x - 9$ (C) $h(x) = x^2 - 81$ (D) $h(x) = x^2 + 81$

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

609. Given two functions from the source object, determine the equation for the combined function with simplified result $+ 10$.

(A) $+ 10$ (B) $- 10$ (C) $+ 3$ (D) $- 13$

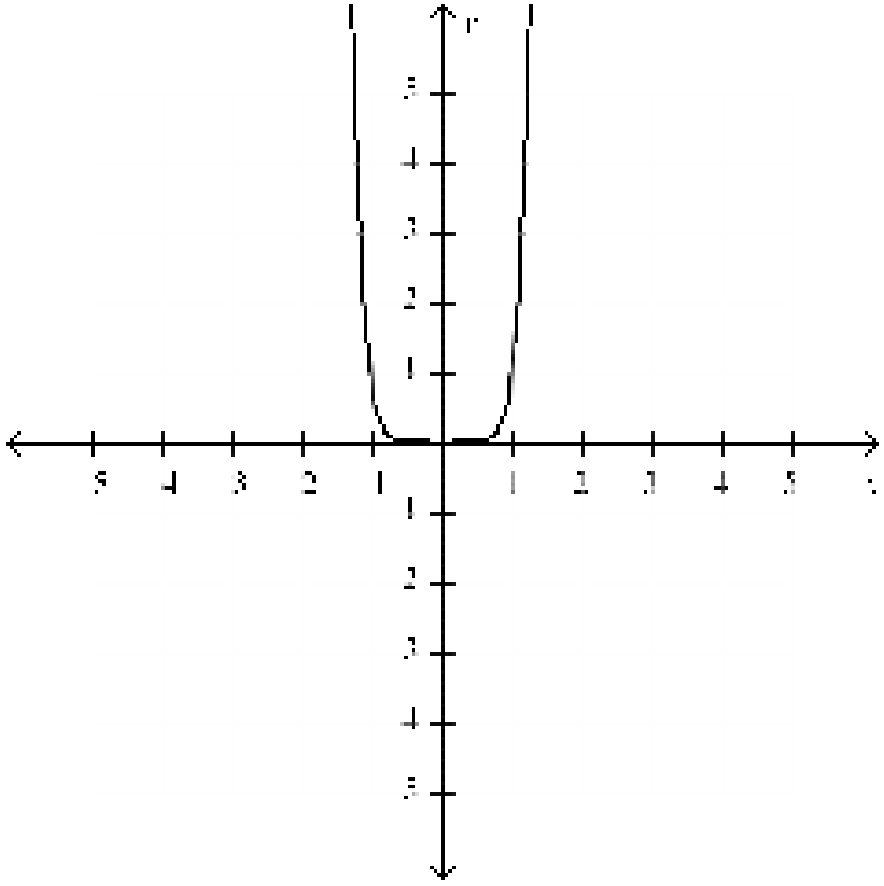
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

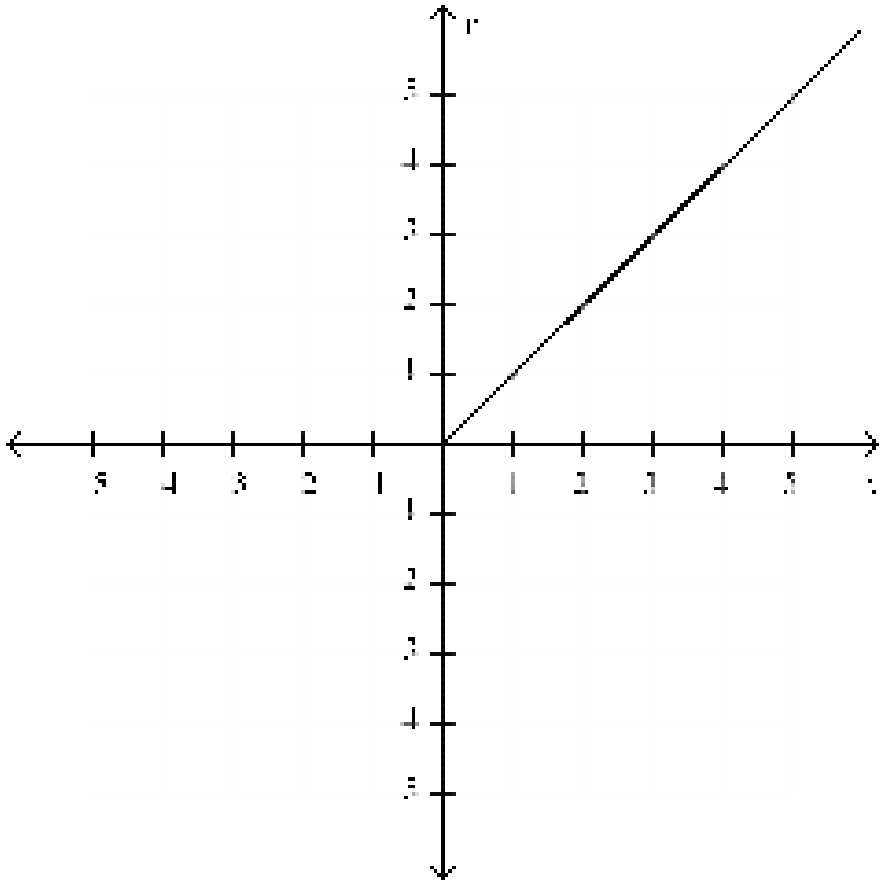
610. Given the functions $8 1 8 0 8 0 8$ and $0 8 1 8 0 8 8,$ the graph of the combined function 1 most likely resembles Recovered graph functions: $x^(8); 1/x; x*sin(x); x*cos(x); x*tan(x); x^((8)+1); (8)*(8)^x; -x$. Rendered ExamView reference:

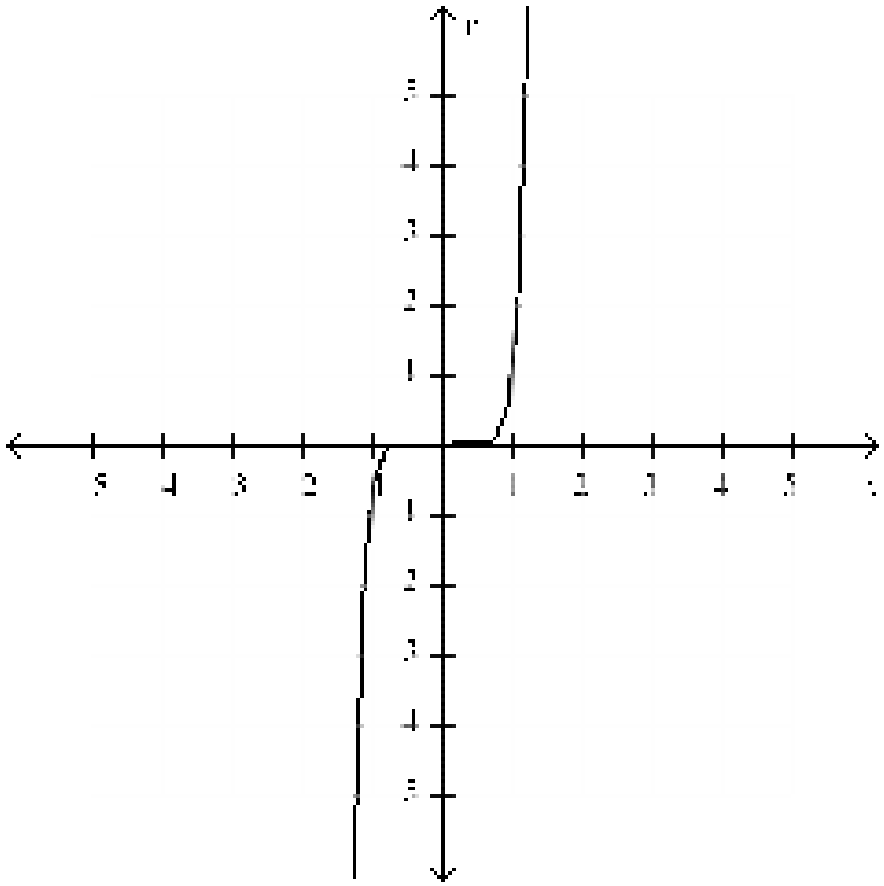
$$f(x) = x^{8}$$

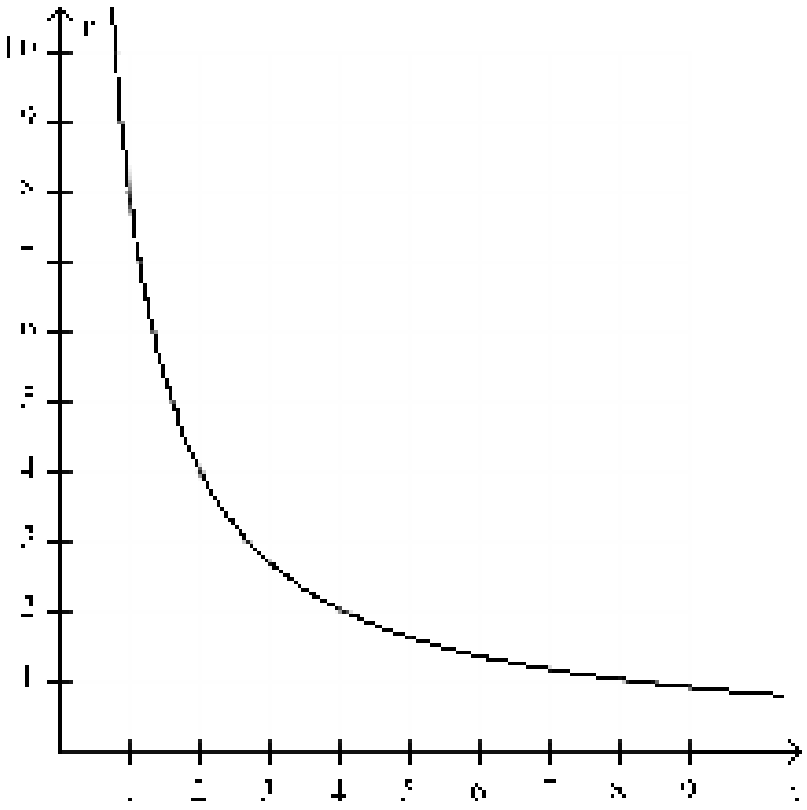
$$g(x) = \sqrt[8]{x}$$

$$h(x) = f(g(x))$$









(A) graph of x^(8) (B) graph of 1/x (C) graph of x*sin(x) (D) graph of x*cos(x)

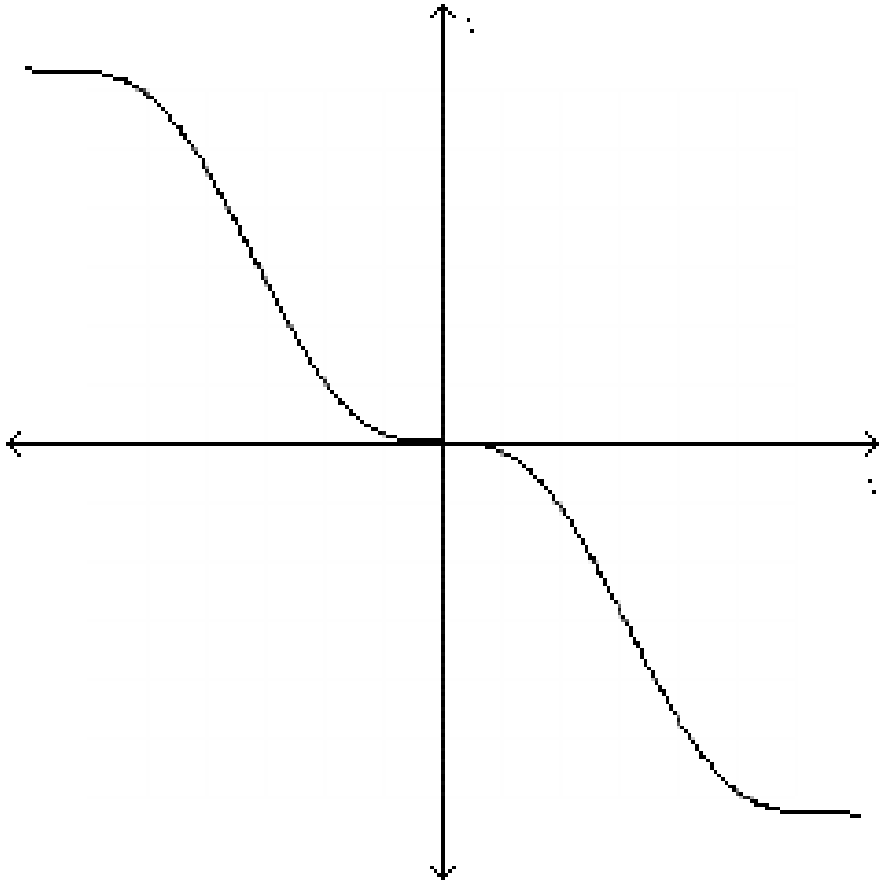
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

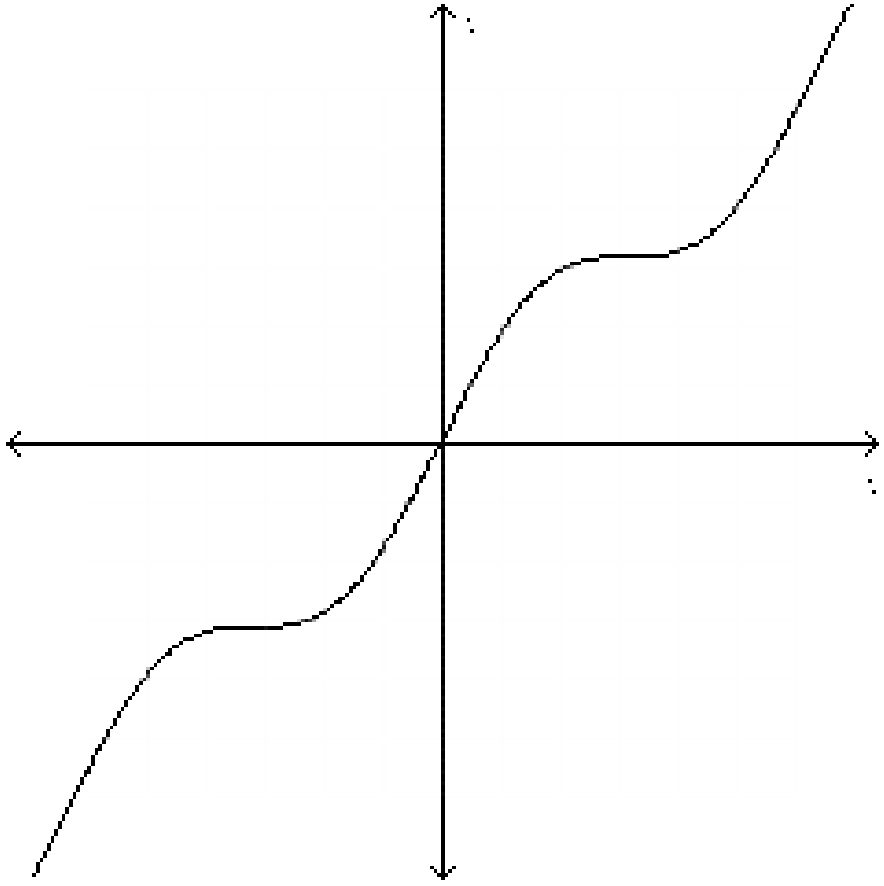
611. Given the functions 2 and cos, a graph of the combined function range($−1,1)$ most likely resembles Recovered graph functions: $sin(x)+x; sin(x)-x; cos(x)+x; cos(x)-x$. Rendered ExamView reference:

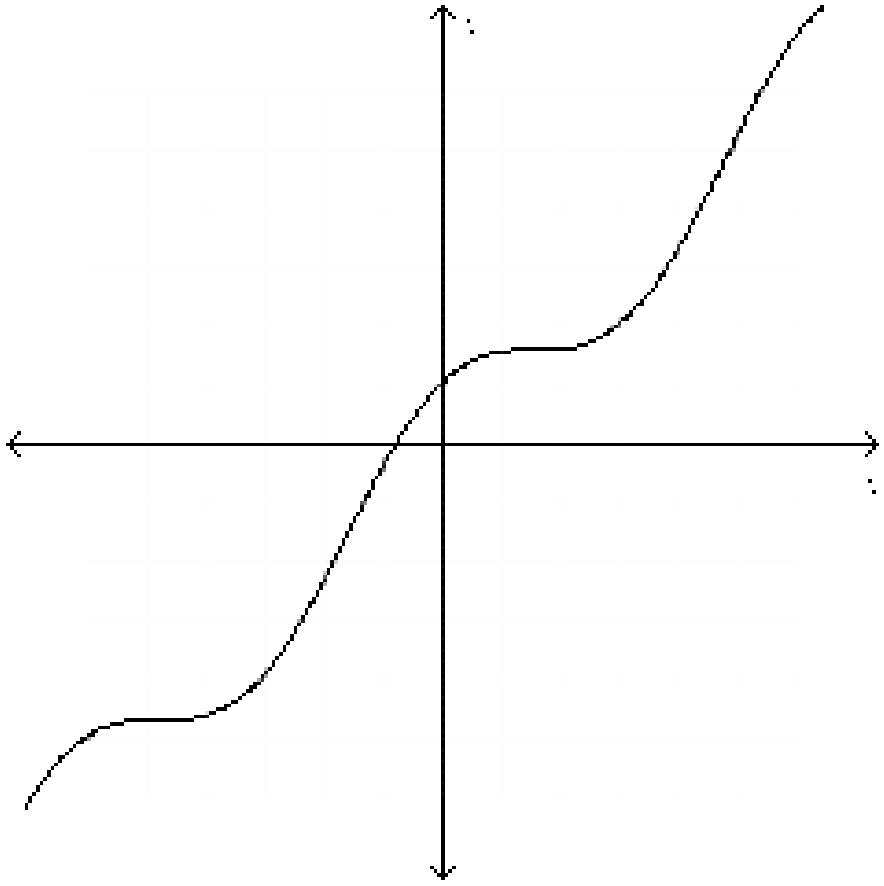
$$f(x) = \cos x$$

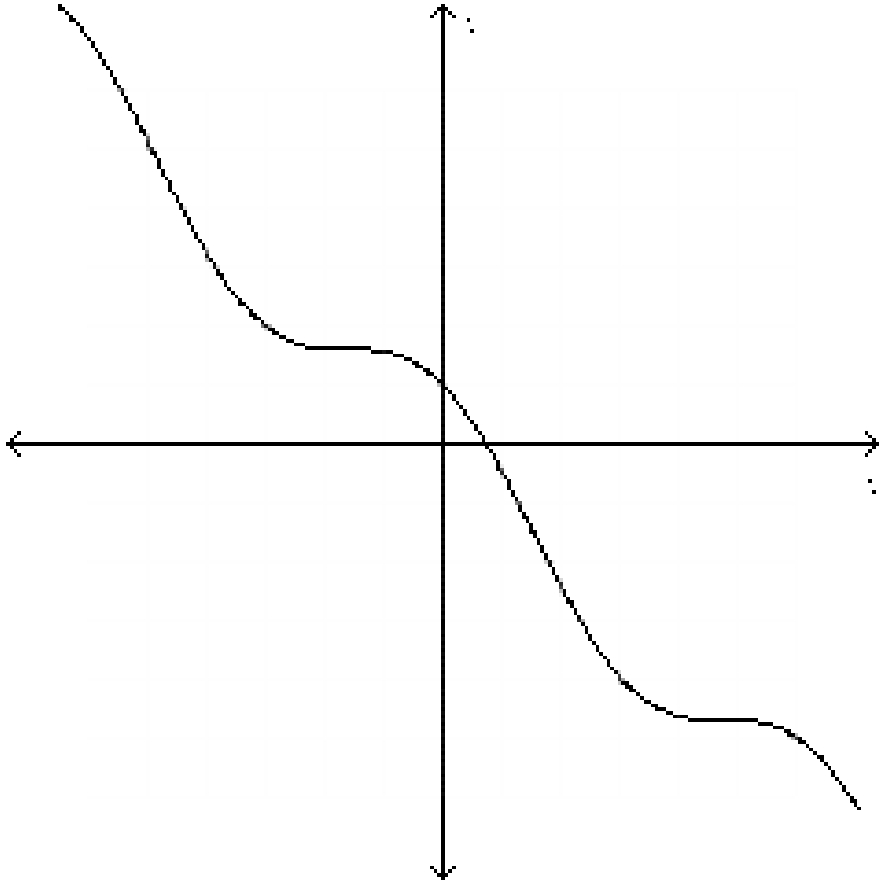
$$g(x) = x$$

$$h(x) = f(x) - g(x)$$









(A) graph of sin(x)+x (B) graph of sin(x)-x (C) graph of cos(x)+x (D) graph of cos(x)-x

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

612. Given the source functions with parameters $a = 9$ and $b = 9$, what is the range of the composite function?

(A) $y <= 0$ (B) cannot be determined (C) $y >= 0$ (D) $−9 <= y <= 81$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

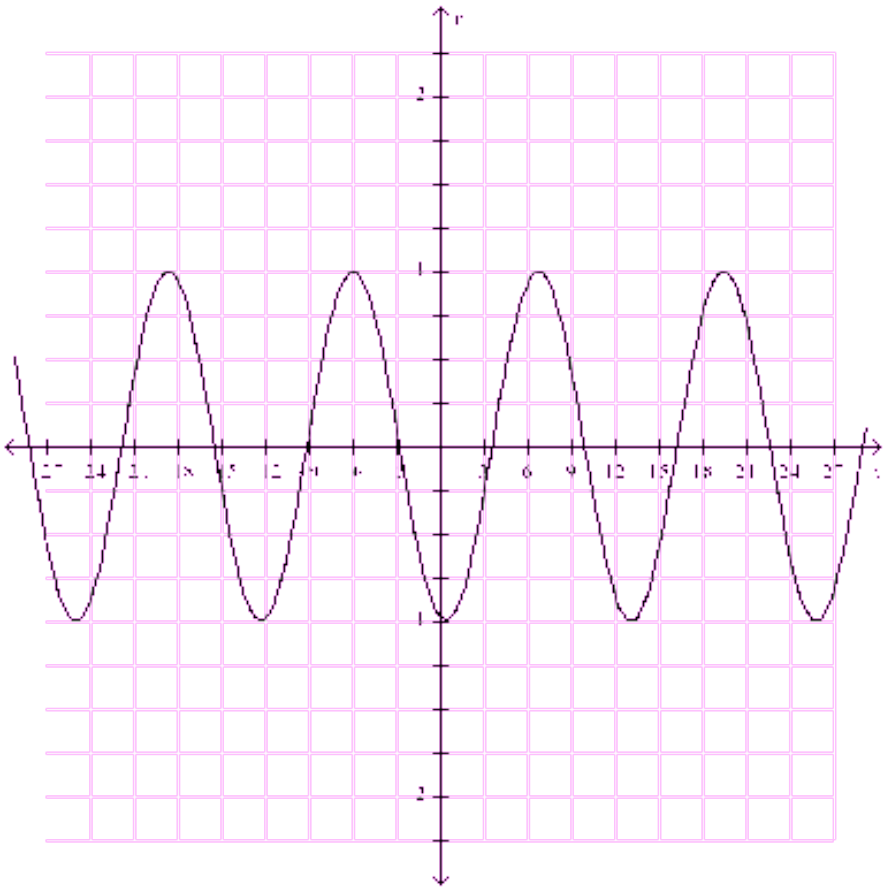
613. What equation represents 1? Recovered graph functions: $cos(((−4)/(−8))*(x+(9))); sin(((−4)/(−8))*(x+(9)))$. Rendered ExamView reference:

$$h(x) = f(g(x))$$

$$f(x) = \sin x$$

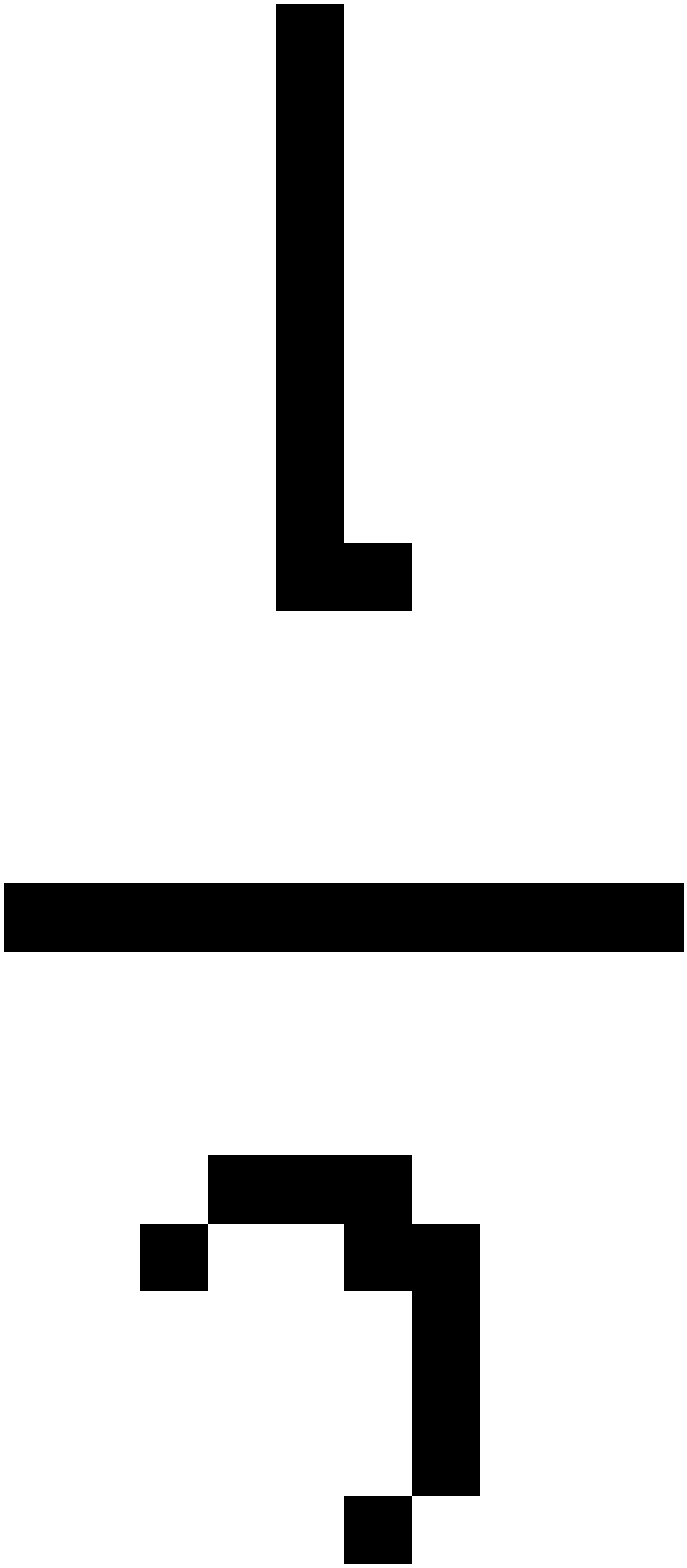
$$g(x)$$

$$g(x) = a(x + b)$$



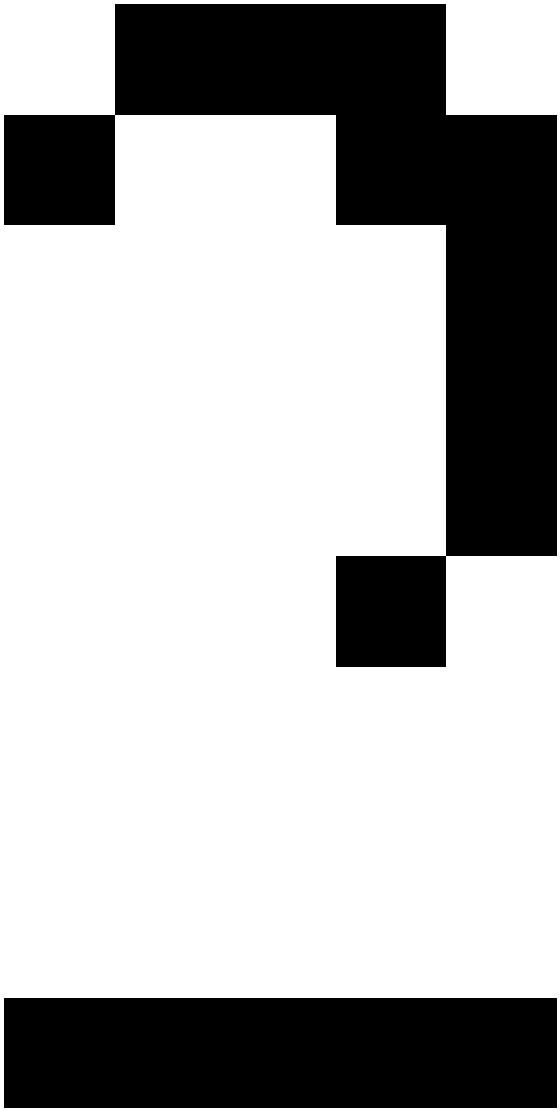
$$g(x)$$

$$g(x) =$$



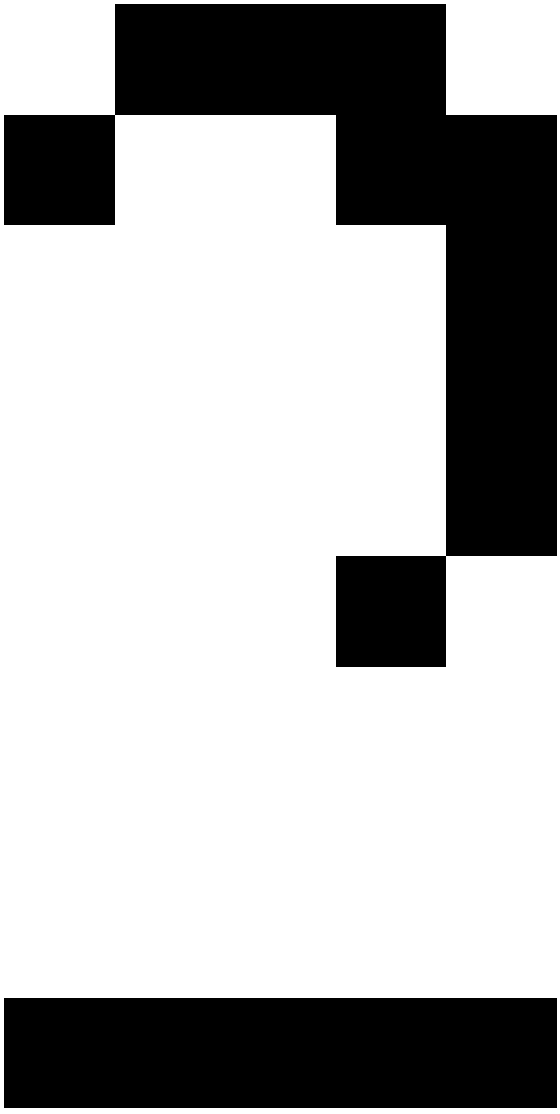
$$(x+9)$$

$$g(x) =$$



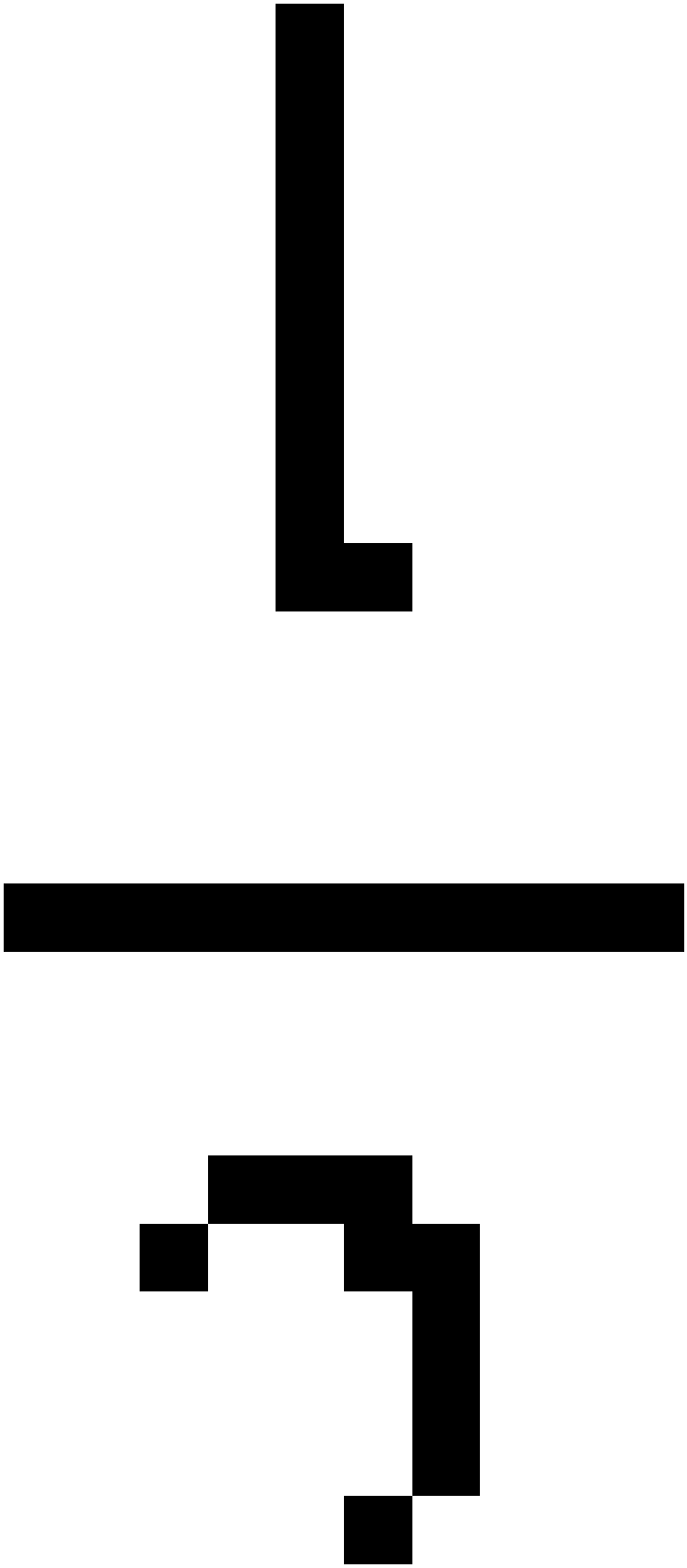
$$(x+9)$$

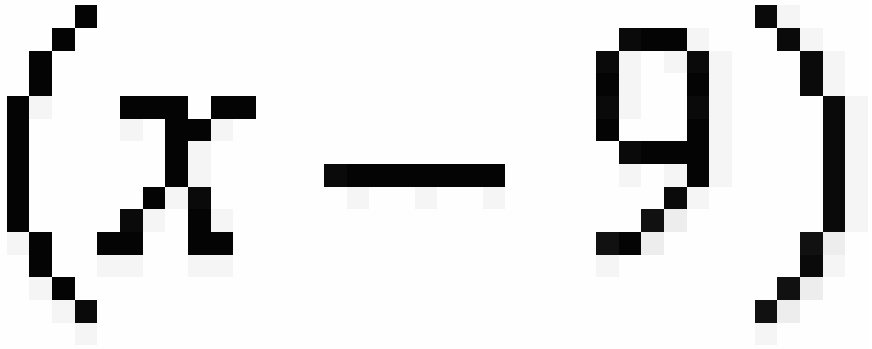
$$g(x) =$$



$$(x-9)$$

$$g(x) =$$





(A) 2 (B) – 9 (C) $1/2$ (D) – 9

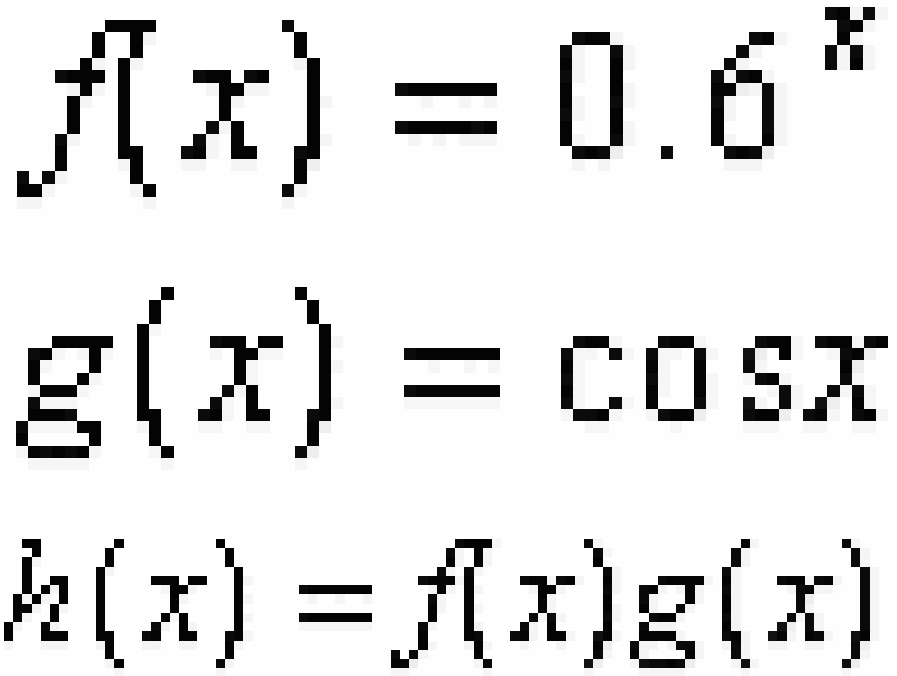
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

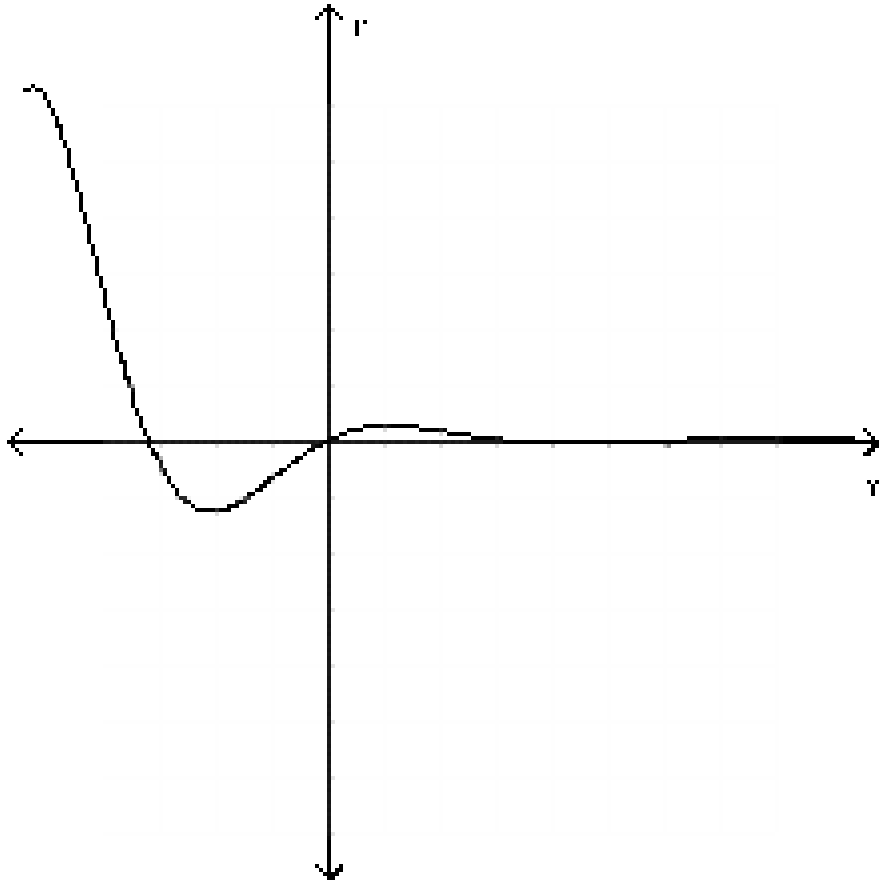
614. Given the functions sin and $2,$ what is the value of $g(x) = a + x$?

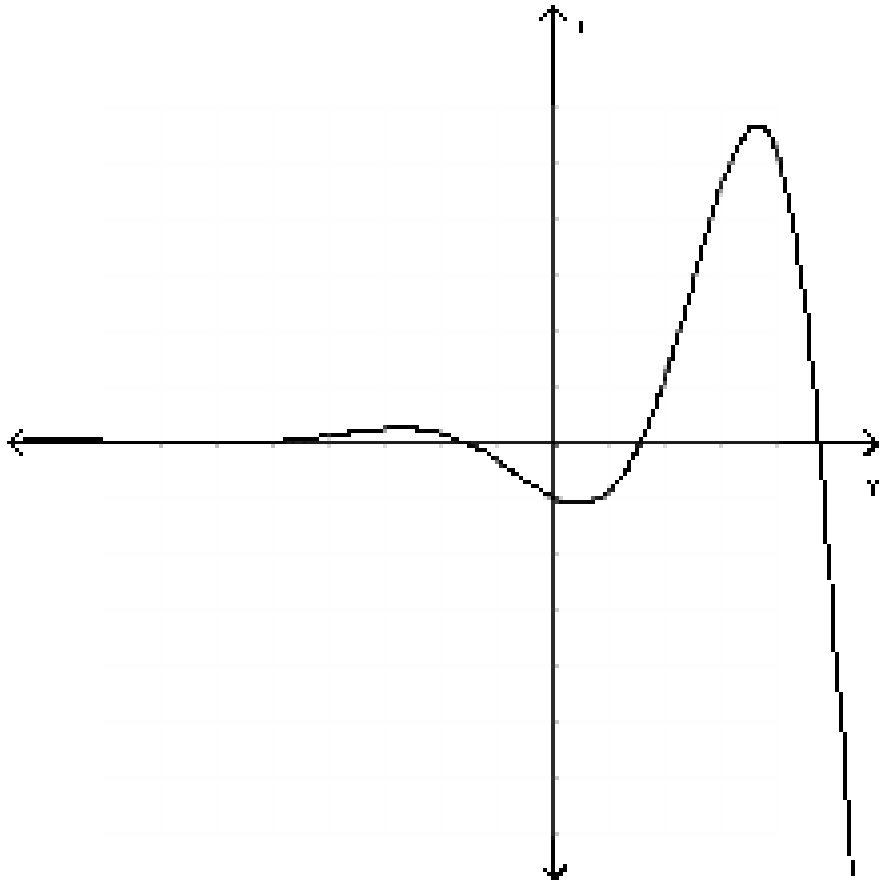
(A) $1/2$ (B) $1/3$ (C) 1

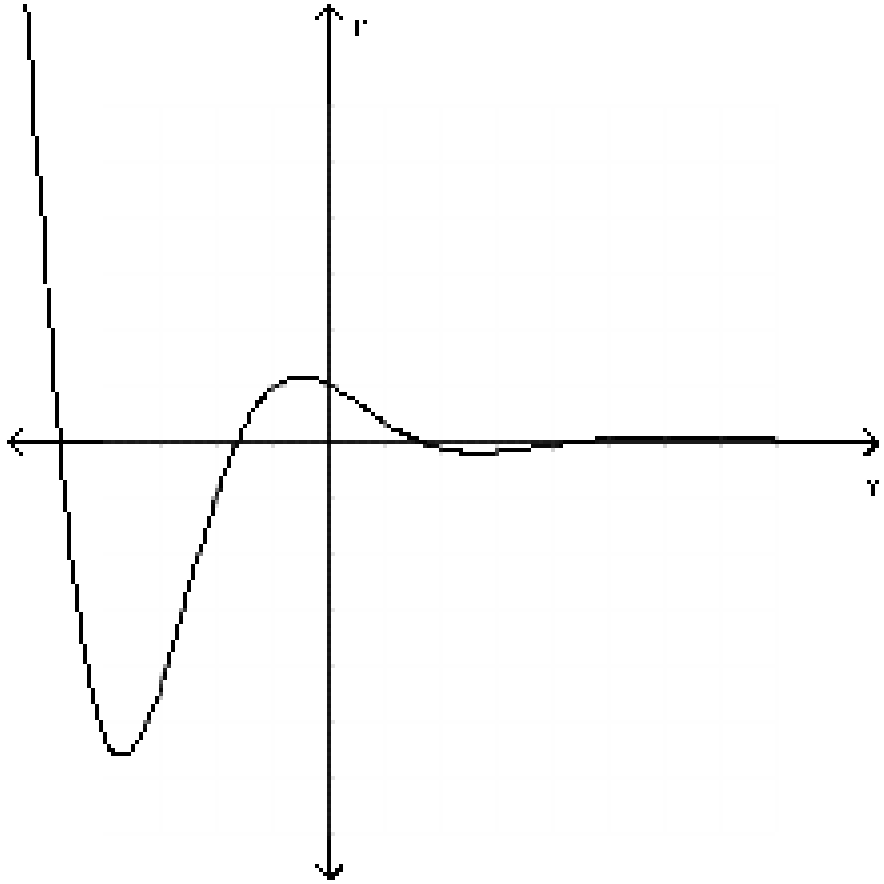
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

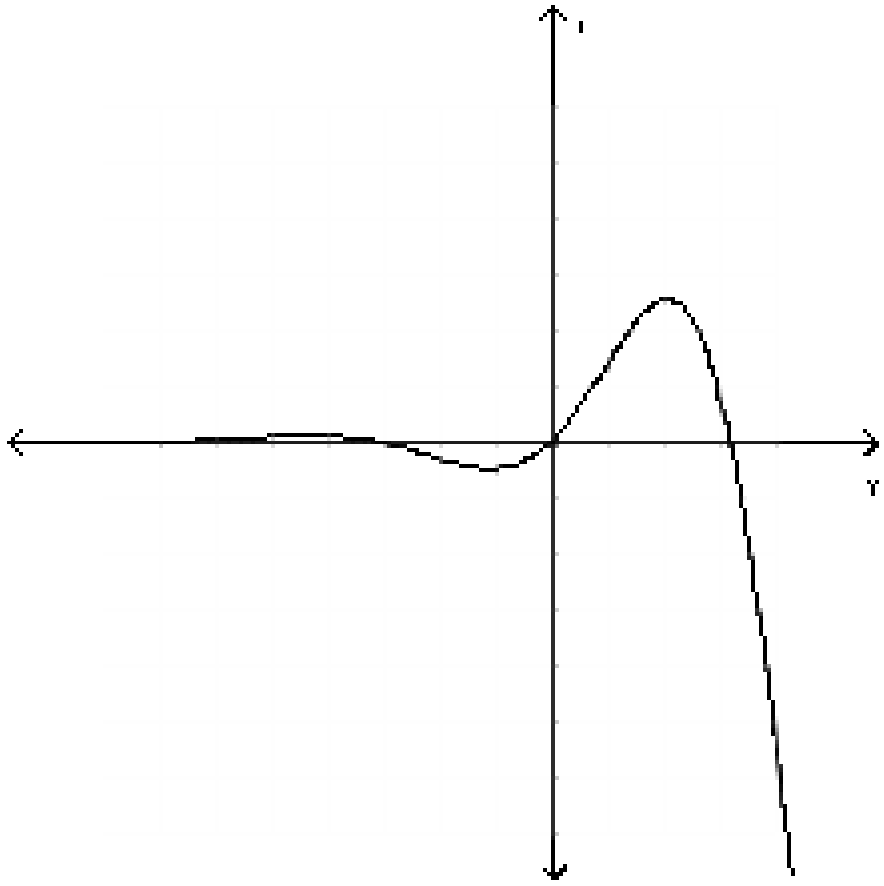
615. Given the functions 0 and $0.6,$ the graph of the combined function cos most likely resembles Recovered graph functions: (($0.6)^x)*sin(x); ((0.6)^x)*cos(x); -((0.6)^-x)*sin(x); -((0.6)^-x)*cos(x); ((0.6)^(-x))*sin(x); ((0.6)^-x)*cos(x)$. Rendered ExamView reference:




Y
X


Y
X





(A) graph of ((0.6)^x)*sin(x)

(B) graph of ((0.6)^x)*cos(x)

(C) graph of -((0.6)^-x)*sin(x)

(D) graph of -((0.6)^-x)*cos(x)

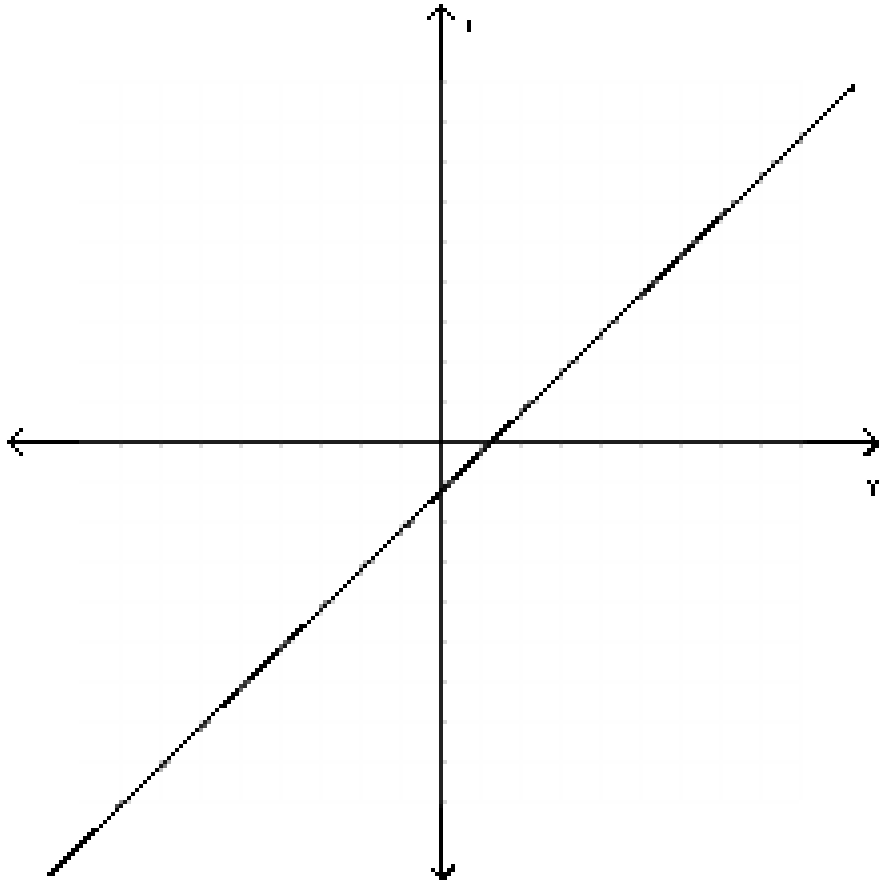
ANS: B **TOP:** bnk | mcq **TYPE:** mcq

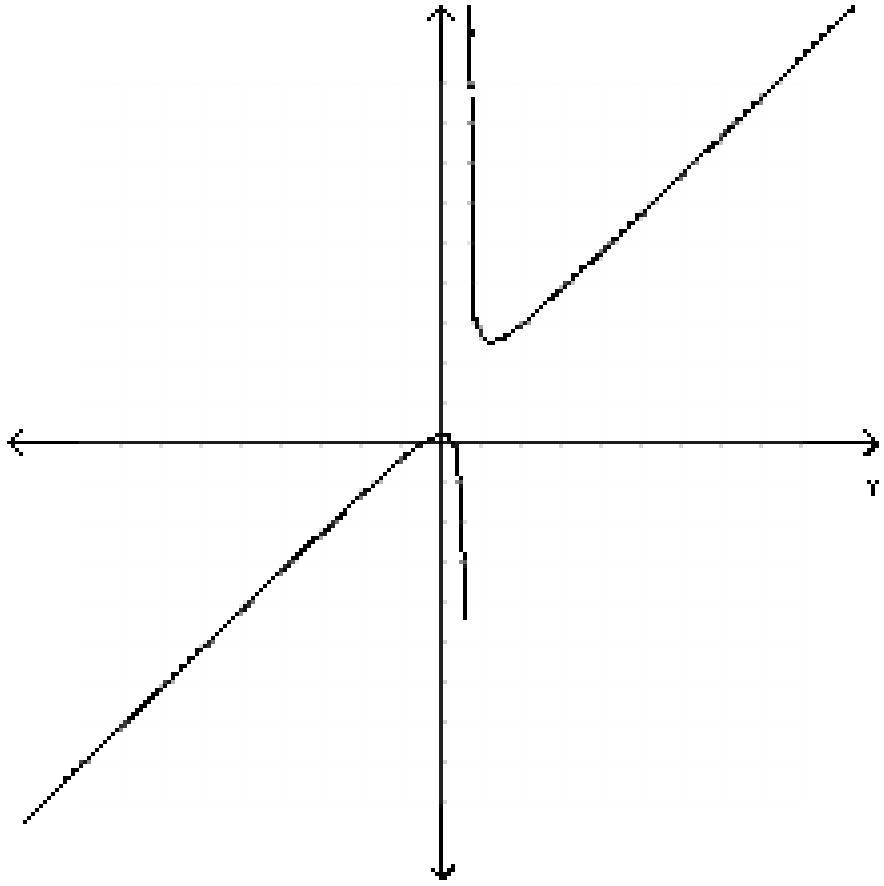
616. Given the functions 4 and $4, a$ graph of the combined function 4 most likely resembles Recovered graph functions: $x-(4); (x^2-(4))/(x-(4)); (4)/(x-(4)); -(x^2-(4))/(x-(4))$. Rendered ExamView reference:

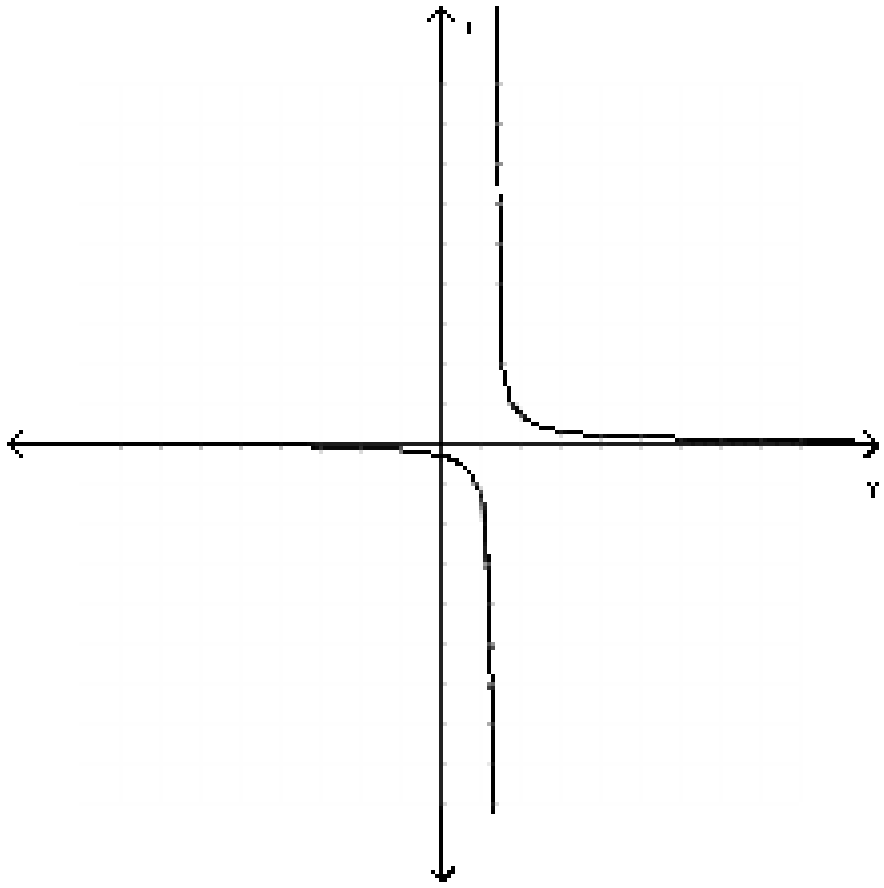
$$f(x) = x^2 - 4$$

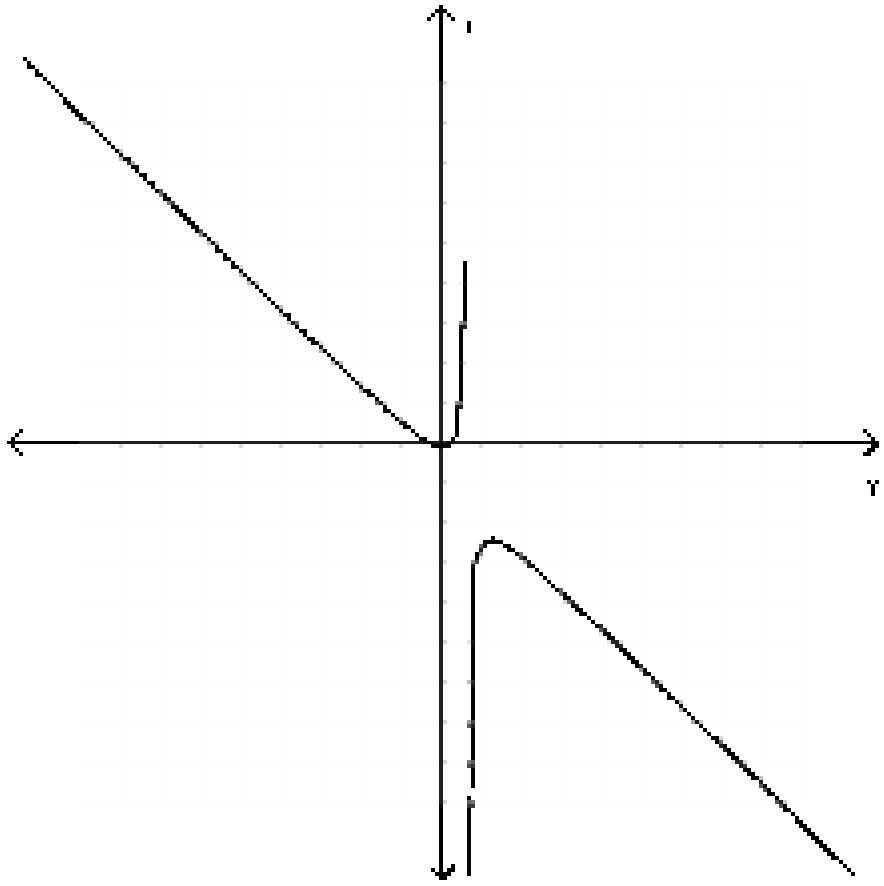
$$g(x) = x - 4$$

$$h_2(x) = \frac{f(x)}{g(x)}$$


I
Y


I
Y





(A) graph of x-(4)

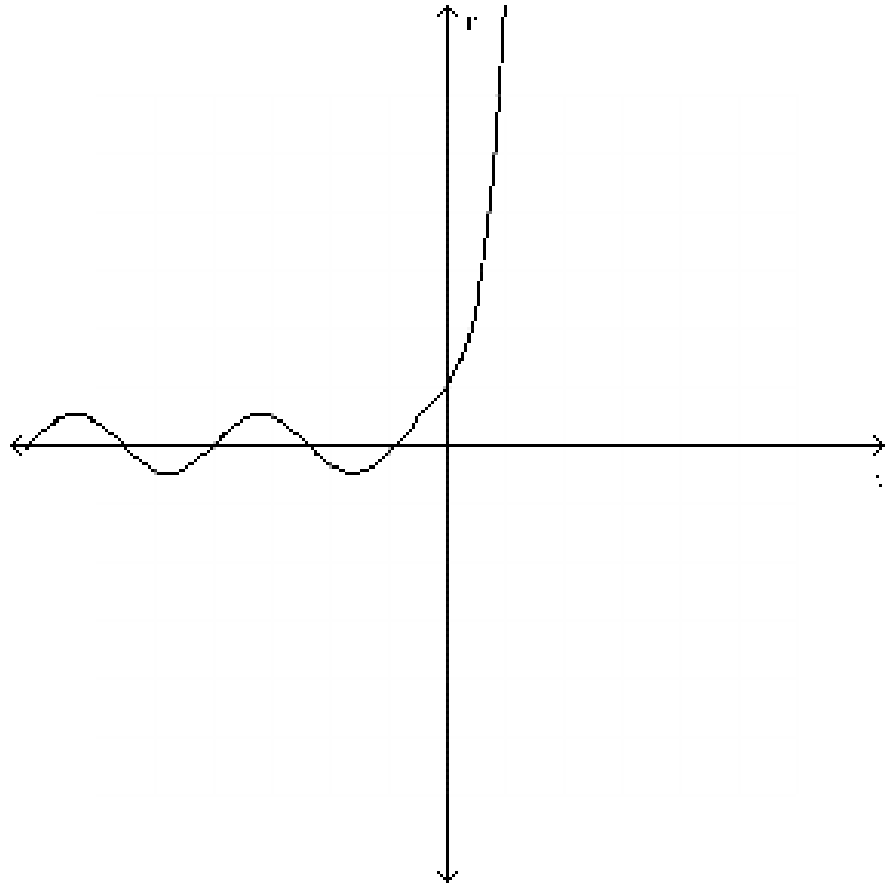
(B) graph of (x^2-(4))/(x-(4))

(C) graph of (4)/(x-(4))

(D) graph of -(x^2-(4))/(x-(4))

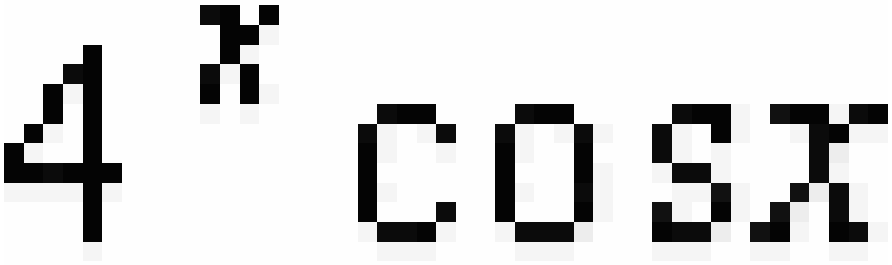
ANS: C **TOP:** bnk | mcq **TYPE:** mcq

617. An equation for the graph shown is most likely 0 Recovered graph functions: ($4)^x+sin(x); (4)^x+cos(x)$. Rendered ExamView reference:



$4^x + \cos x$

$4^x + \sin x$

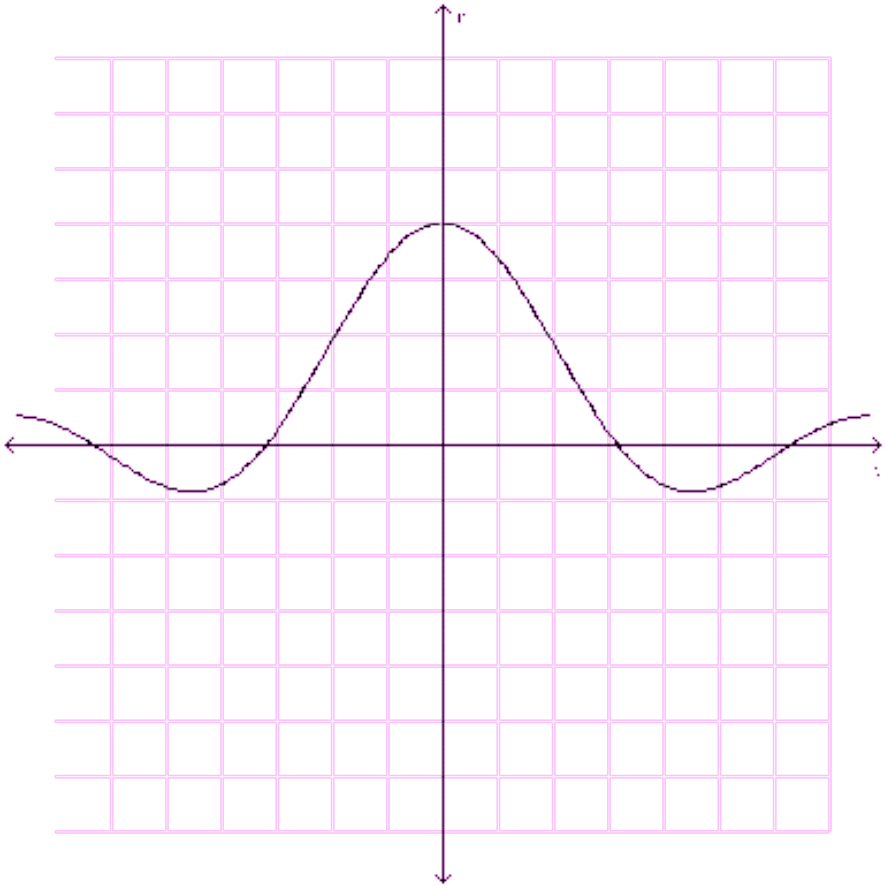




(A) 4 (B) cos (C) 4 (D) sin

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

618. An equation for the graph shown is most likely 1 Recovered graph functions: ($sin(x))/x; (cos(x))/x$. Rendered ExamView reference:



$$f(x) = \frac{\cos x}{x}$$

$$\frac{\dot{\sin x}}{\cos x}$$

$$2^{\dot{\sin x}}$$

$$f(x) = \frac{\sin x}{x}$$

(A) 1 (B) 1 (C) sin (D) sin

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

619. Given the functions $+ 2$ and – $2,$ what is the domain of the combined function $+ 7$?

(A) – 2 (B) + 7 (C) $−2$ (D) $−9$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

620. Given the functions 8 and $7,$ determine the domain of the combined function –7.

(A) 8 (B) 7 (C) $−7$ (D) 7

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

621. Given the functions 7 and – $54,$ determine the domain of the combined function 7.

(A) $−9$ (B) 6 (C) 9 (D) $−6$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

622. Given the functions 5 and $4,$ what is the domain of the composite function –4?

(A) 4 (B) $−4$ (C) 4 (D) 5

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

623. Given the functions 7 and $7,$ determine the range of the combined function sin.

(A) $−14$ (B) 0 (C) $−7$ (D) 7

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

624. Given the functions 3 and $3,$ what is the simplified form of – 8?

(A) 3 (B) – 8 (C) 3 (D) $−3$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

625. If –7 and –$7,$ what are the restrictions on the domain for the combined function 5?

(A) 9 (B) 81 (C) 81 (D) 3

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

626. If the source functions are divided, what is the simplified combined function with denominator restriction $x != + 14$?

(A) $x != 1/5$ (B) $x != 5$ (C) $x != 25$ (D) $x != + 14$

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

627. Given 9 and $7,$ determine 5.

(A) 45 (B) 36 (C) 6

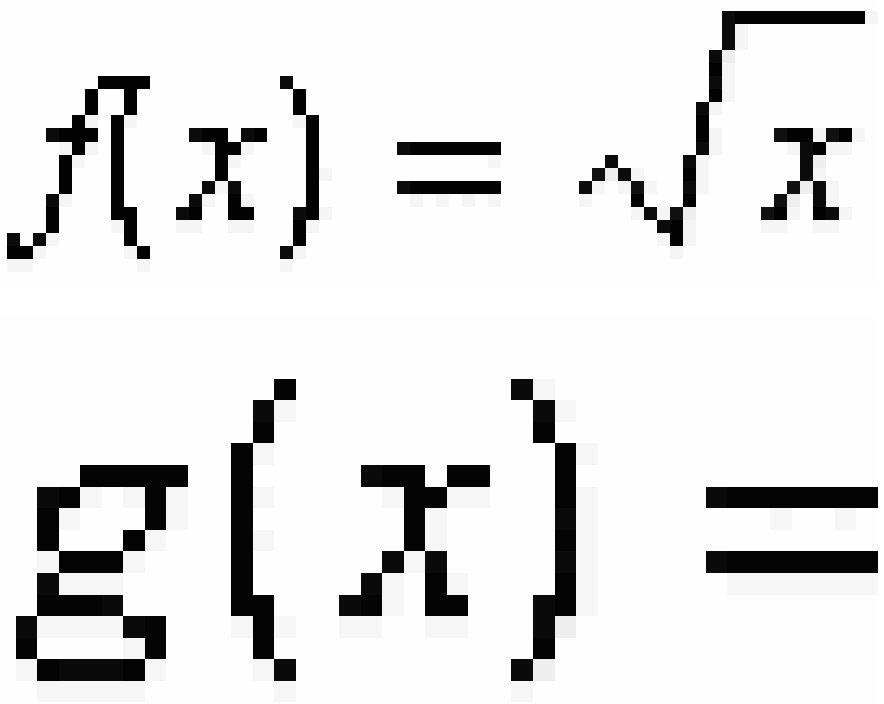
ANS: A **TOP:** bnk | mcq **TYPE:** mcq

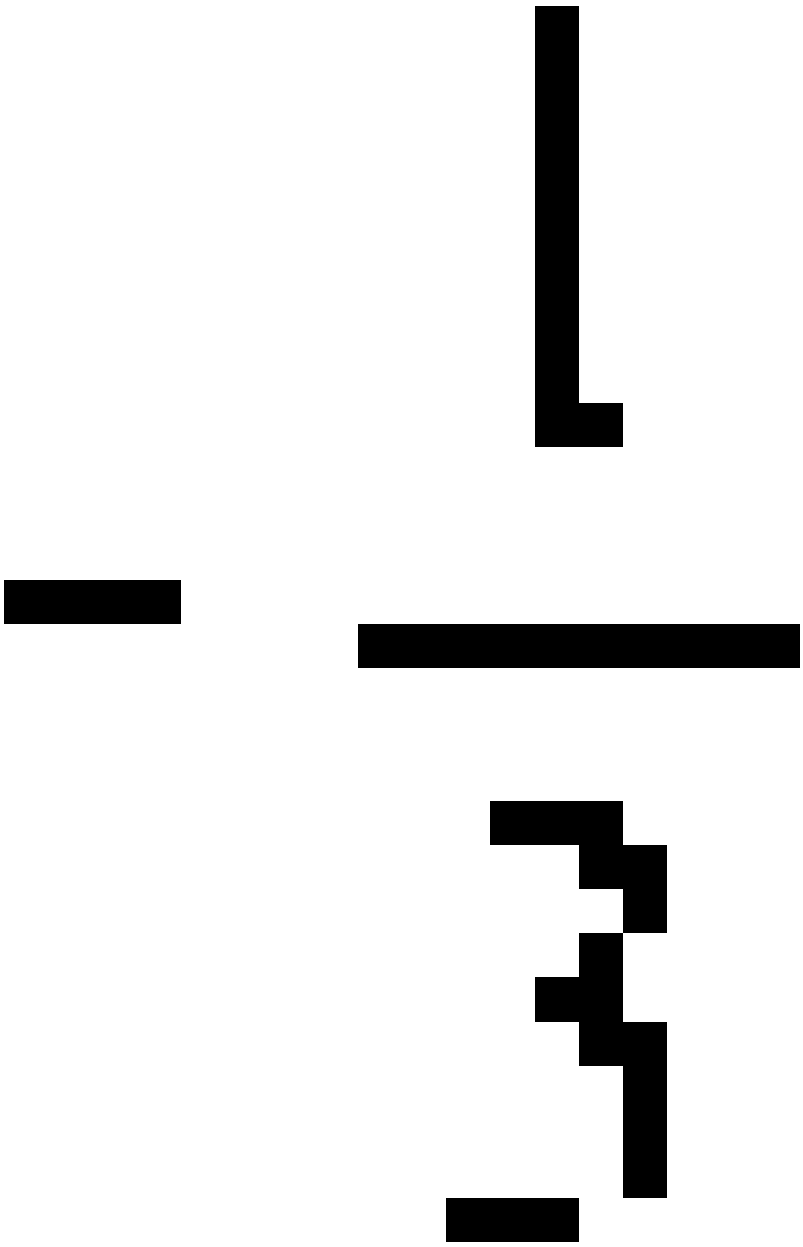
628. Given the function – $5,$ determine the value of –1.

(A) $−8$ (B) 11 (C) 16

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

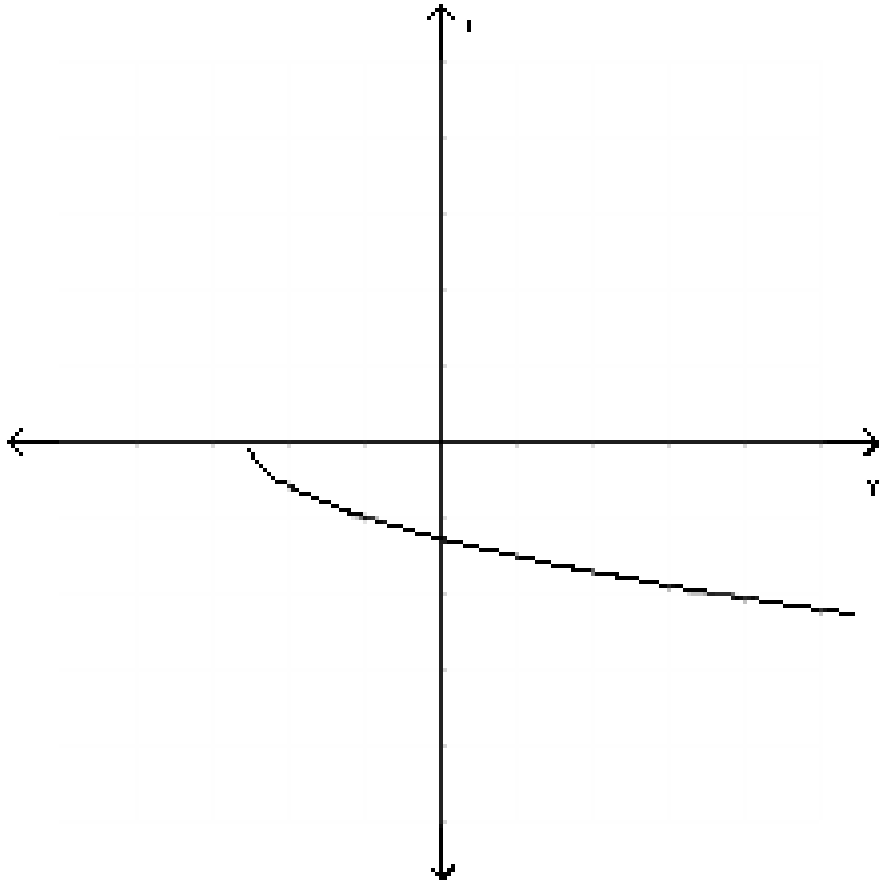
629. Given the functions $−1/3$ and $−1/3 −1/3$ – $5,$ which of the following is most likely the graph of $-sqrt(((2)/(−6))*(-x+(−5)))$? Rendered ExamView reference:

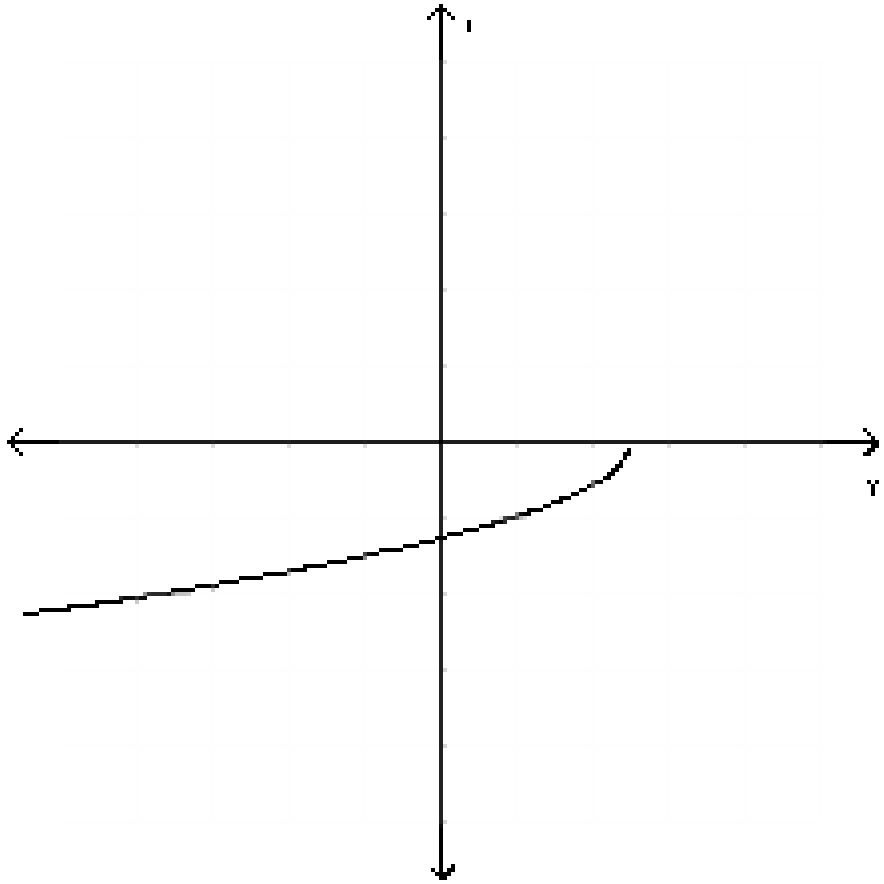


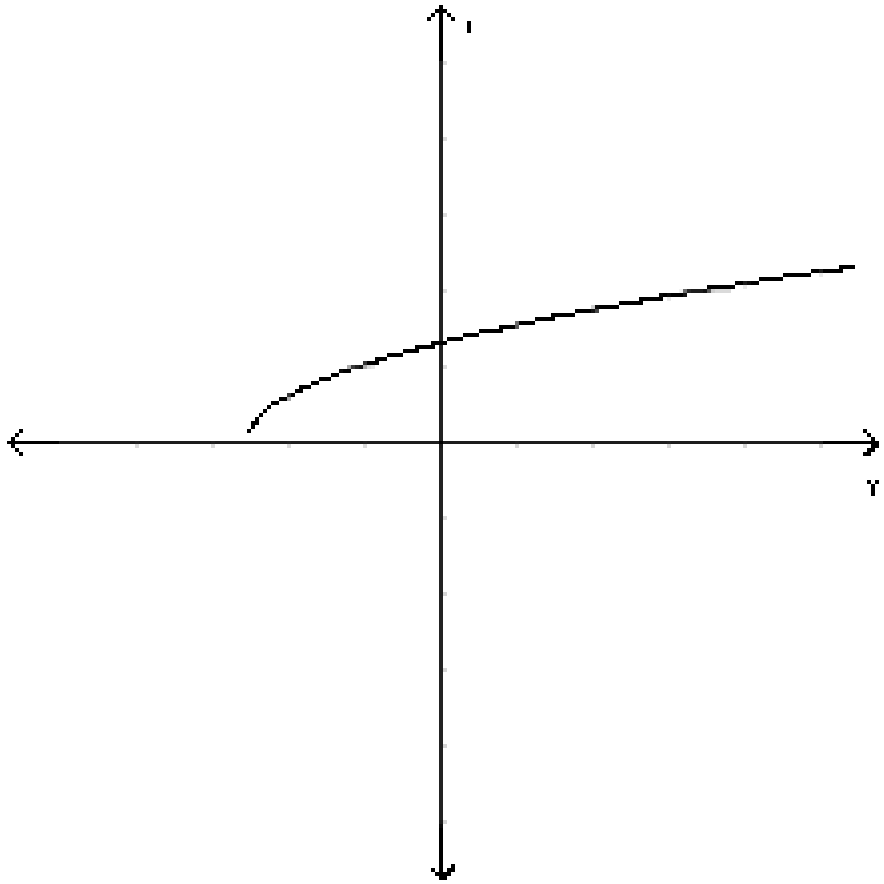


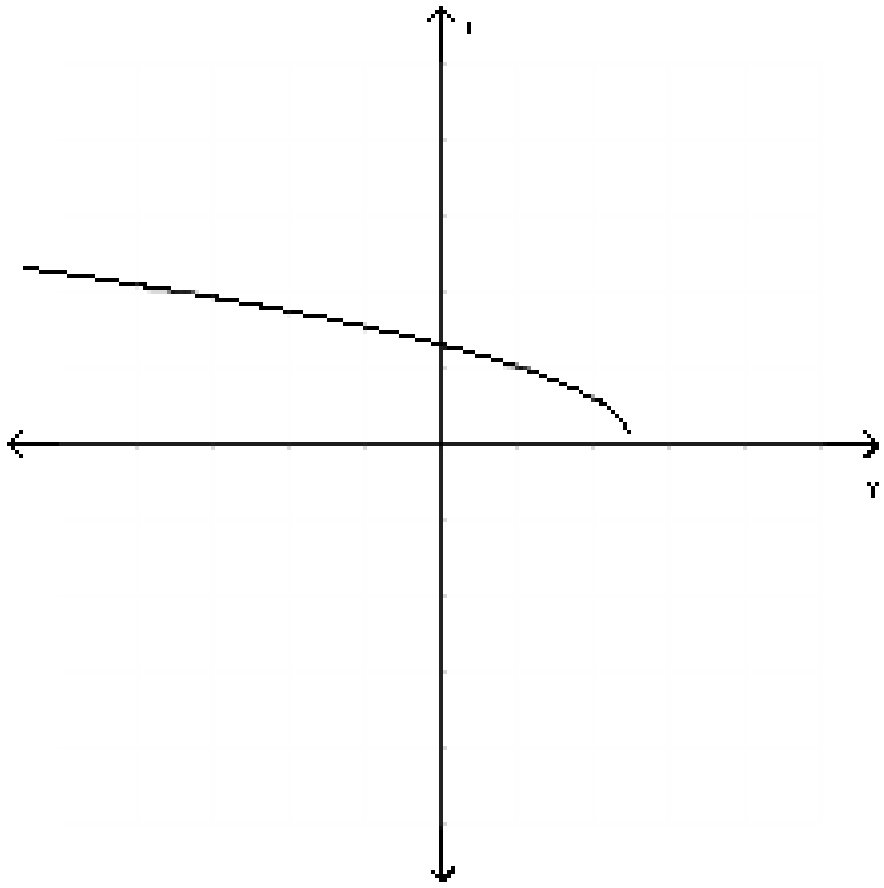
$$(x-5)$$

$$y = f(g(x))$$




y
x


y
x



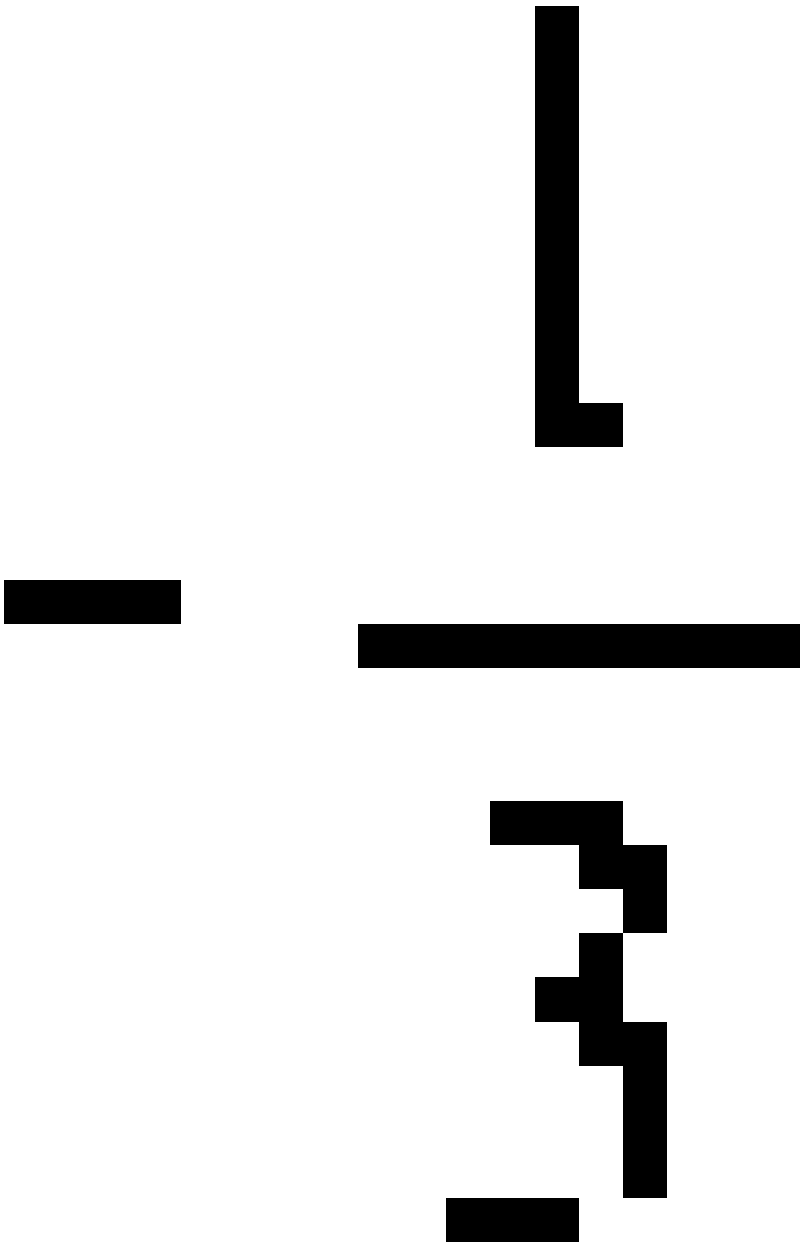
(A) $−1/3$ (B) $−1/3$ (C) $−1/3$ (D) – 5

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

630. Given the functions $+ 4$ and $+ 4,$ which of the following is most likely the graph of $+ 4$? Recovered graph functions: $-log(-x+(4)); log(x+(4)); -log(x+(4)); log(-x+(4))$. Rendered ExamView reference:

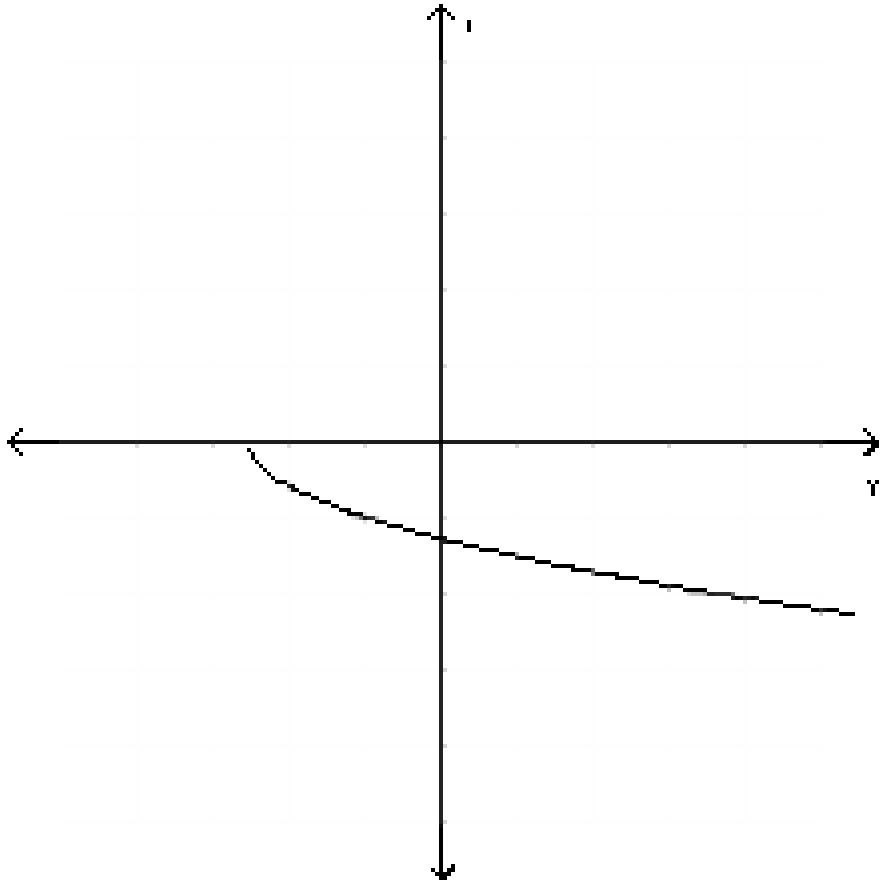
$$f(x) = \sqrt{x}$$

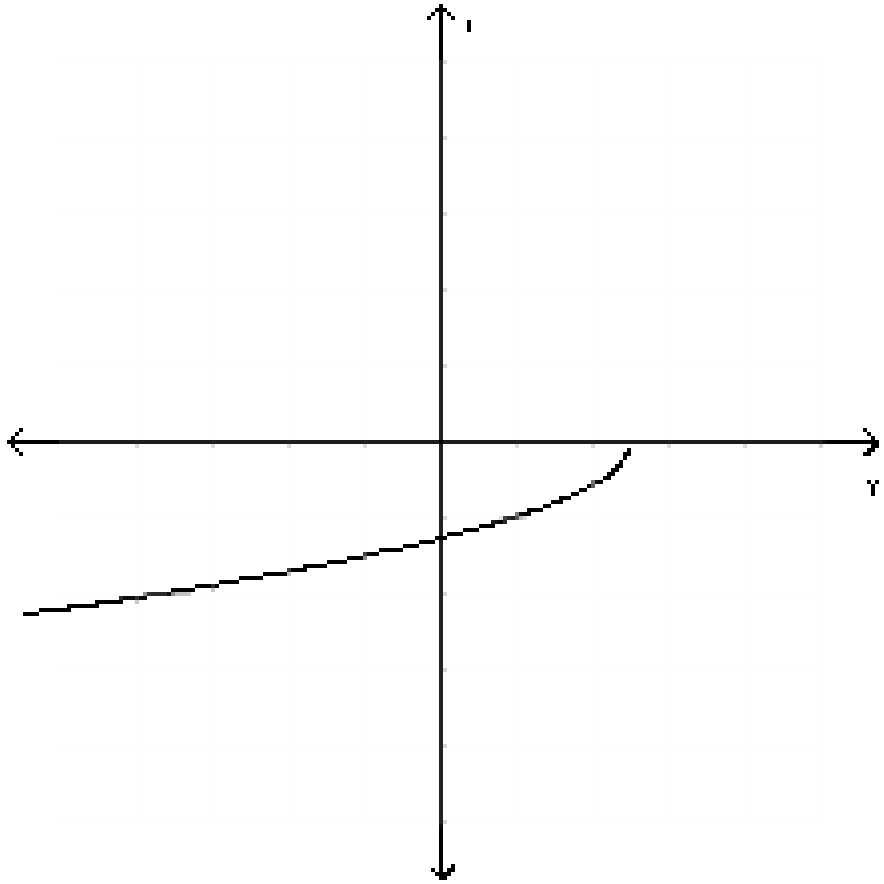
$$g(x) =$$

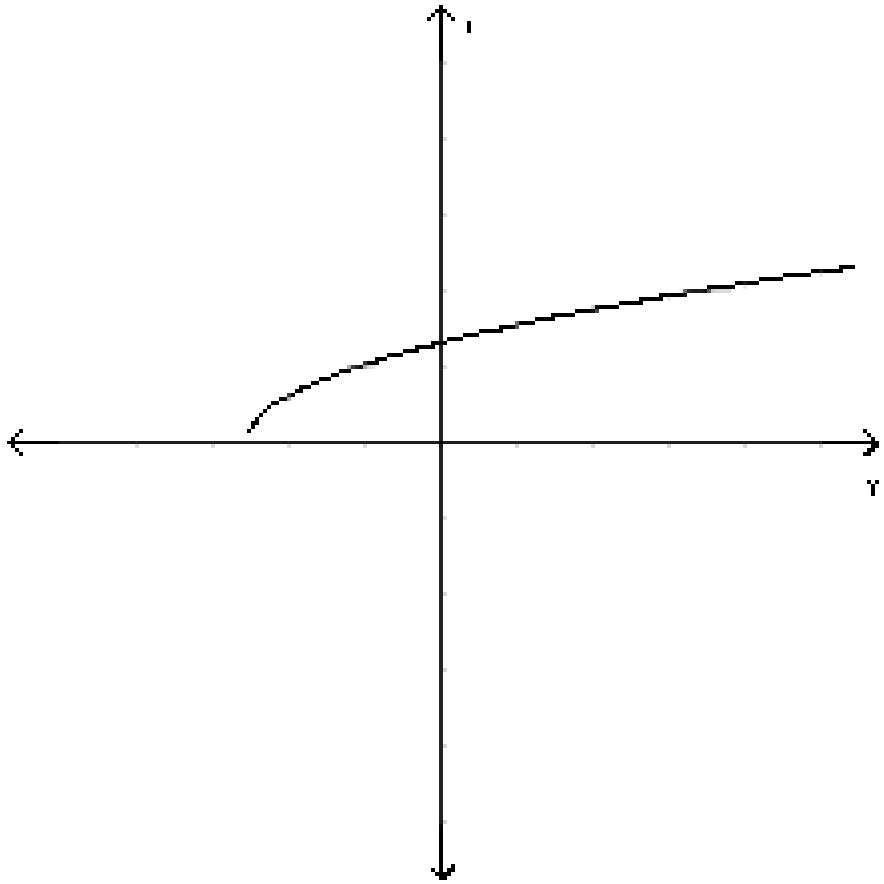


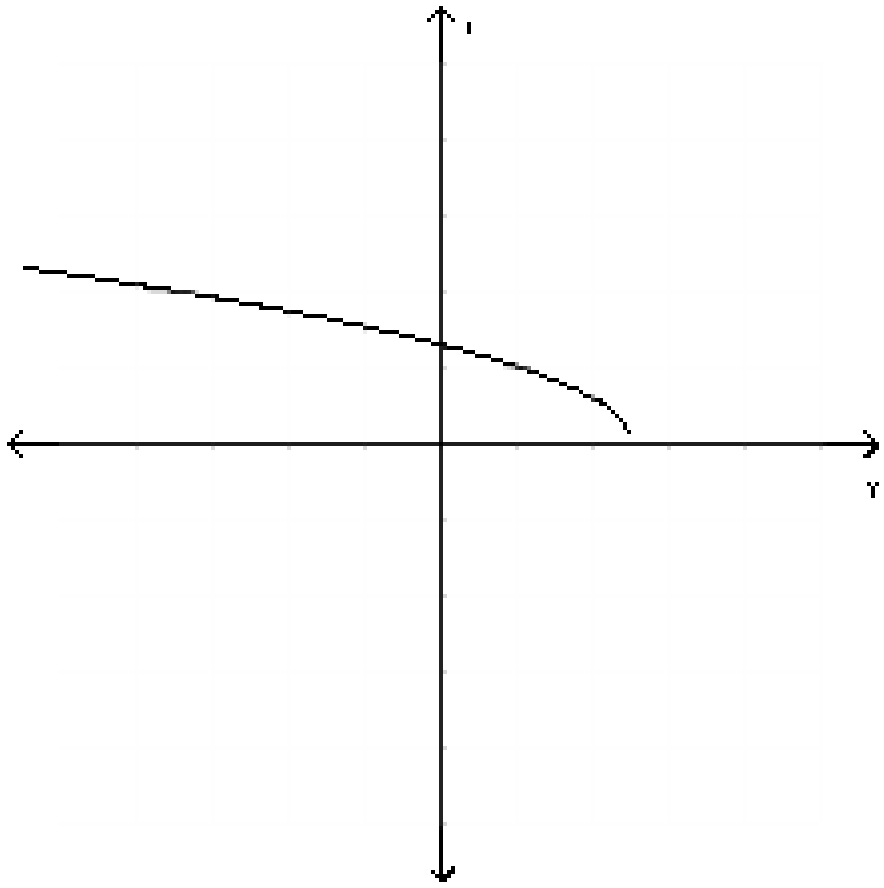
$$(x-5)$$

$$y = f(g(x))$$




y
x


y
x



(A) graph of -log(-x+(4)) (B) graph of log(x+(4)) (C) graph of -log(x+(4)) (D) graph of log(-x+(4))

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

631. $h(x) = f(x) + g(x)$

TOP: bnk | short_answer **TYPE:** short_answer

632. $h(x) = f(x) - g(x)$

TOP: bnk | short_answer **TYPE:** short_answer

633. $h(x) = f(x) g(x)$

TOP: bnk | short_answer **TYPE:** short_answer

634. $h(x) = f(x) / g(x)$

TOP: bnk | short_answer **TYPE:** short_answer

635. $h(x) = f(g(x))$

TOP: bnk | short_answer **TYPE:** short_answer

636. $h(x) = g(f(x))$

TOP: bnk | short_answer **TYPE:** short_answer

637. If $f(x)$ is a quadratic function with positive coefficients and $g(x)$ is a linear function with positive coefficients, then $(f + g)(x)$ is a quadratic function.

ANS: quadratic **TOP:** bnk | problem **TYPE:** problem

638. If $f(x)$ is a linear function and $g(x)$ is a linear function, then $(f g)(x)$ is a quadratic function.

ANS: quadratic **TOP:** bnk | problem **TYPE:** problem

639. $abs(a)>1$ and $abs(b)>1$ and $b+a<0 (a =$ – $4, b =$ – $3, c = 7, c = 7,$ domain $= 1/7, b =$ – $3, b =$ – $3,$ domain $= 1/7)$

TOP: bnk | problem **TYPE:** problem

640. $abs(a)>1$ and $abs(b)>1 (a =$ – $8, b = + 4)$

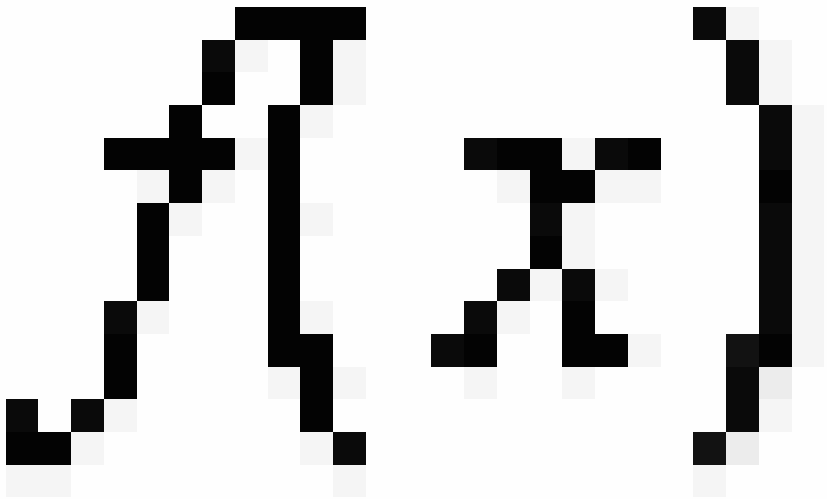
ANS: 8 **TOP:** bnk | problem **TYPE:** problem

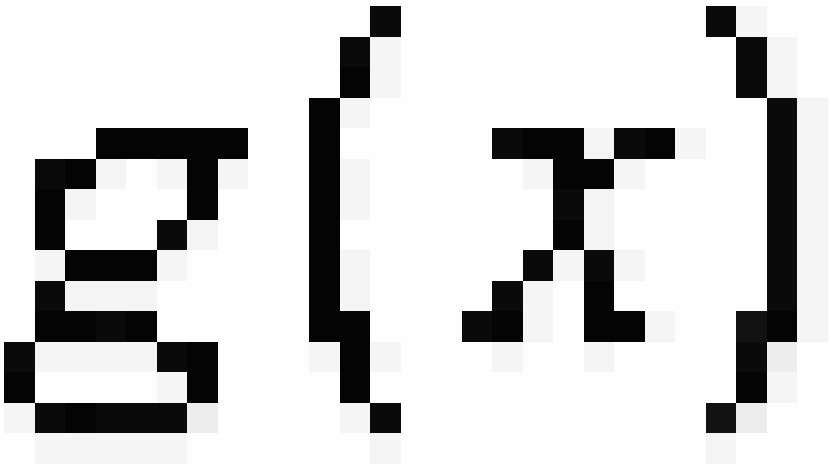
641. If $7,$ and $cot,$ then $7 =$ _____ and

$$h(x) = f(g(x))$$

$=$ _____. Rendered ExamView reference:

$$h(x) = 7\cot^2 x$$





ANS: tan **TOP:** bnk | problem **TYPE:** problem

642. Given the functions $f(x) = x + 1$ and $g(x) = x^2 + 3x + 1$, determine a simplified equation for $h(x) = f(x) + g(x)$.

TOP: bnk | short_answer **TYPE:** short_answer

643. Given the functions $f(x) = x^2 + 3$ and $g(x) = x^3 - x$, determine a simplified equation for $h(x) = f(x) - g(x)$.

TOP: bnk | short_answer **TYPE:** short_answer

644. Given the functions $f(x) = x^2 - 7$ and $g(x) = 2 - x^3$, determine a simplified equation for $h(x) = f(x) g(x)$.

TOP: bnk | short_answer **TYPE:** short_answer

645. Given the functions $f(x) = x^2 - 4$ and $g(x) = x^2 - 3x + 2$, determine a simplified equation for $h(x) = f(x) / g(x)$.

TOP: bnk | short_answer **TYPE:** short_answer

646. (a) Graph the functions $sin(x)$ and [Graph diagram: Graph: $log(x)$ | x: [$0.5, 10.0$], y: [$−0.3, 1.0$]] on the grid.
(b) Graph the function $log(x)+sin(x)$ on the same grid. [Graph diagram: Graph: $sin(x)$ | x: [$−10.0, 10.0$], y: [$ −1.0, 1.0$]]

TOP: bnk | problem **TYPE:** problem

647. Determine the domain and range of the function $y = sin(x) x^2 - x - 6$.

TOP: bnk | short_answer **TYPE:** short_answer

648. Given the functions $f(x) = cos(x)$ and $g(x) = 2^(x) + 5$, determine the value of $g(f(pi))$.

TOP: bnk | short_answer **TYPE:** short_answer

649. (a) Graph the functions $sin(x)*(2^(x^2+1))$ and [Graph diagram: Graph: $sin(x)*(2^(x^2+1))$ | x: [$−4.0, 4.0$], y: [$−99195.6, 99195.6$]].
(b) Use the graphs to graph the function $2^(x^2+1)$. [Graph diagram: Graph: $sin(x)$ | x: [$−10.0, 10.0$], y: [$−1.0, 1.0$]]

TOP: bnk | problem **TYPE:** problem

650. Determine the equation(s) of the vertical asymptote(s) of the function $y = 0.9^(x) 2x 2 - 3$.

TOP: bnk | short_answer **TYPE:** short_answer

651. Given the functions $f(x) = x^2 - 7$ and

$$f(x) = x^2 - 7$$

, what is the value of

$$g(x) = 2 - x^3$$

? Rendered ExamView reference:

$$f(g(2))$$

TOP: bnk | short_answer **TYPE:** short_answer

652. Given the functions $f(x) = x^2 + 3x + 2$ and $g(x) = x + 1$, determine a simplified equation for $k x = f(g(x))$.

TOP: bnk | short_answer **TYPE:** short_answer

653. Given the functions $f(x) = 1 / 1 - x^2$ and $g(x) = sin(x)$, determine the equation for $h(x) = f(g(x))$.

TOP: bnk | short_answer **TYPE:** short_answer

654. Given the functions $f(x) = x + 2$ and $g(x) = x^2$, what values of x result in $f(g(x)) = g(f(x))$?

TOP: bnk | short_answer **TYPE:** short_answer

655. Given the functions $f(x) = sqrt(x)$ and $g(x) = x + 1$, determine all possible values of x for which $f(g(x)) = g(f(x))$.

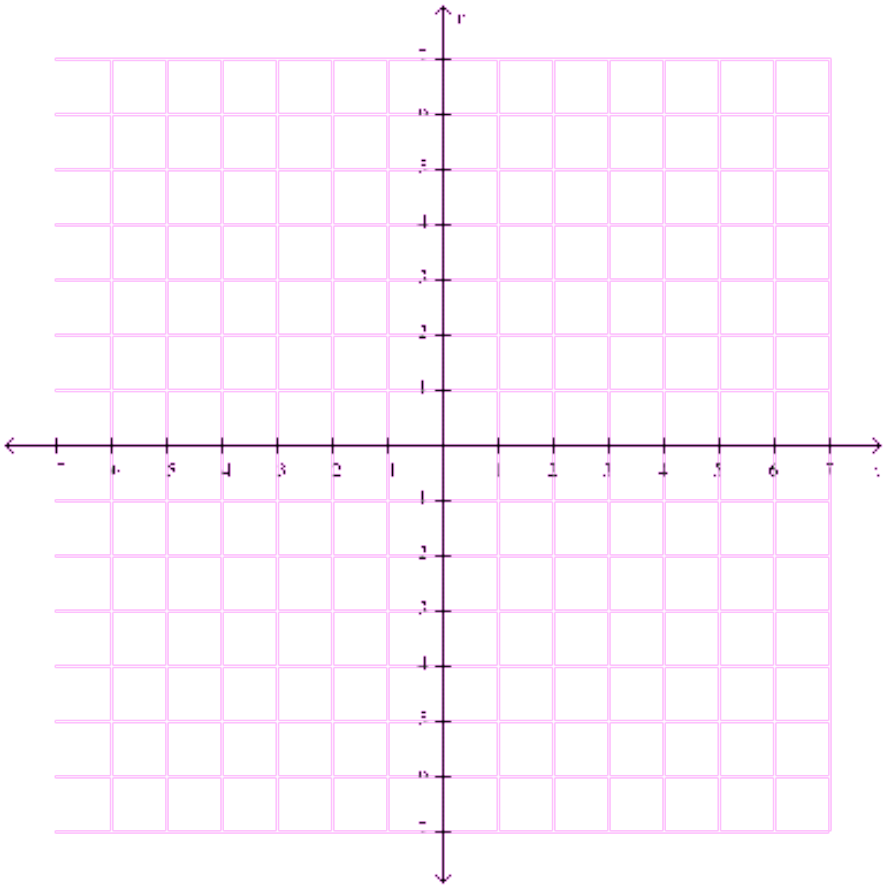
TOP: bnk | short_answer **TYPE:** short_answer

656. Given the functions ($x^2-(3*x)−10)/(x^2-5*x)$ and [Graph diagram: Graph: $(x^2-(3*x)−10)/(x^2-5*x)$ | x: [$−10.0, 10.0$], y: [$−3.0, 5.0$]], graph the function $(x^2-(3*x)−10)/(x^2-5*x)$. Label all intercepts and identify any asymptotes and$/$or points of discontinuity. [Graph diagram: Graph: $(x^2-(3*x)−10)/(x^2-5*x)$ | x: [$−10.0, 10.0$], y: [$−3.0, 5.0$]] Rendered ExamView reference:

$$f(x) = x^2 - 3x - 10$$

$$g(x) = x^2 - 5x$$

$$h(x) = \frac{f(x)}{g(x)}$$

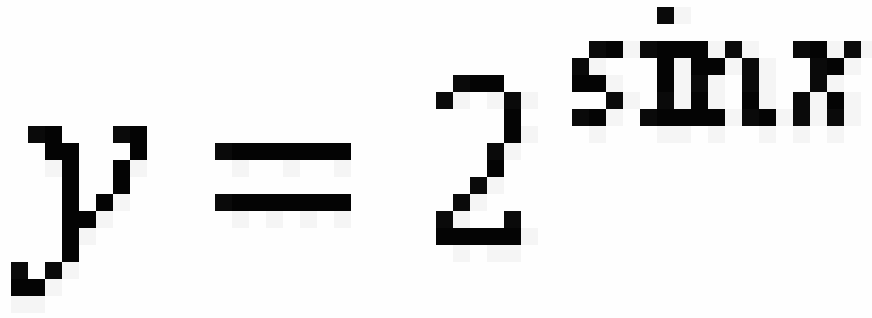


TOP: bnk | short_answer **TYPE:** short_answer

657. What are the domain and range of the function $y = 2^(sin(x))$, where x is in radians?

TOP: bnk | short_answer **TYPE:** short_answer

658. Given that $f (g (x))$ and



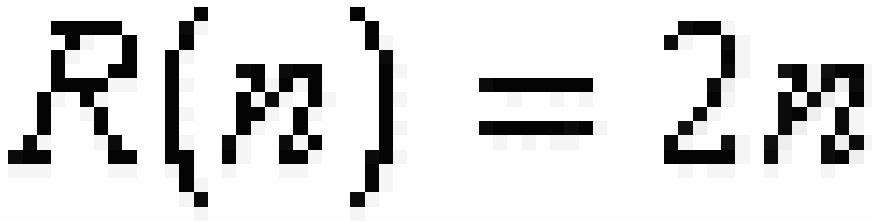
, what are the domain and range of the function $f(g(x))$?

TOP: bnk | short_answer **TYPE:** short_answer

659. Determine the equation for the composite function $y = f(f(x))$ given that $f(x) = 1 x$.

TOP: bnk | short_answer **TYPE:** short_answer

660. The revenue function for a school group selling n bookmarks is given by $R n = 2n$, and the total cost function is given by

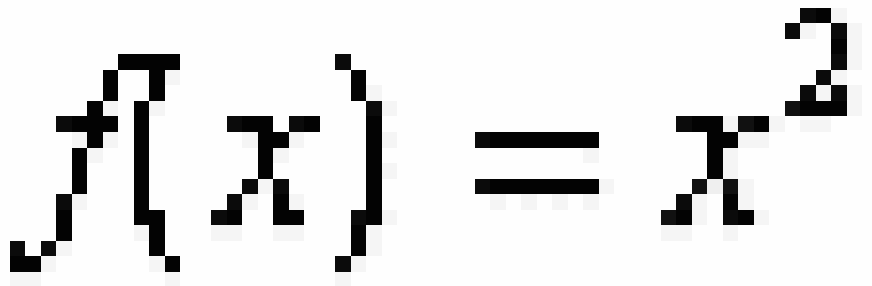


. Determine the number of bookmarks that need to be sold for the school group to break even. Rendered ExamView reference:



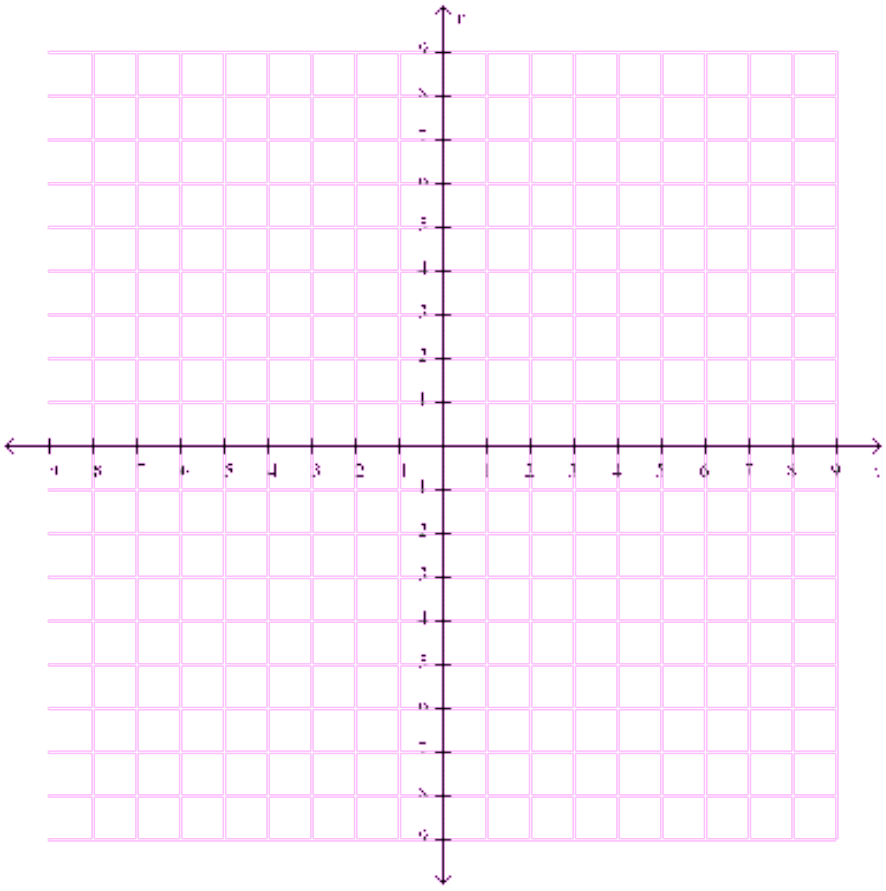
TOP: bnk | short_answer **TYPE:** short_answer

661. Given the functions x^2-3 and [Graph diagram: Graph: x^2-3 | x: [$−10.0, 10.0$], y: [$−3.0, 97.0$]], sketch the graph of the combined function x^2-3. [Graph diagram: Graph: x^2-3 | x: [$−10.0, 10.0$], y: [$−3.0, 97.0$]] Rendered ExamView reference:



$$g(x) = 3$$

$$y = f(x) - g(x)$$



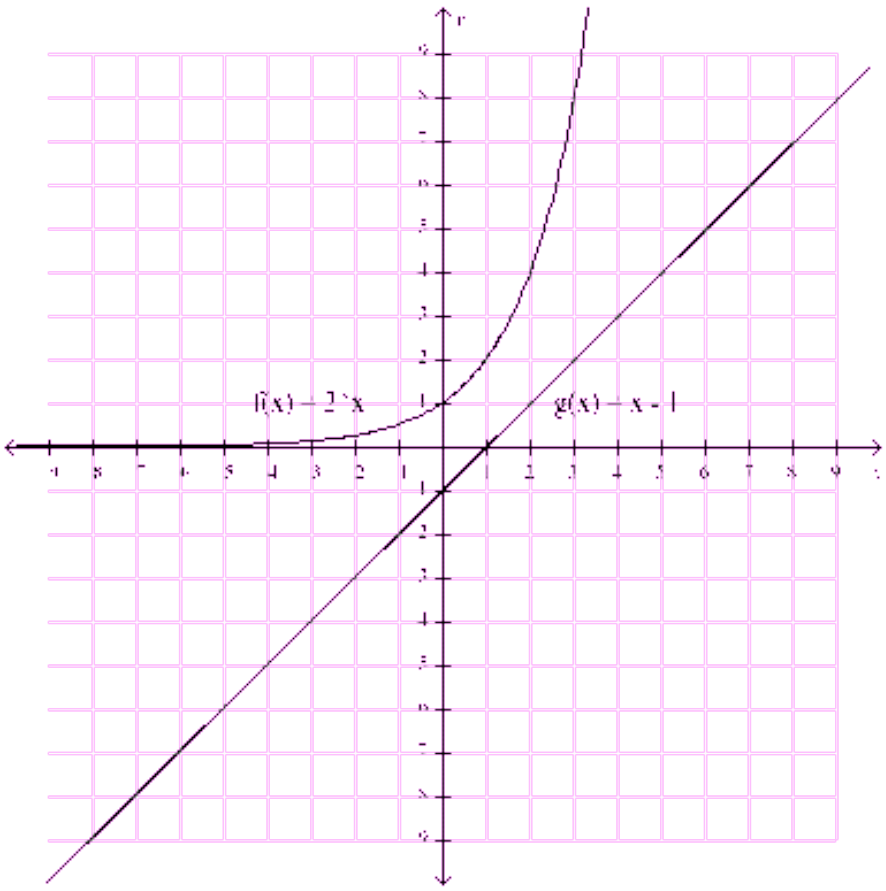
TOP: bnk | short_answer **TYPE:** short_answer

662. Given the graphs of the functions 2^x and [Graph diagram: Graph: $x-1$ | x: [$−10.0, 10.0$], y: [$−11.0, 9.0$]], sketch the graph of the combined function $2^(x-1)$. [Graph diagram: Graph: 2^x | x: [$−10.0, 10.0$], y: [$0.0, 1024.0$]] Rendered ExamView reference:

$$f(x) = 2^x$$

$$g(x) = x - 1$$

$$y = f(g(x))$$



TOP: bnk | short_answer **TYPE:** short_answer

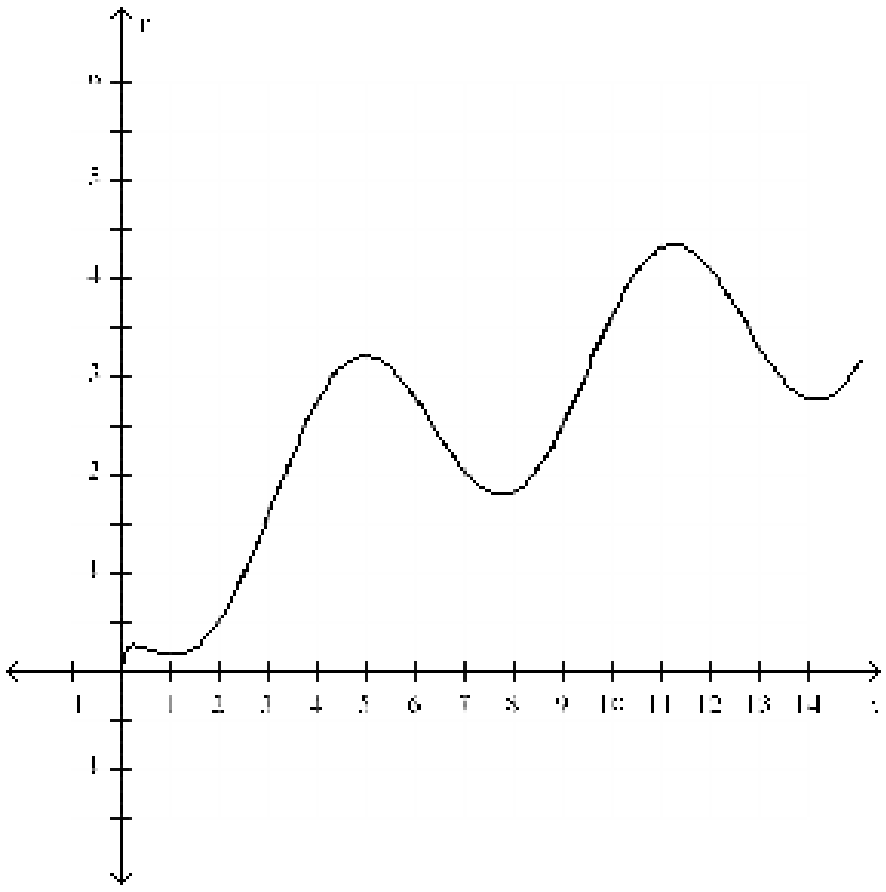
663. The revenue function for a company selling n key chains is given by $R n = 2n$, and the total cost function is given by $C n = 100 + 0.80n$. Determine the simplified equation for $P n = R n - C n = 2n - 100 + 0.80n = 1.2n - 100$, the profit function for the company.

TOP: bnk | short_answer **TYPE:** short_answer

664. Use the graph of the combined function $sqrt(x)-sin(x)$ to solve the inequality [Graph diagram: Graph: $sqrt(x)-sin(x)$ | x: [$0.0, 10.0$], y: [$0.0, 3.7$]]. [Graph diagram: Graph: $sqrt(x)-sin(x)$ | x: [$0.0, 10.0$], y: [$0.0, 3.7$]] Rendered ExamView reference:

$$f(x) = \sqrt{x} - \sin x$$

$$\sin x \geq \sqrt{x}$$

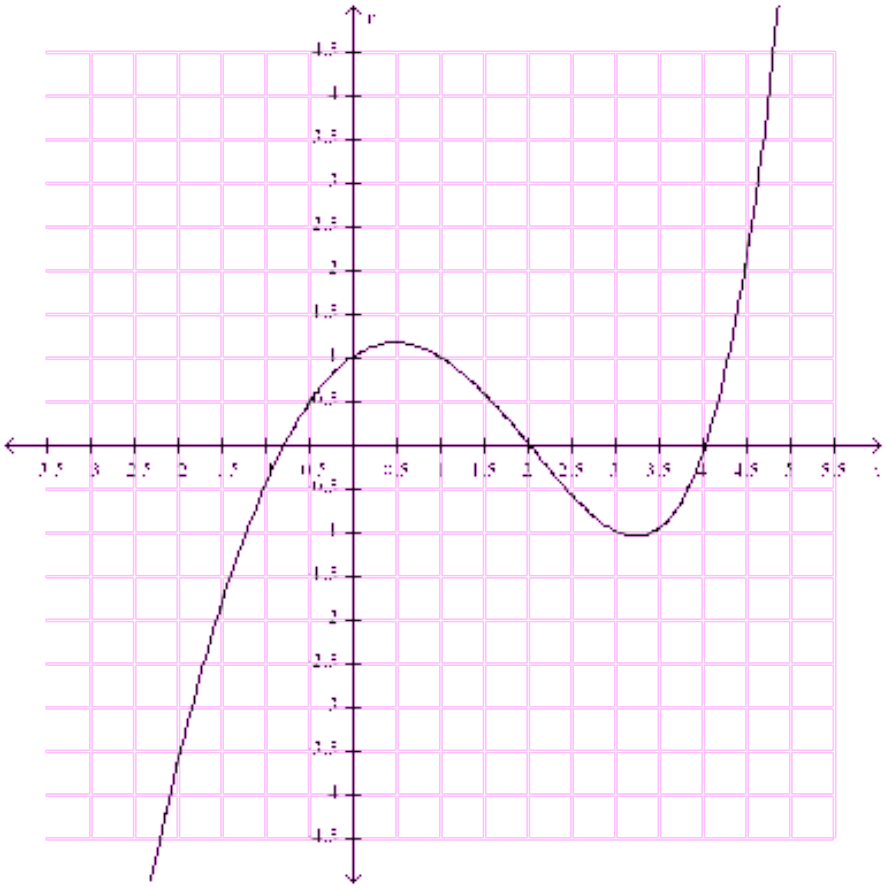


TOP: bnk | short_answer **TYPE:** short_answer

665. Use the graph of the combined function 2^x-x^2 to determine an approximate solution to the inequality [Graph diagram: Graph: 2^x-x^2 | x: [$−10.0, 10.0$], y: [$−100.0, 924.0$]]. [Graph diagram: Graph: 2^x-x^2 | x: [$−10.0, 10.0$], y: [$−100.0, 924.0$]] Rendered ExamView reference:

$$f(x) = 2^x - x^2$$

$$2^x > x^2$$



TOP: bnk | short_answer **TYPE:** short_answer

666. Jenny and Jimmy are a married couple who work at the same store. Jimmy s total weekly salary, in dollars, if he sells x items is given by [embedded object], and Jenny s total weekly salary, in dollars, if she sells x items is given by [DIAGRAM NEEDED].

(a) Assuming that they sell the same number of items in a week, what is the minimum number of items they have to sell so that Jenny s weekly salary is at least $100 more than Jimmy s?

(b) Assuming that they sell the same number of items in a week, what is the minimum number of items they each need to sell to make their combined weekly salary greater than $1000?

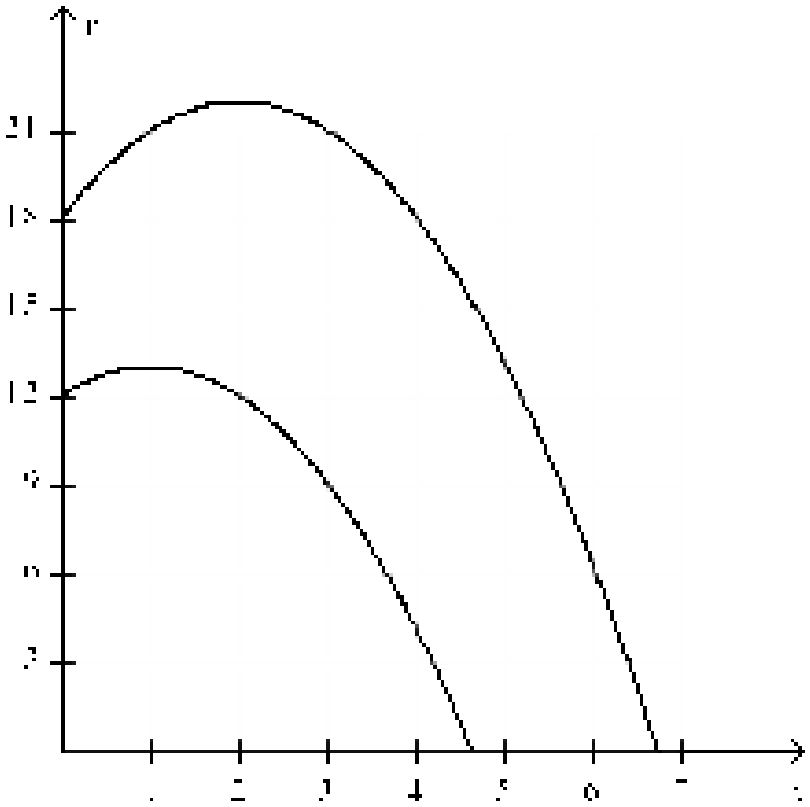
TOP: bnk | problem **TYPE:** problem

667. The revenue function for a company selling n coffee mugs can be modelled by the function $R n = 10n$, and the total cost function can be modelled by the function $C n = 360 + 6n$.

(a) Determine a simplified equation for $P n = R n - C n = 10n - 360 + 6n = 4n - 360$, the profit function for the company.
(b) How many coffee mugs must be sold to make a profit of $1000?

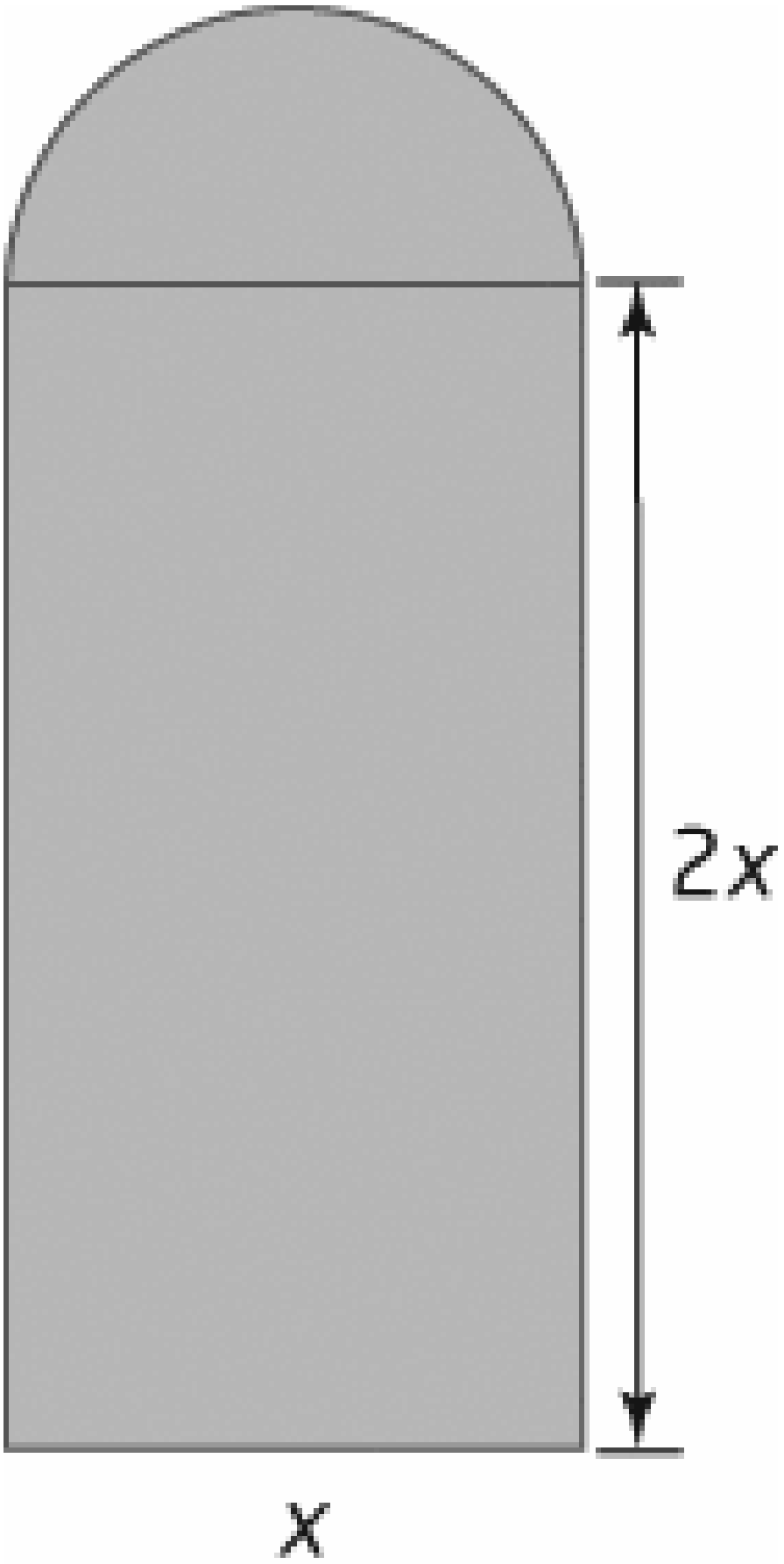
TOP: bnk | problem **TYPE:** problem

668. The heights, h, of two balls, in metres, for a horizontal distance of x metres are shown in the graph. $−1*((x-2)^2)+22$ What was the difference in height of the two balls when the horizontal distance was $0 m$? Rendered ExamView reference:



TOP: bnk | short_answer **TYPE:** short_answer

669. The dimensions of a window are shown. [image omitted: asset-1] a) What function in simplest form represents the area of the entire window? b) If the width, x, of the window is $1.2 m,$ what is the total area of the window, to the nearest tenth of a square metre? Rendered ExamView reference:



TOP: bnk | problem **TYPE:** problem

670. A vehicle depreciates \$20\$% in value in the first year and \$10\$% in each year after the first.

(a) Write a function, $c = 0.8c$, to describe the value of the vehicle after 1 year in terms of the original cost, c.
(b) Write a function, $n c = 0.8c 0.9^(n)$, to describe the value after n years in terms of the original cost, c.
(c) If the original value of the vehicle is $$45 000,$ what is its value after 5 years?

TOP: bnk | problem **TYPE:** problem

671. For the following question(s), assume that x is in radians, if applicable. (d $= -, p = 2, d = -, d = -, d = -, d = -, d = -, d = -, d = -, d = -, d = -)$

TOP: bnk | problem **TYPE:** problem

672. What is the value of 6 !?

(A) 46 656 (B) 21 (C) 720 (D) 36

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

673. The number of different ways that 9 bikes can be locked in a bike rack is

(A) 3 628 800 (B) 40 320 (C) 20 160 (D) 362 880

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

674. Evaluate 15.

(A) 32 432 400 (B) 6435 (C) 163 459 296 000 (D) 259 459 200

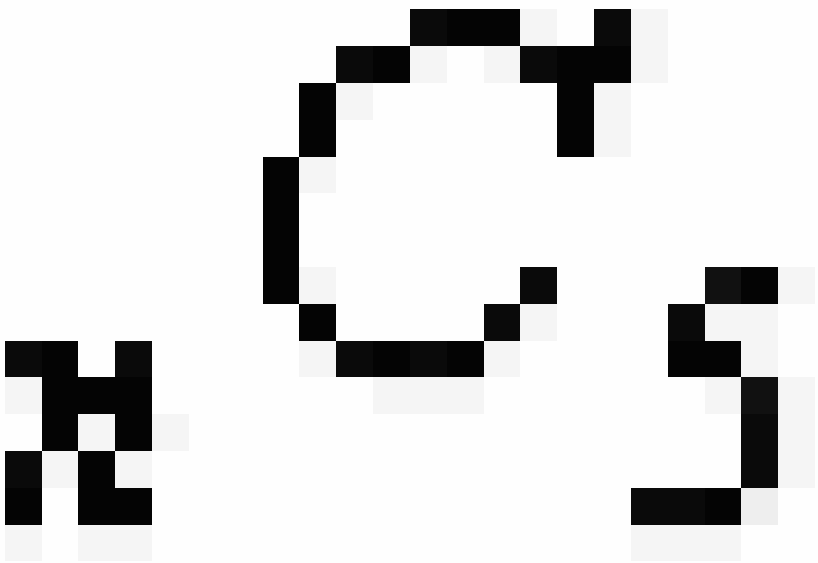
ANS: D **TOP:** bnk | mcq **TYPE:** mcq

675. Solve for the variable: $5 2$

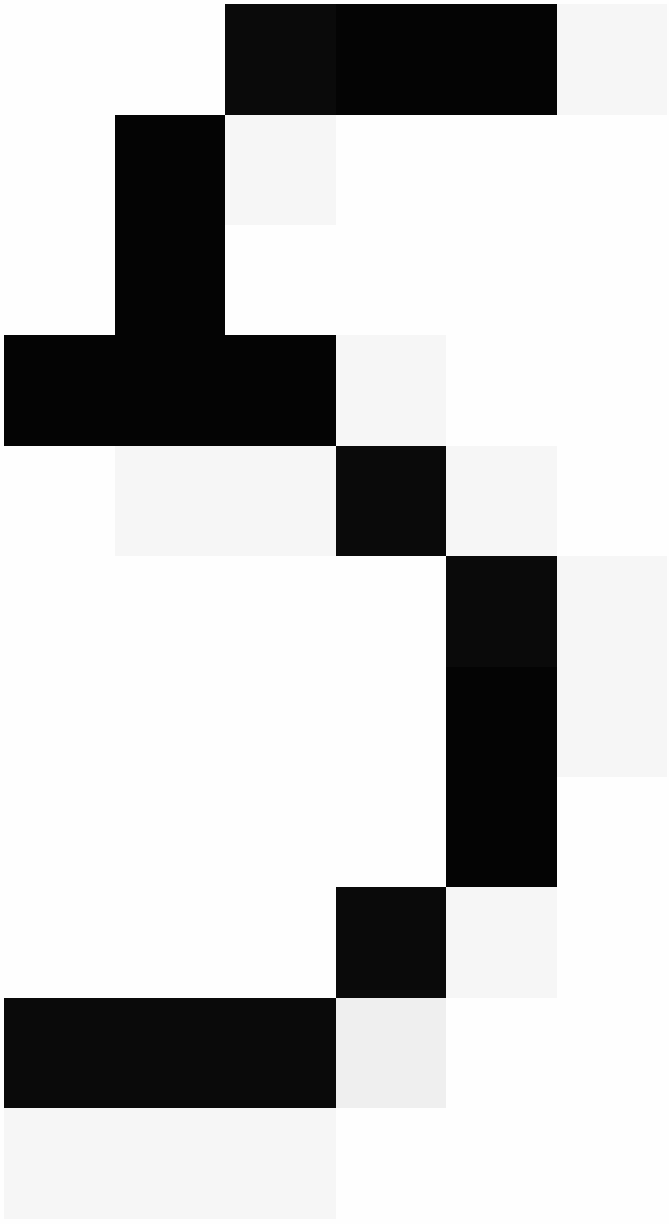
(A) 20 (B) 60 (C) 7 (D) 3

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

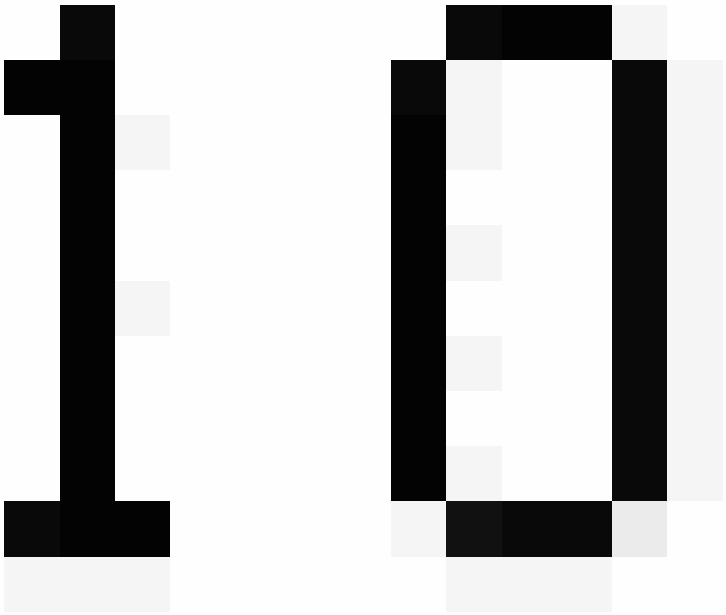
676. Solve for the variable: $X + X$



$=$



Rendered ExamView reference:



(A) 10 (B) 5 (C) 15 (D) 6

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

677. An orchestra has 4 violinists, 4 cellists, and 4 harpists. Assume that the players of each instrument have to sit together, but they can sit in any position in their own group. In how many ways can the conductor seat the members of the orchestra in a line?

(A) 144 (B) 24 (C) 72 (D) 1728

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

678. Evaluate the expression 14.

(A) 726 485 760 (B) 2002 (C) 240 240 (D) 126

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

679. Evaluate the expression 14.

(A) 2184 (B) 794 976 (C) 87 178 291 200 (D) 239 500 800

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

680. Solve for n in the expression 8.

(A) 8 (B) 9 (C) 16

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

681. In how many ways can the coach pick the people to go?

(A) 726 485 760 (B) 630 (C) 2002 (D) 126

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

682. Which expression can be used to calculate the number of ways can she load songs on to the MP3 player so that it is full?

(A) 750 (B) 1500 (C) 800 (D) 1450

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

683. For a mock United Nations, 6 boys and 7 girls are to be chosen. If there are 12 boys and 9 girls to choose from, how many groups are possible?

(A) 846 720 (B) 960 (C) 33 264 (D) 120 708 403 200

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

684. If there are 20 items on the menu, and Jenni orders 7 items and Hari orders 11 items, which expression represents the total number of choices between them?

(A) 11 (B) 20 (C) 9 (D) 13

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

685. Determine the coefficient, a, for the term 5 of the binomial expansion of 7.

(A) 35 (B) 792 (C) 60 (D) 3 991 680

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

686. For which of the following terms is $a = 55$ in the expansion of $(x + y)^11$?

(A) 18 (B) 22 (C) 55 (D) 19958400

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

687. Determine the 5 th term in the expansion of 12.

(A) $−7920$ (B) 202 752 (C) 7920

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

688. Which row do the numbers come from?

(A) 34 (B) 18 (C) 16

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

689. How many terms are there in the expansion of 2?

(A) 6 (B) 13 (C) 11

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

690. Which of the following has 29 terms in its binomial expansion?

(A) 30 (B) $−4$ (C) – 2 (D) 31

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

691. A scout troop is arranged in a circle for an opening ceremony. If there are 9 scouts, in how many unique ways can they stand around the circle?

(A) 5040 (B) 81 (C) 40 320 (D) 362 880

ANS: B **TOP:** bnk | mcq **TYPE:** mcq

692. A scout troop is arranged in a circle for an opening ceremony with its leaders. Assuming there are 7 troop leaders who stand together for the ceremony and 39 scouts in total, including the troop leaders, in how many unique ways can they be arranged around the circle?

(A) 1.326179418146e+039 (B) 2.03978820812e+046 (C) 1.376375309123e+043 (D) 1521

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

693. Suppose that 10 non$-$collinear points are plotted on a plane. Which expression represents the number of triangles that can be formed using these points?

(A) \$10 P 3\$ (B) \$7 C 3\$ (C) \$10 C 3\$ (D) \$10 C 7\$

ANS: C **TOP:** bnk | mcq **TYPE:** mcq

694. The leadership committee at a high school has \$4\$ grade \$10\$ students, \$2\$ grade \$11\$ students, and \$6\$ grade \$12\$ students. This year, \$12\$ grade \$10, 8\$ grade \$11,\$ and \$10\$ grade \$12\$ students applied for the committee. How many ways are there to select the committee?

(A) 2 910 600 (B) 733 (C) 100 590 336 000 (D) 163 136

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

695. If there are \$36\$ handshakes in total, how many players are at the tournament?

(A) 18 (B) 8 (C) 10 (D) 9

ANS: D **TOP:** bnk | mcq **TYPE:** mcq

696. The number of ways he could select which items to buy is

(A) 19 19 19 (B) 19 19 0 (C) 19 1 19 (D) 19 19 19

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

697. An abbreviation for products of successive positive integers, The arrangements of objects in a definite order, The selection of objects in which order is not important, The pattern from which the binomial theorem can be derived The selection of objects in which order is not important \$0 X - 2\$ is called \$a(n)\$ _____.

TOP: bnk | short_answer **TYPE:** short_answer

698. \$14\$ can be written in factorial notation as _____.

ANS: 14 **TOP:** bnk | problem **TYPE:** problem

699. \$20\$ can be written in factorial notation as _____.

ANS: 20 **TOP:** bnk | problem **TYPE:** problem

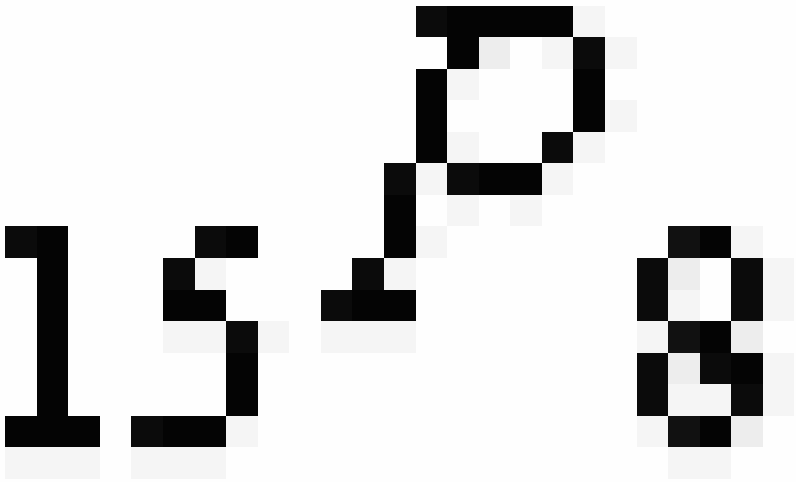
700. The expansion of the binomial \$10\$ includes the coefficients in row _____ of Pascal s triangle.

ANS: 10 **TOP:** bnk | problem **TYPE:** problem

701. The sum of the numbers in the \$13\$ th row in Pascal s triangle is _____.

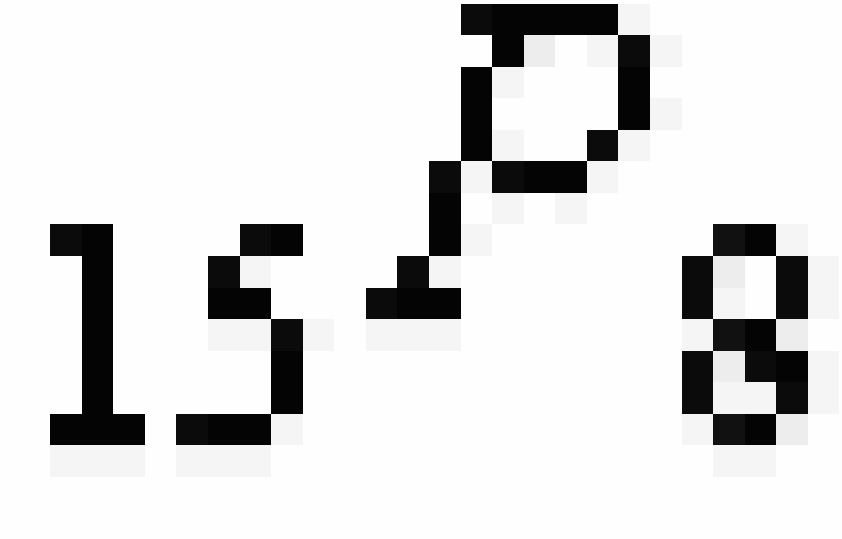
ANS: 4096 **TOP:** bnk | short_answer **TYPE:** short_answer

702.



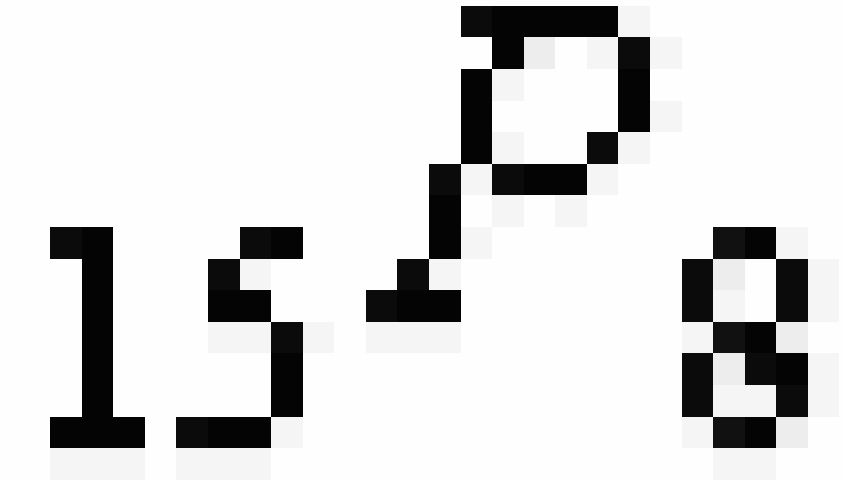
TOP: bnk | short_answer **TYPE:** short_answer

703.



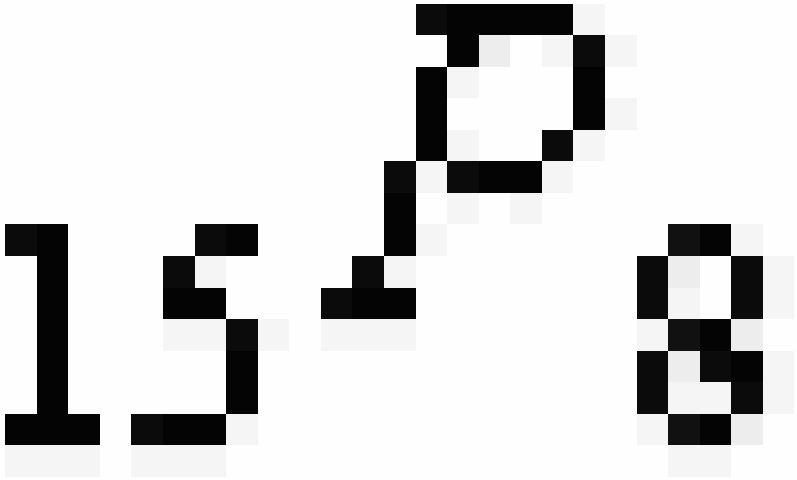
TOP: bnk | short_answer **TYPE:** short_answer

704.



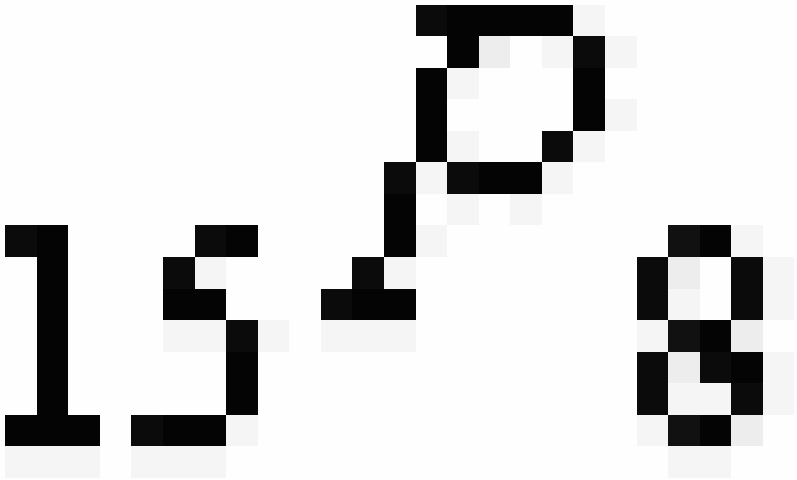
TOP: bnk | short_answer **TYPE:** short_answer

705.



TOP: bnk | short_answer **TYPE:** short_answer

706.



TOP: bnk | short_answer **TYPE:** short_answer

707. Josie goes to the local submarine sandwich shop. She can choose from 21 types of sandwiches. There are 3 types of cheese, 8 types of vegetables, and 7 types of sauce. Assuming that Josie only chooses one type of cheese, one vegetable, and one sauce for her sandwich, how many choices does she have?

TOP: bnk | short_answer **TYPE:** short_answer

708. Sun Eui and Pattie are arguing about the number of ways the letters in PERMUTATION and COMBINATION can be rearranged. Sun Eui believes that there are more ways to rearrange PERMUTATION, while Pattie thinks that COMBINATION has more unique arrangements. Which person is correct?

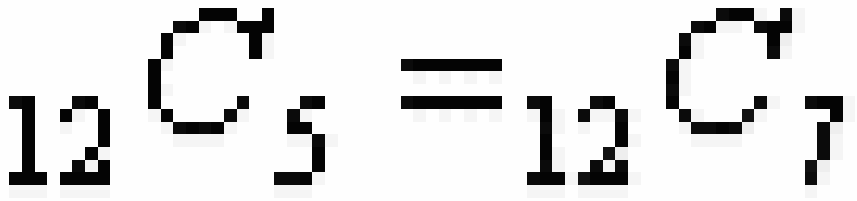
TOP: bnk | short_answer **TYPE:** short_answer

709. The number of different routes can be calculated using the following permutation: 924 Joe can take one of 924 different routes to travel from home to school.
[image omitted: asset-1]

(A) 924 (B) 924 (C) 924 (D) 924

ANS: A **TOP:** bnk | mcq **TYPE:** mcq

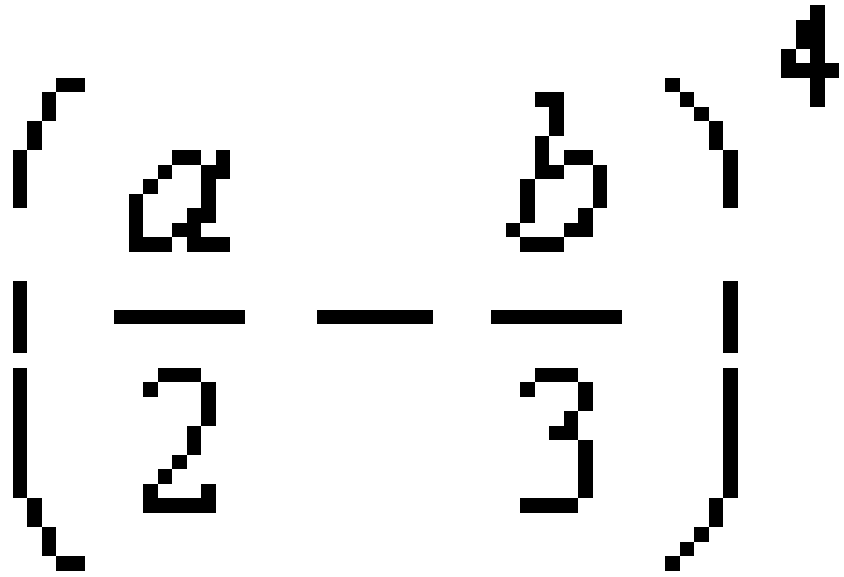
710. Show that



.

TOP: bnk | short_answer **TYPE:** short_answer

711. Use the binomial theorem to expand



.

TOP: bnk | short_answer **TYPE:** short_answer

712. There are 5 men and 8 women available for a $5-$person committee. If the committee must have at least one man and one woman, how many possible committees are there?

TOP: bnk | problem **TYPE:** problem

713. Simplify the expression $n(4n 2 + 6n + 2) = 8n 3 + 12n 2 + 4n$.

TOP: bnk | short_answer **TYPE:** short_answer

714. A neon sign with the words Espresso Coffee on it has 5 letters burnt out. In how many ways can you select 3 good letters and 2 burnt$-$out letters?

TOP: bnk | short_answer **TYPE:** short_answer

715. A math teacher is preparing a quiz for all of the students in grade 12. She wants to give each student the same questions, but have each student s questions appear in a different order. If there are 128 students in the grade 12 class, what is the least number of questions the quiz must contain so everyone gets a test with the questions in a different order.

TOP: bnk | short_answer **TYPE:** short_answer

716. Solve for n: $n(10 P^2) = 10 C$.

TOP: bnk | short_answer **TYPE:** short_answer

717. A bingo card has 5 columns. The tops of the columns are labelled with the letters B, I, N, G, and O. Each column contains exactly five numbers, except for the N column, which has a free space in the centre. No numbers repeat on the card. The B column contains numbers from 1 to $15,$ the I column from 16 to $30,$ the N column from 31 to $45, $ the G column from 46 to $60,$ and the O column from 61 to 75.

(a) How many unique bingo cards can be printed?
(b) If every Canadian were able to count one card per second, how many years would it take to count all of the cards? Assume the Canadian population is about $34 000 000$.
(c) Suppose the cards are printed on card stock 0.3 mm thick. How many kilometres high would a stack of all of the possible cards be?

TOP: bnk | problem **TYPE:** problem

718. A standard deck of playing cards contains 4 suits (spades, clubs, diamonds, and hearts), each with 13 cards.

(a) How many different $5-$card hands are possible?
(b) How many different $5-$card hands with only black cards are possible?
(c) How many different $5-$card hands are possible containing at least 3 black cards?

TOP: bnk | problem **TYPE:** problem

719. Expand and simplify:

$$(x+y)^6 - (x-y)^6$$

.

TOP: bnk | short_answer **TYPE:** short_answer

720. The number of different permutations using all of the $1-$digit numbers of a set is given by [DIAGRAM NEEDED].

(a) What is the smallest number that can be created that meets these conditions? Explain your reasoning.
(b) What is the difference between the largest number and the smallest number? Explain your reasoning.

TOP: bnk | problem **TYPE:** problem

721. Arrange the following provincial capitals from the least to the greatest number of arrangements of letters: Fredericton, Charlottetown, Edmonton, Victoria, and Winnipeg.

TOP: bnk | short_answer **TYPE:** short_answer

722. A to T on each tumbler.

(a) What is the probability that Matthew chooses WATER as the combination?
(b) How many possible combinations are there if the letters can be repeated?
(c) If it takes $12 s$ to test one combination, how long will it take to check all of the combinations in part b) if Matthew forgets his combination? Express your answer in years and months.

TOP: bnk | problem **TYPE:** problem

723. The chorus of a play has 17 females and 13 males. The director wishes to meet with 6 of them to discuss the upcoming production.

(a) How many selections are possible?
(b) How many selections are possible if the group consists of three females and three males?
(c) One of the male students is named Ajay. How many six$-$member selections consisting of Ajay, two other males, and three females are possible?

TOP: bnk | problem **TYPE:** problem

724. A group of police officers are planning a bike ride in Alberta to raise money for cancer research. They are planning to go to three areas of the province. They want to minimize their travel between areas with possible visits to each of the following communities: Edmonton Area Calgary Area Lethbridge Area St. Albert Spruce Grove Stony Plain Sherwood Park a) If the group goes to 7 communities in total, how many different schedules can be created if there are no restrictions on which communities they visit in each area?

(a) If the group must visit all of the communities in the Edmonton Area, 3 in the Calgary area, and 2 in the Lethbridge area, how many different schedules can be created?
(b) If the group decides that Sherwood Park and Coaldale should not be on the schedule due to distance, how many different schedules can be created if the group visits all of the communities in the Edmonton area, 2 in the Calgary area, and 2 in the Lethbridge area?

TOP: bnk | problem **TYPE:** problem

725. To win the grand prize in lottery A, a player must select all six of the winning numbers drawn from the numbers 1 to 49. To win in lottery B, a player must select all seven of the winning numbers drawn from 1 to 49. Bernadette argues that the chances of randomly selecting the winning number for lottery A are seven times as good as winning for lottery B. Create an argument to agree or disagree with this statement.

TOP: bnk | problem **TYPE:** problem

726. Tanya goes to a fast food stand at the beach. There are 4 types of burgers, 3 sizes of French fries, and either orange pop or root beer to drink.

(a) Create a tree diagram to show the possible choices of lunch if one of each item can be selected.
(b) In how many ways can Tanya buy 2 burgers, 2 fries, and 2 drinks for her and her friend?
(c) If Tanya does not like root beer and her friend does not like orange pop, how many possible choices are there?

[image omitted: asset-2]

TOP: bnk | problem **TYPE:** problem

727. On a Saturday, Charlie has to go to the library to study for a few hours, and then to the school to play a volleyball game. [image omitted: asset-3] a) How many routes are there for Charlie to go from home to the library if she only moves south and east?

(a) How many routes are there for her to go from home to school moving only south and east?
(b) Assuming Charlie moves south and east going from home to school and north and west going from school to home, how many routes are there for her to complete the round trip?
(c) If Charlie could walk each route from home to school in $40 min,$ how long would it take her and her 22 classmates to walk all of the routes? Consider only the route from home to school, not the round trip.

TOP: bnk | problem **TYPE:** problem

728. The principal asked Ms. Jackson to prepare a display for the upcoming parent–teacher evening.

(a) In how many ways can 20 of the projects be chosen?
(b) The principal asks to set the 4 regional winners out of the 20 selected near the entrance of the room. In how many ways can the projects be displayed? Leave your answer in factorial form.

TOP: bnk | problem **TYPE:** problem

729. How many different options are possible if

(a) students must complete three of the six questions in the first part and two of the three questions in the second part?
(b) students must answer three of the questions in the first part and at least two of the three questions in the second part?

TOP: bnk | problem **TYPE:** problem

730. Seven students (Catherine, Jim, Jerome, Lucia, Lisa, Melinda, and Zyoji) are entered in a debate contest.

(a) In how many ways can they give their arguments if the following conditions must be met?
(b) Lucia must go second.

(i) Lisa must go first and Zyoji must go second to last.

(ii) All of the boys go first followed by the girls.
(iii) All of the girls go first followed by the boys.

(a) Females and males alternate with a female going first.
(b) Are there any generalizations you can make from the solutions to part a)i) to v)?

TOP: bnk | problem **TYPE:** problem

731. Simplify. a) $(500 C 400)/(500 C 100)$ b) $(500 P 400)/(500 P 100) dot 1/(400 P 300)$

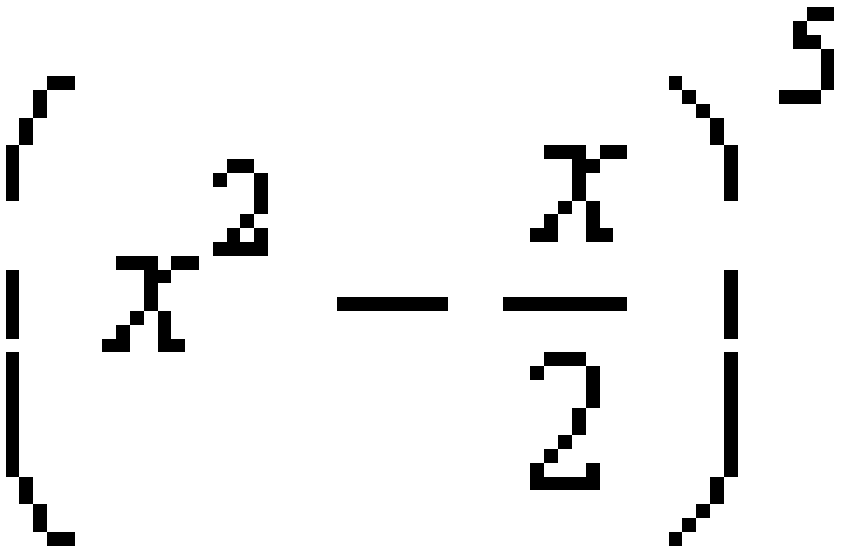
TOP: bnk | problem **TYPE:** problem

732. The lottery will be called Lotto $5-45$.To win a prize, a player must pick numbers from 1 to 45. The number of correct numbers chosen determines the amount won. Correct Numbers Probability of Winning $1000 $$10$ Free Ticket a) Complete the table.

(a) To win a lottery jackpot in Lotto Fab, a player must chose six numbers from 1 to 49 without repetition. Are the chances of winning the jackpot in Lotto $5-45$ better than in Lotto Fab?
(b) In Lotto Fab, a player also has to choose two correct numbers to win a free ticket. Which lottery has the better chance of winning a free ticket?

TOP: bnk | problem **TYPE:** problem

733. Expand using the binomial theorem:



.

TOP: bnk | short_answer **TYPE:** short_answer

734. The first term is of the form $- -,$ or anxn. Since the first term of the expanded binomial is $16x4, a = 2$ and n $= 4$. The last term is in the form nCn(by)n, or bnyn. Since the last term of the expanded binomial is $81y4,$ and the signs alternate between $+$ and, b $= 3$. Thus, $-$.

TOP: bnk | problem **TYPE:** problem

735. Which of the following binomial expansions has the greater coefficient: the 12th term in the expansion of

$$(2x-2)^{13}$$

or the 8th term in the expansion of

$$(3x-2)^{10}$$

?

TOP: bnk | short_answer **TYPE:** short_answer

736. (a) If all of the letters in the word ROCKING are used, how many arrangements can be made with all of the vowels together?
(b) Which word has the greater number of different arrangements: CATCHPHRASE or MISSISSIPPI?
(c) Is it possible to create a word whose letters can be arranged with exactly 20 possible different arrangements? If so, explain how.

TOP: bnk | problem **TYPE:** problem